

# **Sequential Parameter Optimization Cookbook**

**A guide for scikit-learn and PyTorch**

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# Preface

This is a Quarto book.

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**Part I.**

**Optimization**





# 1. Aircraft Wing Weight Example

## i Note

- This section is based on chapter 1.3 “A ten-variable weight function” in Forrester, Sóbester, and Keane (2008).
- The following Python packages are imported:

```
import math
import matplotlib.pyplot as plt
import numpy as np
from mpl_toolkits.axes_grid1 import ImageGrid
```

## 1.1. AWWE Equation

- Example from Forrester, Sóbester, and Keane (2008)
- Understand the **weight** of an unpainted light aircraft wing as a function of nine design and operational parameters:

$$W = 0.036 S_W^{0.758} \times W_{fw}^{0.0035} \left( \frac{A}{\cos^2 \Lambda} \right)^{0.6} \times q^{0.006} \times \lambda^{0.04} \\ \times \left( \frac{100 R_{tc}}{\cos \Lambda} \right)^{-0.3} \times (N_z W_{dg})^{0.49}$$

## 1.2. AWWE Parameters and Equations (Part 1)

Table 1.1.: Aircraft Wing Weight Parameters

| Symbol   | Parameter                   | Baseline | Minimum | Maximum |
|----------|-----------------------------|----------|---------|---------|
| $S_W$    | Wing area ( $ft^2$ )        | 174      | 150     | 200     |
| $W_{fw}$ | Weight of fuel in wing (lb) | 252      | 220     | 300     |

### 1. Aircraft Wing Weight Example

| Symbol    | Parameter                                | Baseline | Minimum | Maximum |
|-----------|------------------------------------------|----------|---------|---------|
| $A$       | Aspect ratio                             | 7.52     | 6       | 10      |
| $\Lambda$ | Quarter-chord sweep (deg)                | 0        | -10     | 10      |
| $q$       | Dynamic pressure at cruise ( $lb/ft^2$ ) | 34       | 16      | 45      |
| $\lambda$ | Taper ratio                              | 0.672    | 0.5     | 1       |
| $R_{tc}$  | Aerofoil thickness to chord ratio        | 0.12     | 0.08    | 0.18    |
| $N_z$     | Ultimate load factor                     | 3.8      | 2.5     | 6       |
| $W_{dg}$  | Flight design gross weight (lb)          | 2000     | 1700    | 2500    |
| $W_p$     | paint weight ( $lb/ft^2$ )               | 0.064    | 0.025   | 0.08    |

The study begins with a baseline Cessna C172 Skyhawk Aircraft as its reference point. It aims to investigate the impact of wing area and fuel weight on the overall weight of the aircraft. Two crucial parameters in this analysis are the aspect ratio ( $A$ ), defined as the ratio of the wing's length to the average chord (thickness of the airfoil), and the taper ratio ( $\lambda$ ), which represents the ratio of the maximum to the minimum thickness of the airfoil or the maximum to minimum chord.

It's important to note that the equation used in this context is not a computer simulation but will be treated as one for the purpose of illustration. This approach involves employing a true mathematical equation, even if it's considered unknown, as a useful tool for generating realistic settings to test the methodology. The functional form of this equation was derived by "calibrating" known physical relationships to curves obtained from existing aircraft data, as referenced in Raymer (2006). Essentially, it acts as a surrogate for actual measurements of aircraft weight.

Examining the mathematical properties of the AWWE (Aircraft Weight With Wing Area and Fuel Weight Equation), it is evident that the response is highly nonlinear concerning its inputs. While it's common to apply the logarithm to simplify equations with complex exponents, even when modeling the logarithm, which transforms powers into slope coefficients and products into sums, the response remains nonlinear due to the presence of trigonometric terms. Given the combination of nonlinearity and high input dimension, simple linear and quadratic response surface approximations are likely to be inadequate for this analysis.

## 1.3. Goals: Understanding and Optimization

The primary goals of this study revolve around understanding and optimization:

1. **Understanding:** One of the straightforward objectives is to gain a deep understanding of the input-output relationships in this context. Given the global perspective implied by this setting, it becomes evident that a more sophisticated

### 1.3. Goals: Understanding and Optimization

model is almost necessary. At this stage, let's focus on this specific scenario to establish a clear understanding.

2. **Optimization:** Another application of this analysis could be optimization. There may be an interest in minimizing the weight of the aircraft, but it's likely that there will be constraints in place. For example, the presence of wings with a nonzero area is essential for the aircraft to be capable of flying. In situations involving (constrained) optimization, a global perspective and, consequently, the use of flexible modeling are vital.

The provided Python code serves as a genuine computer implementation that “solves” a mathematical model. It accepts arguments encoded in the unit cube, with defaults used to represent baseline settings, as detailed in the table labeled as Table 1.1. To map values from the interval  $[a, b]$  to the interval  $[0, 1]$ , the following formula can be employed:

$$y = f(x) = \frac{x - a}{b - a}. \quad (1.1)$$

To reverse this mapping and obtain the original values, the formula

$$g(y) = a + (b - a)y \quad (1.2)$$

can be used. The function `wingwt()` expects inputs from the unit cube, which are then transformed back to their original scales using Equation 1.2. The function is defined as follows:

```
def wingwt(Sw=0.48, Wfw=0.4, A=0.38, L=0.5, q=0.62, l=0.344, Rtc=0.4, Nz=0.37, Wdg=0.38):
    # put coded inputs back on natural scale
    Sw = Sw * (200 - 150) + 150
    Wfw = Wfw * (300 - 220) + 220
    A = A * (10 - 6) + 6
    L = (L * (10 - (-10)) - 10) * np.pi/180
    q = q * (45 - 16) + 16
    l = l * (1 - 0.5) + 0.5
    Rtc = Rtc * (0.18 - 0.08) + 0.08
    Nz = Nz * (6 - 2.5) + 2.5
    Wdg = Wdg*(2500 - 1700) + 1700
    # calculation on natural scale
    W = 0.036 * Sw**0.758 * Wfw**0.0035 * (A/np.cos(L)**2)**0.6 * q**0.006
    W = W * l**0.04 * (100*Rtc/np.cos(L))**(-0.3) * (Nz*Wdg)**(0.49)
    return(W)
```

## 1.4. Properties of the Python “Solver”

The compute time required by the “wingwt” solver is extremely short and can be considered trivial in terms of computational resources. The approximation error is exceptionally small, effectively approaching machine precision, which indicates the high accuracy of the solver’s results.

To simulate time-consuming evaluations, a deliberate delay is introduced by incorporating a `sleep(3600)` command, which effectively synthesizes a one-hour execution time for a particular evaluation.

Moving on to the AWWE visualization, plotting in two dimensions is considerably simpler than dealing with nine dimensions. To aid in creating visual representations, the code provided below establishes a grid within the unit square to facilitate the generation of sliced visuals. This involves generating a “meshgrid” as outlined in the code.

```
x = np.linspace(0, 1, 3)
y = np.linspace(0, 1, 3)
X, Y = np.meshgrid(x, y)
zp = zip(np.ravel(X), np.ravel(Y))
list(zp)
```

```
[(np.float64(0.0), np.float64(0.0)),
 (np.float64(0.5), np.float64(0.0)),
 (np.float64(1.0), np.float64(0.0)),
 (np.float64(0.0), np.float64(0.5)),
 (np.float64(0.5), np.float64(0.5)),
 (np.float64(1.0), np.float64(0.5)),
 (np.float64(0.0), np.float64(1.0)),
 (np.float64(0.5), np.float64(1.0)),
 (np.float64(1.0), np.float64(1.0))]
```

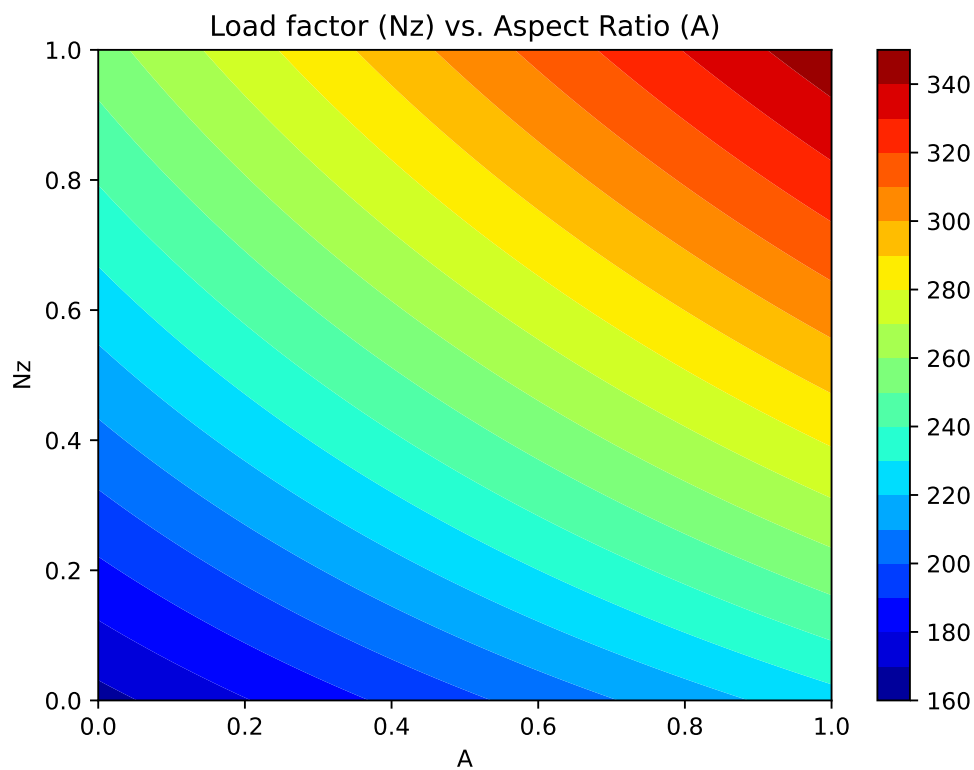
The coding used to transform inputs from natural units is largely a matter of taste, so long as it’s easy to undo for reporting back on original scales

```
%matplotlib inline
# plt.style.use('seaborn-white')
x = np.linspace(0, 1, 100)
y = np.linspace(0, 1, 100)
X, Y = np.meshgrid(x, y)
```

## 1.5. Plot 1: Load Factor ( $N_z$ ) and Aspect Ratio ( $A$ )

We will vary  $N_z$  and  $A$ , with other inputs fixed at their baseline values.

```
z = wingwt(A = X, Nz = Y)
fig = plt.figure(figsize=(7., 5.))
plt.contourf(X, Y, z, 20, cmap='jet')
plt.xlabel("A")
plt.ylabel("Nz")
plt.title("Load factor (Nz) vs. Aspect Ratio (A)")
plt.colorbar()
plt.show()
```



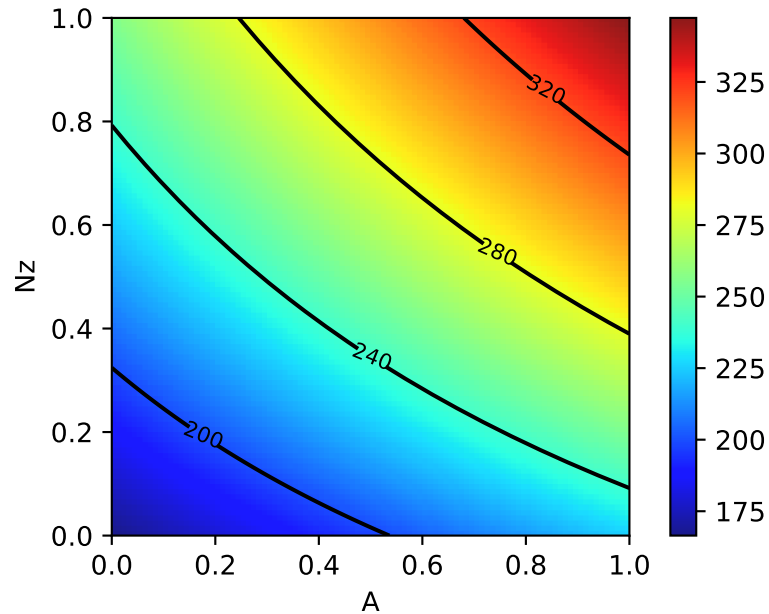
Contour plots can be refined, e.g., by adding explicit contour lines as shown in the following figure.

```
contours = plt.contour(X, Y, z, 4, colors='black')
plt.clabel(contours, inline=True, fontsize=8)
```

### 1. Aircraft Wing Weight Example

```
plt.xlabel("A")
plt.ylabel("Nz")

plt.imshow(z, extent=[0, 1, 0, 1], origin='lower',
           cmap='jet', alpha=0.9)
plt.colorbar()
```



The interpretation of the AWWE plot can be summarized as follows:

- The figure displays the weight response as a function of two variables,  $N_z$  and  $A$ , using an image-contour plot.
- The slight curvature observed in the contours suggests an interaction between these two variables.
- Notably, the range of outputs depicted in the figure, spanning from approximately 160 to 320, nearly encompasses the entire range of outputs observed from various input settings within the full 9-dimensional input space.
- The plot indicates that aircraft wings tend to be heavier when the aspect ratios ( $A$ ) are high.
- This observation aligns with the idea that wings are designed to withstand and accommodate high gravitational forces ( $g$ -forces, large  $N_z$ ), and there may be a compounding effect where larger values of  $N_z$  contribute to increased wing weight.

### 1.6. Plot 2: Taper Ratio and Fuel Weight

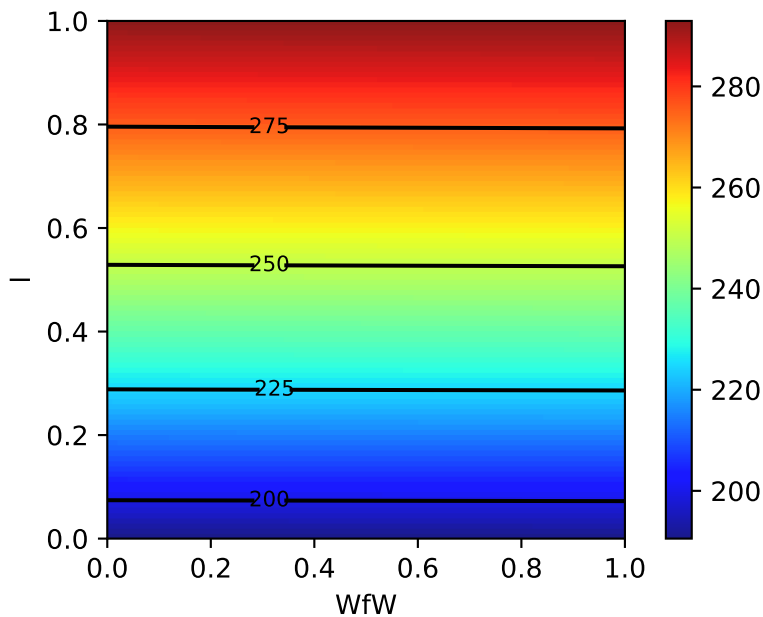
- It's plausible that this phenomenon is related to the design considerations of fighter jets, which cannot have the efficient and lightweight glider-like wings typically found in other types of aircraft.

## 1.6. Plot 2: Taper Ratio and Fuel Weight

- The same experiment for two other inputs, e.g., taper ratio  $\lambda$  and fuel weight  $W_{fw}$

```
z = wingwt(Wfw = X, Nz = Y)
contours = plt.contour(X, Y, z, 4, colors='black')
plt.clabel(contours, inline=True, fontsize=8)
plt.xlabel("WfW")
plt.ylabel("l")

plt.imshow(z, extent=[0, 1, 0, 1], origin='lower',
           cmap='jet', alpha=0.9)
plt.colorbar();
```



- Interpretation of Taper Ratio ( $l$ ) and Fuel Weight ( $W_{fw}$ )
  - Apparently, neither input has much effect on wing weight:

### 1. Aircraft Wing Weight Example

- \* with  $\lambda$  having a marginally greater effect, covering less than 4 percent of the span of weights observed in the  $A \times N_z$  plane
- There's no interaction evident in  $\lambda \times W_{fw}$

## 1.7. The Big Picture: Combining all Variables

```
p1 = ["Sw", "Wfw", "A", "L", "q", "l", "Rtc", "Nz", "Wdg"]
```

```
Z = []
Zlab = []
l = len(p1)
# lc = math.comb(l,2)
for i in range(l):
    for j in range(i+1, l):
        # for j in range(l):
            # print(p1[i], p1[j])
            d = {p1[i]: X, p1[j]: Y}
            Z.append(wingwt(**d))
            Zlab.append([p1[i], p1[j]])
```

Now we can generate all 36 combinations, e.g., our first example is combination  $p = 19$ .

```
p = 19
Zlab[p]
```

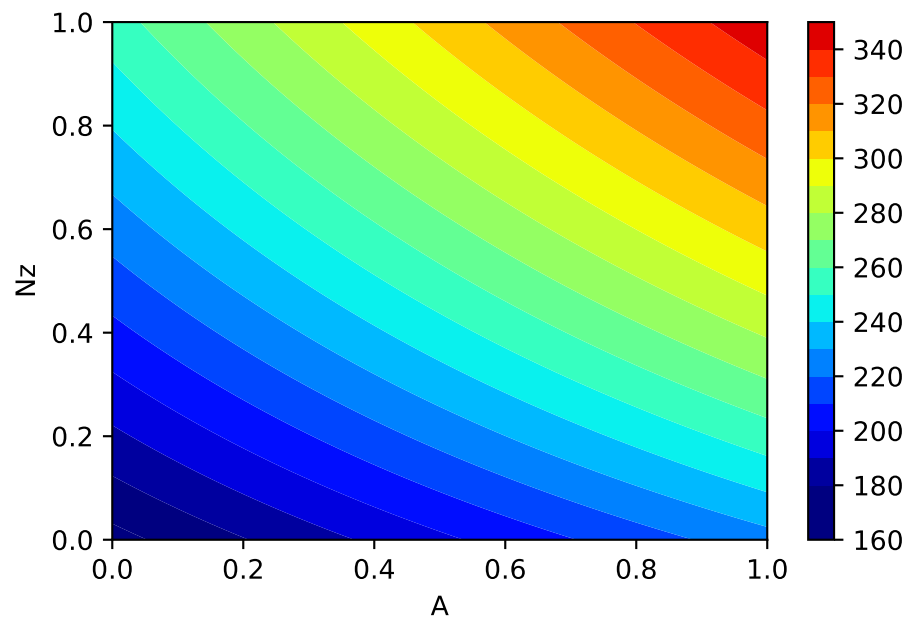
```
['A', 'Nz']
```

To help interpret outputs from experiments such as this one—to level the playing field when comparing outputs from other pairs of inputs—code below sets up a color palette that can be re-used from one experiment to the next. We use the arguments `vmin=180` and `vmax=360` to implement comparability

```
plt.contourf(X, Y, Z[p], 20, cmap='jet', vmin=180, vmax=360)
plt.xlabel(Zlab[p][0])
plt.ylabel(Zlab[p][1])
plt.colorbar()
```



### 1.7. The Big Picture: Combining all Variables



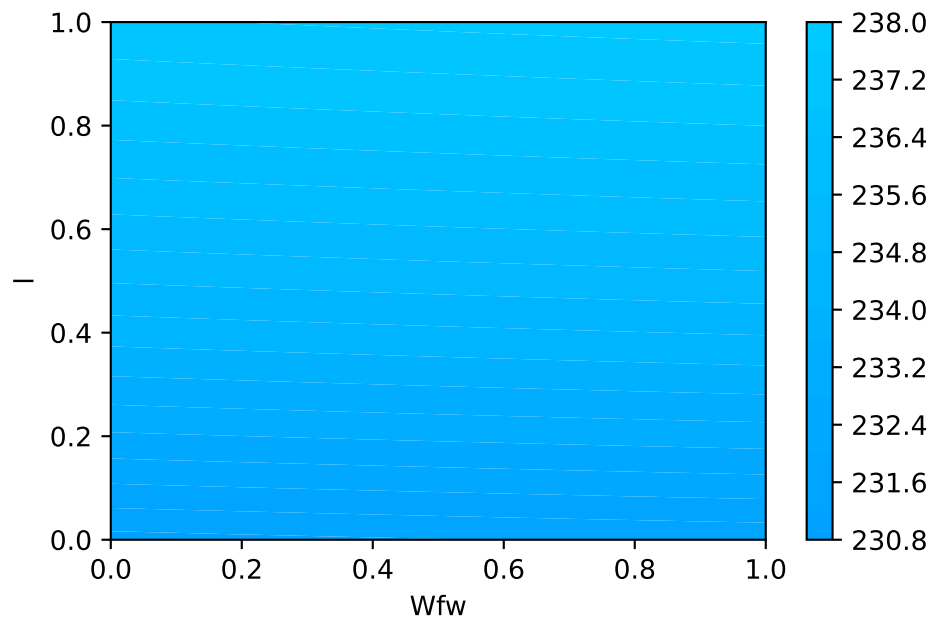
- Let's plot the second example, taper ratio  $\lambda$  and fuel weight  $W_{fw}$
- This is combination 11:

```
p = 11
Zlab[p]
```

```
['Wfw', '1']
```

```
plt.contourf(X, Y, Z[p], 20, cmap='jet', vmin=180, vmax=360)
plt.xlabel(Zlab[p][0])
plt.ylabel(Zlab[p][1])
plt.colorbar()
```

### 1. Aircraft Wing Weight Example



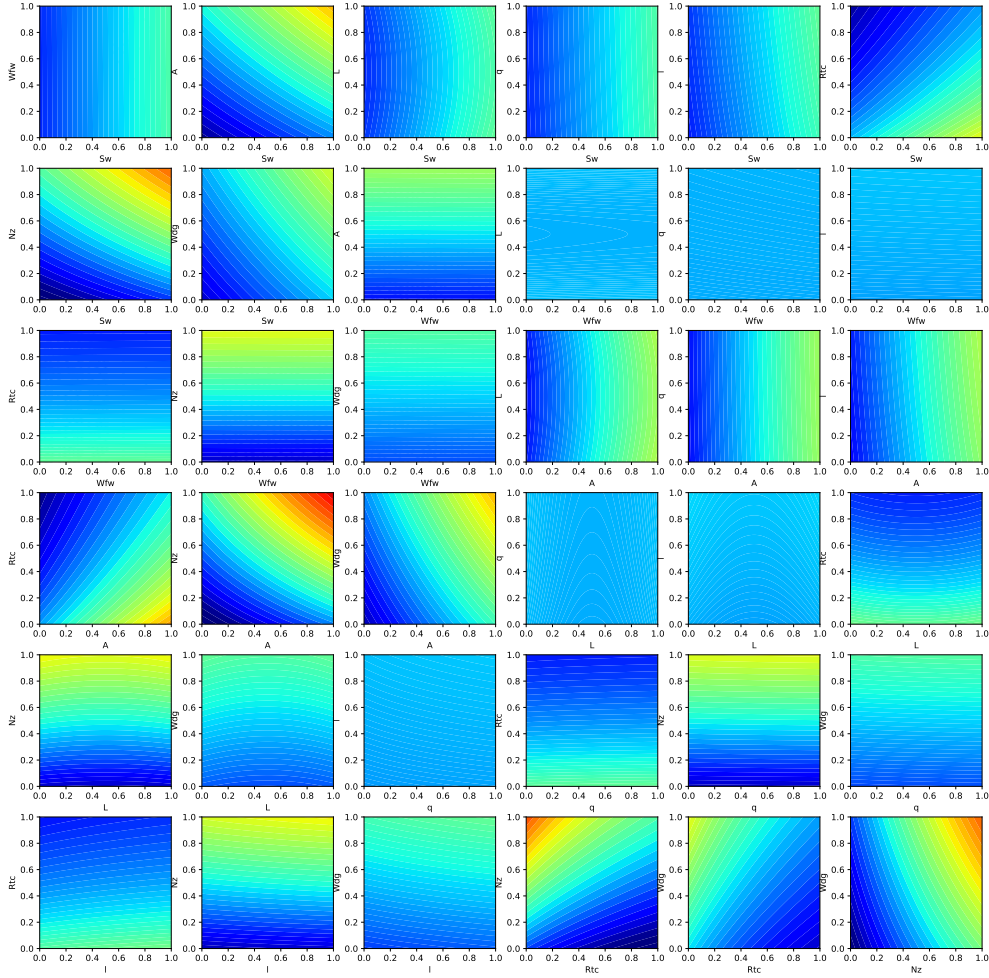
- Using a global colormap indicates that these variables have minor effects on the wing weight.
- Important factors can be detected by visual inspection
- Plotting the Big Picture: we can plot all 36 combinations in one figure.

```
fig = plt.figure(figsize=(20., 20.))
grid = ImageGrid(fig, 111, # similar to subplot(111)
                  nrows_ncols=(6,6), # creates 2x2 grid of axes
                  axes_pad=0.5, # pad between axes in inch.
                  share_all=True,
                  label_mode="all",
                  )

i = 0
for ax, im in zip(grid, Z):
    # Iterating over the grid returns the Axes.
    ax.set_xlabel(Zlab[i][0])
    ax.set_ylabel(Zlab[i][1])
    # ax.set_title(Zlab[i][1] + " vs. " + Zlab[i][0])
    ax.contourf(X, Y, im, 30, cmap = "jet", vmin = 180, vmax = 360)
    i = i + 1

plt.show()
```

## 1.8. AWWE Landscape



## 1.8. AWWE Landscape

- Our Observations

1. The load factor  $N_z$ , which determines the magnitude of the maximum aerodynamic load on the wing, is very active and involved in interactions with other variables.
  - Classic example: the interaction of  $N_z$  with the aspect ratio  $A$  indicates a heavy wing for high aspect ratios and large  $g$ -forces
  - This is the reason why highly manoeuvrable fighter jets cannot have very efficient, glider wings)

### 1. Aircraft Wing Weight Example

2. Aspect ratio  $A$  and airfoil thickness to chord ratio  $R_{tc}$  have nonlinear interactions.
  3. Most important variables:
    - Ultimate load factor  $N_z$ , wing area  $S_w$ , and flight design gross weight  $W_{dg}$ .
  4. Little impact: dynamic pressure  $q$ , taper ratio  $l$ , and quarter-chord sweep  $L$ .
- Expert Knowledge
    - Aircraft designers know that the overall weight of the aircraft and the wing area must be kept to a minimum
    - the latter usually dictated by constraints such as required stall speed, landing distance, turn rate, etc.

## 1.9. Summary of the First Experiments

- First, we considered two pairs of inputs, out of 36 total pairs
- Then, the “Big Picture”:
  - For each pair we evaluated `wingwt` 10,000 times
- Doing the same for all pairs would require 360K evaluations:
  - not a reasonable number with a real computer simulation that takes any non-trivial amount of time to evaluate
  - Only 1s per evaluation: > 100 hours
- Many solvers take minutes/hours/days to execute a single run
- And: three-way interactions?
- Consequence: a different strategy is needed

## 1.10. Exercise

### 1.10.1. Adding Paint Weight

- Paint weight is not considered.
- Add Paint Weight  $W_p$  to formula (the updated formula is shown below) and update the functions and plots in the notebook.

$$W = 0.036 S_w^{0.758} \times W_{fw}^{0.0035} \times \left( \frac{A}{\cos^2 \Lambda} \right)^{0.6} \times q^{0.006} \times \lambda^{0.04} \\ \times \left( \frac{100 R_{tc}}{\cos \Lambda} \right)^{-0.3} \times (N_z W_{dg})^{0.49} + S_w W_p$$

## 1.11. Jupyter Notebook

### **i** Note

- The Jupyter-Notebook of this lecture is available on GitHub in the Hyperparameter-Tuning-Cookbook Repository



**Part II.**

**Numerical Methods**





## 2. Simulation and Surrogate Modeling

- We will consider the interplay between
  - mathematical models,
  - numerical approximation,
  - simulation,
  - computer experiments, and
  - field data
- Experimental design will play a key role in our developments, but not in the classical regression and response surface methodology sense
- Challenging real-data/real-simulation examples benefiting from modern surrogate modeling methodology
- We will consider the classical, response surface methodology (RSM) approach, and then move on to more modern approaches
- All approaches are based on surrogates

### 2.1. Surrogates

- Gathering data is **expensive**, and sometimes getting exactly the data you want is impossible or unethical
- **Surrogate**: substitute for the real thing
- In statistics, draws from predictive equations derived from a fitted model can act as a surrogate for the data-generating mechanism
- Benefits of the surrogate approach:
  - Surrogate could represent a cheaper way to explore relationships, and entertain “what ifs?”
  - Surrogates favor faithful yet pragmatic reproduction of dynamics:
    - \* interpretation,
    - \* establishing causality, or
    - \* identification
  - Many numerical simulators are **deterministic**, whereas field observations are noisy or have measurement error

## 2. *Simulation and Surrogate Modeling*

### 2.1.1. Costs of Simulation

- Computer simulations are generally cheaper (but not always!) than physical observation
- Some computer simulations can be just as expensive as field experimentation, but computer modeling is regarded as easier because:
  - the experimental apparatus is better understood
  - more aspects may be controlled.

### 2.1.2. Mathematical Models and Meta-Models

- Use of mathematical models leveraging numerical solvers has been commonplace for some time
- Mathematical models became more complex, requiring more resources to simulate/solve numerically
- Practitioners increasingly relied on **meta-models** built off of limited simulation campaigns

### 2.1.3. Surrogates = Trained Meta-models

- Data collected via expensive computer evaluations tuned flexible functional forms that could be used in lieu of further simulation to
  - save money or computational resources;
  - cope with an inability to perform future runs (expired licenses, off-line or over-impacted supercomputers)
- Trained meta-models became known as **surrogates**

### 2.1.4. Computer Experiments

- **Computer experiment**: design, running, and fitting meta-models.
  - Like an ordinary statistical experiment, except the data are generated by computer codes rather than physical or field observations, or surveys
- **Surrogate modeling** is statistical modeling of computer experiments

### 2.1.5. Limits of Mathematical Modeling

- Mathematical biologists, economists and others had reached the limit of equilibrium-based mathematical modeling with cute closed-form solutions
- **Stochastic simulations replace deterministic solvers** based on FEM, Navier–Stokes or Euler methods
- Agent-based simulation models are used to explore predator-prey (Lotka–Volterra) dynamics, spread of disease, management of inventory or patients in health insurance markets
- Consequence: the distinction between surrogate and statistical model is all but gone

### 2.1.6. Why Computer Simulations are Necessary

- You can't seed a real community with Ebola and watch what happens
- If there's (real) field data, say on a historical epidemic, further experimentation may be almost entirely limited to the mathematical and computer modeling side
- Classical statistical methods offer little guidance

### 2.1.7. Simulation Requirements

- Simulation should
  - enable rich **diagnostics** to help criticize that models
  - **understanding** its sensitivity to inputs and other configurations
  - providing the ability to **optimize** and
  - refine both **automatically** and with expert intervention
- And it has to do all that while remaining **computationally tractable**
- One perspective is so-called **response surface methods** (RSMs):
- a poster child from industrial statistics' heyday, well before information technology became a dominant industry

## 2.2. Applications of Surrogate Models

The four most common usages of surrogate models are:

- **Augmenting Expensive Simulations:** Surrogate models act as a 'curve fit' to approximate the results of expensive simulation codes, enabling predictions without rerunning the primary source. This provides significant speed improvements while maintaining useful accuracy.

## 2. *Simulation and Surrogate Modeling*

- **Calibration of Predictive Codes:** Surrogates bridge the gap between simpler, faster but less accurate models and more accurate, slower models. This multi-fidelity approach allows for improved accuracy without the full computational expense.
- **Handling Noisy or Missing Data:** Surrogates smooth out random or systematic errors in experimental or computational data, filling gaps and revealing overall trends while filtering out extraneous details.
- **Data Mining and Insight Generation:** Surrogates help identify functional relationships between variables and their impact on results. They enable engineers to focus on critical variables and visualize data trends effectively.

### 2.3. DACE and RSM

Mathematical models implemented in computer codes are used to circumvent the need for expensive field data collection. These models are particularly useful when dealing with highly nonlinear response surfaces, high signal-to-noise ratios (which often involve deterministic evaluations), and a global scope. As a result, a new approach is required in comparison to Response Surface Methodology (RSM), which is discussed in Section 5.1.

With the improvement in computing power and simulation fidelity, researchers gain higher confidence and a better understanding of the dynamics in physical, biological, and social systems. However, the expansion of configuration spaces and increasing input dimensions necessitates more extensive designs. High-performance computing (HPC) allows for thousands of runs, whereas previously only tens were possible. This shift towards larger models and training data presents new computational challenges.

Research questions for DACE (Design and Analysis of Computer Experiments) include how to design computer experiments that make efficient use of computation and how to meta-model computer codes to save on simulation effort. The choice of surrogate model for computer codes significantly impacts the optimal experiment design, and the preferred model-design pairs can vary depending on the specific goal.

The combination of computer simulation, design, and modeling with field data from similar real-world experiments introduces a new category of computer model tuning problems. The ultimate goal is to automate these processes to the greatest extent possible, allowing for the deployment of HPC with minimal human intervention.

One of the remaining differences between RSM and DACE lies in how they handle noise. DACE employs replication, a technique that would not be used in a deterministic setting, to separate signal from noise. Traditional RSM is best suited for situations where a substantial proportion of the variability in the data is due to noise, and where the acquisition of data values can be severely limited. Consequently, RSM is better suited for a different class of problems, aligning with its intended purposes.

## 2.4. Updating a Surrogate Model

Two very good texts on computer experiments and surrogate modeling are Santner, Williams, and Notz (2003) and Forrester, Sóbester, and Keane (2008). The former is the canonical reference in the statistics literature and the latter is perhaps more popular in engineering.

**Example 2.1** (Example: DACE and RSM). Imagine you are a chemical engineer tasked with optimizing a chemical process to maximize yield. You can control temperature and pressure, but repeated experiments show variability in yield due to inconsistencies in raw materials.

- Using RSM: You would use RSM to design a series of experiments varying temperature and pressure. You would then fit a response surface (a mathematical model) to the data, helping you understand how changes in temperature and pressure affect yield. Using this model, you can identify optimal conditions for maximizing yield despite the noise.
- Using DACE: If instead you use a computational model to simulate the chemical process and want to account for numerical noise or uncertainty in model parameters, you might use DACE. You would run simulations at different conditions, possibly repeating them to assess variability and build a surrogate model that accurately predicts yields, which can be optimized to find the best conditions.

### 2.3.1. Noise Handling in RSM and DACE

Noise in RSM: In experimental settings, noise often arises due to variability in experimental conditions, measurement errors, or other uncontrollable factors. This noise can significantly affect the response variable,  $Y$ . Replication is a standard procedure for handling noise in RSM. In the context of computer experiments, noise might not be present in the traditional sense since simulations can be deterministic. However, variability can arise from uncertainty in input parameters or model inaccuracies. DACE predominantly utilizes advanced interpolation to construct accurate models of deterministic data, sometimes considering statistical noise modeling if needed.

## 2.4. Updating a Surrogate Model

A surrogate model is updated by incorporating new data points, known as infill points, into the model to improve its accuracy and predictive capabilities. This process is iterative and involves the following steps:

- Identify Regions of Interest: The surrogate model is analyzed to determine areas where it is inaccurate or where further exploration is needed. This could be regions with high uncertainty or areas where the model predicts promising results (e.g., potential optima).

## 2. Simulation and Surrogate Modeling

- **Select Infill Points:** Infill points are new data points chosen based on specific criteria, such as:
- **Exploitation:** Sampling near predicted optima to refine the solution. **Exploration:** Sampling in regions of high uncertainty to improve the model globally. **Balanced Approach:** Combining exploitation and exploration to ensure both local and global improvements.
- **Evaluate the True Function:** The true function (e.g., a simulation or experiment) is evaluated at the selected infill points to obtain their corresponding outputs.
- **Update the Surrogate Model:** The surrogate model is retrained or updated using the new data, including the infill points, to improve its accuracy.
- **Repeat:** The process is repeated until the model meets predefined accuracy criteria or the computational budget is exhausted.

**Definition 2.1** (Infill Points). Infill points are strategically chosen new data points added to the surrogate model. They are selected to:

- Reduce uncertainty in the model.
- Improve predictions in regions of interest.
- Enhance the model's ability to identify optima or trends.

The selection of infill points is often guided by infill criteria, such as:

- **Expected Improvement (EI):** Maximizing the expected improvement over the current best solution.
- **Uncertainty Reduction:** Sampling where the model's predictions have high variance.
- **Probability of Improvement (PI):** Sampling where the probability of improving the current best solution is highest.

The iterative infill-points updating process ensures that the surrogate model becomes increasingly accurate and useful for optimization or decision-making tasks.

## 3. Sampling Plans

### **i** Note

- This section is based on chapter 1 in Forrester, Sóbester, and Keane (2008).
- The following Python packages are imported:

```
import numpy as np
import pandas as pd
import numpy as np
from typing import Tuple, Optional
import matplotlib.pyplot as plt
from spotpython.utils.sampling import rlh
from spotpython.utils.effects import screening_plot, screeningplan
from spotpython.fun.objectivefunctions import Analytical
from spotpython.utils.sampling import (fullfactorial, bestlh,
    jd, mm, mmphi, mmsort, perturb, mmlhs, phisort, mmphi_intensive)
from spotpython.design.poor import Poor
from spotpython.design.clustered import Clustered
from spotpython.design.sobol import Sobol
from spotpython.design.spacefilling import SpaceFilling
from spotpython.design.random import Random
from spotpython.design.grid import Grid
```

### 3.1. Ideas and Concepts

**Definition 3.1** (Sampling Plan). In the context of computer experiments, the term sampling plan refers to the set of input values, say  $X$ , at which the computer code is evaluated.

The goal of a sampling plan is to efficiently explore the input space to understand the behavior of the computer code and build a surrogate model that accurately represents the code's behavior. Traditionally, Response Surface Methodology (RSM) has been used to design sampling plans for computer experiments. These sampling plans are

### 3. Sampling Plans

based on procedures that generate points by means of a rectangular grid or a factorial design.

However, more recently, Design and Analysis of Computer Experiments (DACE) has emerged as a more flexible and powerful approach for designing sampling plans.

Engineering design often requires the construction of a surrogate model  $\hat{f}$  to approximate the expensive response of a black-box function  $f$ . The function  $f(x)$  represents a continuous metric (e.g., quality, cost, or performance) defined over a design space  $D \subset \mathbb{R}^k$ , where  $x$  is a  $k$ -dimensional vector of design variables. Since evaluating  $f$  is costly, only a sparse set of samples is used to construct  $\hat{f}$ , which can then provide inexpensive predictions for any  $x \in D$ .

The process involves:

- Sampling discrete observations:
- Using these samples to construct an approximation  $\hat{f}$ .
- Ensuring the surrogate model is well-posed, meaning it is mathematically valid and can generalize predictions effectively.

A sampling plan

$$X = \{x^{(i)} \in D | i = 1, \dots, n\}$$

determines the spatial arrangement of observations. While some models require a minimum number of data points  $n$ , once this threshold is met, a surrogate model can be constructed to approximate  $f$  efficiently.

A well-posed model does not always perform well because its ability to generalize depends heavily on the sampling plan used to collect data. If the sampling plan is poorly designed, the model may fail to capture critical behaviors in the design space. For example:

- Extreme Sampling: Measuring performance only at the extreme values of parameters may miss important behaviors in the center of the design space, leading to incomplete understanding.
- Uneven Sampling: Concentrating samples in certain regions while neglecting others forces the model to extrapolate over unsampled areas, potentially resulting in inaccurate or misleading predictions. Additionally, in some cases, the data may come from external sources or be limited in scope, leaving little control over the sampling plan. This can further restrict the model's ability to generalize effectively.



### 3.1.1. The ‘Curse of Dimensionality’ and How to Avoid It

The “curse of dimensionality” refers to the exponential increase in computational complexity and data requirements as the number of dimensions (variables) in a problem grows. For a one-dimensional space, sampling  $n$  locations may suffice for accurate predictions. In high-dimensional spaces, the amount of data needed to maintain the same level of accuracy or coverage increases dramatically. For example, if a one-dimensional space requires  $n$  samples for a certain accuracy, a  $k$ -dimensional space would require  $n^k$  samples. This makes tasks like optimization, sampling, and modeling computationally expensive and often impractical in high-dimensional settings.

**Example 3.1** (Example: Curse of Dimensionality). Consider a simple example where we want to model the cost of a car tire based on its wheel diameter. If we have one variable (wheel diameter), we might need 10 simulations to get a good estimate of the cost. Now, if we add 8 more variables (e.g., tread pattern, rubber type, etc.), the number of simulations required increases to  $10^8$  (10 million). This is because the number of combinations of design variables grows exponentially with the number of dimensions. This means that the computational budget required to evaluate all combinations of design variables becomes infeasible. In this case, it would take 11,416 years to complete the simulations, making it impractical to explore the design space fully.

### 3.1.2. Physical versus Computational Experiments

Physical experiments are prone to experimental errors from three main sources:

- Human error: Mistakes made by the experimenter.
- Random error: Measurement inaccuracies that vary unpredictably.
- Systematic error: Consistent bias due to flaws in the experimental setup.

The key distinction is repeatability: systematic errors remain constant across repetitions, while random errors vary.

Computational experiments, on the other hand, are deterministic and free from random errors. However, they are still affected by:

- Human error: Bugs in code or incorrect boundary conditions.
- Systematic error: Biases from model simplifications (e.g., inviscid flow approximations) or finite resolution (e.g., insufficient mesh resolution).

The term “noise” is used differently in physical and computational contexts. In physical experiments, it refers to random errors, while in computational experiments, it often refers to systematic errors.

### 3. Sampling Plans

Understanding these differences is crucial for designing experiments and applying techniques like Gaussian process-based approximations. For physical experiments, replication mitigates random errors, but this is unnecessary for deterministic computational experiments.

#### 3.1.3. Designing Preliminary Experiments (Screening)

Minimizing the number of design variables  $x_1, x_2, \dots, x_k$  is crucial before modeling the objective function  $f$ . This process, called screening, aims to reduce dimensionality without compromising the analysis. If  $f$  is at least once differentiable over the design domain  $D$ , the partial derivative  $\frac{\partial f}{\partial x_i}$  can be used to classify variables:

- Negligible Variables: If  $\frac{\partial f}{\partial x_i} = 0, \forall x \in D$ , the variable  $x_i$  can be safely neglected.
- Linear Additive Variables: If  $\frac{\partial f}{\partial x_i} = \text{constant} \neq 0, \forall x \in D$ , the effect of  $x_i$  is linear and additive.
- Nonlinear Variables: If  $\frac{\partial f}{\partial x_i} = g(x_i), \forall x \in D$ , where  $g(x_i)$  is a non-constant function,  $f$  is nonlinear in  $x_i$ .
- Interactive Nonlinear Variables: If  $\frac{\partial f}{\partial x_i} = g(x_i, x_j, \dots), \forall x \in D$ , where  $g(x_i, x_j, \dots)$  is a function involving interactions with other variables,  $f$  is nonlinear in  $x_i$  and interacts with  $x_j$ .

Measuring  $\frac{\partial f}{\partial x_i}$  across the entire design space is often infeasible due to limited budgets. The percentage of time allocated to screening depends on the problem: If many variables are expected to be inactive, thorough screening can significantly improve model accuracy by reducing dimensionality. If most variables are believed to impact the objective, focus should shift to modeling instead. Screening is a trade-off between computational cost and model accuracy, and its effectiveness depends on the specific problem context.

##### 3.1.3.1. Estimating the Distribution of Elementary Effects

In order to simplify the presentation of what follows, we make, without loss of generality, the assumption that the design space  $D = [0, 1]^k$ ; that is, we normalize all variables into the unit cube. We shall adhere to this convention for the rest of the book and strongly urge the reader to do likewise when implementing any algorithms described here, as this step not only yields clearer mathematics in some cases but also safeguards against scaling issues.

Before proceeding with the description of the Morris algorithm, we need to define an important statistical concept. Let us restrict our design space  $D$  to a  $k$ -dimensional,  $p$ -level full factorial grid, that is,

$$x_i \in \{0, \frac{1}{p-1}, \frac{2}{p-1}, \dots, 1\}, \quad \text{for } i = 1, \dots, k.$$

### 3.1. Ideas and Concepts

**Definition 3.2** (Elementary Effect). For a given baseline value  $x \in D$ , let  $d_i(x)$  denote the elementary effect of  $x_i$ , where:

$$d_i(x) = \frac{f(x_1, \dots, x_i + \Delta, \dots, x_k) - f(x_1, \dots, x_i - \Delta, \dots, x_k)}{2\Delta}, \quad i = 1, \dots, k, \quad (3.1)$$

where  $\Delta$  is the step size, which is defined as the distance between two adjacent levels in the grid. In other words, we have:

with

$$\Delta = \frac{\xi}{p-1}, \quad \xi \in \mathbb{N}^*, \quad \text{and} \quad x \in D, \quad \text{such that its components } x_i \leq 1 - \Delta.$$

$\Delta$  is the step size. The elementary effect  $d_i(x)$  measures the sensitivity of the function  $f$  to changes in the variable  $x_i$  at the point  $x$ .

Morris's method aims to estimate the parameters of the distribution of elementary effects associated with each variable. A large measure of central tendency indicates that a variable has a significant influence on the objective function across the design space, while a large measure of spread suggests that the variable is involved in interactions or contributes to the nonlinearity of  $f$ . In practice, the sample mean and standard deviation of a set of  $d_i(x)$  values, calculated in different parts of the design space, are used for this estimation.

To ensure efficiency, the preliminary sampling plan  $X$  should be designed so that each evaluation of the objective function  $f$  contributes to the calculation of two elementary effects, rather than just one (as would occur with a naive random spread of baseline  $x$  values and adding  $\Delta$  to one variable). Additionally, the sampling plan should provide a specified number (e.g.,  $r$ ) of elementary effects for each variable, independently drawn with replacement. For a detailed discussion on constructing such a sampling plan, readers are encouraged to consult Morris's original paper (Morris, 1991). Here, we focus on describing the process itself.

The random orientation of the sampling plan  $B$  can be constructed as follows:

- Let  $B$  be a  $(k+1) \times k$  matrix of 0s and 1s, where for each column  $i$ , two rows differ only in their  $i$ -th entries.
- Compute a random orientation of  $B$ , denoted  $B^*$ :

$$B^* = (1_{k+1,k} x^* + (\Delta/2) [(2B - 1_{k+1,k}) D^* + 1_{k+1,k}]) P^*,$$

where:

- $D^*$  is a  $k$ -dimensional diagonal matrix with diagonal elements  $\pm 1$  (equal probability),

### 3. Sampling Plans

- $\mathbf{1}$  is a matrix of 1s,
- $x^*$  is a randomly chosen point in the  $p$ -level design space (limited by  $\Delta$ ),
- $P^*$  is a  $k \times k$  random permutation matrix with one 1 per column and row.

`spotpython` provides a Python implementation to compute  $B^*$ , see <https://github.com/sequential-parameter-optimization/spotPython/blob/main/src/spotpython/utils/effects.py>.

Here is the corresponding code:

```
def randorient(k, p, xi, seed=None):
    # Initialize random number generator with the provided seed
    if seed is not None:
        rng = np.random.default_rng(seed)
    else:
        rng = np.random.default_rng()

    # Step length
    Delta = xi / (p - 1)

    m = k + 1

    # A truncated p-level grid in one dimension
    xs = np.arange(0, 1 - Delta, 1 / (p - 1))
    xsl = len(xs)
    if xsl < 1:
        print(f"xi = {xi}.")
        print(f"p = {p}.")
        print(f"Delta = {Delta}.")
        print(f"p - 1 = {p - 1}.")
        raise ValueError(f"The number of levels xsl is {xsl}, but it must be greater than 1")

    # Basic sampling matrix
    B = np.vstack((np.zeros((1, k)), np.tril(np.ones((k, k)))))

    # Randomization

    # Matrix with +1s and -1s on the diagonal with equal probability
    Dstar = np.diag(2 * rng.integers(0, 2, size=k) - 1)

    # Random base value
    xstar = xs[rng.integers(0, xsl, size=k)]

    # Permutation matrix
    Pstar = np.zeros((k, k))
```

```

rp = rng.permutation(k)
for i in range(k):
    Pstar[i, rp[i]] = 1

# A random orientation of the sampling matrix
Bstar = (np.ones((m, 1)) @ xstar.reshape(1, -1) +
         (Delta / 2) * ((2 * B - np.ones((m, k))) @ Dstar +
                        np.ones((m, k)))) @ Pstar

return Bstar

```

The code following snippet generates a random orientation of a sampling matrix **Bstar** using the **randorient()** function. The input parameters are:

- **k** = 3: The number of design variables (dimensions).
- **p** = 3: The number of levels in the grid for each variable.
- **xi** = 1: A parameter used to calculate the step size Delta.

Step-size calculation is performed as follows:  $\Delta = \text{xi} / (p - 1) = 1 / (3 - 1) = 0.5$ , which determines the spacing between levels in the grid.

Next, random sampling matrix construction is computed:

- A truncated grid is created with levels  $[0, 0.5]$  (based on Delta).
- A basic sampling matrix **B** is constructed, which is a lower triangular matrix with 0s and 1s.

Then, randomization is applied:

- **Dstar**: A diagonal matrix with random entries of +1 or -1.
- **xstar**: A random starting point from the grid.
- **Pstar**: A random permutation matrix.

Random orientation is applied to the basic sampling matrix **B** to create **Bstar**. This involves scaling, shifting, and permuting the rows and columns of **B**.

The final output is the matrix **Bstar**, which represents a random orientation of the sampling plan. Each row corresponds to a sampled point in the design space, and each column corresponds to a design variable.

**Example 3.2** (Random Orientation of the Sampling Matrix in 2-D).

```

k = 2
p = 3
xi = 1
Bstar = randorient(k, p, xi, seed=123)
print(f"Random orientation of the sampling matrix:\n{Bstar}")

```

### 3. Sampling Plans

Random orientation of the sampling matrix:

```
[[0.5 0. ]  
 [0.  0. ]  
 [0.  0.5]]
```

We can visualize the random orientation of the sampling matrix in 2-D as shown in Figure 3.1.

```
plt.figure(figsize=(6, 6))  
plt.scatter(Bstar[:, 0], Bstar[:, 1], color='blue', s=50, label='Hypercube Points')  
for i in range(Bstar.shape[0]):  
    plt.text(Bstar[i, 0] + 0.01, Bstar[i, 1] + 0.01, str(i), fontsize=9)  
plt.xlim(-0.1, 1.1)  
plt.ylim(-0.1, 1.1)  
plt.xlabel('x1')  
plt.ylabel('x2')  
plt.grid()
```

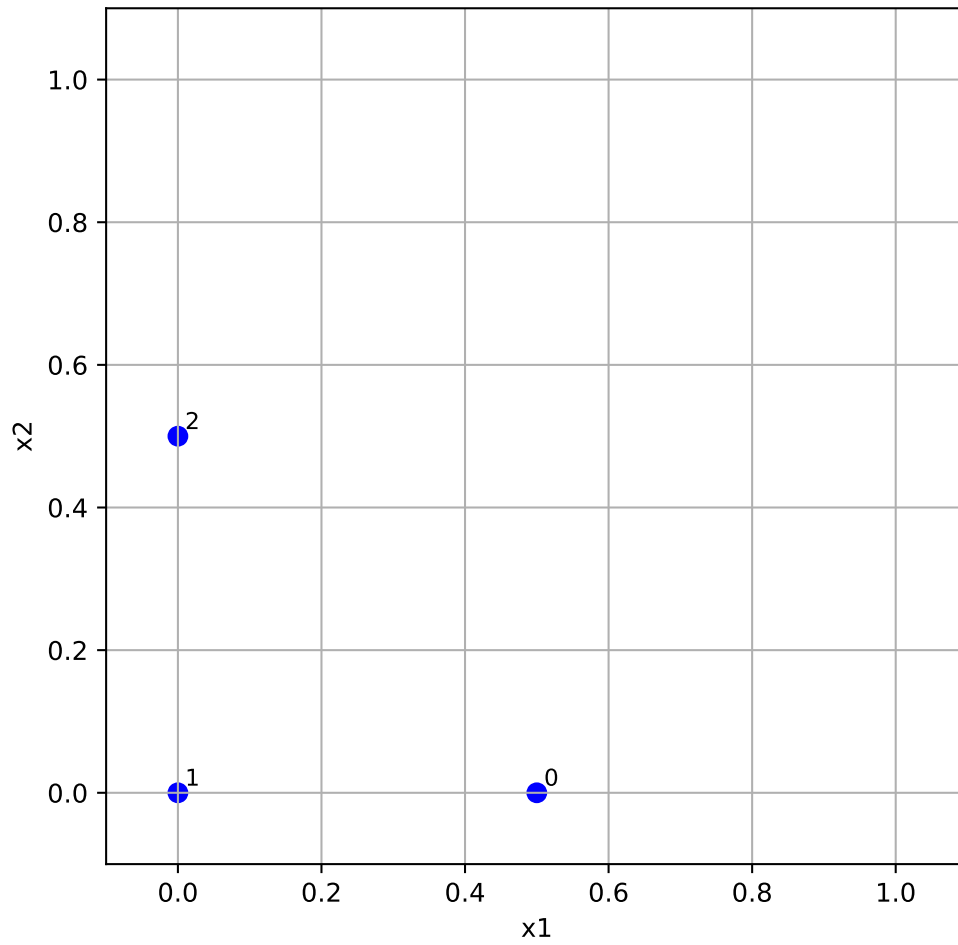


Figure 3.1.: Random orientation of the sampling matrix in 2-D. The labels indicate the row index of the points.

**Example 3.3** (Random Orientation of the Sampling Matrix).

```
k = 3
p = 3
xi = 1
Bstar = randorient(k, p, xi)
print(f"Random orientation of the sampling matrix:\n{Bstar}")
```

Random orientation of the sampling matrix:

### 3. Sampling Plans

```
[[0.  0.  0.5]
 [0.5 0.  0.5]
 [0.5 0.  0. ]
 [0.5 0.5 0. ]]
```

To obtain  $r$  elementary effects for each variable, the screening plan is built from  $r$  random orientations:

$$X = \begin{pmatrix} B_1^* \\ B_2^* \\ \vdots \\ B_r^* \end{pmatrix}$$

The function `screeningplan()` generates a screening plan by calling the `randorient()` function  $r$  times. It creates a list of random orientations and then concatenates them into a single array, which represents the screening plan. The screening plan implementation in Python is as follows (see <https://github.com/sequential-parameter-optimization/spotPython/blob/main/src/spotpython/utils/effects.py>):

```
def screeningplan(k, p, xi, r):
    # Empty list to accumulate screening plan rows
    X = []
    for i in range(r):
        X.append(randorient(k, p, xi))
    # Concatenate list of arrays into a single array
    X = np.vstack(X)
    return X
```

It works like follows:

- The value of the objective function  $f$  is computed for each row of the screening plan matrix  $X$ . These values are stored in a column vector  $t$  of size  $(r * (k + 1)) \times 1$ , where:
  - $r$  is the number of random orientations.
  - $k$  is the number of design variables.

The elementary effects are calculated using the following formula:

- For each random orientation, adjacent rows of the screening plan matrix  $X$  and their corresponding function values from  $t$  are used.
- These values are inserted into Equation 3.1 to compute elementary effects for each variable. An elementary effect measures the sensitivity of the objective function to changes in a specific variable.



### 3.1. Ideas and Concepts

Results can be used for a statistical analysis. After collecting a sample of  $r$  elementary effects for each variable:

- The sample mean (central tendency) is computed to indicate the overall influence of the variable.
- The sample standard deviation (spread) is computed to capture variability, which may indicate interactions or nonlinearity.

The results (sample means and standard deviations) are plotted on a chart for comparison. This helps identify which variables have the most significant impact on the objective function and whether their effects are linear or involve interactions. This is implemented in the function `screening_plot()` in `Python`, which uses the helper function `_screening()` to calculate the elementary effects and their statistics.

```
def _screening(X, fun, xi, p, labels, bounds=None) -> tuple:
    """Helper function to calculate elementary effects for a screening design.

    Args:
        X (np.ndarray): The screening plan matrix, typically structured
            within a  $[0,1]^k$  box.
        fun (object): The objective function to evaluate at each
            design point in the screening plan.
        xi (float): The elementary effect step length factor.
        p (int): Number of discrete levels along each dimension.
        labels (list of str): A list of variable names corresponding to
            the design variables.
        bounds (np.ndarray): A  $2 \times k$  matrix where the first row contains
            lower bounds and the second row contains upper bounds for
            each variable.

    Returns:
        tuple: A tuple containing two arrays:
            - sm: The mean of the elementary effects for each variable.
            - ssd: The standard deviation of the elementary effects for
                each variable.
    """
    k = X.shape[1]
    r = X.shape[0] // (k + 1)

    # Scale each design point
    t = np.zeros(X.shape[0])
    for i in range(X.shape[0]):
        if bounds is not None:
            X[i, :] = bounds[0, :] + X[i, :] * (bounds[1, :] - bounds[0, :])
        t[i] = fun(X[i, :])
```

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```
# Elementary effects
F = np.zeros((k, r))
for i in range(r):
    for j in range(i * (k + 1), i * (k + 1) + k):
        idx = np.where(X[j, :] - X[j + 1, :] != 0)[0][0]
        F[idx, i] = (t[j + 1] - t[j]) / (xi / (p - 1))

# Statistical measures (divide by n)
ssd = np.std(F, axis=1, ddof=0)
sm = np.mean(F, axis=1)
return sm, ssd

def screening_plot(X, fun, xi, p, labels, bounds=None, show=True) -> None:
    """Generates a plot with elementary effect screening metrics.

    This function calculates the mean and standard deviation of the
    elementary effects for a given set of design variables and plots
    the results.

    Args:
        X (np.ndarray):
            The screening plan matrix, typically structured within a  $[0,1]^k$  box.
        fun (object):
            The objective function to evaluate at each design point in the screening p
        xi (float):
            The elementary effect step length factor.
        p (int):
            Number of discrete levels along each dimension.
        labels (list of str):
            A list of variable names corresponding to the design variables.
        bounds (np.ndarray):
            A 2xk matrix where the first row contains lower bounds and
            the second row contains upper bounds for each variable.
        show (bool):
            If True, the plot is displayed. Defaults to True.

    Returns:
        None: The function generates a plot of the results.
    """
    k = X.shape[1]
    sm, ssd = _screening(X=X, fun=fun, xi=xi, p=p, labels=labels, bounds=bounds)
    plt.figure()
    for i in range(k):
```

```
plt.text(sm[i], ssd[i], labels[i], fontsize=10)
plt.axis([min(sm), 1.1 * max(sm), min(ssd), 1.1 * max(ssd)])
plt.xlabel("Sample means")
plt.ylabel("Sample standard deviations")
plt.gca().tick_params(labelsize=10)
plt.grid(True)
if show:
    plt.show()
```

### 3.1.4. Special Considerations When Deploying Screening Algorithms

When implementing the screening algorithm described above, two specific scenarios require special attention:

- **Duplicate Design Points:** If the dimensionality  $k$  of the space is relatively low and you can afford a large number of elementary effects  $r$ , we should be aware of the increased probability of duplicate design points appearing in the sampling plan  $X$ . \*Since the responses at sample points are deterministic, there's no value in evaluating the same point multiple times. Fortunately, this issue is relatively uncommon in practice, as screening high-dimensional spaces typically requires large numbers of elementary effects, which naturally reduces the likelihood of duplicates.
- **Failed Simulations:** Numerical simulation codes occasionally fail to return valid results due to meshing errors, non-convergence of partial differential equation solvers, numerical instabilities, or parameter combinations outside the stable operating range.

From a screening perspective, this is particularly problematic because an entire random orientation  $B^*$  becomes compromised if even a single point within it fails to evaluate properly. Implementing error handling strategies or fallback methods to manage such cases should be considered.

For robust screening studies, monitoring simulation success rates and having contingency plans for failed evaluations are important aspects of the experimental design process.

## 3.2. Analyzing Variable Importance in Aircraft Wing Weight

Let us consider the following analytical expression used as a conceptual level estimate of the weight of a light aircraft wing as discussed in Chapter 1.

### 3. Sampling Plans

```
fun = Analytical()
k = 10
p = 10
xi = 1
r = 25
X = screeningplan(k=k, p=p, xi=xi, r=r) # shape (r x (k+1), k)
value_range = np.array([
    [150, 220, 6, -10, 16, 0.5, 0.08, 2.5, 1700, 0.025],
    [200, 300, 10, 10, 45, 1.0, 0.18, 6.0, 2500, 0.08 ],
])
labels = [
    "S_W", "W_fw", "A", "Lambda",
    "q", "lambda", "tc", "N_z",
    "W_dg", "W_p"
]
screening_plot(
    X=X,
    fun=fun.fun_wingwt,
    bounds=value_range,
    xi=xi,
    p=p,
    labels=labels,
)
```

### 3.2. Analyzing Variable Importance in Aircraft Wing Weight

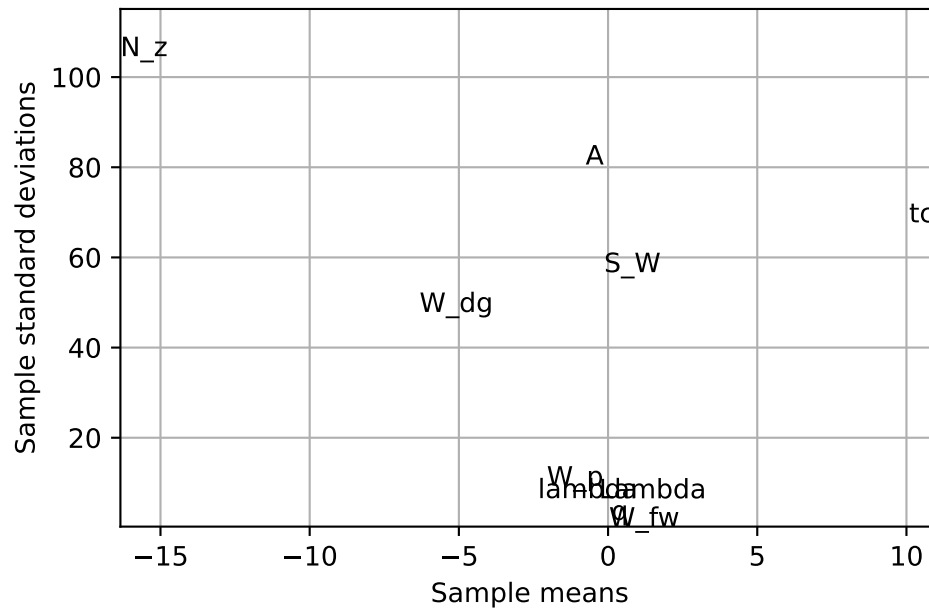


Figure 3.2.: Estimated means and standard deviations of the elementary effects for the 10 design variables of the wing weight function. Example based on Forrester, Sóbester, and Keane (2008).

#### **i** Nondeterministic Results

The code will generate a slightly different screening plan each time, as it uses random orientations of the sampling matrix  $B$ .

Figure 3.2 provides valuable insights into variable activity without requiring domain expertise. The screening study with  $r = 25$  elementary effects reveals distinct patterns in how variables affect wing weight:

- Variables with Minimal Impact: A clearly defined group of variables clusters around the origin - indicating their minimal impact on wing weight:
  - Paint weight ( $W_p$ ) - as expected, contributes little to overall wing weight
  - Dynamic pressure ( $q$ ) - within our chosen range, this has limited effect (essentially representing different cruise altitudes at the same speed)
  - Taper ratio ( $\lambda$ ) and quarter-chord sweep ( $\Lambda$ ) - these geometric parameters have minor influence within the narrow range ( $-10^\circ$  to  $10^\circ$ ) typical of light aircraft
- Variables with Linear Effects:

### 3. Sampling Plans

- While still close to the origin, fuel weight ( $W_{fw}$ ) shows a slightly larger central tendency with very low standard deviation. This indicates moderate importance but minimal involvement in interactions with other variables.
- Variables with Nonlinear/Interactive Effects:
  - Aspect ratio ( $A$ ) and airfoil thickness ratio ( $R_{tc}$ ) show similar importance levels, but their high standard deviations suggest significant nonlinear behavior and interactions with other variables.
- Dominant Variables: The most significant impacts come from:
  - Flight design gross weight ( $W_{dg}$ )
  - Wing area ( $S_W$ )
  - Ultimate load factor ( $N_z$ )

These variables show both large central tendency values and high standard deviations, indicating strong direct effects and complex interactions. The interaction between aspect ratio and load factor is particularly important - high values of both create extremely heavy wings, explaining why highly maneuverable fighter jets cannot use glider-like wing designs.

What makes this screening approach valuable is its ability to identify critical variables without requiring engineering knowledge or expensive modeling. In real-world applications, we rarely have the luxury of creating comprehensive parameter space visualizations, which is precisely why surrogate modeling is needed. After identifying the active variables through screening, we can design a focused sampling plan for these key variables. This forms the foundation for building an accurate surrogate model of the objective function.

When the objective function is particularly expensive to evaluate, we might recycle the runs performed during screening for the actual model fitting step. This is most effective when some variables prove to have no impact at all. However, since completely inactive variables are rare in practice, engineers must carefully balance the trade-off between reusing expensive simulation runs and introducing potential noise into the model.

## 3.3. Designing a Sampling Plan

### 3.3.1. Stratification

A feature shared by all of the approximation models discussed in Forrester, Sóbester, and Keane (2008) is that they are more accurate in the vicinity of the points where we have evaluated the objective function. In later chapters we will delve into the laws that quantify our decaying trust in the model as we move away from a known, sampled point, but for the purposes of the present discussion we shall merely draw the intuitive conclusion that a uniform level of model accuracy throughout the design

### 3.3. Designing a Sampling Plan

space requires a uniform spread of points. A sampling plan possessing this feature is said to be space-filling.

The most straightforward way of sampling a design space in a uniform fashion is by means of a rectangular grid of points. This is the full factorial sampling technique.

Here is the simplified version of a **Python** function that will sample the unit hypercube at all levels in all dimensions, with the  $k$ -vector  $q$  containing the number of points required along each dimension, see <https://github.com/sequential-parameter-optimization/spotPython/blob/main/src/spotpython/utils/sampling.py>.

The variable **Edges** specifies whether we want the points to be equally spaced from edge to edge (**Edges=1**) or we want them to be in the centres of  $n = q_1 \times q_2 \times \dots \times q_k$  bins filling the unit hypercube (for any other value of **Edges**).

```
def fullfactorial(q_param, Edges=1) -> np.ndarray:
    """Generates a full factorial sampling plan in the unit cube.

    Args:
        q (list or np.ndarray):
            A list or array containing the number of points along each dimension (k-vector).
        Edges (int, optional):
            Determines spacing of points. If `Edges=1`, points are equally spaced from edge to edge.
            Otherwise, points will be in the centers of  $n = q[0] \times q[1] \times \dots \times q[k-1]$  bins filling the unit cube.

    Returns:
        (np.ndarray): Full factorial sampling plan as an array of shape (n, k), where n is the total number of points.

    Raises:
        ValueError: If any dimension in `q` is less than 2.
    """
    q_levels = np.array(q_param) # Use a distinct variable for original levels
    if np.min(q_levels) < 2:
        raise ValueError("You must have at least two points per dimension.")

    n = np.prod(q_levels)
    k = len(q_levels)
    X = np.zeros((n, k))

    # q_for_prod_calc is used for calculating repetitions, includes the phantom element.
    # This matches the logic of the user-provided snippet where 'q' was modified.
    q_for_prod_calc = np.append(q_levels, 1)

    for j in range(k): # k is the original number of dimensions
        # current_dim_levels is the number of levels for the current dimension j
        # In the user's snippet, q[j] correctly refers to the original level count
```

### 3. Sampling Plans

```
# as j ranges from 0 to k-1, and q_for_prod_calc[j] = q_levels[j] for this ran
current_dim_levels = q_for_prod_calc[j]

if Edges == 1:
    one_d_slice = np.linspace(0, 1, int(current_dim_levels))
else:
    # Corrected calculation for bin centers
    if current_dim_levels == 1: # Should not be hit if np.min(q_levels) >= 2
        one_d_slice = np.array([0.5])
    else:
        one_d_slice = np.linspace(1 / (2 * current_dim_levels),
                                   1 - 1 / (2 * current_dim_levels),
                                   int(current_dim_levels))

column = np.array([])
# The product q_for_prod_calc[j + 1 : k] correctly calculates
# the product of remaining original dimensions' levels.
num_consecutive_repeats = np.prod(q_for_prod_calc[j + 1 : k])

# This loop structure replicates the logic from the user's snippet
while len(column) < n:
    for ll_idx in range(int(current_dim_levels)): # Iterate through levels of
        val_to_repeat = one_d_slice[ll_idx]
        column = np.append(column, np.ones(int(num_consecutive_repeats)) * va
    X[:, j] = column
return X

q = [3, 2]
X = fullfactorial(q, Edges=0)
print(X)
```

```
[[0.16666667 0.25      ]
 [0.16666667 0.75      ]
 [0.5        0.25      ]
 [0.5        0.75      ]
 [0.83333333 0.25      ]
 [0.83333333 0.75      ]]
```

Figure 3.3 shows the points in the unit hypercube for the case of 3x2 points.

```
X = fullfactorial(q, Edges=1)
print(X)
```



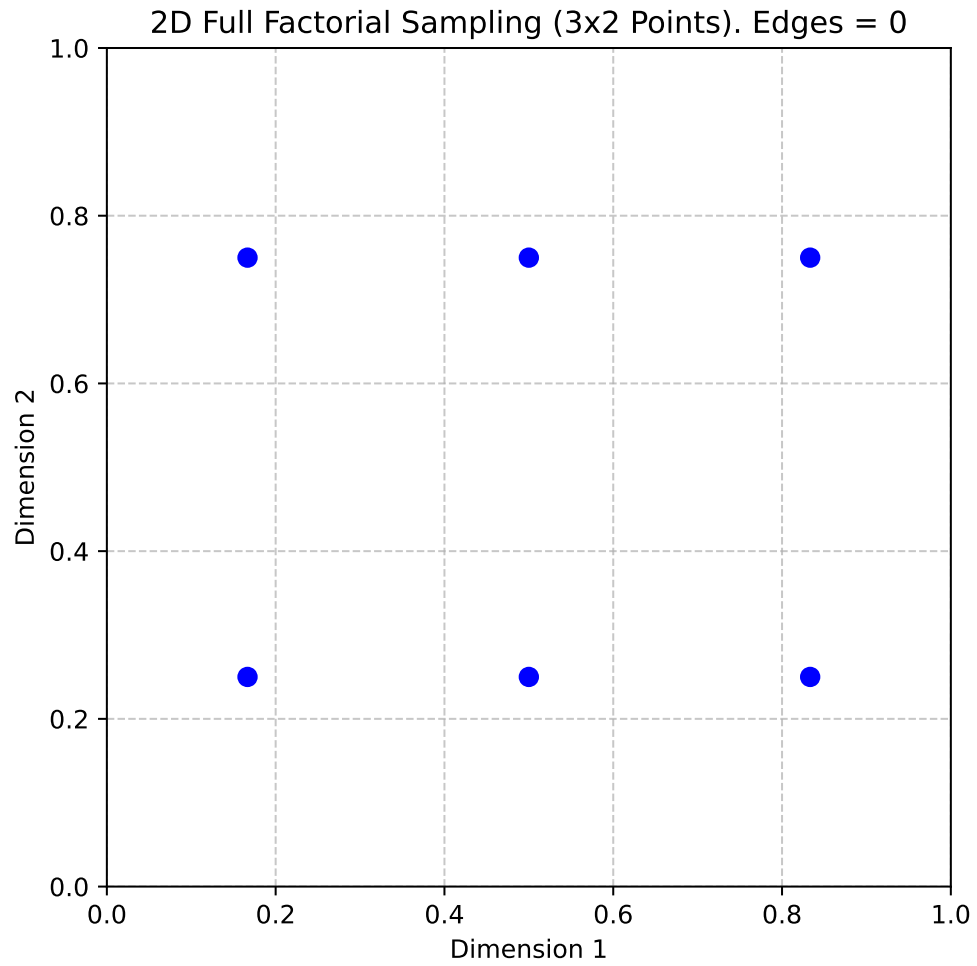


Figure 3.3.: 2D Full Factorial Sampling (3x2 Points). Edges = 0

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```
[[0.  0. ]  
 [0.  1. ]  
 [0.5 0. ]  
 [0.5 1. ]  
 [1.  0. ]  
 [1.  1. ]]
```

Figure 3.4 shows the points in the unit hypercube for the case of 3x2 points with edges.

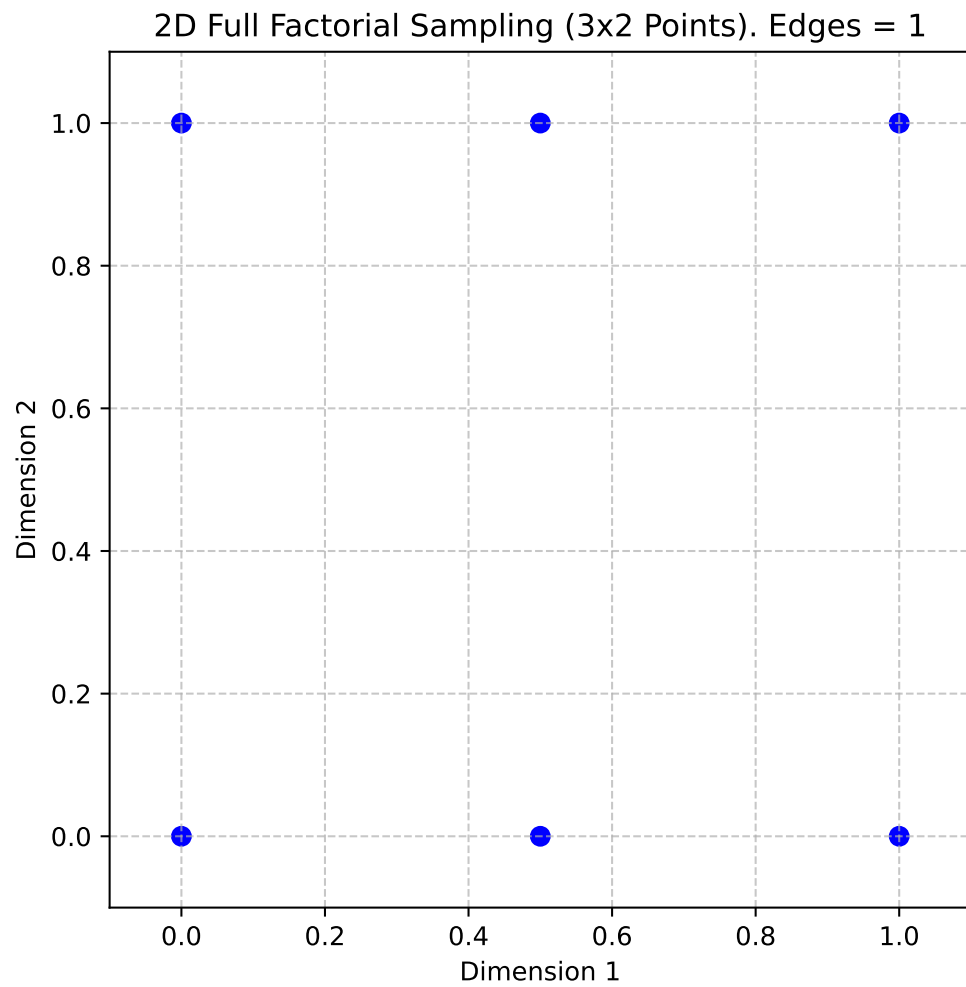


Figure 3.4.: 2D Full Factorial Sampling (3x2 Points). Edges = 1

### 3.3. Designing a Sampling Plan

The full factorial sampling plan method generates a uniform sampling design by creating a grid of points across all dimensions. For example, calling `fullfactorial([3, 4, 5], 1)` produces a three-dimensional sampling plan with 3, 4, and 5 levels along each dimension, respectively. While this approach satisfies the uniformity criterion, it has two significant limitations:

- **Restricted Design Sizes:** The method only works for designs where the total number of points  $n$  can be expressed as the product of the number of levels in each dimension, i.e.,  $n = q_1 \times q_2 \times \dots \times q_k$ .
- **Overlapping Projections:** When the sampling points are projected onto individual axes, sets of points may overlap, reducing the effectiveness of the sampling plan. This can lead to non-uniform coverage in the projections, which may not fully represent the design space.

#### 3.3.2. Latin Squares and Random Latin Hypercubes

To improve the uniformity of projections for any individual variable, the range of that variable can be divided into a large number of equal-sized bins, and random subsamples of equal size can be generated within these bins. This method is called stratified random sampling. Extending this idea to all dimensions results in a stratified sampling plan, commonly implemented using Latin hypercube sampling.

**Definition 3.3** (Latin Squares and Hypercubes). In the context of statistical sampling, a square grid containing sample positions is a Latin square if (and only if) there is only one sample in each row and each column. A Latin hypercube is the generalisation of this concept to an arbitrary number of dimensions, whereby each sample is the only one in each axis-aligned hyperplane containing it

For two-dimensional discrete variables, a Latin square ensures uniform projections. An  $(n \times n)$  Latin square is constructed by filling each row and column with a permutation of  $\{1, 2, \dots, n\}$ , ensuring each number appears only once per row and column.

**Example 3.4** (Latin Square). For  $n = 4$ , a Latin square might look like this:

|   |   |   |   |
|---|---|---|---|
| 2 | 1 | 3 | 4 |
| 3 | 2 | 4 | 1 |
| 1 | 4 | 2 | 3 |
| 4 | 3 | 1 | 2 |

Latin Hypercubes are the multidimensional extension of Latin squares. The design space is divided into equal-sized hypercubes (bins), and one point is placed in each bin. The placement ensures that moving along any axis from an occupied bin does not encounter another occupied bin. This guarantees uniform projections across all dimensions. To construct a Latin hypercube, the following steps are taken:

### 3. Sampling Plans

- Represent the sampling plan as an  $n \times k$  matrix  $X$ , where  $n$  is the number of points and  $k$  is the number of dimensions.
- Fill each column of  $X$  with random permutations of  $\{1, 2, \dots, n\}$ .
- Normalize the plan into the unit hypercube  $[0, 1]^k$ .

This approach ensures multidimensional stratification and uniformity in projections. Here is the code:

```
def rlh(n: int, k: int, edges: int = 0) -> np.ndarray:
    # Initialize array
    X = np.zeros((n, k), dtype=float)

    # Fill with random permutations
    for i in range(k):
        X[:, i] = np.random.permutation(n)

    # Adjust normalization based on the edges flag
    if edges == 1:
        # [X=0..n-1] -> [0..1]
        X = X / (n - 1)
    else:
        # Points at true midpoints
        # [X=0..n-1] -> [0.5/n..(n-0.5)/n]
        X = (X + 0.5) / n

    return X
```

**Example 3.5** (Random Latin Hypercube). The following code can be used to generate a 2D Latin hypercube with 5 points and edges=0:

```
X = rlh(n=5, k=2, edges=0)
print(X)
```

```
[[0.7 0.3]
 [0.1 0.7]
 [0.5 0.1]
 [0.9 0.5]
 [0.3 0.9]]
```

Figure 3.5 shows the points in the unit hypercube for the case of 5 points with edges=0.

**Example 3.6** (Random Latin Hypercube with Edges). The following code can be used to generate a 2D Latin hypercube with 5 points and edges=1:

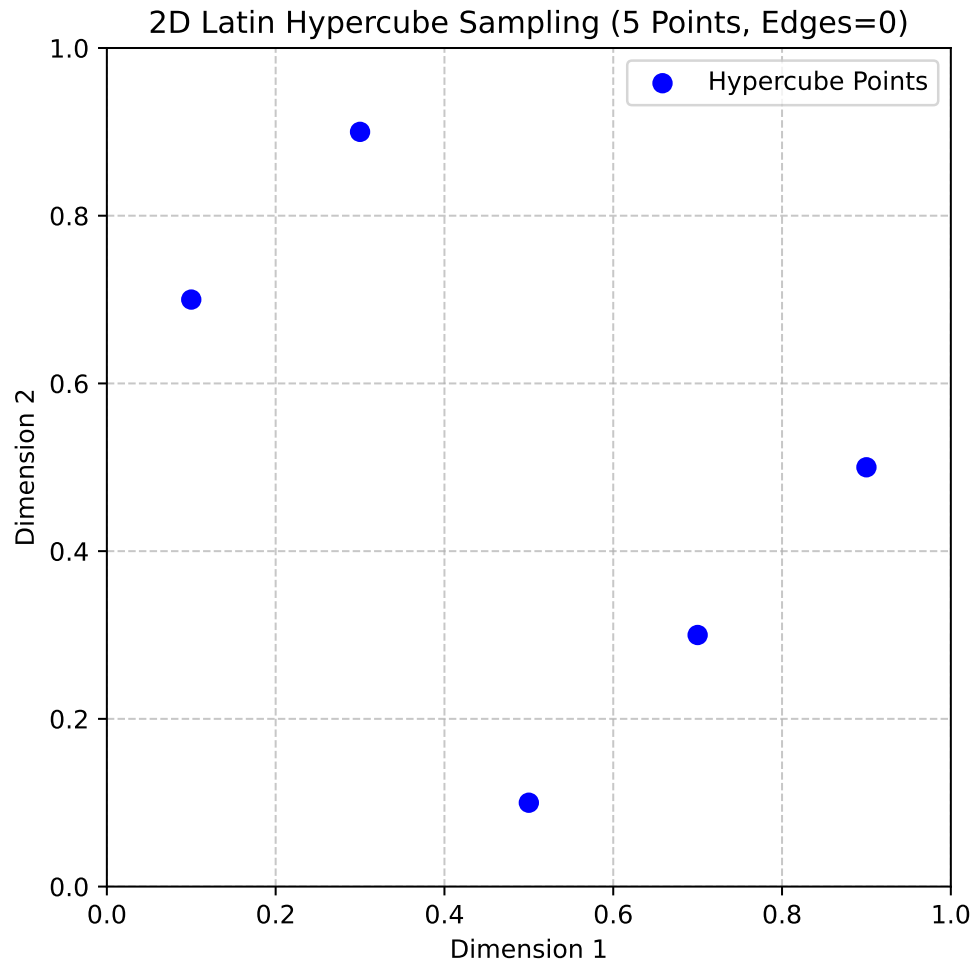


Figure 3.5.: 2D Latin Hypercube Sampling (5 Points, Edges=0)

### 3. Sampling Plans

```
X = r1h(n=5, k=2, edges=1)
print(X)
```

```
[[0.25 1.  ]
 [0.5  0.5 ]
 [1.   0.75]
 [0.75 0.  ]
 [0.   0.25]]
```

Figure 3.6 shows the points in the unit hypercube for the case of 5 points with `edges=1`.

#### 3.3.3. Space-filling Designs: Maximin Plans

A widely adopted measure for assessing the uniformity, or ‘space-fillingness’, of a sampling plan is the maximin metric, initially proposed by Johnson, Moore, and Ylvisaker (1990). This criterion can be formally defined as follows.

Consider a sampling plan  $X$ . Let  $d_1, d_2, \dots, d_m$  represent the unique distances between all possible pairs of points within  $X$ , arranged in ascending order. Furthermore, let  $J_1, J_2, \dots, J_m$  be defined such that  $J_j$  denotes the count of point pairs in  $X$  separated by the distance  $d_j$ .

**Definition 3.4** (Maximin plan). A sampling plan  $X$  is considered a maximin plan if, among all candidate plans, it maximizes the smallest inter-point distance  $d_1$ . Among plans that satisfy this condition, it further minimizes  $J_1$ , the number of pairs separated by this minimum distance.

While this definition is broadly applicable to any collection of sampling plans, our focus is narrowed to Latin hypercube designs to preserve their desirable stratification properties. However, even within this restricted class, Definition 3.4 may identify multiple equivalent maximin designs. To address this, a more comprehensive ‘tie-breaker’ definition, as proposed by Morris and Mitchell (1995), is employed:

**Definition 3.5** (Maximin plan with tie-breaker). A sampling plan  $X$  is designated as the maximin plan if it sequentially optimizes the following conditions: it maximizes  $d_1$ ; among those, it minimizes  $J_1$ ; among those, it maximizes  $d_2$ ; among those, it minimizes  $J_2$ ; and so forth, concluding with minimizing  $J_m$ .

Johnson, Moore, and Ylvisaker (1990) established that the maximin criterion (Definition 3.4) is equivalent to the D-optimality criterion used in linear regression. However, the extended maximin criterion incorporating a tie-breaker (Definition 3.5) is often preferred due to its intuitive nature and practical utility. Given that the sampling plans

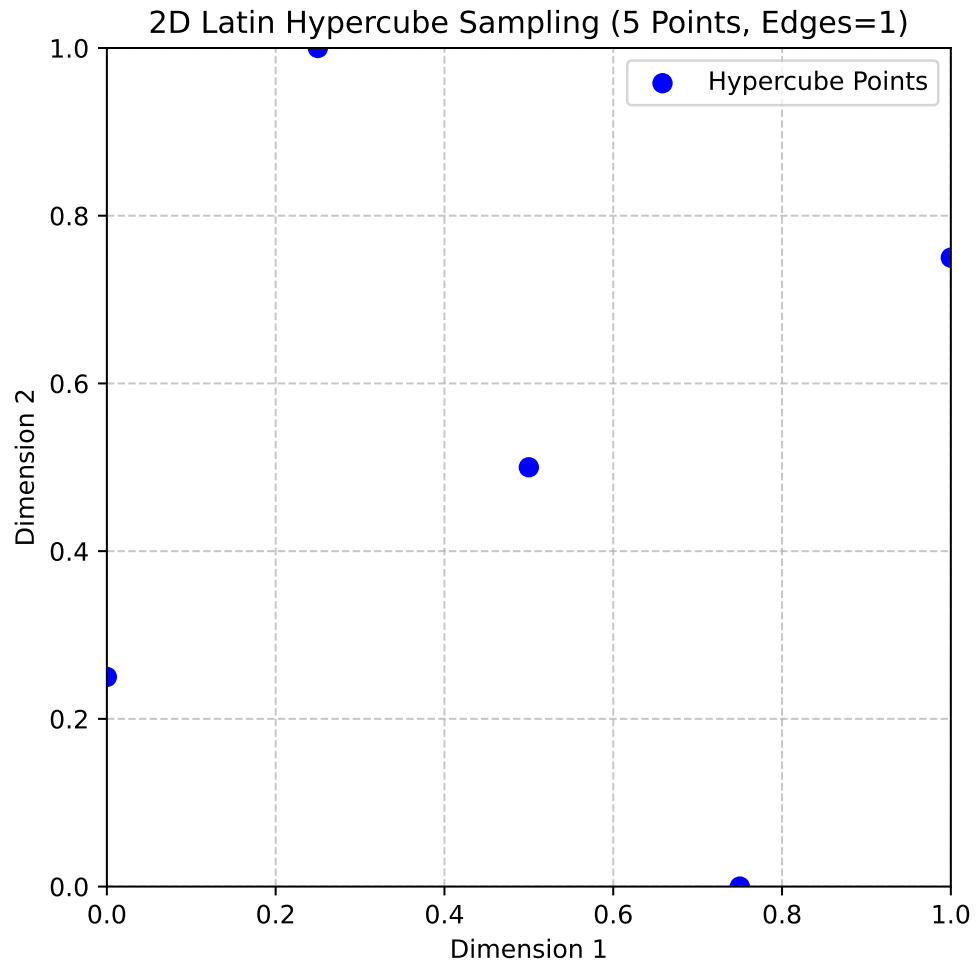


Figure 3.6.: 2D Latin Hypercube Sampling (5 Points, Edges=1)

### 3. Sampling Plans

under consideration make no assumptions about model structure, the latter criterion (Definition 3.5) will be employed.

To proceed, a precise definition of ‘distance’ within these contexts is necessary. The p-norm is the most widely adopted metric for this purpose:

**Definition 3.6** (p-norm). The p-norm of a vector  $\vec{x} = (x_1, x_2, \dots, x_k)$  is defined as:

$$d_p(\vec{x}^{(i_1)}, \vec{x}^{(i_2)}) = \left( \sum_{j=1}^k |x_j^{(i_1)} - x_j^{(i_2)}|^p \right)^{1/p}. \quad (3.2)$$

When  $p = 1$ , Equation 3.2 defines the rectangular distance, occasionally referred to as the Manhattan norm (an allusion to a grid-like city layout). Setting  $p = 2$  yields the Euclidean norm. The existing literature offers limited evidence to suggest the superiority of one norm over the other for evaluating sampling plans when no model structure assumptions are made. It is important to note, however, that the rectangular distance is considerably less computationally demanding. This advantage can be quite significant, particularly when evaluating large sampling plans.

For the computational implementation of Definition 3.5, the initial step involves constructing the vectors  $d_1, d_2, \dots, d_m$  and  $J_1, J_2, \dots, J_m$ . The `jd` function facilitates this task.

#### 3.3.3.1. The Function `jd`

The function `jd` computes the distinct p-norm distances between all pairs of points in a given set and counts their occurrences. It returns two arrays: one for the distinct distances and another for their multiplicities.

```
def jd(X: np.ndarray, p: float = 1.0) -> Tuple[np.ndarray, np.ndarray]:
    """
    Args:
        X (np.ndarray):
            A 2D array of shape (n, d) representing n points
            in d-dimensional space.
        p (float, optional):
            The distance norm to use.
            p=1 uses the Manhattan (L1) norm, while p=2 uses the
            Euclidean (L2) norm. Defaults to 1.0 (Manhattan norm).

    Returns:
        (np.ndarray, np.ndarray):
            A tuple (J, distinct_d), where:
            - distinct_d is a 1D float array of unique,
```



```

        sorted distances between points.
        - J is a 1D integer array that provides
        the multiplicity (occurrence count)
        of each distance in distinct_d.
    """
    n = X.shape[0]

    # Allocate enough space for all pairwise distances
    # (n*(n-1))/2 pairs for an n-point set
    pair_count = n * (n - 1) // 2
    d = np.zeros(pair_count, dtype=float)

    # Fill the distance array
    idx = 0
    for i in range(n - 1):
        for j in range(i + 1, n):
            # Compute the p-norm distance
            d[idx] = np.linalg.norm(X[i] - X[j], ord=p)
            idx += 1

    # Find unique distances and their multiplicities
    distinct_d = np.unique(d)
    J = np.zeros_like(distinct_d, dtype=int)
    for i, val in enumerate(distinct_d):
        J[i] = np.sum(d == val)
    return J, distinct_d

```

**Example 3.7** (The Function `jd`). Consider a small 3-point set in 2D space, with points located at (0,0), (1,1), and (2,2) as shown in Figure 3.7. The distinct distances and their occurrences can be computed using the `jd` function, as shown in the following code:

```

J, distinct_d = jd(X, p=2.0)
print("Distinct distances (d_i):", distinct_d)
print("Occurrences (J_i):", J)

```

```

Distinct distances (d_i): [1.41421356 2.82842712]
Occurrences (J_i): [2 1]

```

### 3.3.4. Memory Management

A computationally intensive part of the calculation performed with the `jd`-function is the creation of the vector  $\vec{d}$  containing all pairwise distances. This is particularly true

### 3. Sampling Plans

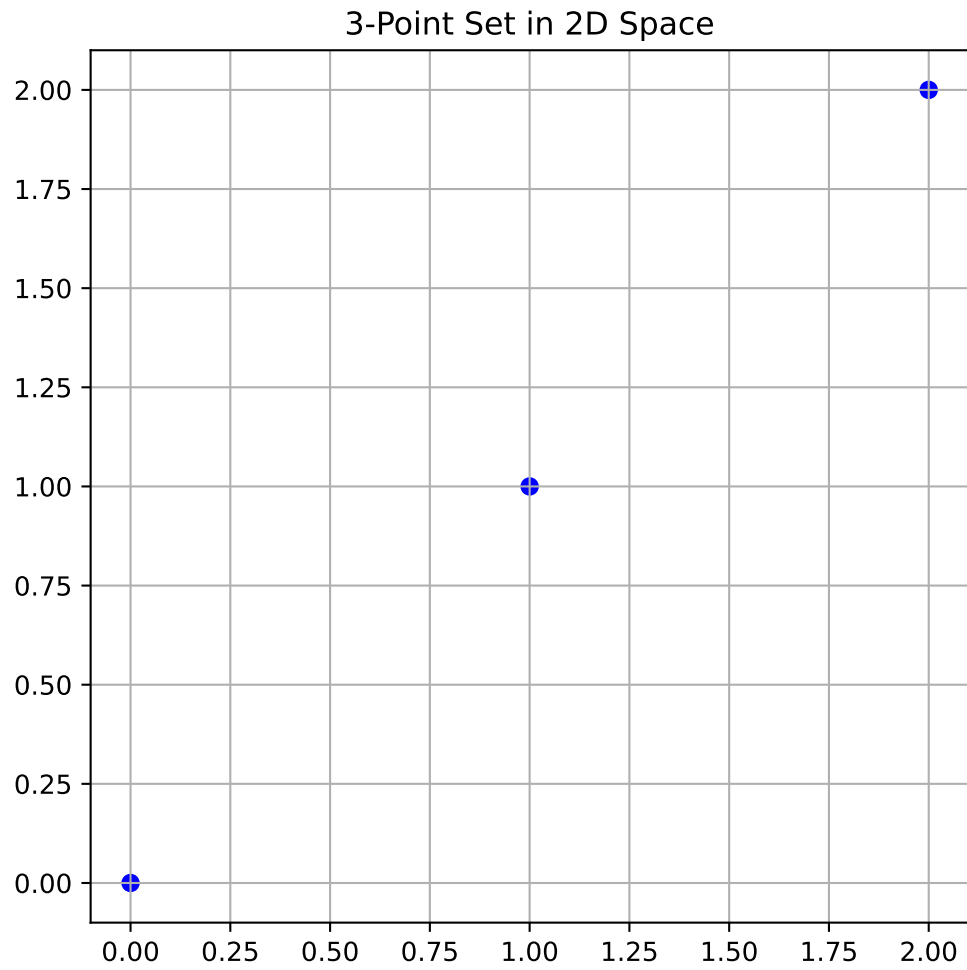


Figure 3.7.: 3-Point Set in 2D Space

### 3.3. Designing a Sampling Plan

for large sampling plans; for instance, a 1000-point plan requires nearly half a million distance calculations.

**Definition 3.7** (Pre-allocation of Memory). Pre-allocation of memory is a programming technique where a fixed amount of memory is reserved for a data structure (like an array or vector) before it is actually filled with data. This is done to avoid the computational overhead associated with dynamic memory allocation, which involves repeatedly requesting and resizing memory as new elements are added.

Consequently, pre-allocating memory for the distance vector  $\vec{d}$  is essential. This necessitates a slightly less direct method for computing the indices of  $\vec{d}$ , rather than appending each new element, which would involve costly dynamic memory allocation.

The implementation of Definition 3.5 is now required. Finding the most space-filling design involves pairwise comparisons. This problem can be approached using a ‘divide and conquer’ strategy, simplifying it to the task of selecting the better of two sampling plans. The function `mm(X1,X2,p)` is designed for this purpose. It returns an index indicating which of the two designs is more space-filling, or 0 if they are equally space-filling, based on the  $p$ -norm for distance computation.

#### 3.3.4.1. The Function `mm`

The function `mm` compares two sampling plans based on the Morris-Mitchell criterion. It uses the `jd` function to compute the distances and multiplicities, constructs vectors for comparison, and determines which plan is more space-filling.

```
def mm(X1: np.ndarray, X2: np.ndarray, p: Optional[float] = 1.0) -> int:
    """
    Args:
        X1 (np.ndarray): A 2D array representing the first sampling plan.
        X2 (np.ndarray): A 2D array representing the second sampling plan.
        p (float, optional): The distance metric. p=1 uses Manhattan (L1) distance,
            while p=2 uses Euclidean (L2). Defaults to 1.0.

    Returns:
        int:
            - 0 if both plans are identical or equally space-filling
            - 1 if X1 is more space-filling
            - 2 if X2 is more space-filling
    """
    X1_sorted = X1[np.lexsort(np.rot90(X1))]
    X2_sorted = X2[np.lexsort(np.rot90(X2))]
    if np.array_equal(X1_sorted, X2_sorted):
        return 0 # Identical sampling plans
```

### 3. Sampling Plans

```
# Compute distance multiplicities for each plan
J1, d1 = jd(X1, p)
J2, d2 = jd(X2, p)
m1, m2 = len(d1), len(d2)

# Construct V1 and V2: alternate distance and negative multiplicity
V1 = np.zeros(2 * m1)
V1[0::2] = d1
V1[1::2] = -J1

V2 = np.zeros(2 * m2)
V2[0::2] = d2
V2[1::2] = -J2

# Trim the longer vector to match the size of the shorter
m = min(m1, m2)
V1 = V1[:m]
V2 = V2[:m]

# Compare element-by-element:
# c[i] = 1 if V1[i] > V2[i], 2 if V1[i] < V2[i], 0 otherwise.
c = (V1 > V2).astype(int) + 2 * (V1 < V2).astype(int)

if np.sum(c) == 0:
    # Equally space-filling
    return 0
else:
    # The first non-zero entry indicates which plan is better
    idx = np.argmax(c != 0)
    return c[idx]
```

**Example 3.8** (The Function `mm`). We can use the `mm` function to compare two sampling plans. The following code creates two 3-point sampling plans in 2D (shown in Figure 3.8) and compares them using the Morris-Mitchell criterion:

```
X1 = np.array([[0.0, 0.0], [0.5, 0.5], [0.0, 1.0], [1.0, 1.0]])
X2 = np.array([[0.1, 0.1], [0.4, 0.6], [0.1, 0.9], [0.9, 0.9]])
```

We can compare which plan has better space-filling (Morris-Mitchell). The output is either 0, 1, or 2 depending on which plan is more space-filling.

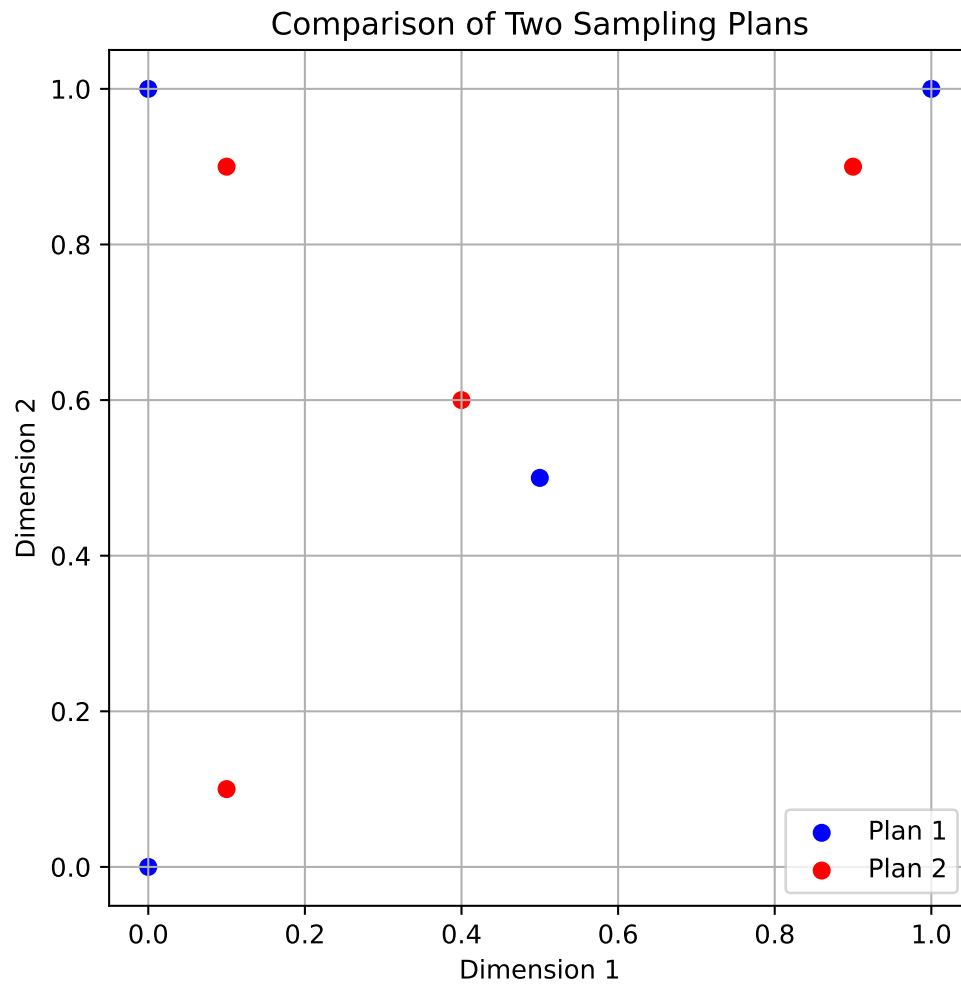


Figure 3.8.: Comparison of Two Sampling Plans

### 3. Sampling Plans

```
better = mm(X1, X2, p=2.0)
print(f"Plan {better} is more space-filling.")
```

Plan 1 is more space-filling.

#### 3.3.4.2. The Function `mmphi`

Searching across a space of potential sampling plans can be accomplished by pairwise comparisons. An optimization algorithm could, in theory, be written with `mm` as the comparative objective. However, experimental evidence (Morris and Mitchell 1995) suggests that the resulting optimization landscape can be quite deceptive, making it difficult to search reliably. This difficulty arises because the comparison process terminates upon finding the first non-zero element in the comparison array `c`. Consequently, the remaining values in the distance  $(d_1, d_2, \dots, d_m)$  and multiplicity  $(J_1, J_2, \dots, J_m)$  arrays are disregarded. These disregarded values, however, might contain potentially useful ‘slope’ information about the global landscape for the optimization process.

To address this, Morris and Mitchell (1995) defined the following scalar-valued criterion function, which is used to rank competing sampling plans. This function, while based on the logic of Definition 3.5, incorporates the complete vectors  $d_1, d_2, \dots, d_m$  and  $J_1, J_2, \dots, J_m$ .

**Definition 3.8** (Morris-Mitchell Criterion). The Morris-Mitchell criterion is defined as:

$$\Phi_q(X) = \left( \sum_{j=1}^m J_j d_j^{-q} \right)^{1/q}, \quad (3.3)$$

where  $X$  is the sampling plan,  $d_j$  is the distance between points,  $J_j$  is the multiplicity of that distance, and  $q$  is a user-defined exponent. The parameter  $q$  can be adjusted to control the influence of smaller distances on the overall metric.

The smaller the value of  $\Phi_q$ , the better the space-filling properties of  $X$  will be.

The function `mmphi` computes the Morris-Mitchell sampling plan quality criterion for a given sampling plan. It takes a 2D array of points and calculates the space-fillingness metric based on the distances between points. This can be implemented in Python as follows:

### 3.3. Designing a Sampling Plan

```
def mmphi(X: np.ndarray,
          q: Optional[float] = 2.0,
          p: Optional[float] = 1.0) -> float:
    """
    Args:
        X (np.ndarray):
            A 2D array representing the sampling plan,
            where each row is a point in
            d-dimensional space (shape: (n, d)).
        q (float, optional):
            Exponent used in the computation of the metric.
            Defaults to 2.0.
        p (float, optional):
            The distance norm to use.
            For example, p=1 is Manhattan (L1),
            p=2 is Euclidean (L2). Defaults to 1.0.

    Returns:
        float:
            The space-fillingness metric  $\Phi_q$ . Larger values typically indicate a more
            space-filling plan according to the Morris-Mitchell criterion.
    """
    # Compute the distance multiplicities: J, and unique distances: d
    J, d = jd(X, p)
    # Summation of  $J[i] * d[i]^{-q}$ , then raised to  $1/q$ 
    # This follows the Morris-Mitchell definition.
     $\Phi_q$  = np.sum(J * (d ** (-q))) ** (1.0 / q)
    return  $\Phi_q$ 
```

**Example 3.9** (The Function `mmphi`). We can use the `mmphi` function to evaluate the space-filling quality of the two sampling plans from Example 3.8. The following code uses these two 3-point sampling plans in 2D and computes their quality using the Morris-Mitchell criterion:

```
# Two simple sampling plans from above
quality1 = mmphi(X1, q=2, p=2)
quality2 = mmphi(X2, q=2, p=2)
print(f"Quality of sampling plan X1: {quality1}")
print(f"Quality of sampling plan X2: {quality2}")
```

```
Quality of sampling plan X1: 2.91547594742265
Quality of sampling plan X2: 3.917162046269215
```

### 3. Sampling Plans

This equation provides a more compact representation of the maximin criterion, but the selection of the  $q$  value is an important consideration. Larger values of  $q$  ensure that terms in the sum corresponding to smaller inter-point distances (the  $d_j$  values, which are sorted in ascending order) have a dominant influence. As a result,  $\Phi_q$  will rank sampling plans in a way that closely emulates the original maximin definition (Definition 3.5). This implies that the optimization landscape might retain the challenging characteristics that the  $\Phi_q$  metric, especially with smaller  $q$  values, is intended to alleviate. Conversely, smaller  $q$  values tend to produce a  $\Phi_q$  landscape that, while not perfectly aligning with the original definition, is generally more conducive to optimization.

To illustrate the relationship between Equation 3.3 and the maximin criterion of Definition 3.5, sets of 50 random Latin hypercubes of varying sizes and dimensionalities were considered by Forrester, Sóbester, and Keane (2008). The correlation plots from this analysis suggest that as the sampling plan size increases, a smaller  $q$  value is needed for the  $\Phi_q$ -based ranking to closely match the ranking derived from Definition 3.5.

Rankings based on both the direct maximin comparison (`mm`) and the  $\Phi_q$  metric (`mmphi`), determined using a simple bubble sort algorithm, are implemented in the Python function `mmsort`.

#### 3.3.4.3. The Function `mmsort`

The function `mmsort` is designed to rank multiple sampling plans based on their space-filling properties using the Morris-Mitchell criterion. It takes a 3D array of sampling plans and returns the indices of the plans sorted in ascending order of their space-filling quality.

```
def mmsort(X3D: np.ndarray, p: Optional[float] = 1.0) -> np.ndarray:
    """
    Args:
        X3D (np.ndarray):
            A 3D NumPy array of shape (n, d, m), where m is the number of
            sampling plans, and each plan is an (n, d) matrix of points.
        p (float, optional):
            The distance metric to use. p=1 for Manhattan (L1), p=2 for
            Euclidean (L2). Defaults to 1.0.

    Returns:
        np.ndarray:
            A 1D integer array of length m that holds the plan indices in
            ascending order of space-filling quality. The first index in the
            returned array corresponds to the most space-filling plan.
    """
    # Number of plans (m)
    m = X3D.shape[2]
```



### 3.3. Designing a Sampling Plan

```
# Create index array (1-based to match original MATLAB convention)
Index = np.arange(1, m + 1)

swap_flag = True
while swap_flag:
    swap_flag = False
    i = 0
    while i < m - 1:
        # Compare plan at Index[i] vs. Index[i+1] using mm()
        # Note: subtract 1 from each index to convert to 0-based array indexing
        if mm(X3D[:, :, Index[i] - 1], X3D[:, :, Index[i + 1] - 1], p) == 2:
            # Swap indices if the second plan is more space-filling
            Index[i], Index[i + 1] = Index[i + 1], Index[i]
            swap_flag = True
        i += 1

return Index
```

**Example 3.10** (The Function `mmsort`). The `mmsort` function can be used to rank multiple sampling plans based on their space-filling properties. The following code demonstrates how to use `mmsort` to compare two 3-point sampling plans in 3D space:

Suppose we have two 3-point sampling plans  $X_1$  and  $X_2$  from above. They are sorted using the Morris-Mitchell criterion with  $p = 2.0$ . For example, the output `[1, 2]` indicates that  $X_1$  is more space-filling than  $X_2$ :

```
X3D = np.stack([X1, X2], axis=2)
ranking = mmsort(X3D, p=2.0)
print(ranking)
```

`[1 2]`

To determine the optimal Latin hypercube for a specific application, a recommended approach by Morris and Mitchell (1995) involves minimizing  $\Phi_q$  for a set of  $q$  values (1, 2, 5, 10, 20, 50, and 100). Subsequently, the best plan from these results is selected based on the actual maximin definition. The `mmsort` function can be utilized for this purpose: a 3D matrix,  $X3D$ , can be constructed where each 2D slice represents the best sampling plan found for each  $\Phi_q$ . Applying `mmsort(X3D, 1)` then ranks these plans according to Definition 3.5, using the rectangular distance metric. The subsequent discussion will address the methods for finding these optimized  $\Phi_q$  designs.

### 3. Sampling Plans

#### 3.3.4.4. The Function phisort

`phisort` only differs from `mmsort` in having  $q$  as an additional argument, as well as the comparison line being:

```
if mmphi(X3D[:, :, Index[i] - 1], q=q, p=p) >
    mmphi(X3D[:, :, Index[i + 1] - 1], q=q, p=p):
```

```
def phisort(X3D: np.ndarray,
            q: Optional[float] = 2.0,
            p: Optional[float] = 1.0) -> np.ndarray:
    """
    Args:
        X3D (np.ndarray):
            A 3D array of shape (n, d, m),
            where m is the number of sampling plans.
        q (float, optional):
            Exponent for the mmphi metric. Defaults to 2.0.
        p (float, optional):
            Distance norm for mmphi.
            p=1 is Manhattan; p=2 is Euclidean.
            Defaults to 1.0.

    Returns:
        np.ndarray:
            A 1D integer array of length m, giving the plan indices in ascending
            order of mmphi. The first index in the returned array corresponds
            to the numerically lowest mmphi value.
    """
    # Number of 2D sampling plans
    m = X3D.shape[2]
    # Create a 1-based index array
    Index = np.arange(1, m + 1)
    # Bubble-sort: plan with lower mmphi() climbs toward the front
    swap_flag = True
    while swap_flag:
        swap_flag = False
        for i in range(m - 1):
            # Retrieve mmphi values for consecutive plans
            val_i = mmphi(X3D[:, :, Index[i] - 1], q=q, p=p)
            val_j = mmphi(X3D[:, :, Index[i + 1] - 1], q=q, p=p)

            # Swap if the left plan's mmphi is larger (i.e. 'worse')
            if val_i > val_j:
```

```

        Index[i], Index[i + 1] = Index[i + 1], Index[i]
        swap_flag = True
    return Index

```

**Example 3.11** (The Function `phisort`). The `phisort` function can be used to rank multiple sampling plans based on the Morris-Mitchell criterion. The following code demonstrates how to use `phisort` to compare two 3-point sampling plans in 3D space:

```

X1 = bestlh(n=5, k=2, population=5, iterations=10)
X2 = bestlh(n=5, k=2, population=15, iterations=20)
X3 = bestlh(n=5, k=2, population=25, iterations=30)
# Map X1 and X2 so that X3D has the two sampling plans
# in X3D[:, :, 0] and X3D[:, :, 1]
X3D = np.array([X1, X2])
print(phisort(X3D))
X3D = np.array([X3, X2])
print(phisort(X3D))

```

```

[2 1]
[2 1]

```

### 3.3.5. Optimizing the Morris-Mitchell Criterion $\Phi_q$

Once a criterion for assessing the quality of a Latin hypercube sampling plan has been established, a systematic method for optimizing this metric across the space of Latin hypercubes is required. This task is non-trivial; as the reader may recall from the earlier discussion on Latin squares, this search space is vast. In fact, its vastness means that for many practical applications, locating the globally optimal solution is often infeasible. Therefore, the objective becomes finding the best possible sampling plan achievable within a specific computational time budget.

This budget is influenced by the computational cost associated with obtaining each objective function value. Determining the optimal allocation of total computational effort—between generating the sampling plan and actually evaluating the objective function at the selected points—remains an open research question. However, it is typical for no more than approximately 5% of the total available time to be allocated to the task of generating the sampling plan itself.

Forrester, Sóbester, and Keane (2008) draw an analogy to the process of devising a revision timetable before an exam. While a well-structured timetable enhances the effectiveness of revision, an excessive amount of the revision time itself should not be consumed by the planning phase.

### 3. Sampling Plans

A significant challenge in devising a sampling plan optimizer is ensuring that the search process remains confined to the space of valid Latin hypercubes. As previously discussed, the defining characteristic of a Latin hypercube  $X$  is that each of its columns represents a permutation of the possible levels for the corresponding variable. Consequently, the smallest modification that can be applied to a Latin hypercube—without compromising its crucial multidimensional stratification property—involves swapping two elements within any single column of  $X$ . A Python implementation for ‘mutating’ a Latin hypercube through such an operation, generalized to accommodate random changes applied to multiple sites, is provided below:

#### 3.3.5.1. The Function `perturb()`

The function `perturb` randomly swaps elements in a Latin hypercube sampling plan. It takes a 2D array representing the sampling plan and performs a specified number of random element swaps, ensuring that the result remains a valid Latin hypercube.

```
def perturb(X: np.ndarray,
            PertNum: Optional[int] = 1) -> np.ndarray:
    """
    Args:
        X (np.ndarray):
            A 2D array (sampling plan) of shape (n, k),
            where each row is a point
            and each column is a dimension.
        PertNum (int, optional):
            The number of element swaps (perturbations)
            to perform. Defaults to 1.

    Returns:
        np.ndarray:
            The perturbed sampling plan,
            identical in shape to the input, with
            one or more random column swaps executed.
    """
    # Get dimensions of the plan
    n, k = X.shape
    if n < 2 or k < 2:
        raise ValueError("Latin hypercubes require at least 2 points and 2 dimensions")
    for _ in range(PertNum):
        # Pick a random column
        col = int(np.floor(np.random.rand() * k))
        # Pick two distinct row indices
        el1, el2 = 0, 0
        while el1 == el2:
```

```

    e11 = int(np.floor(np.random.rand() * n))
    e12 = int(np.floor(np.random.rand() * n))
    # Swap the two selected elements in the chosen column
    X[e11, col], X[e12, col] = X[e12, col], X[e11, col]
return X

```

**Example 3.12** (The Function `perturb()`). The `perturb` function can be used to randomly swap elements in a Latin hypercube sampling plan. The following code demonstrates how to use `perturb` to create a perturbed version of a 4x2 sampling plan:

```

X_original = np.array([[1, 3],[2, 4],[3, 1],[4, 2]])
print("Original Sampling Plan:")
print(X_original)
print("Perturbed Sampling Plan:")
X_perturbed = perturb(X_original, PertNum=1)
print(X_perturbed)

```

Original Sampling Plan:

```

[[1 3]
 [2 4]
 [3 1]
 [4 2]]

```

Perturbed Sampling Plan:

```

[[1 4]
 [2 3]
 [3 1]
 [4 2]]

```

Forrester, Sóbester, and Keane (2008) uses the term ‘mutation’, because this problem lends itself to nature-inspired computation. Morris and Mitchell (1995) use a simulated annealing algorithm, the detailed pseudocode of which can be found in their paper. As an alternative, a method based on evolutionary operation (EVOP) is offered by Forrester, Sóbester, and Keane (2008).

### 3.3.6. Evolutionary Operation

As introduced by Box (1957), evolutionary operation was designed to optimize chemical processes. The current parameters of the reaction would be recorded in a box at the centre of a board, with a series of ‘offspring’ boxes along the edges containing values of the parameters slightly altered with respect to the central, ‘parent’ values. Once the reaction was completed for all of these sets of variable values and the corresponding yields recorded, the contents of the central box would be replaced with that of the

### 3. Sampling Plans

setup with the highest yield and this would then become the parent of a new set of peripheral boxes.

This is generally viewed as a local search procedure, though this depends on the mutation step sizes, that is on the differences between the parent box and its offspring. The longer these steps, the more global is the scope of the search.

For the purposes of the Latin hypercube search, a variable scope strategy is applied. The process starts with a long step length (that is a relatively large number of swaps within the columns) and, as the search progresses, the current best basin of attraction is gradually approached by reducing the step length to a single change.

In each generation the parent is mutated (randomly, using the `perturb` function) a pertnum number of times. The sampling plan that yields the smallest  $\Phi_q$  value (as per the Morris-Mitchell criterion, calculated using `mmphi`) among all offspring and the parent is then selected; in evolutionary computation parlance this selection philosophy is referred to as elitism.

The EVOP based search for space-filling Latin hypercubes is thus a truly evolutionary process: the optimized sampling plan results from the nonrandom survival of random variations.

#### 3.3.7. Putting it all Together

All the pieces of the optimum Latin hypercube sampling process puzzle are now in place: the random hypercube generator as a starting point for the optimization process, the ‘spacefillingness’ metric that needs to be optimized, the optimization engine that performs this task and the comparison function that selects the best of the optima found for the various  $q$ ’s. These pieces just need to be put into a sequence. Here is the Python embodiment of the completed puzzle. It results in a function `bestlh` that uses the function `mmlhs` to find the best Latin hypercube sampling plan for a given set of parameters.

##### 3.3.7.1. The Function `mmlhs`

Performs an evolutionary search (using perturbations) to find a Morris-Mitchell optimal Latin hypercube, starting from an initial plan `X_start`.

This function does the following:

1. Initializes a “best” Latin hypercube (`X_best`) from the provided `X_start`.
2. Iteratively perturbs `X_best` to create offspring.
3. Evaluates the space-fillingness of each offspring via the Morris-Mitchell metric (using `mmphi`).
4. Updates the best plan whenever a better offspring is found.

### 3.3. Designing a Sampling Plan

```
def mmlhs(X_start: np.ndarray,
          population: int,
          iterations: int,
          q: Optional[float] = 2.0,
          plot=False) -> np.ndarray:
    """
    Args:
        X_start (np.ndarray):
            A 2D array of shape (n, k) providing the initial Latin hypercube
            (n points in k dimensions).
        population (int):
            Number of offspring to create in each generation.
        iterations (int):
            Total number of generations to run the evolutionary search.
        q (float, optional):
            The exponent used by the Morris-Mitchell space-filling criterion.
            Defaults to 2.0.
        plot (bool, optional):
            If True, a simple scatter plot of the first two dimensions will be
            displayed at each iteration. Only if k >= 2. Defaults to False.

    Returns:
        np.ndarray:
            A 2D array representing the most space-filling Latin hypercube found
            after all iterations, of the same shape as X_start.
    """
    n = X_start.shape[0]
    if n < 2:
        raise ValueError("Latin hypercubes require at least 2 points")
    k = X_start.shape[1]
    if k < 2:
        raise ValueError("Latin hypercubes are not defined for dim k < 2")
    # Initialize best plan and its metric
    X_best = X_start.copy()
    Phi_best = mmphi(X_best, q=q)
    # After 85% of iterations, reduce the mutation rate to 1
    leveloff = int(np.floor(0.85 * iterations))
    for it in range(1, iterations + 1):
        # Decrease number of mutations over time
        if it < leveloff:
            mutations = int(round(1 + (0.5 * n - 1) * (leveloff - it) / (leveloff - 1)))
        else:
            mutations = 1
        X_improved = X_best.copy()
```

### 3. Sampling Plans

```
Phi_improved = Phi_best
# Create offspring, evaluate, and keep the best
for _ in range(population):
    X_try = perturb(X_best.copy(), mutations)
    Phi_try = mmphi(X_try, q=q)

    if Phi_try < Phi_improved:
        X_improved = X_try
        Phi_improved = Phi_try
# Update the global best if we found a better plan
if Phi_improved < Phi_best:
    X_best = X_improved
    Phi_best = Phi_improved
# Simple visualization of the first two dimensions
if plot and (X_best.shape[1] >= 2):
    plt.clf()
    plt.scatter(X_best[:, 0], X_best[:, 1], marker="o")
    plt.grid(True)
    plt.title(f"Iteration {it} - Current Best Plan")
    plt.pause(0.01)
return X_best
```

**Example 3.13** (The Function `mmlhs`). The `mmlhs` function can be used to optimize a Latin hypercube sampling plan. The following code demonstrates how to use `mmlhs` to optimize a 4x2 Latin hypercube starting from an initial plan:

```
# Suppose we have an initial 4x2 plan
X_start = np.array([[0.1, 0.3], [.1, .4], [.2, .9], [.9, .2]])
print("Initial plan:")
print(X_start)
# Search for a more space-filling plan
X_opt = mmlhs(X_start, population=10, iterations=100, q=2)
print("Optimized plan:")
print(X_opt)
```

```
Initial plan:
[[0.1 0.3]
 [0.1 0.4]
 [0.2 0.9]
 [0.9 0.2]]
Optimized plan:
[[0.9 0.3]
 [0.2 0.4]]
```



### 3.3. Designing a Sampling Plan

```
[0.1 0.9]  
[0.1 0.2]]
```

Figure 3.9 shows the initial and optimized plans in 2D. The blue points represent the initial plan, while the red points represent the optimized plan.

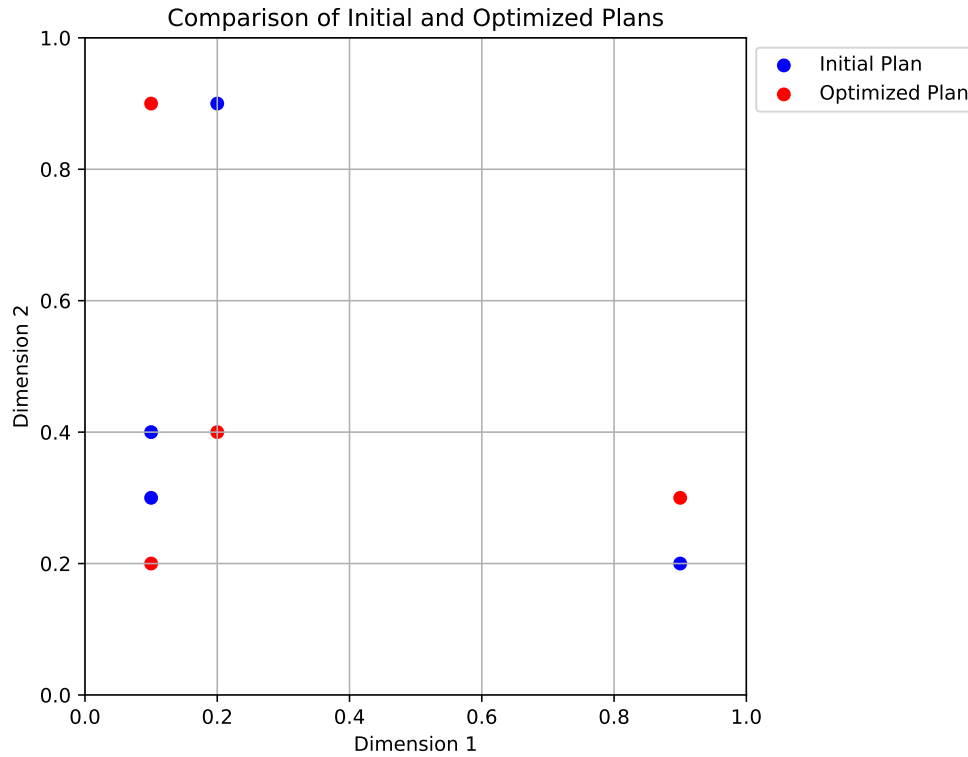


Figure 3.9.: Comparison of the initial and optimized plans in 2D.

#### 3.3.7.2. The Function `bestlh`

Generates an optimized Latin hypercube by evolving the Morris-Mitchell criterion across multiple exponents (q values) and selecting the best plan.

```
def bestlh(n: int,  
           k: int,  
           population: int,  
           iterations: int,  
           p=1,
```

### 3. Sampling Plans

```
        plot=False,
        verbosity=0,
        edges=0,
        q_list=[1, 2, 5, 10, 20, 50, 100]) -> np.ndarray:
"""
Args:
    n (int):
        Number of points required in the Latin hypercube.
    k (int):
        Number of design variables (dimensions).
    population (int):
        Number of offspring in each generation of the evolutionary search.
    iterations (int):
        Number of generations for the evolutionary search.
    p (int, optional):
        The distance norm to use. p=1 for Manhattan (L1), p=2 for Euclidean (L2).
        Defaults to 1 (faster than 2).
    plot (bool, optional):
        If True, a scatter plot of the optimized plan in the first two dimensions
        will be displayed. Only if k>=2. Defaults to False.
    verbosity (int, optional):
        Verbosity level. 0 is silent, 1 prints the best q value found. Defaults to 0.
    edges (int, optional):
        If 1, places centers of the extreme bins at the domain edges ([0,1]).
        Otherwise, bins are fully contained within the domain, i.e. midpoints.
        Defaults to 0.
    q_list (list, optional):
        A list of q values to optimize. Defaults to [1, 2, 5, 10, 20, 50, 100].
        These values are used to evaluate the space-fillingness of the Latin
        hypercube. The best plan is selected based on the lowest mmphi value.

Returns:
    np.ndarray:
        A 2D array of shape (n, k) representing an optimized Latin hypercube.
"""
if n < 2:
    raise ValueError("Latin hypercubes require at least 2 points")
if k < 2:
    raise ValueError("Latin hypercubes are not defined for dim k < 2")

# A list of exponents (q) to optimize

# Start with a random Latin hypercube
X_start = rlh(n, k, edges=edges)
```

### 3.3. Designing a Sampling Plan

```
# Allocate a 3D array to store the results for each q
# (shape: (n, k, number_of_q_values))
X3D = np.zeros((n, k, len(q_list)))

# Evolve the plan for each q in q_list
for i, q_val in enumerate(q_list):
    if verbosity > 0:
        print(f"Now optimizing for q={q_val}...")
    X3D[:, :, i] = mmlhs(X_start, population, iterations, q_val)

# Sort the set of evolved plans according to the Morris-Mitchell criterion
index_order = mmsort(X3D, p=p)

# index_order is a 1-based array of plan indices; the first element is the best
best_idx = index_order[0] - 1
if verbosity > 0:
    print(f"Best lh found using q={q_list[best_idx]}...")

# The best plan in 3D array order
X = X3D[:, :, best_idx]

# Plot the first two dimensions
if plot and (k >= 2):
    plt.scatter(X[:, 0], X[:, 1], c="r", marker="o")
    plt.title(f"Morris-Mitchell optimum plan found using q={q_list[best_idx]}")
    plt.xlabel("x_1")
    plt.ylabel("x_2")
    plt.grid(True)
    plt.show()

return X
```

**Example 3.14** (The Function `bestlh`). The `bestlh` function can be used to generate an optimized Latin hypercube sampling plan. The following code demonstrates how to use `bestlh` to create a 5x2 Latin hypercube with a population of 5 and 10 iterations:

```
Xbestlh= bestlh(n=5, k=2, population=5, iterations=10)
```

Figure 3.10 shows the best Latin hypercube sampling in 2D. The red points represent the optimized plan.

Sorting all candidate plans in ascending order is not strictly necessary - after all, only the best one is truly of interest. Nonetheless, the added computational complexity is

### 3. Sampling Plans

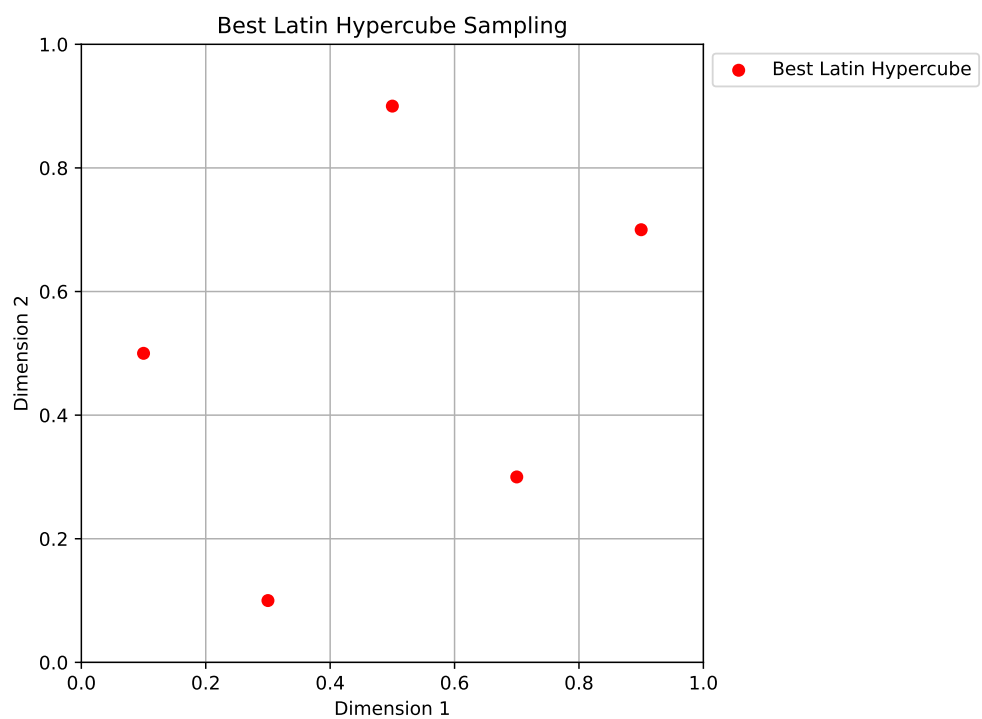


Figure 3.10.: Best Latin Hypercube Sampling

minimal (the vector will only ever contain as many elements as there are candidate  $q$  values, and only an index array is sorted, not the actual repository of plans). This sorting gives the reader the opportunity to compare, if desired, how different choices of  $q$  influence the resulting plans.

## 3.4. Experimental Analysis of the Morris-Mitchell Criterion

Morris-Mitchell Criterion Experimental Analysis

- Number of points: 16, Dimensions: 2
- mmphi parameters:  $q$  (exponent) = 2.0,  $p$  (distance norm) = 2.0 (1=Manhattan, 2=Euclidean)

```
N_POINTS = 16
N_DIM = 2
RANDOM_SEED = 42
q = 2.0
p = 2.0
```

### 3.4.1. Evaluation of Sampling Designs

We generate various sampling designs and evaluate their space-filling properties using the Morris-Mitchell criterion.

```
designs = {}
if int(np.sqrt(N_POINTS))**2 == N_POINTS:
    grid_design = Grid(k=N_DIM)
    designs["Grid (4x4)"] = grid_design.generate_grid_design(points_per_dim=int(np.sqrt(N_POINTS)))
else:
    print(f"Skipping grid design as N_POINTS={N_POINTS} is not a perfect square for a simple 2D grid")

lhs_design = SpaceFilling(k=N_DIM, seed=42)
designs["LHS"] = lhs_design.generate_qms_lhs_design(n_points=N_POINTS)

sobol_design = Sobol(k=N_DIM, seed=42)
designs["Sobol"] = sobol_design.generate_sobol_design(n_points=N_POINTS)

random_design = Random(k=N_DIM)
designs["Random"] = random_design.uniform(n_points=N_POINTS)
```

### 3. Sampling Plans

```
poor_design = Poor(k=N_DIM)
designs["Collinear"] = poor_design.generate_collinear_design(n_points=N_POINTS)

clustered_design = Clustered(k=N_DIM)
designs["Clustered (3 clusters)"] = clustered_design.generate_clustered_design(n_points=N_POINTS)

results = {}

print("Calculating Morris-Mitchell metric (smaller is better):")
for name, X_design in designs.items():
    metric_val = mmpm(X_design, q=q, p=p)
    results[name] = metric_val
    print(f" {name}: {metric_val:.4f}")
```

```
Calculating Morris-Mitchell metric (smaller is better):
Grid (4x4): 20.2617
LHS: 28.1868
Sobol: 29.0216
Random: 44.9176
Collinear: 87.8829
Clustered (3 clusters): 90.3702
```

```
if N_DIM == 2:
    num_designs = len(designs)
    cols = 2
    rows = int(np.ceil(num_designs / cols))
    fig, axes = plt.subplots(rows, cols, figsize=(5 * cols, 5 * rows))
    axes = axes.ravel() # Flatten axes array for easy iteration

    for i, (name, X_design) in enumerate(designs.items()):
        ax = axes[i]
        ax.scatter(X_design[:, 0], X_design[:, 1], s=50, edgecolors='k', alpha=0.7)
        ax.set_title(f"{name} \nmmpm = {results[name]:.3f}", fontsize=10)
        ax.set_xlabel("X1")
        ax.set_ylabel("X2")
        ax.set_xlim(-0.05, 1.05)
        ax.set_ylim(-0.05, 1.05)
        ax.set_aspect('equal', adjustable='box')
        ax.grid(True, linestyle='--', alpha=0.6)

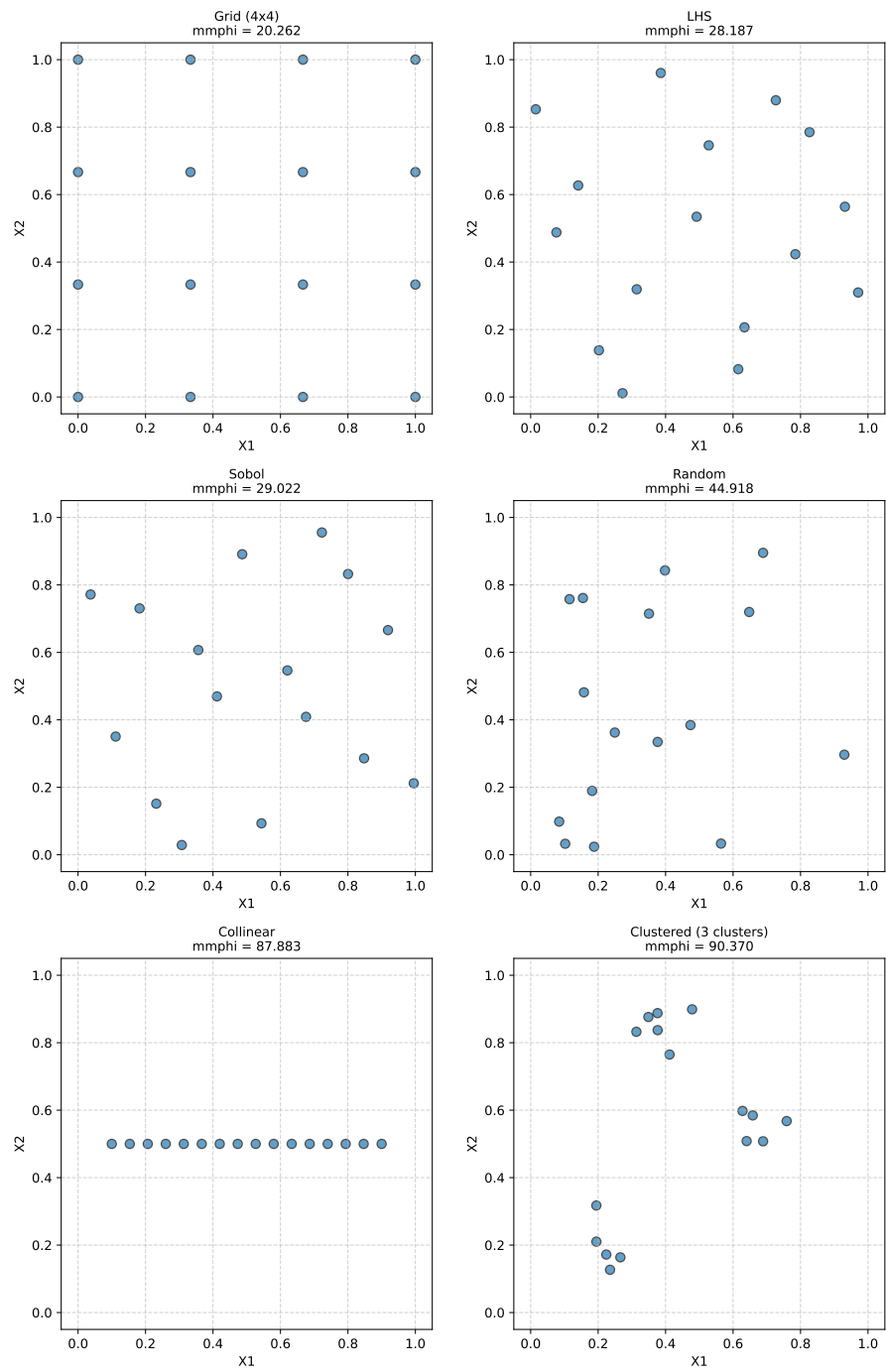
    # Hide any unused subplots
    for j in range(i + 1, len(axes)):
        fig.delaxes(axes[j])
```

### 3.4. Experimental Analysis of the Morris-Mitchell Criterion

```
plt.tight_layout()
plt.suptitle(f"Comparison of 2D Sampling Designs ({N_POINTS} points each)", fontsize=14, y=1.02)
plt.show()
```

### 3. Sampling Plans

Comparison of 2D Sampling Designs (16 points each)





### 3.4.2. Demonstrate the Impact of mmphi Parameters

Demonstrating Impact of mmphi Parameters on 'LHS' Design

```
X_lhs = designs["LHS"]

# 1. Default parameters (already calculated)
print(f"  LHS (q={q}, p={p} Euclidean): {results['LHS']:.4f}")

# 2. Change q (main exponent, literature's p or k)
q_high = 15.0
metric_lhs_q_high = mmphi(X_lhs, q=q_high, p=p)
print(f"  LHS (q={q_high}, p={p} Euclidean): {metric_lhs_q_high:.4f} (Higher q penalizes small distances more)")

# 3. Change p (distance norm, literature's q or m)
p_manhattan = 1.0
metric_lhs_p_manhattan = mmphi(X_lhs, q=q, p=p_manhattan)
print(f"  LHS (q={q}, p={p_manhattan} Manhattan): {metric_lhs_p_manhattan:.4f} (Using L1 distance)")
```

```
LHS (q=2.0, p=2.0 Euclidean): 28.1868
LHS (q=15.0, p=2.0 Euclidean): 8.1573 (Higher q penalizes small distances more)
LHS (q=2.0, p=1.0 Manhattan): 22.0336 (Using L1 distance)
```

### 3.4.3. Morris-Mitchell Criterion: Impact of Adding Points

Impact of adding a point to a 2x2 grid design

```
# Initial 2x2 Grid Design
X_initial = np.array([[0.0, 0.0], [1.0, 0.0], [0.0, 1.0], [1.0, 1.0]])
mmphi_initial = mmphi(X_initial, q=q, p=p)

print(f"Parameters: q (exponent) = {q}, p (distance) = {p} (Euclidean)\n")
print(f"Initial 2x2 Grid Design (4 points):")
print(f"  Points:\n{X_initial}")
print(f"  Morris-Mitchell Criterion (Phi_q): {mmphi_initial:.4f}\n")
```

```
Parameters: q (exponent) = 2.0, p (distance) = 2.0 (Euclidean)
```

```
Initial 2x2 Grid Design (4 points):
```

```
  Points:
[[0. 0.]
 [1. 0.]
```

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```
[0. 1.]
[1. 1.]
Morris-Mitchell Criterion (Phi_q): 2.2361
```

Scenarios for adding a 5th point:

```
scenarios = {
    "Scenario 1: Add to Center": {
        "new_point": np.array([[0.5, 0.5]]),
        "description": "Adding a point in the center of the grid."
    },
    "Scenario 2: Add Close to Existing (Cluster)": {
        "new_point": np.array([[0.1, 0.1]]),
        "description": "Adding a point very close to an existing point (0,0)."
    },
    "Scenario 3: Add on Edge": {
        "new_point": np.array([[0.5, 0.0]]),
        "description": "Adding a point on an edge between (0,0) and (1,0)."
    }
}

results_summary = []
augmented_designs_for_plotting = {"Initial Design": X_initial}

for name, scenario_details in scenarios.items():
    new_point = scenario_details["new_point"]
    X_augmented = np.vstack((X_initial, new_point))
    augmented_designs_for_plotting[name] = X_augmented

    mmphi_augmented = mmphi(X_augmented, q=q, p=p)
    change = mmphi_augmented - mmphi_initial

    print(f"{name}:")
    print(f"  Description: {scenario_details['description']}")
    print(f"  New Point Added: {new_point}")
    # print(f"  Augmented Design (5 points):\n{X_augmented}") # Optional: print full n
    print(f"  Morris-Mitchell Criterion (Phi_q): {mmphi_augmented:.4f}")
    print(f"  Change from Initial Phi_q: {change:+.4f}\n")

    results_summary.append({
        "Scenario": name,
        "Initial Phi_q": mmphi_initial,
        "Augmented Phi_q": mmphi_augmented,
        "Change": change
    })
```

### 3.4. Experimental Analysis of the Morris-Mitchell Criterion

Scenario 1: Add to Center:

Description: Adding a point in the center of the grid.

New Point Added:  $[[0.5 \ 0.5]]$

Morris-Mitchell Criterion ( $\Phi_q$ ): 3.6056

Change from Initial  $\Phi_q$ : +1.3695

Scenario 2: Add Close to Existing (Cluster):

Description: Adding a point very close to an existing point (0,0).

New Point Added:  $[[0.1 \ 0.1]]$

Morris-Mitchell Criterion ( $\Phi_q$ ): 7.6195

Change from Initial  $\Phi_q$ : +5.3834

Scenario 3: Add on Edge:

Description: Adding a point on an edge between (0,0) and (1,0).

New Point Added:  $[[0.5 \ 0. \ ]]$

Morris-Mitchell Criterion ( $\Phi_q$ ): 3.8210

Change from Initial  $\Phi_q$ : +1.5849

```
num_designs = len(augmented_designs_for_plotting)
cols = 2
rows = int(np.ceil(num_designs / cols))

fig, axes = plt.subplots(rows, cols, figsize=(6 * cols, 5 * rows))
axes = axes.ravel()

plot_idx = 0
# Plot initial design first
ax = axes[plot_idx]
ax.scatter(X_initial[:, 0], X_initial[:, 1], s=100, edgecolors='k', alpha=0.7, label="Original Point")
ax.set_title(f"Initial Design\nPhi_q = {mmphi_initial:.3f}", fontsize=10)
ax.set_xlabel("X1")
ax.set_ylabel("X2")
ax.set_xlim(-0.1, 1.1)
ax.set_ylim(-0.1, 1.1)
ax.set_aspect('equal', adjustable='box')
ax.grid(True, linestyle='--', alpha=0.6)
ax.legend(fontsize='small')
plot_idx += 1

# Plot augmented designs
for name, X_design in augmented_designs_for_plotting.items():
    if name == "Initial Design":
```

### 3. Sampling Plans

```
        continue # Already plotted

    ax = axes[plot_idx]
    # Highlight original vs new point
    original_points = X_design[:-1, :]
    new_point = X_design[-1, :].reshape(1,2)

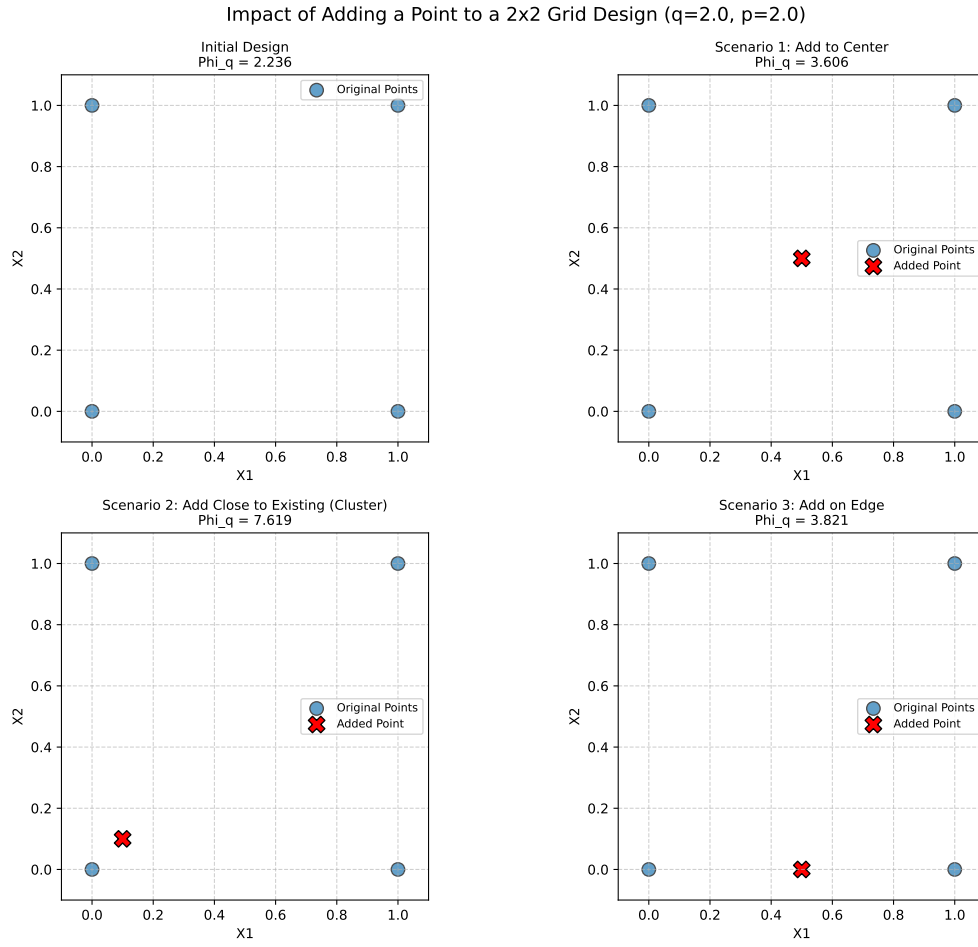
    ax.scatter(original_points[:, 0], original_points[:, 1], s=100, edgecolors='k', a
    ax.scatter(new_point[:, 0], new_point[:, 1], s=150, color='red', edgecolors='k', r

    current_phi_q = next(item['Augmented Phi_q'] for item in results_summary if item[
    ax.set_title(f"{name}\nPhi_q = {current_phi_q:.3f}", fontsize=10)
    ax.set_xlabel("X1")
    ax.set_ylabel("X2")
    ax.set_xlim(-0.1, 1.1)
    ax.set_ylim(-0.1, 1.1)
    ax.set_aspect('equal', adjustable='box')
    ax.grid(True, linestyle='--', alpha=0.6)
    ax.legend(fontsize='small')
    plot_idx +=1

# Hide any unused subplots
for j in range(plot_idx, len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout(rect=[0, 0, 1, 0.96]) # Adjust layout to make space for supitle
plt.suptitle(f"Impact of Adding a Point to a 2x2 Grid Design (q={q}, p={p})", fontsize
plt.show()
```

### 3.4. Experimental Analysis of the Morris-Mitchell Criterion



Summary Table (Conceptual):

| Scenario                                    | Initial $\Phi_q$ | Augmented $\Phi_q$ | Change |
|---------------------------------------------|------------------|--------------------|--------|
| Baseline (2x2 Grid)                         | 2.236            | —                  | —      |
| Scenario 1: Add to Center                   | 2.236            | 3.606              | +1.369 |
| Scenario 2: Add Close to Existing (Cluster) | 2.236            | 7.619              | +5.383 |
| Scenario 3: Add on Edge                     | 2.236            | 3.821              | +1.585 |

### 3.5. A Sample-Size Invariant Version of the Morris-Mitchell Criterion

#### 3.5.1. Comparison of `mmphi()` and `mmphi_intensive()`

The Morris-Mitchell criterion is a widely used metric for evaluating the space-filling properties of Latin hypercube sampling designs. However, it is sensitive to the number of points in the design, which can lead to misleading comparisons between designs with different sample sizes. To address this issue, a sample-size invariant version of the Morris-Mitchell criterion has been proposed. It is available in the `spotpython` package as `mmphi_intensive()`, see [SOURCE].

The functions `mmphi()` and `mmphi_intensive()` both calculate a Morris-Mitchell criterion, but they differ in their normalization, which makes `mmphi_intensive()` invariant to the sample size.

Let  $X$  be a sampling plan with  $n$  points  $\{x_1, x_2, \dots, x_n\}$  in a  $k$ -dimensional space. Let  $d_{ij} = \|x_i - x_j\|_p$  be the  $p$ -norm distance between points  $x_i$  and  $x_j$ . Let  $J_l$  be the multiplicity of the  $l$ -th unique distance  $d_l$  among all pairs of points in  $X$ . Let  $m$  be the total number of unique distances.

##### 1. `mmphi()` (Morris-Mitchell Criterion $\Phi_q$ )

The `mmphi()` function, as defined in the context and implemented in `sampling.py`, calculates the Morris-Mitchell criterion  $\Phi_q$  as:

$$\Phi_q(X) = \left( \sum_{l=1}^m J_l d_l^{-q} \right)^{1/q},$$

where:

- $J_l$  is the number of pairs of points separated by the unique distance  $d_l$ .
- $d_l$  are the unique pairwise distances.
- $q$  is a user-defined exponent (typically  $q > 0$ ).

This formulation is directly based on the sum of inverse powers of distances. The value of  $\Phi_q$  is generally dependent on the number of points  $n$  in the design  $X$ , as the sum  $\sum J_l d_l^{-q}$  will typically increase with more points (and thus more pairs).

##### 2. `mmphi_intensive()` (Intensive Morris-Mitchell Criterion)

The `mmphi_intensive()` function, as implemented in `sampling.py` calculates a sample-size invariant version of the Morris-Mitchell criterion, which will be referred to as  $\Phi_q^I$ . The formula is:

### 3.5. A Sample-Size Invariant Version of the Morris-Mitchell Criterion

$$\Phi_q^I(X) = \left( \frac{1}{M} \sum_{l=1}^m J_l d_l^{-q} \right)^{1/q}$$

where:

- $M = \binom{n}{2} = \frac{n(n-1)}{2}$  is the total number of unique pairs of points in the design  $X$ .
- The other terms  $J_l$ ,  $d_l$ ,  $q$  are the same as in `mmphi()`.

The key mathematical difference is the normalization factor  $\frac{1}{M}$  inside the parentheses before the outer exponent  $1/q$  is applied.

- `mmphi()`: Calculates  $(\text{SumTerm})^{1/q}$ , where  $\text{SumTerm} = \sum J_l d_l^{-q}$ .
- `mmphi_intensive()`: Calculates  $\left(\frac{\text{SumTerm}}{M}\right)^{1/q}$ .

By dividing the sum  $\sum J_l d_l^{-q}$  by  $M$  (the total number of pairs), `mmphi_intensive()` effectively calculates an *average* contribution per pair to the  $-q$ -th power of distance, before taking the  $q$ -th root. This normalization makes the criterion less dependent on the absolute number of points  $n$  and allows for more meaningful comparisons of space-fillingness between designs of different sizes. A smaller value indicates a better (more space-filling) design for both criteria.

#### 3.5.2. Plotting the Two Morris-Mitchell Criteria for Different Sample Sizes

Figure 3.11 shows the comparison of the two Morris-Mitchell criteria for different sample sizes using the `plot_mmphi_vs_n_lhs` function. The red line represents the standard Morris-Mitchell criterion, while the blue line represents the sample-size invariant version. Note the difference in the y-axis scales, which highlights how the sample-size invariant version remains consistent across varying sample sizes.

```
def plot_mmphi_vs_n_lhs(k_dim: int,
                        seed: int,
                        n_min: int = 10,
                        n_max: int = 100,
                        n_step: int = 5,
                        q_phi: float = 2.0,
                        p_phi: float = 2.0):
    """
    Generates LHS designs for varying n, calculates mmphi and mmphi_intensive,
    and plots them against the number of samples (n).

    Args:
        k_dim (int): Number of dimensions for the LHS design.
```

### 3. Sampling Plans

```
seed (int): Random seed for reproducibility.
n_min (int): Minimum number of samples.
n_max (int): Maximum number of samples.
n_step (int): Step size for increasing n.
q_phi (float): Exponent q for the Morris-Mitchell criteria.
p_phi (float): Distance norm p for the Morris-Mitchell criteria.
"""
n_values = list(range(n_min, n_max + 1, n_step))
if not n_values:
    print("Warning: n_values list is empty. Check n_min, n_max, and n_step.")
    return
mmphi_results = []
mmphi_intensive_results = []
lhs_generator = SpaceFilling(k=k_dim, seed=seed)
print(f"Calculating for n from {n_min} to {n_max} with step {n_step}...")
for n_points in n_values:
    if n_points < 2 : # mmphi requires at least 2 points to calculate distances
        print(f"Skipping n={n_points} as it's less than 2.")
        mmphi_results.append(np.nan)
        mmphi_intensive_results.append(np.nan)
        continue
    try:
        X_design = lhs_generator.generate_qms_lhs_design(n_points=n_points)
        phi = mmphi(X_design, q=q_phi, p=p_phi)
        phi_intensive, _, _ = mmphi_intensive(X_design, q=q_phi, p=p_phi)
        mmphi_results.append(phi)
        mmphi_intensive_results.append(phi_intensive)
    except Exception as e:
        print(f"Error calculating for n={n_points}: {e}")
        mmphi_results.append(np.nan)
        mmphi_intensive_results.append(np.nan)

fig, ax1 = plt.subplots(figsize=(9, 6))

color = 'tab:red'
ax1.set_xlabel('Number of Samples (n)')
ax1.set_ylabel('mmphi (Phiq)', color=color)
ax1.plot(n_values, mmphi_results, color=color, marker='o', linestyle='-', label='n')
ax1.tick_params(axis='y', labelcolor=color)
ax1.grid(True, linestyle='--', alpha=0.7)

ax2 = ax1.twinx() # instantiate a second axes that shares the same x-axis
color = 'tab:blue'
ax2.set_ylabel('mmphi_intensive (PhiqI)', color=color) # we already handled the x-axis
```



### 3.5. A Sample-Size Invariant Version of the Morris-Mitchell Criterion

```
ax2.plot(n_values, mmphi_intensive_results, color=color, marker='x', linestyle='--', label='mmphi_intensive')
ax2.tick_params(axis='y', labelcolor=color)

fig.tight_layout() # otherwise the right y-label is slightly clipped
plt.title(f'Morris-Mitchell Criteria vs. Number of Samples (n)\nLHS (k={k_dim}, q={q_phi}, p={p_phi})')
# Add legends
lines, labels = ax1.get_legend_handles_labels()
lines2, labels2 = ax2.get_legend_handles_labels()
ax2.legend(lines + lines2, labels + labels2, loc='best')
plt.show()
```

```
N_DIM = 2
RANDOM_SEED = 42
plot_mmphi_vs_n_lhs(k_dim=N_DIM, seed=RANDOM_SEED, n_min=10, n_max=100, n_step=5)
```

Calculating for n from 10 to 100 with step 5...

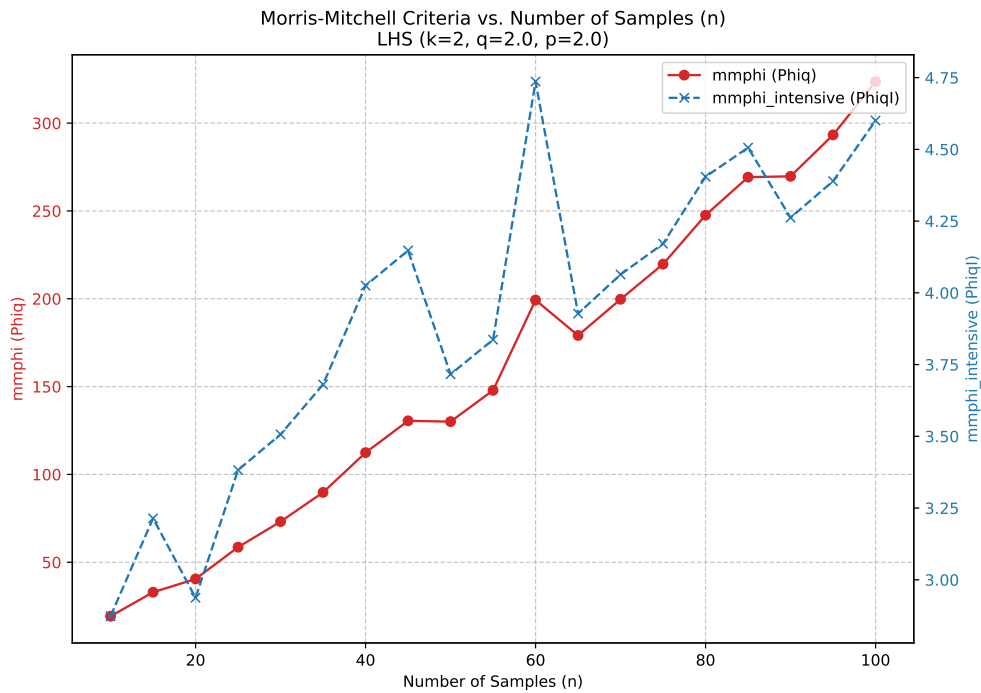


Figure 3.11.: Comparison of the two Morris-Mitchell Criteria for Different Sample Sizes

## 3.6. Jupyter Notebook

### Note

- The Jupyter-Notebook of this lecture is available on GitHub in the Hyperparameter-Tuning-Cookbook Repository

## 4. Constructing a Surrogate

### Note

This section is based on chapter 2 in Forrester, Sóbester, and Keane (2008).

**Definition 4.1** (Black Box Problem). We are trying to learn a mapping that converts the vector  $\vec{x}$  into a scalar output  $y$ , i.e., we are trying to learn a function

$$y = f(x).$$

If function is hidden (“lives in a black box”), so that the physics of the problem is not known, the problem is called a black box problem.

This black box could take the form of either a physical or computer experiment, for example, a finite element code, which calculates the maximum stress ( $\sigma$ ) for given product dimensions ( $\vec{x}$ ).

**Definition 4.2** (Generic Solution). The generic solution method is to collect the output values  $y^{(1)}, y^{(2)}, \dots, y^{(n)}$  that result from a set of inputs  $\vec{x}^{(1)}, \vec{x}^{(2)}, \dots, \vec{x}^{(n)}$  and find a best guess  $\hat{f}(\vec{x})$  for the black box mapping  $f$ , based on these known observations.

### 4.1. Stage One: Preparing the Data and Choosing a Modelling Approach

The first step is the identification, through a small number of observations, of the inputs that have a significant impact on  $f$ ; that is the determination of the shortest design variable vector  $\vec{x} = \{x_1, x_2, \dots, x_k\}^T$  that, by sweeping the ranges of all of its variables, can still elicit most of the behavior the black box is capable of. The ranges of the various design variables also have to be established at this stage.

The second step is to recruit  $n$  of these  $k$ -vectors into a list

$$X = \{\vec{x}^{(1)}, \vec{x}^{(2)}, \dots, \vec{x}^{(n)}\}^T,$$

where each  $\vec{x}^{(i)}$  is a  $k$ -vector. The corresponding responses are collected in a vector such that this represents the design space as thoroughly as possible.

#### 4. Constructing a Surrogate

In the surrogate modeling process, the number of samples  $n$  is often limited, as it is constrained by the computational cost (money and/or time) associated with obtaining each observation.

It is advisable to scale  $\vec{x}$  at this stage into the unit cube  $[0, 1]^k$ , a step that can simplify the subsequent mathematics and prevent multidimensional scaling issues.

We now focus on the attempt to learn  $f$  through data pairs

$$\{(\vec{x}^{(1)}, y^{(1)}), (\vec{x}^{(2)}, y^{(2)}), \dots, (\vec{x}^{(n)}, y^{(n)})\}.$$

This supervised learning process essentially involves searching across the space of possible functions  $\hat{f}$  that would replicate observations of  $f$ . This space of functions is infinite. Any number of hypersurfaces could be drawn to pass through or near the known observations, accounting for experimental error. However, most of these would generalize poorly; they would be practically useless at predicting responses at new sites, which is the ultimate goal.

**Example 4.1** (The Needle(s) in the Haystack Function). An extreme example is the ‘needle(s) in the haystack’ function:

$$f(x) = \begin{cases} y^{(1)}, & \text{if } x = \vec{x}^{(1)} \\ y^{(2)}, & \text{if } x = \vec{x}^{(2)} \\ \vdots & \\ y^{(n)}, & \text{if } x = \vec{x}^{(n)} \\ 0, & \text{otherwise.} \end{cases}$$

While this predictor reproduces all training data, it seems counter-intuitive and unsettling to predict 0 everywhere else for most engineering functions. Although there is a small chance that the function genuinely resembles the equation above and we sampled exactly where the needles are, it is highly unlikely.

There are countless other configurations, perhaps less contrived, that still generalize poorly. This suggests a need for systematic means to filter out nonsensical predictors. In our approach, we embed the structure of  $f$  into the model selection algorithm and search over its parameters to fine-tune the approximation to observations. For instance, consider one of the simplest models,

$$f(x, \vec{w}) = \vec{w}^T \vec{x} + v. \quad (4.1)$$

Learning  $f$  with this model implies that its structure—a hyperplane—is predetermined, and the fitting process involves finding the  $k + 1$  parameters (the slope vector  $\vec{w}$  and the intercept  $v$ ) that best fit the data. This will be accomplished in Stage Two.

Complicating this further is the noise present in observed responses (we assume design vectors  $\vec{x}$  are not corrupted). Here, we focus on learning from such data, which sometimes risks overfitting.

## 4.2. Stage Two: Parameter Estimation and Training

**Definition 4.3** (Overfitting). Overfitting occurs when the model becomes too flexible and captures not only the underlying trend but also the noise in the data.

In the surrogate modeling process, the second stage as described in Section 4.2, addresses this issue of complexity control by estimating the parameters of the fixed structure model. However, foresight is necessary even at the model type selection stage.

Model selection often involves physics-based considerations, where the modeling technique is chosen based on expected underlying responses.

**Example 4.2** (Model Selection). Modeling stress in an elastically deformed solid due to small strains may justify using a simple linear approximation. Without insights into the physics, and if one fails to account for the simplicity of the data, a more complex and excessively flexible model may be incorrectly chosen. Although parameter estimation might still adjust the approximation to become linear, an opportunity to develop a simpler and robust model may be lost.

- Simple linear (or polynomial) models, despite their lack of flexibility, have advantages like applicability in further symbolic computations.
- Conversely, if we incorrectly assume a quadratic process when multiple peaks and troughs exist, the parameter estimation stage will not compensate for an unsuitable model choice. A quadratic model is too rigid to fit a multimodal function, regardless of parameter adjustments.

## 4.2. Stage Two: Parameter Estimation and Training

Assuming that Stage One helped identify the  $k$  critical design variables, acquire the learning data set, and select a generic model structure  $f(\vec{x}, \vec{w})$ , the task now is to estimate parameters  $\vec{w}$  to ensure the model fits the data optimally. Among several estimation criteria, we will discuss two methods here.

**Definition 4.4** (Maximum Likelihood Estimation). Given a set of parameters  $\vec{w}$ , the model  $f(\vec{x}, \vec{w})$  allows computation of the probability of the data set

$$\{(\vec{x}^{(1)}, y^{(1)} \pm \epsilon), (\vec{x}^{(2)}, y^{(2)} \pm \epsilon), \dots, (\vec{x}^{(n)}, y^{(n)} \pm \epsilon)\}$$

resulting from  $f$  (where  $\epsilon$  is a small error margin around each data point).

### **i** Maximum Likelihood Estimation

?@sec-max-likelihood presents a more detailed discussion of the maximum likelihood estimation (MLE) method.

#### 4. Constructing a Surrogate

Taking  $\text{?@eq-likelihood-mvn}$  and assuming errors  $\epsilon$  are independently and normally distributed with standard deviation  $\sigma$ , the probability of the data set is given by:

$$P = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp \left[ -\frac{1}{2\sigma^2} \sum_{i=1}^n (y^{(i)} - f(\vec{x}^{(i)}, \vec{w}))^2 \epsilon \right].$$

Intuitively, this is equivalent to the likelihood of the parameters given the data. Accepting this intuitive relationship as a mathematical one aids in model parameter estimation. This is achieved by maximizing the likelihood or, more conveniently, minimizing the negative of its natural logarithm:

$$\min_{\vec{w}} \sum_{i=1}^n \frac{[y^{(i)} - f(\vec{x}^{(i)}, \vec{w})]^2}{2\sigma^2} + \frac{n}{2} \ln \epsilon. \quad (4.2)$$

If we assume  $\sigma$  and  $\epsilon$  are constants, Equation 4.2 simplifies to the well-known least squares criterion:

$$\min_{\vec{w}} \sum_{i=1}^n [y^{(i)} - f(\vec{x}^{(i)}, \vec{w})]^2.$$

Cross-validation is another method used to estimate model performance.

**Definition 4.5** (Cross-Validation). Cross-validation splits the data randomly into  $q$  roughly equal subsets, and then cyclically removing each subset and fitting the model to the remaining  $q-1$  subsets. A loss function  $L$  is then computed to measure the error between the predictor and the withheld subset for each iteration, with contributions summed over all  $q$  iterations. More formally, if a mapping  $\theta : \{1, \dots, n\} \rightarrow \{1, \dots, q\}$  describes the allocation of the  $n$  training points to one of the  $q$  subsets and  $f^{(-\theta(i))}(\vec{x})$  is the predicted value by removing the subset  $\theta(i)$  (i.e., the subset where observation  $i$  belongs), the cross-validation measure, used as an estimate of prediction error, is:

$$CV = \frac{1}{n} \sum_{i=1}^n L(y^{(i)}, f^{(-\theta(i))}(\vec{x}^{(i)})). \quad (4.3)$$

Introducing the squared error as the loss function and considering our generic model  $f$  still dependent on undetermined parameters, we write Equation 4.3 as:

$$CV = \frac{1}{n} \sum_{i=1}^n [y^{(i)} - f^{(-\theta(i))}(\vec{x}^{(i)})]^2. \quad (4.4)$$

The extent to which Equation 4.4 is an unbiased estimator of true risk depends on  $q$ . It is shown that if  $q = n$ , the leave-one-out cross-validation (LOOCV) measure is

almost unbiased. However, LOOCV can have high variance because subsets are very similar. Hastie, Tibshirani, and Friedman (2017)) suggest using compromise values like  $q = 5$  or  $q = 10$ . Using fewer subsets also reduces the computational cost of the cross-validation process, see also Arlot, Celisse, et al. (2010) and Kohavi (1995).

### 4.3. Stage Three: Model Testing

If there is a sufficient amount of observational data, a random subset should be set aside initially for model testing. Hastie, Tibshirani, and Friedman (2017) recommend setting aside approximately  $0.25n$  of  $\vec{x} \rightarrow y$  pairs for testing purposes. These observations must remain untouched during Stages One and Two, as their sole purpose is to evaluate the testing error—the difference between true and approximated function values at the test sites—once the model has been built. Interestingly, if the main goal is to construct an initial surrogate for seeding a global refinement criterion-based strategy (as discussed in Section 3.2 in Forrester, Sóbester, and Keane (2008)), the model testing phase might be skipped.

It is noted that, ideally, parameter estimation (Stage Two) should also rely on a separate subset. However, observational data is rarely abundant enough to afford this luxury (if the function is cheap to evaluate and evaluation sites are selectable, a surrogate model might not be necessary).

When data are available for model testing and the primary objective is a globally accurate model, using either a root mean square error (RMSE) metric or the correlation coefficient ( $r^2$ ) is recommended. To test the model, a test data set of size  $n_t$  is used alongside predictions at the corresponding locations to calculate these metrics.

The RMSE is defined as follows:

**Definition 4.6** (Root Mean Square Error (RMSE)).

$$\text{RMSE} = \sqrt{\frac{1}{n_t} \sum_{i=1}^{n_t} (y^{(i)} - \hat{y}^{(i)})^2},$$

Ideally, the RMSE should be minimized, acknowledging its limitation by errors in the objective function  $f$  calculation. If the error level is known, like a standard deviation, the aim might be to achieve an RMSE within this value. Often, the target is an RMSE within a specific percentage of the observed data's objective value range.

The squared correlation coefficient  $r$ , see [?@eq-pears-corr](#), between the observed  $y$  and predicted  $\hat{y}$  values can be computed as:

$$r^2 = \left( \frac{\text{cov}(y, \hat{y})}{\sqrt{\text{var}(y)\text{var}(\hat{y})}} \right)^2, \quad (4.5)$$

#### 4. Constructing a Surrogate

Equation 4.5 and can be expanded as:

$$r^2 = \left( \frac{n_t \sum_{i=1}^{n_t} y^{(i)} \hat{y}^{(i)} - \sum_{i=1}^{n_t} y^{(i)} \sum_{i=1}^{n_t} \hat{y}^{(i)}}{\sqrt{\left(n_t \sum_{i=1}^{n_t} (y^{(i)})^2 - \left(\sum_{i=1}^{n_t} y^{(i)}\right)^2\right) \left(n_t \sum_{i=1}^{n_t} (\hat{y}^{(i)})^2 - \left(\sum_{i=1}^{n_t} \hat{y}^{(i)}\right)^2\right)}} \right)^2.$$

The correlation coefficient  $r^2$  does not require scaling the data sets and only compares landscape shapes, not values. An  $r^2 > 0.8$  typically indicates a surrogate with good predictive capability.

The methods outlined provide quantitative assessments of model accuracy, yet visual evaluations can also be insightful. In general, the RMSE will not reach zero but will stabilize around a low value. At this point, the surrogate model is saturated with data, and further additions do not enhance the model globally (though local improvements can occur at newly added points if using an interpolating model).

**Example 4.3** (The Tea and Sugar Analogy). Forrester, Sóbester, and Keane (2008) illustrates this saturation point using a comparison with a cup of tea and sugar. The tea represents the surrogate model, and sugar represents data. Initially, the tea is unsweetened, and adding sugar increases its sweetness. Eventually, a saturation point is reached where no more sugar dissolves, and the tea cannot get any sweeter. Similarly, a more flexible model, like one with additional parameters or employing interpolation rather than regression, can increase the saturation point—akin to making a hotter cup of tea for dissolving more sugar.

### 4.4. Jupyter Notebook

#### Note

- The Jupyter-Notebook of this lecture is available on GitHub in the Hyperparameter-Tuning-Cookbook Repository



## 5. Response Surface Methods

This part deals with numerical implementations of optimization methods. The goal is to understand the implementation of optimization methods and to solve real-world problems numerically and efficiently. We will focus on the implementation of surrogate models, because they are the most efficient way to solve real-world problems.

Starting point is the well-established response surface methodology (RSM). It will be extended to the design and analysis of computer experiments (DACE). The DACE methodology is a modern extension of the response surface methodology. It is based on the use of surrogate models, which are used to replace the real-world problem with a simpler problem. The simpler problem is then solved numerically. The solution of the simpler problem is then used to solve the real-world problem.

### ! Numerical methods: Goals

- Understand implementation of optimization methods
- Solve real-world problems numerically and efficiently

### 5.1. What is RSM?

Response Surface Methods (RSM) refer to a collection of statistical and mathematical tools that are valuable for developing, improving, and optimizing processes. The overarching theme of RSM involves studying how input variables that control a product or process can potentially influence a response that measures performance or quality characteristics.

The advantages of RSM include a rich literature, well-established methods often used in manufacturing, the importance of careful experimental design combined with a well-understood model, and the potential to add significant value to scientific inquiry, process refinement, optimization, and more. However, there are also drawbacks to RSM, such as the use of simple and crude surrogates, the hands-on nature of the methods, and the limitation of local methods.

RSM is related to various fields, including Design of Experiments (DoE), quality management, reliability, and productivity. Its applications are widespread in industry and manufacturing, focusing on designing, developing, and formulating new products and improving existing ones, as well as from laboratory research. RSM is commonly

## 5. Response Surface Methods

applied in domains such as materials science, manufacturing, applied chemistry, climate science, and many others.

An example of RSM involves studying the relationship between a response variable, such as yield ( $y$ ) in a chemical process, and two process variables: reaction time ( $\xi_1$ ) and reaction temperature ( $\xi_2$ ). The provided code illustrates this scenario, following a variation of the so-called “banana function.”

In the context of visualization, RSM offers the choice between 3D plots and contour plots. In a 3D plot, the independent variables  $\xi_1$  and  $\xi_2$  are represented, with  $y$  as the dependent variable.

```
import numpy as np
import matplotlib.pyplot as plt

def fun_rosen(x1, x2):
    b = 10
    return (x1-1)**2 + b*(x2-x1**2)**2

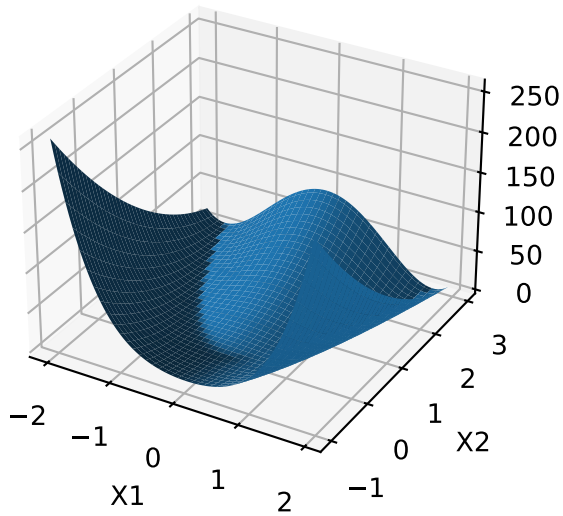
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
x = np.arange(-2.0, 2.0, 0.05)
y = np.arange(-1.0, 3.0, 0.05)
X, Y = np.meshgrid(x, y)
zs = np.array(fun_rosen(np.ravel(X), np.ravel(Y)))
Z = zs.reshape(X.shape)

ax.plot_surface(X, Y, Z)

ax.set_xlabel('X1')
ax.set_ylabel('X2')
ax.set_zlabel('Y')

plt.show()
```

### 5.1. What is RSM?



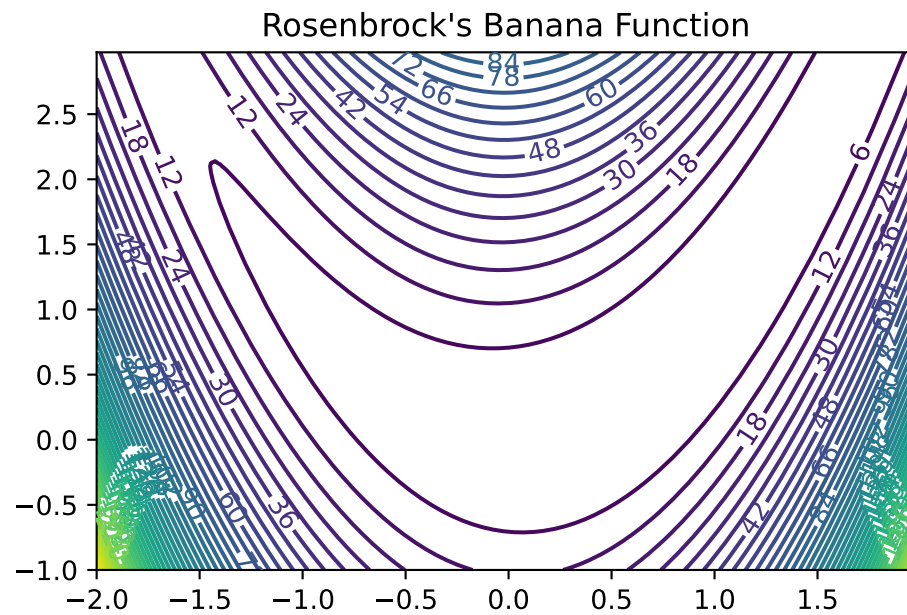
- contour plot example:

- $x_1$  and  $x_2$  are the independent variables
- $y$  is the dependent variable

```
import numpy as np
import matplotlib.cm as cm
import matplotlib.pyplot as plt

delta = 0.025
x1 = np.arange(-2.0, 2.0, delta)
x2 = np.arange(-1.0, 3.0, delta)
X1, X2 = np.meshgrid(x1, x2)
Y = fun_rosen(X1, X2)
fig, ax = plt.subplots()
CS = ax.contour(X1, X2, Y, 50)
ax.clabel(CS, inline=True, fontsize=10)
ax.set_title("Rosenbrock's Banana Function")
```

```
Text(0.5, 1.0, "Rosenbrock's Banana Function")
```



- Visual inspection: yield is optimized near  $(\xi_1, \xi_2)$

#### 5.1.1. Visualization: Problems in Practice

- True response surface is unknown in practice
- When yield evaluation is not as simple as a toy banana function, but a process requiring care to monitor, reconfigure and run, it's far too expensive to observe over a dense grid
- And, measuring yield may be a noisy/inexact process
- That's where stats (RSM) comes in

#### 5.1.2. RSM: Strategies

- RSMs consist of experimental strategies for
- **exploring** the space of the process (i.e., independent/input) variables (above  $\xi_1$  and  $\xi_2$ )
- empirical statistical **modeling** targeted toward development of an appropriate approximating relationship between the response (yield) and process variables local to a study region of interest

### 5.1. What is RSM?

- **optimization** methods for sequential refinement in search of the levels or values of process variables that produce desirable responses (e.g., that maximize yield or explain variation)
- RSM used for fitting an Empirical Model
- True response surface driven by an unknown physical mechanism
- Observations corrupted by noise
- Helpful: fit an empirical model to output collected under different process configurations
- Consider response  $Y$  that depends on controllable input variables  $\xi_1, \xi_2, \dots, \xi_m$
- RSM: Equations of the Empirical Model
  - $Y = f(\xi_1, \xi_2, \dots, \xi_m) + \epsilon$
  - $\mathbb{E}\{Y\} = \eta = f(\xi_1, \xi_2, \dots, \xi_m)$
  - $\epsilon$  is treated as zero mean idiosyncratic noise possibly representing
    - \* inherent variation, or
    - \* the effect of other systems or
    - \* variables not under our purview at this time

#### 5.1.3. RSM: Noise in the Empirical Model

- Typical simplifying assumption:  $\epsilon \sim N(0, \sigma^2)$
- We seek estimates for  $f$  and  $\sigma^2$  from noisy observations  $Y$  at inputs  $\xi$

#### 5.1.4. RSM: Natural and Coded Variables

- Inputs  $\xi_1, \xi_2, \dots, \xi_m$  called **natural variables**:
  - expressed in natural units of measurement, e.g., degrees Celsius, pounds per square inch (psi), etc.
- Transformed to **coded variables**  $x_1, x_2, \dots, x_m$ :
  - to mitigate hassles and confusion that can arise when working with a multitude of scales of measurement
- Typical **Transformations** offering dimensionless inputs  $x_1, x_2, \dots, x_m$ 
  - in the unit cube, or
  - scaled to have a mean of zero and standard deviation of one, are common choices.
- Empirical model becomes  $\eta = f(x_1, x_2, \dots, x_m)$

### 5.1.5. RSM Low-order Polynomials

- Low-order polynomial make the following simplifying Assumptions
  - Learning about  $f$  is lots easier if we make some simplifying approximations
  - Appealing to **Taylor's theorem**, a low-order polynomial in a small, localized region of the input ( $x$ ) space is one way forward
  - Classical RSM:
    - \* disciplined application of **local analysis** and
    - \* **sequential refinement** of locality through conservative extrapolation
  - Inherently a **hands-on process**

## 5.2. First-Order Models (Main Effects Model)

- **First-order model** (sometimes called main effects model) useful in parts of the input space where it's believed that there's little curvature in  $f$ :

$$\eta = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

- For example:

$$\eta = 50 + 8x_1 + 3x_2$$

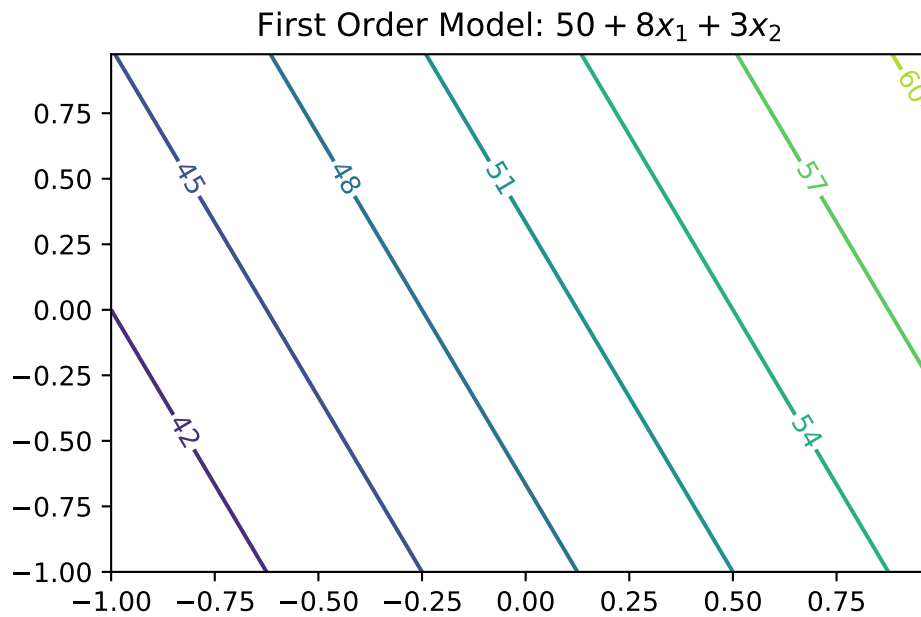
- In practice, such a surface would be obtained by fitting a model to the outcome of a designed experiment
- First-Order Model in python Evaluated on a Grid
- Evaluate model on a grid in a double-unit square centered at the origin
- Coded units are chosen arbitrarily, although one can imagine deploying this approximating function nearby  $x^{(0)} = (0, 0)$

```
def fun_1(x1,x2):  
    return 50 + 8*x1 + 3*x2
```

```
import numpy as np  
import matplotlib.cm as cm  
import matplotlib.pyplot as plt  
  
delta = 0.025  
x1 = np.arange(-1.0, 1.0, delta)  
x2 = np.arange(-1.0, 1.0, delta)  
X1, X2 = np.meshgrid(x1, x2)  
Y = fun_1(X1,X2)  
fig, ax = plt.subplots()  
CS = ax.contour(X1, X2, Y)  
ax.clabel(CS, inline=True, fontsize=10)  
ax.set_title('First Order Model: $50 + 8x_1 + 3x_2$')
```

## 5.2. First-Order Models (Main Effects Model)

```
Text(0.5, 1.0, 'First Order Model: $50 + 8x_1 + 3x_2$')
```



### 5.2.1. First-Order Model Properties

- First-order model in 2d traces out a **plane** in  $y \times (x_1, x_2)$  space
- Only be appropriate for the most trivial of response surfaces, even when applied in a highly localized part of the input space
- Adding **curvature** is key to most applications:
  - First-order model with **interactions** induces limited degree of curvature via different rates of change of  $y$  as  $x_1$  is varied for fixed  $x_2$ , and vice versa:

$$\eta = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_{12}$$

- For example  $\eta = 50 + 8x_1 + 3x_2 - 4x_1x_2$

### 5.2.2. First-order Model with Interactions in python

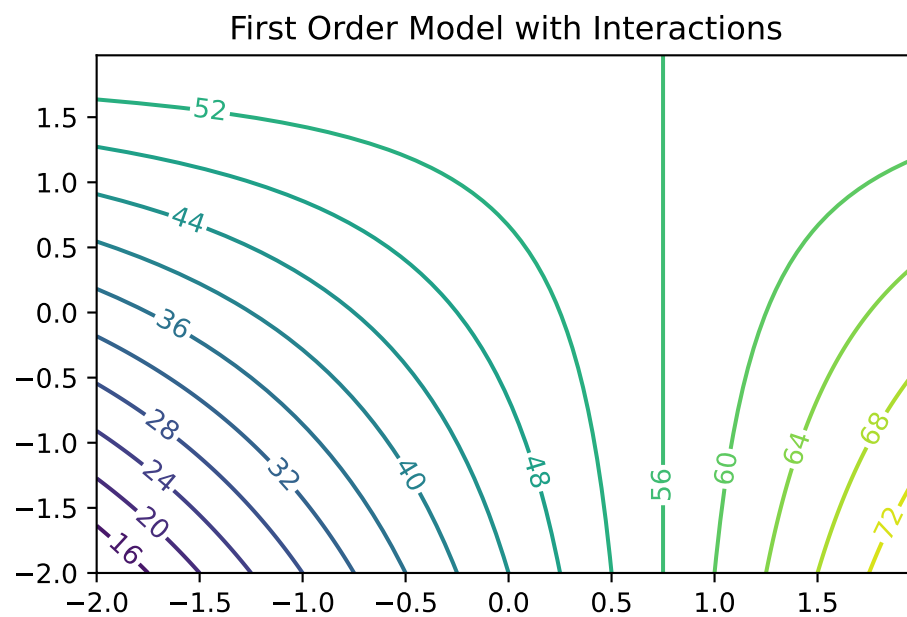
- Code below facilitates evaluations for pairs  $(x_1, x_2)$
- Responses may be observed over a mesh in the same double-unit square

## 5. Response Surface Methods

```
def fun_11(x1,x2):  
    return 50 + 8 * x1 + 3 * x2 - 4 * x1 * x2
```

```
import numpy as np  
import matplotlib.cm as cm  
import matplotlib.pyplot as plt  
  
delta = 0.025  
x1 = np.arange(-2.0, 2.0, delta)  
x2 = np.arange(-2.0, 2.0, delta)  
X1, X2 = np.meshgrid(x1, x2)  
Y = fun_11(X1,X2)  
fig, ax = plt.subplots()  
CS = ax.contour(X1, X2, Y, 20)  
ax.clabel(CS, inline=True, fontsize=10)  
ax.set_title('First Order Model with Interactions')
```

```
Text(0.5, 1.0, 'First Order Model with Interactions')
```





### 5.2.3. Observations: First-Order Model with Interactions

- Mean response  $\eta$  is increasing marginally in both  $x_1$  and  $x_2$ , or conditional on a fixed value of the other until  $x_1$  is 0.75
- Rate of increase slows as both coordinates grow simultaneously since the coefficient in front of the interaction term  $x_1x_2$  is negative
- Compared to the first-order model (without interactions): surface is far more useful locally
- Least squares regressions often flag up significant interactions when fit to data collected on a design far from local optima

## 5.3. Second-Order Models

- Second-order model may be appropriate near local optima where  $f$  would have substantial curvature:

$$\eta = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_{11}x_1^2 + \beta_{22}x_2^2 + \beta_{12}x_1x_2$$

- For example

$$\eta = 50 + 8x_1 + 3x_2 - 7x_1^2 - 3x_2^2 - 4x_1x_2$$

- Implementation of the Second-Order Model as `fun_2()`.

```
def fun_2(x1,x2):
    return 50 + 8 * x1 + 3 * x2 - 7 * x1**2 - 3*x2**2 - 4 * x1 * x2
```

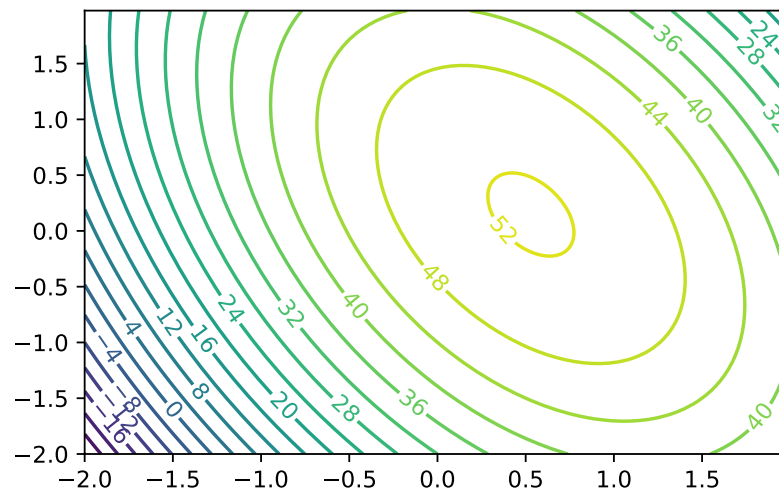
```
import numpy as np
import matplotlib.cm as cm
import matplotlib.pyplot as plt

delta = 0.025
x1 = np.arange(-2.0, 2.0, delta)
x2 = np.arange(-2.0, 2.0, delta)
X1, X2 = np.meshgrid(x1, x2)
Y = fun_2(X1,X2)
fig, ax = plt.subplots()
CS = ax.contour(X1, X2, Y, 20)
ax.clabel(CS, inline=True, fontsize=10)
ax.set_title('Second Order Model with Interactions. Maximum near about $(0.6,0.2)$')
```

```
Text(0.5, 1.0, 'Second Order Model with Interactions. Maximum near about $(0.6,0.2)$')
```

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Second Order Model with Interactions. Maximum near about (0.6, 0.2)



### 5.3.1. Second-Order Models: Properties

- Not all second-order models would have a single stationary point (in RSM jargon called “a simple maximum”)
- In “yield maximizing” setting we’re presuming response surface is **concave** down from a global viewpoint
  - even though local dynamics may be more nuanced
- Exact criteria depend upon the eigenvalues of a certain matrix built from those coefficients
- Box and Draper (2007) provide a diagram categorizing all of the kinds of second-order surfaces in RSM analysis, where finding local maxima is the goal

### 5.3.2. Example: Stationary Ridge

- Example set of coefficients describing what’s called a **stationary ridge** is provided by the code below

```
def fun_ridge(x1, x2):  
    return 80 + 4*x1 + 8*x2 - 3*x1**2 - 12*x2**2 - 12*x1*x2
```

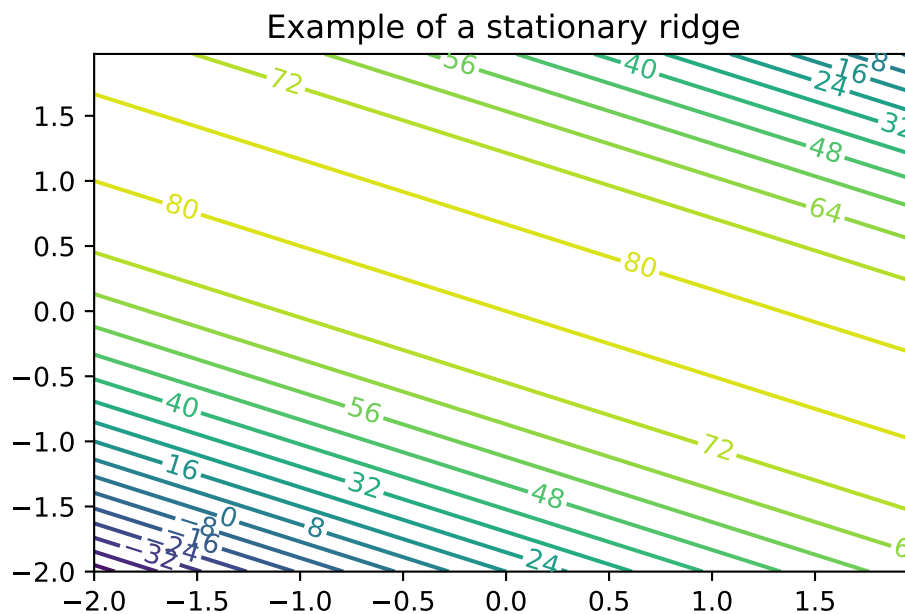
```

import numpy as np
import matplotlib.cm as cm
import matplotlib.pyplot as plt

delta = 0.025
x1 = np.arange(-2.0, 2.0, delta)
x2 = np.arange(-2.0, 2.0, delta)
X1, X2 = np.meshgrid(x1, x2)
Y = fun_ridge(X1, X2)
fig, ax = plt.subplots()
CS = ax.contour(X1, X2, Y, 20)
ax.clabel(CS, inline=True, fontsize=10)
ax.set_title('Example of a stationary ridge')

```

```
Text(0.5, 1.0, 'Example of a stationary ridge')
```



### 5.3.3. Observations: Second-Order Model (Ridge)

- **Ridge:** a whole line of stationary points corresponding to maxima
- Situation means that the practitioner has some flexibility when it comes to optimizing:

## 5. Response Surface Methods

- can choose the precise setting of  $(x_1, x_2)$  either arbitrarily or (more commonly) by consulting some tertiary criteria

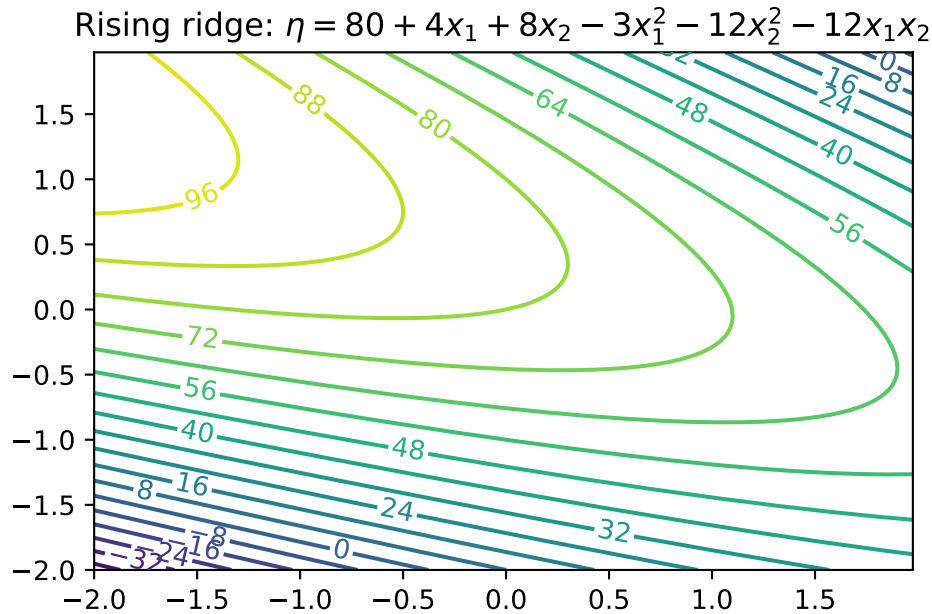
### 5.3.4. Example: Rising Ridge

- An example of a rising ridge is implemented by the code below.

```
def fun_ridge_rise(x1, x2):  
    return 80 - 4*x1 + 12*x2 - 3*x1**2 - 12*x2**2 - 12*x1*x2
```

```
import numpy as np  
import matplotlib.cm as cm  
import matplotlib.pyplot as plt  
  
delta = 0.025  
x1 = np.arange(-2.0, 2.0, delta)  
x2 = np.arange(-2.0, 2.0, delta)  
X1, X2 = np.meshgrid(x1, x2)  
Y = fun_ridge_rise(X1, X2)  
fig, ax = plt.subplots()  
CS = ax.contour(X1, X2, Y, 20)  
ax.clabel(CS, inline=True, fontsize=10)  
ax.set_title('Rising ridge:  $\eta = 80 + 4x_1 + 8x_2 - 3x_1^2 - 12x_2^2 - 12x_1x_2$ ')
```

```
Text(0.5, 1.0, 'Rising ridge:  $\eta = 80 + 4x_1 + 8x_2 - 3x_1^2 - 12x_2^2 - 12x_1x_2$ ')
```



### 5.3.5. Summary: Rising Ridge

- The stationary point is remote to the study region
- A continuum of (local) stationary points along any line going through the 2d space, excepting one that lies directly on the ridge
- Although estimated response will increase while moving along the axis of symmetry toward its stationary point, this situation indicates
  - either a poor fit by the approximating second-order function, or
  - that the study region is not yet precisely in the vicinity of a local optima—often both.

### 5.3.6. Falling Ridge

- Inversion of a rising ridge is a falling ridge
- Similarly indicating one is far from local optima, except that the response decreases as you move toward the stationary point
- Finding a falling ridge system can be a back-to-the-drawing-board affair.

### 5.3.7. Saddle Point

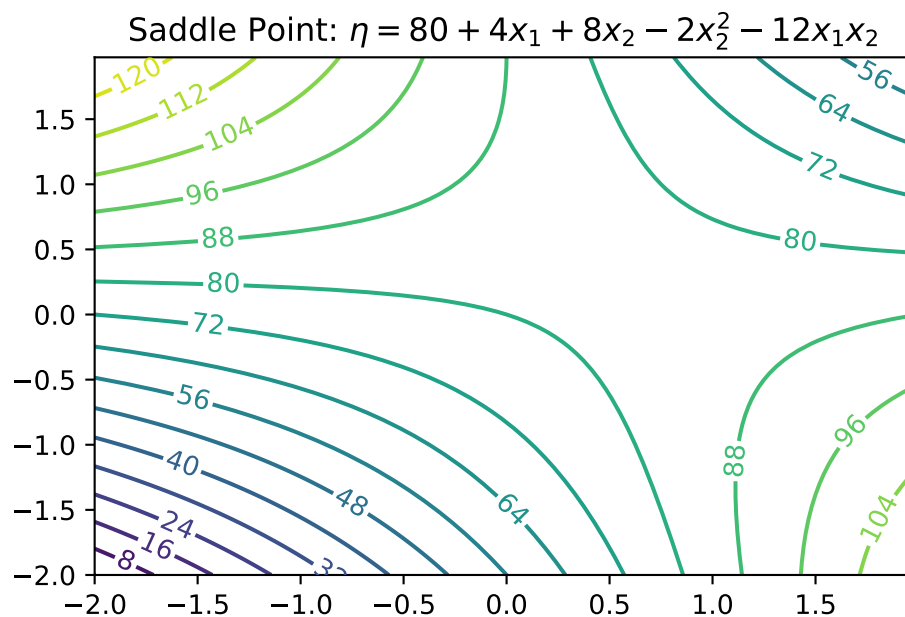
- Finally, we can get what's called a saddle or minimax system.

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```
def fun_saddle(x1, x2):  
    return 80 + 4*x1 + 8*x2 - 2*x2**2 - 12*x1*x2
```

```
import numpy as np  
import matplotlib.cm as cm  
import matplotlib.pyplot as plt  
  
delta = 0.025  
x1 = np.arange(-2.0, 2.0, delta)  
x2 = np.arange(-2.0, 2.0, delta)  
X1, X2 = np.meshgrid(x1, x2)  
Y = fun_saddle(X1, X2)  
fig, ax = plt.subplots()  
CS = ax.contour(X1, X2, Y, 20)  
ax.clabel(CS, inline=True, fontsize=10)  
ax.set_title('Saddle Point:  $\eta = 80 + 4x_1 + 8x_2 - 2x_2^2 - 12x_1x_2$ ')
```

```
Text(0.5, 1.0, 'Saddle Point:  $\eta = 80 + 4x_1 + 8x_2 - 2x_2^2 - 12x_1x_2$ ')
```



### 5.3.8. Interpretation: Saddle Points

- Likely further data collection, and/or outside expertise, is needed before determining a course of action in this situation

### 5.3.9. Summary: Ridge Analysis

- Finding a simple maximum, or stationary ridge, represents ideals in the spectrum of second-order approximating functions
- But getting there can be a bit of a slog
- Using models fitted from data means uncertainty due to noise, and therefore uncertainty in the type of fitted second-order model
- A ridge analysis attempts to offer a principled approach to navigating uncertainties when one is seeking local maxima
- The two-dimensional setting exemplified above is convenient for visualization, but rare in practice
- Complications compound when studying the effect of more than two process variables

## 5.4. General RSM Models

- General **first-order model** on  $m$  process variables  $x_1, x_2, \dots, x_m$  is

$$\eta = \beta_0 + \beta_1 x_1 + \dots + \beta_m x_m$$

- General **second-order model** on  $m$  process variables

$$\eta = \beta_0 + \sum_{j=1}^m x_j + \sum_{j=1}^m x_j^2 + \sum_{j=2}^m \sum_{k=1}^{j-1} \beta_{kj} x_k x_j.$$

### 5.4.1. Ordinary Least Squares

- Inference from data is carried out by **ordinary least squares** (OLS)
- For an excellent review including R examples, see Sheather (2009)
- OLS and maximum likelihood estimators (MLEs) are in the typical Gaussian linear modeling setup basically equivalent

## 5.5. General Linear Regression

We are considering a model, which can be written in the form

$$Y = X\beta + \epsilon,$$

where  $Y$  is an  $(n \times 1)$  vector of observations (responses),  $X$  is an  $(n \times p)$  matrix of known form,  $\beta$  is a  $(1 \times p)$  vector of unknown parameters, and  $\epsilon$  is an  $(n \times 1)$  vector of errors. Furthermore,  $E(\epsilon) = 0$ ,  $Var(\epsilon) = \sigma^2 I$  and the  $\epsilon_i$  are uncorrelated.

Using the normal equations

$$(X'X)b = X'Y,$$

the solution is given by

$$b = (X'X)^{-1}X'Y.$$

**Example 5.1** (Linear Regression).

```
import numpy as np
n = 8
X = np.linspace(0, 2*np.pi, n, endpoint=False).reshape(-1,1)
print(np.round(X, 2))
y = np.sin(X)
print(np.round(y, 2))
# fit an OLS model to the data, predict the response based on the 100 x values
m = 100
x = np.linspace(0, 2*np.pi, m, endpoint=False).reshape(-1,1)
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X, y)
y_pred = model.predict(x)
# visualize the data and the fitted model
import matplotlib.pyplot as plt
plt.scatter(X, y, color='black')
plt.plot(x, y_pred, color='blue', linewidth=1)
# add the ground truth (sine function) in orange
plt.plot(x, np.sin(x), color='orange', linewidth=1)
plt.show()
```

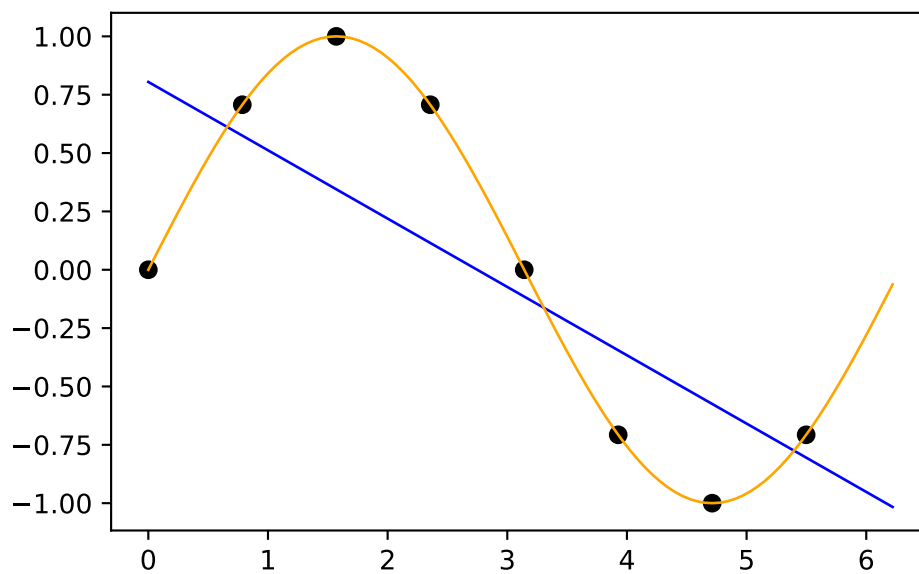
```
[[0.  ]
 [0.79]
 [1.57]
 [2.36]
```



```

[3.14]
[3.93]
[4.71]
[5.5 ]]
[[ 0. ]
 [ 0.71]
 [ 1.  ]
 [ 0.71]
 [ 0.  ]
 [-0.71]
 [-1.  ]
 [-0.71]]

```



## 5.6. Designs

- Important: Organize the data collection phase of a response surface study carefully
- **Design:** choice of  $x$ 's where we plan to observe  $y$ 's, for the purpose of approximating  $f$
- Analyses and designs need to be carefully matched
- When using a first-order model, some designs are preferred over others
- When using a second-order model to capture curvature, a different sort of design is appropriate

## 5. Response Surface Methods

- Design choices often contain features enabling modeling assumptions to be challenged
  - e.g., to check if initial impressions are supported by the data ultimately collected

### 5.6.1. Different Designs

- **Screening designs:** determine which variables matter so that subsequent experiments may be smaller and/or more focused
- Then there are designs tailored to the form of model (first- or second-order, say) in the screened variables
- And then there are more designs still

## 5.7. RSM Experimentation

### 5.7.1. First Step

- RSM-based experimentation begins with a **first-order model**, possibly with interactions
- Presumption: current process operating **far from optimal** conditions
- Collect data and apply **method of steepest ascent** (gradient) on fitted surfaces to move to the optimum

### 5.7.2. Second Step

- Eventually, if all goes well after several such carefully iterated refinements, **second-order models** are used on appropriate designs in order to zero-in on ideal operating conditions
- Careful analysis of the fitted surface:
  - Ridge analysis with further refinement using gradients of, and
  - standard errors associated with, the fitted surfaces, and so on

### 5.7.3. Third Step

- Once the practitioner is satisfied with the full arc of
  - design(s),
  - fit(s), and
  - decision(s):

- A small experiment called **confirmation test** may be performed to check if the predicted optimal settings are realizable in practice

## 5.8. RSM: Review and General Considerations

- First Glimpse, RSM seems sensible, and pretty straightforward as quantitative statistics-based analysis goes
- But: RSM can get complicated, especially when input dimensions are not very low
- Design considerations are particularly nuanced, since the goal is to obtain reliable estimates of main effects, interaction, and curvature while minimizing sampling effort/expense
- RSM Downside: Inefficiency
  - Despite intuitive appeal, several RSM downsides become apparent upon reflection
  - Problems in practice
  - Stepwise nature of sequential decision making is inefficient:
    - \* Not obvious how to re-use or update analysis from earlier phases, or couple with data from other sources/related experiments
- RSM Downside: Locality
  - In addition to being local in experiment-time (stepwise approach), it's local in experiment-space
  - Balance between
    - \* exploration (maybe we're barking up the wrong tree) and
    - \* exploitation (let's make things a little better) is modest at best
- RSM Downside: Expert Knowledge
  - Interjection of expert knowledge is limited to hunches about relevant variables (i.e., the screening phase), where to initialize search, how to design the experiments
  - Yet at the same time classical RSMs rely heavily on constant examination throughout stages of modeling and design and on the instincts of seasoned practitioners
- RSM Downside: Replicability
  - Parallel analyses, conducted according to the same best intentions, rarely lead to the same designs, model fits and so on
  - Sometimes that means they lead to different conclusions, which can be cause for concern

## 5. Response Surface Methods

### 5.8.1. Historical Considerations about RSM

- In spite of those criticisms, however, there was historically little impetus to revise the status quo
- Classical RSM was comfortable in its skin, consistently led to improvements or compelling evidence that none can reasonably be expected
- But then in the late 20th century came an explosive expansion in computational capability, and with it a means of addressing many of those downsides

### 5.8.2. Status Quo

- Nowadays, field experiments and statistical models, designs and optimizations are coupled with mathematical models
- Simple equations are not regarded as sufficient to describe real-world systems anymore
- Physicists figured that out fifty years ago; industrial engineers followed, biologists, social scientists, climate scientists and weather forecasters, etc.
- Systems of equations are required, solved over meshes (e.g., finite elements), or stochastically interacting agents
- Goals for those simulation experiments are as diverse as their underlying dynamics
- Optimization of systems is common, e.g., to identify worst-case scenarios

### 5.8.3. The Role of Statistics

- Solving systems of equations, or interacting agents, requires computing
- Statistics involved at various stages:
  - choosing the mathematical model
  - solving by stochastic simulation (Monte Carlo)
  - designing the computer experiment
  - smoothing over idiosyncrasies or noise
  - finding optimal conditions, or
  - calibrating mathematical/computer models to data from field experiments

### 5.8.4. New RSM is needed: DACE

- Classical RSMs are not well-suited to any of those tasks, because
  - they lack the fidelity required to model these data
  - their intended application is too local
  - they're also too hands-on.

- Once computers are involved, a natural inclination is to automate—to remove humans from the loop and set the computer running on the analysis in order to maximize computing throughput, or minimize idle time
- **Design and Analysis of Computer Experiments** as a modern extension of RSM
- Experimentation is changing due to advances in machine learning
- **Gaussian process** (GP) regression is the canonical surrogate model
- Origins in geostatistics (gold mining)
- Wide applicability in contexts where prediction is king
- Machine learners exposed GPs as powerful predictors for all sorts of tasks:
  - from regression to classification,
  - active learning/sequential design,
  - reinforcement learning and optimization,
  - latent variable modeling, and so on

## 5.9. Exercises

1. Generate 3d Plots for the Contour Plots in this notebook.
2. Write a `plot_3d` function, that takes the objective function `fun` as an argument.
  - It should provide the following interface: `plot_3d(fun)`.
3. Write a `plot_contour` function, that takes the objective function `fun` as an argument:
  - It should provide the following interface: `plot_contour(fun)`.
4. Consider further arguments that might be useful for both function, e.g., ranges, size, etc.

## 5.10. Jupyter Notebook

### Note

- The Jupyter-Notebook of this lecture is available on GitHub in the Hyperparameter-Tuning-Cookbook Repository



## 6. Polynomial Models

### Note

- This section is based on chapter 2.2 in Forrester, Sóbester, and Keane (2008).
- The following Python packages are imported:

```
import numpy as np
import matplotlib.pyplot as plt
```

### 6.1. Fitting a Polynomial

We will consider one-variable cases, i.e.,  $k = 1$ , first.

Let us consider the scalar-valued function  $f : \mathbb{R} \rightarrow \mathbb{R}$  observed according to the sampling plan  $X = \{x^{(1)}, x^{(2)} \dots, x^{(n)}\}^T$ , yielding the responses  $\vec{y} = \{y^{(1)}, y^{(2)}, \dots, y^{(n)}\}^T$ .

A polynomial approximation of  $f$  of order  $m$  can be written as:

$$\hat{f}(x, m, \vec{w}) = \sum_{i=0}^m w_i x^i.$$

In the spirit of the earlier discussion of maximum likelihood parameter estimation, we seek to estimate  $w = w_0, w_1, \dots, w_m^T$  through a least squares solution of:

$$\Phi \vec{w} = \vec{y}$$

where  $\Phi$  is the Vandermonde matrix:

$$\Phi = \begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^m \\ 1 & x_2 & x_2^2 & \dots & x_2^m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^m \end{bmatrix}.$$

The maximum likelihood estimate of  $w$  is given by:

## 6. Polynomial Models

$$\vec{w} = (\Phi^T \Phi)^{-1} \Phi^T y,$$

where  $\Phi^+ = (\Phi^T \Phi)^{-1} \Phi^T$  is the Moore-Penrose pseudo-inverse of  $\Phi$  (see Section 9.3).

The polynomial approximation of order  $m$  is essentially a truncated Taylor series expansion. While higher values of  $m$  yield more accurate approximations, they risk overfitting the noise in the data.

To prevent this, we estimate  $m$  using cross-validation. This involves minimizing the cross-validation error over a discrete set of possible orders  $m$  (e.g.,  $m \in 1, 2, \dots, 15$ ).

For each  $m$ , the data is split into  $q$  subsets. The model is trained on  $q - 1$  subsets, and the error is computed on the left-out subset. This process is repeated for all subsets, and the cross-validation error is summed. The order  $m$  with the smallest cross-validation error is chosen.

## 6.2. Polynomial Fitting in Python

### 6.2.1. Fitting the Polynomial

```
from sklearn.model_selection import KFold
def polynomial_fit(X, Y, max_order=15, q=5):
    """
    Fits a one-variable polynomial to one-dimensional data using cross-validation.

    Args:
        X (array-like): Training data vector (independent variable).
        Y (array-like): Training data vector (dependent variable).
        max_order (int): Maximum polynomial order to consider. Default is 15.
        q (int): Number of cross-validation folds. Default is 5.

    Returns:
        best_order (int): The optimal polynomial order.
        coeff (array): Coefficients of the best-fit polynomial.
        mnstd (tuple): Normalization parameters (mean, std) for X.
    """
    X = np.array(X)
    Y = np.array(Y)
    n = len(X)
    # Normalize X
    mnstd = (np.mean(X), np.std(X))
    X_norm = (X - mnstd[0]) / mnstd[1]
```



```

# Cross-validation setup
kf = KFold(n_splits=q, shuffle=True, random_state=42)
cross_val_errors = np.zeros(max_order)
for order in range(1, max_order + 1):
    fold_errors = []
    for train_idx, val_idx in kf.split(X_norm):
        X_train, X_val = X_norm[train_idx], X_norm[val_idx]
        Y_train, Y_val = Y[train_idx], Y[val_idx]
        # Fit polynomial
        coeff = np.polyfit(X_train, Y_train, order)
        # Predict on validation set
        Y_pred = np.polyval(coeff, X_val)
        # Compute mean squared error
        mse = np.mean((Y_val - Y_pred) ** 2)
        fold_errors.append(mse)
    cross_val_errors[order - 1] = np.mean(fold_errors)
# Find the best order
best_order = np.argmin(cross_val_errors) + 1
# Fit the best polynomial on the entire dataset
best_coeff = np.polyfit(X_norm, Y, best_order)
return best_order, best_coeff, mnstd

```

### 6.2.2. Explaining the $k$ -fold Cross-Validation

The line

```

kf = KFold(n_splits=q, shuffle=True, random_state=42)

```

initializes a  $k$ -Fold cross-validator object from the `sklearn.model_selection` library. The `n_splits` parameter specifies the number of folds. The data will be divided into  $q$  parts. In each iteration of the cross-validation, one part will be used as the validation set, and the remaining  $q-1$  parts will be used as the training set.

The `kf.split` method takes the dataset `X_norm` as input and yields pairs of index arrays for each fold: \* `train_idx`: In each iteration, `train_idx` is an array containing the indices of the data points that belong to the training set for that specific fold. \* `val_idx`: Similarly, `val_idx` is an array containing the indices of the data points that belong to the validation (or test) set for that specific fold.

The loop will run  $q$  times (the number of splits). In each iteration, a different fold serves as the validation set, while the other  $q-1$  folds form the training set.

Here's a Python example to demonstrate the values of `train_idx` and `val_idx`:

## 6. Polynomial Models

```
# Sample data (e.g., X_norm)
X_norm = np.array([0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0])
print(f"Original data indices: {np.arange(len(X_norm))}\n")
# Number of splits (folds)
q = 3 # Let's use 3 folds for this example
# Initialize KFold
kf = KFold(n_splits=q, shuffle=True, random_state=42)
# Iterate through the splits and print the indices
fold_number = 1
for train_idx, val_idx in kf.split(X_norm):
    print(f"--- Fold {fold_number} ---")
    print(f"Train indices: {train_idx}")
    print(f"Validation indices: {val_idx}")
    print(f"Training data for this fold: {X_norm[train_idx]}")
    print(f"Validation data for this fold: {X_norm[val_idx]}\n")
    fold_number += 1
```

Original data indices: [0 1 2 3 4 5 6 7 8 9]

--- Fold 1 ---

Train indices: [2 3 4 6 7 9]

Validation indices: [0 1 5 8]

Training data for this fold: [0.3 0.4 0.5 0.7 0.8 1. ]

Validation data for this fold: [0.1 0.2 0.6 0.9]

--- Fold 2 ---

Train indices: [0 1 3 4 5 6 8]

Validation indices: [2 7 9]

Training data for this fold: [0.1 0.2 0.4 0.5 0.6 0.7 0.9]

Validation data for this fold: [0.3 0.8 1. ]

--- Fold 3 ---

Train indices: [0 1 2 5 7 8 9]

Validation indices: [3 4 6]

Training data for this fold: [0.1 0.2 0.3 0.6 0.8 0.9 1. ]

Validation data for this fold: [0.4 0.5 0.7]

### 6.2.3. Making Predictions

To make predictions, we can use the coefficients. The data is standardized around its mean in the polynomial function, which is why the vector `mnstd` is required. The coefficient vector is computed based on the normalized data, and this must be taken into account if further analytical calculations are performed on the fitted model.

## 6.2. Polynomial Fitting in Python

The polynomial approximation of  $C_D$  is:

$$C_D(x) = w_8x^8 + w_7x^7 + \dots + w_1x + w_0,$$

where  $x$  is normalized as:

$$\bar{x} = \frac{x - \mu(X)}{\sigma(X)}$$

```
def predict_polynomial_fit(X, coeff, mnstd):  
    """  
    Generates predictions for the polynomial fit.  
  
    Args:  
        X (array-like): Original independent variable data.  
        coeff (array): Coefficients of the best-fit polynomial.  
        mnstd (tuple): Normalization parameters (mean, std) for X.  
  
    Returns:  
        tuple: De-normalized predicted X values and corresponding Y predictions.  
    """  
    # Normalize X  
    X_norm = (X - mnstd[0]) / mnstd[1]  
  
    # Generate predictions  
    X_pred = np.linspace(min(X_norm), max(X_norm), 100)  
    Y_pred = np.polyval(coeff, X_pred)  
  
    # De-normalize X for plotting  
    X_pred_original = X_pred * mnstd[1] + mnstd[0]  
  
    return X_pred_original, Y_pred
```

### 6.2.4. Plotting the Results

```
def plot_polynomial_fit(X, Y, X_pred_original, Y_pred, best_order, y_true=None):  
    """  
    Visualizes the polynomial fit.  
  
    Args:  
        X (array-like): Original independent variable data.
```

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```
Y (array-like): Original dependent variable data.
X_pred_original (array): De-normalized predicted X values.
Y_pred (array): Predicted Y values.
y_true (array): True Y values.
best_order (int): The optimal polynomial order.
"""
plt.scatter(X, Y, label="Training Data", color="grey", marker="o")
plt.plot(X_pred_original, Y_pred, label=f"Order {best_order} Polynomial", color="blue")
if y_true is not None:
    plt.plot(X, y_true, label="True Function", color="blue", linestyle="--")
plt.title(f"Polynomial Fit (Order {best_order})")
plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.show()
```

### 6.3. Example One: Aerofoil Drag

The circles in Figure 6.1 represent 101 drag coefficient values obtained through a numerical simulation by iterating each member of a family of aerofoils towards a target lift value (see the Appendix, Section A.3 in Forrester, Sóbester, and Keane (2008)). The members of the family have different shapes, as determined by the sampling plan:

$$X = x_1, x_2, \dots, x_{101}$$

The responses are:

$$C_D = \{C_D^{(1)}, C_D^{(2)}, \dots, C_D^{(101)}\}$$

These responses are corrupted by “noise,” which are deviations of the systematic variety caused by small changes in the computational mesh from one design to the next.

The original data is measured in natural units, i.e., from  $-0.3$  unit to  $0.1$  unit. The data is normalized to the range of 0 to 1 for the computation with the `aerofoilcd` function. The data is then fitted with a polynomial of order  $m$ . To obtain the best polynomial through this data, the following Python code can be used:

```
from spotpython.surrogate.functions.forr08a import aerofoilcd
import numpy as np
import matplotlib.pyplot as plt
X = np.linspace(-0.3, 0.1, 101)
# normalize the data so that it will be in the range of 0 to 1
```

### 6.3. Example One: Aerofoil Drag

```
a = np.min(X)
b = np.max(X)
X_cod = (X - a) / (b - a)
y = aerofoilcd(X_cod)
best_order, best_coeff, mnstd = polynomial_fit(X, y)
X_pred_original, Y_pred = predict_polynomial_fit(X, best_coeff, mnstd)

plot_polynomial_fit(X, y, X_pred_original, Y_pred, best_order)
```

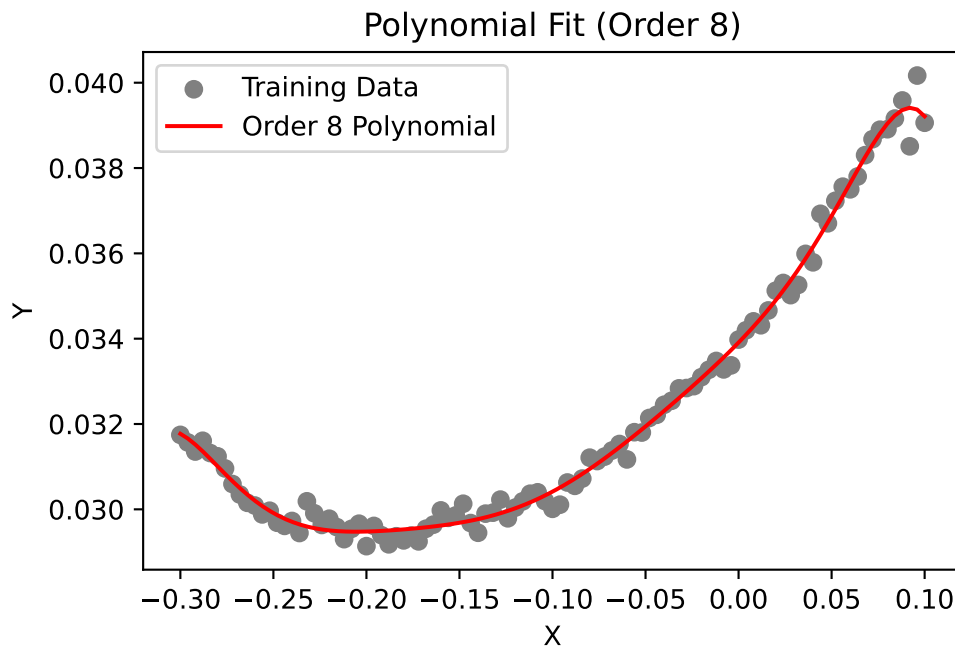


Figure 6.1.: Aerofoil drag data

Figure 6.1 shows an eighth-order polynomial fitted through the aerofoil drag data. The order was selected via cross-validation, and the coefficients were determined through likelihood maximization. Results, i.e, the best polynomial order and coefficients, are printed in the console. The coefficients are stored in the vector `best_coeff`, which contains the coefficients of the polynomial in descending order. The first element is the coefficient of  $x^8$ , and the last element is the constant term. The vector `mnstd`, containing the mean and standard deviation of  $X$ , is:

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```
print(f"Best polynomial order: {best_order}\n")
print(f"Coefficients (starting with w0):\n {best_coeff}\n")
print(f"Normalization parameters (mean, std):\n {mnstd}\n")
```

Best polynomial order: 8

Coefficients (starting with w0):

```
[-0.00022964 -0.00014636  0.00116742  0.00052988 -0.0016912  -0.00047398
  0.00244373  0.00270342  0.03041508]
```

Normalization parameters (mean, std):

```
(np.float64(-0.09999999999999999), np.float64(0.11661903789690602))
```

### 6.4. Example Two: A Multimodal Test Case

Let us consider the one-variable test function:

$$f(x) = (6x - 2)^2 \sin(12x - 4).$$

```
import numpy as np
from spotpython.surrogate.functions.forr08a import onevar
X = np.linspace(0, 1, 51)
y_true = onevar(X)
# initialize random seed
np.random.seed(42)
y = y_true + np.random.normal(0, 1, len(X))*1.1
best_order, best_coeff, mnstd = polynomial_fit(X, y)
X_pred_original, Y_pred = predict_polynomial_fit(X, best_coeff, mnstd)
```

```
plot_polynomial_fit(X, y, X_pred_original, Y_pred, best_order, y_true=y_true)
```

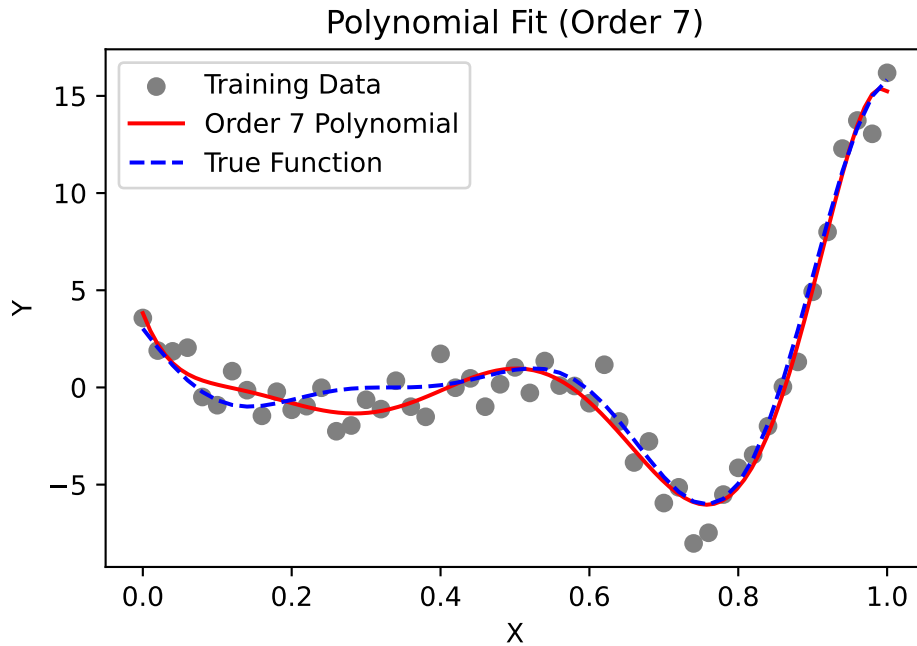


Figure 6.2.: Onevar function

This function, depicted by the dotted line in Figure 6.2, has local minima of different depths, which can be deceptive to some surrogate-based optimization procedures. Here, we use it as an example of a multimodal function for polynomial fitting.

We generate the training data (depicted by circles in Figure 6.2) by adding normally distributed noise to the function. Figure 6.2 shows a seventh-order polynomial fitted through the noisy data. This polynomial was selected as it minimizes the cross-validation metric.

## 6.5. Extending to Multivariable Polynomial Models

While the examples above focus on the one-variable case, real-world engineering problems typically involve multiple input variables. For  $k$ -variable problems, polynomial approximation becomes significantly more complex but follows the same fundamental principles. For a  $k$ -dimensional input space, the polynomial approximation can be expressed as:

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$$\hat{f}(\vec{x}) = \sum_{i=1}^N w_i \phi_i(\vec{x}),$$

where  $\phi_i(\vec{x})$  represents multivariate basis functions, and  $N$  is the total number of terms in the polynomial. Unlike the univariate case, these basis functions include all possible combinations of variables up to the selected polynomial order  $m$ , which might result in a “basis function explosion” as the number of variables increases.

For a third-order polynomial ( $m = 3$ ) with three variables ( $k = 3$ ), the complete set of basis functions would include 20 terms:

$$\text{Constant term: } 1 \tag{6.1}$$

$$\text{First-order terms: } x_1, x_2, x_3 \tag{6.2}$$

$$\text{Second-order terms: } x_1^2, x_2^2, x_3^2, x_1x_2, x_1x_3, x_2x_3 \tag{6.3}$$

$$\text{Third-order terms: } x_1^3, x_2^3, x_3^3, x_1^2x_2, x_1^2x_3, x_2^2x_1, x_2^2x_3, x_3^2x_1, x_3^2x_2, x_1x_2x_3 \tag{6.4}$$

The total number of terms grows combinatorially as  $N = \binom{k+m}{m}$ , which quickly becomes prohibitive as dimensionality increases. For example, a 10-variable cubic polynomial requires  $\binom{13}{3} = 286$  coefficients! This exponential growth creates three interrelated challenges:

- **Model Selection:** Determining the appropriate polynomial order  $m$  that balances complexity with generalization ability
- **Coefficient Estimation:** Computing the potentially large number of weights  $\vec{w}$  while avoiding numerical instability
- **Term Selection:** Identifying which specific basis functions should be included, as many may be irrelevant to the response

Several techniques have been developed to address these challenges:

- Regularization methods (LASSO, ridge regression) that penalize model complexity
- Stepwise regression algorithms that incrementally add or remove terms
- Dimension reduction techniques that project the input space to lower dimensions
- Orthogonal polynomials that improve numerical stability for higher-order models

These limitations of polynomial models in higher dimensions motivate the exploration of more flexible surrogate modeling approaches like Radial Basis Functions and Kriging, which we’ll examine in subsequent sections.



## 6.6. Jupyter Notebook

### **i** Note

- The Jupyter-Notebook of this lecture is available on GitHub in the Hyperparameter-Tuning-Cookbook Repository



## 7. Radial Basis Function Models

### Note

- This section is based on chapter 2.3 in Forrester, Sóbester, and Keane (2008).
- The following Python packages are imported:

```
import numpy as np
import matplotlib.pyplot as plt
```

### 7.1. Radial Basis Function Models

Scientists and engineers frequently tackle complex functions by decomposing them into a “vocabulary” of simpler, well-understood basic functions. These fundamental building blocks possess properties that make them easier to analyze mathematically and implement computationally. We explored this concept earlier with multivariable polynomials, where complex behaviors were modeled using combinations of polynomial terms such as  $1$ ,  $x_1$ ,  $x_2$ ,  $x_1^2$ , and  $x_1x_2$ . This approach is not limited to polynomials; it extends to various function classes, including trigonometric functions, exponential functions, and even more complex structures.

While Fourier analysis—perhaps the most widely recognized example of this approach—excels at representing periodic phenomena through sine and cosine functions, the focus in Forrester, Sóbester, and Keane (2008) is broader. They aim to approximate arbitrary smooth, continuous functions using strategically positioned basis functions. Specifically, radial basis function (RBF) models employ symmetrical basis functions centered at selected points distributed throughout the design space. These basis functions have the unique property that their output depends only on the distance from their center point.

First, we give a definition of the Euclidean distance, which is the most common distance measure used in RBF models.

**Definition 7.1** (Euclidean Distance). The Euclidean distance between two points in a  $k$ -dimensional space is defined as:

## 7. Radial Basis Function Models

$$\|\vec{x} - \vec{c}\| = \sqrt{\sum_{i=1}^k (x_i - c_i)^2}, \quad (7.1)$$

where:

- $\vec{x} = (x_1, x_2, \dots, x_d)$  is the first point,
- $\vec{c} = (c_1, c_2, \dots, c_d)$  is the second point, and
- $k$  is the number of dimensions.

The Euclidean distance measure represents the straight-line distance between two points in Euclidean space.

Using the Euclidean distance, we can define the radial basis function (RBF) model.

**Definition 7.2** (Radial Basis Function (RBF)). Mathematically, a radial basis function  $\psi$  can be expressed as:

$$\psi(\vec{x}) = \psi(\|\vec{x} - \vec{c}\|), \quad (7.2)$$

where  $\vec{x}$  is the input vector,  $\vec{c}$  is the center of the function, and  $\|\vec{x} - \vec{c}\|$  denotes the Euclidean distance between  $\vec{x}$  and  $\vec{c}$ .

In the context of RBFs, the Euclidean distance calculation determines how much influence a particular center point  $\vec{c}$  has on the prediction at point  $\vec{x}$ .

We will first examine interpolating RBF models, which assume noise-free data and pass exactly through all training points. This approach provides an elegant mathematical foundation before we consider more practical scenarios where data contains measurement or process noise.

### 7.1.1. Fitting Noise-Free Data

Let us consider the scalar valued function  $f$  observed without error, according to the sampling plan  $X = \{\vec{x}^{(1)}, \vec{x}^{(2)}, \dots, \vec{x}^{(n)}\}^T$ , yielding the responses  $\vec{y} = \{y^{(1)}, y^{(2)}, \dots, y^{(n)}\}^T$ .

For a given set of  $n_c$  centers  $\vec{c}^{(i)}$ , we would like to express the RBF model as a linear combination of the basis functions centered at these points. The goal is to find the weights  $\vec{w}$  that minimize the error between the predicted and observed values. Thus, we seek a radial basis function approximation to  $f$  of the fixed form:

$$\hat{f}(\vec{x}) = \sum_{i=1}^{n_c} w_i \psi(\|\vec{x} - \vec{c}^{(i)}\|), \quad (7.3)$$

where

- $w_i$  are the weights of the  $n_c$  basis functions,
- $\vec{c}^{(i)}$  are the  $n_c$  centres of the basis functions, and
- $\psi$  is a radial basis function.

The notation  $\|\cdot\|$  denotes the Euclidean distance between two points in the design space as defined in Equation 7.1.

#### 7.1.1.1. Selecting Basis Functions: From Fixed to Parametric Forms

When implementing a radial basis function model, we initially have one undetermined parameter per basis function: the weight applied to each function's output. This simple parameterization remains true when we select from several standard fixed-form basis functions, such as:

- Linear ( $\psi(r) = r$ ): The simplest form, providing a response proportional to distance
- Cubic ( $\psi(r) = r^3$ ): Offers stronger emphasis on points farther from the center
- Thin plate spline ( $\psi(r) = r^2 \ln r$ ): Models the physical bending of a thin sheet, providing excellent smoothness properties

While these fixed basis functions are computationally efficient, they offer limited flexibility in how they generalize across the design space. For more adaptive modeling power, we can employ parametric basis functions that introduce additional tunable parameters:

- Gaussian ( $\psi(r) = e^{-r^2/(2\sigma^2)}$ ): Produces bell-shaped curves with  $\sigma$  controlling the width of influence
- Multiquadric ( $\psi(r) = (r^2 + \sigma^2)^{1/2}$ ): Provides broader coverage with less localized effects
- Inverse multiquadric ( $\psi(r) = (r^2 + \sigma^2)^{-1/2}$ ): Offers sharp peaks near centers with asymptotic behavior

The parameter  $\sigma$  in these functions serves as a shape parameter that controls how rapidly the function's influence decays with distance. This added flexibility enables significantly better generalization, particularly when modeling complex responses, though at the cost of a more involved parameter estimation process requiring optimization of both weights and shape parameters.

**Example 7.1** (Gaussian RBF). Using the general definition of a radial basis function (Equation 7.2), we can express the Gaussian RBF as:

$$\psi(\vec{x}) = \exp\left(-\frac{\|\vec{x} - \vec{c}\|^2}{2\sigma^2}\right) = \exp\left(-\frac{\sum_{j=1}^k (x_j - c_j)^2}{2\sigma^2}\right) \quad (7.4)$$

where:

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- $\vec{x}$  is the input vector,
- $\vec{c}$  is the center vector,
- $\|\vec{x} - \vec{c}\|$  is the Euclidean distance between the input and center, and
- $\sigma$  is the width parameter that controls how quickly the function's response diminishes with distance from the center.

The Gaussian RBF produces a bell-shaped response that reaches its maximum value of 1 when  $\vec{x} = \vec{c}$  and asymptotically approaches zero as the distance increases. The parameter  $\sigma$  determines how “localized” the response is—smaller values create a narrower peak with faster decay, while larger values produce a broader, more gradual response across the input space. Figure 7.1 shows the Gaussian RBF for different values of  $\sigma$  in an one-dimensional space. The center of the RBF is set at 0, and the width parameter  $\sigma$  varies to illustrate how it affects the shape of the function.

```
def gaussian_rbf(x, center, sigma):  
    """  
    Compute the Gaussian Radial Basis Function.  
  
    Args:  
        x (ndarray): Input points  
        center (float): Center of the RBF  
        sigma (float): Width parameter  
  
    Returns:  
        ndarray: RBF values  
    """  
    return np.exp(-((x - center)**2) / (2 * sigma**2))
```

The sum of Gaussian RBFs can be visualized by summing the individual Gaussian RBFs centered at different points as shown in Figure 7.2. The following code snippet demonstrates how to create this plot showing the sum of three Gaussian RBFs with different centers and a common width parameter  $\sigma$ .

### 7.1.1.2. The Interpolation Condition: Elegant Solutions Through Linear Systems

A remarkable property of radial basis function models is that regardless of which basis functions we choose—parametric or fixed—determining the weights  $\vec{w}$  remains straightforward through interpolation. The core principle is elegantly simple: we require our model to exactly reproduce the observed data points:

$$\hat{f}(\vec{x}^{(i)}) = y^{(i)}, \quad i = 1, 2, \dots, n. \quad (7.5)$$

This constraint produces one of the most powerful aspects of RBF modeling: while the system in Equation 7.5 is linear with respect to the weights  $\vec{w}$ , the resulting predictor  $\hat{f}$

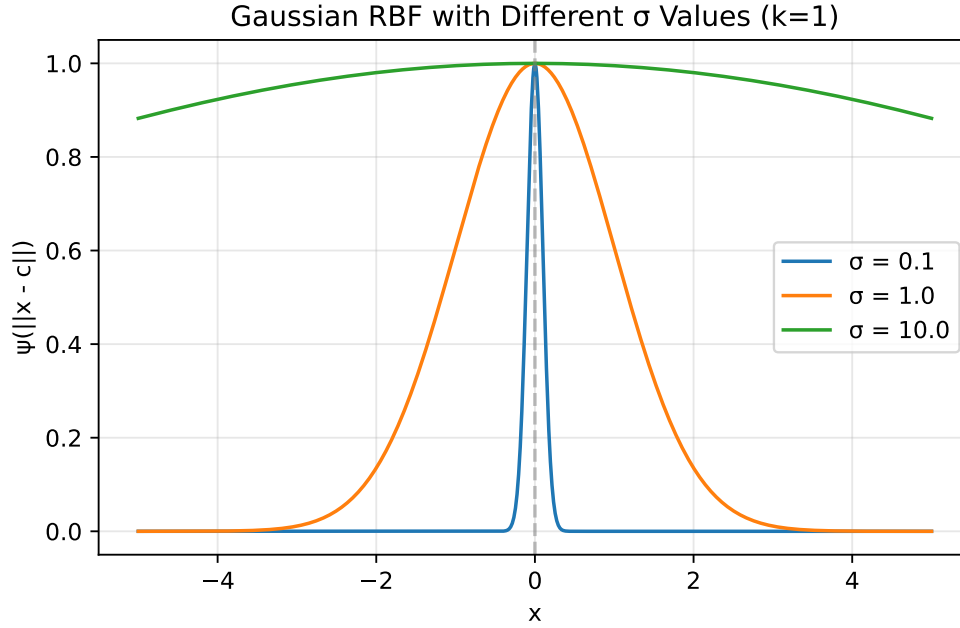


Figure 7.1.: Gaussian RBF

can capture highly nonlinear relationships in the data. The RBF approach transforms a complex nonlinear modeling problem into a solvable linear algebra problem.

For a unique solution to exist, we require that the number of basis functions equals the number of data points ( $n_c = n$ ). The standard practice, which greatly simplifies implementation, is to center each basis function at a training data point, setting  $\vec{c}^{(i)} = \vec{x}^{(i)}$  for all  $i = 1, 2, \dots, n$ . This choice allows us to express the interpolation condition as a compact matrix equation:

$$\Psi \vec{w} = \vec{y}.$$

Here,  $\Psi$  represents the Gram matrix (also called the design matrix or kernel matrix), whose elements measure the similarity between data points:

$$\Psi_{i,j} = \psi(\|\vec{x}^{(i)} - \vec{x}^{(j)}\|), \quad i, j = 1, 2, \dots, n.$$

The solution for the weight vector becomes:

$$\vec{w} = \Psi^{-1} \vec{y}.$$

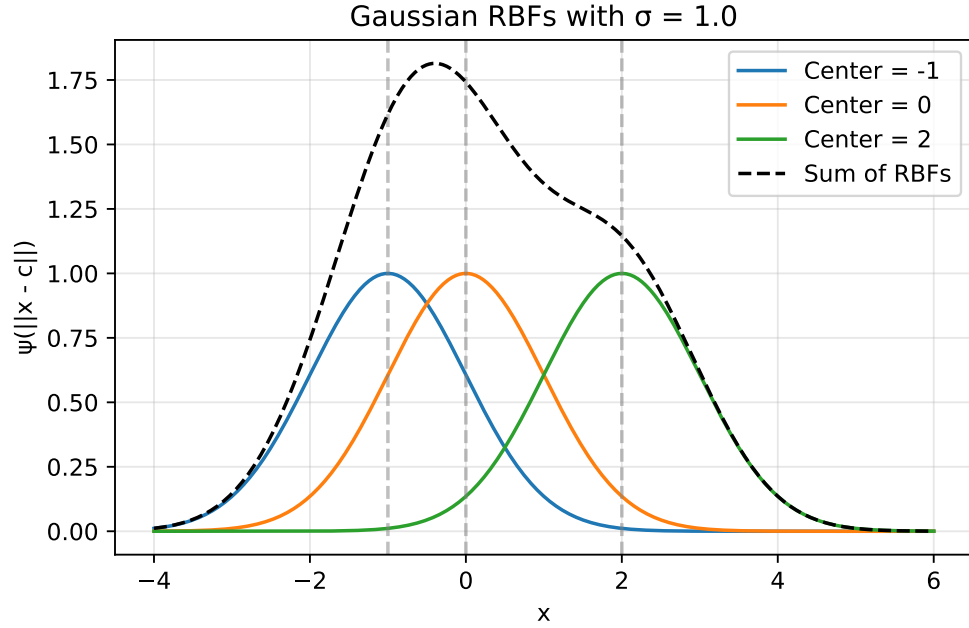


Figure 7.2.: Sum of Gaussian RBFs

This matrix inversion step is the computational core of the RBF model fitting process, and the numerical properties of this operation depend critically on the chosen basis function. Different basis functions produce Gram matrices with distinct conditioning properties, directly affecting both computational stability and the model's generalization capabilities.

### 7.1.2. Numerical Stability Through Positive Definite Matrices

A significant advantage of Gaussian and inverse multiquadric basis functions lies in their mathematical guarantees. Vapnik (1998) demonstrated that these functions always produce symmetric positive definite Gram matrices when using strictly positive definite kernels (see Section 9.4), which is a critical property for numerical reliability. Unlike other basis functions that may lead to ill-conditioned systems, these functions ensure the existence of unique, stable solutions.

This positive definiteness enables the use of Cholesky factorization, which offers substantial computational advantages over standard matrix inversion techniques. The Cholesky approach reduces the computational cost (reducing from  $O(n^3)$  to roughly  $O(n^3/3)$ ) while significantly improving numerical stability when handling the inevitable rounding errors in floating-point arithmetic. This robustness to numerical issues explains why



Gaussian and inverse multiquadric basis functions remain the preferred choice in many practical RBF implementations.

Furthermore, the positive definiteness guarantee provides theoretical assurances about the model's interpolation properties—ensuring that the RBF interpolant exists and is unique for any distinct set of centers. This mathematical foundation gives practitioners confidence in the method's reliability, particularly for complex engineering applications where model stability is paramount.

The computational advantage stems from how a symmetric positive definite matrix  $\Psi$  can be efficiently decomposed into the product of an upper triangular matrix  $U$  and its transpose:

$$\Psi = U^T U.$$

This decomposition transforms the system

$$\Psi \vec{w} = \vec{y}$$

into

$$U^T U \vec{w} = \vec{y},$$

which can be solved through two simpler triangular systems:

- First solve  $U^T \vec{v} = \vec{y}$  for the intermediate vector  $\vec{v}$
- Then solve  $U \vec{w} = \vec{v}$  for the desired weights  $\vec{w}$

In Python implementations, this process is elegantly handled using NumPy's or SciPy's Cholesky decomposition functions, followed by specialized solvers that exploit the triangular structure:

```
from scipy.linalg import cholesky, cho_solve
# Compute the Cholesky factorization
L = cholesky(Psi, lower=True) # L is the lower triangular factor
weights = cho_solve((L, True), y) # Efficient solver for (L L^T)w = y
```

### 7.1.3. Ill-Conditioning

An important numerical consideration in RBF modeling is that points positioned extremely close to each other in the input space  $X$  can lead to severe ill-conditioning of the Gram matrix (Micchelli 1986). This ill-conditioning manifests as nearly linearly dependent rows and columns in  $\Psi$ , potentially causing the Cholesky factorization to fail.

While this problem rarely arises with initial space-filling experimental designs (such as Latin Hypercube or quasi-random sequences), it frequently emerges during sequential

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optimization processes that adaptively add infill points in promising regions. As these clusters of points concentrate in areas of high interest, the condition number of the Gram matrix deteriorates, jeopardizing numerical stability.

Several mitigation strategies exist: regularization through ridge-like penalties (modifying the standard RBF interpolation problem by adding a penalty term to the diagonal of the Gram matrix. This creates a literal “ridge” along the diagonal of the matrix), removing nearly coincident points, clustering, or applying more sophisticated approaches. One theoretically elegant solution involves augmenting non-conditionally positive definite basis functions with polynomial terms (Keane and Nair 2005). This technique not only improves conditioning but also ensures polynomial reproduction properties, enhancing the approximation quality for certain function classes while maintaining numerical stability.

Beyond determining  $\vec{w}$ , there is, of course, the additional task of estimating any other parameters introduced via the basis functions. A typical example is the  $\sigma$  of the Gaussian basis function, usually taken to be the same for all basis functions, though a different one can be selected for each centre, as is customary in the case of the Kriging basis function, to be discussed shortly (once again, we trade additional parameter estimation complexity versus increased flexibility and, hopefully, better generalization).

### 7.1.4. Parameter Optimization: A Two-Level Approach

When building RBF models, we face two distinct parameter estimation challenges:

- Determining the weights ( $\vec{w}$ ): These parameters ensure our model precisely reproduces the training data. For any fixed basis function configuration, we can calculate these weights directly through linear algebra as shown earlier.
- Optimizing shape parameters (like  $\sigma$  in Gaussian RBF): These parameters control how the model generalizes to new, unseen data. Unlike weights, there’s no direct formula to find their optimal values.

To address this dual challenge, we employ a nested optimization strategy (inner and outer levels):

#### 7.1.4.1. Inner Level ( $\vec{w}$ )

For each candidate value of shape parameters (e.g.,  $\sigma$ ), we determine the corresponding optimal weights  $\vec{w}$  by solving the linear system. The `estim_weights()` method implements the inner level optimization by calculating the optimal weights  $\vec{w}$  for a given shape parameter ( $\sigma$ ):

```

def estim_weights(self):
    # [...]

    # Construct the Phi (Psi) matrix
    self.Phi = np.zeros((n, n))
    for i in range(n):
        for j in range(i+1):
            self.Phi[i, j] = self.basis(d[i, j], self.sigma)
            self.Phi[j, i] = self.Phi[i, j]

    # Calculate weights using appropriate method
    if self.code == 4 or self.code == 6:
        # Use Cholesky factorization for Gaussian or inverse multiquadric
        try:
            L = cholesky(self.Phi, lower=True)
            self.weights = cho_solve((L, True), self.y)
            self.success = True
        except np.linalg.LinAlgError:
            # Error handling...
    else:
        # Use direct solve for other basis functions
        try:
            self.weights = np.linalg.solve(self.Phi, self.y)
            self.success = True
        except np.linalg.LinAlgError:
            # Error handling...

    return self

```

This method:

- Creates the Gram matrix (Phi) based on distances between points
- Solves the linear system  $\Psi \vec{w} = \vec{y}$  for weights
- Uses appropriate numerical methods based on the basis function type (Cholesky factorization or direct solve)

#### 7.1.4.2. Outer Level ( $\sigma$ )

We use cross-validation to evaluate how well the model generalizes with different shape parameter values. The outer level optimization is implemented within the `fit()` method, where cross-validation is used to evaluate different  $\sigma$  values:

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```
def fit(self):
    if self.code < 4:
        # Fixed basis function, only w needs estimating
        self.estim_weights()
    else:
        # Basis function requires a sigma, estimate first using cross-validation
        # [...]

        # Generate candidate sigma values
        sigmas = np.logspace(-2, 2, 30)

        # Setup cross-validation (determine number of folds)
        # [...]

        cross_val = np.zeros(len(sigmas))

        # For each candidate sigma value
        for sig_index, sigma in enumerate(sigmas):
            print(f"Computing cross-validation metric for Sigma={sigma:.4f}...")

            # Perform k-fold cross-validation
            for j in range(len(from_idx)):
                # Create and fit model on training subset
                temp_model = Rbf(
                    X=X_orig[xs_temp],
                    y=y_orig[xs_temp],
                    code=self.code
                )
                temp_model.sigma = sigma

                # Call inner level optimization
                temp_model.estim_weights()

                # Evaluate on held-out data
                # [...]

            # Select best sigma based on cross-validation performance
            min_cv_index = np.argmin(cross_val)
            best_sig = sigmas[min_cv_index]

        # Use the best sigma for final model
        self.sigma = best_sig
        self.estim_weights() # Call inner level again with optimal sigma
```

The outer level:

- Generates a range of candidate  $\sigma$  values
- For each  $\sigma$ , performs k-fold cross-validation:
  - Creates models on subsets of the data
  - Calls the inner level method (`estim_weights()`) to determine weights
  - Evaluates prediction quality on held-out data
- Selects the  $\sigma$  that minimizes cross-validation error
- Performs a final call to the inner level method with the optimal  $\sigma$

This two-level approach is particularly critical for parametric basis functions (Gaussian, multiquadric, etc.), where the wrong choice of shape parameter could lead to either overfitting (too much flexibility) or underfitting (too rigid). Cross-validation provides an unbiased estimate of how well different parameter choices will perform on new data, helping us balance the trade-off between fitting the training data perfectly and generalizing well.

## 7.2. Python Implementation of the RBF Model

Section 7.2 shows a Python implementation of this parameter estimation process (based on a cross-validation routine), which will represent the surrogate, once its parameters have been estimated. The model building process is very simple.

Instead of using a dictionary for bookkeeping, we implement a Python class `Rbf` that encapsulates all the necessary data and functionality. The class stores the sampling plan  $X$  as the `X` attribute and the corresponding  $n$ -vector of responses  $y$  as the `y` attribute. The `code` attribute specifies the type of basis function to be used. After fitting the model, the class will also contain the estimated parameter values  $\bar{w}$  and, if a parametric basis function is used,  $\sigma$ . These are stored in the `weights` and `sigma` attributes respectively.

Finally, a note on prediction error estimation. We have already indicated that the guarantee of a positive definite  $\Psi$  is one of the advantages of Gaussian radial basis functions. They also possess another desirable feature: it is relatively easy to estimate their prediction error at any  $\bar{x}$  in the design space. Additionally, the expectation function of the improvement in minimum (or maximum) function value with respect to the minimum (or maximum) known so far can also be calculated quite easily, both of these features being very useful when the optimization of  $f$  is the goal of the surrogate modelling process.

## 7. Radial Basis Function Models

### 7.2.1. The Rbf Class

The Rbf class implements the Radial Basis Function model. It encapsulates all the data and methods needed for fitting the model and making predictions.

```
import numpy as np
from scipy.linalg import cholesky, cho_solve
import numpy.random as rnd

class Rbf:
    """Radial Basis Function model implementation.

    Attributes:
        X (ndarray): The sampling plan (input points).
        y (ndarray): The response vector.
        code (int): Type of basis function to use.
        weights (ndarray, optional): The weights vector (set after fitting).
        sigma (float, optional): Parameter for parametric basis functions.
        Phi (ndarray, optional): The Gram matrix (set during fitting).
        success (bool, optional): Flag indicating successful fitting.
    """

    def __init__(self, X=None, y=None, code=3):
        """Initialize the RBF model.

        Args:
            X (ndarray, optional):
                The sampling plan.
            y (ndarray, optional):
                The response vector.
            code (int, optional):
                Type of basis function.
                Default is 3 (thin plate spline).
        """
        self.X = X
        self.y = y
        self.code = code
        self.weights = None
        self.sigma = None
        self.Phi = None
        self.success = None

    def basis(self, r, sigma=None):
        """Compute the value of the basis function.
```

## 7.2. Python Implementation of the RBF Model

```
Args:
    r (float): Radius (distance)
    sigma (float, optional): Parameter for parametric basis functions

Returns:
    float: Value of the basis function
"""
# Use instance sigma if not provided
if sigma is None and hasattr(self, 'sigma'):
    sigma = self.sigma

if self.code == 1:
    # Linear function
    return r
elif self.code == 2:
    # Cubic
    return r**3
elif self.code == 3:
    # Thin plate spline
    if r < 1e-200:
        return 0
    else:
        return r**2 * np.log(r)
elif self.code == 4:
    # Gaussian
    return np.exp(-(r**2)/(2*sigma**2))
elif self.code == 5:
    # Multi-quadric
    return (r**2 + sigma**2)**0.5
elif self.code == 6:
    # Inverse Multi-Quadric
    return (r**2 + sigma**2)**(-0.5)
else:
    raise ValueError("Invalid basis function code")

def estim_weights(self):
    """Estimates the basis function weights if sigma is known or not required.

    Returns:
        self: The updated model instance
    """
    # Check if sigma is required but not provided
    if self.code > 3 and self.sigma is None:
        raise ValueError("The basis function requires a sigma parameter")
```

## 7. Radial Basis Function Models

```
# Number of points
n = len(self.y)

# Build distance matrix
d = np.zeros((n, n))
for i in range(n):
    for j in range(i+1):
        d[i, j] = np.linalg.norm(self.X[i] - self.X[j])
        d[j, i] = d[i, j]

# Construct the Phi (Psi) matrix
self.Phi = np.zeros((n, n))
for i in range(n):
    for j in range(i+1):
        self.Phi[i, j] = self.basis(d[i, j], self.sigma)
        self.Phi[j, i] = self.Phi[i, j]

# Calculate weights using appropriate method
if self.code == 4 or self.code == 6:
    # Use Cholesky factorization for Gaussian or inverse multiquadric
    try:
        L = cholesky(self.Phi, lower=True)
        self.weights = cho_solve((L, True), self.y)
        self.success = True
    except np.linalg.LinAlgError:
        print("Cholesky factorization failed.")
        print("Two points may be too close together.")
        self.weights = None
        self.success = False
else:
    # Use direct solve for other basis functions
    try:
        self.weights = np.linalg.solve(self.Phi, self.y)
        self.success = True
    except np.linalg.LinAlgError:
        self.weights = None
        self.success = False

return self

def fit(self):
    """Estimates the parameters of the Radial Basis Function model.

    Returns:
```



## 7.2. Python Implementation of the RBF Model

```
self: The updated model instance
"""
if self.code < 4:
    # Fixed basis function, only w needs estimating
    self.estim_weights()
else:
    # Basis function also requires a sigma, estimate first
    # Save original model data
    X_orig = self.X.copy()
    y_orig = self.y.copy()

    # Direct search between 10^-2 and 10^2
    sigmas = np.logspace(-2, 2, 30)

    # Number of cross-validation subsets
    if len(self.X) < 6:
        q = 2
    elif len(self.X) < 15:
        q = 3
    elif len(self.X) < 50:
        q = 5
    else:
        q = 10

    # Number of sample points
    n = len(self.X)

    # X split into q randomly selected subsets
    xs = rnd.permutation(n)
    full_xs = xs.copy()

    # The beginnings of the subsets...
    from_idx = np.arange(0, n, n//q)
    if from_idx[-1] >= n:
        from_idx = from_idx[:-1]

    # ...and their ends
    to_idx = np.zeros_like(from_idx)
    for i in range(len(from_idx) - 1):
        to_idx[i] = from_idx[i+1] - 1
    to_idx[-1] = n - 1

    cross_val = np.zeros(len(sigmas))
```

## 7. Radial Basis Function Models

```
# Cycling through the possible values of Sigma
for sig_index, sigma in enumerate(sigmas):
    print(f"Computing cross-validation metric for Sigma={sigma:.4f}...")

    cross_val[sig_index] = 0

    # Model fitting to subsets of the data
    for j in range(len(from_idx)):
        removed = xs[from_idx[j]:to_idx[j]+1]
        xs_temp = np.delete(xs, np.arange(from_idx[j], to_idx[j]+1))

        # Create a temporary model for CV
        temp_model = Rbf(
            X=X_orig[xs_temp],
            y=y_orig[xs_temp],
            code=self.code
        )
        temp_model.sigma = sigma

        # Sigma and subset chosen, now estimate w
        temp_model.estim_weights()

        if temp_model.weights is None:
            cross_val[sig_index] = 1e20
            xs = full_xs.copy()
            break

        # Compute vector of predictions at the removed sites
        pr = np.zeros(len(removed))
        for jj, idx in enumerate(removed):
            pr[jj] = temp_model.predict(X_orig[idx])

        # Calculate cross-validation error
        cross_val[sig_index] += np.sum((y_orig[removed] - pr)**2) / len(r

    xs = full_xs.copy()

    # Now attempt Cholesky on the full set, in case the subsets could
    # be fitted correctly, but the complete X could not
    temp_model = Rbf(
        X=X_orig,
        y=y_orig,
        code=self.code
    )
```

```

        temp_model.sigma = sigma
        temp_model.estim_weights()

        if temp_model.weights is None:
            cross_val[sig_index] = 1e20
            print("Failed to fit complete sample data.")

        # Find the best sigma
        min_cv_index = np.argmin(cross_val)
        best_sig = sigmas[min_cv_index]

        # Set the best sigma and recompute weights
        print(f"Selected sigma={best_sig:.4f}")
        self.sigma = best_sig
        self.estim_weights()

    return self

def predict(self, x):
    """Calculates the value of the Radial Basis Function surrogate model at x.

    Args:
        x (ndarray): Point at which to make prediction

    Returns:
        float: Predicted value
    """
    # Calculate distances to all sample points
    d = np.zeros(len(self.X))
    for k in range(len(self.X)):
        d[k] = np.linalg.norm(x - self.X[k])

    # Calculate basis function values
    phi = np.zeros(len(self.X))
    for k in range(len(self.X)):
        phi[k] = self.basis(d[k], self.sigma)

    # Calculate prediction
    y = np.dot(phi, self.weights)
    return y

```

### 7.3. RBF Example: The One-Dimensional $\sin$ Function

```

import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import cholesky, cho_solve

# Define the data points for fitting
x_centers = np.array([np.pi/2, np.pi, 3*np.pi/2]).reshape(-1, 1) # Centers for RBFs
y_values = np.sin(x_centers.flatten()) # Sine values at these points

# Create and fit the RBF model
rbf_model = Rbf(X=x_centers, y=y_values, code=4) # Code 4 is Gaussian RBF
rbf_model.sigma = 1.0 # Set sigma parameter directly
rbf_model.estimate_weights() # Calculate optimal weights

# Print the weights
print("RBF model weights:", rbf_model.weights)

# Create a grid for visualization
x_grid = np.linspace(0, 2*np.pi, 1000).reshape(-1, 1)
y_true = np.sin(x_grid.flatten()) # True sine function

# Generate predictions using the RBF model
y_pred = np.zeros(len(x_grid))
for i in range(len(x_grid)):
    y_pred[i] = rbf_model.predict(x_grid[i])

# Calculate individual basis functions for visualization
basis_funcs = np.zeros((len(x_grid), len(x_centers)))
for i in range(len(x_grid)):
    for j in range(len(x_centers)):
        # Calculate distance
        distance = np.linalg.norm(x_grid[i] - x_centers[j])
        # Compute basis function value scaled by its weight
        basis_funcs[i, j] = rbf_model.basis(distance, rbf_model.sigma) * rbf_model.weights[j]

# Plot the results
plt.figure(figsize=(6, 4))

# Plot the true sine function
plt.plot(x_grid, y_true, 'k-', label='True sine function', linewidth=2)

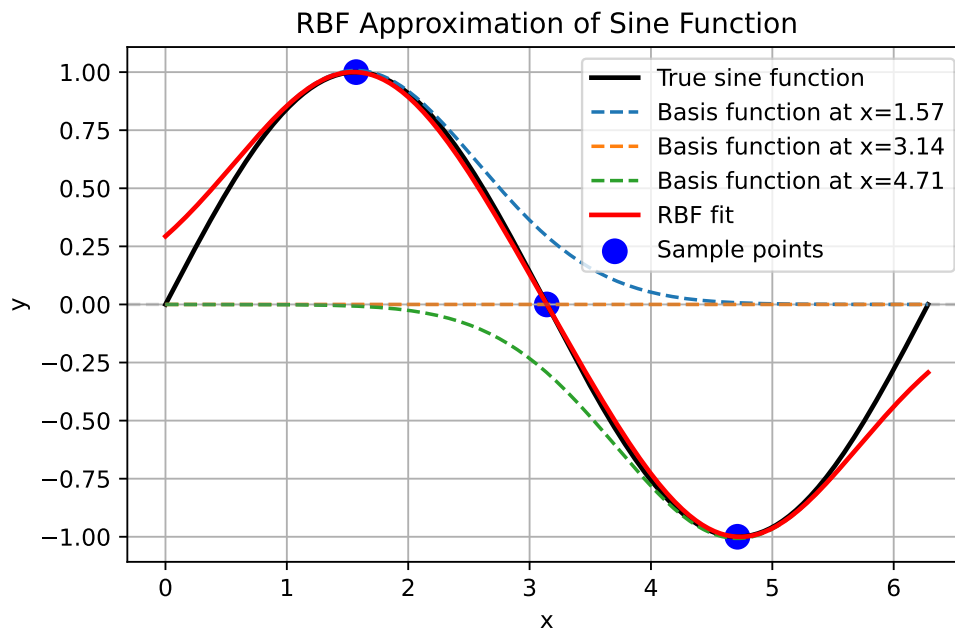
# Plot individual basis functions

```

### 7.3. RBF Example: The One-Dimensional $\sin$ Function

```
for i in range(len(x_centers)):  
    plt.plot(x_grid, basis_funcs[:, i], '--',  
             label=f'Basis function at x={x_centers[i][0]:.2f}')  
  
# Plot the RBF fit (sum of basis functions)  
plt.plot(x_grid, y_pred, 'r-', label='RBF fit', linewidth=2)  
  
# Plot the sample points  
plt.scatter(x_centers, y_values, color='blue', s=100, label='Sample points')  
  
# Add horizontal line at y=0  
plt.axhline(y=0, color='gray', linestyle='--', alpha=0.3)  
  
plt.title('RBF Approximation of Sine Function')  
plt.xlabel('x')  
plt.ylabel('y')  
plt.grid(True)  
plt.legend()  
plt.tight_layout()  
plt.show()
```

RBF model weights: [ 1.00724398e+00 2.32104414e-16 -1.00724398e+00]



## 7.4. RBF Example: The Two-Dimensional dome Function

The `dome` function is an example of a test function that can be used to evaluate the performance of the Radial Basis Function model. It is a simple mathematical function defined over a two-dimensional space.

```
def dome(x) -> float:
    """
    Dome test function.

    Args:
        x (ndarray): Input vector (1D array of length 2)

    Returns:
        float: Function value

    Examples:
        dome(np.array([0.5, 0.5]))
    """
    return np.sum(1 - (2*x - 1)**2) / len(x)
```

The following code demonstrates how to use the Radial Basis Function model to approximate a function. It generates a Latin Hypercube sample, computes the objective function values, estimates the model parameters, and plots the results.

```
def generate_rbf_data(n_samples=10, grid_points=41):
    """
    Generates data for RBF visualization.

    Args:
        n_samples (int): Number of samples for the RBF model
        grid_points (int): Number of grid points for prediction

    Returns:
        tuple: (rbf_model, X, Y, Z, Z_0) - Model and grid data for plotting
    """
    from spotpython.utils.sampling import bestlh as best_lh
    # Generate sampling plan
    X_samples = best_lh(n_samples, 2, population=10, iterations=100)
    # Compute objective function values
    y_samples = np.zeros(len(X_samples))
    for i in range(len(X_samples)):
        y_samples[i] = dome(X_samples[i])
    # Create and fit RBF model
```

#### 7.4. RBF Example: The Two-Dimensional dome Function

```
rbf_model = Rbf(X=X_samples, y=y_samples, code=3) # Thin plate spline
rbf_model.fit()
# Generate grid for prediction
x = np.linspace(0, 1, grid_points)
y = np.linspace(0, 1, grid_points)
X, Y = np.meshgrid(x, y)
Z_0 = np.zeros_like(X)
Z = np.zeros_like(X)

# Evaluate model at grid points
for i in range(len(x)):
    for j in range(len(y)):
        Z_0[j, i] = dome(np.array([x[i], y[j]]))
        Z[j, i] = rbf_model.predict(np.array([x[i], y[j]]))

return rbf_model, X, Y, Z, Z_0

def plot_rbf_results(rbf_model, X, Y, Z, Z_0=None, n_contours=10):
    """
    Plots RBF approximation results.

    Args:
        rbf_model (Rbf): Fitted RBF model
        X (ndarray): Grid X-coordinates
        Y (ndarray): Grid Y-coordinates
        Z (ndarray): RBF model predictions
        Z_0 (ndarray, optional): True function values for comparison
        n_contours (int): Number of contour levels to plot
    """
    import matplotlib.pyplot as plt

    plt.figure(figsize=(10, 8))

    # Plot the contour
    contour = plt.contour(X, Y, Z, n_contours)

    if Z_0 is not None:
        contour_0 = plt.contour(X, Y, Z_0, n_contours, colors='k', linestyle='dashed')

    # Plot the sample points
    plt.scatter(rbf_model.X[:, 0], rbf_model.X[:, 1],
                c='r', marker='o', s=50)

    plt.title('RBF Approximation (Thin Plate Spline)')
```

## 7. Radial Basis Function Models

```
plt.xlabel('x1')
plt.ylabel('x2')
plt.colorbar(label='f(x1, x2)')
plt.show()
```

Figure 7.3 shows the contour plots of the underlying function  $f(x_1, x_2) = 0.5[-(2x_1 - 1)^2 - (2x_2 - 1)^2]$  and its thin plate spline radial basis function approximation, along with the 10 points of a Morris-Mitchell optimal Latin hypercube sampling plan (obtained via `best_lh()`).

```
rbf_model, X, Y, Z, Z_0 = generate_rbf_data(n_samples=10, grid_points=41)
plot_rbf_results(rbf_model, X, Y, Z, Z_0)
```

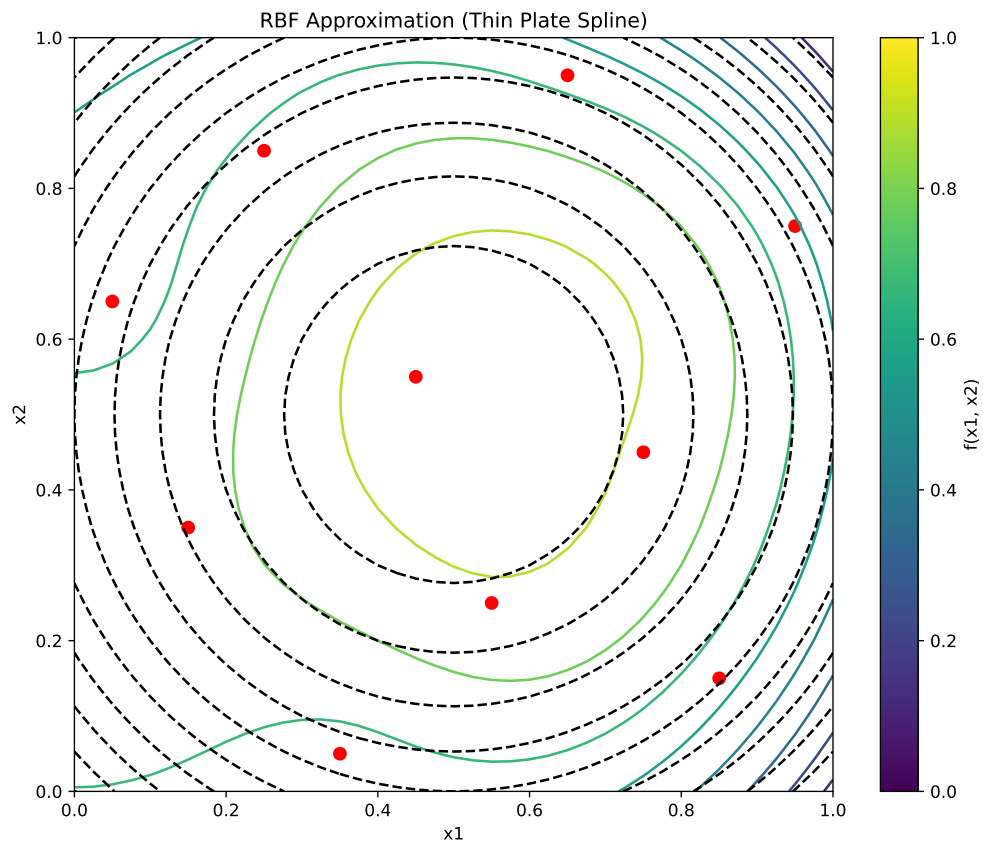


Figure 7.3.: RBF Approximation.



### 7.4.1. The Connection Between RBF Models and Neural Networks

Radial basis function models share a profound architectural similarity with artificial neural networks, specifically with what's known as RBF networks. This connection provides valuable intuition about how RBF models function. A radial basis function model can be viewed as a specialized neural network with the following structure:

- Input Layer: Receives the feature vector  $\vec{x}$
- Hidden Layer: Contains neurons (basis functions) that compute radial distances
- Output Layer: Produces a weighted sum of the hidden unit activations

Unlike traditional neural networks that use dot products followed by nonlinear activation functions, RBF networks measure the distance between inputs and learned center points. This distance is then transformed by the radial basis function.

Mathematically, the equivalence between RBF models and RBF networks can be expressed as follows:

The RBF model equation:

$$\hat{f}(\vec{x}) = \sum_{i=1}^{n_c} w_i \psi(\|\vec{x} - \vec{c}^{(i)}\|)$$

directly maps to the following neural network components:

- $\vec{x}$ : Input vector
- $\vec{c}^{(i)}$ : Center vectors for each hidden neuron
- $\psi(\cdot)$ : Activation function (Gaussian, inverse multiquadric, etc.)
- $w_i$ : Output weights
- $\hat{f}(\vec{x})$ : Network output

**Example 7.2** (Comparison of RBF Networks and Traditional Neural Networks). Consider approximating a simple 1D function  $f(x) = \sin(2\pi x)$  over the interval  $[0, 1]$ :

The neural network approach would use multiple layers with neurons computing  $\sigma(w \cdot x + b)$ . It would require a large number of neurons and layers to capture the sine wave's complexity. The network would learn both weights and biases, making it less interpretable.

The RBF network approach, on the other hand, places basis functions at strategic points (e.g., 5 evenly spaced centers). Each neuron computes  $\psi(\|x - c_i\|)$  (e.g., using Gaussian RBF). The output layer combines these values with learned weights. If we place Gaussian RBFs with  $\sigma = 0.15$  at 0.1, 0.3, 0.5, 0.7, 0.9, each neuron responds strongly when the input is close to its center and weakly otherwise. The network can then learn weights that, when multiplied by these response patterns and summed, closely approximate the sine function.

## 7. Radial Basis Function Models

This locality property gives RBF networks a notable advantage: they offer more interpretable internal representations and often require fewer neurons for certain types of function approximation compared to traditional multilayer perceptrons.

The key insight is that while standard neural networks create complex decision boundaries through compositions of hyperplanes, RBF networks directly model functions using a set of overlapping “bumps” positioned strategically in the input space.

### 7.5. Radial Basis Function Models for Noisy Data

When the responses  $\vec{y} = y^{(1)}, y^{(2)}, \dots, y^{(n)T}$  contain measurement or simulation noise, the standard RBF interpolation approach can lead to overfitting—the model captures both the underlying function and the random noise. This compromises generalization performance on new data points. Two principal strategies address this challenge:

#### 7.5.1. Ridge Regularization Approach

The most straightforward solution involves introducing regularization through the parameter  $\lambda$  (Poggio and Girosi 1990). This is implemented by adding  $\lambda$  to the diagonal elements of the Gram matrix, creating a “ridge” that improves numerical stability. Mathematically, the weights are determined by:

$$\vec{w} = (\Psi + \lambda I)^{-1} \vec{y},$$

where  $I$  is an  $n \times n$  identity matrix. This regularized solution balances two competing objectives:

- fitting the training data accurately versus
- keeping the magnitude of weights controlled to prevent overfitting.

Theoretically, optimal performance is achieved when  $\lambda$  equals the variance of the noise in the response data  $\vec{y}$  (Keane and Nair 2005). Since this information is rarely available in practice,  $\lambda$  is typically estimated through cross-validation alongside other model parameters.

#### 7.5.2. Reduced Basis Approach

An alternative strategy involves reducing  $m$ , the number of basis functions. This might result in a non-square  $\Psi$  matrix. With a non-square  $\Psi$  matrix, the weights are found through least squares minimization:

$$\vec{w} = (\Psi^T \Psi)^{-1} \Psi^T \vec{y}$$

This approach introduces an important design decision: which subset of points should serve as basis function centers? Several selection strategies exist:

- Clustering methods that identify representative points
- Greedy algorithms that sequentially select influential centers
- Support vector regression techniques (discussed elsewhere in the literature)

Additional parameters such as the width parameter  $\sigma$  in Gaussian bases can be optimized through cross-validation to minimize generalization error estimates.

The ridge regularization and reduced basis approaches can be combined, allowing for a flexible modeling framework, though at the cost of a more complex parameter estimation process. This hybrid approach often yields superior results for highly noisy datasets or when the underlying function has varying complexity across the input space.

The broader challenge of building accurate models from noisy observations is examined comprehensively in the context of Kriging models, which provide a statistical framework for explicitly modeling both the underlying function and the noise process.

## 7.6. Jupyter Notebook

### **i** Note

- The Jupyter-Notebook of this lecture is available on GitHub in the Hyperparameter-Tuning-Cookbook Repository



## 8. Kriging (Gaussian Process Regression)

### Note

- This section is based on chapter 2.4 in Forrester, Sóbester, and Keane (2008).
- The following Python packages are imported:

```
import matplotlib.pyplot as plt
import numpy as np
from numpy import (array, zeros, power, ones, exp, multiply,
                  eye, linspace, spacing, sqrt, arange,
                  append, ravel)
from numpy.linalg import cholesky, solve
from scipy.spatial.distance import squareform, pdist, cdist
```

### 8.1. From Gaussian RBF to Kriging Basis Functions

Kriging can be explained using the concept of radial basis functions (RBFs), which were introduced in Chapter 7. An RBF is a real-valued function whose value depends only on the distance from a certain point, called the center, usually in a multidimensional space. The basis function is a function of the distance between the point  $\vec{x}$  and the center  $\vec{x}^{(i)}$ . Other names for basis functions are *kernel* or *covariance* functions.

**Definition 8.1** (The Kriging Basis Functions). Kriging (also known as Gaussian Process Regression) uses  $k$ -dimensional basis functions of the form

$$\psi^{(i)}(\vec{x}) = \psi(\vec{x}^{(i)}, \vec{x}) = \exp \left( - \sum_{j=1}^k \theta_j |x_j^{(i)} - x_j|^{p_j} \right), \quad (8.1)$$

where  $\vec{x}$  and  $\vec{x}^{(i)}$  denote the  $k$ -dim vector  $\vec{x} = (x_1, \dots, x_k)^T$  and  $\vec{x}^{(i)} = (x_1^{(i)}, \dots, x_k^{(i)})^T$ , respectively.

□

## 8. Kriging (Gaussian Process Regression)

Kriging uses a specialized basis function that offers greater flexibility than standard RBFs. Examining Equation 8.1, we can observe how Kriging builds upon and extends the Gaussian basis concept. The key enhancements of Kriging over Gaussian RBF can be summarized as follows:

- Dimension-specific width parameters: While a Gaussian RBF uses a single width parameter  $1/\sigma^2$ , Kriging employs a vector  $\vec{\theta} = (\theta_1, \theta_2, \dots, \theta_k)^T$ . This allows the model to automatically adjust its sensitivity to each input dimension, effectively performing automatic feature relevance determination.
- Flexible smoothness control: The Gaussian RBF fixes the exponent at 2, producing uniformly smooth functions. In contrast, Kriging's dimension-specific exponents  $\vec{p} = (p_1, p_2, \dots, p_k)^T$  (typically with  $p_j \in [1, 2]$ ) enable precise control over smoothness properties in each dimension.
- Unifying framework: When all exponents are set to  $p_j = 2$  and all width parameters are equal ( $\theta_j = 1/\sigma^2$  for all  $j$ ), the Kriging basis function reduces exactly to the Gaussian RBF. This makes Gaussian RBF a special case within the more general Kriging framework.

These enhancements make Kriging particularly well-suited for engineering problems where variables may operate at different scales or exhibit varying degrees of smoothness across dimensions. For now, we will only consider Kriging interpolation. We will cover Kriging regression later.

### 8.2. Building the Kriging Model

Consider sample data  $X$  and  $\vec{y}$  from  $n$  locations that are available in matrix form:  $X$  is a  $(n \times k)$  matrix, where  $k$  denotes the problem dimension and  $\vec{y}$  is a  $(n \times 1)$  vector. We want to find an expression for predicted values at a new point  $\vec{x}$ , denoted as  $\hat{y}$ .

We start with an abstract, not really intuitive concept: The observed responses  $\vec{y}$  are considered as if they are from a stochastic process, which will be denoted as

$$\begin{pmatrix} Y(\vec{x}^{(1)}) \\ \vdots \\ Y(\vec{x}^{(n)}) \end{pmatrix}. \quad (8.2)$$

The set of random vectors from Equation 8.2 (also referred to as a *random field*) has a mean of  $\vec{1}\mu$ , which is a  $(n \times 1)$  vector. The random vectors are correlated with each other using the basis function expression from Equation 8.1:

$$\text{cor}(Y(\vec{x}^{(i)}), Y(\vec{x}^{(l)})) = \exp\left(-\sum_{j=1}^k \theta_j |x_j^{(i)} - x_j^{(l)}|^{p_j}\right). \quad (8.3)$$

Using Equation 8.3, we can compute the  $(n \times n)$  correlation matrix  $\Psi$  of the observed sample data as shown in Equation 8.4,

## 8.2. Building the Kriging Model

$$\Psi = \begin{pmatrix} \text{cor}(Y(\vec{x}^{(1)}), Y(\vec{x}^{(1)})) & \dots & \text{cor}(Y(\vec{x}^{(1)}), Y(\vec{x}^{(n)})) \\ \vdots & \ddots & \vdots \\ \text{cor}(Y(\vec{x}^{(n)}), Y(\vec{x}^{(1)})) & \dots & \text{cor}(Y(\vec{x}^{(n)}), Y(\vec{x}^{(n)})) \end{pmatrix}, \quad (8.4)$$

and a covariance matrix as shown in Equation 8.5,

$$\text{Cov}(Y, Y) = \sigma^2 \Psi. \quad (8.5)$$

This assumed correlation between the sample data reflects our expectation that an engineering function will behave in a certain way and it will be smoothly and continuous.

*Remark 8.1* (Note on Stochastic Processes). See [?@sec-random-samples-gp](#) for a more detailed discussion on realizations of stochastic processes.

□

We now have a set of  $n$  random variables ( $\mathbf{Y}$ ) that are correlated with each other as described in the  $(n \times n)$  correlation matrix  $\Psi$ , see Equation 8.4. The correlations depend on the absolute distances in dimension  $j$  between the  $i$ -th and the  $l$ -th sample point  $|x_j^{(i)} - x_j^{(l)}|$  and the corresponding parameters  $p_j$  and  $\theta_j$  for dimension  $j$ . The correlation is intuitive, because when

- two points move close together, then  $|x_j^{(i)} - x_j| \rightarrow 0$  and  $\exp(-|x_j^{(i)} - x_j|^{p_j}) \rightarrow 1$  (these points show very close correlation and  $Y(x_j^{(i)}) = Y(x_j)$ ).
- two points move far apart, then  $|x_j^{(i)} - x_j| \rightarrow \infty$  and  $\exp(-|x_j^{(i)} - x_j|^{p_j}) \rightarrow 0$  (these points show very low correlation).

**Example 8.1** (Correlations for different  $p_j$ ). Three different correlations are shown in Figure 8.1:  $p_j = 0.1, 1, 2$ . The smoothness parameter  $p_j$  affects the correlation:

- With  $p_j = 0.1$ , there is basically no immediate correlation between the points and there is a near discontinuity between the points  $Y(\vec{x}_j^{(i)})$  and  $Y(\vec{x}_j)$ .
- With  $p_j = 2$ , the correlation is more smooth and we have a continuous gradient through  $x_j^{(i)} - x_j$ .

Reducing  $p_j$  increases the rate at which the correlation initially drops with distance. This is shown in Figure 8.1.

□

**Example 8.2** (Correlations for different  $\theta$ ). Figure 8.2 visualizes the correlation between two points  $Y(\vec{x}_j^{(i)})$  and  $Y(\vec{x}_j)$  for different values of  $\theta$ . The parameter  $\theta$  can be seen as a *width* parameter:

## 8. Kriging (Gaussian Process Regression)

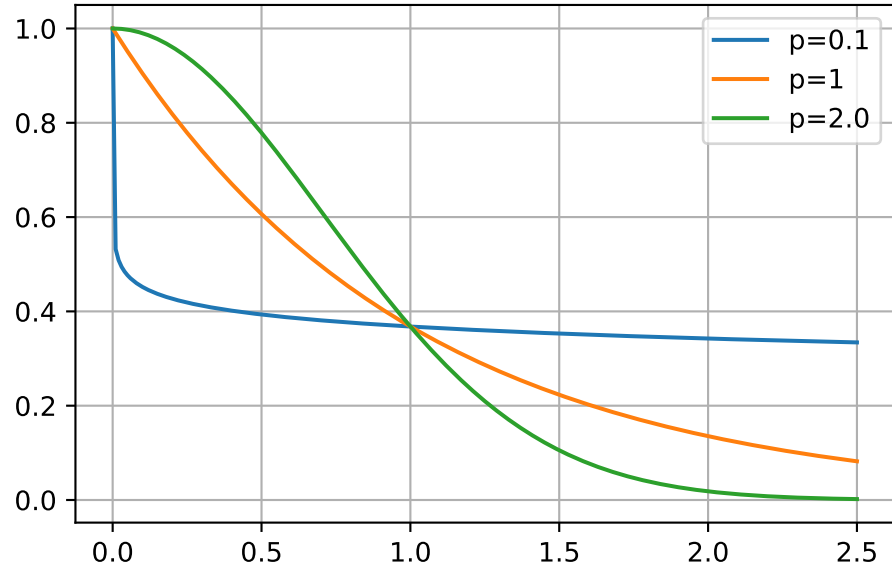


Figure 8.1.: Correlations with varying  $p$ .  $\theta$  set to 1.

- low  $\theta_j$  means that all points will have a high correlation, with  $Y(x_j)$  being similar across the sample.
- high  $\theta_j$  means that there is a significant difference between the  $Y(x_j)$ 's.
- $\theta_j$  is a measure of how *active* the function we are approximating is.
- High  $\theta_j$  indicate important parameters, see Figure 8.2.

□

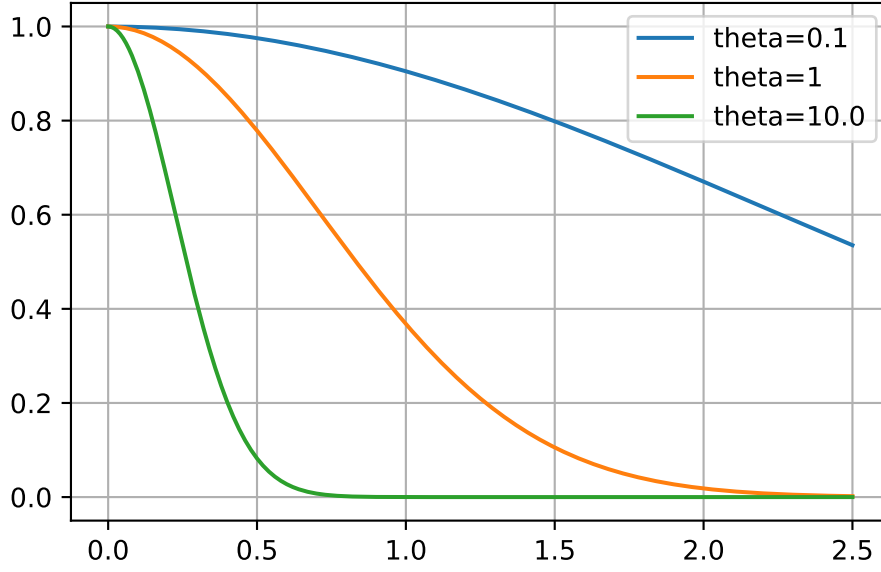
Considering the activity parameter  $\theta$  is useful in high-dimensional problems where it is difficult to visualize the design landscape and the effect of the variable is unknown. By examining the elements of the vector  $\theta$ , we can identify the most important variables and focus on them. This is a crucial step in the optimization process, as it allows us to reduce the dimensionality of the problem and focus on the most important variables.

**Example 8.3** (The Correlation Matrix (Detailed Computation)). Let  $n = 4$  and  $k = 3$ . The sample plan is represented by the following matrix  $X$ :

$$X = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \\ x_{41} & x_{42} & x_{43} \end{pmatrix}$$

To compute the elements of the matrix  $\Psi$ , the following  $k$  (one for each of the  $k$  dimensions)  $(n, n)$ -matrices have to be computed:



Figure 8.2.: Correlations with varying  $\theta$ .  $p$  set to 2.

- For  $k = 1$ , i.e., the first column of  $X$ :

$$D_1 = \begin{pmatrix} x_{11} - x_{11} & x_{11} - x_{21} & x_{11} - x_{31} & x_{11} - x_{41} \\ x_{21} - x_{11} & x_{21} - x_{21} & x_{21} - x_{31} & x_{21} - x_{41} \\ x_{31} - x_{11} & x_{31} - x_{21} & x_{31} - x_{31} & x_{31} - x_{41} \\ x_{41} - x_{11} & x_{41} - x_{21} & x_{41} - x_{31} & x_{41} - x_{41} \end{pmatrix}$$

- For  $k = 2$ , i.e., the second column of  $X$ :

$$D_2 = \begin{pmatrix} x_{12} - x_{12} & x_{12} - x_{22} & x_{12} - x_{32} & x_{12} - x_{42} \\ x_{22} - x_{12} & x_{22} - x_{22} & x_{22} - x_{32} & x_{22} - x_{42} \\ x_{32} - x_{12} & x_{32} - x_{22} & x_{32} - x_{32} & x_{32} - x_{42} \\ x_{42} - x_{12} & x_{42} - x_{22} & x_{42} - x_{32} & x_{42} - x_{42} \end{pmatrix}$$

- For  $k = 3$ , i.e., the third column of  $X$ :

$$D_3 = \begin{pmatrix} x_{13} - x_{13} & x_{13} - x_{23} & x_{13} - x_{33} & x_{13} - x_{43} \\ x_{23} - x_{13} & x_{23} - x_{23} & x_{23} - x_{33} & x_{23} - x_{43} \\ x_{33} - x_{13} & x_{33} - x_{23} & x_{33} - x_{33} & x_{33} - x_{43} \\ x_{43} - x_{13} & x_{43} - x_{23} & x_{43} - x_{33} & x_{43} - x_{43} \end{pmatrix}$$

Since the matrices are symmetric and the main diagonals are zero, it is sufficient to

### 8. Kriging (Gaussian Process Regression)

compute the following matrices:

$$D_1 = \begin{pmatrix} 0 & x_{11} - x_{21} & x_{11} - x_{31} & x_{11} - x_{41} \\ 0 & 0 & x_{21} - x_{31} & x_{21} - x_{41} \\ 0 & 0 & 0 & x_{31} - x_{41} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$D_2 = \begin{pmatrix} 0 & x_{12} - x_{22} & x_{12} - x_{32} & x_{12} - x_{42} \\ 0 & 0 & x_{22} - x_{32} & x_{22} - x_{42} \\ 0 & 0 & 0 & x_{32} - x_{42} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$D_3 = \begin{pmatrix} 0 & x_{13} - x_{23} & x_{13} - x_{33} & x_{13} - x_{43} \\ 0 & 0 & x_{23} - x_{33} & x_{23} - x_{43} \\ 0 & 0 & 0 & x_{33} - x_{43} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

We will consider  $p_l = 2$ . The differences will be squared and multiplied by  $\theta_i$ , i.e.:

$$D_1 = \theta_1 \begin{pmatrix} 0 & (x_{11} - x_{21})^2 & (x_{11} - x_{31})^2 & (x_{11} - x_{41})^2 \\ 0 & 0 & (x_{21} - x_{31})^2 & (x_{21} - x_{41})^2 \\ 0 & 0 & 0 & (x_{31} - x_{41})^2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$D_2 = \theta_2 \begin{pmatrix} 0 & (x_{12} - x_{22})^2 & (x_{12} - x_{32})^2 & (x_{12} - x_{42})^2 \\ 0 & 0 & (x_{22} - x_{32})^2 & (x_{22} - x_{42})^2 \\ 0 & 0 & 0 & (x_{32} - x_{42})^2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$D_3 = \theta_3 \begin{pmatrix} 0 & (x_{13} - x_{23})^2 & (x_{13} - x_{33})^2 & (x_{13} - x_{43})^2 \\ 0 & 0 & (x_{23} - x_{33})^2 & (x_{23} - x_{43})^2 \\ 0 & 0 & 0 & (x_{33} - x_{43})^2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The sum of the three matrices  $D = D_1 + D_2 + D_3$  will be calculated next:

$$\begin{pmatrix} 0 & \theta_1(x_{11} - x_{21})^2 + \theta_2(x_{12} - x_{22})^2 + \theta_3(x_{13} - x_{23})^2 & \theta_1(x_{11} - x_{31})^2 + \theta_2(x_{12} - x_{32})^2 + \theta_3(x_{13} - x_{33})^2 & \theta_1(x_{11} - x_{41})^2 + \theta_2(x_{12} - x_{42})^2 + \theta_3(x_{13} - x_{43})^2 \\ 0 & 0 & \theta_1(x_{21} - x_{31})^2 + \theta_2(x_{22} - x_{32})^2 + \theta_3(x_{23} - x_{33})^2 & \theta_1(x_{21} - x_{41})^2 + \theta_2(x_{22} - x_{42})^2 + \theta_3(x_{23} - x_{43})^2 \\ 0 & 0 & 0 & \theta_1(x_{31} - x_{41})^2 + \theta_2(x_{32} - x_{42})^2 + \theta_3(x_{33} - x_{43})^2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Finally,

$$\Psi = \exp(-D)$$

## 8.2. Building the Kriging Model

is computed.

Next, we will demonstrate how this computation can be implemented in Python. We will consider four points in three dimensions and compute the correlation matrix  $\Psi$  using the basis function from Equation 8.1. These points are placed at the origin, at the unit vectors, and at the points (100,100,100) and (101,100,100). So, they form two clusters: one at the origin and one at (100,100,100).

```
theta = np.array([1,2,3])
X = np.array([ [1,0,0], [0,1,0], [100, 100, 100], [101, 100, 100]])
X
```

```
array([[ 1,  0,  0],
       [ 0,  1,  0],
       [100, 100, 100],
       [101, 100, 100]])
```

```
def build_Psi(X, theta):
    n = X.shape[0]
    k = X.shape[1]
    D = zeros((k, n, n))
    for l in range(k):
        for i in range(n):
            for j in range(i, n):
                D[l, i, j] = theta[l]*(X[i,l] - X[j,l])**2
    D = sum(D)
    D = D + D.T
    return exp(-D)
```

```
Psi = build_Psi(X, theta)
Psi
```

```
array([[1.          , 0.04978707, 0.          , 0.          ],
       [0.04978707, 1.          , 0.          , 0.          ],
       [0.          , 0.          , 1.          , 0.36787944],
       [0.          , 0.          , 0.36787944, 1.          ]])
```

□

**Example 8.4** (Example: The Correlation Matrix (Using Existing Functions)). The same result as computed in Example 8.3 can be obtained with existing python functions, e.g., from the package `scipy`.

## 8. Kriging (Gaussian Process Regression)

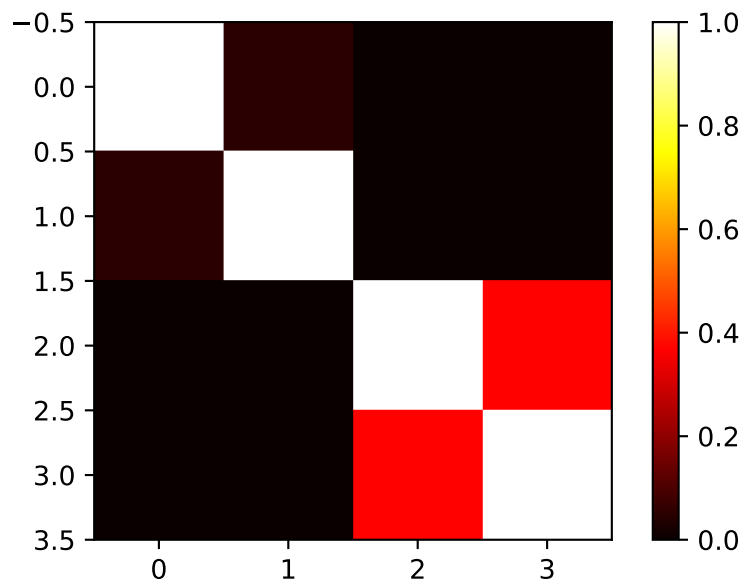


Figure 8.3.: Correlation matrix  $\Psi$ .

```
def build_Psi(X, theta, eps=sqrt(spacing(1))):
    return exp(- squareform(pdist(X,
        metric='sqeuclidean',
        out=None,
        w=theta))) + multiply(eye(X.shape[0]),
        eps)

Psi = build_Psi(X, theta, eps=.0)
Psi
```

```
array([[1.          , 0.04978707, 0.          , 0.          ],
       [0.04978707, 1.          , 0.          , 0.          ],
       [0.          , 0.          , 1.          , 0.36787944],
       [0.          , 0.          , 0.36787944, 1.          ]])
```

The condition number of the correlation matrix  $\Psi$  is a measure of how well the matrix can be inverted. A high condition number indicates that the matrix is close to singular, which can lead to numerical instability in computations involving the inverse of the matrix, see Section 9.2.

```
np.linalg.cond(Psi)
```

```
np.float64(2.163953413738652)
```

□

### 8.3. MLE to estimate $\theta$ and $p$

#### 8.3.1. The Log-Likelihood

Until now, the observed data  $\vec{y}$  was not used. We know what the correlations mean, but how do we estimate the values of  $\theta_j$  and where does our observed data  $y$  come in? To estimate the values of  $\vec{\theta}$  and  $\vec{p}$ , they are chosen to maximize the likelihood of  $\vec{y}$ ,

$$L = L(Y(\vec{x}^{(1)}), \dots, Y(\vec{x}^{(n)}) | \mu, \sigma) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp \left[ -\frac{\sum_{i=1}^n (Y(\vec{x}^{(i)}) - \mu)^2}{2\sigma^2} \right], \quad (8.6)$$

where  $\mu$  is the mean of the observed data  $\vec{y}$  and  $\sigma$  is the standard deviation of the errors  $\epsilon$ , which can be expressed in terms of the sample data

$$L = \frac{1}{(2\pi\sigma^2)^{n/2} |\vec{\Psi}|^{1/2}} \exp \left[ -\frac{(\vec{y} - \vec{1}\mu)^T \vec{\Psi}^{-1} (\vec{y} - \vec{1}\mu)}{2\sigma^2} \right]. \quad (8.7)$$

*Remark 8.2.* The transition from Equation 8.6 to Equation 8.7 reflects a shift from assuming independent errors in the observed data to explicitly modeling the *correlation structure* between the observed responses, which is a key aspect of the stochastic process framework used in methods like Kriging. It can be explained as follows:

1. **Initial Likelihood Expression (assuming independent errors):** Equation 8.6 is an expression for the likelihood of the data set, which is based on the assumption that the errors  $\epsilon$  are *independently randomly distributed according to a normal distribution with standard deviation  $\sigma$* . This form is characteristic of the likelihood of  $n$  independent observations  $Y(\vec{x}^{(i)})$ , each following a normal distribution with mean  $\mu$  and variance  $\sigma^2$ .
2. **Using Vector Notation.** The sum in the exponent, i.e.,

$$\sum_{i=1}^n (Y(\vec{x}^{(i)}) - \mu)^2$$

is equivalent to

$$(\vec{y} - \vec{1}\mu)^T (\vec{y} - \vec{1}\mu),$$

assuming  $Y(\vec{x}^{(i)}) = y^{(i)}$  and using vector notation for  $\vec{y}$  and  $\vec{1}\mu$ .

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3. **Assuming Independent Observations:** Equation 8.6 assumes that the observations are independent, which means that the covariance between any two observations  $Y(\vec{x}^{(i)})$  and  $Y(\vec{x}^{(l)})$  is zero for  $i \neq l$ . In this case, the covariance matrix of the observations would be a diagonal matrix with  $\sigma^2$  along the diagonal (i.e.,  $\sigma^2 I$ ), where  $I$  is the identity matrix.
4. **Stochastic Process and Correlation:** In the context of Kriging, the observed responses  $\vec{y}$  are considered as if they are from a *stochastic process* or *random field*. This means the random variables  $Y(\vec{x}^{(i)})$  at different locations  $\vec{x}^{(i)}$  are not independent, but they correlated with each other. This correlation is described by an  $(n \times n)$  **correlation matrix**  $\Psi$ , which is used instead of  $\sigma^2 I$ . The strength of the correlation between two points  $Y(\vec{x}^{(i)})$  and  $Y(\vec{x}^{(l)})$  depends on the distance between them and model parameters  $\theta_j$  and  $p_j$ .
5. **Multivariate Normal Distribution:** When random variables are correlated, their joint probability distribution is generally described by a *multivariate distribution*. Assuming the stochastic process follows a Gaussian process, the joint distribution of the observed responses  $\vec{y}$  is a **multivariate normal distribution**. A multivariate normal distribution for a vector  $\vec{Y}$  with mean vector  $\vec{\mu}$  and covariance matrix  $\Sigma$  has a probability density function given by:

$$p(\vec{y}) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp \left[ -\frac{1}{2} (\vec{y} - \vec{\mu})^T \Sigma^{-1} (\vec{y} - \vec{\mu}) \right].$$

6. **Connecting the Expressions:** In the stochastic process framework, the following holds:

- The mean vector of the observed data  $\vec{y}$  is  $\vec{1}\mu$ .
- The covariance matrix  $\Sigma$  is constructed by considering both the variance  $\sigma^2$  and the correlations  $\Psi$ .
- The covariance between  $Y(\vec{x}^{(i)})$  and  $Y(\vec{x}^{(l)})$  is  $\sigma^2 \text{cor}(Y(\vec{x}^{(i)}), Y(\vec{x}^{(l)}))$ .
- Therefore, the covariance matrix is  $\Sigma = \sigma^2 \vec{\Psi}$ .
- Substituting  $\vec{\mu} = \vec{1}\mu$  and  $\Sigma = \sigma^2 \vec{\Psi}$  into the multivariate normal PDF formula, we get:

$$\Sigma^{-1} = (\sigma^2 \vec{\Psi})^{-1} = \frac{1}{\sigma^2} \vec{\Psi}^{-1}$$

and

$$|\Sigma| = |\sigma^2 \vec{\Psi}| = (\sigma^2)^n |\vec{\Psi}|.$$

The PDF becomes:

$$p(\vec{y}) = \frac{1}{\sqrt{(2\pi)^n (\sigma^2)^n |\vec{\Psi}|}} \exp \left[ -\frac{1}{2} (\vec{y} - \vec{1}\mu)^T \left( \frac{1}{\sigma^2} \vec{\Psi}^{-1} \right) (\vec{y} - \vec{1}\mu) \right]$$

and simplifies to:

$$p(\vec{y}) = \frac{1}{(2\pi\sigma^2)^{n/2} |\vec{\Psi}|^{1/2}} \exp \left[ -\frac{1}{2\sigma^2} (\vec{y} - \vec{1}\mu)^T \vec{\Psi}^{-1} (\vec{y} - \vec{1}\mu) \right].$$

### 8.3. MLE to estimate $\theta$ and $p$

This is the **likelihood of the sample data**  $\vec{y}$  given the parameters  $\mu$ ,  $\sigma$ , and the correlation structure defined by the parameters within  $\vec{\Psi}$  (i.e.,  $\vec{\theta}$  and  $\vec{p}$ ).

In summary, the Equation 8.6 represents the likelihood under a simplified assumption of independent errors, whereas Equation 8.7 is the likelihood derived from the assumption that the observed data comes from a **multivariate normal distribution** where observations are correlated according to the matrix  $\vec{\Psi}$ . Equation 8.7, using the sample data vector  $\vec{y}$  and the correlation matrix  $\vec{\Psi}$ , properly accounts for the dependencies between data points inherent in the stochastic process model. Maximizing this likelihood is how the correlation parameters  $\vec{\theta}$  and  $\vec{p}$  are estimated in Kriging.

□

Equation 8.7 can be formulated as the log-likelihood:

$$\ln(L) = -\frac{n}{2} \ln(2\pi\sigma) - \frac{1}{2} \ln |\vec{\Psi}| - \frac{(\vec{y} - \vec{1}\mu)^T \vec{\Psi}^{-1} (\vec{y} - \vec{1}\mu)}{2\sigma^2}. \quad (8.8)$$

#### 8.3.2. Differentiation with Respect to $\mu$

Looking at the log-likelihood function, only the last term depends on  $\mu$ :

$$\frac{1}{2\sigma^2} (\vec{y} - \vec{1}\mu)^T \vec{\Psi}^{-1} (\vec{y} - \vec{1}\mu)$$

To differentiate this with respect to the scalar  $\mu$ , we can use matrix calculus rules.

Let  $\mathbf{v} = \vec{y} - \vec{1}\mu$ .  $\vec{y}$  is a constant vector with respect to  $\mu$ , and  $\vec{1}\mu$  is a vector whose derivative with respect to the scalar  $\mu$  is  $\vec{1}$ . So,  $\frac{\partial \mathbf{v}}{\partial \mu} = -\vec{1}$ .

The term is in the form  $\mathbf{v}^T \mathbf{A} \mathbf{v}$ , where  $\mathbf{A} = \vec{\Psi}^{-1}$  is a symmetric matrix. The derivative of  $\mathbf{v}^T \mathbf{A} \mathbf{v}$  with respect to  $\mathbf{v}$  is  $2\mathbf{A} \mathbf{v}$  as explained in Remark 8.3.

*Remark 8.3* (Derivative of a Quadratic Form). Consider the derivative of  $\mathbf{v}^T \mathbf{A} \mathbf{v}$  with respect to  $\mathbf{v}$ :

- The derivative of a scalar function  $f(\mathbf{v})$  with respect to a vector  $\mathbf{v}$  is a vector (the gradient).
- For a quadratic form  $\mathbf{v}^T \mathbf{A} \mathbf{v}$ , where  $\mathbf{A}$  is a matrix and  $\mathbf{v}$  is a vector, the general formula for the derivative with respect to  $\mathbf{v}$  is  $\frac{\partial}{\partial \mathbf{v}} (\mathbf{v}^T \mathbf{A} \mathbf{v}) = \mathbf{A} \mathbf{v} + \mathbf{A}^T \mathbf{v}$ . (This is a standard result in matrix calculus and explained in Equation 9.1).
- Since  $\mathbf{A} = \vec{\Psi}^{-1}$  is a symmetric matrix, its transpose  $\mathbf{A}^T$  is equal to  $\mathbf{A}$ .
- Substituting  $\mathbf{A}^T = \mathbf{A}$  into the general derivative formula, we get  $\mathbf{A} \mathbf{v} + \mathbf{A} \mathbf{v} = 2\mathbf{A} \mathbf{v}$ .

□

### 8. Kriging (Gaussian Process Regression)

Using the chain rule for differentiation with respect to the scalar  $\mu$ :

$$\frac{\partial}{\partial \mu}(\mathbf{v}^T \mathbf{A} \mathbf{v}) = 2 \left( \frac{\partial \mathbf{v}}{\partial \mu} \right)^T \mathbf{A} \mathbf{v}$$

Substituting  $\frac{\partial \mathbf{v}}{\partial \mu} = -\vec{1}$  and  $\mathbf{v} = \vec{y} - \vec{1}\mu$ :

$$\frac{\partial}{\partial \mu}(\vec{y} - \vec{1}\mu)^T \vec{\Psi}^{-1}(\vec{y} - \vec{1}\mu) = 2(-\vec{1})^T \vec{\Psi}^{-1}(\vec{y} - \vec{1}\mu) = -2\vec{1}^T \vec{\Psi}^{-1}(\vec{y} - \vec{1}\mu)$$

Now, differentiate the full log-likelihood term depending on  $\mu$ :

$$\frac{\partial}{\partial \mu} \left( -\frac{1}{2\sigma^2}(\vec{y} - \vec{1}\mu)^T \vec{\Psi}^{-1}(\vec{y} - \vec{1}\mu) \right) = -\frac{1}{2\sigma^2} (-2\vec{1}^T \vec{\Psi}^{-1}(\vec{y} - \vec{1}\mu)) = \frac{1}{\sigma^2} \vec{1}^T \vec{\Psi}^{-1}(\vec{y} - \vec{1}\mu)$$

Setting this to zero for maximization gives:

$$\frac{1}{\sigma^2} \vec{1}^T \vec{\Psi}^{-1}(\vec{y} - \vec{1}\mu) = 0.$$

Rearranging gives:

$$\vec{1}^T \vec{\Psi}^{-1}(\vec{y} - \vec{1}\mu) = 0.$$

Solving for  $\mu$  gives:

$$\vec{1}^T \vec{\Psi}^{-1} \vec{y} = \mu \vec{1}^T \vec{\Psi}^{-1} \vec{1}.$$

#### 8.3.3. Differentiation with Respect to $\sigma$

Let  $\nu = \sigma^2$  for simpler differentiation notation. The log-likelihood becomes:

$$\ln(L) = C_1 - \frac{n}{2} \ln(\nu) - \frac{(\vec{y} - \vec{1}\mu)^T \vec{\Psi}^{-1}(\vec{y} - \vec{1}\mu)}{2\nu},$$

where  $C_1 = -\frac{n}{2} \ln(2\pi) - \frac{1}{2} \ln |\vec{\Psi}|$  is a constant with respect to  $\nu = \sigma^2$ .

We differentiate with respect to  $\nu$ :

$$\frac{\partial \ln(L)}{\partial \nu} = \frac{\partial}{\partial \nu} \left( -\frac{n}{2} \ln(\nu) \right) + \frac{\partial}{\partial \nu} \left( -\frac{(\vec{y} - \vec{1}\mu)^T \vec{\Psi}^{-1}(\vec{y} - \vec{1}\mu)}{2\nu} \right).$$

The first term's derivative is straightforward:

$$\frac{\partial}{\partial \nu} \left( -\frac{n}{2} \ln(\nu) \right) = -\frac{n}{2} \cdot \frac{1}{\nu} = -\frac{n}{2\sigma^2}.$$



### 8.3. MLE to estimate $\theta$ and $p$

For the second term, let  $C_2 = (\vec{y} - \vec{1}\mu)^T \vec{\Psi}^{-1} (\vec{y} - \vec{1}\mu)$ . This term is constant with respect to  $\sigma^2$ . The derivative is:

$$\frac{\partial}{\partial \nu} \left( -\frac{C_2}{2\nu} \right) = -\frac{C_2}{2} \frac{\partial}{\partial \nu} (\nu^{-1}) = -\frac{C_2}{2} (-\nu^{-2}) = \frac{C_2}{2\nu^2} = \frac{(\vec{y} - \vec{1}\mu)^T \vec{\Psi}^{-1} (\vec{y} - \vec{1}\mu)}{2(\sigma^2)^2}.$$

Combining the derivatives, the gradient of the log-likelihood with respect to  $\sigma^2$  is:

$$\frac{\partial \ln(L)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{(\vec{y} - \vec{1}\mu)^T \vec{\Psi}^{-1} (\vec{y} - \vec{1}\mu)}{2(\sigma^2)^2}.$$

Setting this to zero for maximization gives:

$$-\frac{n}{2\sigma^2} + \frac{(\vec{y} - \vec{1}\mu)^T \vec{\Psi}^{-1} (\vec{y} - \vec{1}\mu)}{2(\sigma^2)^2} = 0.$$

#### 8.3.4. Results of the Optimizations

Optimization of the log-likelihood by taking derivatives with respect to  $\mu$  and  $\sigma$  results in

$$\hat{\mu} = \frac{\vec{1}^T \vec{\Psi}^{-1} \vec{y}^T}{\vec{1}^T \vec{\Psi}^{-1} \vec{1}^T} \quad (8.9)$$

and

$$\hat{\sigma}^2 = \frac{(\vec{y} - \vec{1}\mu)^T \vec{\Psi}^{-1} (\vec{y} - \vec{1}\mu)}{n}. \quad (8.10)$$

#### 8.3.5. The Concentrated Log-Likelihood Function

Combining the equations, i.e., substituting Equation 8.9 and Equation 8.10 into Equation 8.8 leads to the concentrated log-likelihood function:

$$\ln(L) \approx -\frac{n}{2} \ln(\hat{\sigma}) - \frac{1}{2} \ln |\vec{\Psi}|. \quad (8.11)$$

*Remark 8.4* (The Concentrated Log-Likelihood).

- The first term in Equation 8.11 requires information about the measured point (observations)  $y_i$ .
- To maximize  $\ln(L)$ , optimal values of  $\vec{\theta}$  and  $\vec{p}$  are determined numerically, because the function (Equation 8.11) is not differentiable.

□

### 8.3.6. Optimizing the Parameters $\vec{\theta}$ and $\vec{p}$

The concentrated log-likelihood function is very quick to compute. We do not need a statistical model, because we are only interested in the maximum likelihood estimate (MLE) of  $\theta$  and  $p$ . Optimizers such as Nelder-Mead, Conjugate Gradient, or Simulated Annealing can be used to determine optimal values for  $\theta$  and  $p$ . After the optimization, the correlation matrix  $\Psi$  is build with the optimized  $\theta$  and  $p$  values. This is best (most likely) Kriging model for the given data  $y$ .

Observing Figure 8.2, there's significant change between  $\theta = 0.1$  and  $\theta = 1$ , just as there is between  $\theta = 1$  and  $\theta = 10$ . Hence, it is sensible to search for  $\theta$  on a logarithmic scale. Suitable search bounds typically range from  $10^{-3}$  to  $10^2$ , although this is not a stringent requirement. Importantly, the scaling of the observed data does not affect the values of  $\hat{\theta}$ , but the scaling of the design space does. Therefore, it is advisable to consistently scale variable ranges between zero and one to ensure consistency in the degree of activity  $\hat{\theta}_j$  represents across different problems.

### 8.3.7. Correlation and Covariance Matrices Revisited

The covariance matrix  $\Sigma$  is constructed by considering both the variance  $\sigma^2$  and the correlation matrix  $\Psi$ . They are related as follows:

1. **Covariance vs. Correlation:** Covariance is a measure of the joint variability of two random variables, while correlation is a standardized measure of this relationship, ranging from -1 to 1. The relationship between covariance and correlation for two random variables  $X$  and  $Y$  is given by  $\text{cor}(X, Y) = \text{cov}(X, Y) / (\sigma_X \sigma_Y)$ , where  $\sigma_X$  and  $\sigma_Y$  are their standard deviations.
2. **The Covariance Matrix  $\Sigma$ :** The **covariance matrix**  $\Sigma$  (or  $\text{Cov}(Y, Y)$  for the vector  $\vec{Y}$ ) captures the **pairwise covariances** between all elements of the vector of observed responses.
3. **Connecting  $\sigma^2$  and  $\Psi$  to  $\Sigma$ :** In the Kriging framework described, the variance of each observation is often assumed to be constant,  $\sigma^2$ . The covariance between any two observations  $Y(\vec{x}^{(i)})$  and  $Y(\vec{x}^{(l)})$  is given by  $\sigma^2$  multiplied by their correlation. That is,

$$\text{cov}(Y(\vec{x}^{(i)}), Y(\vec{x}^{(l)})) = \sigma^2 \text{cor}(Y(\vec{x}^{(i)}), Y(\vec{x}^{(l)})).$$

This relationship holds for *all* pairs of points. When expressed in matrix form, the covariance matrix  $\Sigma$  is the product of the variance  $\sigma^2$  (a scalar) and the correlation matrix  $\Psi$ :

$$\Sigma = \sigma^2 \Psi.$$

In essence, the correlation matrix  $\Psi$  defines the *structure* or *shape* of the dependencies between the data points based on their locations. The parameter  $\sigma^2$  acts as a **scaling factor** that converts these unitless correlation values (which are between -1 and 1) into

#### 8.4. Implementing an MLE of the Model Parameters

actual covariance values with units of variance, setting the overall level of variability in the system.

So,  $\sigma^2$  tells us about the general spread or variability of the underlying process, while  $\Psi$  tells you *how* that variability is distributed and how strongly points are related to each other based on their positions. Together, they completely define the covariance structure of your observed data in the multivariate normal distribution used in Kriging.

### 8.4. Implementing an MLE of the Model Parameters

The matrix algebra necessary for calculating the likelihood is the most computationally intensive aspect of the Kriging process. It is crucial to ensure that the code implementation is as efficient as possible.

Given that  $\Psi$  (our correlation matrix) is symmetric, only half of the matrix needs to be computed before adding it to its transpose. When calculating the log-likelihood, several matrix inversions are required. The fastest approach is to conduct one Cholesky factorization and then apply backward and forward substitution for each inverse.

The Cholesky factorization is applicable only to positive-definite matrices, which  $\Psi$  generally is. However, if  $\Psi$  becomes nearly singular, such as when the  $\vec{x}^{(i)}$ 's are densely packed, the Cholesky factorization might fail. In these cases, one could employ an LU-decomposition, though the result might be unreliable. When  $\Psi$  is near singular, the best course of action is to either use regression techniques or, as we do here, assign a poor likelihood value to parameters generating the near singular matrix, thus diverting the MLE search towards better-conditioned  $\Psi$  matrices.

When working with correlation matrices, increasing the values on the main diagonal of a matrix will increase the absolute value of its determinant. A critical numerical consideration in calculating the concentrated log-likelihood is that for poorly conditioned matrices,  $\det(\Psi)$  approaches zero, leading to potential numerical instability. To address this issue, it is advisable to calculate  $\ln(|\Psi|)$  in Equation 8.11 using twice the sum of the logarithms of the diagonal elements of the Cholesky factorization. This approach provides a more numerically stable method for computing the log-determinant, as the Cholesky decomposition  $\Psi = LL^T$  allows us to express  $\ln(|\Psi|) = 2 \sum_{i=1}^n \ln(L_{ii})$ , avoiding the direct computation of potentially very small determinant values.

### 8.5. Kriging Prediction

We will use the Kriging correlation  $\Psi$  to predict new values based on the observed data. The presentation follows the approach described in Forrester, Sóbester, and Keane (2008) and Bartz et al. (2022).

## 8. Kriging (Gaussian Process Regression)

Main idea for prediction is that the new  $Y(\vec{x})$  should be consistent with the old sample data  $X$ . For a new prediction  $\hat{y}$  at  $\vec{x}$ , the value of  $\hat{y}$  is chosen so that it maximizes the likelihood of the sample data  $X$  and the prediction, given the (optimized) correlation parameter  $\vec{\theta}$  and  $\vec{p}$  from above. The observed data  $\vec{y}$  is augmented with the new prediction  $\hat{y}$  which results in the augmented vector  $\vec{\tilde{y}} = (\vec{y}^T, \hat{y})^T$ . A vector of correlations between the observed data and the new prediction is defined as

$$\vec{\psi} = \begin{pmatrix} \text{cor}(Y(\vec{x}^{(1)}), Y(\vec{x})) \\ \vdots \\ \text{cor}(Y(\vec{x}^{(n)}), Y(\vec{x})) \end{pmatrix} = \begin{pmatrix} \psi^{(1)} \\ \vdots \\ \psi^{(n)} \end{pmatrix}.$$

**Definition 8.2** (The Augmented Correlation Matrix). The augmented correlation matrix is constructed as

$$\tilde{\Psi} = \begin{pmatrix} \vec{\Psi} & \vec{\psi} \\ \vec{\psi}^T & 1 \end{pmatrix}.$$

□

The log-likelihood of the augmented data is

$$\ln(L) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\hat{\sigma}^2) - \frac{1}{2} \ln |\tilde{\Psi}| - \frac{(\vec{\tilde{y}} - \vec{1}\hat{\mu})^T \tilde{\Psi}^{-1} (\vec{\tilde{y}} - \vec{1}\hat{\mu})}{2\hat{\sigma}^2}, \quad (8.12)$$

where  $\vec{1}$  is a vector of ones and  $\hat{\mu}$  and  $\hat{\sigma}^2$  are the MLEs from Equation 8.9 and Equation 8.10. Only the last term in Equation 8.12 depends on  $\hat{y}$ , so we need only consider this term in the maximization. Details can be found in Forrester, Sóbester, and Keane (2008). Finally, the MLE for  $\hat{y}$  can be calculated as

$$\hat{y}(\vec{x}) = \hat{\mu} + \vec{\psi}^T \tilde{\Psi}^{-1} (\vec{\tilde{y}} - \vec{1}\hat{\mu}). \quad (8.13)$$

Equation 8.13 reveals two important properties of the Kriging predictor:

- **Basis functions:** The basis function impacts the vector  $\vec{\psi}$ , which contains the  $n$  correlations between the new point  $\vec{x}$  and the observed locations. Values from the  $n$  basis functions are added to a mean base term  $\mu$  with weightings

$$\vec{w} = \tilde{\Psi}^{(-1)} (\vec{\tilde{y}} - \vec{1}\hat{\mu}).$$

- **Interpolation:** The predictions interpolate the sample data. When calculating the prediction at the  $i$ th sample point,  $\vec{x}^{(i)}$ , the  $i$ th column of  $\tilde{\Psi}^{-1}$  is  $\vec{\psi}$ , and  $\vec{\psi} \tilde{\Psi}^{-1}$  is the  $i$ th unit vector. Hence,

$$\hat{y}(\vec{x}^{(i)}) = y^{(i)}.$$

## 8.6. Kriging Example: Sinusoid Function

Toy example in 1d where the response is a simple sinusoid measured at eight equally spaced  $x$ -locations in the span of a single period of oscillation.

### 8.6.1. Calculating the Correlation Matrix $\Psi$

The correlation matrix  $\Psi$  is based on the pairwise squared distances between the input locations. Here we will use  $n = 8$  sample locations and  $\theta$  is set to 1.0.

```
n = 8
X = np.linspace(0, 2*np.pi, n, endpoint=False).reshape(-1,1)
print(np.round(X, 2))
```

```
[[0. ]
 [0.79]
 [1.57]
 [2.36]
 [3.14]
 [3.93]
 [4.71]
 [5.5 ]]
```

Evaluate at sample points

```
y = np.sin(X)
print(np.round(y, 2))
```

```
[[ 0. ]
 [ 0.71]
 [ 1. ]
 [ 0.71]
 [ 0. ]
 [-0.71]
 [-1. ]
 [-0.71]]
```

We have the data points shown in Table 8.1.

## 8. Kriging (Gaussian Process Regression)

Table 8.1.: Data points for the sinusoid function

| $x$  | $y$   |
|------|-------|
| 0.0  | 0.0   |
| 0.79 | 0.71  |
| 1.57 | 1.0   |
| 2.36 | 0.71  |
| 3.14 | 0.0   |
| 3.93 | -0.71 |
| 4.71 | -1.0  |
| 5.5  | -0.71 |

The data points are visualized in Figure 8.4.

```
plt.plot(X, y, "bo")
plt.title(f"Sin(x) evaluated at {n} points")
plt.grid()
plt.show()
```

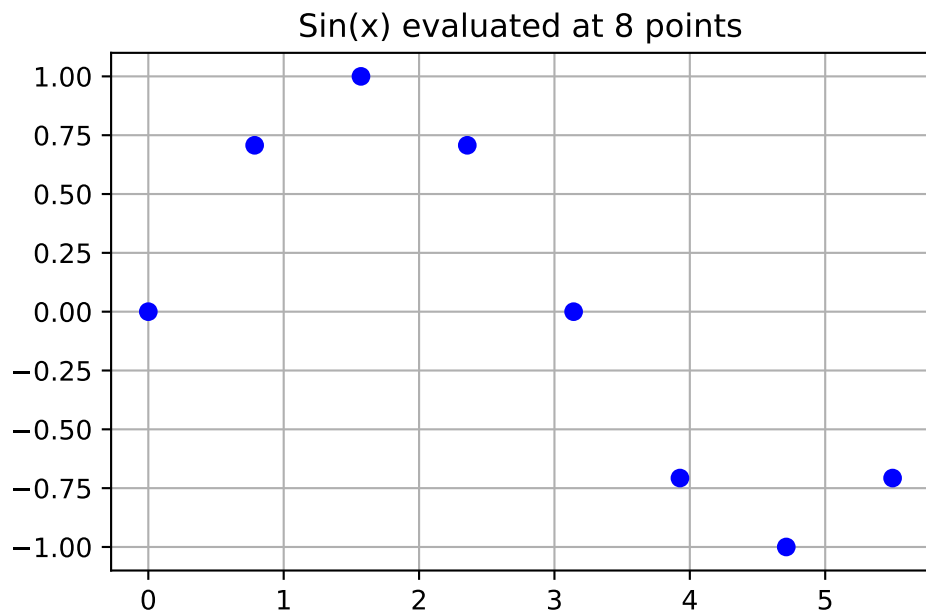


Figure 8.4.: Sin(x) evaluated at 8 points.

### 8.6.2. Computing the $\Psi$ Matrix

We will use the `build_Psi` function from Example 8.4 to compute the correlation matrix  $\Psi$ .  $\theta$  should be an array of one value, because we are only working in one dimension ( $k = 1$ ).

```
theta = np.array([1.0])
Psi = build_Psi(X, theta)
print(np.round(Psi, 2))
```

```
[[1.  0.54 0.08 0.  0.  0.  0.  0. ]
 [0.54 1.  0.54 0.08 0.  0.  0.  0. ]
 [0.08 0.54 1.  0.54 0.08 0.  0.  0. ]
 [0.  0.08 0.54 1.  0.54 0.08 0.  0. ]
 [0.  0.  0.08 0.54 1.  0.54 0.08 0. ]
 [0.  0.  0.  0.08 0.54 1.  0.54 0.08]
 [0.  0.  0.  0.  0.08 0.54 1.  0.54]
 [0.  0.  0.  0.  0.  0.08 0.54 1. ]]
```

Figure 8.5 visualizes the  $(8, 8)$  correlation matrix  $\Psi$ .

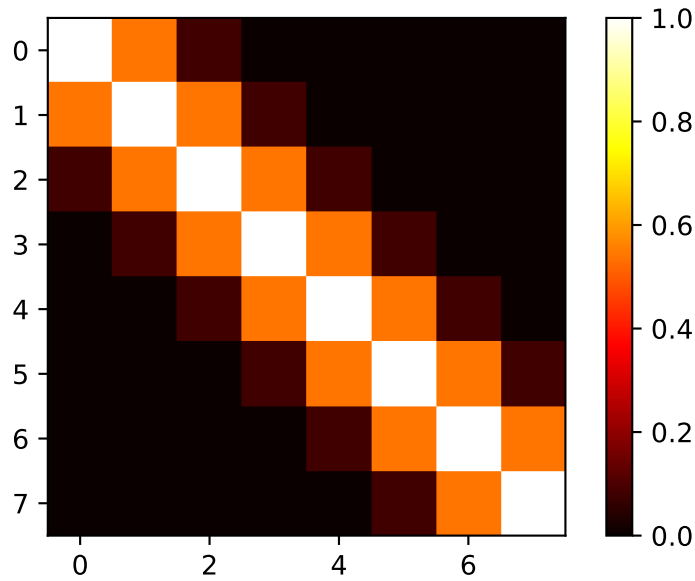


Figure 8.5.: Correlation matrix  $\Psi$  for the sinusoid function.

## 8. Kriging (Gaussian Process Regression)

### 8.6.3. Selecting the New Locations

We would like to predict at  $m = 100$  new locations (or testign locations) in the interval  $[0, 2\pi]$ . The new locations are stored in the variable  $\mathbf{x}$ .

```
m = 100
x = np.linspace(0, 2*np.pi, m, endpoint=False).reshape(-1,1)
```

### 8.6.4. Computing the $\psi$ Vector

Distances between testing locations  $x$  and training data locations  $X$ .

```
def build_psi(X, x, theta, eps=sqrt(spacing(1))):
    n = X.shape[0]
    k = X.shape[1]
    m = x.shape[0]
    psi = zeros((n, m))
    theta = theta * ones(k)
    D = zeros((n, m))
    D = cdist(x.reshape(-1, k),
              X.reshape(-1, k),
              metric='sqeuclidean',
              out=None,
              w=theta)
    psi = exp(-D)
    # return psi transpose to be consistent with the literature
    print(f"Dimensions of psi: {psi.T.shape}")
    return(psi.T)

psi = build_psi(X, x, theta)
```

Dimensions of psi: (8, 100)

Figure 8.6 visualizes the (8,100) prediction matrix  $\psi$ .

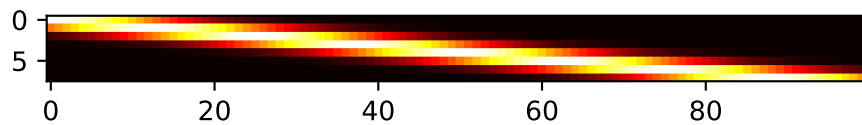


Figure 8.6.: Visualization of the prediction matrix  $\psi$



### 8.6.5. Predicting at New Locations

Computation of the predictive equations.

```
U = cholesky(Psi).T
one = np.ones(n).reshape(-1,1)
mu = (one.T.dot(solve(U, solve(U.T, y)))) / one.T.dot(solve(U, solve(U.T, one)))
f = mu * ones(m).reshape(-1,1) + psi.T.dot(solve(U, solve(U.T, y - one * mu)))
print(f"Dimensions of f: {f.shape}")
```

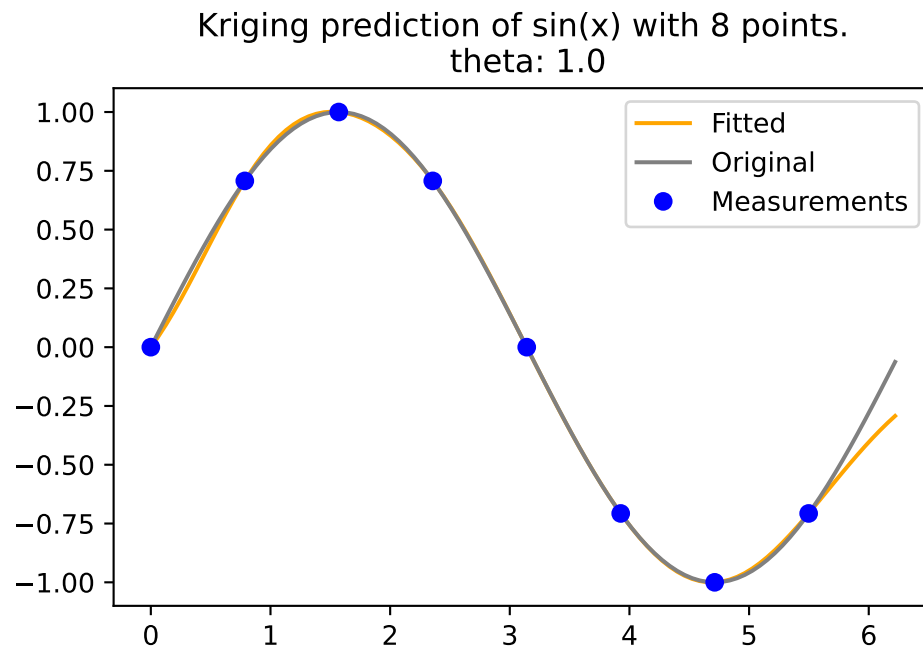
Dimensions of f: (100, 1)

To compute  $f$ , Equation 8.13 is used.

### 8.6.6. Visualization

```
plt.plot(x, f, color = "orange", label="Fitted")
plt.plot(x, np.sin(x), color = "grey", label="Original")
plt.plot(X, y, "bo", label="Measurements")
plt.title("Kriging prediction of sin(x) with {} points.\n theta: {}".format(n, theta[0]))
plt.legend(loc='upper right')
plt.show()
```

## 8. Kriging (Gaussian Process Regression)



### 8.6.7. The Complete Python Code for the Example

Here is the self-contained Python code for direct use in a notebook:

```
import numpy as np
import matplotlib.pyplot as plt
from numpy import (array, zeros, power, ones, exp, multiply, eye, linspace, spacing, s
from numpy.linalg import cholesky, solve
from scipy.spatial.distance import squareform, pdist, cdist

# --- 1. Kriging Basis Functions (Defining the Correlation) ---
# The core of Kriging uses a specialized basis function for correlation:
#  $\psi(x^{(i)}, x) = \exp(-\sum_{j=1}^k \theta_{j} |x_j^{(i)} - x_j|^{p_j})$ 
# For this 1D example ( $k=1$ ), and with  $p_j=2$  (squared Euclidean distance implicit from
# and  $\theta_j = \theta$  (a single value), it simplifies.

def build_Psi(X, theta, eps=sqrt(spacing(1))):
    """
    Computes the correlation matrix Psi based on pairwise squared Euclidean distances
    between input locations, scaled by theta.
    Adds a small epsilon to the diagonal for numerical stability (nugget effect).
```

## 8.6. Kriging Example: Sinusoid Function

```
"""
# Calculate pairwise squared Euclidean distances (D) between points in X
D = squareform(pdist(X, metric='sqeuclidean', out=None, w=theta))
# Compute Psi = exp(-D)
Psi = exp(-D)
# Add a small value to the diagonal for numerical stability (nugget)
# This is often done in Kriging implementations, though a regression method
# with a 'nugget' parameter (Lambda) is explicitly mentioned for noisy data later.
# The source code snippet for build_Psi explicitly includes `multiply(eye(X.shape), eps)`.
# FIX: Use X.shape to get the number of rows for the identity matrix
Psi += multiply(eye(X.shape[0]), eps) # Corrected line
return Psi

def build_psi(X_train, x_predict, theta):
    """
    Computes the correlation vector (or matrix) psi between new prediction locations
    and training data locations.
    """
    # Calculate pairwise squared Euclidean distances (D) between prediction points (x_predict)
    # and training points (X_train).
    # `cdist` computes distances between each pair of the two collections of inputs.
    D = cdist(x_predict, X_train, metric='sqeuclidean', out=None, w=theta)
    # Compute psi = exp(-D)
    psi = exp(-D)
    return psi.T # Return transpose to be consistent with literature (n x m or n x 1)

# --- 2. Data Points for the Sinusoid Function Example ---
# The example uses a 1D sinusoid measured at eight equally spaced x-locations [153, Table 9.1].
n = 8 # Number of sample locations
X_train = np.linspace(0, 2 * np.pi, n, endpoint=False).reshape(-1, 1) # Generate x-locations
y_train = np.sin(X_train) # Corresponding y-values (sine of x)

print("--- Training Data (X_train, y_train) ---")
print("x values:\n", np.round(X_train, 2))
print("y values:\n", np.round(y_train, 2))
print("-" * 40)

# Visualize the data points
plt.figure(figsize=(8, 5))
plt.plot(X_train, y_train, "bo", label=f"Measurements ({n} points)")
plt.title(f"Sin(x) evaluated at {n} points")
plt.xlabel("x")
plt.ylabel("sin(x)")
plt.grid(True)
```

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```
plt.legend()
plt.show()

# --- 3. Calculating the Correlation Matrix (Psi) ---
# Psi is based on pairwise squared distances between input locations.
# theta is set to 1.0 for this 1D example.
theta = np.array([1.0])
Psi = build_Psi(X_train, theta)

print("\n--- Computed Correlation Matrix (Psi) ---")
print("Dimensions of Psi:", Psi.shape) # Should be (8, 8)
print("First 5x5 block of Psi:\n", np.round(Psi[:5,:5], 2))
print("-" * 40)

# --- 4. Selecting New Locations (for Prediction) ---
# We want to predict at m = 100 new locations in the interval [0, 2*pi].
m = 100 # Number of new locations
x_predict = np.linspace(0, 2 * np.pi, m, endpoint=True).reshape(-1, 1)

print("\n--- New Locations for Prediction (x_predict) ---")
print(f"Number of prediction points: {m}")
print("First 5 prediction points:\n", np.round(x_predict[:5], 2).flatten())
print("-" * 40)

# --- 5. Computing the psi Vector ---
# This vector contains correlations between each of the n observed data points
# and each of the m new prediction locations.
psi = build_psi(X_train, x_predict, theta)

print("\n--- Computed Prediction Correlation Matrix (psi) ---")
print("Dimensions of psi:", psi.shape) # Should be (8, 100)
print("First 5x5 block of psi:\n", np.round(psi[:5,:5], 2))
print("-" * 40)

# --- 6. Predicting at New Locations (Kriging Prediction) ---
# The Maximum Likelihood Estimate (MLE) for y_hat is calculated using the formula:
# y_hat(x) = mu_hat + psi.T @ Psi_inv @ (y - 1 * mu_hat) [p. 2 of previous response, a
# Matrix inversion is efficiently performed using Cholesky factorization.

# Step 6a: Cholesky decomposition of Psi
U = cholesky(Psi).T # Note: `cholesky` in numpy returns lower triangular L, we need U

# Step 6b: Calculate mu_hat (estimated mean)
# mu_hat = (one.T @ Psi_inv @ y) / (one.T @ Psi_inv @ one) [p. 2 of previous response]
```

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```
one = np.ones(n).reshape(-1, 1) # Vector of ones
mu_hat = (one.T @ solve(U, solve(U.T, y_train))) / (one.T @ solve(U, solve(U.T, one)))
mu_hat = mu_hat.item() # Extract scalar value

print("\n--- Kriging Prediction Calculation ---")
print(f"Estimated mean (mu_hat): {np.round(mu_hat, 4)}")

# Step 6c: Calculate predictions f (y_hat) at new locations
# f = mu_hat * ones(m) + psi.T @ Psi_inv @ (y - one * mu_hat)
f_predict = mu_hat * np.ones(m).reshape(-1, 1) + psi.T @ solve(U, solve(U.T, y_train - one * mu_hat))

print(f"Dimensions of predicted values (f_predict): {f_predict.shape}") # Should be (100, 1)
print("First 5 predicted f values:\n", np.round(f_predict[:5], 2).flatten())
print("-" * 40)

# --- 7. Visualization ---
# Plot the original sinusoid function, the measured points, and the Kriging predictions.

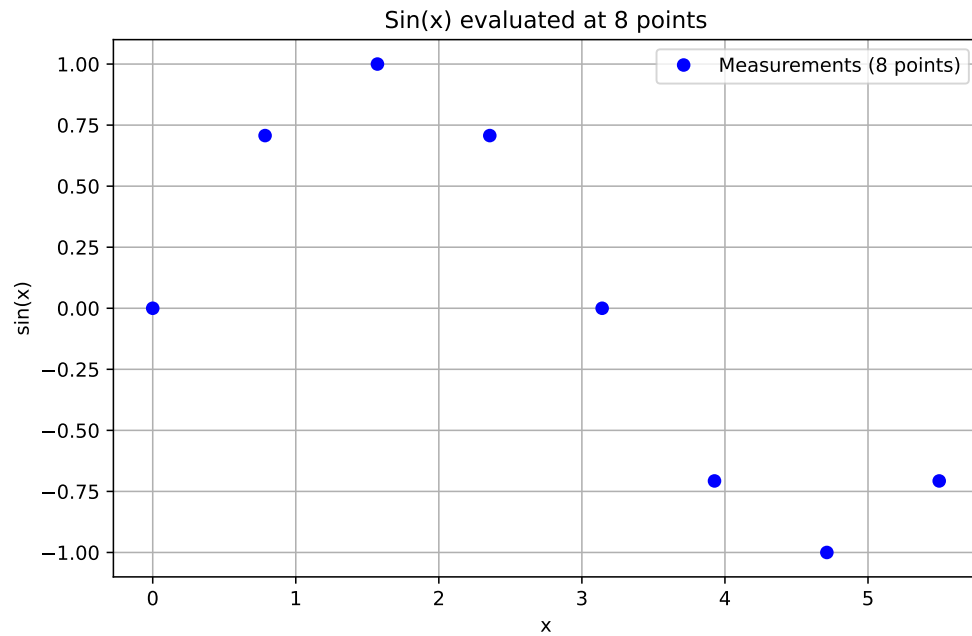
plt.figure(figsize=(10, 6))
plt.plot(x_predict, f_predict, color="orange", label="Kriging Prediction")
plt.plot(x_predict, np.sin(x_predict), color="grey", linestyle='--', label="True Sinusoid Function")
plt.plot(X_train, y_train, "bo", markersize=8, label="Measurements")
plt.title(f"Kriging prediction of sin(x) with {n} points. (theta: {theta})")
plt.xlabel("x")
plt.ylabel("y")
plt.legend(loc='upper right')
plt.grid(True)
plt.show()

--- Training Data (X_train, y_train) ---
x values:
[[0. ]
 [0.79]
 [1.57]
 [2.36]
 [3.14]
 [3.93]
 [4.71]
 [5.5 ]]
y values:
[[ 0. ]
 [ 0.71]
 [ 1. ]
 [ 0.71]
```

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```
[ 0. ]
[-0.71]
[-1. ]
[-0.71]]
```

---



```
--- Computed Correlation Matrix (Psi) ---
Dimensions of Psi: (8, 8)
First 5x5 block of Psi:
[[1.  0.54 0.08 0.  0. ]
 [0.54 1.  0.54 0.08 0. ]
 [0.08 0.54 1.  0.54 0.08]
 [0.  0.08 0.54 1.  0.54]
 [0.  0.  0.08 0.54 1.  ]]
```

---

```
--- New Locations for Prediction (x_predict) ---
Number of prediction points: 100
First 5 prediction points:
[0.  0.06 0.13 0.19 0.25]
```

---

## 8.6. Kriging Example: Sinusoid Function

--- Computed Prediction Correlation Matrix (psi) ---

Dimensions of psi: (8, 100)

First 5x5 block of psi:

```
[[1.   1.   0.98 0.96 0.94]
 [0.54 0.59 0.65 0.7  0.75]
 [0.08 0.1  0.12 0.15 0.18]
 [0.   0.01 0.01 0.01 0.01]
 [0.   0.   0.   0.   0.  ]]
```

-----

--- Kriging Prediction Calculation ---

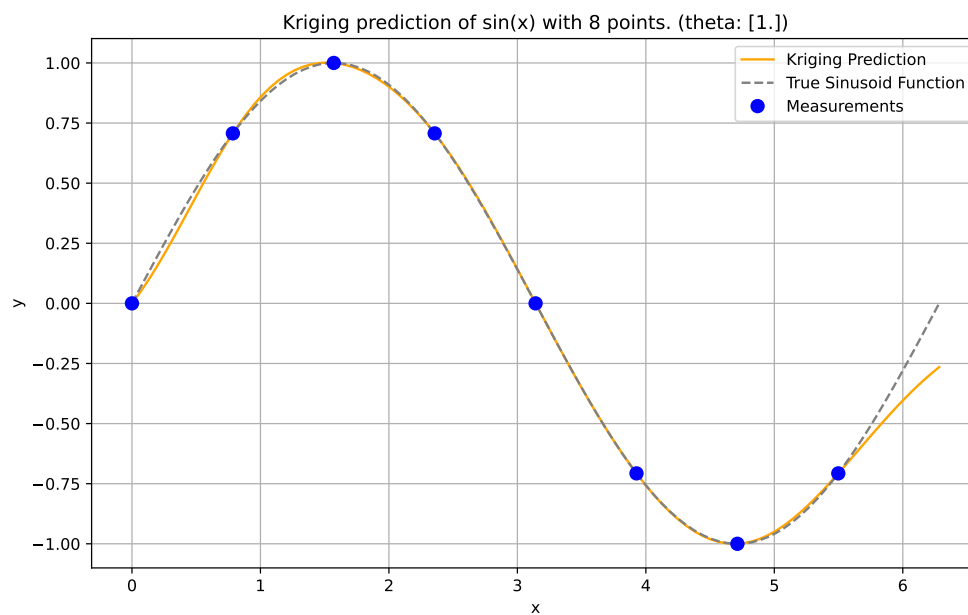
Estimated mean ( $\mu_{\text{hat}}$ ): -0.0499

Dimensions of predicted values ( $f_{\text{predict}}$ ): (100, 1)

First 5 predicted f values:

```
[0.   0.05 0.1  0.15 0.21]
```

-----



## 8.7. Jupyter Notebook

**i** Note

- The Jupyter-Notebook of this lecture is available on GitHub in the Hyperparameter-Tuning-Cookbook Repository



## 9. Matrices

### 9.1. Derivatives of Quadratic Forms

We present a step-by-step derivation of the general formula

$$\frac{\partial}{\partial \mathbf{v}}(\mathbf{v}^T \mathbf{A} \mathbf{v}) = \mathbf{A} \mathbf{v} + \mathbf{A}^T \mathbf{v}. \quad (9.1)$$

1. Define the components. Let  $\mathbf{v}$  be a vector of size  $n \times 1$ , and let  $\mathbf{A}$  be a matrix of size  $n \times n$ .
2. Write out the quadratic form in summation notation. The product  $\mathbf{v}^T \mathbf{A} \mathbf{v}$  is a scalar. It can be expanded and be rewritten as a double summation:

$$\mathbf{v}^T \mathbf{A} \mathbf{v} = \sum_{i=1}^n \sum_{j=1}^n v_i a_{ij} v_j.$$

3. Calculate the partial derivative with respect to a component  $v_k$ : The derivative of the scalar  $\mathbf{v}^T \mathbf{A} \mathbf{v}$  with respect to the vector  $\mathbf{v}$  is the gradient vector, whose  $k$ -th component is  $\frac{\partial}{\partial v_k}(\mathbf{v}^T \mathbf{A} \mathbf{v})$ . We need to find  $\frac{\partial}{\partial v_k} \left( \sum_{i=1}^n \sum_{j=1}^n v_i a_{ij} v_j \right)$ . Consider the terms in the summation that involve  $v_k$ . A term  $v_i a_{ij} v_j$  involves  $v_k$  if  $i = k$  or  $j = k$  (or both).
  - Terms where  $i = k$ :  $v_k a_{kj} v_j$ . The derivative with respect to  $v_k$  is  $a_{kj} v_j$ .
  - Terms where  $j = k$ :  $v_i a_{ik} v_k$ . The derivative with respect to  $v_k$  is  $v_i a_{ik}$ .
  - The term where  $i = k$  and  $j = k$ :  $v_k a_{kk} v_k = a_{kk} v_k^2$ . Its derivative with respect to  $v_k$  is  $2a_{kk} v_k$ . Notice this term is included in both cases above when  $i = k$  and  $j = k$ . When  $i = k$ , the term is  $v_k a_{kk} v_k$ , derivative is  $a_{kk} v_k$ . When  $j = k$ , the term is  $v_k a_{kk} v_k$ , derivative is  $v_k a_{kk}$ . Summing these two gives  $2a_{kk} v_k$ .
4. Let's differentiate the sum  $\sum_{i=1}^n \sum_{j=1}^n v_i a_{ij} v_j$  with respect to  $v_k$ :

$$\frac{\partial}{\partial v_k} \left( \sum_{i=1}^n \sum_{j=1}^n v_i a_{ij} v_j \right) = \sum_{i=1}^n \sum_{j=1}^n \frac{\partial}{\partial v_k} (v_i a_{ij} v_j).$$

5. The partial derivative  $\frac{\partial}{\partial v_k}(v_i a_{ij} v_j)$  is non-zero only if  $i = k$  or  $j = k$ .

## 9. Matrices

- If  $i = k$  and  $j \neq k$ :  $\frac{\partial}{\partial v_k}(v_k a_{kj} v_j) = a_{kj} v_j$ .
  - If  $i \neq k$  and  $j = k$ :  $\frac{\partial}{\partial v_k}(v_i a_{ik} v_k) = v_i a_{ik}$ .
  - If  $i = k$  and  $j = k$ :  $\frac{\partial}{\partial v_k}(v_k a_{kk} v_k) = \frac{\partial}{\partial v_k}(a_{kk} v_k^2) = 2a_{kk} v_k$ .
6. So, the partial derivative is the sum of derivatives of all terms involving  $v_k$ :  
 $\frac{\partial}{\partial v_k}(\mathbf{v}^T \mathbf{A} \mathbf{v}) = \sum_{j \neq k} (a_{kj} v_j) + \sum_{i \neq k} (v_i a_{ik}) + (2a_{kk} v_k)$ .
7. We can rewrite this by including the  $i = k, j = k$  term back into the summations:  
 $\sum_{j \neq k} (a_{kj} v_j) + a_{kk} v_k + \sum_{i \neq k} (v_i a_{ik}) + v_k a_{kk}$  (since  $v_k a_{kk} = a_{kk} v_k$ )  $= \sum_{j=1}^n a_{kj} v_j + \sum_{i=1}^n v_i a_{ik}$ .
8. Convert back to matrix/vector notation: The first summation  $\sum_{j=1}^n a_{kj} v_j$  is the  $k$ -th component of the matrix-vector product  $\mathbf{A} \mathbf{v}$ . The second summation  $\sum_{i=1}^n v_i a_{ik}$  can be written as  $\sum_{i=1}^n a_{ik} v_i$ . Recall that the element in the  $k$ -th row and  $i$ -th column of the transpose matrix  $\mathbf{A}^T$  is  $(A^T)_{ki} = a_{ik}$ . So,  $\sum_{i=1}^n a_{ik} v_i = \sum_{i=1}^n (A^T)_{ki} v_i$ , which is the  $k$ -th component of the matrix-vector product  $\mathbf{A}^T \mathbf{v}$ .
9. Assemble the gradient vector: The  $k$ -th component of the gradient  $\frac{\partial}{\partial \mathbf{v}}(\mathbf{v}^T \mathbf{A} \mathbf{v})$  is  $(\mathbf{A} \mathbf{v})_k + (\mathbf{A}^T \mathbf{v})_k$ . Since this holds for all  $k = 1, \dots, n$ , the gradient vector is the sum of the two vectors  $\mathbf{A} \mathbf{v}$  and  $\mathbf{A}^T \mathbf{v}$ . Therefore, the general formula for the derivative is  $\frac{\partial}{\partial \mathbf{v}}(\mathbf{v}^T \mathbf{A} \mathbf{v}) = \mathbf{A} \mathbf{v} + \mathbf{A}^T \mathbf{v}$ .

## 9.2. The Condition Number

A small value, `eps`, can be passed to the function `build_Psi` to improve the condition number. For example, `eps=sqrt(spacing(1))` can be used. The numpy function `spacing()` returns the distance between a number and its nearest adjacent number.

The condition number of a matrix is a measure of its sensitivity to small changes in its elements. It is used to estimate how much the output of a function will change if the input is slightly altered.

A matrix with a low condition number is well-conditioned, which means its behavior is relatively stable, while a matrix with a high condition number is ill-conditioned, meaning its behavior is unstable with respect to numerical precision.

```
import numpy as np

# Define a well-conditioned matrix (low condition number)
A = np.array([[1, 0.1], [0.1, 1]])
print("Condition number of A: ", np.linalg.cond(A))

# Define an ill-conditioned matrix (high condition number)
```

```
B = np.array([[1, 0.99999999], [0.99999999, 1]])
print("Condition number of B: ", np.linalg.cond(B))
```

```
Condition number of A: 1.2222222222222225
Condition number of B: 200000000.57495335
```

## 9.3. The Moore-Penrose Pseudoinverse

### 9.3.1. Definitions

The Moore-Penrose pseudoinverse is a generalization of the inverse matrix for non-square or singular matrices. It is computed as

$$A^+ = (A^*A)^{-1}A^*,$$

where  $A^*$  is the conjugate transpose of  $A$ .

It satisfies the following properties:

1.  $AA^+A = A$
2.  $A^+AA^+ = A^+$
3.  $(AA^+)^* = AA^+$ .
4.  $(A^+A)^* = A^+A$
5.  $A^+ = (A^*)^+$
6.  $A^+ = A^T$  if  $A$  is a square matrix and  $A$  is invertible.

The pseudoinverse can be computed using Singular Value Decomposition (SVD).

### 9.3.2. Implementation in Python

```
import numpy as np
from numpy.linalg import pinv
A = np.array([[1, 2], [3, 4], [5, 6]])
print(f"Matrix A:\n {A}")
A_pseudo_inv = pinv(A)
print(f"Moore-Penrose Pseudoinverse:\n {A_pseudo_inv}")
```

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Matrix A:

```
[[1 2]
 [3 4]
 [5 6]]
```

Moore-Penrose Pseudoinverse:

```
[[ -1.33333333  -0.33333333   0.66666667]
 [  1.08333333   0.33333333  -0.41666667]]
```

## 9.4. Strictly Positive Definite Kernels

### 9.4.1. Definition

**Definition 9.1** (Strictly Positive Definite Kernel). A kernel function  $k(x, y)$  is called strictly positive definite if for any finite collection of distinct points  $x_1, x_2, \dots, x_n$  in the input space and any non-zero vector of coefficients  $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$ , the following inequality holds:

$$\sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j k(x_i, x_j) > 0. \quad (9.2)$$

In contrast, a kernel function  $k(x, y)$  is called positive definite (but not strictly) if the “>” sign is replaced by “ $\geq$ ” in the above inequality.

### 9.4.2. Connection to Positive Definite Matrices

The connection between strictly positive definite kernels and positive definite matrices lies in the Gram matrix construction:

- When we evaluate a kernel function  $k(x, y)$  at all pairs of data points in our sample, we construct the Gram matrix  $K$  where  $K_{ij} = k(x_i, x_j)$ .
- If the kernel function  $k$  is strictly positive definite, then for any set of distinct points, the resulting Gram matrix will be symmetric positive definite.

A symmetric matrix is positive definite if and only if for any non-zero vector  $\alpha$ , the quadratic form  $\alpha^T K \alpha > 0$ , which directly corresponds to the kernel definition above.

### 9.4.3. Connection to RBF Models

For RBF models, the kernel function is the radial basis function itself:

$$k(x, y) = \psi(\|x - y\|).$$

The Gaussian RBF kernel  $\psi(r) = e^{-r^2/(2\sigma^2)}$  is strictly positive definite in  $\mathbb{R}^n$  for any dimension  $n$ . The inverse multiquadric kernel  $\psi(r) = (r^2 + \sigma^2)^{-1/2}$  is also strictly positive definite in any dimension.

This mathematical property guarantees that the interpolation problem has a unique solution (the weight vector  $\vec{w}$  is uniquely determined). The linear system  $\Psi\vec{w} = \vec{y}$  can be solved reliably using Cholesky decomposition. The RBF interpolant exists and is unique for any distinct set of centers.

## 9.5. Cholesky Decomposition and Positive Definite Matrices

We consider the definiteness of a matrix, before discussing the Cholesky decomposition.

**Definition 9.2** (Positive Definite Matrix). A symmetric matrix  $A$  is positive definite if all its eigenvalues are positive.

**Example 9.1** (Positive Definite Matrix). Given a symmetric matrix  $A = \begin{pmatrix} 9 & 4 \\ 4 & 9 \end{pmatrix}$ , the eigenvalues of  $A$  are  $\lambda_1 = 13$  and  $\lambda_2 = 5$ . Since both eigenvalues are positive, the matrix  $A$  is positive definite.

**Definition 9.3** (Negative Definite, Positive Semidefinite, and Negative Semidefinite Matrices). Similarly, a symmetric matrix  $A$  is negative definite if all its eigenvalues are negative. It is positive semidefinite if all its eigenvalues are non-negative, and negative semidefinite if all its eigenvalues are non-positive.

The covariance matrix must be positive definite for a multivariate normal distribution for a couple of reasons:

- Semidefinite vs Definite: A covariance matrix is always symmetric and positive semidefinite. However, for a multivariate normal distribution, it must be positive definite, not just semidefinite. This is because a positive semidefinite matrix can have zero eigenvalues, which would imply that some dimensions in the distribution have zero variance, collapsing the distribution in those dimensions. A positive definite matrix has all positive eigenvalues, ensuring that the distribution has positive variance in all dimensions.

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- Invertibility: The multivariate normal distribution's probability density function involves the inverse of the covariance matrix. If the covariance matrix is not positive definite, it may not be invertible, and the density function would be undefined.

In summary, the covariance matrix being positive definite ensures that the multivariate normal distribution is well-defined and has positive variance in all dimensions.

The definiteness of a matrix can be checked by examining the eigenvalues of the matrix. If all eigenvalues are positive, the matrix is positive definite.

```
import numpy as np

def is_positive_definite(matrix):
    return np.all(np.linalg.eigvals(matrix) > 0)

matrix = np.array([[9, 4], [4, 9]])
print(is_positive_definite(matrix)) # Outputs: True
```

True

However, a more efficient way to check the definiteness of a matrix is through the Cholesky decomposition.

**Definition 9.4** (Cholesky Decomposition). For a given symmetric positive-definite matrix  $A \in \mathbb{R}^{n \times n}$ , there exists a unique lower triangular matrix  $L \in \mathbb{R}^{n \times n}$  with positive diagonal elements such that:

$$A = LL^T.$$

Here,  $L^T$  denotes the transpose of  $L$ .

**Example 9.2** (Cholesky decomposition using `numpy`). `linalg.cholesky` computes the Cholesky decomposition of a matrix, i.e., it computes a lower triangular matrix  $L$  such that  $LL^T = A$ . If the matrix is not positive definite, an error (`LinAlgError`) is raised.

```
import numpy as np

# Define a Hermitian, positive-definite matrix
A = np.array([[9, 4], [4, 9]])

# Compute the Cholesky decomposition
L = np.linalg.cholesky(A)
```

```
print("L = \n", L)
print("L*LT = \n", np.dot(L, L.T))
```

```
L =
[[3.          0.          ]
 [1.33333333  2.68741925]]
L*LT =
[[9.  4.]
 [4.  9.]]
```

**Example 9.3** (Cholesky Decomposition). Given a symmetric positive-definite matrix  $A = \begin{pmatrix} 9 & 4 \\ 4 & 9 \end{pmatrix}$ , the Cholesky decomposition computes the lower triangular matrix  $L$  such that  $A = LL^T$ . The matrix  $L$  is computed as:

$$L = \begin{pmatrix} 3 & 0 \\ 4/3 & 2 \end{pmatrix},$$

so that

$$LL^T = \begin{pmatrix} 3 & 0 \\ 4/3 & \sqrt{65}/3 \end{pmatrix} \begin{pmatrix} 3 & 4/3 \\ 0 & \sqrt{65}/3 \end{pmatrix} = \begin{pmatrix} 9 & 4 \\ 4 & 9 \end{pmatrix} = A.$$

An efficient implementation of the definiteness-check based on Cholesky is already available in the `numpy` library. It provides the `np.linalg.cholesky` function to compute the Cholesky decomposition of a matrix. This more efficient `numpy`-approach can be used as follows:

```
import numpy as np

def is_pd(K):
    try:
        np.linalg.cholesky(K)
        return True
    except np.linalg.linalg.LinAlgError as err:
        if 'Matrix is not positive definite' in err.message:
            return False
        else:
            raise
matrix = np.array([[9, 4], [4, 9]])
print(is_pd(matrix)) # Outputs: True
```

True

### 9.5.1. Example of Cholesky Decomposition

We consider dimension  $k = 1$  and  $n = 2$  sample points. The sample points are located at  $x_1 = 1$  and  $x_2 = 5$ . The response values are  $y_1 = 2$  and  $y_2 = 10$ . The correlation parameter is  $\theta = 1$  and  $p$  is set to 1. Using Equation 8.1, we can compute the correlation matrix  $\Psi$ :

$$\Psi = \begin{pmatrix} 1 & e^{-1} \\ e^{-1} & 1 \end{pmatrix}.$$

To determine MLE as in Equation 8.13, we need to compute  $\Psi^{-1}$ :

$$\Psi^{-1} = \frac{e}{e^2 - 1} \begin{pmatrix} e & -1 \\ -1 & e \end{pmatrix}.$$

Cholesky-decomposition of  $\Psi$  is recommended to compute  $\Psi^{-1}$ . Cholesky decomposition is a decomposition of a positive definite symmetric matrix into the product of a lower triangular matrix  $L$ , a diagonal matrix  $D$  and the transpose of  $L$ , which is denoted as  $L^T$ . Consider the following example:

$$\begin{aligned} LDL^T &= \begin{pmatrix} 1 & 0 \\ l_{21} & 1 \end{pmatrix} \begin{pmatrix} d_{11} & 0 \\ 0 & d_{22} \end{pmatrix} \begin{pmatrix} 1 & l_{21} \\ 0 & 1 \end{pmatrix} = \\ &= \begin{pmatrix} d_{11} & 0 \\ d_{11}l_{21} & d_{22} \end{pmatrix} \begin{pmatrix} 1 & l_{21} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} d_{11} & d_{11}l_{21} \\ d_{11}l_{21} & d_{11}l_{21}^2 + d_{22} \end{pmatrix}. \end{aligned} \quad (9.3)$$

Using Equation 9.3, we can compute the Cholesky decomposition of  $\Psi$ :

1.  $d_{11} = 1$ ,
2.  $l_{21}d_{11} = e^{-1} \Rightarrow l_{21} = e^{-1}$ , and
3.  $d_{11}l_{21}^2 + d_{22} = 1 \Rightarrow d_{22} = 1 - e^{-2}$ .

The Cholesky decomposition of  $\Psi$  is

$$\Psi = \begin{pmatrix} 1 & 0 \\ e^{-1} & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 - e^{-2} \end{pmatrix} \begin{pmatrix} 1 & e^{-1} \\ 0 & 1 \end{pmatrix} = LDL^T$$

Some programs use  $U$  instead of  $L$ . The Cholesky decomposition of  $\Psi$  is

$$\Psi = LDL^T = U^T D U.$$

Using

$$\sqrt{D} = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1 - e^{-2}} \end{pmatrix},$$



we can write the Cholesky decomposition of  $\Psi$  without a diagonal matrix  $D$  as

$$\Psi = \begin{pmatrix} 1 & 0 \\ e^{-1} & \sqrt{1-e^{-2}} \end{pmatrix} \begin{pmatrix} 1 & e^{-1} \\ 0 & \sqrt{1-e^{-2}} \end{pmatrix} = U^T U.$$

### 9.5.2. Inverse Matrix Using Cholesky Decomposition

To compute the inverse of a matrix using the Cholesky decomposition, you can follow these steps:

1. Decompose the matrix  $A$  into  $L$  and  $L^T$ , where  $L$  is a lower triangular matrix and  $L^T$  is the transpose of  $L$ .
2. Compute  $L^{-1}$ , the inverse of  $L$ .
3. The inverse of  $A$  is then  $(L^{-1})^T L^{-1}$ .

Please note that this method only applies to symmetric, positive-definite matrices.

The inverse of the matrix  $\Psi$  from above is:

$$\Psi^{-1} = \frac{e}{e^2 - 1} \begin{pmatrix} e & -1 \\ -1 & e \end{pmatrix}.$$

Here's an example of how to compute the inverse of a matrix using Cholesky decomposition in Python:

```
import numpy as np
from scipy.linalg import cholesky, inv
E = np.exp(1)

# Psi is a symmetric, positive-definite matrix
Psi = np.array([[1, 1/E], [1/E, 1]])
L = cholesky(Psi, lower=True)
L_inv = inv(L)
# The inverse of A is (L^-1)^T * L^-1
Psi_inv = np.dot(L_inv.T, L_inv)

print("Psi:\n", Psi)
print("Psi Inverse:\n", Psi_inv)
```

```
Psi:
[[1.          0.36787944]
 [0.36787944  1.        ]]
Psi Inverse:
[[ 1.15651764 -0.42545906]
 [-0.42545906  1.15651764]]
```

## 9.6. Nyström Approximation

### 9.6.1. What's the Big Idea?

Imagine you have a huge, detailed map of a country. Working with the full, high-resolution map is slow and takes up a lot of computer memory. The Nyström method is like creating a smaller-scale summary map by only looking at a few key, representative locations.

In machine learning, we often work with a **kernel matrix** (or Gram matrix), which tells us how similar every pair of data points is to each other. For very large datasets, this matrix can become massive, making it computationally expensive to store and process.

The Nyström method provides an efficient way to create a **low-rank approximation** of this large kernel matrix. In simple terms, it finds a “simpler” version of the matrix that captures its most important properties without needing to compute or store the whole thing.

---

### 9.6.2. How Does It Work?

The core idea is to select a small, random subset of the columns of the full kernel matrix and use them to reconstruct the entire matrix. Let's say our full kernel matrix is  $K$ .

1. **Sample:** Randomly select  $l$  columns from the  $n$  total columns of  $K$ . Let  $C$  be the  $n \times l$  matrix of these sampled columns.
2. **Intersect:** Take the rows of  $C$  corresponding to the sampled column indices to form the  $l \times l$  matrix  $W$ .
3. **Approximate:** Using  $C$  and  $W$ , calculate the Nyström approximation  $\tilde{K}$  of  $K$ :

$$\tilde{K} \approx CW^+C^T$$

where  $W^+$  is the pseudoinverse of  $W$ .

---

### 9.6.3. Example

Suppose we have 4 data points and the full kernel matrix  $K$  is:

$$K = \begin{pmatrix} 9 & 6 & 3 & 1 \\ 6 & 4 & 2 & 0.5 \\ 3 & 2 & 1 & 0.25 \\ 1 & 0.5 & 0.25 & 0.1 \end{pmatrix}$$

Let's approximate it by sampling 2 columns ( $l = 2$ ):

1. **Sample:** Pick the 1st and 3rd columns:

$$C = \begin{pmatrix} 9 & 3 \\ 6 & 2 \\ 3 & 1 \\ 1 & 0.25 \end{pmatrix}$$

2. **Intersect:** Take the 1st and 3rd rows from  $C$  to form  $W$ :

$$W = \begin{pmatrix} 9 & 3 \\ 3 & 1 \end{pmatrix}$$

3. **Approximate:** Suppose the pseudoinverse of  $W$  is:

$$W^+ = \begin{pmatrix} 0.09 & -0.27 \\ -0.27 & 0.81 \end{pmatrix}$$

Then,

$$\tilde{K} = CW^+C^T = \begin{pmatrix} 9 & 6 & 3 & 0.675 \\ 6 & 4 & 2 & 0.45 \\ 3 & 2 & 1 & 0.225 \\ 0.675 & 0.45 & 0.225 & 0.05 \end{pmatrix}$$

$\tilde{K}$  is a good approximation of the original  $K$ , especially in the top-left portion.

---

### 9.6.4. Why Is This Useful?

- **Speed:** The Nyström method is much faster than computing the full kernel matrix. The complexity is roughly  $O(l^2n)$  instead of  $O(n^2d)$  (where  $d$  is the number of features).
- **Scalability:** It allows kernel methods (like SVM or Kernel PCA) to be used on much larger datasets.

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- **Feature Mapping:** The method can be used to project new data points into the same feature space for prediction tasks.

The quality of the approximation depends on the columns you sample. Uniform random sampling is common and often effective, but more advanced techniques exist to select more informative columns.

### 9.6.5. Applying the Nyström Approximation: How Nyström Approximation Helps Kriging

Kriging can significantly benefit from the Nyström approximation, especially when dealing with large datasets. Kriging is a spatial interpolation method used to estimate values at unmeasured locations based on observed points. It relies on a **covariance matrix** (often denoted as  $\mathbf{K}$ ) that describes the spatial correlation between all observed data points.

#### The Problem with Standard Kriging:

The main computational challenge in Kriging is solving for the weights needed for prediction, which requires **inverting the covariance matrix  $\mathbf{K}$** . For  $n$  data points,  $\mathbf{K}$  is an  $n \times n$  matrix, and inverting it has computational complexity  $O(n^3)$ . This becomes impractical for large datasets.

#### The Nyström Solution:

Since the covariance matrix in Kriging is a type of kernel matrix, we can use the Nyström method to create a low-rank approximation,  $\tilde{\mathbf{K}}$ . Instead of inverting the full matrix, we use the **Woodbury matrix identity** on the Nyström approximation, allowing us to efficiently compute  $\tilde{\mathbf{K}}^{-1}$  without forming the full matrix. This reduces computational complexity to roughly  $O(l^2n)$ , where  $l$  is the number of sampled columns.

In summary, Nyström makes Kriging feasible for large-scale problems by replacing expensive matrix inversion with a faster, memory-efficient approximation.

---

### 9.6.6. Example: Predicting Temperature with Nyström-Kriging

Suppose we have temperature readings from 100 weather stations ( $n=100$ ) and want to predict the temperature at a new location.

#### Data:

- Observed Locations ( $\mathbf{X}$ ): 100 coordinate pairs
- Observed Temperatures ( $\mathbf{y}$ ): 100 values
- Prediction Location ( $\mathbf{x}^*$ ): Coordinates of the new location

**9.6.6.1. Step 1: Nyström Approximation of the Covariance Matrix**

1. **Sample Representative Points:** Randomly select 1=10 stations as landmarks.
2. **Compute C and W:**
  - **C:** Covariance between all 100 stations and the 10 landmarks (100x10 matrix)
  - **W:** Covariance among the 10 landmarks (10x10 matrix)

Nyström approximation:  $\tilde{K} = CW^+C^T$

**9.6.6.2. Step 2: Modeling and Prediction**

Standard Kriging prediction:

$$y(x^*) = \mathbf{k}^{*T} \mathbf{K}^{-1} \mathbf{y}$$

where  $\mathbf{k}^{*T}$  is the covariance vector between the prediction location and all observed locations.

Nyström-Kriging prediction:

$$y(x^*) \approx \mathbf{k}^{*T}(\text{fast\_approx\_inverse}(\mathbf{C}, \mathbf{W}))\mathbf{y}$$

**Prediction Steps:**

1. Calculate  $\mathbf{k}^{*T}$ : Covariance between new location and all stations.
2. Approximate the inverse term using the Woodbury identity with **C** and **W**.
3. Make the prediction: Take the dot product of  $\mathbf{k}^{*T}$  and the weights vector.

This yields an accurate prediction efficiently, enabling rapid mapping for large regions.

**9.6.7. Details: Woodbury Matrix Identity for Avoiding the Big Inversion**

First, what is the **Woodbury matrix identity**? It's a mathematical rule that tells you how to find the inverse of a matrix that's been modified slightly. Its most useful form is for a "low-rank update":

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$$

This looks complicated, but the core idea is simple:

- If you have a matrix  $A$  that is **easy to invert** (like a diagonal matrix).
- And you add a low-rank matrix to it (the  $UCV$  part, where  $C$  is small).

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- You can find the new inverse without directly inverting the big  $(A + UCV)$  matrix. Instead, you only need to invert the much smaller matrix in the middle of the formula:  $(C^{-1} + VA^{-1}U)$ .

### How does this apply to the Nyström approximation?

In many machine learning and Kriging applications, we don't just need the kernel matrix  $\tilde{K}$ , but a "regularized" version,  $(\lambda I + \tilde{K})$ , where  $\lambda I$  is a diagonal matrix that helps prevent overfitting. We need to find the inverse of this:

$$(\lambda I + \tilde{K})^{-1}$$

Substituting the Nyström formula  $\tilde{K} = CW^+C^T$ , we get:

$$(\lambda I + CW^+C^T)^{-1}$$

This expression fits the Woodbury identity perfectly!

- $A = \lambda I$  (very easy to invert:  $A^{-1} = \frac{1}{\lambda}I$ )
- $U = C$  (our  $n \times l$  matrix)
- $C$  (middle matrix)  $= W^+$  (our small  $l \times l$  matrix)
- $V = C^T$  (our  $l \times n$  matrix)

By plugging these into the Woodbury formula, we get an expression for the inverse that only requires inverting a small  $l \times l$  matrix. This means we never have to build the full  $n \times n$  matrix  $\tilde{K}$  or invert it directly. This is the source of the massive speed-up.

---

### 9.6.8. The Example: Step-by-Step

Let's reuse our 4-point example and show both the slow way and the fast Woodbury way.

**Recall our matrices:**

- $C = \begin{pmatrix} 9 & 3 \\ 6 & 2 \\ 3 & 1 \\ 1 & 0.25 \end{pmatrix}$
- $W^+ = \begin{pmatrix} 0.09 & -0.27 \\ -0.27 & 0.81 \end{pmatrix}$
- Let's use a regularization value  $\lambda = 0.1$ .

**9.6.8.1. Method 1: The Slow Way (Forming the full matrix)**

1. **Construct  $\tilde{K}$ :** First, we explicitly calculate the full  $4 \times 4$  Nyström approximation  $\tilde{K} = CW^+C^T$ .

$$\tilde{K} = \begin{pmatrix} 9 & 6 & 3 & 0.675 \\ 6 & 4 & 2 & 0.45 \\ 3 & 2 & 1 & 0.225 \\ 0.675 & 0.45 & 0.225 & 0.05 \end{pmatrix}$$

2. **Add the regularization:** Now we compute  $(\lambda I + \tilde{K})$ .

$$(\lambda I + \tilde{K}) = \begin{pmatrix} 9.1 & 6 & 3 & 0.675 \\ 6 & 4.1 & 2 & 0.45 \\ 3 & 2 & 1.1 & 0.225 \\ 0.675 & 0.45 & 0.225 & 0.15 \end{pmatrix}$$

3. **Invert the  $4 \times 4$  matrix:** This is the expensive step. The result is:

$$(\lambda I + \tilde{K})^{-1} \approx \begin{pmatrix} 9.85 & -14.78 & -0.07 & 0.27 \\ -14.78 & 22.22 & 0.09 & -0.41 \\ -0.07 & 0.09 & 0.91 & -0.03 \\ 0.27 & -0.41 & -0.03 & 6.67 \end{pmatrix}$$

This works for our tiny  $4 \times 4$  example, but it would be computationally infeasible if  $n$  was 10,000.

**9.6.8.2. Method 2: The Fast Way (Using Woodbury Identity)**

We use the Woodbury formula to get the same result without ever creating a  $4 \times 4$  matrix. The formula simplifies to:

$$(\lambda I + \tilde{K})^{-1} = \frac{1}{\lambda} I - \frac{1}{\lambda^2} C \left( W + \frac{1}{\lambda} C^T C \right)^{-1} C^T$$

1. **Compute the small  $2 \times 2$  pieces:**

- $C^T C = \begin{pmatrix} 127 & 42.25 \\ 42.25 & 14.0625 \end{pmatrix}$
- $W = \begin{pmatrix} 9 & 3 \\ 3 & 1 \end{pmatrix}$

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- The matrix to invert is  $W + \frac{1}{0.1}C^TC = W + 10 \cdot (C^TC)$ , which is:

$$\begin{pmatrix} 9 & 3 \\ 3 & 1 \end{pmatrix} + \begin{pmatrix} 1270 & 422.5 \\ 422.5 & 140.625 \end{pmatrix} = \begin{pmatrix} 1279 & 425.5 \\ 425.5 & 141.625 \end{pmatrix}$$

2. **Invert the small  $2 \times 2$  matrix:** This is the only inversion we need, and it's extremely fast.

$$(W + \frac{1}{\lambda}C^TC)^{-1} \approx \begin{pmatrix} 0.22 & -0.66 \\ -0.66 & 1.99 \end{pmatrix}$$

3. **Combine the results:** Now we plug this small inverse back into the full formula. The rest is just matrix multiplication, no more inversions.

- First, calculate the middle term:  $M = \frac{1}{\lambda^2}C(\dots)^{-1}C^T$ . This will result in a  $4 \times 4$  matrix.
- Then, calculate the final result:  $\frac{1}{\lambda}I - M$ .

After performing these multiplications, you will get the **exact same  $4 \times 4$  inverse matrix** as in the slow method.

The crucial difference is that the most expensive operation—the matrix inversion—was performed on a tiny  $2 \times 2$  matrix instead of a  $4 \times 4$  one. For a large-scale problem, this is the difference between a calculation that takes seconds and one that could take hours or even be impossible.

## 9.7. Extending spotpython's Kriging Surrogate with Nyström Approximation for Enhanced Scalability

### 9.7.1. Introduction: Overcoming the Scalability Challenge in Kriging for Sequential Optimization

The Sequential Parameter Optimization Toolbox (spotpython) is a framework for hyperparameter tuning and black-box optimization based on Sequential Model-Based Optimization (SMBO). At the core of SMBO lies a surrogate model that approximates the true, expensive objective. Kriging (Gaussian Process regression) is a premier choice because it provides both predictions and a principled measure of uncertainty. This uncertainty enables a balance between exploration and exploitation. In each SMBO iteration, the Kriging model is updated with new evaluations, refining its approximation and proposing the next points.

Standard Kriging requires constructing and inverting an  $n \times n$  covariance matrix, where  $n$  is the number of data points. Matrix inversion scales as  $O(n^3)$ . During SMBO,  $n$  can reach hundreds or thousands; refitting the surrogate each iteration becomes



prohibitively expensive. This cubic scaling is the key obstacle to applying Kriging at larger scales.

We integrate the Nyström method into the `spotpython` Kriging class. The Nyström method yields a low-rank approximation of a symmetric positive semidefinite (SPSD) kernel matrix by selecting  $l \ll n$  “landmark” points. It approximates the full  $n \times n$  covariance while requiring inversion of only an  $l \times l$  matrix, reducing fitting cost from  $O(n^3)$  to  $O(nl^2)$ . This makes Kriging viable even when the number of function evaluations is large.

### 9.7.2. Report Objectives and Structure

- Review theoretical foundations of Kriging and Nyström approximation
- Present documented Python code updates for Kriging (as in `kriging.py`)
- Explain changes to `__init__`, `fit`, and `predict`
- Show how mixed variable types are preserved via `build_Psi` and `build_psi_vec`
- Provide practical usage guidance and a formal complexity analysis

---

## 9.8. Theoretical Foundations: The Nyström–Kriging Framework

### 9.8.1. A Primer on Kriging (Gaussian Process Regression)

Kriging models  $f(x)$  as a Gaussian Process with mean function  $m(\cdot)$  and covariance (kernel)  $k(\cdot, \cdot)$ . For training inputs  $X = \{x_1, \dots, x_n\}$  and observations  $y = \{y_1, \dots, y_n\}$ :

$$y \sim \mathcal{N}(m(X), K(X, X) + \sigma_n^2 I)$$

For a new point  $x_*$ :

$$\mu(x_*) = k(x_*, X) [K(X, X) + \sigma_n^2 I]^{-1} y$$

$$\sigma^2(x_*) = k(x_*, x_*) - k(x_*, X) [K(X, X) + \sigma_n^2 I]^{-1} k(X, x_*)$$

The challenge is inverting the  $n \times n$  matrix  $K(X, X) + \sigma_n^2 I$ .

### 9.8.2. The Nyström Method for Low-Rank Kernel Approximation

Select  $l$  landmark points  $X_m \subset X$ . Let: -  $C = K_{nm} = K(X, X_m) \in \mathbb{R}^{n \times l}$  -  $W = K_{mm} = K(X_m, X_m) \in \mathbb{R}^{l \times l}$  Then the Nyström approximation is:

$$\tilde{K}_{nn} = C W^+ C^\top = K_{nm} K_{mm}^+ K_{mn}$$

where  $W^+$  is the pseudoinverse of  $W$ . The approximation has rank  $\leq l$ .

### 9.8.3. Justification for Landmark Selection

Uniform sampling without replacement is an effective and inexpensive strategy for selecting landmarks across varied datasets and kernels.

---

## 9.9. Implementation: A Scalable Kriging Class for spotpython

### 9.9.1. Updated kriging.py with Nyström Approximation (excerpt)

```
"""
Kriging surrogate with optional Nyström approximation.
"""
import numpy as np
from scipy.spatial.distance import cdist
from scipy.linalg import cholesky, cho_solve, solve_triangular

class Kriging:
    def __init__(self, fun_control, n_theta=None, theta=None, p=2.0,
                 corr="squared_exponential", isotropic=False,
                 approximation="None", n_landmarks=100):
        self.fun_control = fun_control
        self.dim = self.fun_control["lower"].shape
        self.p = p
        self.corr = corr
        self.isotropic = isotropic
        self.approximation = approximation
        self.n_landmarks = n_landmarks
        self.factor_mask = self.fun_control["var_type"] == "factor"
```

### 9.9. Implementation: A Scalable Kriging Class for spotpython

```
self.ordered_mask = ~self.factor_mask
self.n_theta = 1 if isotropic else (n_theta or self.dim)
self.theta = np.full(self.n_theta, 0.1) if theta is None else theta
self.X_, self.y_, self.L_, self.alpha_ = None, None, None, None
self.landmarks_, self.W_cho_, self.nystrom_alpha_ = None, None, None

def fit(self, X, y):
    self.X_, self.y_ = X, y
    n_samples = X.shape[0]
    if self.approximation.lower() == "nystroem" and n_samples > self.n_landmarks:
        return self._fit_nystrom(X, y)
    return self._fit_standard(X, y)

def _fit_standard(self, X, y):
    Psi = self.build_Psi(X, X)
    Psi[np.diag_indices_from(Psi)] += 1e-8
    try:
        self.L_ = cholesky(Psi, lower=True)
        self.alpha_ = cho_solve((self.L_, True), y)
    except np.linalg.LinAlgError:
        self.L_ = None
        self.alpha_ = np.linalg.pinv(Psi) @ y

def _fit_nystrom(self, X, y):
    n_samples = X.shape[0]
    idx = np.random.choice(n_samples, self.n_landmarks, replace=False)
    self.landmarks_ = X[idx, :]
    W = self.build_Psi(self.landmarks_, self.landmarks_) + 1e-8 * np.eye(self.n_landmarks)
    C = self.build_Psi(X, self.landmarks_)
    try:
        self.W_cho_ = cholesky(W, lower=True)
        self.nystrom_alpha_ = cho_solve((self.W_cho_, True), C.T @ y)
    except np.linalg.LinAlgError:
        self.W_cho_ = None
        self._fit_standard(X, y)

def predict(self, X_star):
    if self.approximation.lower() == "nystroem" and self.landmarks_ is not None:
        return self._predict_nystrom(X_star)
    return self._predict_standard(X_star)

def _predict_standard(self, X_star):
    psi = self.build_Psi(X_star, self.X_)
    y_pred = psi @ self.alpha_
```

```

    if self.L_ is not None:
        v = solve_triangular(self.L_, psi.T, lower=True)
        y_mse = 1.0 - np.sum(v**2, axis=0)
    else:
        Psi = self.build_Psi(self.X_, self.X_) + 1e-8 * np.eye(self.X_.shape[0])
        pi_Psi = np.linalg.pinv(Psi)
        y_mse = 1.0 - np.sum((psi @ pi_Psi) * psi, axis=1)
    y_mse[y_mse < 0] = 0
    return y_pred, y_mse.reshape(-1, 1)

def _predict_nystrom(self, X_star):
    psi_star_m = self.build_Psi(X_star, self.landmarks_)
    y_pred = psi_star_m @ self.nystrom_alpha_
    if self.W_cho_ is not None:
        v = cho_solve((self.W_cho_, True), psi_star_m.T)
        quad = np.sum(psi_star_m * v.T, axis=1)
        y_mse = 1.0 - quad
    else:
        y_mse = np.ones(X_star.shape[0])
    y_mse[y_mse < 0] = 0
    return y_pred, y_mse.reshape(-1, 1)

def build_Psi(self, X1, X2):
    n1 = X1.shape[0]
    Psi = np.zeros((n1, X2.shape[0]))
    for i in range(n1):
        Psi[i, :] = self.build_psi_vec(X1[i, :], X2)
    return Psi

def build_psi_vec(self, x, X_):
    theta10 = np.full(self.dim, 10**self.theta) if self.isotropic else 10**self.t
    D = np.zeros(X_.shape[0])
    if self.ordered_mask.any():
        Xo = X_[ :, self.ordered_mask]
        xo = x[self.ordered_mask]
        D += cdist(xo.reshape(1, -1), Xo, metric="sqeuclidean",
                    w=theta10[self.ordered_mask]).ravel()
    if self.factor_mask.any():
        Xf = X_[ :, self.factor_mask]
        xf = x[self.factor_mask]
        D += cdist(xf.reshape(1, -1), Xf, metric="hamming",
                    w=theta10[self.factor_mask]).ravel() * self.factor_mask.sum()
    return np.exp(-D) if self.corr == "squared_exponential" else np.exp(-(D**self

```

## 9.10. Implementation Details

### 9.10.1. Architectural Enhancements to `init`

- New argument `approximation="None"` for backward-compatible selection between exact Kriging and Nyström
- New argument `n_landmarks` (default 100) controls the number of inducing points when using Nyström
- State attributes for both exact and Nyström paths are maintained separately

### 9.10.2. The `fit()` Method: A Dual-Pathway Approach

- Dispatcher selecting exact or Nyström pathway
- The Nyström fit Pathway (`_fit_nystrom`):
  - Landmark selection via uniform sampling without replacement
  - Core matrices:
    - \*  $W = K_{mm}$  (landmark-landmark)
    - \*  $C = K_{nm}$  (data-landmark)
  - Cholesky factorization of  $W$  (with jitter) for stability
  - Pre-computation:  $\alpha_{nys} = W^{-1}C^T y$  via `cho_solve`
- The Standard fit Pathway (`_fit_standard`):
  - Full  $\Psi$  construction, Cholesky decomposition, and solve for  $\alpha$
  - Fallback to pseudoinverse if Cholesky fails

### 9.10.3. The `predict()` Method: Conditional Prediction Logic

- Routes to Nyström or standard prediction path based on fitted model state
- The Nyström predict Pathway (`_predict_nystrom`):
  - Cross-covariance  $\psi$  between test points and landmarks
  - Mean:  $\psi \cdot \alpha_{nys}$
  - Variance: uses `cho_solve` with  $W$  Cholesky; non-negative clipping
- The Standard predict Pathway (`_predict_standard`):
  - Cross-covariance with all training points
  - Mean from  $\alpha$ ; variance via triangular solves or pseudoinverse fallback

#### 9.10.4. Critical Detail: Preserving Mixed Variable Type Functionality

The Significance of `build_psi_vec`:

- Mixed spaces: continuous (ordered) and categorical (factor) variables
- Distances:
  - Weighted squared Euclidean for ordered variables
  - Weighted Hamming for factors
- Anisotropic kernel via per-dimension length-scales  $\theta$
- Nyström path reuses `build_Psi`  $\rightarrow$  `build_psi_vec`, preserving mixed-type handling

#### 9.10.5. Seamless Integration into the Nyström Workflow

All covariance computations ( $W$ ,  $C$ , predictive cross-covariance) use `build_Psi`, ensuring identical handling for mixed variable types in both standard and Nyström modes.

### 9.11. Jupyter Notebook

#### Note

- The Jupyter-Notebook of this lecture is available on GitHub in the Hyperparameter-Tuning-Cookbook Repository

## 10. Infill Criteria

In the context of computer experiments and surrogate modeling, a sampling plan refers to the set of input values, often denoted as  $X$ , at which a computer code is evaluated. The primary objective of a sampling plan is to efficiently explore the input space to understand the behavior of the computer code and to construct a surrogate model that accurately represents that behavior. Historically, Response Surface Methodology (RSM) provided methods for designing such plans, often based on rectangular grids or factorial designs. More recently, Design and Analysis of Computer Experiments (DACE) has emerged as a more flexible and powerful approach for this purpose.

A surrogate model, or  $\hat{f}$ , is built to approximate the expensive response of a black-box function  $f(x)$ . Since evaluating  $f$  is costly, only a sparse set of samples is used to construct  $\hat{f}$ , which can then provide inexpensive predictions for any point in the design space. However, as a surrogate model is inherently an approximation of the true function, its accuracy and predictive capabilities can be significantly improved by incorporating new data points, known as infill points. Infill points are strategically chosen to either reduce uncertainty, improve predictions in specific regions of interest, or enhance the model's ability to identify optima or trends.

The process of updating a surrogate model with infill points is iterative. It typically involves:

- **Identifying Regions of Interest:** Analyzing the current surrogate model to determine areas where it is inaccurate, has high uncertainty, or predicts promising results (e.g., potential optima).
- **Selecting Infill Points:** Choosing new data points based on specific criteria that balance different objectives.
- **Evaluating the True Function:** Running the actual simulation or experiment at the selected infill points to obtain their corresponding outputs.
- **Updating the Surrogate Model:** Retraining or updating the surrogate model using the new, augmented dataset.
- **Repeating:** Iterating this process until the model meets predefined accuracy criteria or the computational budget is exhausted.

## 10.1. Balancing Exploitation and Exploration

A crucial aspect of selecting infill points is navigating the inherent trade-off between exploitation and exploration.

**Definition 10.1** (Exploitation). Exploitation refers to sampling near predicted optima to refine the solution. This strategy aims to rapidly converge on a good solution by focusing computational effort where the surrogate model suggests the best values might lie.

**Definition 10.2** (Exploration). Exploration involves sampling in regions of high uncertainty to improve the global accuracy of the model. This approach ensures that the model is well-informed across the entire design space, preventing it from getting stuck in local optima.

Forrester, Sóbester, and Keane (2008) emphasizes that effective infill criteria are designed to combine both exploitation and exploration.

## 10.2. Expected Improvement (EI)

Expected Improvement (EI) is one of the most influential and widely-used infill criteria. Formalized by Jones, Schonlau, and Welch (1998) and building upon the work of Moćkus (1974), EI provides a mathematically elegant framework that naturally balances exploitation and exploration. Rather than simply picking the point with the best predicted value (pure exploitation) or the point with the highest uncertainty (pure exploration), EI asks a more nuanced question: “How much improvement over the current best solution can we *expect* to gain by evaluating the true function at a new point  $x$ ?”

The Expected Improvement,  $EI(x)$ , can be calculated using the following formula:

$$EI(x) = \sigma(x) [Z\Phi(Z) + \phi(Z)]$$

where:

- $\mu(x)$  (or  $\hat{y}(x)$ ) is the Kriging prediction (mean of the stochastic process) at a new, unobserved point  $x$ .
- $\sigma(x)$  (or  $\hat{s}(x)$ ) is the estimated standard deviation (square root of the variance  $\hat{s}^2(x)$ ) of the prediction at point  $x$ .
- $f_{best}$  (or  $y_{min}$ ) is the best (minimum, for minimization problems) observed function value found so far.
- $Z = \frac{f_{best} - \mu(x)}{\sigma(x)}$  is the standardized improvement.



### 10.3. Expected Improvement in the Hyperparameter Tuning Cookbook (Python Implementation)

- $\Phi(Z)$  is the cumulative distribution function (CDF) of the standard normal distribution.
- $\phi(Z)$  is the probability density function (PDF) of the standard normal distribution.

If  $\sigma(x) = 0$  (meaning there is no uncertainty at point  $x$ , typically because it's an already sampled point), then  $EI(x) = 0$ , reflecting the intuition that no further improvement can be expected at a known point. A maximization of Expected Improvement as an infill criterion will eventually lead to the global optimum.

The elegance of the EI formula lies in its combination of two distinct terms:

- **Exploitation Term:**  $(f_{best} - \mu(x))\Phi(Z)$ . This part of the formula contributes more when the predicted value  $\mu(x)$  is significantly lower (better) than the current best observed value  $f_{best}$ . It is weighted by the probability  $\Phi(Z)$  that the true function value at  $x$  will indeed be an improvement over  $f_{best}$ .
- **Exploration Term:**  $\sigma(x)\phi(Z)$ . This term becomes larger when there is high uncertainty ( $\sigma(x)$  is large) in the model's prediction at  $x$ . It accounts for the potential of discovering unexpectedly good values in areas that have not been thoroughly explored, even if the current mean prediction there is not the absolute best.

Expected Improvement offers several significant practical benefits:

- **Automatic Balance:** It inherently balances exploitation and exploration without requiring any manual adjustment of weights or parameters.
- **Scale Invariance:** EI is relatively insensitive to the scaling of the objective function, making it robust across various problem types.
- **Theoretical Foundation:** It is underpinned by a strong theoretical basis derived from decision theory and information theory.
- **Efficient Optimization:** The smooth and differentiable nature of the EI function allows for efficient optimization using gradient-based algorithms to find the next infill point.
- **Proven Performance:** EI has demonstrated consistent and strong performance in numerous real-world applications across various domains.

## 10.3. Expected Improvement in the Hyperparameter Tuning Cookbook (Python Implementation)

Within the context of the Hyperparameter Tuning Cookbook, Expected Improvement serves a critical role in Sequential Model-Based Optimization. It systematically guides the selection of which hyperparameter configurations to evaluate next, facilitating the efficient utilization of computational resources. By intelligently balancing the need to exploit promising regions and explore uncertain areas, EI helps identify optimal

## 10. Infill Criteria

hyperparameters with a reduced number of expensive model training runs. This provides a principled and automated method for navigating complex hyperparameter spaces without extensive manual intervention.

While the foundational concepts in Forrester, Sóbester, and Keane (2008) are often illustrated with MATLAB code, the Hyperparameter Tuning Cookbook emphasizes and provides implementations in Python. The `spotpython` package, consistent with the Cookbook's approach, provides a Python implementation of Expected Improvement within its Kriging class. For minimization problems, `spotpython` typically calculates and returns the negative Expected Improvement, aligning with standard optimization algorithm conventions. Furthermore, to enhance numerical stability and mitigate issues when EI values are very small, `spotpython` often works with a logarithmic transformation of EI and incorporates a small epsilon value.

### 10.4. Jupyter Notebook

#### Note

- The Jupyter-Notebook of this lecture is available on GitHub in the Hyperparameter-Tuning-Cookbook Repository

## **Part III.**

# **Sequential Parameter Optimization Toolbox (SPOT)**



## **11. SpotOptim Step-by-Step Optimization Process**



## 12. Introduction

This document provides a comprehensive step-by-step explanation of the optimization process in the `SpotOptim` class. We'll use the 2-dimensional Rosenbrock function as our example and cover all methods involved in each stage of optimization.

### Topics Covered:

- Standard optimization workflow
- Handling noisy functions with repeats
- Handling function evaluation failures (NaN/inf values)





## 13. Setup and Test Functions

Let's start by defining our test functions, including variants with noise and occasional failures. The two-dimensional Rosenbrock function is defined as:

$$f(x, y) = (a - x)^2 + b(y - x^2)^2$$

where typically ( $a = 1$ ) and ( $b = 100$ ). The generalized form for  $n$  dimensions is:

$$f(X) = \sum_{i=1}^{N-1} [100 \cdot (x_{i+1} - x_i^2)^2 + (1 - x_i)^2]$$

The documentation can be found here: [DOC](#)

```
import numpy as np
from spotoptim import SpotOptim
from spotoptim.function import rosenbrock
import warnings
warnings.filterwarnings('ignore')
```

Set random seed for reproducibility:

```
np.random.seed(42)
```

Based on the standard Rosenbrock function, we define two variants:

1. the noisy Rosenbrock function:

```
def rosenbrock_noisy(X, noise_std=0.1):
    """
    Rosenbrock with Gaussian noise for testing noisy optimization.
    """
    X = np.atleast_2d(X)
    base_values = rosenbrock(X)
    noise = np.random.normal(0, noise_std, size=base_values.shape)
    return base_values + noise
```

and 2. the Rosenbrock function with occasional failures:

### 13. Setup and Test Functions

```
def rosenbrock_with_failures(X, failure_prob=0.15):
    """
    Rosenbrock that occasionally returns NaN to simulate evaluation failures.
    """
    X = np.atleast_2d(X)
    values = rosenbrock(X)

    # Randomly inject failures
    for i in range(len(values)):
        if np.random.random() < failure_prob:
            values[i] = np.nan

    return values
```

## 14. Standard Optimization Workflow

Let's trace through a complete optimization run to understand each step.

### 14.1. Phase: Initialization and Setup

When you create a `SpotOptim` instance, several initialization steps occur during the `__init__()` method, see DOC.

1. Objective function (`fun`) and bounds are stored.
2. Acquisition function (`acquisition`) is stored (default is `y`).
3. Determine if noise handling is active (based on `repeats_initial` and `repeats_surrogate`)
4. For dimensions with tuple bounds (factor variables), internal integer mappings are created and bounds are replaced with  $(0, n\_levels-1)$ . This is handled by `_process_factor_bounds()`. So, e.g., `[('red', 'green', 'blue')]` is mapped to the integer interval `[(0, 2)]` internally.
5. The dimension of the problem (`n_dim`) is inferred from the bounds.
6. If `var_type` is not provided, it defaults to all “float” (continuous) variables, except for factor variables which are set to “factor” via `_process_factor_bounds()`.
7. Default variable names (`var_name`) are set if not provided.
8. Default variable transformations (`var_transform`) are set if not provided.
9. Transformations are applied to bounds based on `var_transform` settings. Natural bounds are stored in `_original_lower` and `_original_upper`.
10. Dimension reduction by identifying fixed dimensions (if any) is performed via `_setup_dimension_reduction()`.
11. The surrogate is initialized (default: Gaussian Process with Matérn kernel) as follows:

```
kernel = ConstantKernel(  
    constant_value=1.0, constant_value_bounds=(1e-3, 1e3)  
) * Matern(length_scale=1.0, length_scale_bounds=(1e-2, 1e2), nu=2.5)  
self.surrogate = GaussianProcessRegressor(  
    kernel=kernel,  
    n_restarts_optimizer=10,  
    normalize_y=True,
```

## 14. Standard Optimization Workflow

```
    random_state=self.seed,  
)
```

12. The Design generator is initialized (default: Latin Hypercube Sampling).
13. The storage for results ins initialized. This includes the following attributes:

- `X_`: Evaluated design points
- `y_`: Corresponding function values
- `y_mo`: For multi-objective functions, stores all objectives
- `best_x_`: Best point found so far
- `best_y_`: Best function value found so far
- `n_iter_`: Number of iterations completed
- `mean_X`, `mean_y`, `var_y`: For noisy functions, mean and variance tracking
- `min_mean_X`: Best mean point for noisy functions
- `min_mean_y`: Best mean value for noisy functions
- `min_var_y`: Variance at best mean point
- `min_X`: Best point found (deterministic)
- `min_y`: Best function value found (deterministic)
- `counter`: Total number of function evaluations
- `success_rate`: Ratio of successful evaluations
- `success_counter`: Count of successful evaluations
- `window_size`: For moving average of success rate
- `success_history`: History of success/failure for evaluations

```
# Create optimizer  
opt = SpotOptim(  
    fun=rosenbrock,  
    bounds=[(-2, 2), (1, 2)],  
    n_initial=10,  
    max_iter=30,  
    verbose=True,  
    seed=42,  
    var_trans=["id", "log"]  
)
```

TensorBoard logging disabled

## 14.2. Phase: Initial Design Generation

### 14.2.1. Method: `get_initial_design()`

This method generates or processes the initial sample points. It is manually called here for demonstration (normally done inside `optimize()`).

**What happens in `get_initial_design()`:**

1. If `X0=None`: Generate space-filling design using Latin Hypercube Sampling (LHS)
2. If `x0` (starting point) provided: Include it as first point and transform user's points to internal scale
3. Apply dimension reduction if configured
4. Round integer/factor variables

```
X0 = opt.get_initial_design(X0=None)

print(f"Generated {len(X0)} initial design points using Latin Hypercube Sampling")
print(f"Design shape: {X0.shape}")
print(f"\nFirst 3 points (internal scale):")
print(X0[:3])
```

```
Generated 10 initial design points using Latin Hypercube Sampling
Design shape: (10, 2)
```

```
First 3 points (internal scale):
[[-1.10958242  0.45478229]
 [ 1.65656083  0.228921  ]
 [-1.63767094  0.55620747]]
```

Note, because `bounds` were set to `[(-2, 2), (-2, 2)]` and `n_initial=10`, the generated points are within this range and the design has two-dimensional shape (10, 2).

- Print initial design in original scale and in internal scale:

```
X0_original = opt._inverse_transform_X(X0)
print(f"\nFirst 3 points (original scale):")
print(X0_original[:3])
print(f"\nFirst 3 points (internal scale):")
print(X0[:3])
```

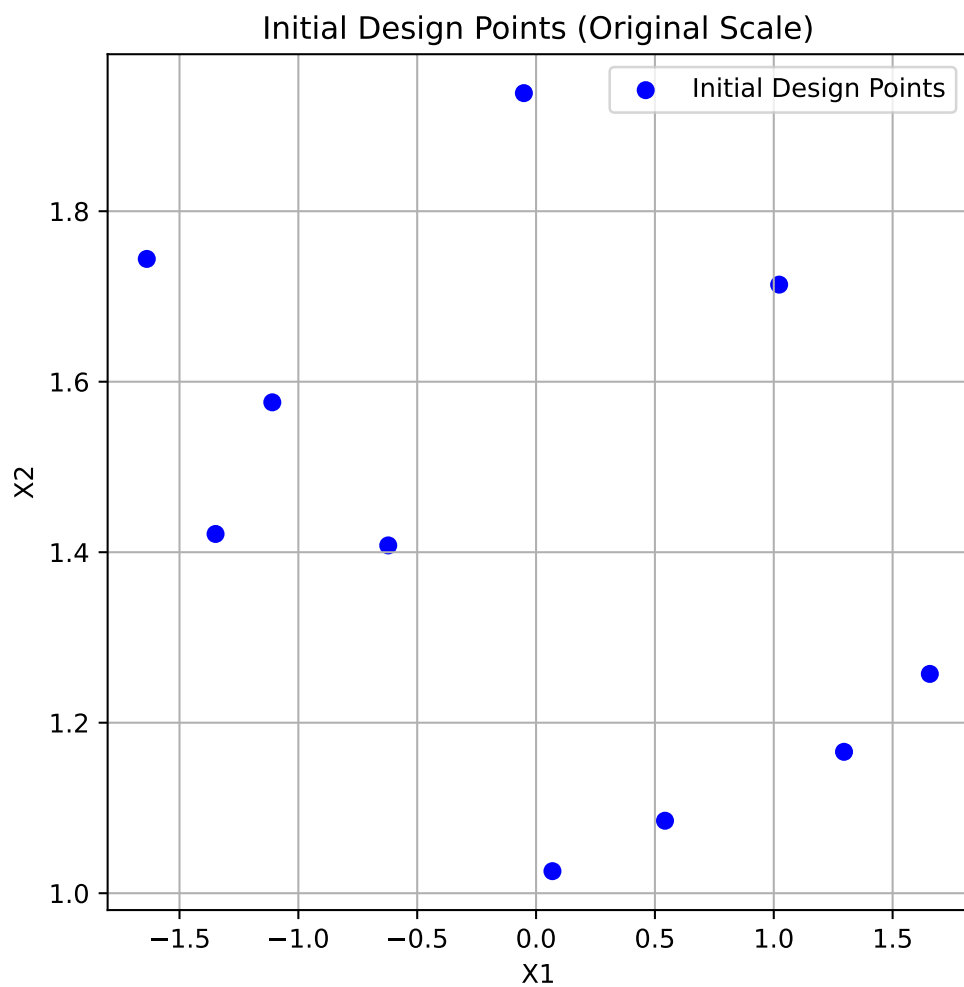
```
First 3 points (original scale):
[[-1.10958242  1.57583027]
 [ 1.65656083  1.25724272]
 [-1.63767094  1.7440456  ]]
```

```
First 3 points (internal scale):
[[-1.10958242  0.45478229]
 [ 1.65656083  0.228921  ]
 [-1.63767094  0.55620747]]
```

#### 14. Standard Optimization Workflow

- Plot initial design in original scale:

```
import matplotlib.pyplot as plt
plt.figure(figsize=(6, 6))
plt.scatter(X0_original[:, 0], X0_original[:, 1], c='blue', label='Initial Design Points')
plt.xlabel("X1")
plt.ylabel("X2")
plt.title("Initial Design Points (Original Scale)")
plt.legend()
plt.grid(True)
plt.show()
```



## 14.3. Phase 3: Initial Design Curation

### 14.3.1. Method: `_curate_initial_design()`

This method ensures we have sufficient unique points and handles repeats.

**What happens in `_curate_initial_design()`:**

1. Remove duplicate points (can occur after rounding integers)
2. Generate additional points if duplicates reduced count below `n_initial`
3. Repeat each point `repeats_initial` times if  $> 1$  (for noisy functions)

```
X0_curated = opt._curate_initial_design(X0)
print(f"Curated design shape: {X0_curated.shape}")
print(f"Unique points: {len(np.unique(X0_curated, axis=0))}")
print(f"Total points (with repeats): {len(X0_curated)}")
```

```
Curated design shape: (10, 2)
Unique points: 10
Total points (with repeats): 10
```

## 14.4. Phase 4: Initial Design Evaluation

### 14.4.1. Method: `_evaluate_function()`

Evaluates the objective function at all initial design points. The points are converted back to the original scale for evaluation.

**What happens in `_evaluate_function()`:**

1. Convert points from internal to original scale
2. Call objective function with batch of points
3. Convert multi-objective to single-objective if needed
4. Return array of function values

```
X0_original = opt._inverse_transform_X(X0_curated)
y0 = opt._evaluate_function(X0_curated)

print(f"Evaluated {len(y0)} points")
print(f"Function values shape: {y0.shape}")
print(f"First 5 points with function values:")
for i in range(min(5, len(y0))):
    print(f"  Point {i+1}: {X0_original[i]} → f(x)={y0[i]:.6f}")
```

## 14. Standard Optimization Workflow

```
print(f"\nBest initial value: {np.min(y0):.6f}")
print(f"Worst initial value: {np.max(y0):.6f}")
print(f"Mean initial value: {np.mean(y0):.6f}")
```

Evaluated 10 points

Function values shape: (10,)

First 5 points with function values:

```
Point 1: [-1.10958242  1.57583027] → f(x)=16.329192
Point 2: [1.65656083  1.25724272] → f(x)=221.533421
Point 3: [-1.63767094  1.7440456 ] → f(x)=94.926795
Point 4: [1.29554412  1.1658592 ] → f(x)=26.360696
Point 5: [-0.05124545  1.93852778] → f(x)=375.876648
```

Best initial value: 16.329192

Worst initial value: 375.876648

Mean initial value: 107.571208

## 14.5. Phase 5: Handling Failed Evaluations

### 14.5.1. Method: `_handle_NA_initial_design(X0,y0)`

Removes points that returned NaN or inf values. In contrast to later phases, no penalties are applied here; invalid points of the initial design are simply removed.

What happens in `_handle_NA_initial_design()`:

1. Identify NaN/inf values in function evaluations
2. Remove corresponding design points
3. Return cleaned arrays and original count
4. Note: No penalties applied in initial design - invalid points simply removed

```
n_before = len(y0)
X0_clean, y0_clean, n_evaluated = opt._handle_NA_initial_design(X0_curated, y0)

print(f"Points before filtering: {n_before}")
print(f"Points after filtering: {len(y0_clean)}")
print(f"Removed: {n_before - len(y0_clean)} NaN/inf values")
print(f"\nAll remaining values finite: {np.all(np.isfinite(y0_clean))}")
```

Points before filtering: 10

Points after filtering: 10

Removed: 0 NaN/inf values



All remaining values finite: True

## 14.6. Phase 6: Validation Check

### 14.6.1. Method: `_check_size_initial_design(y0, n_evaluated)`

Ensures we have enough valid points to continue. The minimum required is the smaller of:

- typical minimum for surrogate fitting (3 for multi-dimensional, 2 for 1D), or
- what the user requested (`n_initial`).

What happens in `_check_size_initial_design()`:

1. Check if at least 1 valid point exists
2. Raise error if all initial evaluations failed
3. Print warnings if many points were invalid

```
try:
    opt._check_size_initial_design(y0_clean, n_evaluated)
    print(f" Validation passed: {len(y0_clean)} valid points available")
    print(f" Minimum required: 1 point")
    print(f" Original evaluated: {n_evaluated} points")
except ValueError as e:
    print(f" Validation failed: {e}")
```

```
Validation passed: 10 valid points available
Minimum required: 1 point
Original evaluated: 10 points
```

## 14.7. Phase 7: Storage Initialization

### 14.7.1. Method: `_init_storage()`

Store evaluated points and initialize tracking variables.

What happens during storage initialization:

1. `_init_storage()`: Initialize statistics tracking variables.
  - Initialize storage (as done in `optimize()`)

## 14. Standard Optimization Workflow

```
# Initialize storage and statistics using the new _init_storage() method
opt._init_storage(X0_clean, y0_clean)
```

```
print(f"X_ (evaluated points): shape {opt.X_.shape}")
print(f"y_ (function values): shape {opt.y_.shape}")
print(f"n_iter_ (iterations): {opt.n_iter_}")
```

```
X_ (evaluated points): shape (10, 2)
y_ (function values): shape (10,)
n_iter_ (iterations): 0
```

## 14.8. Phase: Statistics Update

### 14.8.1. Method: `update_stats()`

Update statistics like means and variances for noisy evaluations.

\*\*\* What happens in `update_stats()`:\*\*\*

1. If noise is set (i.e., `repeats_initial > 1`):
  - Compute mean and variance of function values for each unique point
  - Store in `mean_X`, `mean_y`, and `var_y`

```
print(f"\nStatistics updated:")
if opt.noise:
    print(f" - mean_X: {opt.mean_X.shape}")
    print(f" - mean_y: {opt.mean_y.shape}")
    print(f" - var_y: {opt.var_y.shape}")
else:
    print(f" - No noise tracking (repeats=1)")
```

```
Statistics updated:
- No noise tracking (repeats=1)
```

## 14.9. Phase: Log initial Design to TensorBoard

### 14.9.1. Method: `_log_initial_design_tensorboard()`

This method logs the initial design points and their evaluations to TensorBoard for visualization.

## 14.10. Phase: Initial Best Point

### 14.10.1. Method: `_get_best_xy_initial_design()`

Identify and report the best point from initial design.

What happens in `_get_best_xy_initial_design()`:

1. Find minimum value in `y_` (or `mean_y` if noise)
2. Store as `best_y_`
3. Store corresponding point as `best_x_`
4. Print progress if verbose

```
opt._get_best_xy_initial_design()

print(f"Best point found: {opt.best_x_}")
print(f"Best value found: {opt.best_y_:.6f}")
print(f"\nOptimum location: [1, 1]")
print(f"Optimum value: 0")
print(f"Current gap: {opt.best_y_ - 0:.6f}")
```

```
Initial best: f(x) = 16.329192
Best point found: [-1.10958242  1.57583027]
Best value found: 16.329192
```

```
Optimum location: [1, 1]
Optimum value: 0
Current gap: 16.329192
```

## 14.11. Illustration of the Initial Design Phase Results

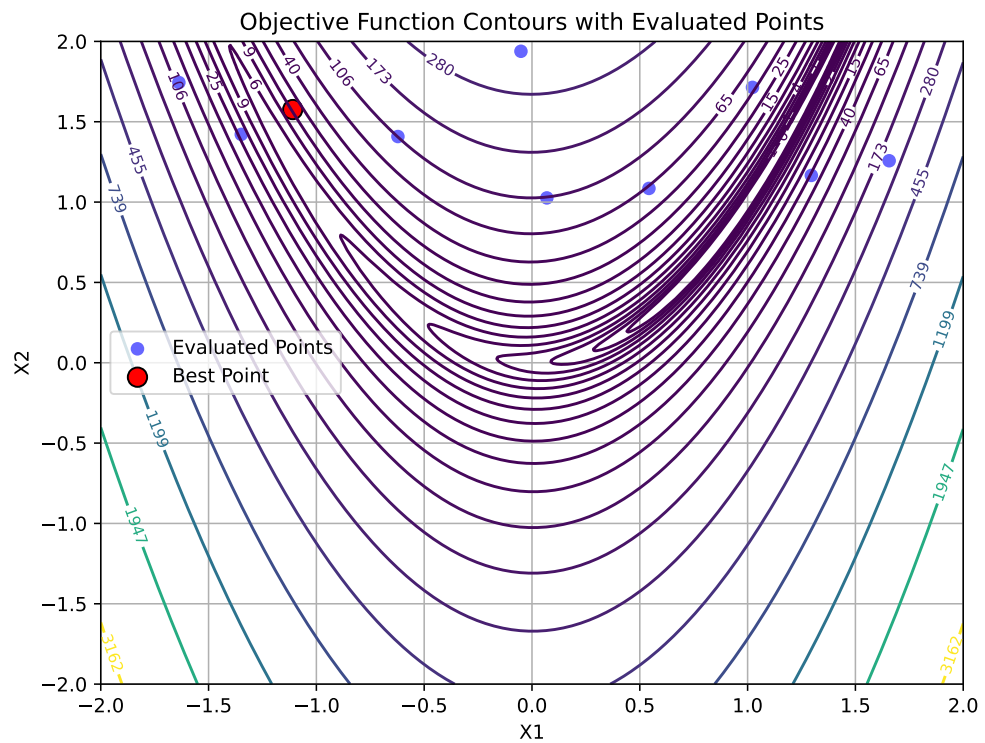
To visualize the results, we generate a contour plot with contour lines of the objective function and mark the best point. The other evaluated points are shown as well:

```
# Create a grid of points for contour plotting
x = np.linspace(-2, 2, 400)
y = np.linspace(-2, 2, 400)
X, Y = np.meshgrid(x, y)
grid_points = np.column_stack([X.ravel(), Y.ravel()])
Z = rosenbrock(grid_points).reshape(X.shape)
plt.figure(figsize=(8, 6))
# Contour plot
```

#### 14. Standard Optimization Workflow

```
contour = plt.contour(X, Y, Z, levels=np.logspace(-0.5, 3.5, 20), cmap='viridis')
plt.clabel(contour, inline=True, fontsize=8)

# Plot evaluated points from the initial design phase:
plt.scatter(opt.X[:, 0], opt.X[:, 1], c='blue', label='Evaluated Points', alpha=0.6)
# Mark best point
plt.scatter(opt.best_x[0], opt.best_x[1], c='red', s=100, label='Best Point', edgecolor='black')
plt.xlabel('X1')
plt.ylabel('X2')
plt.title('Objective Function Contours with Evaluated Points')
plt.legend()
plt.grid(True)
plt.show()
```



## 15. Sequential Optimization Loop

After initialization, the main optimization loop begins. Each iteration follows these steps:

### 15.1. Step: Surrogate Model Fitting

#### 15.1.1. Method: `_fit_scheduler()`

Fit surrogate model using appropriate data based on noise handling.

\*\*\* What happens in `_fit_scheduler()`:\*\*\*

First, the method transforms the input data `X` to the internal scale used by the optimizer. Then, it decides which data to use for fitting the surrogate model based on whether noise handling is enabled:

1. If `noise` is set (i.e., `repeats_surrogate > 1`):
  - Fit surrogate using mean points (`mean_X`, `mean_y`)
2. Else:
  - Fit surrogate using all evaluated points (`X_`, `y_`)

`_fit_scheduler()` then calls `_fit_surrogate()` with the selected data.

#### 15.1.2. Method: `_fit_surrogate()`

Fit a surrogate model (Gaussian Process) to current data.

What happens in `_fit_surrogate()`:

1. If `max_surrogate_points` set and exceeded: Select subset of points using the `_selection_dispatcher()` method.
2. Fit surrogate model using `surrogate.fit(X, y)`
3. Surrogate learns the function landscape and uncertainty

## 15. Sequential Optimization Loop

```
X_for_surrogate = opt._transform_X(opt.X_)
opt._fit_surrogate(X_for_surrogate, opt.y_)

print(f"Surrogate fitted with {len(opt.y_)} points")
print(f"Surrogate type: {type(opt.surrogate).__name__}")
print(f"Kernel: {opt.surrogate.kernel_}")
```

```
Surrogate fitted with 10 points
Surrogate type: GaussianProcessRegressor
Kernel: 1.06**2 * Matern(length_scale=0.355, nu=2.5)
```

### 15.1.3. Method: `_selection_dispatcher()`

What happens in `_selection_dispatcher()`:

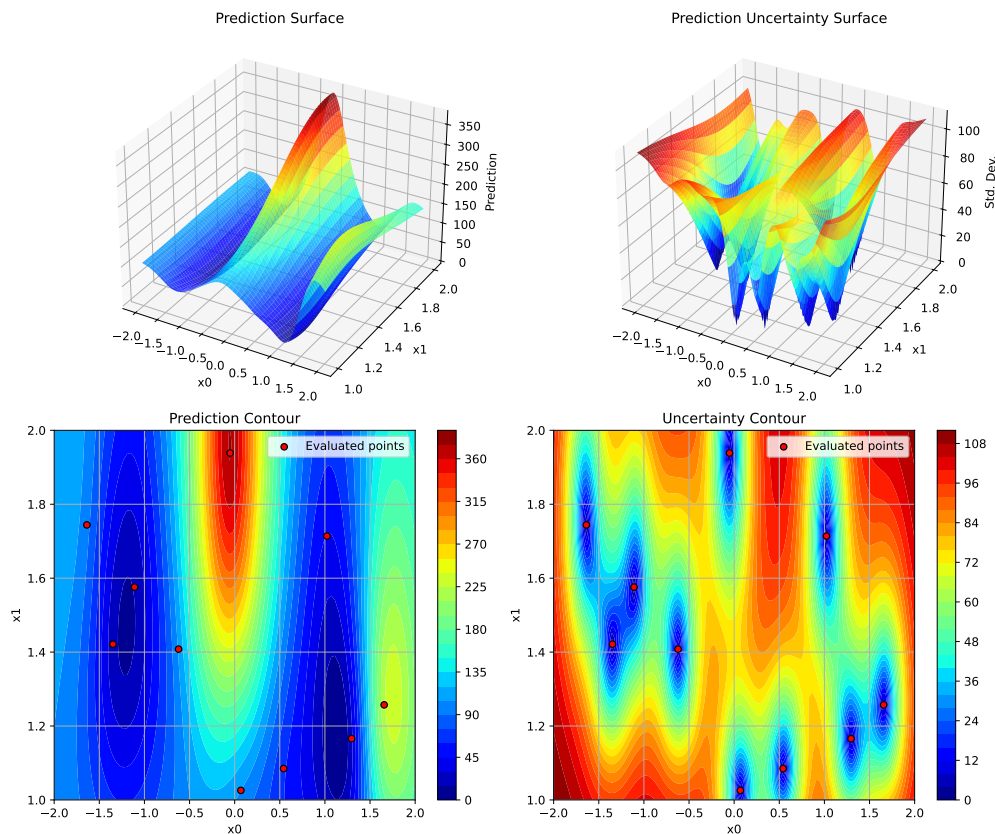
1. If `max_surrogate_points` is set and exceeded:
  - Select a subset of points from `X_` and `y_` for fitting the surrogate model.
  - Strategies may include random sampling, clustering, or other heuristics to reduce the dataset size while preserving important information.

#### Surrogate Model Selection:

- Default: Gaussian Process with Matérn kernel
- Provides: Mean prediction  $\mu(x)$  and uncertainty  $\sigma(x)$
- Can be replaced with Random Forest, Kriging, etc.
- The fitted surrogate can be plotted with the following code:

```
opt.plot_surrogate()
```

## 15.1. Step: Surrogate Model Fitting



### 15.1.4. Step: Apply OCBA

### 15.1.5. Method: `_apply_ocba()`

What happens in `_apply_ocba()`:

1. Compute the optimality criteria for each point in the design space.
2. Select the most promising points based on the criteria.
3. Update the surrogate model with the selected points.

```
X_ocba = opt._apply_ocba()
if X_ocba is not None:
    print(f"OCBA selected {X_ocba.shape[0]} points for re-evaluation")
    print(f"OCBA points shape: {X_ocba.shape}")
else:
    print("OCBA not applied (noise=False or ocba_delta=0)")
```

## 15. Sequential Optimization Loop

OCBA not applied (noise=False or ocba\_delta=0)

### 15.2. Step: Predict with Uncertainty

#### 15.2.1. Method: `_predict_with_uncertainty()`

Make predictions at new locations with uncertainty estimates.

What happens in `_predict_with_uncertainty()`:

1. Call `surrogate.predict(X, return_std=True)`
2. Returns mean ( $\mu$ ) and standard deviation ( $\sigma$ )
3. Used by acquisition function to balance exploitation (low  $\sigma$ ) and exploration (high  $\sigma$ )

Here we test prediction at a few points:

```
X_test = np.array([[0.5, 0.5], [1.0, 1.0], [-0.5, 0.5]])
mu, sigma = opt._predict_with_uncertainty(X_test)

print(f"Test points: {X_test.shape}")
print(f"\nPredictions (  $\mu$   $\pm$   $\sigma$  ):")
for i, (x, m, s) in enumerate(zip(X_test, mu, sigma)):
    print(f"  x={x}  $\rightarrow$   $\mu$ ={m:.4f},  $\sigma$ ={s:.4f}")
```

Test points: (3, 2)

```
Predictions (  $\mu$   $\pm$   $\sigma$  ):
  x=[0.5 0.5]  $\rightarrow$   $\mu$ =135.9904,  $\sigma$ =96.1193
  x=[1. 1.]  $\rightarrow$   $\mu$ =97.6979,  $\sigma$ =104.5930
  x=[-0.5 0.5]  $\rightarrow$   $\mu$ =169.1890,  $\sigma$ =64.4389
```

### 15.3. Step: Next Point Suggestion

#### 15.3.1. Method: `_suggest_next_point()`

Optimize acquisition function to find next evaluation point.

What happens in `_suggest_next_point()`:

1. Use `differential_evolution` to optimize acquisition function
2. Find point that maximizes EI (or minimizes predicted value for 'y')



3. Apply rounding for integer/factor variables
4. Check distance to existing points (avoid duplicates)
5. If duplicate or too close: Use fallback strategy
6. Return suggested point

```
x_next = opt._suggest_next_point()
print(f"Next point suggested: {x_next}")
```

Next point suggested: [1.13866324 0.15750508]

Predict at suggested point:

```
mu_next, sigma_next = opt._predict_with_uncertainty(x_next.reshape(1, -1))
print(f"\nPrediction at suggested point:")
print(f"    (x_next) = {mu_next[0]:.4f}")
print(f"    (x_next) = {sigma_next[0]:.4f}")
```

Prediction at suggested point:

```
(x_next) = 1.3255
(x_next) = 50.4602
```

Fallback Strategies (if acquisition optimization fails):

- 'best': Re-evaluate current best point
- 'mean': Select point with best predicted mean
- 'random': Random point in search space

## 15.4. Step: Acquisition Function Evaluation

### 15.4.1. Method: `_acquisition_function()`

Compute acquisition function value to guide search. The acquisition function is used as an objective function by the optimizer on the surrogate, e.g., differential evolution. It determines where to sample next.

```
# Evaluate acquisition at test points
print(f"Acquisition type: {opt.acquisition} (Expected Improvement)")
print(f"\nAcquisition values at test points:")

for x in X_test:
    acq_val = opt._acquisition_function(x)
    print(f"    x={x} → acq={acq_val:.6f}")
```

## 15. Sequential Optimization Loop

Acquisition type: y (Expected Improvement)

Acquisition values at test points:

x=[0.5 0.5] → acq=135.990446

x=[1. 1.] → acq=97.697886

x=[-0.5 0.5] → acq=169.189001

Acquisition function can be used to balance:

- exploitation, i.e., low predicted mean  $\mu(x)$  and
- exploration, i.e., high uncertainty  $\sigma(x)$

**Acquisition Functions Available:**

1. **Expected Improvement (EI)** - Default, best balance

$$\text{EI}(x) = (f^* - \mu(x))\Phi(Z) + \sigma(x)\phi(Z)$$

$$\text{where } Z = \frac{f^* - \mu(x)}{\sigma(x)}$$

2. **Probability of Improvement (PI)** - More exploitative

$$\text{PI}(x) = \Phi\left(\frac{f^* - \mu(x)}{\sigma(x)}\right)$$

3. **Mean ('y')** - Pure exploitation

$$\text{acq}(x) = \mu(x)$$

## 15.5. Step: Update Repeats for Infill Points

### 15.5.1. Method: `_update_repeats_infill_points()`

Repeat the infill point for noisy function evaluation if `repeats_surrogate > 1`.

**What happens in `_update_repeats_infill_points()`:**

1. Takes the suggested next point (`x_next`)
2. If `repeats_surrogate > 1`: Creates multiple copies for repeated evaluation
3. Otherwise: Returns point as 2D array (shape:  $1 \times n_{\text{features}}$ )
4. Returns array ready for function evaluation

```
x_next_repeated = opt._update_repeats_infill_points(x_next)
print(f"Shape before repeating: {x_next.shape}")
print(f"Shape after repeating: {x_next_repeated.shape}")
print(f"Number of evaluations planned: {x_next_repeated.shape[0]}")
```

Shape before repeating: (2,)  
Shape after repeating: (1, 2)  
Number of evaluations planned: 1

## 15.6. Append OCBA Points to Infill Points

Combines OCBA selected points with the next suggested point for evaluation.

```
if X_ocba is not None:  
    x_next_repeated = np.append(X_ocba, x_next_repeated, axis=0)
```

## 15.7. Step: Evaluation of New Points

### 15.7.1. Method: `_evaluate_function()` (again)

Evaluate the objective function at the suggested points.

```
x_next_2d = x_next.reshape(1, -1)  
y_next = opt._evaluate_function(x_next_2d)  
  
print(f"Evaluated point: {x_next}")  
print(f"Function value: {y_next[0]:.6f}")  
print(f"Current best: {opt.best_y:.6f}")
```

Evaluated point: [1.13866324 0.15750508]  
Function value: 1.606003  
Current best: 16.329192

## 15.8. Step: Handle Failed Evaluations (Sequential)

### 15.8.1. Method: `_handle_NA_new_points()`

Handle NaN/inf values in new evaluations with penalty approach.

What happens in `_handle_NA_new_points()`:

1. **Apply penalty** to NaN/inf values (unlike initial design)
  - $\text{Penalty} = \max(\text{history}) + 3 \times \text{std}(\text{history}) + \text{random noise}$

## 15. Sequential Optimization Loop

2. **Remove** remaining invalid values after penalty
3. **Return None** if all evaluations failed → skip iteration
4. **Continue** if any valid evaluations exist

```
x_clean, y_clean = opt._handle_NA_new_points(x_next_2d, y_next)

if x_clean is not None:
    print(f" Valid evaluations: {len(y_clean)}")
    print(f" All values finite: {np.all(np.isfinite(y_clean))}")
else:
    print(f" All evaluations failed - iteration would be skipped")
```

```
Valid evaluations: 1
All values finite: True
```

### Why penalties in sequential phase?

- Preserves optimization history
- Allows surrogate to learn “bad” regions
- Random noise prevents identical penalties

## 15.9. Step: Update Success Rate

### 15.9.1. Method: `_update_success_rate(y0)`

Update success rate BEFORE updating storage (initial design - all should be successes since starting from scratch)

What happens in `_update_success_rate()`:

1. Calculate success rate as ratio of valid evaluations to total evaluated

```
opt._update_success_rate(y0_clean)
print(f"\nSuccess rate updated: {opt.success_rate:.2%} (valid evaluations / total evals)")
```

```
Success rate updated: 0.00% (valid evaluations / total evaluations)
```

## 15.10. Step: Update Storage

### 15.10.1. Internal updates

Add new evaluations to storage.

```
opt._update_storage(x_next_repeated, y_next)
```

## 15.11. Update Statistics

Update statistics after new evaluations.

### 15.11.1. What happens in `update_stats()`:

Always updated:

- `min_y`: Best (minimum) function value
- `min_X`: Design point with best value
- `counter`: Total number of evaluations

For noisy functions only (if `noise=True`):

- `mean_X`: Unique design points
- `mean_y`: Mean values per design point
- `var_y`: Variance per design point

```
opt.update_stats()
print(f"Basic statistics:")
print(f"  min_y: {opt.min_y:.6f}")
print(f"  min_X: {opt.min_X}")
print(f"  counter: {opt.counter}")

if opt.noise:
    print(f"\nNoise statistics:")
    print(f"  mean_X shape: {opt.mean_X.shape}")
    print(f"  mean_y shape: {opt.mean_y.shape}")
    print(f"  var_y shape: {opt.var_y.shape}")
else:
    print(f"\nNoise handling: disabled (deterministic function)")
```

## 15. Sequential Optimization Loop

```
Basic statistics:
  min_y: 1.606003
  min_X: [1.13866324 1.1705867 ]
  counter: 11
```

Noise handling: disabled (deterministic function)

### 15.12. Step: Update Best Solution

#### 15.12.1. Method: `_update_best_main_loop()`

Update the best solution if improvement found.

```
best_before = opt.best_y_
opt._update_best_main_loop(x_clean, y_clean)

print(f"Best before: {best_before:.6f}")
print(f"Best after: {opt.best_y_:.6f}")

if opt.best_y_ < best_before:
    print(f"\n New best found!")
    print(f"  Location: {opt.best_x_}")
    print(f"  Value: {opt.best_y_:.6f}")
else:
    print(f"\n Best unchanged")
```

```
Iteration 0: New best f(x) = 1.606003
Best before: 16.329192
Best after: 1.606003
```

```
New best found!
Location: [1.13866324 1.1705867 ]
Value: 1.606003
```

#### Loop Termination Conditions:

The optimization continues until:

1. `len(y_) >= max_iter` (reached evaluation budget), OR
2. `elapsed_time >= max_time` (reached time limit)

## 16. Complete Optimization Example

Now let's run a complete optimization to see all steps in action:

```
# Create fresh optimizer
opt_complete = SpotOptim(
    fun=rosenbrock,
    bounds=[(-2, 2), (-2, 2)],
    n_initial=8,
    max_iter=25,
    acquisition='ei',
    verbose=False, # Set to False for cleaner output
    seed=42
)

# Run optimization
result = opt_complete.optimize()

print(f"\nOptimization Result:")
print(f"{'='*70}")
print(f"Best point found: {result.x}")
print(f"Best value: {result.fun:.6f}")
print(f"True optimum: [1.0, 1.0]")
print(f"True minimum: 0.0")
print(f"Gap to optimum: {result.fun:.6f}")
print(f"\nFunction evaluations: {result.nfev}")
print(f"Sequential iterations: {result.nit}")
print(f"Success: {result.success}")
print(f"Message: {result.message}")
```

Optimization Result:

```
=====
Best point found: [1.08776853 1.15560806]
Best value: 0.084058
True optimum: [1.0, 1.0]
True minimum: 0.0
Gap to optimum: 0.084058
```

## *16. Complete Optimization Example*

Function evaluations: 25  
Sequential iterations: 17  
Success: True  
Message: Optimization terminated: maximum evaluations (25) reached



## 17. Noisy Functions with Repeats

When dealing with noisy objective functions, SpotOptim can evaluate each point multiple times and track statistics.

### 17.1. Configuration for Noisy Functions

```
NOISE_STD = 10.0
print("\n" + "="*70)
print("NOISY FUNCTION OPTIMIZATION")
print("="*70)
print(f"\nConfiguration for noisy optimization with noise std = {NOISE_STD}:")

# Wrapper to add noise
def rosenbrock_noisy_wrapper(X):
    return rosenbrock_noisy(X, noise_std=NOISE_STD)

opt_noisy = SpotOptim(
    fun=rosenbrock_noisy_wrapper,
    bounds=[(-2, 2), (-2, 2)],
    n_initial=6,
    max_iter=20,
    repeats_initial=3,      # Evaluate each initial point 3 times
    repeats_surrogate=2,    # Evaluate each sequential point 2 times
    verbose=False,
    seed=42
)

print("Configuration for noisy optimization:")
print(f"  repeats_initial: {opt_noisy.repeats_initial}")
print(f"  repeats_surrogate: {opt_noisy.repeats_surrogate}")
print(f"  noise: {opt_noisy.noise}")
```

## 17. Noisy Functions with Repeats

### NOISY FUNCTION OPTIMIZATION

```
Configuration for noisy optimization with noise std = 10.0:
Configuration for noisy optimization:
  repeats_initial: 3
  repeats_surrogate: 2
  noise: True
```

## 17.2. Noisy Optimization Workflow Differences

### 17.2.1. Modified Initial Design

With `repeats_initial > 1`:

```
result_noisy = opt_noisy.optimize()

print(f"\nInitial design with repeats:")
print(f"  n_initial = {opt_noisy.n_initial}")
print(f"  repeats_initial = {opt_noisy.repeats_initial}")
print(f"  Total initial evaluations: {opt_noisy.n_initial * opt_noisy.repeats_initial}")

print(f"\nStatistics tracked:")
print(f"  mean_X shape: {opt_noisy.mean_X.shape} (unique points)")
print(f"  mean_y shape: {opt_noisy.mean_y.shape} (mean values)")
print(f"  var_y shape: {opt_noisy.var_y.shape} (variances)")

print(f"\nExample statistics for first point:")
idx = 0
print(f"  Point: {opt_noisy.mean_X[idx]}")
print(f"  Mean value: {opt_noisy.mean_y[idx]:.4f}")
print(f"  Variance: {opt_noisy.var_y[idx]:.4f}")
print(f"  Std dev: {np.sqrt(opt_noisy.var_y[idx]):.4f}")
```

```
Initial design with repeats:
  n_initial = 6
  repeats_initial = 3
  Total initial evaluations: 18
```

Statistics tracked:

```
mean_X shape: (7, 2) (unique points)
mean_y shape: (7,) (mean values)
var_y shape: (7,) (variances)
```

Example statistics for first point:

```
Point: [-1.90573195 -1.79824535]
Mean value: 2960.5138
Variance: 68.6148
Std dev: 8.2834
```

### Key Differences with Noise:

1. **Repeated Evaluations:** Each point evaluated multiple times
2. **Statistics Tracking:**
  - `mean_X`: Unique evaluation locations
  - `mean_y`: Mean function values at each location
  - `var_y`: Variance of function values
  - `n_eval`: Number of evaluations per location
3. **Surrogate Fitting:** Uses `mean_y` instead of `y_`
4. **Best Selection:** Based on `mean_y` not individual `y_`

### 17.2.2. Update Statistics Method

```
print(f"After optimization:")
print(f"  Total evaluations: {len(opt_noisy.y_)}")
print(f"  Unique points: {len(opt_noisy.mean_X)}")
print(f"  Average repeats per point: {len(opt_noisy.y_) / len(opt_noisy.mean_X):.2f}")

print(f"\nVariance statistics:")
print(f"  Mean variance: {np.mean(opt_noisy.var_y):.6f}")
print(f"  Max variance: {np.max(opt_noisy.var_y):.6f}")
print(f"  Min variance: {np.min(opt_noisy.var_y):.6f}")
```

After optimization:

```
Total evaluations: 20
Unique points: 7
Average repeats per point: 2.86
```

Variance statistics:

```
Mean variance: 43.630970
Max variance: 94.990756
Min variance: 6.357449
```



## 18. Optimal Computing Budget Allocation (OCBA)

OCBA intelligently allocates additional evaluations to distinguish between competing solutions.

### 18.1. OCBA Configuration

```
print("\n" + "="*70)
print("OPTIMAL COMPUTING BUDGET ALLOCATION (OCBA)")
print("="*70)

opt_ocba = SpotOptim(
    fun=rosenbrock_noisy_wrapper,
    bounds=[(-2, 2), (-2, 2)],
    n_initial=8,
    max_iter=30,
    repeats_initial=2,
    repeats_surrogate=1,
    ocba_delta=3,          # Allocate 3 additional evaluations per iteration
    verbose=False,
    seed=42
)

print("OCBA Configuration:")
print(f"  ocba_delta: {opt_ocba.ocba_delta}")
print(f"  Purpose: Intelligently re-evaluate existing points")
print(f"  Benefit: Better distinguish between similar solutions")
```

```
=====
OPTIMAL COMPUTING BUDGET ALLOCATION (OCBA)
=====
OCBA Configuration:
```

## 18. Optimal Computing Budget Allocation (OCBA)

ocba\_delta: 3  
Purpose: Intelligently re-evaluate existing points  
Benefit: Better distinguish between similar solutions

### 18.2. OCBA Method

#### 18.2.1. Method: `_apply_ocba()`

```
result_ocba = opt_ocba.optimize()

print(f"\nOCBA applied during optimization")
print(f"  Final total evaluations: {result_ocba.nfev}")
print(f"  Expected without OCBA: ~{opt_ocba.n_initial * opt_ocba.repeats_initial + result_ocba.nfev}")
print(f"  Additional OCBA evaluations: ~{result_ocba.nit * opt_ocba.ocba_delta}")

print(f"\nOCBA intelligently allocated extra evaluations to:")
print(f"  - Current best candidate (confirm it's truly best)")
print(f"  - Close competitors (distinguish between similar solutions)")
print(f"  - High-variance points (reduce uncertainty)")
```

```
OCBA applied during optimization
  Final total evaluations: 30
  Expected without OCBA: ~27
  Additional OCBA evaluations: ~33
```

```
OCBA intelligently allocated extra evaluations to:
  - Current best candidate (confirm it's truly best)
  - Close competitors (distinguish between similar solutions)
  - High-variance points (reduce uncertainty)
```

#### OCBA Algorithm:

1. Identify best solution (lowest mean value)
2. Calculate allocation ratios based on:
  - Distance to best solution
  - Variance of each solution
3. Allocate `ocba_delta` additional evaluations
4. Returns points to re-evaluate
5. Requires: 3 points with variance  $> 0$

## 18.2. OCBA Method

**OCBA Activation Conditions:** - `noise = True` (repeats  $> 1$ ) - `ocba_delta`  $> 0$  -  
At least 3 design points exist - All points have variance  $> 0$





## 19. Handling Function Evaluation Failures

SpotOptim robustly handles functions that occasionally fail (return NaN/inf).

### 19.1. Example with Failures

```
opt_failures = SpotOptim(
    fun=rosenbrock_with_failures,
    bounds=[(-2, 2), (-2, 2)],
    n_initial=12,
    max_iter=35,
    verbose=False,
    seed=42
)

result_failures = opt_failures.optimize()

print(f"\nOptimization with ~15% random failure rate:")
print(f"  Function evaluations: {result_failures.nfev}")
print(f"  Sequential iterations: {result_failures.nit}")
print(f"  Success: {result_failures.success}")

# Count how many values are non-finite in raw evaluations
n_total = len(opt_failures.y_)
n_finite = np.sum(np.isfinite(opt_failures.y_))
print(f"\nEvaluation statistics:")
print(f"  Total evaluations: {n_total}")
print(f"  Finite values: {n_finite}")
print(f"  Note: Failed evaluations handled transparently")
```

```
Optimization with ~15% random failure rate:
  Function evaluations: 35
```

## 19. Handling Function Evaluation Failures

Sequential iterations: 27  
Success: True

Evaluation statistics:  
Total evaluations: 35  
Finite values: 35  
Note: Failed evaluations handled transparently

## 19.2. Failure Handling in Initial Design

### 19.2.1. Method: `_handle_NA_initial_design()`

```
print("Initial design failure handling:")
print("  1. Identify NaN/inf values")
print("  2. Remove invalid points entirely")
print("  3. Continue with valid points only")
print("  4. No penalties applied")
print("  5. Require at least 1 valid point")

print("\nRationale:")
print("  - Initial design should be clean")
print("  - Invalid regions identified naturally")
print("  - Surrogate trained on good data only")
```

Initial design failure handling:

1. Identify NaN/inf values
2. Remove invalid points entirely
3. Continue with valid points only
4. No penalties applied
5. Require at least 1 valid point

Rationale:

- Initial design should be clean
- Invalid regions identified naturally
- Surrogate trained on good data only

## 19.3. Failure Handling in Sequential Phase

### 19.3.1. Method: `_handle_NA_new_points()`

```
print("Sequential phase failure handling:")
print("  1. Apply penalty to NaN/inf values")
print("    - Penalty = max(history) + 3*std(history)")
print("    - Add random noise to avoid duplicates")
print("  2. Remove remaining invalid values")
print("  3. Skip iteration if all evaluations failed")
print("  4. Continue if any valid evaluations")

print("\nPenalty approach benefits:")
print("  Preserves optimization history")
print("  Surrogate learns to avoid bad regions")
print("  Better exploration-exploitation balance")
print("  More robust convergence")
```

Sequential phase failure handling:

1. Apply penalty to NaN/inf values
  - Penalty =  $\max(\text{history}) + 3 \times \text{std}(\text{history})$
  - Add random noise to avoid duplicates
2. Remove remaining invalid values
3. Skip iteration if all evaluations failed
4. Continue if any valid evaluations

Penalty approach benefits:

- Preserves optimization history
- Surrogate learns to avoid bad regions
- Better exploration-exploitation balance
- More robust convergence

## 19.4. Penalty Application

### 19.4.1. Method: `_apply_penalty_NA()`

Let's demonstrate penalty calculation:

## 19. Handling Function Evaluation Failures

```
# Simulate historical values
y_history_sim = np.array([10.0, 15.0, 8.0, 12.0, 20.0, 9.0])
y_new_sim = np.array([7.0, np.nan, 11.0, np.inf])

print("Historical values:", y_history_sim)
print("New evaluations:", y_new_sim)

# Apply penalty
y_repaired = opt_failures._apply_penalty_NA(
    y_new_sim,
    y_history=y_history_sim,
    penalty_value=None, # Compute adaptively
    sd=0.1
)

print(f"\nAfter penalty application:", y_repaired)
print(f"All finite: {np.all(np.isfinite(y_repaired))}")

# Show penalty calculation
max_hist = np.max(y_history_sim)
std_hist = np.std(y_history_sim, ddof=1)
penalty_base = max_hist + 3 * std_hist

print(f"\nPenalty calculation:")
print(f"  max(history) = {max_hist:.2f}")
print(f"  std(history) = {std_hist:.2f}")
print(f"  Base penalty = {max_hist:.2f} + 3×{std_hist:.2f} = {penalty_base:.2f}")
print(f"  Actual penalty = {penalty_base:.2f} + noise")
```

Historical values: [10. 15. 8. 12. 20. 9.]

New evaluations: [ 7. nan 11. inf]

After penalty application: [ 7. 33.62137308 11. 33.5370057 ]

All finite: True

Penalty calculation:

max(history) = 20.00

std(history) = 4.50

Base penalty = 20.00 + 3×4.50 = 33.51

Actual penalty = 33.51 + noise

## 20. Complete Method Summary

### 20.1. Methods Called During `optimize()`

#### 20.1.1. Preparation Phase

1. `get_initial_design()` - Generate/process initial sample points
2. `_curate_initial_design()` - Remove duplicates, handle repeats
3. `_evaluate_function()` - Evaluate objective function
4. `_handle_NA_initial_design()` - Remove NaN/inf from initial design
5. `_check_size_initial_design()` - Validate sufficient points
6. `_init_storage()` - Initialize storage (`X_`, `y_`, `n_iter_`)
7. `update_stats()` - Compute mean/variance for noisy functions
8. `_init_tensorboard()` - Log initial design to TensorBoard (if enabled)
9. `_get_best_xy_initial_design()` - Identify initial best

#### 20.1.2. Sequential Optimization Loop (each iteration)

10. `_fit_scheduler()` - Select data and fit surrogate based on noise handling
  - Internally calls `_transform_X()` - Transform to internal scale
  - Internally calls `_fit_surrogate()` - Fit Gaussian Process to data
11. `_apply_ocba()` - OCBA allocation (if enabled)
12. `_suggest_next_point()` - Optimize acquisition function
  - Internally calls `_acquisition_function()`
  - Internally calls `_predict_with_uncertainty()`
13. `_update_repeats_infill_points()` - Repeat suggested point for noisy functions
14. `_evaluate_function()` - Evaluate at suggested point(s)
15. `_handle_NA_new_points()` - Handle failures with penalties
  - Internally calls `_apply_penalty_NA()`
  - Internally calls `_remove_nan()`
16. `_update_success_rate()` - Update success tracking
17. `_update_storage()` - Append new evaluations to storage
  - Internally calls `_inverse_transform_X()` - Convert back to original scale

## 20. Complete Method Summary

18. `update_stats()` - Update mean/variance statistics
19. `_write_tensorboard_hparams()` - Log hyperparameters (if enabled)
20. `_write_tensorboard_scalars()` - Log scalar metrics (if enabled)
21. `_update_best_main_loop()` - Update best solution

### 20.1.3. Finalization

22. `to_all_dim()` - Expand to full dimensions (if dimensionality reduction used)
23. `_determine_termination()` - Determine termination reason
24. `_close_tensorboard_writer()` - Close logging (if enabled)
25. `_map_to_factor_values()` - Convert factors back to strings
26. Return `OptimizeResult` object

## 20.2. Helper Methods Used

- `_generate_initial_design()` - LHS generation
- `_repair_non_numeric()` - Round integer/factor variables
- `_select_new()` - Check for duplicate points
- `_handle_acquisition_failure()` - Fallback strategies
- `to_red_dim()` - Dimension reduction (if enabled)
- `_selection_dispatcher()` - Subset selection for large datasets

## 21. Termination Conditions

### 21.1. Method: `_determine_termination()`

```
print("\n" + "="*70)
print("TERMINATION CONDITIONS")
print("="*70)
print("\nMethod: _determine_termination()")
print("-" * 70)

print("Optimization terminates when:")
print("  1. len(y_) >= max_iter (evaluation budget exhausted)")
print("  2. elapsed_time >= max_time (time limit reached)")
print("  3. Whichever comes first")

print(f"\nExample from previous run:")
print(f"  Message: {result_failures.message}")
print(f"  Evaluations: {result_failures.nfev}/{opt_failures.max_iter}")
print(f"  Iterations: {result_failures.nit}")
```

```
=====
TERMINATION CONDITIONS
=====
```

```
Method: _determine_termination()
-----
```

```
Optimization terminates when:
```

1. len(y\_) >= max\_iter (evaluation budget exhausted)
2. elapsed\_time >= max\_time (time limit reached)
3. Whichever comes first

```
Example from previous run:
```

```
  Message: Optimization terminated: maximum evaluations (35) reached
  Evaluations: 35/35
  Iterations: 27
```





## 22. Performance Comparison on Noisy Functions

Let's compare standard, noisy, and noisy+OCBA optimization:

```
print("\n" + "="*70)
print("PERFORMANCE COMPARISON")
print("="*70)

# Standard optimization
opt_std = SpotOptim(
    fun=rosenbrock_noisy_wrapper,
    bounds=[(-2, 2), (-2, 2)],
    n_initial=8,
    max_iter=50,
    repeats_initial=1,
    repeats_surrogate=1,
    verbose=False,
    seed=10
)
result_std = opt_std.optimize()

# With repeats (no OCBA)
opt_rep = SpotOptim(
    fun=rosenbrock_noisy_wrapper,
    bounds=[(-2, 2), (-2, 2)],
    n_initial=8,
    max_iter=50,
    repeats_initial=2,
    repeats_surrogate=1,
    ocba_delta=0,
    verbose=False,
    seed=10
)
result_rep = opt_rep.optimize()

# With repeats + OCBA
```

## 22. Performance Comparison on Noisy Functions

```
opt_ocba_comp = SpotOptim(
    fun=rosenbrock_noisy_wrapper,
    bounds=[(-2, 2), (-2, 2)],
    n_initial=8,
    max_iter=50,
    repeats_initial=1,
    repeats_surrogate=1,
    ocba_delta=1,
    verbose=False,
    seed=10
)
result_ocba_comp = opt_ocba_comp.optimize()

# Evaluate the found optima with the TRUE (noise-free) Rosenbrock function
# to get the actual quality of the solutions
print(f"\n" + "="*70)
print("TRUE FUNCTION VALUE EVALUATION")
print("="*70)
print("\nRe-evaluating found optima with noise-free Rosenbrock function:")
print("-" * 70)

# Evaluate each solution with the TRUE function (no noise)
true_val_std = rosenbrock(result_std.x.reshape(1, -1))[0]
true_val_rep = rosenbrock(result_rep.x.reshape(1, -1))[0]
true_val_ocba = rosenbrock(result_ocba_comp.x.reshape(1, -1))[0]

print(f"\nStandard (no repeats):")
print(f"  Found x: {result_std.x}")
print(f"  Noisy value (from optimization): {result_std.fun:.6f}")
print(f"  TRUE value (noise-free): {true_val_std:.6f}")

print(f"\nWith repeats (no OCBA):")
print(f"  Found x: {result_rep.x}")
print(f"  Noisy value (from optimization): {result_rep.fun:.6f}")
print(f"  TRUE value (noise-free): {true_val_rep:.6f}")

print(f"\nWith repeats + OCBA:")
print(f"  Found x: {result_ocba_comp.x}")
print(f"  Noisy value (from optimization): {result_ocba_comp.fun:.6f}")
print(f"  TRUE value (noise-free): {true_val_ocba:.6f}")

print(f"\n" + "="*70)
print("PERFORMANCE COMPARISON SUMMARY")
print("="*70)
```

```

print(f"\nComparison on noisy Rosenbrock (={NOISE_STD}):")
print(f"{'Method':<30} {'Noisy Value':<15} {'TRUE Value':<15} {'Evaluations':<15}")
print(f"{'-'*75}")
print(f"{'Standard (no repeats)':<30} {result_std.fun:<15.6f} {true_val_std:<15.6f} {result_std.nfev:<15}")
print(f"{'With repeats (no OCBA)':<30} {result_rep.fun:<15.6f} {true_val_rep:<15.6f} {result_rep.nfev:<15}")
print(f"{'With repeats + OCBA':<30} {result_ocba_comp.fun:<15.6f} {true_val_ocba:<15.6f} {result_ocba_comp.nfev:<15}")
print(f"{'-'*75}")
print(f"{'True optimum':<30} {'0.0':<15} {'0.0':<15}")

print("\nKey observations:")
print(" • Noisy values (from optimization) are misleading due to noise")
print(" • TRUE values show actual solution quality")
print(" • Repeats help find better solutions despite noise")
print(" • OCBA focuses evaluations on distinguishing best solutions")
print(" • More evaluations generally lead to better TRUE performance")

```

---

#### PERFORMANCE COMPARISON

---



---

#### TRUE FUNCTION VALUE EVALUATION

---

Re-evaluating found optima with noise-free Rosenbrock function:

---

Standard (no repeats):

Found x: [-0.02303646 -0.12724366]  
 Noisy value (from optimization): -14.055814  
 TRUE value (noise-free): 2.679232

With repeats (no OCBA):

Found x: [-0.18461315 0.02269759]  
 Noisy value (from optimization): -12.282442  
 TRUE value (noise-free): 1.416269

With repeats + OCBA:

Found x: [-0.06715147 -0.03562076]  
 Noisy value (from optimization): -16.585110  
 TRUE value (noise-free): 1.299855

---

## 22. Performance Comparison on Noisy Functions

### PERFORMANCE COMPARISON SUMMARY

=====

Comparison on noisy Rosenbrock (=10.0):

| Method                 | Noisy Value | TRUE Value | Evaluations |
|------------------------|-------------|------------|-------------|
| -----                  |             |            |             |
| Standard (no repeats)  | -14.055814  | 2.679232   | 50          |
| With repeats (no OCBA) | -12.282442  | 1.416269   | 50          |
| With repeats + OCBA    | -16.585110  | 1.299855   | 50          |
| -----                  |             |            |             |
| True optimum           | 0.0         | 0.0        |             |

Key observations:

- Noisy values (from optimization) are misleading due to noise
- TRUE values show actual solution quality
- Repeats help find better solutions despite noise
- OCBA focuses evaluations on distinguishing best solutions
- More evaluations generally lead to better TRUE performance

## 23. Summary

### 23.1. Complete Workflow Diagram

#### SPOTOPTIM WORKFLOW

##### INITIALIZATION PHASE

```
get_initial_design()
    Generate LHS or process user design

_curate_initial_design()
    Remove duplicates, add repeats

_evaluate_function()
    Evaluate objective function

_handle_NA_initial_design()
    Remove NaN/inf points

_check_size_initial_design()
    Validate sufficient points

_init_storage()
    Initialize X_, y_, n_iter_

update_stats()
    Compute mean/variance (if noise)

_init_tensorboard() [if enabled]
    Log initial design to TensorBoard

_get_best_xy_initial_design()
    Identify initial best
```

### 23. Summary

SEQUENTIAL OPTIMIZATION LOOP (until max\_iter or max\_time)

```
_fit_scheduler()
    _transform_X() - Transform to internal scale
    _fit_surrogate() - Fit GP to current data

_apply_ocba() [if enabled]
    Allocate additional evaluations

_suggest_next_point()
    _acquisition_function()
    _predict_with_uncertainty()
    Optimize to find next point

_update_repeats_infill_points()
    Repeat point if repeats_surrogate > 1

_evaluate_function()
    Evaluate at suggested point(s)

_handle_NA_new_points()
    _apply_penalty_NA()
    _remove_nan()

_update_success_rate()
    Track evaluation success

_update_storage()
    Append new evaluations (with scale conversion)

update_stats()
    Update mean/variance

_write_tensorboard_hparams() [if enabled]
    Log hyperparameters

_write_tensorboard_scalars() [if enabled]
    Log scalar metrics

_update_best_main_loop()
    Update best if improved
```

FINALIZATION

```

to_all_dim() [if dimensionality reduction used]
    Expand to full dimensions

_determine_termination()
    Set termination message

_close_tensorboard_writer() [if enabled]
    Close TensorBoard logging

_map_to_factor_values() [if factors used]
    Convert factors back to strings

Return OptimizeResult

```

## 23.2. Key Concepts

### 23.2.1. 1. Initial Design

- **Latin Hypercube Sampling:** Space-filling design for efficient exploration
- **Curation:** Remove duplicates from integer/factor rounding
- **Failure Handling:** Remove invalid points, no penalties

### 23.2.2. 2. Surrogate Model

- **Default:** Gaussian Process with Matérn kernel
- **Provides:** Mean  $\hat{y}(x)$  and uncertainty  $\hat{\sigma}(x)$  predictions
- **Purpose:** Learn function landscape with limited evaluations

### 23.2.3. 3. Acquisition Function

- **EI:** Expected Improvement (default) - best balance
- **PI:** Probability of Improvement - more exploitative
- **Mean:** Pure exploitation of surrogate predictions

### 23.2.4. 4. Noise Handling

- **Repeats:** Evaluate each point multiple times
- **Statistics:** Track mean and variance
- **OCBA:** Intelligently allocate additional evaluations

## 23. Summary

### 23.2.5. 5. Failure Handling

- **Initial Phase:** Remove invalid points
- **Sequential Phase:** Apply adaptive penalties with noise
- **Robustness:** Continue optimization despite failures

## 23.3. Best Practices

1. **For deterministic functions:**
  - Use default settings (no repeats)
  - Acquisition = 'ei'
  - Focus on `n_initial` and `max_iter`
2. **For noisy functions:**
  - Set `repeats_initial` = 2
  - Set `repeats_surrogate` = 1
  - Consider OCBA with `ocba_delta` = 2
3. **For unreliable functions:**
  - SpotOptim handles failures automatically
  - No special configuration needed
  - Penalties guide search away from bad regions
4. **For expensive functions:**
  - Increase `n_initial` (better initial model)
  - Use 'ei' acquisition (best sample efficiency)
  - Consider `max_time` limit

## 23.4. Conclusion

SpotOptim provides a robust and flexible framework for surrogate-based optimization with:

Efficient space-filling initial designs (LHS)  
Powerful Gaussian Process surrogate models  
Smart acquisition functions (EI, PI, Mean)  
Automatic noise handling with statistics  
Intelligent budget allocation (OCBA)  
Robust failure handling  
Comprehensive progress tracking

The modular design allows easy customization while maintaining robust defaults for most use cases.



## 24. SpotOptim Internal Methods Examples

### 24.1. Setup: Import Required Packages

First, we import all necessary packages for the examples.

```
# Core imports
import numpy as np
import time
from scipy.optimize import OptimizeResult

# SpotOptim
from spotoptim import SpotOptim

# Surrogate models
from sklearn.gaussian_process import GaussianProcessRegressor
from sklearn.gaussian_process.kernels import RBF, ConstantKernel

# Set random seed for reproducibility
np.random.seed(42)

print("All packages imported successfully!")
```

All packages imported successfully!

### 24.2. 1. Main Optimization Method: `optimize()`

The `optimize()` method is the main entry point for running the optimization process. It coordinates all other methods in the optimization workflow:

1. **Initial Design Phase:** `get_initial_design()`, `_curate_initial_design()`, `_handle_NA_initial_design()`, `_check_size_initial_design()`, `_get_best_xy_initial_design()`
2. **Main Loop:** Surrogate fitting, OCBA application, point suggestion, evaluation
3. **Termination:** `_determine_termination()`

## 24. SpotOptim Internal Methods Examples

Let's see a complete optimization example:

```
# Define a simple quadratic function
def sphere(X):
    """Sphere function:  $f(x) = \sum(x^2)$ """
    X = np.atleast_2d(X)
    return np.sum(X**2, axis=1)

# Create optimizer
opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    n_initial=5,
    max_iter=20,
    verbose=True
)

# Run optimization
result = opt.optimize()

print(f"\nBest point found: {result.x}")
print(f"Best value: {result.fun:.6f}")
print(f"Total evaluations: {result.nfev}")
print(f"Sequential iterations: {result.nit}")
print(f"Success: {result.success}")
print(f"Message: {result.message}")
```

```
TensorBoard logging disabled
Initial best:  $f(x) = 7.852365$ 
Iteration 1:  $f(x) = 26.076731$ 
Iteration 2:  $f(x) = 11.151488$ 
Iteration 3: New best  $f(x) = 7.138214$ 
Iteration 4: New best  $f(x) = 3.009422$ 
Iteration 5: New best  $f(x) = 0.617891$ 
Iteration 6: New best  $f(x) = 0.073844$ 
Iteration 7: New best  $f(x) = 0.005029$ 
Iteration 8: New best  $f(x) = 0.001957$ 
Iteration 9:  $f(x) = 0.002422$ 
Iteration 10: New best  $f(x) = 0.000011$ 
Iteration 11: New best  $f(x) = 0.000000$ 
Iteration 12:  $f(x) = 0.000000$ 
Iteration 13: New best  $f(x) = 0.000000$ 
Iteration 14:  $f(x) = 0.000000$ 
Iteration 15:  $f(x) = 0.000000$ 
```

```

Best point found: [ 8.82207046e-05 -3.76165030e-04]
Best value: 0.000000
Total evaluations: 20
Sequential iterations: 15
Success: True
Message: Optimization terminated: maximum evaluations (20) reached

```

## 24.3. 2. Initial Design Methods

These methods handle the creation and validation of the initial design of experiments.

### 24.3.1. 2.1 `get_initial_design()`

**Purpose:** Generate or process initial design points

**Used by:** `optimize()` method at the start

**Calls:** `_generate_initial_design()` if `X0` is None

This method handles three scenarios: 1. Generate LHS design when `X0=None` 2. Include starting point `x0` if provided 3. Transform user-provided `X0`

```

# Example 1: Generate default LHS design
opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    n_initial=10,
    seed=42
)
X0 = opt.get_initial_design()
print(f"Generated LHS design shape: {X0.shape}")
print(f"First 3 points:\n{X0[:3]}")

# Example 2: With starting point x0
opt_x0 = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    n_initial=10,
    x0=[0.0, 0.0],
    seed=42
)
X0_with_x0 = opt_x0.get_initial_design()
print(f"\nDesign with x0, first point: {X0_with_x0[0]}")

```

## 24. SpotOptim Internal Methods Examples

```
# Example 3: Provide custom initial design
X0_custom = np.array([[0, 0], [1, 1], [2, 2]])
X0_processed = opt.get_initial_design(X0_custom)
print(f"\nCustom design shape: {X0_processed.shape}")
```

Generated LHS design shape: (10, 2)

First 3 points:

```
[[-2.77395605  1.56112156]
 [ 4.14140208 -1.69736803]
 [-4.09417735  3.02437765]]
```

Design with x0, first point: [0. 0.]

Custom design shape: (3, 2)

### 24.3.2. 2.2 \_curate\_initial\_design()

**Purpose:** Remove duplicates and ensure sufficient unique points

**Used by:** optimize() after get\_initial\_design()

**Handles:** Duplicate removal, point generation, repetition for noisy functions

```
# Example 1: Remove duplicates
opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    n_initial=10,
    seed=42
)
X0_with_dups = np.array([[1, 2], [1, 2], [3, 4], [3, 4], [5, 6]])
X0_curated = opt._curate_initial_design(X0_with_dups)
print(f"Original points: {len(X0_with_dups)}")
print(f"After removing duplicates: {len(X0_curated)}")

# Example 2: With repeats for noisy functions
opt_repeat = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    n_initial=5,
    repeats_initial=3, # Repeat each point 3 times
    seed=42
)
X0 = np.array([[1, 2], [3, 4], [5, 6], [7, 8], [9, 10]])
```

```
X0_repeated = opt_repeat._curate_initial_design(X0)
print(f"\nOriginal points: {len(X0)}")
print(f"After repeating (3x): {len(X0_repeated)}")
```

```
Original points: 5
After removing duplicates: 10
```

```
Original points: 5
After repeating (3x): 15
```

### 24.3.3. 2.3 \_handle\_NA\_initial\_design()

**Purpose:** Remove NaN/inf values from initial design evaluations

**Used by:** `optimize()` after evaluating initial design

**Returns:** Cleaned arrays and original count

```
opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    n_initial=10,
    seed=42
)

# Simulate initial design with some NaN values
X0 = np.array([[1, 2], [3, 4], [5, 6]])
y0 = np.array([5.0, np.nan, np.inf])

X0_clean, y0_clean, n_eval = opt._handle_NA_initial_design(X0, y0)

print(f"Original evaluations: {n_eval}")
print(f"Valid points remaining: {X0_clean.shape[0]}")
print(f"Clean X0:\n{X0_clean}")
print(f"Clean y0: {y0_clean}")
```

```
Original evaluations: 3
Valid points remaining: 1
Clean X0:
[[1 2]]
Clean y0: [5.]
```

#### 24.3.4. 2.4 \_check\_size\_initial\_design()

**Purpose:** Validate sufficient points for surrogate fitting

**Used by:** optimize() after handling NaN values

**Raises:** ValueError if insufficient points

```
opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    n_initial=10,
    seed=42
)

# Example 1: Sufficient points - no error
y0_sufficient = np.array([1.0, 2.0, 3.0, 4.0, 5.0])
try:
    opt._check_size_initial_design(y0_sufficient, n_evaluated=10)
    print(" Sufficient points - validation passed")
except ValueError as e:
    print(f" Error: {e}")

# Example 2: Insufficient points - raises error
y0_insufficient = np.array([1.0]) # Only 1 point, need at least 3 for 2D
try:
    opt._check_size_initial_design(y0_insufficient, n_evaluated=10)
    print(" Validation passed")
except ValueError as e:
    print(f" Expected error: {e}")
```

Sufficient points - validation passed

Expected error: Insufficient valid initial design points: only 1 finite value(s) out

#### 24.3.5. 2.5 \_get\_best\_xy\_initial\_design()

**Purpose:** Determine and store the best point from initial design

**Used by:** optimize() after initial design evaluation

**Updates:** self.best\_x\_ and self.best\_y\_ attributes

```
opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    n_initial=5,
    verbose=True,
```

```

    seed=42
)

# Simulate initial design (normally done in optimize())
opt.X_ = np.array([[1, 2], [0, 0], [2, 1]])
opt.y_ = np.array([5.0, 0.0, 5.0])

opt._get_best_xy_initial_design()

print(f"\nBest x from initial design: {opt.best_x_}")
print(f"Best y from initial design: {opt.best_y_}")

```

TensorBoard logging disabled  
Initial best: f(x) = 0.000000

Best x from initial design: [0 0]  
Best y from initial design: 0.0

## 24.4. 3. Surrogate Model Methods

These methods handle surrogate model operations during the optimization loop.

### 24.4.1. 3.1 `_fit_surrogate()`

**Purpose:** Fit surrogate model to data

**Used by:** `optimize()` in main loop

**Calls:** `_selection_dispatcher()` if `max_surrogate_points` exceeded

```

opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    max_surrogate_points=10,
    seed=42
)

# Generate some training data
X = np.random.rand(50, 2) * 10 - 5 # 50 points in [-5, 5]
y = np.sum(X**2, axis=1)

# Fit surrogate (will select 10 best points)
opt._fit_surrogate(X, y)

```

## 24. SpotOptim Internal Methods Examples

```
print(f"Surrogate fitted successfully!")
print(f"Surrogate model: {type(opt.surrogate).__name__}")
```

```
Surrogate fitted successfully!
Surrogate model: GaussianProcessRegressor
```

### 24.4.2. 3.2 `_predict_with_uncertainty()`

**Purpose:** Predict with uncertainty estimates, handling surrogates without `return_std`

**Used by:** `_acquisition_function()` and `plot_surrogate()`

**Returns:** Predictions and standard deviations

```
opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    surrogate=GaussianProcessRegressor(),
    seed=42
)

# Train surrogate
X_train = np.array([[0, 0], [1, 1], [2, 2]])
y_train = np.array([0, 2, 8])
opt._fit_surrogate(X_train, y_train)

# Predict on new points
X_test = np.array([[1.5, 1.5], [3.0, 3.0]])
preds, stds = opt._predict_with_uncertainty(X_test)

print(f"Test points:\n{X_test}")
print(f"Predictions: {preds}")
print(f"Uncertainties: {stds}")
```

```
Test points:
[[1.5 1.5]
 [3.  3. ]]
Predictions: [5.66013019 3.0829763 ]
Uncertainties: [0.31263595 0.92017596]
```

### 24.4.3. 3.3 `_acquisition_function()`

**Purpose:** Compute acquisition function value

**Used by:** `_suggest_next_point()` for optimization



**Calls:** `_predict_with_uncertainty()`

**Supports:** Expected Improvement (EI), Probability of Improvement (PI), Mean prediction

```
opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    surrogate=GaussianProcessRegressor(),
    acquisition='ei',
    seed=42
)

# Setup
X_train = np.array([[0, 0], [1, 1], [2, 2]])
y_train = np.array([0, 2, 8])
opt._fit_surrogate(X_train, y_train)
opt.y_ = y_train # Needed for acquisition function

# Evaluate acquisition function
x_eval = np.array([1.5, 1.5])
acq_value = opt._acquisition_function(x_eval)

print(f"Point to evaluate: {x_eval}")
print(f"Acquisition function value (EI): {acq_value:.6f}")
print(f"(Lower is better for minimization)")
```

```
Point to evaluate: [1.5 1.5]
Acquisition function value (EI): -0.000000
(Lower is better for minimization)
```

#### 24.4.4. 3.4 `_suggest_next_point()`

**Purpose:** Suggest next point to evaluate using acquisition function optimization

**Used by:** `optimize()` in main loop

**Calls:** `_acquisition_function()`, `_handle_acquisition_failure()` if needed

**Handles:** Integer/factor rounding, duplicate avoidance

```
opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    surrogate=GaussianProcessRegressor(),
    acquisition='ei',
    seed=42
```

```

)

# Setup
X_train = np.array([[0, 0], [1, 1], [2, 2]])
y_train = np.array([0, 2, 8])
opt._fit_surrogate(X_train, y_train)
opt.X_ = X_train
opt.y_ = y_train

# Suggest next point
x_next = opt._suggest_next_point()

print(f"Next point to evaluate: {x_next}")
print(f"Expected to be between known points or in unexplored regions")

```

```

Next point to evaluate: [-1.6288683  1.99062025]
Expected to be between known points or in unexplored regions

```

## 24.5. 4. OCBA Method (Optimal Computing Budget Allocation)

### 24.5.1. 4.1 \_apply\_ocba()

**Purpose:** Apply OCBA for noisy functions to determine which points to re-evaluate

**Used by:** optimize() in main loop when noise=True and ocba\_delta > 0

**Returns:** Points to re-evaluate or None

```

# Example with OCBA enabled
opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    repeats_initial=2, # Enable noise handling
    ocba_delta=5,
    verbose=True,
    seed=42
)

# Simulate optimization state
opt.mean_X = np.array([[1, 2], [0, 0], [2, 1], [1, 1]])
opt.mean_y = np.array([5.0, 0.1, 5.0, 2.0])
opt.var_y = np.array([0.1, 0.05, 0.15, 0.08])

```

```

X_ocba = opt._apply_ocba()

if X_ocba is not None:
    print(f"\nOCBA returned {X_ocba.shape[0]} points for re-evaluation")
else:
    print("\nOCBA: No re-evaluation needed")

```

```

TensorBoard logging disabled
OCBA: Adding 5 re-evaluation(s)

```

```

OCBA returned 5 points for re-evaluation

```

## 24.6. 5. NaN/Inf Handling Methods

### 24.6.1. 5.1 \_apply\_penalty\_NA()

**Purpose:** Replace NaN and infinite values with penalty plus random noise

**Used by:** `_handle_NA_new_points()` and indirectly by `optimize()`

**Algorithm:**  $\text{penalty} = \max(\text{finite\_y}) + 3 * \text{std}(\text{finite\_y}) + \text{noise}$

```

opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5)],
    seed=42
)

# Historical values (used for penalty computation)
y_hist = np.array([1.0, 2.0, 3.0, 5.0])

# New evaluations with NaN/inf
y_new = np.array([4.0, np.nan, np.inf])

y_clean = opt._apply_penalty_NA(y_new, y_history=y_hist)

print(f"Original: {y_new}")
print(f"After penalty: {y_clean}")
print(f"All finite: {np.all(np.isfinite(y_clean))}")
print(f"Penalty values > max(hist): {y_clean[1] > np.max(y_hist)}")

```

```

Original: [ 4. nan inf]

```

```

After penalty: [ 4.          9.97584134 10.24603729]

```

## 24. SpotOptim Internal Methods Examples

All finite: True  
Penalty values > max(hist): True

### 24.6.2. 5.2 \_remove\_nan()

**Purpose:** Remove rows where y contains NaN or inf values  
**Used by:** optimize() after function evaluations  
**Returns:** Cleaned arrays

```
opt = SpotOptim(  
    fun=sphere,  
    bounds=[(-5, 5)],  
    seed=42  
)  
  
X = np.array([[1, 2], [3, 4], [5, 6]])  
y = np.array([1.0, np.nan, np.inf])  
  
X_clean, y_clean = opt._remove_nan(X, y, stop_on_zero_return=False)  
  
print(f"Original X shape: {X.shape}")  
print(f"Clean X shape: {X_clean.shape}")  
print(f"Clean X:\n{X_clean}")  
print(f"Clean y: {y_clean}")
```

```
Original X shape: (3, 2)  
Clean X shape: (1, 2)  
Clean X:  
[[1 2]]  
Clean y: [1.]
```

### 24.6.3. 5.3 \_handle\_NA\_new\_points()

**Purpose:** Handle NaN/inf values in new evaluation points during main loop  
**Used by:** optimize() after evaluating new points  
**Calls:** \_apply\_penalty\_NA() and \_remove\_nan()  
**Returns:** None, None if all evaluations invalid (skip iteration)

```
opt = SpotOptim(  
    fun=sphere,  
    bounds=[(-5, 5), (-5, 5)],  
    n_initial=5,
```

```

        verbose=True,
        seed=42
    )

    # Simulate optimization state
    opt.y_ = np.array([1.0, 2.0, 3.0]) # Historical values
    opt.n_iter_ = 1

    # Case 1: Some valid values
    print("Case 1: Some valid evaluations")
    x_next = np.array([[1, 2], [3, 4], [5, 6]])
    y_next = np.array([5.0, np.nan, 10.0])
    x_clean, y_clean = opt._handle_NA_new_points(x_next, y_next)
    print(f"Valid points remaining: {x_clean.shape[0] if x_clean is not None else 0}")

    # Case 2: All invalid - should return None
    print("\nCase 2: All invalid evaluations")
    x_all_bad = np.array([[1, 2], [3, 4]])
    y_all_bad = np.array([np.nan, np.inf])
    x_clean, y_clean = opt._handle_NA_new_points(x_all_bad, y_all_bad)
    print(f"Result: {'None (skip iteration)' if x_clean is None else 'Valid points'}")

```

TensorBoard logging disabled

Case 1: Some valid evaluations

Warning: Found 1 NaN/inf value(s), replacing with adaptive penalty (max + 3\*std = 6.0000)

Valid points remaining: 3

Case 2: All invalid evaluations

Warning: Found 2 NaN/inf value(s), replacing with adaptive penalty (max + 3\*std = 6.0000)

Result: Valid points

## 24.7. 6. Main Loop Update Methods

### 24.7.1. 6.1 \_update\_best\_main\_loop()

**Purpose:** Update best solution found during main optimization loop

**Used by:** optimize() after each iteration

**Updates:** self.best\_x\_ and self.best\_y\_ if improvement found

```

opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],

```

```

    n_initial=5,
    verbose=True,
    seed=42
)

# Simulate optimization state
opt.n_iter_ = 1
opt.best_x_ = np.array([1.0, 1.0])
opt.best_y_ = 2.0

# Case 1: New best found
print("Case 1: New best found")
x_new = np.array([[0.1, 0.1], [0.5, 0.5]])
y_new = np.array([0.02, 0.5])
opt._update_best_main_loop(x_new, y_new)
print(f"Updated best_y: {opt.best_y_}\n")

# Case 2: No improvement
print("Case 2: No improvement")
opt.n_iter_ = 2
x_no_improve = np.array([[1.5, 1.5]])
y_no_improve = np.array([4.5])
opt._update_best_main_loop(x_no_improve, y_no_improve)
print(f"Best_y unchanged: {opt.best_y_}")

```

```

TensorBoard logging disabled
Case 1: New best found
Iteration 1: New best f(x) = 0.020000
Updated best_y: 0.02

Case 2: No improvement
Iteration 2: f(x) = 4.500000
Best_y unchanged: 0.02

```

## 24.8. 7. Termination Method

### 24.8.1. 7.1 \_determine\_termination()

**Purpose:** Determine termination reason for optimization

**Used by:** optimize() at the end

**Checks:** Max iterations, time limit, or successful completion

```

opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=20,
    max_time=10.0,
    seed=42
)

# Case 1: Maximum evaluations reached
print("Case 1: Maximum evaluations reached")
opt.y_ = np.zeros(20)
start_time = time.time()
msg = opt._determine_termination(start_time)
print(f"Message: {msg}\n")

# Case 2: Time limit exceeded (simulated)
print("Case 2: Time limit exceeded")
opt.y_ = np.zeros(10)
start_time = time.time() - 700 # 11.67 minutes ago
msg = opt._determine_termination(start_time)
print(f"Message: {msg}\n")

# Case 3: Successful completion
print("Case 3: Successful completion")
opt.y_ = np.zeros(10)
start_time = time.time()
msg = opt._determine_termination(start_time)
print(f"Message: {msg}")

```

Case 1: Maximum evaluations reached

Message: Optimization terminated: maximum evaluations (20) reached

Case 2: Time limit exceeded

Message: Optimization terminated: time limit (10.00 min) reached

Case 3: Successful completion

Message: Optimization finished successfully

## 24.9. 8. Utility Methods

### 24.9.1. 8.1 \_select\_new()

**Purpose:** Select rows from A that are not in X (avoid duplicate evaluations)

**Used by:** `_suggest_next_point()` to ensure new points are different from evaluated points

```
opt = SpotOptim(  
    fun=sphere,  
    bounds=[(-5, 5)],  
    seed=42  
)  
  
A = np.array([[1, 2], [3, 4], [5, 6]])  
X = np.array([[3, 4], [7, 8]])  
  
new_A, is_new = opt._select_new(A, X)  
  
print(f"Candidate points A:\n{A}")  
print(f"\nKnown points X:\n{X}")  
print(f"\nNew points from A:\n{new_A}")  
print(f"\nIs new mask: {is_new}")
```

Candidate points A:

```
[[1 2]  
 [3 4]  
 [5 6]]
```

Known points X:

```
[[3 4]  
 [7 8]]
```

New points from A:

```
[[1 2]  
 [5 6]]
```

Is new mask: [ True False True]

### 24.9.2. 8.2 \_repair\_non\_numeric()

**Purpose:** Round non-numeric values (int, factor) to integers

**Used by:** Various methods to ensure integer/factor variables have valid values



```

opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    var_type=['int', 'float'],
    seed=42
)

X = np.array([[1.2, 2.5], [3.7, 4.1], [5.9, 6.8]])
X_repaired = opt._repair_non_numeric(X.copy(), opt.var_type)

print(f"Original X:\n{X}")
print(f"\nRepaired X (first column rounded to int):\n{X_repaired}")

```

Original X:

```

[[1.2 2.5]
 [3.7 4.1]
 [5.9 6.8]]

```

Repaired X (first column rounded to int):

```

[[1. 2.5]
 [4. 4.1]
 [6. 6.8]]

```

### 24.9.3. 8.3 \_map\_to\_factor\_values()

**Purpose:** Map internal integer values to original factor strings

**Used by:** optimize() when preparing results for user

**Handles:** Factor (categorical) variables

```

opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), ('red', 'green', 'blue')],
    var_type=['float', 'factor'],
    seed=42
)

# Internal representation (integers for factors)
X_internal = np.array([[1.0, 0], [2.0, 1], [3.0, 2]])
X_mapped = opt._map_to_factor_values(X_internal)

print(f"Internal representation:\n{X_internal}")
print(f"\nMapped to factor values:\n{X_mapped}")

```

## 24. SpotOptim Internal Methods Examples

Internal representation:

```
[[1. 0.]  
 [2. 1.]  
 [3. 2.]]
```

Mapped to factor values:

```
[[1.0 'red']  
 [2.0 'green']  
 [3.0 'blue']]
```

## 24.10. 9. Integration Examples

### 24.10.1. 9.1 Optimization with Custom Initial Design

```
opt = SpotOptim(  
    fun=sphere,  
    bounds=[(-5, 5), (-5, 5)],  
    max_iter=15,  
    seed=42,  
    verbose=True  
)  
  
# Provide custom initial design  
X0 = np.array([[0, 0], [1, 1], [2, 2], [-1, -1], [-2, -2]])  
result = opt.optimize(X0=X0)  
  
print(f"\nBest point: {result.x}")  
print(f"Best value: {result.fun:.6f}")
```

```
TensorBoard logging disabled  
Note: Initial design size (5) is smaller than requested (10) due to NaN/inf values  
Initial best: f(x) = 0.000000  
Iteration 1: f(x) = 0.000000  
Iteration 2: f(x) = 0.000000  
Iteration 3: f(x) = 0.000000  
    Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 4: f(x) = 7.878776  
Iteration 5: f(x) = 0.000000  
Iteration 6: f(x) = 0.099268  
Iteration 7: f(x) = 0.003575
```

Iteration 8:  $f(x) = 0.003187$   
 Iteration 9:  $f(x) = 0.000019$   
 Iteration 10:  $f(x) = 0.000001$

Best point: [0 0]  
 Best value: 0.000000

### 24.10.2. 9.2 Optimization with Starting Point

```
opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=15,
    n_initial=5,
    x0=[0.5, 0.5], # Starting point near optimum
    seed=42,
    verbose=True
)

result = opt.optimize()

print(f"\nBest point: {result.x}")
print(f"Best value: {result.fun:.6f}")
```

Starting point x0 validated and processed successfully.  
 Original scale: [0.5 0.5]  
 Internal scale: [0.5 0.5]  
 TensorBoard logging disabled  
 Including starting point x0 in initial design as first evaluation.  
 Initial best:  $f(x) = 0.500000$   
 Iteration 1:  $f(x) = 0.691041$   
 Iteration 2: New best  $f(x) = 0.149351$   
 Iteration 3: New best  $f(x) = 0.015439$   
 Iteration 4: New best  $f(x) = 0.000530$   
 Iteration 5: New best  $f(x) = 0.000070$   
 Iteration 6: New best  $f(x) = 0.000004$   
 Iteration 7: New best  $f(x) = 0.000003$   
 Iteration 8: New best  $f(x) = 0.000003$   
 Iteration 9: New best  $f(x) = 0.000003$   
 Iteration 10:  $f(x) = 0.000003$

Best point: [0.00169077 0.0001854 ]  
 Best value: 0.000003

**24.10.3. 9.3 Optimization with Integer Variables**

```

opt = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    var_type=['int', 'int'],
    max_iter=20,
    n_initial=10,
    seed=42,
    verbose=True
)

result = opt.optimize()

print(f"\nBest point: {result.x}")
print(f"Best value: {result.fun:.6f}")
print(f"Point has integers: {np.all(result.x == np.round(result.x))}")

```

```

TensorBoard logging disabled
Initial best: f(x) = 4.000000
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 1: f(x) = 29.000000
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 2: New best f(x) = 1.000000
Iteration 3: New best f(x) = 0.000000
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 4: f(x) = 25.000000
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 5: f(x) = 9.000000
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 6: f(x) = 5.000000
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 7: f(x) = 13.000000
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 8: f(x) = 13.000000
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback

```

```

Acquisition failure: Using random space-filling design as fallback.
Iteration 9: f(x) = 10.000000
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 10: f(x) = 8.000000

Best point: [ 0. -0.]
Best value: 0.000000
Point has integers: True

```

#### 24.10.4. 9.4 Noisy Function Optimization with OCBA

```

# Noisy sphere function
def noisy_sphere(X):
    X = np.atleast_2d(X)
    return np.sum(X**2, axis=1) + np.random.normal(0, 0.1, X.shape[0])

opt = SpotOptim(
    fun=noisy_sphere,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=25,
    n_initial=10,
    repeats_initial=2, # Enable noise handling
    ocba_delta=5, # Enable OCBA with 5 re-evaluations
    seed=42,
    verbose=True
)

result = opt.optimize()

print(f"\nBest point: {result.x}")
print(f"Best value: {result.fun:.6f}")
print(f>Note: OCBA re-evaluated promising points to reduce uncertainty")

```

```

TensorBoard logging disabled
Initial best: f(x) = 2.471535, mean best: f(x) = 2.526017
  OCBA: Adding 5 re-evaluation(s)
Iteration 1: New best f(x) = 2.375017, mean best: f(x) = 2.375017

Best point: [-1.5514404  0.10417656]
Best value: 2.375017
Note: OCBA re-evaluated promising points to reduce uncertainty

```

**24.10.5. 9.5 Optimization with NaN Handling**

```

# Function that sometimes returns NaN
def sometimes_nan(X):
    X = np.atleast_2d(X)
    y = np.sum(X**2, axis=1)
    # Return NaN for large values (penalty will be applied)
    y[y > 100] = np.nan
    return y

opt = SpotOptim(
    fun=sometimes_nan,
    bounds=[(-10, 10), (-10, 10)],
    max_iter=20,
    n_initial=10,
    seed=42,
    verbose=True
)

result = opt.optimize()

print(f"\nBest point: {result.x}")
print(f"Best value: {result.fun:.6f}")
print(f"Optimization succeeded despite NaN values: {result.success}")

```

```

TensorBoard logging disabled
Warning: 1 initial design point(s) returned NaN/inf and will be ignored (reduced from
Note: Initial design size (9) is smaller than requested (10) due to NaN/inf values
Initial best: f(x) = 9.683226
Iteration 1: New best f(x) = 8.496828
Iteration 2: New best f(x) = 4.452328
Iteration 3: New best f(x) = 0.174337
Iteration 4: New best f(x) = 0.038871
Iteration 5: New best f(x) = 0.000492
Iteration 6: f(x) = 0.000649
Iteration 7: New best f(x) = 0.000114
Iteration 8: New best f(x) = 0.000003
Iteration 9: New best f(x) = 0.000002
Iteration 10: f(x) = 0.000002
Iteration 11: f(x) = 0.000002

Best point: [-0.00082295 -0.00113022]
Best value: 0.000002
Optimization succeeded despite NaN values: True

```

## 24.11. 10. Method Relationship Diagram

Here's how all the methods relate to each other in the optimization workflow:

```
optimize()
```

```
  Initial Design Phase
```

```
    get_initial_design()
      _generate_initial_design() [if X0 is None]
    _curate_initial_design()
    _evaluate_function()
    _handle_NA_initial_design()
    _check_size_initial_design()
    _get_best_xy_initial_design()
```

```
  Main Optimization Loop (while not terminated)
```

```
    _fit_surrogate()
      _selection_dispatcher() [if max_surrogate_points exceeded]

    _apply_ocba() [if noise=True and ocba_delta > 0]

    _suggest_next_point()
      _acquisition_function()
        _predict_with_uncertainty()
      _repair_non_numeric()
      _select_new()
      _handle_acquisition_failure() [if needed]

    _evaluate_function()

    _handle_NA_new_points()
      _apply_penalty_NA()
      _remove_nan()

    _update_best_main_loop()
```

```
  Termination Phase
```

```
    _determine_termination()
    _map_to_factor_values() [for results]
```

## 24.12. 11. Summary

This notebook demonstrated all major methods in SpotOptim with executable examples:

**Core Flow:** 1. **Initial Design:** Generate/process → Curate → Evaluate → Handle NaN → Validate → Get best 2. **Main Loop:** Fit surrogate → Apply OCBA → Suggest point → Evaluate → Handle NaN → Update best 3. **Termination:** Determine reason → Prepare results

**Key Features:** - Automatic handling of NaN/inf values with penalties - Support for noisy functions with OCBA re-evaluation - Integer and factor variable support - Flexible initial design (LHS, custom, or with starting point) - Multiple acquisition functions (EI, PI, mean) - Termination by max iterations or time limit

All examples can be run independently and demonstrate the modular design of SpotOptim!



## 25. Benchmarking SpotOptim with Sklearn Kriging (Matern Kernel) on 6D Rosenbrock and 10D Michalewicz Functions

### **i** Note

These test functions were used during the Dagstuhl Seminar 25451 Bayesian Optimisation (Nov 02 – Nov 07, 2025), see [here](#).

This notebook demonstrates the use of `SpotOptim` with sklearn's Gaussian Process Regressor as a surrogate model.

### 25.1. SpotOptim with Sklearn Kriging in 6 Dimensions: Rosenbrock Function

This section demonstrates how to use the `SpotOptim` class with sklearn's Gaussian Process Regressor (using Matern kernel) as a surrogate on the 6-dimensional Rosenbrock function. We use a maximum of 100 function evaluations.

```
import warnings
warnings.filterwarnings("ignore")
import json
import numpy as np
from spotoptim import SpotOptim
from spotoptim.function import rosenbrock
```

#### 25.1.1. Define the 6D Rosenbrock Function

## 25. Benchmarking SpotOptim with Sklearn Kriging (Matern Kernel) on 6D Rosenbrock and 10D Michale

```
dim = 6
lower = np.full(dim, -2.0)
upper = np.full(dim, 2.0)
bounds = list(zip(lower, upper))
fun = rosenbrock
max_iter = 100
```

### 25.1.2. Set up SpotOptim Parameters

```
n_initial = dim
seed = 321
```

### 25.1.3. Sklearn Gaussian Process Regressor as Surrogate

```
from sklearn.gaussian_process import GaussianProcessRegressor
from sklearn.gaussian_process.kernels import Matern, ConstantKernel

# Use a Matern kernel instead of the standard RBF kernel
kernel = ConstantKernel(1.0, (1e-2, 1e12)) * Matern(
    length_scale=1.0,
    length_scale_bounds=(1e-4, 1e2),
    nu=2.5
)
surrogate = GaussianProcessRegressor(kernel=kernel, n_restarts_optimizer=100)

# Create SpotOptim instance with sklearn surrogate
opt_rosen = SpotOptim(
    fun=fun,
    bounds=bounds,
    n_initial=n_initial,
    max_iter=max_iter,
    surrogate=surrogate,
    seed=seed,
    verbose=1
)

# Run optimization
result_rosen = opt_rosen.optimize()
```

TensorBoard logging disabled

### 25.1. SpotOptim with Sklearn Kriging in 6 Dimensions: Rosenbrock Function

Initial best:  $f(x) = 321.834153$   
Iteration 1:  $f(x) = 3523.044340$   
Iteration 2:  $f(x) = 1535.228427$   
Iteration 3:  $f(x) = 326.971590$   
Iteration 4: New best  $f(x) = 179.355172$   
Iteration 5: New best  $f(x) = 147.215519$   
Iteration 6: New best  $f(x) = 126.877055$   
Iteration 7: New best  $f(x) = 106.907342$   
Iteration 8: New best  $f(x) = 77.684036$   
Iteration 9: New best  $f(x) = 67.637609$   
Iteration 10:  $f(x) = 70.981266$   
Iteration 11: New best  $f(x) = 66.962496$   
Iteration 12: New best  $f(x) = 66.891979$   
Iteration 13: New best  $f(x) = 63.394184$   
Iteration 14: New best  $f(x) = 53.785852$   
Iteration 15: New best  $f(x) = 53.437961$   
Iteration 16: New best  $f(x) = 52.729913$   
Iteration 17: New best  $f(x) = 51.454280$   
Iteration 18: New best  $f(x) = 48.568475$   
Iteration 19: New best  $f(x) = 48.513252$   
Iteration 20:  $f(x) = 48.647011$   
Iteration 21: New best  $f(x) = 47.077931$   
Iteration 22: New best  $f(x) = 47.055506$   
Iteration 23: New best  $f(x) = 45.080807$   
Iteration 24: New best  $f(x) = 42.673929$   
Iteration 25:  $f(x) = 42.811671$   
Iteration 26: New best  $f(x) = 42.082128$   
Iteration 27:  $f(x) = 43.313655$   
Iteration 28: New best  $f(x) = 39.429855$   
Iteration 29:  $f(x) = 39.690236$   
Iteration 30: New best  $f(x) = 37.726827$   
Iteration 31: New best  $f(x) = 36.722261$   
Iteration 32: New best  $f(x) = 35.431487$   
Iteration 33: New best  $f(x) = 21.265386$   
Iteration 34:  $f(x) = 21.552542$   
Iteration 35:  $f(x) = 21.994144$   
Iteration 36: New best  $f(x) = 19.171772$   
Iteration 37:  $f(x) = 20.274342$   
Iteration 38: New best  $f(x) = 15.390622$   
Iteration 39: New best  $f(x) = 13.850035$   
Iteration 40: New best  $f(x) = 13.111535$   
Iteration 41:  $f(x) = 13.226898$   
Iteration 42: New best  $f(x) = 12.852206$   
Iteration 43: New best  $f(x) = 12.513956$   
Iteration 44: New best  $f(x) = 10.567543$

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Iteration 45: New best  $f(x)$  = 10.109755  
Iteration 46: New best  $f(x)$  = 8.805230  
Iteration 47:  $f(x)$  = 8.946592  
Iteration 48: New best  $f(x)$  = 8.399776  
Iteration 49: New best  $f(x)$  = 8.047033  
Iteration 50: New best  $f(x)$  = 7.940044  
Iteration 51: New best  $f(x)$  = 7.708221  
Iteration 52: New best  $f(x)$  = 6.813571  
Iteration 53: New best  $f(x)$  = 6.755807  
Iteration 54: New best  $f(x)$  = 6.686436  
Iteration 55: New best  $f(x)$  = 6.622111  
Iteration 56: New best  $f(x)$  = 5.974368  
Iteration 57:  $f(x)$  = 6.019614  
Iteration 58: New best  $f(x)$  = 5.869981  
Iteration 59: New best  $f(x)$  = 5.610049  
Iteration 60: New best  $f(x)$  = 5.510559  
Iteration 61:  $f(x)$  = 5.511503  
Iteration 62: New best  $f(x)$  = 5.472302  
Iteration 63: New best  $f(x)$  = 5.387460  
Iteration 64: New best  $f(x)$  = 5.234365  
Iteration 65: New best  $f(x)$  = 5.169011  
Iteration 66:  $f(x)$  = 5.171038  
Iteration 67: New best  $f(x)$  = 5.121998  
Iteration 68: New best  $f(x)$  = 5.016568  
Iteration 69: New best  $f(x)$  = 4.946240  
Iteration 70: New best  $f(x)$  = 4.944222  
Iteration 71: New best  $f(x)$  = 4.934907  
Iteration 72: New best  $f(x)$  = 4.931688  
Iteration 73: New best  $f(x)$  = 4.896228  
Iteration 74:  $f(x)$  = 4.923178  
Iteration 75: New best  $f(x)$  = 4.854686  
Iteration 76:  $f(x)$  = 4.874955  
Iteration 77: New best  $f(x)$  = 4.832856  
Iteration 78:  $f(x)$  = 4.861280  
Iteration 79:  $f(x)$  = 4.836985  
Iteration 80:  $f(x)$  = 4.896492  
Iteration 81: New best  $f(x)$  = 4.601213  
Iteration 82: New best  $f(x)$  = 4.499819  
Iteration 83:  $f(x)$  = 4.515361  
Iteration 84: New best  $f(x)$  = 4.447565  
Iteration 85:  $f(x)$  = 4.474841  
Iteration 86: New best  $f(x)$  = 4.388453  
Iteration 87:  $f(x)$  = 4.413906  
Iteration 88: New best  $f(x)$  = 4.315661  
Iteration 89: New best  $f(x)$  = 4.294393

### 25.1. SpotOptim with Sklearn Kriging in 6 Dimensions: Rosenbrock Function

```
Iteration 90: f(x) = 4.297369
Iteration 91: New best f(x) = 4.271036
Iteration 92: New best f(x) = 4.218094
Iteration 93: New best f(x) = 4.186742
Iteration 94: f(x) = 4.208967
```

```
print(f"[6D] Sklearn Kriging: min y = {result_rosen.fun:.4f} at x = {result_rosen.x}")
print(f"Number of function evaluations: {result_rosen.nfev}")
print(f"Number of iterations: {result_rosen.nit}")
```

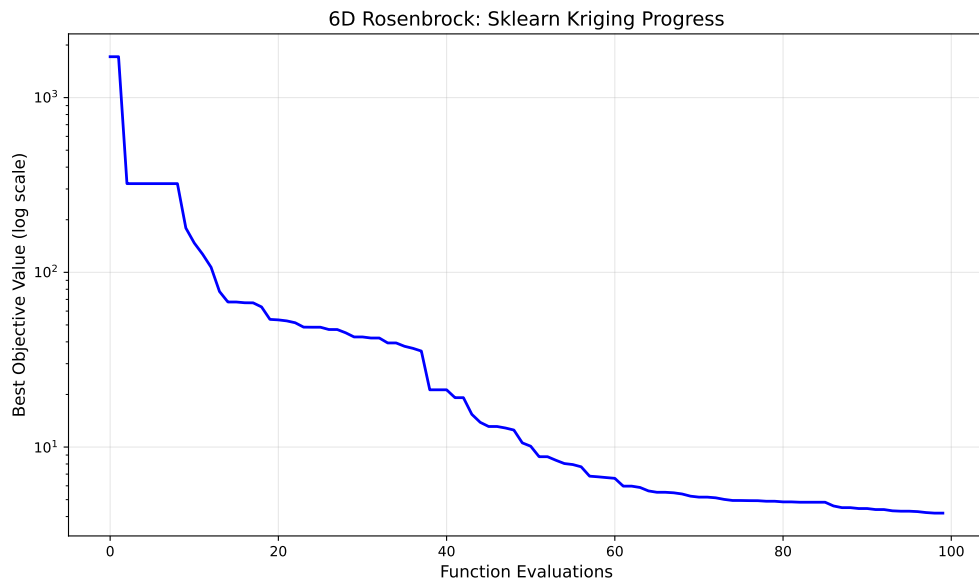
```
[6D] Sklearn Kriging: min y = 4.1867 at x = [0.38230954 0.13023048 0.0100721 0.00537277 0.02792473
Number of function evaluations: 100
Number of iterations: 94
```

#### 25.1.4. Visualize Optimization Progress

```
import matplotlib.pyplot as plt

# Plot the optimization progress
plt.figure(figsize=(10, 6))
plt.semilogy(np.minimum.accumulate(opt_rosen.y_), 'b-', linewidth=2)
plt.xlabel('Function Evaluations', fontsize=12)
plt.ylabel('Best Objective Value (log scale)', fontsize=12)
plt.title('6D Rosenbrock: Sklearn Kriging Progress', fontsize=14)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

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### 25.1.5. Evaluation of Multiple Repeats

To perform 30 repeats and collect statistics:

```
# Perform 30 independent runs
n_repeats = 30
results = []

print(f"Running {n_repeats} independent optimizations...")
for i in range(n_repeats):
    kernel_i = ConstantKernel(1.0, (1e-2, 1e12)) * Matern(
        length_scale=1.0,
        length_scale_bounds=(1e-4, 1e2),
        nu=2.5
    )
    surrogate_i = GaussianProcessRegressor(kernel=kernel_i, n_restarts_optimizer=100)

    opt_i = SpotOptim(
        fun=fun,
        bounds=bounds,
        n_initial=n_initial,
        max_iter=max_iter,
        surrogate=surrogate_i,
        seed=seed + i, # Different seed for each run
```

```

        verbose=0
    )

    result_i = opt_i.optimize()
    results.append(result_i.fun)

    if (i + 1) % 10 == 0:
        print(f" Completed {i + 1}/{n_repeats} runs")

# Compute statistics
mean_result = np.mean(results)
std_result = np.std(results)
min_result = np.min(results)
max_result = np.max(results)

print(f"\nResults over {n_repeats} runs:")
print(f" Mean of best values: {mean_result:.6f}")
print(f" Std of best values: {std_result:.6f}")
print(f" Min of best values: {min_result:.6f}")
print(f" Max of best values: {max_result:.6f}")

```

## 25.2. SpotOptim with Sklearn Kriging in 10 Dimensions: Michalewicz Function

This section demonstrates how to use the `SpotOptim` class with sklearn's Gaussian Process Regressor (using Matern kernel) as a surrogate on the 10-dimensional Michalewicz function. We use a maximum of 300 function evaluations.

### 25.2.1. Define the 10D Michalewicz Function

```

from spotoptim.function import michalewicz

dim = 10
lower = np.full(dim, 0.0)
upper = np.full(dim, np.pi)
bounds = list(zip(lower, upper))
fun = michalewicz
max_iter = 300

```

### 25.2.2. Set up SpotOptim Parameters

```
n_initial = dim
seed = 321
```

### 25.2.3. Sklearn Gaussian Process Regressor as Surrogate

```
from sklearn.gaussian_process import GaussianProcessRegressor
from sklearn.gaussian_process.kernels import Matern, ConstantKernel

# Use a Matern kernel instead of the standard RBF kernel
kernel = ConstantKernel(1.0, (1e-2, 1e12)) * Matern(
    length_scale=1.0,
    length_scale_bounds=(1e-4, 1e2),
    nu=2.5
)
surrogate = GaussianProcessRegressor(kernel=kernel, n_restarts_optimizer=100)

# Create SpotOptim instance with sklearn surrogate
opt_micha = SpotOptim(
    fun=fun,
    bounds=bounds,
    n_initial=n_initial,
    max_iter=max_iter,
    surrogate=surrogate,
    seed=seed,
    verbose=1
)

# Run optimization
result_micha = opt_micha.optimize()
```

```
TensorBoard logging disabled
Initial best: f(x) = -1.909129
Iteration 1: f(x) = -0.471949
Iteration 2: New best f(x) = -2.778323
Iteration 3: f(x) = -1.576468
Iteration 4: f(x) = -1.695763
Iteration 5: New best f(x) = -3.210542
Iteration 6: f(x) = -2.820449
Iteration 7: f(x) = -2.982458
```



## 25.2. SpotOptim with Sklearn Kriging in 10 Dimensions: Michalewicz Function

Iteration 8: New best  $f(x) = -3.366940$   
Iteration 9:  $f(x) = -3.201182$   
Iteration 10: New best  $f(x) = -3.432987$   
Iteration 11: New best  $f(x) = -3.528900$   
Iteration 12:  $f(x) = -3.505738$   
Iteration 13: New best  $f(x) = -3.563138$   
Iteration 14:  $f(x) = -3.548380$   
Iteration 15:  $f(x) = -3.555799$   
Iteration 16: New best  $f(x) = -3.564169$   
Iteration 17: New best  $f(x) = -3.574578$   
Iteration 18:  $f(x) = -3.565124$   
Iteration 19: New best  $f(x) = -3.720213$   
Iteration 20: New best  $f(x) = -3.852760$   
Iteration 21: New best  $f(x) = -4.102168$   
Iteration 22: New best  $f(x) = -4.172580$   
Iteration 23:  $f(x) = -4.164055$   
Iteration 24: New best  $f(x) = -4.507811$   
Iteration 25: New best  $f(x) = -5.052626$   
Iteration 26:  $f(x) = -4.998613$   
Iteration 27: New best  $f(x) = -5.059262$   
Iteration 28: New best  $f(x) = -5.070557$   
Iteration 29:  $f(x) = -5.031981$   
Iteration 30:  $f(x) = -5.057151$   
Iteration 31: New best  $f(x) = -5.164258$   
Iteration 32: New best  $f(x) = -5.491789$   
Iteration 33:  $f(x) = -5.179530$   
Iteration 34: New best  $f(x) = -5.740814$   
Iteration 35: New best  $f(x) = -5.868411$   
Iteration 36: New best  $f(x) = -6.086901$   
Iteration 37: New best  $f(x) = -6.136311$   
Iteration 38:  $f(x) = -6.102396$   
Iteration 39: New best  $f(x) = -6.173755$   
Iteration 40: New best  $f(x) = -6.176473$   
Iteration 41: New best  $f(x) = -6.225432$   
Iteration 42: New best  $f(x) = -6.316170$   
Iteration 43: New best  $f(x) = -6.396362$   
Iteration 44: New best  $f(x) = -6.422816$   
Iteration 45: New best  $f(x) = -6.482548$   
Iteration 46: New best  $f(x) = -6.485573$   
Iteration 47: New best  $f(x) = -6.502783$   
Iteration 48:  $f(x) = -6.484543$   
Iteration 49: New best  $f(x) = -6.717403$   
Iteration 50: New best  $f(x) = -6.862241$   
Iteration 51:  $f(x) = -6.850652$   
Iteration 52: New best  $f(x) = -6.924121$

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Iteration 53:  $f(x) = -6.920386$   
Iteration 54:  $f(x) = -6.900331$   
Iteration 55: New best  $f(x) = -6.951201$   
Iteration 56:  $f(x) = -6.948062$   
Iteration 57: New best  $f(x) = -6.951959$   
Iteration 58: New best  $f(x) = -6.974895$   
Iteration 59: New best  $f(x) = -6.979618$   
Iteration 60:  $f(x) = -6.972484$   
Iteration 61: New best  $f(x) = -6.984013$   
Iteration 62:  $f(x) = -6.976510$   
Iteration 63: New best  $f(x) = -7.005045$   
Iteration 64: New best  $f(x) = -7.020101$   
Iteration 65: New best  $f(x) = -7.061052$   
Iteration 66: New best  $f(x) = -7.066736$   
Iteration 67: New best  $f(x) = -7.087157$   
Iteration 68: New best  $f(x) = -7.110231$   
Iteration 69: New best  $f(x) = -7.115771$   
Iteration 70: New best  $f(x) = -7.129976$   
Iteration 71: New best  $f(x) = -7.131349$   
Iteration 72: New best  $f(x) = -7.132763$   
Iteration 73: New best  $f(x) = -7.149804$   
Iteration 74: New best  $f(x) = -7.169279$   
Iteration 75: New best  $f(x) = -7.203193$   
Iteration 76:  $f(x) = -7.196497$   
Iteration 77: New best  $f(x) = -7.247552$   
Iteration 78: New best  $f(x) = -7.273261$   
Iteration 79: New best  $f(x) = -7.273538$   
Iteration 80:  $f(x) = -7.272606$   
Iteration 81: New best  $f(x) = -7.308913$   
Iteration 82: New best  $f(x) = -7.422864$   
Iteration 83: New best  $f(x) = -7.449577$   
Iteration 84: New best  $f(x) = -7.477545$   
Iteration 85: New best  $f(x) = -7.502946$   
Iteration 86: New best  $f(x) = -7.511144$   
Iteration 87: New best  $f(x) = -7.511382$   
Iteration 88:  $f(x) = -7.511314$   
Iteration 89: New best  $f(x) = -7.518416$   
Iteration 90:  $f(x) = -7.516321$   
Iteration 91: New best  $f(x) = -7.523059$   
Iteration 92: New best  $f(x) = -7.524466$   
Iteration 93: New best  $f(x) = -7.529184$   
Iteration 94: New best  $f(x) = -7.535648$   
Iteration 95:  $f(x) = -7.526547$   
Iteration 96: New best  $f(x) = -7.599863$   
Iteration 97:  $f(x) = -7.599101$

## 25.2. SpotOptim with Sklearn Kriging in 10 Dimensions: Michalewicz Function

Iteration 98: New best  $f(x)$  = -7.616954  
Iteration 99: New best  $f(x)$  = -7.619664  
Iteration 100: New best  $f(x)$  = -7.620009  
Iteration 101: New best  $f(x)$  = -7.620361  
Iteration 102: New best  $f(x)$  = -7.620561  
Iteration 103: New best  $f(x)$  = -7.621385  
Iteration 104: New best  $f(x)$  = -7.624569  
Iteration 105: New best  $f(x)$  = -7.634927  
Iteration 106: New best  $f(x)$  = -7.636352  
Iteration 107:  $f(x)$  = -7.636333  
Iteration 108: New best  $f(x)$  = -7.638109  
Iteration 109: New best  $f(x)$  = -7.641929  
Iteration 110: New best  $f(x)$  = -7.645940  
Iteration 111: New best  $f(x)$  = -7.647977  
Iteration 112:  $f(x)$  = -7.647736  
Iteration 113: New best  $f(x)$  = -7.648136  
Iteration 114: New best  $f(x)$  = -7.648444  
Iteration 115: New best  $f(x)$  = -7.655937  
Iteration 116: New best  $f(x)$  = -7.661437  
Iteration 117: New best  $f(x)$  = -7.662063  
Iteration 118: New best  $f(x)$  = -7.664864  
Iteration 119:  $f(x)$  = -7.664831  
Iteration 120: New best  $f(x)$  = -7.665473  
Iteration 121: New best  $f(x)$  = -7.665692  
Iteration 122: New best  $f(x)$  = -7.666090  
Iteration 123:  $f(x)$  = -7.665928  
Iteration 124: New best  $f(x)$  = -7.668592  
Iteration 125: New best  $f(x)$  = -7.669017  
Iteration 126: New best  $f(x)$  = -7.669081  
Iteration 127: New best  $f(x)$  = -7.669198  
Iteration 128:  $f(x)$  = -7.668825  
Iteration 129: New best  $f(x)$  = -7.669871  
Iteration 130: New best  $f(x)$  = -7.669985  
Iteration 131: New best  $f(x)$  = -7.670393  
Iteration 132: New best  $f(x)$  = -7.676082  
Iteration 133: New best  $f(x)$  = -7.676643  
Iteration 134: New best  $f(x)$  = -7.676648  
Iteration 135:  $f(x)$  = -7.676602  
Iteration 136: New best  $f(x)$  = -7.676838  
Iteration 137:  $f(x)$  = -7.676783  
Iteration 138: New best  $f(x)$  = -7.677634  
Iteration 139: New best  $f(x)$  = -7.677754  
Iteration 140: New best  $f(x)$  = -7.678084  
Iteration 141:  $f(x)$  = -7.678069  
Iteration 142: New best  $f(x)$  = -7.678445

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Iteration 143: New best  $f(x)$  = -7.678526  
Iteration 144: New best  $f(x)$  = -7.678787  
Iteration 145: New best  $f(x)$  = -7.678883  
Iteration 146:  $f(x)$  = -7.678385

Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Iteration 147:  $f(x)$  = -2.167836  
Iteration 148: New best  $f(x)$  = -7.679202  
Iteration 149: New best  $f(x)$  = -7.679605  
Iteration 150: New best  $f(x)$  = -7.679712  
Iteration 151: New best  $f(x)$  = -7.679783  
Iteration 152:  $f(x)$  = -7.679779  
Iteration 153: New best  $f(x)$  = -7.679800  
Iteration 154: New best  $f(x)$  = -7.679810  
Iteration 155: New best  $f(x)$  = -7.679843  
Iteration 156: New best  $f(x)$  = -7.680107  
Iteration 157: New best  $f(x)$  = -7.680676  
Iteration 158: New best  $f(x)$  = -7.680814  
Iteration 159: New best  $f(x)$  = -7.680818  
Iteration 160: New best  $f(x)$  = -7.680818  
Iteration 161:  $f(x)$  = -7.680816  
Iteration 162: New best  $f(x)$  = -7.680834  
Iteration 163: New best  $f(x)$  = -7.680890  
Iteration 164:  $f(x)$  = -7.680869  
Iteration 165: New best  $f(x)$  = -7.681039  
Iteration 166: New best  $f(x)$  = -7.681141  
Iteration 167: New best  $f(x)$  = -7.681142  
Iteration 168: New best  $f(x)$  = -7.681160  
Iteration 169: New best  $f(x)$  = -7.681395  
Iteration 170: New best  $f(x)$  = -7.681459  
Iteration 171: New best  $f(x)$  = -7.681479  
Iteration 172: New best  $f(x)$  = -7.681567  
Iteration 173: New best  $f(x)$  = -7.681627  
Iteration 174: New best  $f(x)$  = -7.681637  
Iteration 175: New best  $f(x)$  = -7.681637  
Iteration 176: New best  $f(x)$  = -7.681646  
Iteration 177: New best  $f(x)$  = -7.681687  
Iteration 178:  $f(x)$  = -7.681677  
Iteration 179: New best  $f(x)$  = -7.681743  
Iteration 180: New best  $f(x)$  = -7.681837  
Iteration 181: New best  $f(x)$  = -7.681876  
Iteration 182:  $f(x)$  = -7.681875  
Iteration 183: New best  $f(x)$  = -7.681878  
Iteration 184: New best  $f(x)$  = -7.681882  
Iteration 185: New best  $f(x)$  = -7.681895

## 25.2. SpotOptim with Sklearn Kriging in 10 Dimensions: Michalewicz Function

Iteration 186: New best  $f(x)$  = -7.681924  
Iteration 187: New best  $f(x)$  = -7.681954  
Iteration 188: New best  $f(x)$  = -7.681964  
Iteration 189: New best  $f(x)$  = -7.681967  
Iteration 190: New best  $f(x)$  = -7.681968  
Iteration 191:  $f(x)$  = -7.681960  
Iteration 192: New best  $f(x)$  = -7.681983  
Iteration 193: New best  $f(x)$  = -7.681993  
Iteration 194:  $f(x)$  = -7.681992  
Iteration 195: New best  $f(x)$  = -7.681995  
Iteration 196:  $f(x)$  = -7.681995  
Iteration 197: New best  $f(x)$  = -7.681996  
Iteration 198:  $f(x)$  = -7.681996  
Iteration 199:  $f(x)$  = -7.604434  
Iteration 200:  $f(x)$  = -7.681995  
Iteration 201:  $f(x)$  = -7.681995  
Iteration 202:  $f(x)$  = -7.681995  
Iteration 203:  $f(x)$  = -7.681994  
Iteration 204:  $f(x)$  = -7.681995  
Iteration 205:  $f(x)$  = -7.681995  
Iteration 206: New best  $f(x)$  = -7.681996  
Iteration 207: New best  $f(x)$  = -7.681997  
Iteration 208: New best  $f(x)$  = -7.681999  
Iteration 209:  $f(x)$  = -7.681998  
Iteration 210: New best  $f(x)$  = -7.682000  
Iteration 211: New best  $f(x)$  = -7.682002  
Iteration 212:  $f(x)$  = -7.682002  
Iteration 213:  $f(x)$  = -7.682001  
Iteration 214:  $f(x)$  = -7.682002  
Iteration 215:  $f(x)$  = -7.682001  
Iteration 216:  $f(x)$  = -7.681999  
Iteration 217:  $f(x)$  = -7.682001  
Iteration 218: New best  $f(x)$  = -7.682002  
Iteration 219: New best  $f(x)$  = -7.682003  
Iteration 220:  $f(x)$  = -7.682002  
Iteration 221:  $f(x)$  = -7.682002  
Iteration 222:  $f(x)$  = -7.682003  
Iteration 223:  $f(x)$  = -7.682002  
Iteration 224:  $f(x)$  = -7.682000  
Iteration 225:  $f(x)$  = -7.682002  
Iteration 226:  $f(x)$  = -7.682003  
Iteration 227:  $f(x)$  = -7.682002  
Iteration 228:  $f(x)$  = -7.682001  
Iteration 229:  $f(x)$  = -7.682002  
Iteration 230:  $f(x)$  = -7.682003

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Iteration 231:  $f(x) = -7.682002$   
Iteration 232: New best  $f(x) = -7.682003$   
Iteration 233:  $f(x) = -7.654023$   
Iteration 234:  $f(x) = -7.682001$   
Iteration 235:  $f(x) = -7.682003$   
Iteration 236:  $f(x) = -7.682002$   
Iteration 237:  $f(x) = -7.682003$   
Iteration 238: New best  $f(x) = -7.682004$   
Iteration 239:  $f(x) = -7.682001$   
Iteration 240:  $f(x) = -7.682002$   
Iteration 241:  $f(x) = -7.682002$   
Iteration 242:  $f(x) = -7.682002$   
Iteration 243:  $f(x) = -7.682003$   
Iteration 244:  $f(x) = -7.682003$   
Iteration 245:  $f(x) = -7.682003$   
Iteration 246:  $f(x) = -7.682003$   
Iteration 247:  $f(x) = -7.682003$   
Iteration 248:  $f(x) = -7.682003$   
Iteration 249:  $f(x) = -7.682003$   
Iteration 250:  $f(x) = -7.682003$   
Iteration 251:  $f(x) = -7.682003$   
Iteration 252:  $f(x) = -7.682003$   
Iteration 253:  $f(x) = -7.682003$   
Iteration 254:  $f(x) = -7.682003$   
Iteration 255:  $f(x) = -7.682003$   
Iteration 256:  $f(x) = -7.682003$   
Iteration 257:  $f(x) = -7.682003$   
Iteration 258:  $f(x) = -7.682004$   
Iteration 259:  $f(x) = -7.682002$   
Iteration 260:  $f(x) = -7.682003$   
Iteration 261:  $f(x) = -7.682003$   
Iteration 262:  $f(x) = -7.682004$   
Iteration 263:  $f(x) = -7.682002$   
Iteration 264:  $f(x) = -7.682003$   
Iteration 265:  $f(x) = -7.682003$   
Iteration 266:  $f(x) = -7.682004$   
Iteration 267:  $f(x) = -7.682003$   
Iteration 268:  $f(x) = -7.682003$   
Iteration 269:  $f(x) = -7.682004$   
Iteration 270:  $f(x) = -7.682003$   
Iteration 271:  $f(x) = -7.682003$   
Iteration 272:  $f(x) = -7.682003$   
Iteration 273:  $f(x) = -7.682003$   
Iteration 274:  $f(x) = -7.682003$   
Iteration 275:  $f(x) = -7.682004$

## 25.2. SpotOptim with Sklearn Kriging in 10 Dimensions: Michalewicz Function

```
Iteration 276: f(x) = -7.682004
Iteration 277: f(x) = -7.682003
Iteration 278: f(x) = -7.682003
Iteration 279: f(x) = -7.682003
Iteration 280: f(x) = -7.682003
Iteration 281: f(x) = -7.682001
Iteration 282: f(x) = -7.682003
Iteration 283: f(x) = -7.682003
Iteration 284: f(x) = -7.682003
Iteration 285: f(x) = -7.682003
Iteration 286: f(x) = -7.682004
Iteration 287: New best f(x) = -7.682004
Iteration 288: New best f(x) = -7.682004
Iteration 289: f(x) = -7.682004
Iteration 290: f(x) = -7.682004
```

```
print(f"[10D] Sklearn Kriging: min y = {result_micha.fun:.4f} at x = {result_micha.x}")
print(f"Number of function evaluations: {result_micha.nfev}")
print(f"Number of iterations: {result_micha.nit}")
```

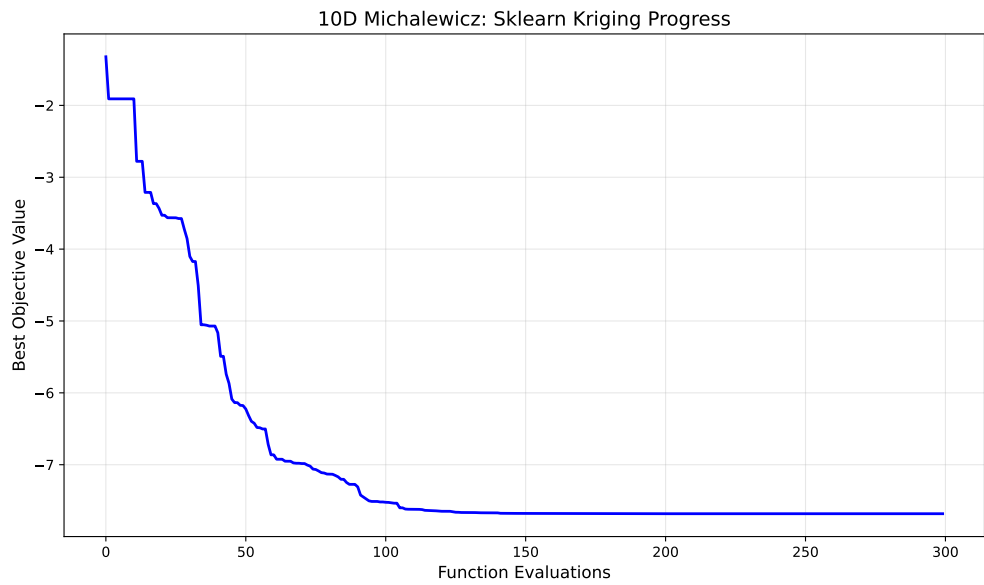
```
[10D] Sklearn Kriging: min y = -7.6820 at x = [2.20291795 2.71155004 2.21933224 2.48204296 2.6271818
2.22103812 1.36057728 1.28283097 1.21703192]
Number of function evaluations: 300
Number of iterations: 290
```

### 25.2.4. Visualize Optimization Progress

```
import matplotlib.pyplot as plt

# Plot the optimization progress
plt.figure(figsize=(10, 6))
plt.plot(np.minimum.accumulate(opt_micha.y_), 'b-', linewidth=2)
plt.xlabel('Function Evaluations', fontsize=12)
plt.ylabel('Best Objective Value', fontsize=12)
plt.title('10D Michalewicz: Sklearn Kriging Progress', fontsize=14)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

## 25. Benchmarking SpotOptim with Sklearn Kriging (Matern Kernel) on 6D Rosenbrock and 10D Michalewicz



### 25.2.5. Evaluation of Multiple Repeats

To perform 30 repeats and collect statistics:

```
# Perform 30 independent runs
n_repeats = 30
results = []

print(f"Running {n_repeats} independent optimizations...")
for i in range(n_repeats):
    kernel_i = ConstantKernel(1.0, (1e-2, 1e12)) * Matern(
        length_scale=1.0,
        length_scale_bounds=(1e-4, 1e2),
        nu=2.5
    )
    surrogate_i = GaussianProcessRegressor(kernel=kernel_i, n_restarts_optimizer=100)

    opt_i = SpotOptim(
        fun=fun,
        bounds=bounds,
        n_initial=n_initial,
        max_iter=max_iter,
        surrogate=surrogate_i,
        seed=seed + i, # Different seed for each run
```



```

        verbose=0
    )

    result_i = opt_i.optimize()
    results.append(result_i.fun)

    if (i + 1) % 10 == 0:
        print(f" Completed {i + 1}/{n_repeats} runs")

# Compute statistics
mean_result = np.mean(results)
std_result = np.std(results)
min_result = np.min(results)
max_result = np.max(results)

print(f"\nResults over {n_repeats} runs:")
print(f" Mean of best values: {mean_result:.6f}")
print(f" Std of best values: {std_result:.6f}")
print(f" Min of best values: {min_result:.6f}")
print(f" Max of best values: {max_result:.6f}")

```

## 25.3. Comparison: SpotOptim vs SpotPython

The **SpotOptim** package provides a scipy-compatible interface for Bayesian optimization with the following key features:

1. **Scipy-compatible API:** Returns **OptimizeResult** objects that work seamlessly with scipy's optimization ecosystem
2. **Custom Surrogates:** Supports any sklearn-compatible surrogate model (as demonstrated with **GaussianProcessRegressor**)
3. **Flexible Interface:** Simplified parameter specification with **bounds**, **n\_initial**, and **max\_iter**
4. **Analytical Test Functions:** Built-in test functions (**rosenbrock**, **ackley**, **michalewicz**) for benchmarking

The main differences from **spotpython** are:

- **SpotOptim:** Uses **bounds**, **n\_initial**, **max\_iter** parameters with scipy-style interface
- **SpotPython:** Uses **fun\_control**, **design\_control**, **surrogate\_control** with more complex configuration

Both packages support custom surrogates and provide powerful Bayesian optimization capabilities.

## 25.4. Summary

This notebook demonstrated how to:

1. Use **SpotOptim** with sklearn's Gaussian Process Regressor (Matern kernel) as a surrogate
2. Optimize 6D Rosenbrock function with 100 evaluations
3. Optimize 10D Michalewicz function with 300 evaluations
4. Visualize optimization progress
5. Perform multiple independent runs for statistical analysis

The results show that **SpotOptim** with sklearn surrogates provides effective Bayesian optimization for challenging benchmark functions.

## 25.5. Jupyter Notebook

### Note

- This Quarto document is part of the `spotoptim` package benchmarking suite
- Source available at: [spotoptim GitHub Repository](#)

## 26. Reproducibility in SpotOptim

### 26.1. Introduction

SpotOptim provides full support for reproducible optimization runs through the `seed` parameter. This is essential for:

- **Scientific research:** Ensuring experiments can be replicated
- **Debugging:** Reproducing specific optimization behaviors
- **Benchmarking:** Fair comparison between different configurations
- **Production:** Consistent results in deployed applications

When you specify a seed, SpotOptim guarantees that running the same optimization multiple times will produce identical results. Without a seed, each run explores the search space differently, which can be useful for robustness testing.

### 26.2. Basic Usage

#### 26.2.1. Making Optimization Reproducible

To ensure reproducible results, simply specify the `seed` parameter when creating the optimizer:

```
import numpy as np
from spotoptim import SpotOptim

def sphere(X):
    """Simple sphere function:  $f(x) = \sum(x^2)$ """
    return np.sum(X**2, axis=1)

# Reproducible optimization
optimizer = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=30,
    n_initial=15,
```

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```
seed=42, # This ensures reproducibility
verbose=True
)

result = optimizer.optimize()
print(f"Best solution: {result.x}")
print(f"Best value: {result.fun}")
```

```
TensorBoard logging disabled
Initial best: f(x) = 5.542803
Iteration 1: New best f(x) = 0.001070
Iteration 2: New best f(x) = 0.000089
Iteration 3: New best f(x) = 0.000066
Iteration 4: New best f(x) = 0.000036
Iteration 5: New best f(x) = 0.000001
Iteration 6: New best f(x) = 0.000000
Iteration 7: f(x) = 0.000000
Iteration 8: f(x) = 0.000000
Iteration 9: f(x) = 0.000000
Iteration 10: f(x) = 0.000000
Iteration 11: f(x) = 0.000000
Iteration 12: f(x) = 0.000000
Iteration 13: f(x) = 0.000000
Iteration 14: f(x) = 0.000000
Iteration 15: New best f(x) = 0.000000
Best solution: [3.31436760e-04 4.18312302e-05]
Best value: 1.1160017787260647e-07
```

**Key Point:** Running this code multiple times (even on different days or machines) will always produce the same result.

### 26.2.2. Running Independent Experiments

If you don't specify a seed, each optimization run will explore the search space differently:

```
# Non-reproducible: different results each time
optimizer = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=30,
    n_initial=15
```

```

    # No seed specified
)

result = optimizer.optimize()
# Results will vary between runs

```

This is useful when you want to: - Explore different regions of the search space - Test the robustness of your results - Run multiple independent optimization attempts

## 26.3. Practical Examples

### 26.3.1. Example 1: Comparing Different Configurations

When comparing different optimizer settings, use the same seed for fair comparison:

```

import numpy as np
from spotoptim import SpotOptim

def rosenbrock(X):
    """Rosenbrock function"""
    x = X[:, 0]
    y = X[:, 1]
    return (1 - x)**2 + 100 * (y - x**2)**2

# Configuration 1: More initial points
opt1 = SpotOptim(
    fun=rosenbrock,
    bounds=[(-2, 2), (-2, 2)],
    max_iter=50,
    n_initial=20,
    seed=42 # Same seed for fair comparison
)
result1 = opt1.optimize()

# Configuration 2: Fewer initial points, more iterations
opt2 = SpotOptim(
    fun=rosenbrock,
    bounds=[(-2, 2), (-2, 2)],
    max_iter=50,
    n_initial=10,
    seed=42 # Same seed
)

```

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```
result2 = opt2.optimize()

print(f"Config 1 (more initial): {result1.fun:.6f}")
print(f"Config 2 (fewer initial): {result2.fun:.6f}")
```

```
Config 1 (more initial): 0.002494
Config 2 (fewer initial): 0.011538
```

### 26.3.2. Example 2: Reproducible Research Experiment

For scientific papers or reports, always use a fixed seed and document it:

```
import numpy as np
from spotoptim import SpotOptim

def rastrigin(X):
    """Rastrigin function (multimodal)"""
    A = 10
    n = X.shape[1]
    return A * n + np.sum(X**2 - A * np.cos(2 * np.pi * X), axis=1)

# Documented seed for reproducibility
RANDOM_SEED = 12345

optimizer = SpotOptim(
    fun=rastrigin,
    bounds=[(-5.12, 5.12), (-5.12, 5.12), (-5.12, 5.12)],
    max_iter=100,
    n_initial=30,
    seed=RANDOM_SEED,
    verbose=True
)

result = optimizer.optimize()

print(f"\nExperiment Results (seed={RANDOM_SEED}):")
print(f"Best solution: {result.x}")
print(f"Best value: {result.fun}")
print(f"Iterations: {result.nit}")
print(f"Function evaluations: {result.nfev}")

# These results can now be cited in a paper
```

```
TensorBoard logging disabled
Initial best: f(x) = 20.392774
Iteration 1: f(x) = 31.367736
Iteration 2: f(x) = 30.822371
Iteration 3: f(x) = 28.761963
Iteration 4: f(x) = 25.355390
Iteration 5: f(x) = 22.087398
Iteration 6: f(x) = 23.609642
Iteration 7: f(x) = 27.024927
Iteration 8: f(x) = 23.154774
Iteration 9: New best f(x) = 8.927550
Iteration 10: New best f(x) = 8.899624
Iteration 11: f(x) = 19.831800
Iteration 12: f(x) = 19.323148
Iteration 13: New best f(x) = 8.692652
Iteration 14: New best f(x) = 8.244681
Iteration 15: f(x) = 8.404445
Iteration 16: f(x) = 17.268172
Iteration 17: f(x) = 17.250262
Iteration 18: f(x) = 16.989215
Iteration 19: f(x) = 22.061741
Iteration 20: New best f(x) = 8.227940
Iteration 21: New best f(x) = 2.732909
Iteration 22: New best f(x) = 2.236454
Iteration 23: New best f(x) = 2.001414
Iteration 24: f(x) = 2.001821
Iteration 25: f(x) = 2.002706
Iteration 26: f(x) = 24.677487
Iteration 27: f(x) = 25.445950
Iteration 28: f(x) = 22.323886
Iteration 29: New best f(x) = 1.990146
Iteration 30: New best f(x) = 1.989946
Iteration 31: f(x) = 30.157363
Iteration 32: f(x) = 13.980500
Iteration 33: f(x) = 26.257931
Iteration 34: f(x) = 26.086756
Iteration 35: f(x) = 16.955502
Iteration 36: f(x) = 86.774141
Iteration 37: f(x) = 42.005362
Iteration 38: New best f(x) = 1.989929
Iteration 39: f(x) = 1.989930
Iteration 40: New best f(x) = 1.989929
Iteration 41: f(x) = 69.717061
Iteration 42: New best f(x) = 1.989929
Iteration 43: f(x) = 66.932192
```

## 26. Reproducibility in SpotOptim

```
Iteration 44: f(x) = 50.905256
Iteration 45: f(x) = 1.989930
Iteration 46: f(x) = 46.800517
Iteration 47: f(x) = 67.429406
Iteration 48: f(x) = 60.775651
Iteration 49: f(x) = 1.989930
Iteration 50: f(x) = 1.989929
Iteration 51: f(x) = 1.989930
Iteration 52: f(x) = 1.989929
Iteration 53: f(x) = 1.989930
Iteration 54: f(x) = 1.989929
Iteration 55: f(x) = 1.989931
Iteration 56: f(x) = 1.989930
Iteration 57: f(x) = 1.989929
Iteration 58: f(x) = 16.986975
Iteration 59: f(x) = 14.520119
Iteration 60: f(x) = 86.774141
Iteration 61: f(x) = 1.989931
Iteration 62: f(x) = 76.830975
Iteration 63: f(x) = 59.126187
Iteration 64: f(x) = 1.989929
Iteration 65: f(x) = 13.047213
Iteration 66: New best f(x) = 1.989929
Iteration 67: New best f(x) = 1.989928
Iteration 68: f(x) = 1.989931
Iteration 69: f(x) = 1.989930
Iteration 70: f(x) = 1.989928
```

```
Experiment Results (seed=12345):
Best solution: [ 9.94876607e-01 -9.94866368e-01  1.89299459e-04]
Best value: 1.9899282459113223
Iterations: 70
Function evaluations: 100
```

### 26.3.3. Example 3: Multiple Independent Runs

To test robustness, run the same optimization with different seeds:

```
import numpy as np
from spotoptim import SpotOptim

def ackley(X):
    """Ackley function"""
    a = 20
```



```

b = 0.2
c = 2 * np.pi
n = X.shape[1]

sum_sq = np.sum(X**2, axis=1)
sum_cos = np.sum(np.cos(c * X), axis=1)

return -a * np.exp(-b * np.sqrt(sum_sq / n)) - np.exp(sum_cos / n) + a + np.e

# Run 5 independent optimizations
results = []
seeds = [42, 123, 456, 789, 1011]

for seed in seeds:
    optimizer = SpotOptim(
        fun=ackley,
        bounds=[(-5, 5), (-5, 5)],
        max_iter=40,
        n_initial=20,
        seed=seed,
        verbose=False
    )
    result = optimizer.optimize()
    results.append(result.fun)
    print(f"Run with seed {seed:4d}: f(x) = {result.fun:.6f}")

# Analyze robustness
print(f"\nBest result: {min(results):.6f}")
print(f"Worst result: {max(results):.6f}")
print(f"Mean: {np.mean(results):.6f}")
print(f"Std dev: {np.std(results):.6f}")

```

```

Run with seed 42: f(x) = 0.000907
Run with seed 123: f(x) = 0.001394
Run with seed 456: f(x) = 0.001941
Run with seed 789: f(x) = 0.000616
Run with seed 1011: f(x) = 0.003029

```

```

Best result: 0.000616
Worst result: 0.003029
Mean: 0.001578
Std dev: 0.000854

```

**26.3.4. Example 4: Reproducible Initial Design**

The seed ensures that even the initial design points are reproducible:

```
import numpy as np
from spotoptim import SpotOptim

def simple_quadratic(X):
    return np.sum((X - 1)**2, axis=1)

# Create two optimizers with same seed
opt1 = SpotOptim(
    fun=simple_quadratic,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=25,
    n_initial=10,
    seed=999
)

opt2 = SpotOptim(
    fun=simple_quadratic,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=25,
    n_initial=10,
    seed=999 # Same seed
)

# Run both optimizations
result1 = opt1.optimize()
result2 = opt2.optimize()

# Verify identical results
print("Initial design points are identical:",
      np.allclose(opt1.X_[:10], opt2.X_[:10]))
print("All evaluated points are identical:",
      np.allclose(opt1.X_, opt2.X_))
print("All function values are identical:",
      np.allclose(opt1.y_, opt2.y_))
print("Best solutions are identical:",
      np.allclose(result1.x, result2.x))
```

```
Initial design points are identical: True
All evaluated points are identical: True
All function values are identical: True
Best solutions are identical: True
```

### 26.3.5. Example 5: Custom Initial Design with Seed

Even when providing a custom initial design, the seed ensures reproducible subsequent iterations:

```
import numpy as np
from spotoptim import SpotOptim

def beale(X):
    """Beale function"""
    x = X[:, 0]
    y = X[:, 1]
    term1 = (1.5 - x + x * y)**2
    term2 = (2.25 - x + x * y**2)**2
    term3 = (2.625 - x + x * y**3)**2
    return term1 + term2 + term3

# Custom initial design (e.g., from previous knowledge)
X_start = np.array([
    [0.0, 0.0],
    [1.0, 1.0],
    [2.0, 2.0],
    [-1.0, -1.0]
])

# Run twice with same seed and initial design
opt1 = SpotOptim(
    fun=beale,
    bounds=[(-4.5, 4.5), (-4.5, 4.5)],
    max_iter=30,
    n_initial=10,
    seed=777
)
result1 = opt1.optimize(X0=X_start)

opt2 = SpotOptim(
    fun=beale,
    bounds=[(-4.5, 4.5), (-4.5, 4.5)],
    max_iter=30,
    n_initial=10,
    seed=777 # Same seed
)
result2 = opt2.optimize(X0=X_start)
```

## 26. Reproducibility in SpotOptim

```
print("Results are identical:", np.allclose(result1.x, result2.x))
print(f"Best value: {result1.fun:.6f}")
```

```
Results are identical: True
Best value: 3.201102
```

## 26.4. Advanced Topics

### 26.4.1. Seed and Noisy Functions

When optimizing noisy functions with repeated evaluations, the seed ensures reproducible noise:

```
import numpy as np
from spotoptim import SpotOptim

def noisy_sphere(X):
    """Sphere function with Gaussian noise"""
    base = np.sum(X**2, axis=1)
    noise = np.random.normal(0, 0.1, size=base.shape)
    return base + noise

optimizer = SpotOptim(
    fun=noisy_sphere,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=40,
    n_initial=20,
    repeats_initial=3, # 3 evaluations per point
    repeats_surrogate=2,
    seed=42 # Ensures same noise pattern
)

result = optimizer.optimize()
print(f"Best mean value: {optimizer.min_mean_y:.6f}")
print(f"Variance at best: {optimizer.min_var_y:.6f}")
```

```
Best mean value: 0.124029
Variance at best: 0.004129
```

**Important:** With the same seed, even the noise will be identical across runs!

### 26.4.2. Different Seeds for Different Exploration

Use different seeds to explore different regions systematically:

```
import numpy as np
from spotoptim import SpotOptim

def griewank(X):
    """Griewank function"""
    sum_sq = np.sum(X**2 / 4000, axis=1)
    prod_cos = np.prod(np.cos(X / np.sqrt(np.arange(1, X.shape[1] + 1))), axis=1)
    return sum_sq - prod_cos + 1

# Systematic exploration with different seeds
best_overall = float('inf')
best_seed = None

for seed in range(10, 20): # Seeds 10-19
    optimizer = SpotOptim(
        fun=griewank,
        bounds=[(-600, 600), (-600, 600)],
        max_iter=50,
        n_initial=25,
        seed=seed
    )
    result = optimizer.optimize()

    if result.fun < best_overall:
        best_overall = result.fun
        best_seed = seed

    print(f"Seed {seed}: f(x) = {result.fun:.6f}")

print(f"\nBest result with seed {best_seed}: {best_overall:.6f}")
```

```
Seed 10: f(x) = 4.597121
Seed 11: f(x) = 1.946733
Seed 12: f(x) = 6.030579
Seed 13: f(x) = 0.851170
Seed 14: f(x) = 0.154832
Seed 15: f(x) = 0.044427
Seed 16: f(x) = 8.343419
Seed 17: f(x) = 0.547492
Seed 18: f(x) = 2.121439
```

## 26. Reproducibility in SpotOptim

Seed 19:  $f(x) = 0.257388$

Best result with seed 15: 0.044427

## 26.5. Best Practices

### 26.5.1. 1. Always Use Seeds for Production Code

```
# Good: Reproducible
optimizer = SpotOptim(fun=objective, bounds=bounds, seed=42)

# Risky: Non-reproducible
optimizer = SpotOptim(fun=objective, bounds=bounds)
```

### 26.5.2. 2. Document Your Seeds

```
# Configuration for experiment reported in Section 4.2
EXPERIMENT_SEED = 2024
MAX_ITERATIONS = 100

optimizer = SpotOptim(
    fun=my_objective,
    bounds=my_bounds,
    max_iter=MAX_ITERATIONS,
    seed=EXPERIMENT_SEED
)
```

### 26.5.3. 3. Use Different Seeds for Different Experiments

```
# Different experiments should use different seeds
BASELINE_SEED = 100
EXPERIMENT_A_SEED = 200
EXPERIMENT_B_SEED = 300
```

#### 26.5.4. 4. Test Robustness Across Multiple Seeds

```
# Run same optimization with multiple seeds
for seed in [42, 123, 456, 789, 1011]:
    optimizer = SpotOptim(fun=objective, bounds=bounds, seed=seed)
    result = optimizer.optimize()
# Analyze results
```

## 26.6. What the Seed Controls

The `seed` parameter ensures reproducibility by controlling:

1. **Initial Design Generation:** Latin Hypercube Sampling produces the same initial points
2. **Surrogate Model:** Gaussian Process random initialization is identical
3. **Acquisition Optimization:** Differential evolution explores the same candidates
4. **Random Sampling:** Any random exploration uses the same random numbers

This guarantees that the entire optimization pipeline is deterministic and reproducible.

## 26.7. Common Questions

**Q: Can I use `seed=0`?**

A: Yes, any integer (including 0) is a valid seed.

**Q: Will different Python versions give the same results?**

A: Generally yes, but minor numerical differences may occur due to underlying library changes. Use the same environment for exact reproducibility.

**Q: Does the seed affect the objective function?**

A: No, the seed only affects SpotOptim's internal random processes. If your objective function has its own randomness, you'll need to control that separately.

**Q: How do I choose a good seed value?**

A: Any integer works. Common choices are 42, 123, or dates (e.g., 20241112). What matters is consistency, not the specific value.

## 26.8. Summary

- Use `seed` parameter for reproducible optimization
- Same seed → identical results (every time)
- No seed → different results (random exploration)
- Essential for research, debugging, and production
- Document your seeds for transparency
- Test robustness with multiple different seeds



## 27. Optimizing the Aircraft Wing Weight Example

### **i** Note

- This section demonstrates optimization of the Aircraft Wing Weight Example (AWWE) function
- We compare three optimization methods:
  - **SpotOptim**: Bayesian optimization with Gaussian Process surrogate
  - **Nelder-Mead**: Derivative-free simplex method from `scipy.optimize`
  - **BFGS**: Quasi-Newton method from `scipy.optimize`
- The following Python packages are imported:

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from scipy.optimize import minimize
from spotoptim import SpotOptim
import time
import pprint
```

### 27.1. The AWWE Objective Function

We use the same AWWE function from Chapter 1, which models the weight of an unpainted light aircraft wing. The function accepts inputs in the unit cube  $[0, 1]^9$  and returns the wing weight.

```
def wingwt(x):
    """
    Aircraft Wing Weight function.

    Args:
        x: array-like of 9 values in [0,1]
```

## 27. Optimizing the Aircraft Wing Weight Example

```
[Sw, Wfw, A, L, q, l, Rtc, Nz, Wdg]

Returns:
    Wing weight (scalar)
    """
    # Ensure x is a 2D array for batch evaluation
    x = np.atleast_2d(x)

    # Transform from unit cube to natural scales
    Sw = x[:, 0] * (200 - 150) + 150
    Wfw = x[:, 1] * (300 - 220) + 220
    A = x[:, 2] * (10 - 6) + 6
    L = (x[:, 3] * (10 - (-10)) - 10) * np.pi/180
    q = x[:, 4] * (45 - 16) + 16
    l = x[:, 5] * (1 - 0.5) + 0.5
    Rtc = x[:, 6] * (0.18 - 0.08) + 0.08
    Nz = x[:, 7] * (6 - 2.5) + 2.5
    Wdg = x[:, 8] * (2500 - 1700) + 1700

    # Calculate weight on natural scale
    W = 0.036 * Sw**0.758 * Wfw**0.0035 * (A/np.cos(L)**2)**0.6 * q**0.006
    W = W * l**0.04 * (100*Rtc/np.cos(L))**(-0.3) * (Nz*Wdg)**(0.49)

    return W.ravel()

# Wrapper for scipy.optimize (expects 1D input, returns scalar)
def wingwt_scipy(x):
    return float(wingwt(x.reshape(1, -1))[0])
```

## 27.2. Baseline Configuration

The baseline Cessna C172 Skyhawk configuration (coded in unit cube):

```
baseline_coded = np.array([0.48, 0.4, 0.38, 0.5, 0.62, 0.344, 0.4, 0.37, 0.38])
baseline_weight = wingwt(baseline_coded)[0]
print(f"Baseline wing weight: {baseline_weight:.2f} lb")
```

Baseline wing weight: 233.91 lb

## 27.3. Optimization Setup

We'll optimize the AWWE function starting from the baseline configuration using three different methods:

1. **SpotOptim**: Bayesian optimization (good for expensive black-box functions)
2. **Nelder-Mead**: Derivative-free simplex method (robust but can be slow)
3. **BFGS**: Quasi-Newton method (fast but requires smooth functions)

```
# Starting point (baseline configuration)
x0 = baseline_coded.copy()

# Bounds for all methods (unit cube)
bounds = [(0, 1)] * 9

# Number of function evaluations budget
max_evals = 30

print(f"Starting point: {x0}")
print(f"Starting weight: {baseline_weight:.2f} lb")
```

```
Starting point: [0.48  0.4   0.38  0.5   0.62  0.344 0.4   0.37  0.38 ]
Starting weight: 233.91 lb
```

## 27.4. Method 1: SpotOptim (Surrogate Model Based Optimization)

```
# Start timing
start_time = time.time()

# Configure SpotOptim
optimizer_spot = SpotOptim(
    fun=wingwt,
    bounds=bounds,
    x0=None,
    max_iter=max_evals,
    n_initial=10, # Initial design points
    var_name=['Sw', 'Wfw', 'A', 'L', 'q', 'l', 'Rtc', 'Nz', 'Wdg'],
    acquisition='y', # ei: Expected Improvement
    max_surrogate_points=100,
    seed=42,
```

## 27. Optimizing the Aircraft Wing Weight Example

```
verbose=True,  
tensorboard_log=True,  
tensorboard_clean=True  
)
```

Removed old TensorBoard logs: runs/spotoptim\_20251212\_224946  
Cleaned 1 old TensorBoard log directory  
TensorBoard logging enabled: runs/spotoptim\_20251212\_224958

### 27.5. Design Table

```
pprint.pprint(optimizer_spot.print_design_table())
```

|  | name | type  | lower  | upper  | default | trans |  |
|--|------|-------|--------|--------|---------|-------|--|
|  | Sw   | float | 0.0000 | 1.0000 | 0.5000  | -     |  |
|  | Wfw  | float | 0.0000 | 1.0000 | 0.5000  | -     |  |
|  | A    | float | 0.0000 | 1.0000 | 0.5000  | -     |  |
|  | L    | float | 0.0000 | 1.0000 | 0.5000  | -     |  |
|  | q    | float | 0.0000 | 1.0000 | 0.5000  | -     |  |
|  | l    | float | 0.0000 | 1.0000 | 0.5000  | -     |  |
|  | Rtc  | float | 0.0000 | 1.0000 | 0.5000  | -     |  |
|  | Nz   | float | 0.0000 | 1.0000 | 0.5000  | -     |  |
|  | Wdg  | float | 0.0000 | 1.0000 | 0.5000  | -     |  |

### 27.6. Run optimization

```
result_spot = optimizer_spot.optimize()
```

Initial best:  $f(x) = 212.219714$   
Iteration 1: New best  $f(x) = 142.977744$   
Iteration 2: New best  $f(x) = 140.904175$   
Iteration 3: New best  $f(x) = 120.465251$   
Iteration 4:  $f(x) = 121.575842$   
Iteration 5: New best  $f(x) = 120.101053$   
Iteration 6: New best  $f(x) = 119.997018$   
Iteration 7:  $f(x) = 120.018134$

## 27.6. Run optimization

```
Iteration 8: New best f(x) = 119.958665
Iteration 9: f(x) = 120.386522
Iteration 10: New best f(x) = 119.691338
Iteration 11: New best f(x) = 119.530080
Iteration 12: f(x) = 119.539469
Iteration 13: New best f(x) = 119.503677
Iteration 14: f(x) = 119.503691
Iteration 15: New best f(x) = 119.503672
Iteration 16: f(x) = 119.503718
Iteration 17: f(x) = 119.503685
Iteration 18: f(x) = 119.503693
Iteration 19: f(x) = 119.503726
Iteration 20: f(x) = 119.503749
TensorBoard writer closed. View logs with: tensorboard --logdir=runs/spotoptim_20251212_224958
```

```
# End timing
spot_time = time.time() - start_time

print(f"\nSpotOptim Results:")
print(f"  Best weight: {result_spot.fun:.4f} lb")
print(f"  Function evaluations: {result_spot.nfev}")
print(f"  Time elapsed: {spot_time:.2f} seconds")
print(f"  Success: {result_spot.success}")
```

```
SpotOptim Results:
  Best weight: 119.5037 lb
  Function evaluations: 30
  Time elapsed: 6.29 seconds
  Success: True
```

```
optimizer_spot.print_best()
```

Best Solution Found:

```
-----
Sw: 0.0000
Wfw: 0.0000
A: 0.0000
L: 0.5000
q: 0.0000
l: 0.0000
Rtc: 1.0000
```

## 27. Optimizing the Aircraft Wing Weight Example

```
Nz: 0.0000
Wdg: 0.0000
Objective Value: 119.5037
Total Evaluations: 30
```

### 27.7. Result Table

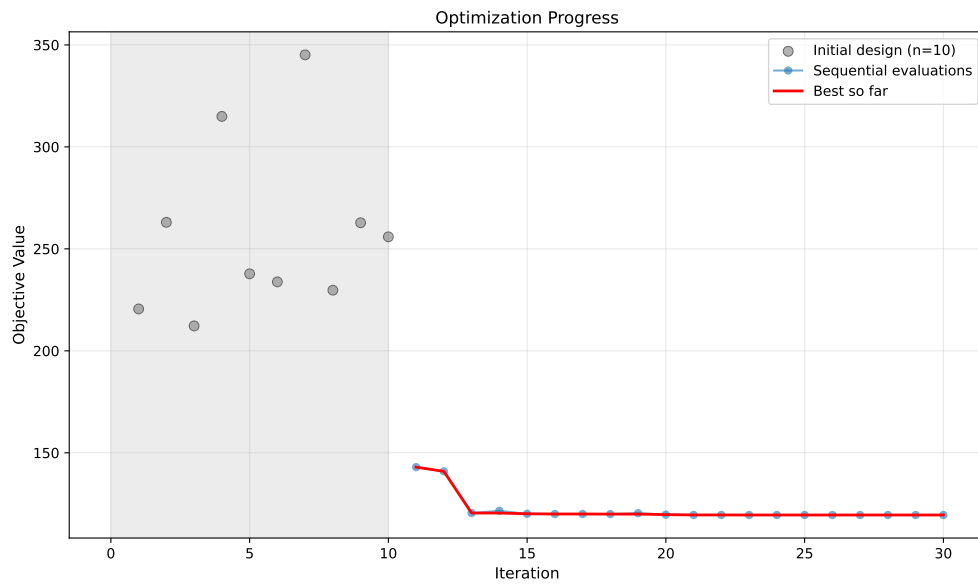
```
pprint.pprint(optimizer_spot.print_results_table(show_importance=True))
```

```
('| name | type | lower | upper | tuned | trans | importance | '
'stars | \n'
'|-----|-----|-----|-----|-----|-----|-----|
|-----| \n'
'| Sw | float | 0.0000 | 1.0000 | 0.0000 | - | 9.52 | '
'* | \n'
'| Wfw | float | 0.0000 | 1.0000 | 0.0000 | - | 12.76 | '
'* | \n'
'| A | float | 0.0000 | 1.0000 | 0.0000 | - | 14.11 | '
'* | \n'
'| L | float | 0.0000 | 1.0000 | 0.5000 | - | 4.30 | '
'* | \n'
'| q | float | 0.0000 | 1.0000 | 0.0000 | - | 6.10 | '
'* | \n'
'| l | float | 0.0000 | 1.0000 | 0.0000 | - | 11.05 | '
'* | \n'
'| Rtc | float | 0.0000 | 1.0000 | 1.0000 | - | 13.87 | '
'* | \n'
'| Nz | float | 0.0000 | 1.0000 | 0.0000 | - | 14.03 | '
'* | \n'
'| Wdg | float | 0.0000 | 1.0000 | 0.0000 | - | 14.27 | '
'* | \n'
'\n'
'Interpretation: ***: >95%, **: >50%, *: >1%, .: >0.1%')
```

### 27.8. Progress of the Optimization

```
optimizer_spot.plot_progress(log_y=False)
```

## 27.9. Contour Plots of Most Important Hyperparameters



## 27.9. Contour Plots of Most Important Hyperparameters

```
optimizer_spot.plot_important_hyperparameter_contour(max_imp=3)
```

Plotting surrogate contours for top 3 most important parameters:

Wdg: importance = 14.27% (type: float)

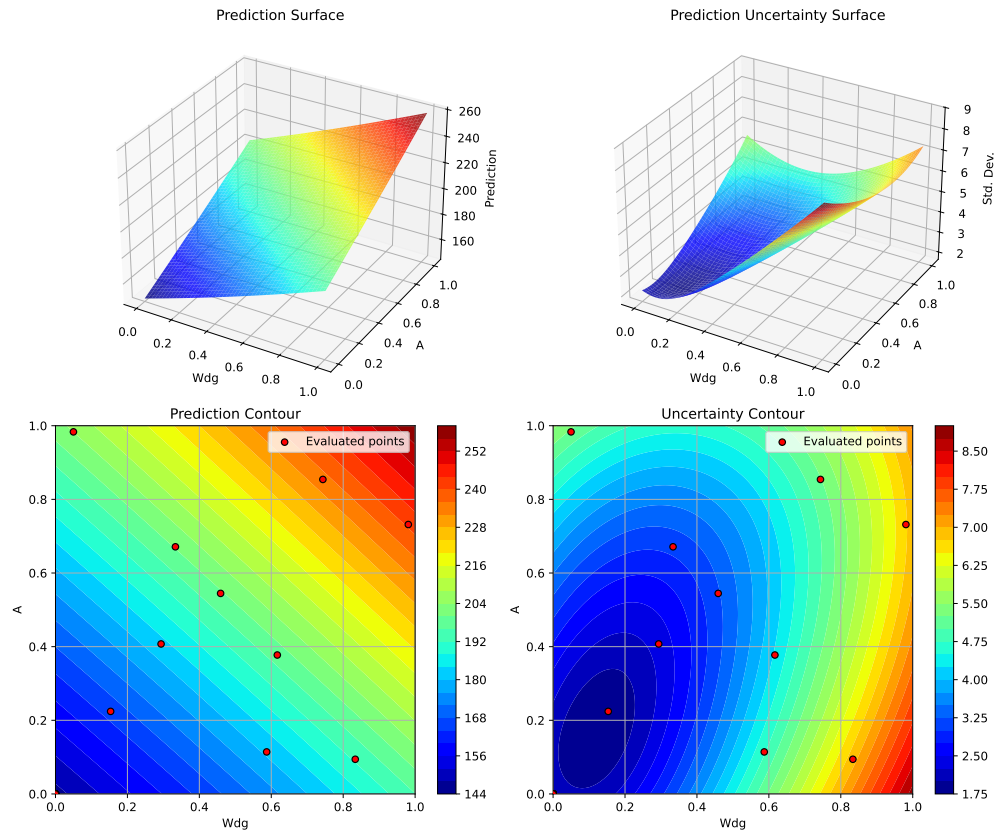
A: importance = 14.11% (type: float)

Nz: importance = 14.03% (type: float)

Generating 3 surrogate plots...

Plotting Wdg vs A

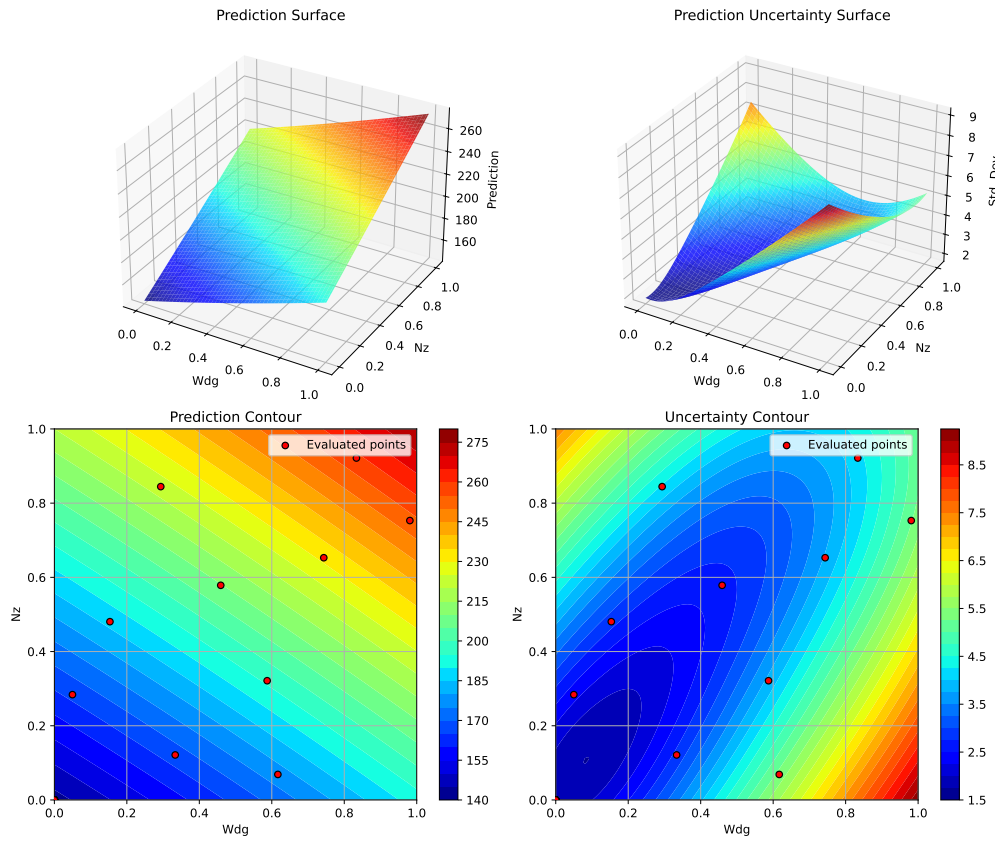
## 27. Optimizing the Aircraft Wing Weight Example



Plotting Wdg vs Nz

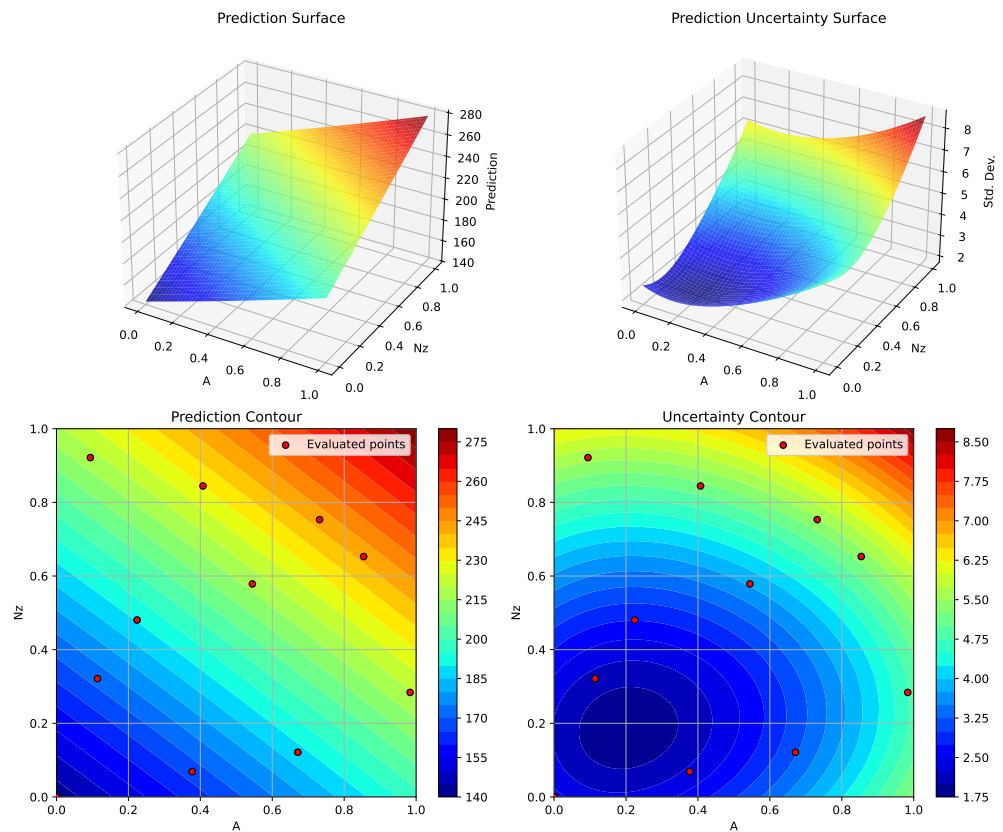


## 27.9. Contour Plots of Most Important Hyperparameters



Plotting A vs Nz

## 27. Optimizing the Aircraft Wing Weight Example



### 27.10. Method 2: Nelder-Mead Simplex

```
print("\n" + "=" * 60)
print("Running Nelder-Mead Simplex...")
print("=" * 60)

# Start timing
start_time = time.time()

# Run optimization
result_nm = minimize(
    wingwt_scipy,
    x0=x0,
    method='Nelder-Mead',
```

### 27.11. Method 3: BFGS (Quasi-Newton)

```
        bounds=bounds,
        options={'maxfev': max_evals, 'disp': True}
    )

    # End timing
    nm_time = time.time() - start_time

    print(f"\nNelder-Mead Results:")
    print(f"   Best weight: {result_nm.fun:.4f} lb")
    print(f"   Function evaluations: {result_nm.nfev}")
    print(f"   Time elapsed: {nm_time:.2f} seconds")
    print(f"   Success: {result_nm.success}")
```

```
=====
Running Nelder-Mead Simplex...
=====
```

```
Nelder-Mead Results:
  Best weight: 220.5449 lb
  Function evaluations: 30
  Time elapsed: 0.00 seconds
  Success: False
```

## 27.11. Method 3: BFGS (Quasi-Newton)

```
print("\n" + "=" * 60)
print("Running BFGS (Quasi-Newton)...")
print("=" * 60)

# Start timing
start_time = time.time()

# Run optimization
result_bfgs = minimize(
    wingwt_scipy,
    x0=x0,
    method='L-BFGS-B', # Bounded BFGS
    bounds=bounds,
    options={'maxfun': max_evals, 'disp': True})
```

## 27. Optimizing the Aircraft Wing Weight Example

```
)

# End timing
bfgs_time = time.time() - start_time

print(f"\nBFGS Results:")
print(f"  Best weight: {result_bfgs.fun:.4f} lb")
print(f"  Function evaluations: {result_bfgs.nfev}")
print(f"  Time elapsed: {bfgs_time:.2f} seconds")
print(f"  Success: {result_bfgs.success}")
```

```
=====
Running BFGS (Quasi-Newton)...
=====
```

```
BFGS Results:
  Best weight: 119.5037 lb
  Function evaluations: 60
  Time elapsed: 0.00 seconds
  Success: False
```

### 27.12. Comparison of Results

```
# Create comparison DataFrame
comparison = pd.DataFrame({
    'Method': ['Baseline', 'SpotOptim', 'Nelder-Mead', 'BFGS'],
    'Best Weight (lb)': [
        baseline_weight,
        result_spot.fun,
        result_nm.fun,
        result_bfgs.fun
    ],
    'Improvement (%)': [
        0.0,
        (baseline_weight - result_spot.fun) / baseline_weight * 100,
        (baseline_weight - result_nm.fun) / baseline_weight * 100,
        (baseline_weight - result_bfgs.fun) / baseline_weight * 100
    ],
    'Function Evals': [
        1,
```

```

        result_spot.nfev,
        result_nm.nfev,
        result_bfgs.nfev
    ],
    'Time (s)': [
        0.0,
        spot_time,
        nm_time,
        bfgs_time
    ],
    'Success': [
        True,
        result_spot.success,
        result_nm.success,
        result_bfgs.success
    ]
})

print("\n" + "=" * 80)
print("OPTIMIZATION COMPARISON")
print("=" * 80)
print(comparison.to_string(index=False))
print("=" * 80)

```

```

=====
OPTIMIZATION COMPARISON
=====

```

| Method      | Best Weight (lb) | Improvement (%) | Function Evals | Time (s) | Success |
|-------------|------------------|-----------------|----------------|----------|---------|
| Baseline    | 233.908405       | 0.000000        | 1              | 0.000000 | True    |
| SpotOptim   | 119.503672       | 48.910057       | 30             | 6.294193 | True    |
| Nelder-Mead | 220.544928       | 5.713124        | 30             | 0.000861 | False   |
| BFGS        | 119.503672       | 48.910057       | 60             | 0.001520 | False   |

```

=====

```

## 27.13. Visualization: Convergence Plots

### 27.13.1. SpotOptim Convergence

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))
```

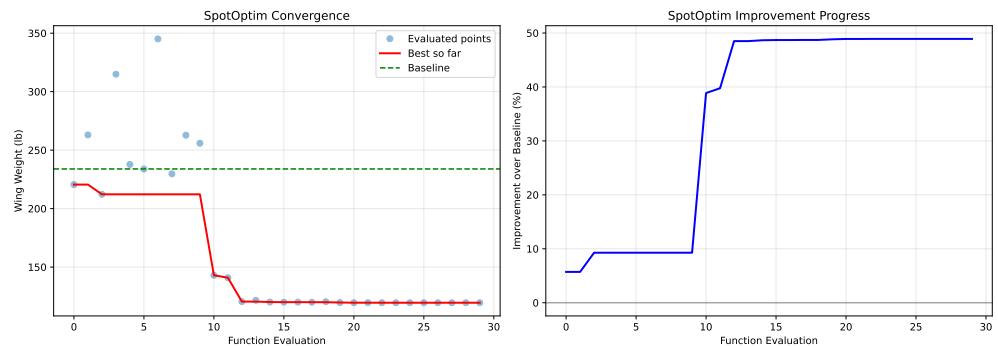
## 27. Optimizing the Aircraft Wing Weight Example

```
# Plot 1: Best value over iterations
y_history = optimizer_spot.y_
best_so_far = np.minimum.accumulate(y_history)

ax1.plot(range(len(y_history)), y_history, 'o', alpha=0.5, label='Evaluated points')
ax1.plot(range(len(best_so_far)), best_so_far, 'r-', linewidth=2, label='Best so far')
ax1.axhline(y=baseline_weight, color='g', linestyle='--', label='Baseline')
ax1.set_xlabel('Function Evaluation')
ax1.set_ylabel('Wing Weight (lb)')
ax1.set_title('SpotOptim Convergence')
ax1.legend()
ax1.grid(True, alpha=0.3)

# Plot 2: Improvement over baseline
improvement = (baseline_weight - best_so_far) / baseline_weight * 100
ax2.plot(range(len(improvement)), improvement, 'b-', linewidth=2)
ax2.set_xlabel('Function Evaluation')
ax2.set_ylabel('Improvement over Baseline (%)')
ax2.set_title('SpotOptim Improvement Progress')
ax2.grid(True, alpha=0.3)
ax2.axhline(y=0, color='k', linestyle='-', alpha=0.3)

plt.tight_layout()
plt.show()
```



### 27.14. Optimal Parameter Values

Let's examine the optimal parameter values found by each method:

```

# Parameter names
param_names = ['Sw', 'Wfw', 'A', 'L', 'q', 'l', 'Rtc', 'Nz', 'Wdg']

# Transform from unit cube to natural scales
def decode_params(x):
    scales = [
        (150, 200),      # Sw
        (220, 300),      # Wfw
        (6, 10),          # A
        (-10, 10),        # L (degrees)
        (16, 45),          # q
        (0.5, 1),          # l
        (0.08, 0.18),     # Rtc
        (2.5, 6),          # Nz
        (1700, 2500)      # Wdg
    ]
    decoded = []
    for i, (low, high) in enumerate(scales):
        decoded.append(x[i] * (high - low) + low)
    return decoded

# Create comparison table
baseline_decoded = decode_params(baseline_coded)
spot_decoded = decode_params(result_spot.x)
nm_decoded = decode_params(result_nm.x)
bfgs_decoded = decode_params(result_bfgs.x)

param_comparison = pd.DataFrame({
    'Parameter': param_names,
    'Baseline': baseline_decoded,
    'SpotOptim': spot_decoded,
    'Nelder-Mead': nm_decoded,
    'BFGS': bfgs_decoded
})

print("\n" + "=" * 100)
print("OPTIMAL PARAMETER VALUES (Natural Scale)")
print("=" * 100)
print(param_comparison.to_string(index=False))
print("=" * 100)

```

```
=====
OPTIMAL PARAMETER VALUES (Natural Scale)
```

## 27. Optimizing the Aircraft Wing Weight Example

| Parameter | Baseline | SpotOptim   | Nelder-Mead | BFGS    |
|-----------|----------|-------------|-------------|---------|
| Sw        | 174.000  | 150.000000  | 173.923947  | 150.00  |
| Wfw       | 252.000  | 220.000000  | 254.534988  | 220.00  |
| A         | 7.520    | 6.000000    | 7.484111    | 6.00    |
| L         | 0.000    | 0.000531    | -0.126137   | 0.00    |
| q         | 33.980   | 16.000000   | 34.156818   | 16.00   |
| l         | 0.672    | 0.500000    | 0.674613    | 0.50    |
| Rtc       | 0.120    | 0.180000    | 0.124872    | 0.18    |
| Nz        | 3.795    | 2.500000    | 3.427058    | 2.50    |
| Wdg       | 2004.000 | 1700.000000 | 2028.938270 | 1700.00 |

### 27.15. Analysis of Optimal Solutions

```
# Calculate percentage changes from baseline
changes_spot = [(spot_decoded[i] - baseline_decoded[i]) / baseline_decoded[i] * 100
                 for i in range(len(param_names))]
changes_nm = [(nm_decoded[i] - baseline_decoded[i]) / baseline_decoded[i] * 100
              for i in range(len(param_names))]
changes_bfgs = [(bfgs_decoded[i] - baseline_decoded[i]) / baseline_decoded[i] * 100
                for i in range(len(param_names))]

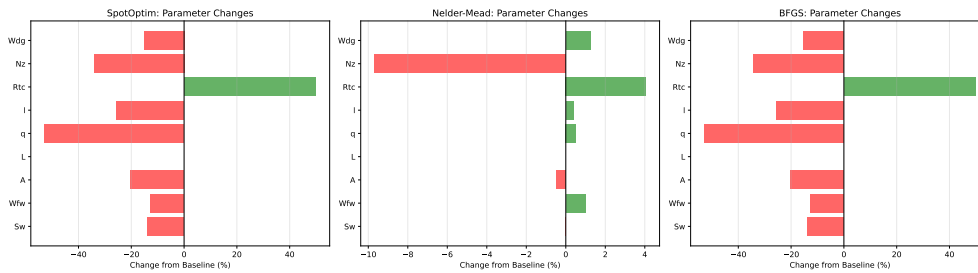
# Visualization
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

for ax, changes, method in zip(axes,
                                [changes_spot, changes_nm, changes_bfgs],
                                ['SpotOptim', 'Nelder-Mead', 'BFGS']):
    colors = ['red' if c < 0 else 'green' for c in changes]
    ax.barh(param_names, changes, color=colors, alpha=0.6)
    ax.axvline(x=0, color='black', linestyle='-', linewidth=0.5)
    ax.set_xlabel('Change from Baseline (%)')
    ax.set_title(f'{method}: Parameter Changes')
    ax.grid(True, alpha=0.3, axis='x')

plt.tight_layout()
plt.show()
```



## 27.16. Key Insights from Optimal Solutions



## 27.16. Key Insights from Optimal Solutions

```
# Find parameters with largest changes for each method
def analyze_changes(decoded, baseline_decoded, method_name):
    changes = {param_names[i]: decoded[i] - baseline_decoded[i]
               for i in range(len(param_names))}
    sorted_changes = sorted(changes.items(), key=lambda x: abs(x[1]), reverse=True)

    print(f"\n{n{method_name}} - Top 5 Parameter Changes:")
    print("-" * 50)
    for param, change in sorted_changes[:5]:
        idx = param_names.index(param)
        pct = change / baseline_decoded[idx] * 100
        print(f" {param:5s}: {change:+8.2f} ({pct:+6.1f}%)")

analyze_changes(spot_decoded, baseline_decoded, "SpotOptim")
analyze_changes(nm_decoded, baseline_decoded, "Nelder-Mead")
analyze_changes(bfgs_decoded, baseline_decoded, "BFGS")
```

SpotOptim - Top 5 Parameter Changes:

```
-----
Wdg  : -304.00 ( -15.2%)
Wfw  : -32.00 ( -12.7%)
Sw   : -24.00 ( -13.8%)
q    : -17.98 ( -52.9%)
A    :  -1.52 ( -20.2%)
```

Nelder-Mead - Top 5 Parameter Changes:

```
-----
Wdg  : +24.94 ( +1.2%)
Wfw  : +2.53 ( +1.0%)
```

## 27. Optimizing the Aircraft Wing Weight Example

```
Nz   :   -0.37 (  -9.7%)
q     :   +0.18 ( +0.5%)
L     :   -0.13 ( -inf%)
```

BFGS - Top 5 Parameter Changes:

```
-----
Wdg   :  -304.00 ( -15.2%)
Wfw   :  -32.00 ( -12.7%)
Sw    :  -24.00 ( -13.8%)
q     :  -17.98 ( -52.9%)
A     :   -1.52 ( -20.2%)
```

## 27.17. Method Efficiency Comparison

```
# Calculate efficiency metrics
efficiency = pd.DataFrame({
    'Method': ['SpotOptim', 'Nelder-Mead', 'BFGS'],
    'Weight Reduction (lb)': [
        baseline_weight - result_spot.fun,
        baseline_weight - result_nm.fun,
        baseline_weight - result_bfgs.fun
    ],
    'Evals to Best': [
        np.argmin(optimizer_spot.y_) + 1,
        result_nm.nfev,
        result_bfgs.nfev
    ],
    'Time per Eval (ms)': [
        spot_time / result_spot.nfev * 1000,
        nm_time / result_nm.nfev * 1000,
        bfgs_time / result_bfgs.nfev * 1000
    ]
})

print("\n" + "=" * 80)
print("METHOD EFFICIENCY METRICS")
print("=" * 80)
print(efficiency.to_string(index=False))
print("=" * 80)
```

## METHOD EFFICIENCY METRICS

| Method      | Weight Reduction (lb) | Evals to Best | Time per Eval (ms) |
|-------------|-----------------------|---------------|--------------------|
| SpotOptim   | 114.404734            | 25            | 209.806434         |
| Nelder-Mead | 13.363478             | 30            | 0.028706           |
| BFGS        | 114.404734            | 60            | 0.025332           |

## 27.18. Visualization: 2D Slices of Optimal Solutions

Let's visualize how the optimal solutions compare in the most important 2D subspaces:

```
# Create 2D slices showing optimal points
fig, axes = plt.subplots(2, 2, figsize=(14, 12))

# Important parameter pairs based on sensitivity analysis
pairs = [
    (7, 2), # Nz vs A (load factor vs aspect ratio)
    (0, 8), # Sw vs Wdg (wing area vs gross weight)
    (7, 0), # Nz vs Sw (load factor vs wing area)
    (2, 6)  # A vs Rtc (aspect ratio vs thickness ratio)
]

for ax, (i, j) in zip(axes.flat, pairs):
    # Create meshgrid for contour plot
    x_range = np.linspace(0, 1, 50)
    y_range = np.linspace(0, 1, 50)
    X, Y = np.meshgrid(x_range, y_range)

    # Evaluate function on grid (fixing other parameters at baseline)
    Z = np.zeros_like(X)
    for ii in range(X.shape[0]):
        for jj in range(X.shape[1]):
            point = baseline_coded.copy()
            point[i] = X[ii, jj]
            point[j] = Y[ii, jj]
            Z[ii, jj] = wingwt(point)[0]

    # Plot contours
    contour = ax.contourf(X, Y, Z, levels=20, cmap='viridis', alpha=0.6)
    ax.contour(X, Y, Z, levels=10, colors='black', alpha=0.3, linewidths=0.5)

# Plot optimal points
```

## 27. Optimizing the Aircraft Wing Weight Example

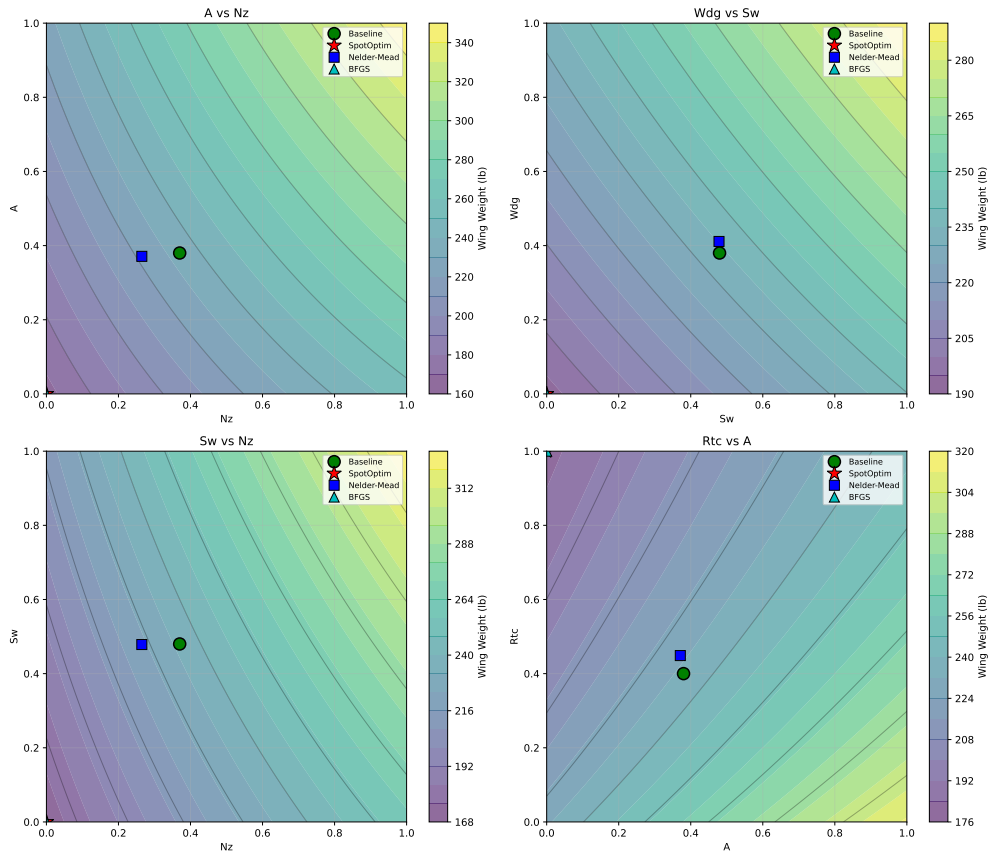
```
ax.plot(baseline_coded[i], baseline_coded[j], 'go', markersize=12,
        label='Baseline', markeredgewidth=1.5)
ax.plot(result_spot.x[i], result_spot.x[j], 'r*', markersize=15,
        label='SpotOptim', markeredgewidth=1)
ax.plot(result_nm.x[i], result_nm.x[j], 'bs', markersize=10,
        label='Nelder-Mead', markeredgewidth=1)
ax.plot(result_bfgs.x[i], result_bfgs.x[j], 'c^', markersize=10,
        label='BFGS', markeredgewidth=1)

ax.set_xlabel(param_names[i])
ax.set_ylabel(param_names[j])
ax.set_title(f'{param_names[j]} vs {param_names[i]}')
ax.legend(loc='best', fontsize=8)
ax.grid(True, alpha=0.3)

# Add colorbar
plt.colorbar(contour, ax=ax, label='Wing Weight (lb)')

plt.tight_layout()
plt.show()
```

## 27.19. Conclusion



## 27.19. Conclusion

This analysis demonstrates the application of different optimization methods to the Aircraft Wing Weight Example. Key takeaways:

1. **SpotOptim** provides efficient global optimization with good exploration of the design space
2. **Nelder-Mead** offers robust derivative-free optimization but may require more evaluations
3. **BFGS** converges quickly for smooth problems but can get trapped in local minima

For aircraft design problems with expensive simulations, Bayesian optimization (SpotOptim) offers the best balance of efficiency and solution quality, making it particularly suitable for real-world engineering applications.

## 27.20. Jupyter Notebook

### Note

- The Jupyter-Notebook of this lecture is available on GitHub in the Hyperparameter-Tuning-Cookbook Repository

## 28. Surrogate Model Selection in SpotOptim

### Note

- This section demonstrates how to select and configure different surrogate models in SpotOptim
- We compare various surrogate options:
  - **Gaussian Process** with different kernels (Matern, RBF, Rational Quadratic)
  - **SpotOptim Kriging** model
  - **Random Forest** regressor
  - **XGBoost** regressor
  - **Support Vector Regression** (SVR)
  - **Gradient Boosting** regressor
- All methods are evaluated on the Aircraft Wing Weight Example (AWWE) function
- We visualize the fitted surrogates and compare optimization performance

### 28.1. Introduction

Surrogate models are the heart of Bayesian optimization. SpotOptim supports any scikit-learn compatible regressor, allowing you to choose the surrogate that best fits your problem characteristics:

- **Gaussian Processes:** Provide uncertainty estimates, smooth interpolation, good for continuous functions. Support Expected Improvement (EI) acquisition.
- **Kriging:** Similar to GP but with customizable correlation functions. Supports EI acquisition.
- **Random Forests:** Robust to noise, handle discontinuities. Don't provide uncertainty, so use `acquisition='y'` (greedy).
- **XGBoost:** Excellent for high-dimensional problems, fast training and prediction. Use `acquisition='y'`.

## 28. Surrogate Model Selection in SpotOptim

- **SVR:** Good for high-dimensional problems with smooth structure. Use `acquisition='y'`.
- **Gradient Boosting:** Strong performance on structured problems. Use `acquisition='y'`.

### ! Acquisition Functions and Uncertainty

Models that provide uncertainty estimates (Gaussian Process, Kriging) work with all acquisition functions: 'ei' (Expected Improvement), 'pi' (Probability of Improvement), and 'y' (greedy).

Tree-based and other models (Random Forest, XGBoost, SVR, Gradient Boosting) don't provide uncertainty estimates by default, so they should use `acquisition='y'` for greedy optimization. SpotOptim automatically handles this gracefully.

## 28.2. Setup and Imports

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from spotoptim import SpotOptim
from spotoptim.surrogate import Kriging
import time
import warnings
warnings.filterwarnings('ignore')

# Scikit-learn models
from sklearn.gaussian_process import GaussianProcessRegressor
from sklearn.gaussian_process.kernels import (
    Matern, RBF, ConstantKernel, WhiteKernel, RationalQuadratic
)
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.svm import SVR
```

- XGBoost (if available)

```
try:
    import xgboost as xgb
    XGBOOST_AVAILABLE = True
except ImportError:
    XGBOOST_AVAILABLE = False
```



```
print("XGBoost not available. Install with: pip install xgboost")

print("Libraries imported successfully!")
```

Libraries imported successfully!

## 28.3. The AWE Objective Function

We use the Aircraft Wing Weight Example function, which models the weight of an unpainted light aircraft wing. The function accepts inputs in the unit cube  $[0, 1]^9$  and returns the wing weight in pounds.

```
def wingwt(x):
    """
    Aircraft Wing Weight function.

    Args:
        x: array-like of 9 values in [0,1]
           [Sw, Wfw, A, L, q, l, Rtc, Nz, Wdg]

    Returns:
        Wing weight (scalar)
    """
    # Ensure x is a 2D array for batch evaluation
    x = np.atleast_2d(x)

    # Transform from unit cube to natural scales
    Sw = x[:, 0] * (200 - 150) + 150      # Wing area (ft2)
    Wfw = x[:, 1] * (300 - 220) + 220      # Fuel weight (lb)
    A = x[:, 2] * (10 - 6) + 6             # Aspect ratio
    L = (x[:, 3] * (10 - (-10)) - 10) * np.pi/180 # Sweep angle (rad)
    q = x[:, 4] * (45 - 16) + 16           # Dynamic pressure (lb/ft2)
    l = x[:, 5] * (1 - 0.5) + 0.5         # Taper ratio
    Rtc = x[:, 6] * (0.18 - 0.08) + 0.08  # Root thickness/chord
    Nz = x[:, 7] * (6 - 2.5) + 2.5        # Ultimate load factor
    Wdg = x[:, 8] * (2500 - 1700) + 1700  # Design gross weight (lb)

    # Calculate weight on natural scale
    W = 0.036 * Sw**0.758 * Wfw**0.0035 * (A/np.cos(L)**2)**0.6 * q**0.006
    W = W * l**0.04 * (100*Rtc/np.cos(L))**(-0.3) * (Nz*Wdg)**(0.49)

    return W.ravel()
```

```
# Problem setup
bounds = [(0, 1)] * 9
param_names = ['Sw', 'Wfw', 'A', 'L', 'q', 'l', 'Rtc', 'Nz', 'Wdg']
max_iter = 30
n_initial = 10
seed = 42

print(f"Problem dimension: {len(bounds)}")
print(f"Optimization budget: {max_iter} evaluations")
```

```
Problem dimension: 9
Optimization budget: 30 evaluations
```

## 28.4. 1. Default Surrogate: Gaussian Process with Matern Kernel

SpotOptim's default surrogate is a Gaussian Process with a Matern kernel ( $\nu=2.5$ ), which provides twice-differentiable sample paths and good performance for most optimization problems.

```
print("=" * 80)
print("1. DEFAULT: Gaussian Process with Matern  $\nu=2.5$  Kernel")
print("=" * 80)

start_time = time.time()

# Default GP (no surrogate specified)
optimizer_default = SpotOptim(
    fun=wingwt,
    bounds=bounds,
    max_iter=max_iter,
    n_initial=n_initial,
    var_name=param_names,
    acquisition='ei',
    seed=seed,
    verbose=False
)

result_default = optimizer_default.optimize()
time_default = time.time() - start_time
```

## 28.4. 1. Default Surrogate: Gaussian Process with Matern Kernel

```
print(f"\nResults:")
print(f"  Best weight: {result_default.fun:.4f} lb")
print(f"  Function evaluations: {result_default.nfev}")
print(f"  Time: {time_default:.2f}s")
print(f"  Success: {result_default.success}")

# Store for comparison
results_comparison = [{
    'Surrogate': 'GP Matern =2.5 (Default)',
    'Best Weight': result_default.fun,
    'Evaluations': result_default.nfev,
    'Time (s)': time_default,
    'Success': result_default.success
}]
```

```
=====
1. DEFAULT: Gaussian Process with Matern =2.5 Kernel
=====
```

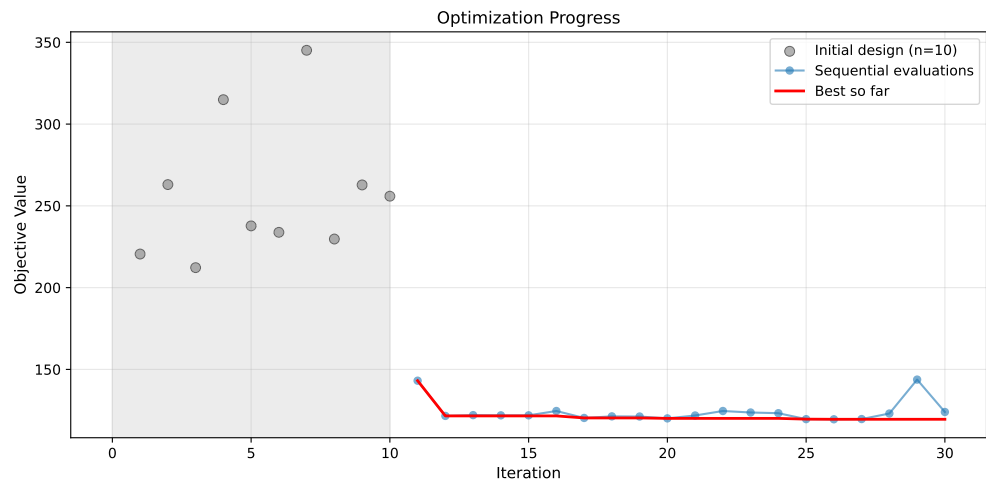
Results:

```
Best weight: 119.5041 lb
Function evaluations: 30
Time: 42.92s
Success: True
```

### 28.4.1. Visualization: Default Surrogate

```
# Plot convergence
optimizer_default.plot_progress(log_y=False, figsize=(10, 5))
```

## 28. Surrogate Model Selection in SpotOptim



```
# Plot most important hyperparameters
optimizer_default.plot_important_hyperparameter_contour(max_imp=3)
plt.suptitle('Default GP Matern =2.5: Most Important Parameters', y=1.02)
plt.show()
```

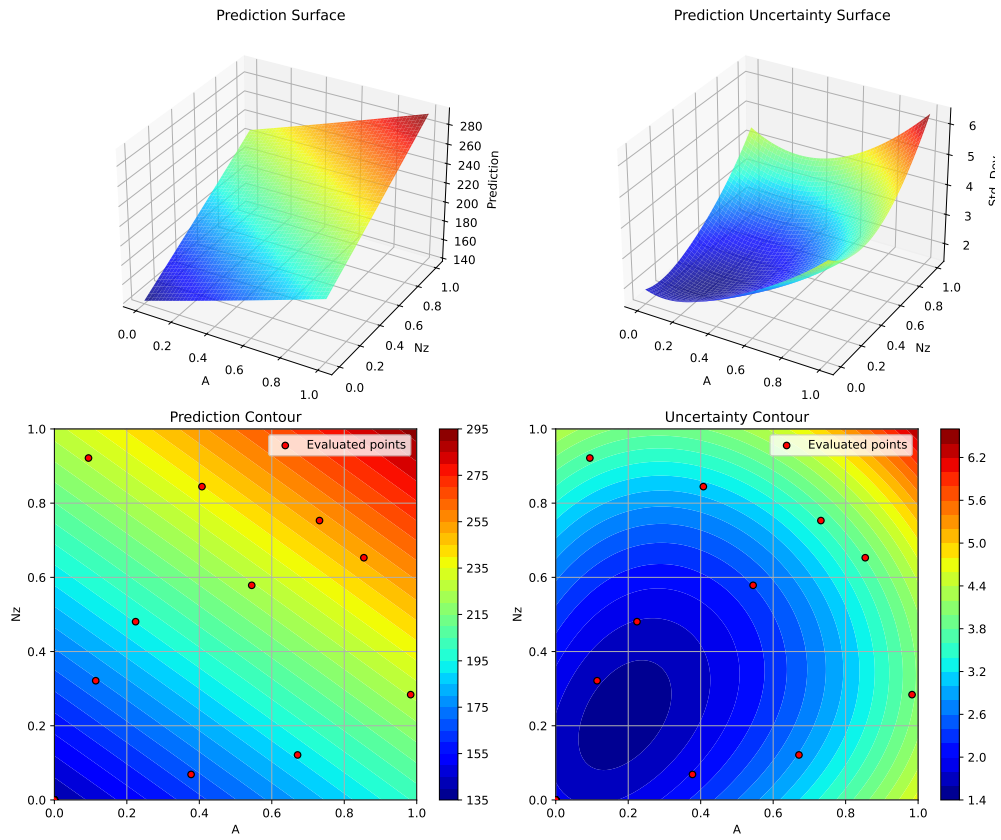
Plotting surrogate contours for top 3 most important parameters:

A: importance = 19.62% (type: float)  
Nz: importance = 19.50% (type: float)  
Rtc: importance = 19.29% (type: float)

Generating 3 surrogate plots...

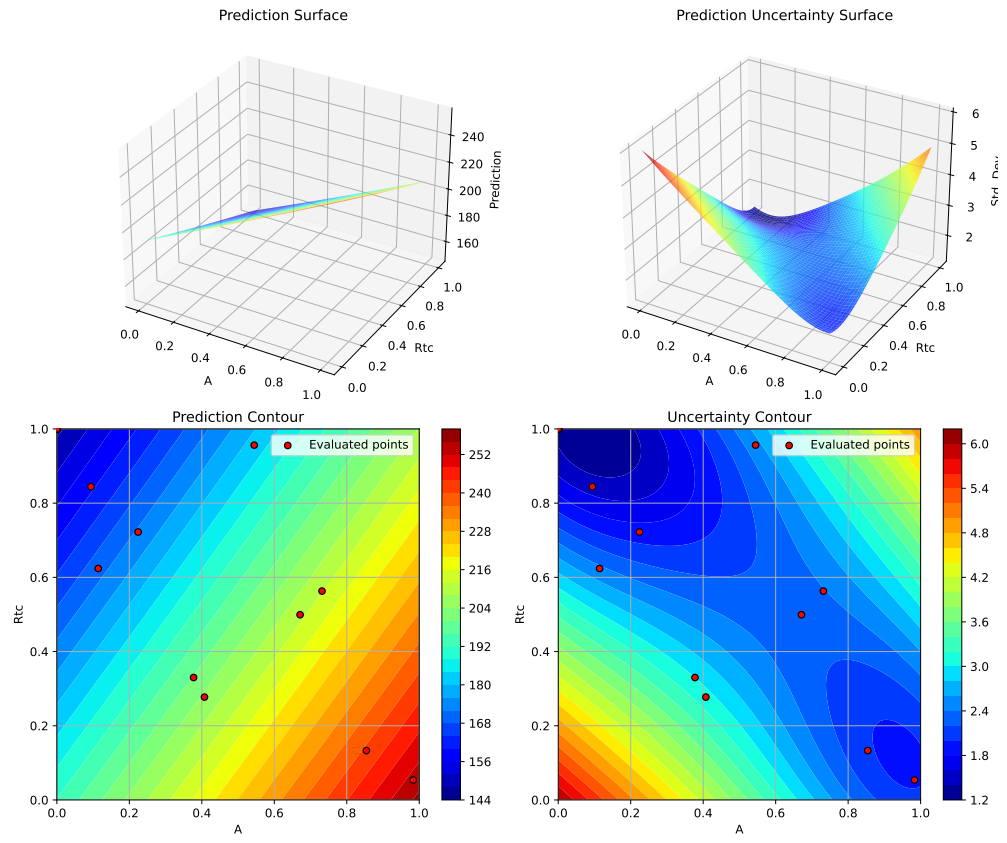
Plotting A vs Nz

### 28.4. 1. Default Surrogate: Gaussian Process with Matern Kernel



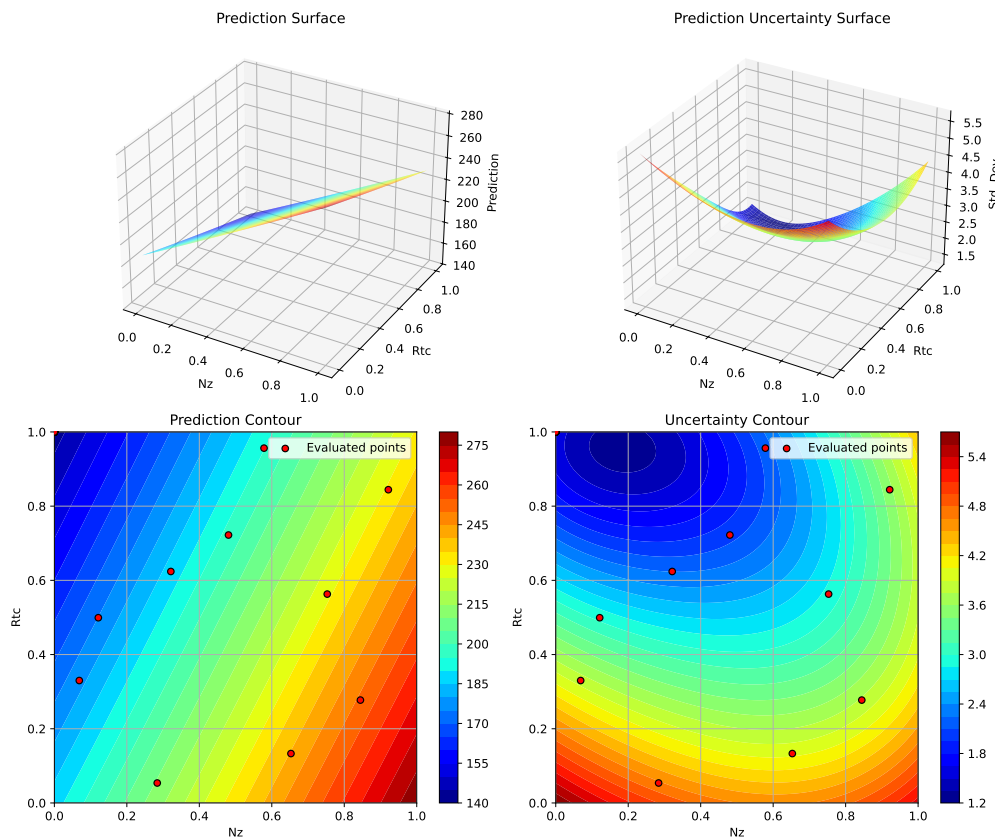
Plotting  $A$  vs  $Rtc$

## 28. Surrogate Model Selection in SpotOptim



Plotting  $N_z$  vs  $R_{tc}$

## 28.5. 2. Gaussian Process with RBF (Radial Basis Function) Kernel



<Figure size 1650x1050 with 0 Axes>

## 28.5. 2. Gaussian Process with RBF (Radial Basis Function) Kernel

The RBF kernel (also called squared exponential) produces infinitely differentiable sample paths, resulting in very smooth predictions.

```
print("=" * 80)
print("2. Gaussian Process with RBF Kernel")
print("=" * 80)

start_time = time.time()
```

## 28. Surrogate Model Selection in SpotOptim

```
# Configure GP with RBF kernel
kernel_rbf = ConstantKernel(1.0, (1e-3, 1e3)) * RBF(
    length_scale=1.0,
    length_scale_bounds=(1e-2, 1e2)
)

gp_rbf = GaussianProcessRegressor(
    kernel=kernel_rbf,
    n_restarts_optimizer=10,
    normalize_y=True,
    random_state=seed
)

optimizer_rbf = SpotOptim(
    fun=wingwt,
    bounds=bounds,
    surrogate=gp_rbf,
    max_iter=max_iter,
    n_initial=n_initial,
    var_name=param_names,
    acquisition='ei',
    seed=seed,
    verbose=False
)

result_rbf = optimizer_rbf.optimize()
time_rbf = time.time() - start_time

print(f"\nResults:")
print(f"  Best weight: {result_rbf.fun:.4f} lb")
print(f"  Function evaluations: {result_rbf.nfev}")
print(f"  Time: {time_rbf:.2f}s")
print(f"  Success: {result_rbf.success}")

results_comparison.append({
    'Surrogate': 'GP RBF',
    'Best Weight': result_rbf.fun,
    'Evaluations': result_rbf.nfev,
    'Time (s)': time_rbf,
    'Success': result_rbf.success
})
```

---

### 2. Gaussian Process with RBF Kernel



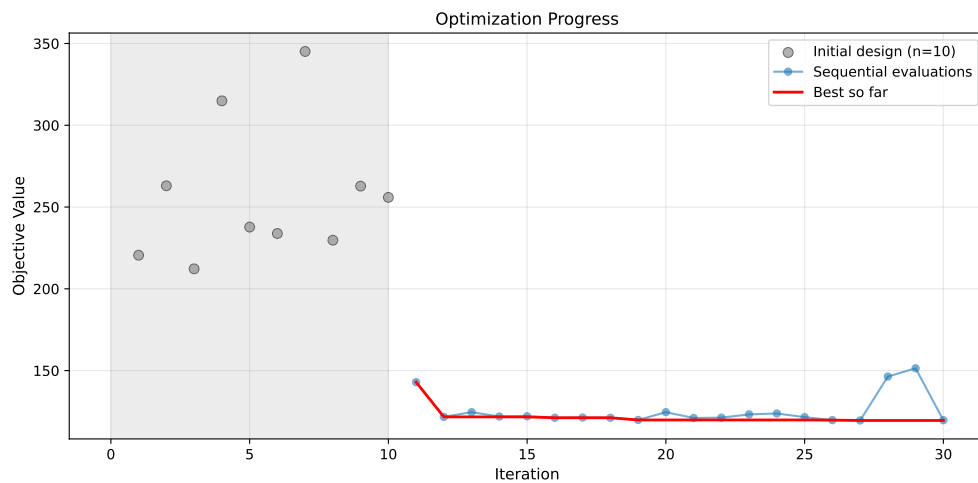
## 28.5. 2. Gaussian Process with RBF (Radial Basis Function) Kernel

Results:

Best weight: 119.5037 lb  
Function evaluations: 30  
Time: 48.77s  
Success: True

### 28.5.1. Visualization: RBF Kernel

```
optimizer_rbf.plot_progress(log_y=False, figsize=(10, 5))
```



```
optimizer_rbf.plot_important_hyperparameter_contour(max_imp=3)
plt.suptitle('GP RBF Kernel: Most Important Parameters', y=1.02)
plt.show()
```

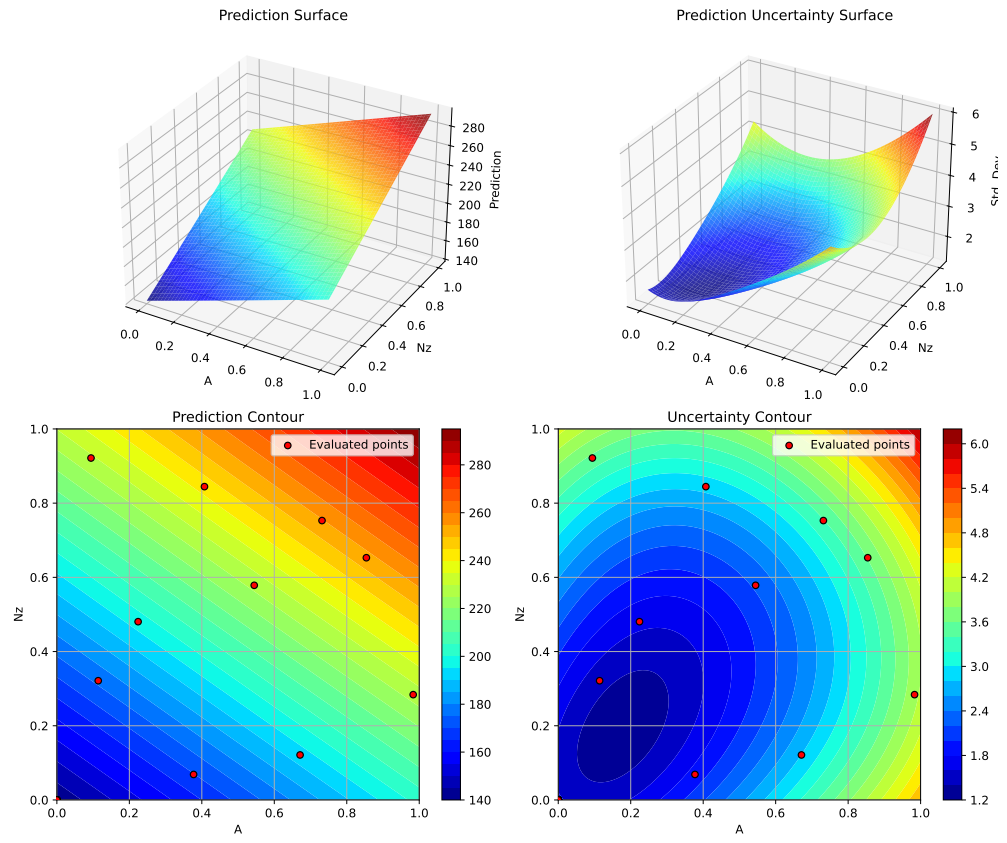
Plotting surrogate contours for top 3 most important parameters:

A: importance = 19.40% (type: float)  
Nz: importance = 19.27% (type: float)  
Rtc: importance = 19.06% (type: float)

Generating 3 surrogate plots...

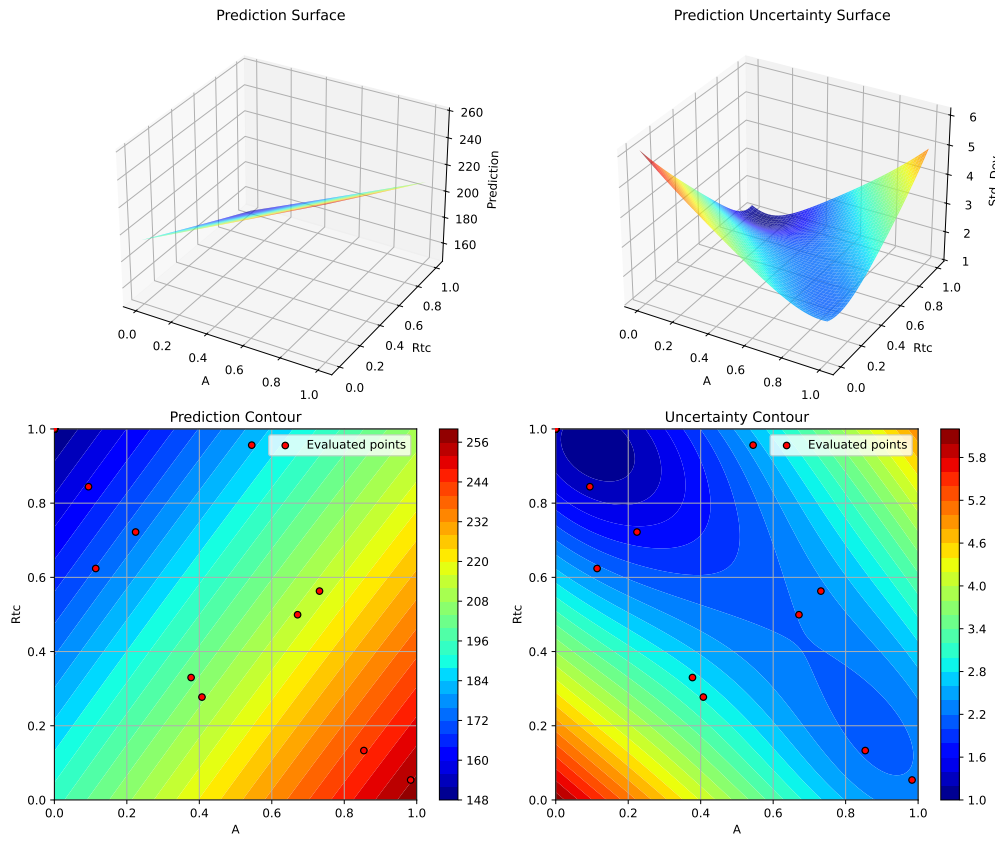
Plotting A vs Nz

## 28. Surrogate Model Selection in SpotOptim



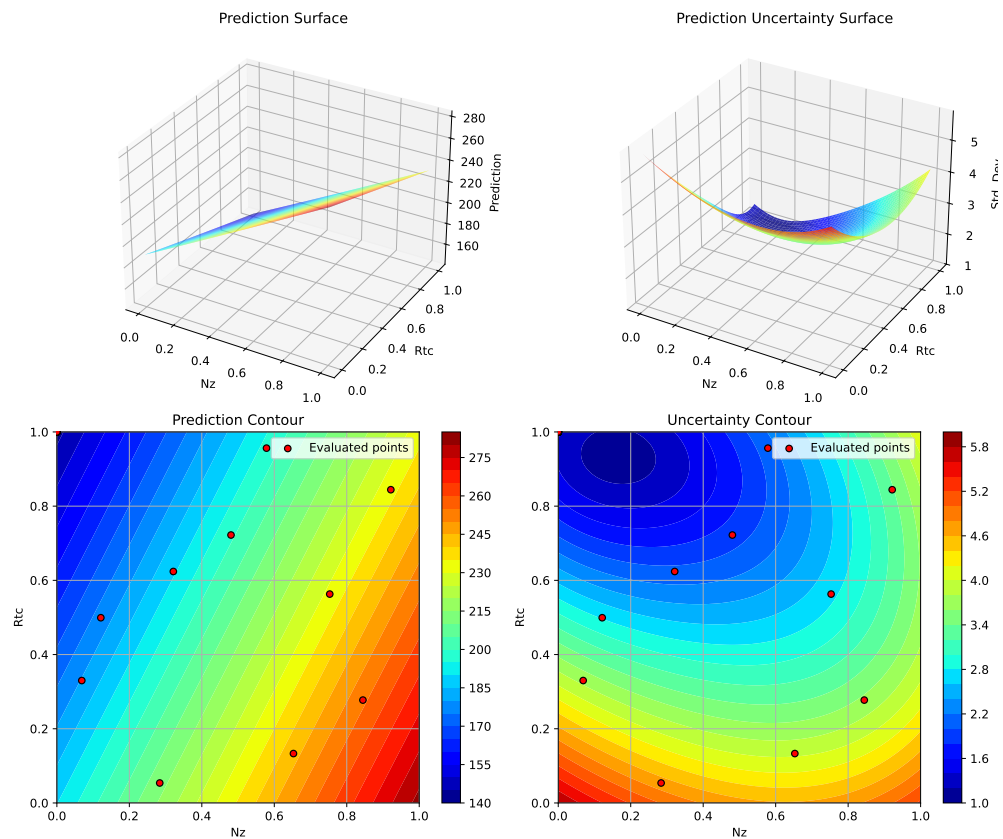
Plotting  $A$  vs  $R_{tc}$

## 28.5. 2. Gaussian Process with RBF (Radial Basis Function) Kernel



Plotting  $N_z$  vs  $R_{tc}$

## 28. Surrogate Model Selection in SpotOptim



<Figure size 1650x1050 with 0 Axes>

### 28.6. 3. Gaussian Process with Matern $\nu=1.5$ Kernel

The Matern  $\nu=1.5$  kernel produces once-differentiable sample paths, allowing for more flexible (less smooth) fits than  $\nu=2.5$ .

```
print("=" * 80)
print("3. Gaussian Process with Matern  $\nu=1.5$  Kernel")
print("=" * 80)

start_time = time.time()

# Configure GP with Matern  $\nu=1.5$ 
kernel_matern15 = ConstantKernel(1.0, (1e-3, 1e3)) * Matern(
```

```

        length_scale=1.0,
        length_scale_bounds=(1e-2, 1e2),
        nu=1.5
    )

    gp_matern15 = GaussianProcessRegressor(
        kernel=kernel_matern15,
        n_restarts_optimizer=10,
        normalize_y=True,
        random_state=seed
    )

    optimizer_matern15 = SpotOptim(
        fun=wingwt,
        bounds=bounds,
        surrogate=gp_matern15,
        max_iter=max_iter,
        n_initial=n_initial,
        var_name=param_names,
        acquisition='ei',
        seed=seed,
        verbose=False
    )

    result_matern15 = optimizer_matern15.optimize()
    time_matern15 = time.time() - start_time

    print(f"\nResults:")
    print(f"  Best weight: {result_matern15.fun:.4f} lb")
    print(f"  Function evaluations: {result_matern15.nfev}")
    print(f"  Time: {time_matern15:.2f}s")
    print(f"  Success: {result_matern15.success}")

    results_comparison.append({
        'Surrogate': 'GP Matern  $\nu=1.5$ ',
        'Best Weight': result_matern15.fun,
        'Evaluations': result_matern15.nfev,
        'Time (s)': time_matern15,
        'Success': result_matern15.success
    })

```

```

=====
3. Gaussian Process with Matern  $\nu=1.5$  Kernel
=====

```

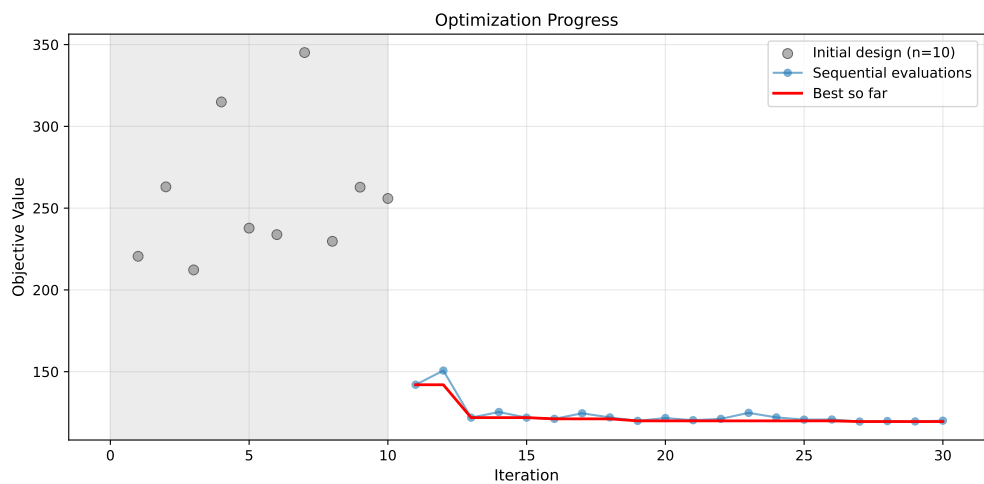
## 28. Surrogate Model Selection in SpotOptim

Results:

Best weight: 119.5056 lb  
Function evaluations: 30  
Time: 41.16s  
Success: True

### 28.6.1. Visualization: Matern $\nu=1.5$ Kernel

```
optimizer_matern15.plot_progress(log_y=False, figsize=(10, 5))
```



```
optimizer_matern15.plot_important_hyperparameter_contour(max_imp=3)  
plt.suptitle('GP Matern  $\nu=1.5$  Kernel: Most Important Parameters', y=1.02)  
plt.show()
```

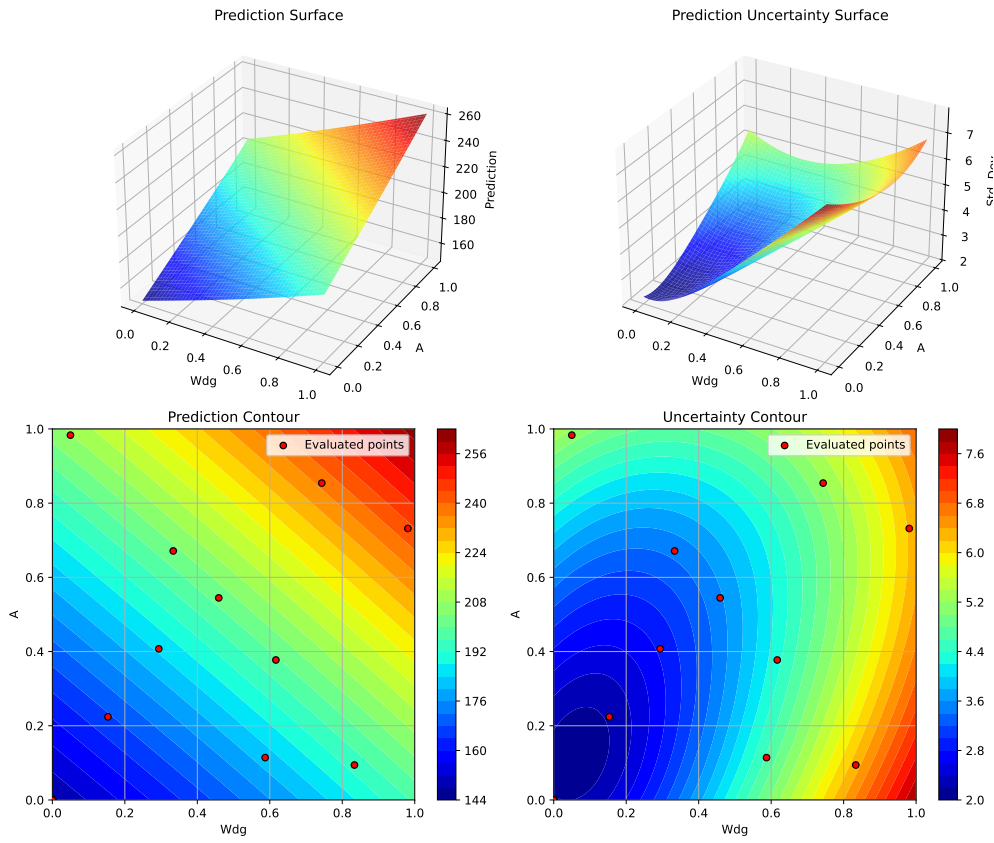
Plotting surrogate contours for top 3 most important parameters:

Wdg: importance = 18.62% (type: float)  
A: importance = 18.41% (type: float)  
Nz: importance = 18.30% (type: float)

Generating 3 surrogate plots...

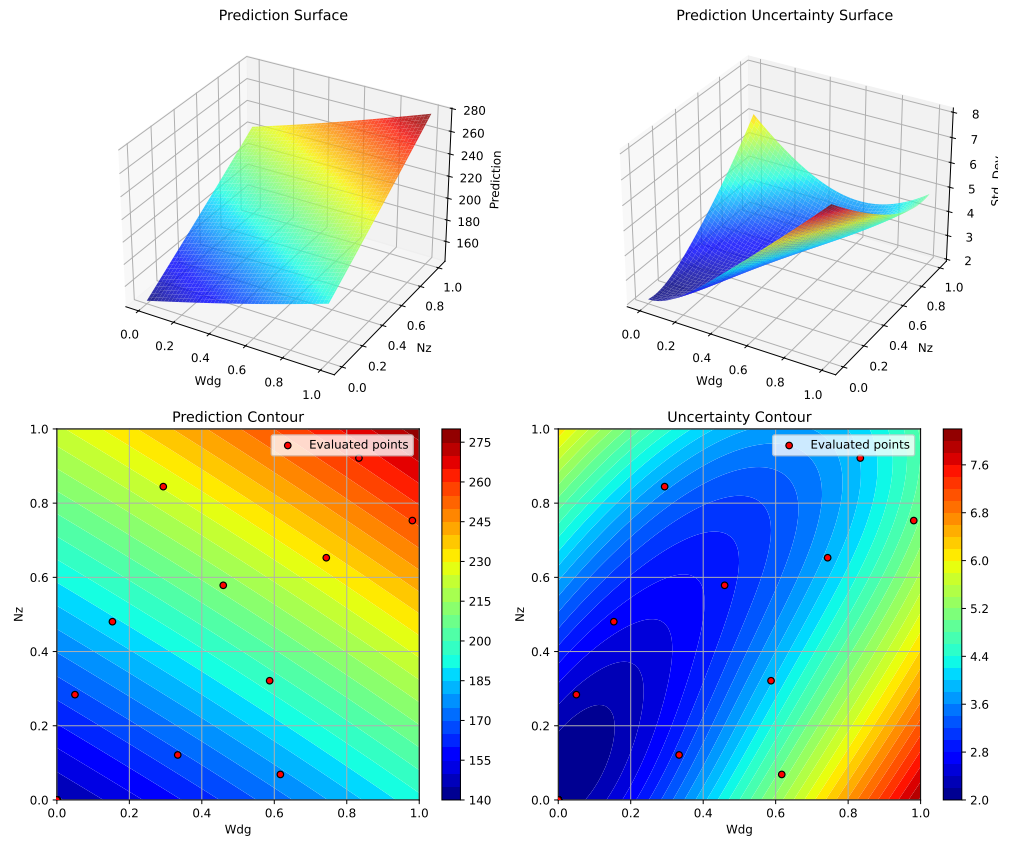
Plotting Wdg vs A

28.6. 3. Gaussian Process with Matern  $\nu=1.5$  Kernel



Plotting Wdg vs Nz

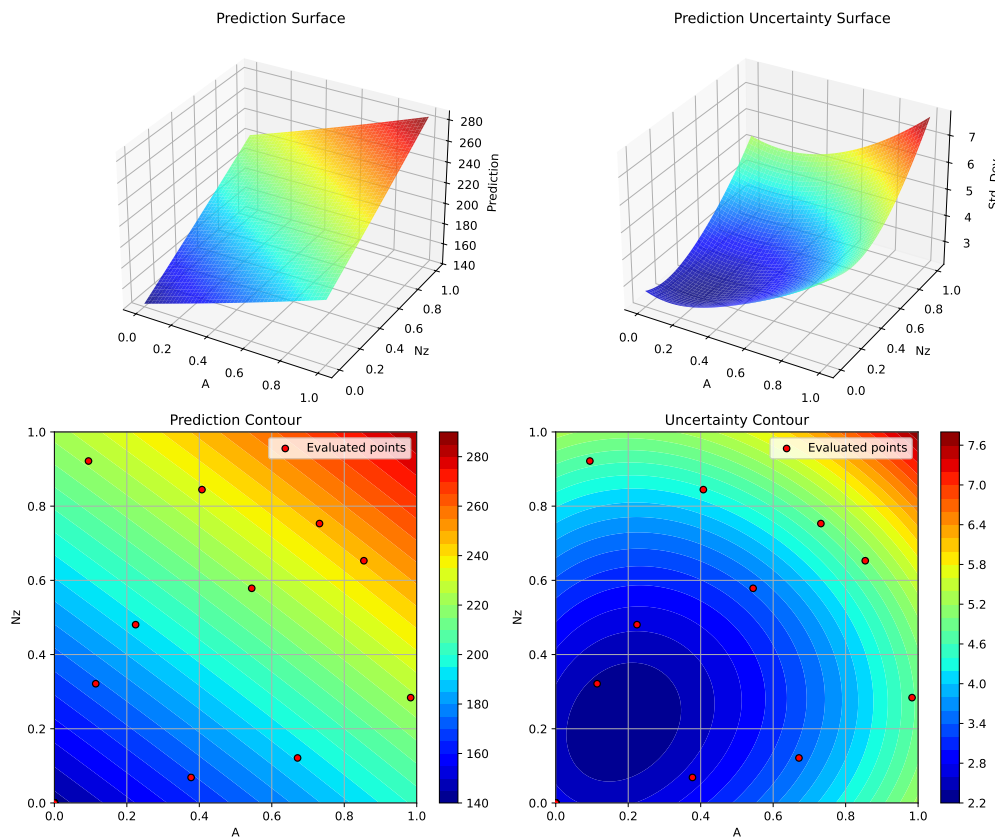
## 28. Surrogate Model Selection in SpotOptim



Plotting A vs Nz



## 28.7. 4. Gaussian Process with Rational Quadratic Kernel



<Figure size 1650x1050 with 0 Axes>

## 28.7. 4. Gaussian Process with Rational Quadratic Kernel

The Rational Quadratic kernel is a scale mixture of RBF kernels with different length scales, providing more flexibility than a single RBF.

```
print("=" * 80)
print("4. Gaussian Process with Rational Quadratic Kernel")
print("=" * 80)

start_time = time.time()
```

## 28. Surrogate Model Selection in SpotOptim

```
# Configure GP with Rational Quadratic kernel
kernel_rq = ConstantKernel(1.0, (1e-3, 1e3)) * RationalQuadratic(
    length_scale=1.0,
    alpha=1.0,
    length_scale_bounds=(1e-2, 1e2),
    alpha_bounds=(1e-2, 1e2)
)

gp_rq = GaussianProcessRegressor(
    kernel=kernel_rq,
    n_restarts_optimizer=10,
    normalize_y=True,
    random_state=seed
)

optimizer_rq = SpotOptim(
    fun=wingwt,
    bounds=bounds,
    surrogate=gp_rq,
    max_iter=max_iter,
    n_initial=n_initial,
    var_name=param_names,
    acquisition='ei',
    seed=seed,
    verbose=False
)

result_rq = optimizer_rq.optimize()
time_rq = time.time() - start_time

print(f"\nResults:")
print(f"  Best weight: {result_rq.fun:.4f} lb")
print(f"  Function evaluations: {result_rq.nfev}")
print(f"  Time: {time_rq:.2f}s")
print(f"  Success: {result_rq.success}")

results_comparison.append({
    'Surrogate': 'GP Rational Quadratic',
    'Best Weight': result_rq.fun,
    'Evaluations': result_rq.nfev,
    'Time (s)': time_rq,
    'Success': result_rq.success
})
```

---

4. Gaussian Process with Rational Quadratic Kernel

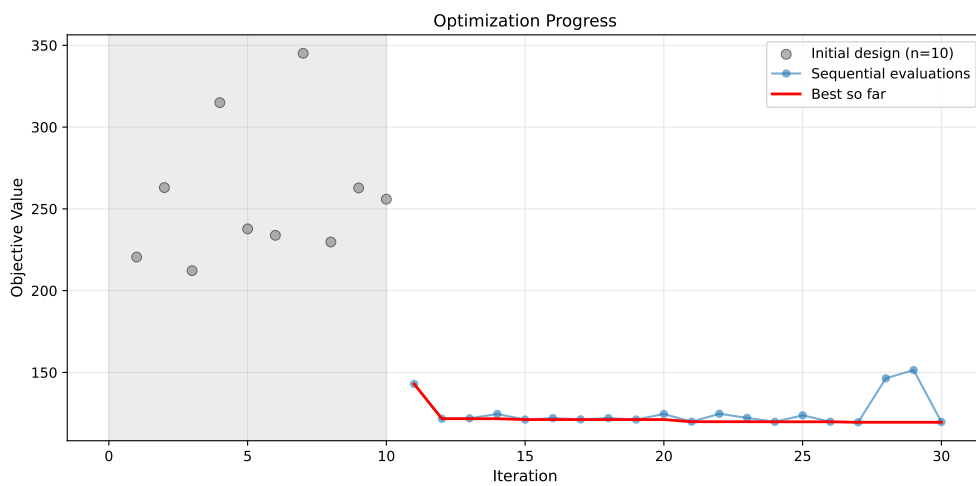
---

Results:

Best weight: 119.5061 lb  
Function evaluations: 30  
Time: 47.90s  
Success: True

## 28.7.1. Visualization: Rational Quadratic Kernel

```
optimizer_rq.plot_progress(log_y=False, figsize=(10, 5))
```



```
optimizer_rq.plot_important_hyperparameter_contour(max_imp=3)
plt.suptitle('GP Rational Quadratic Kernel: Most Important Parameters', y=1.02)
plt.show()
```

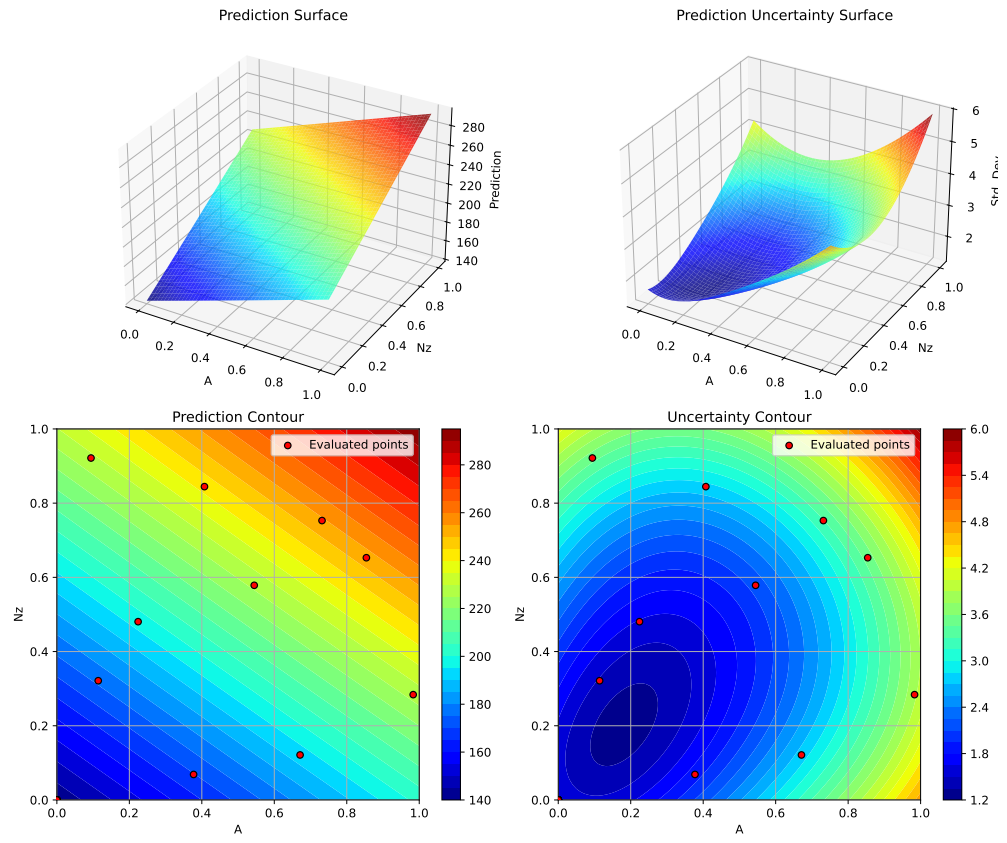
Plotting surrogate contours for top 3 most important parameters:

A: importance = 18.94% (type: float)  
Nz: importance = 18.82% (type: float)  
Rtc: importance = 18.61% (type: float)

Generating 3 surrogate plots...

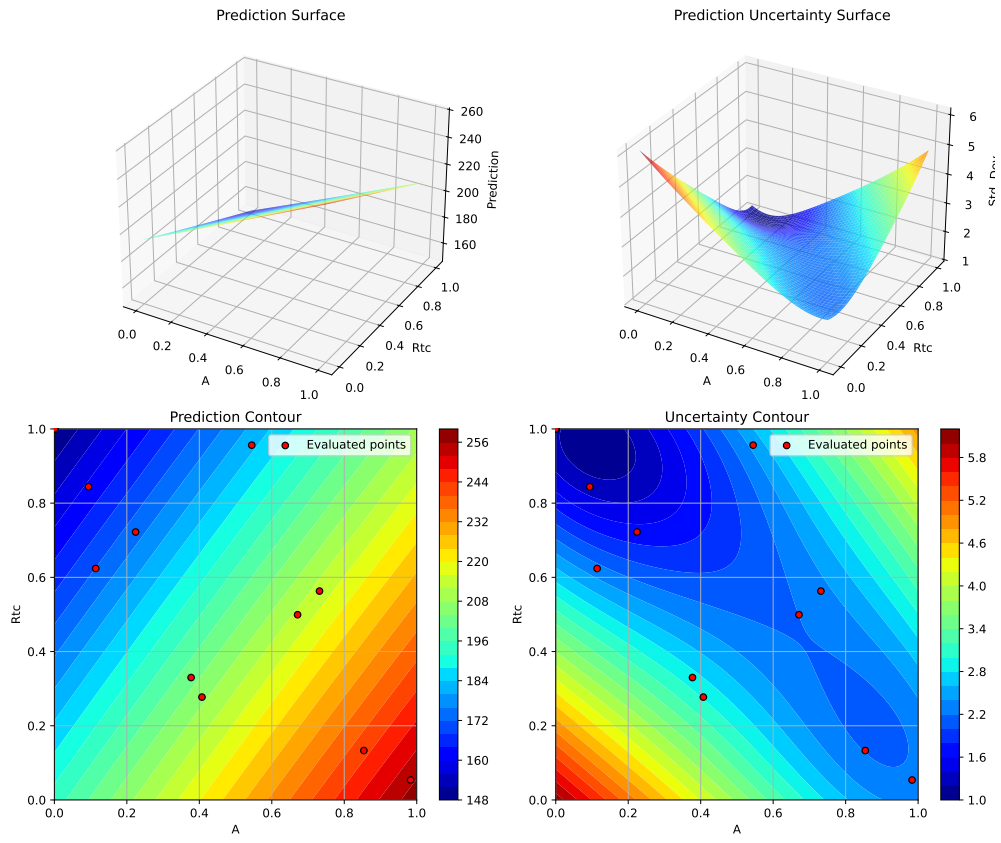
Plotting A vs Nz

## 28. Surrogate Model Selection in SpotOptim



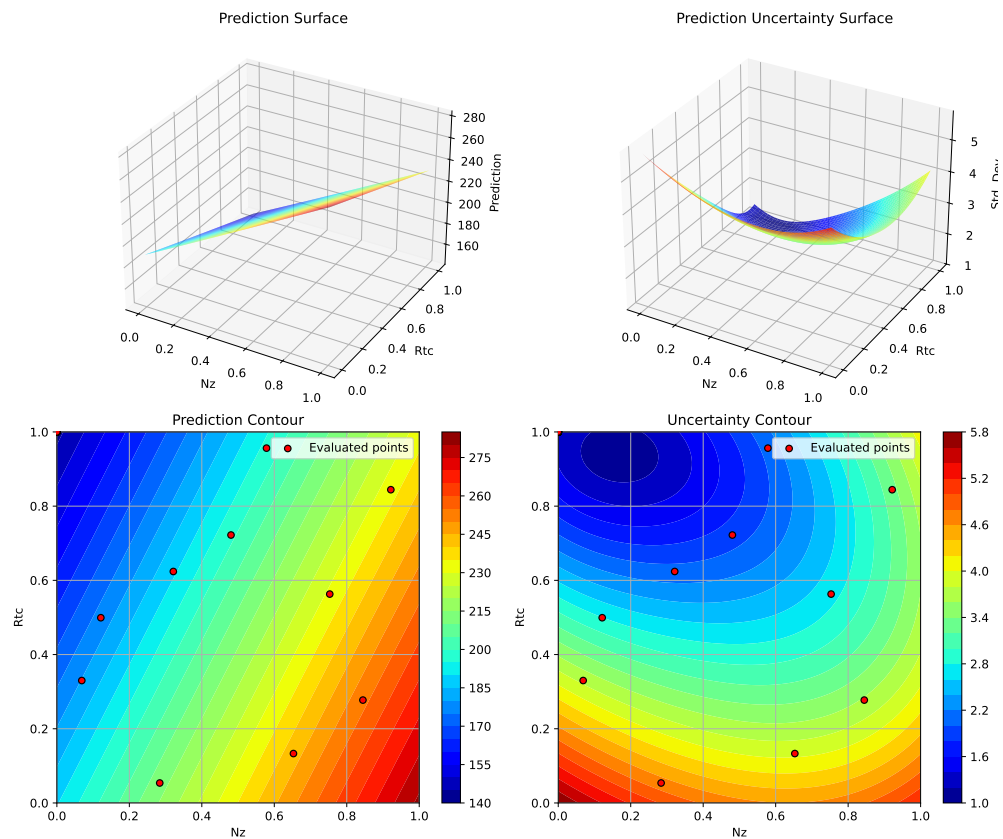
Plotting  $A$  vs  $R_{tc}$

## 28.7. 4. Gaussian Process with Rational Quadratic Kernel



Plotting  $N_z$  vs  $R_{tc}$

## 28. Surrogate Model Selection in SpotOptim



<Figure size 1650x1050 with 0 Axes>

### 28.8. 5. SpotOptim Kriging Model

SpotOptim includes its own Kriging implementation optimized for sequential design. It uses Gaussian correlation function and optimizes hyperparameters via differential evolution.

```
print("=" * 80)
print("5. SpotOptim Kriging Model")
print("=" * 80)

start_time = time.time()

# Configure Kriging model
```

```

kriging_model = Kriging(
    noise=1e-10,      # Regularization parameter
    kernel='gauss',   # Gaussian/RBF kernel
    n_theta=None,     # Auto: use number of dimensions
    min_theta=-3.0,   # Min log10(theta) bound
    max_theta=2.0,    # Max log10(theta) bound
    seed=seed
)

optimizer_kriging = SpotOptim(
    fun=wingwt,
    bounds=bounds,
    surrogate=kriging_model,
    max_iter=max_iter,
    n_initial=n_initial,
    var_name=param_names,
    acquisition='ei',
    seed=seed,
    verbose=False
)

result_kriging = optimizer_kriging.optimize()
time_kriging = time.time() - start_time

print(f"\nResults:")
print(f"  Best weight: {result_kriging.fun:.4f} lb")
print(f"  Function evaluations: {result_kriging.nfev}")
print(f"  Time: {time_kriging:.2f}s")
print(f"  Success: {result_kriging.success}")

results_comparison.append({
    'Surrogate': 'SpotOptim Kriging',
    'Best Weight': result_kriging.fun,
    'Evaluations': result_kriging.nfev,
    'Time (s)': time_kriging,
    'Success': result_kriging.success
})

```

---

## 5. SpotOptim Kriging Model

---

Results:

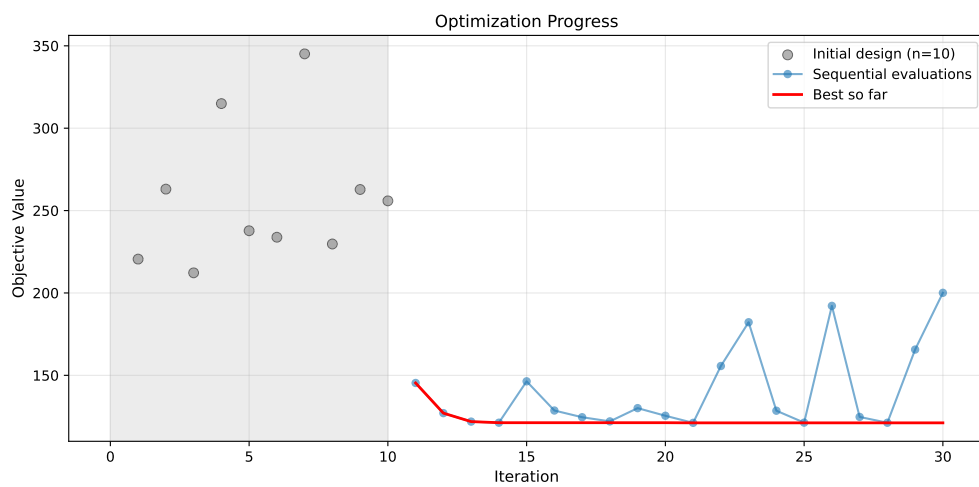
Best weight: 121.1616 lb

## 28. Surrogate Model Selection in SpotOptim

Function evaluations: 30  
Time: 55.24s  
Success: True

### 28.8.1. Visualization: Kriging Model

```
optimizer_kriging.plot_progress(log_y=False, figsize=(10, 5))
```



```
optimizer_kriging.plot_important_hyperparameter_contour(max_imp=3)
plt.suptitle('SpotOptim Kriging: Most Important Parameters', y=1.02)
plt.show()
```

Plotting surrogate contours for top 3 most important parameters:

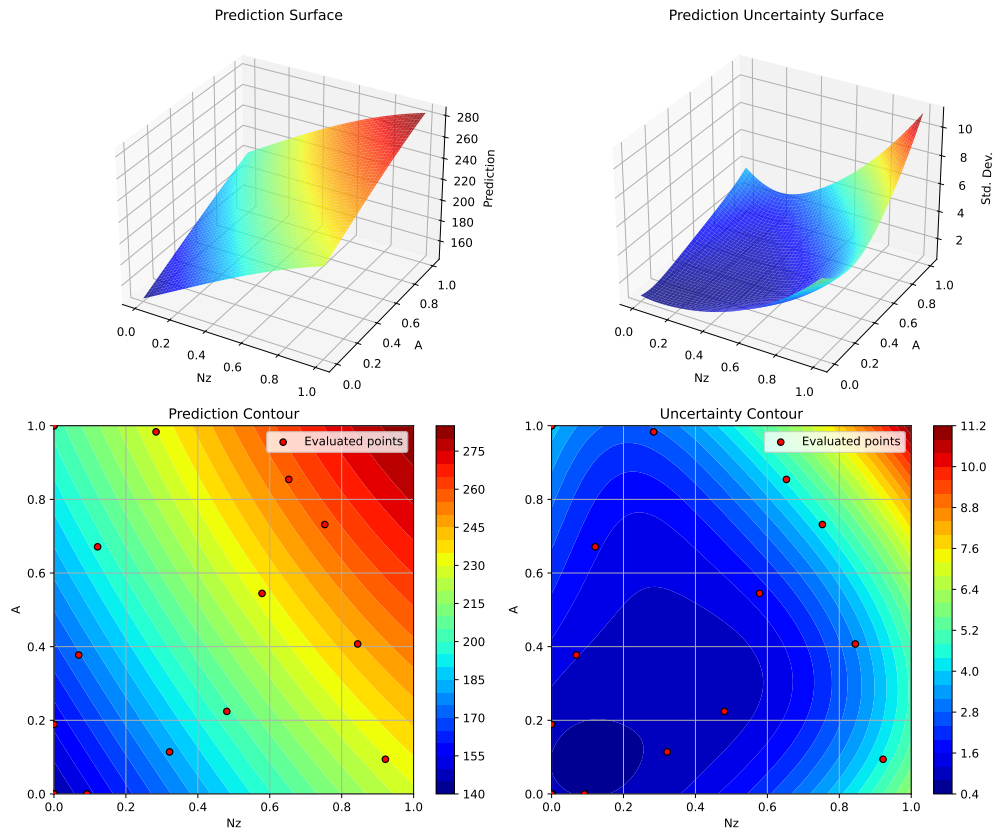
Nz: importance = 19.13% (type: float)  
A: importance = 16.01% (type: float)  
Wdg: importance = 14.70% (type: float)

Generating 3 surrogate plots...

Plotting Nz vs A

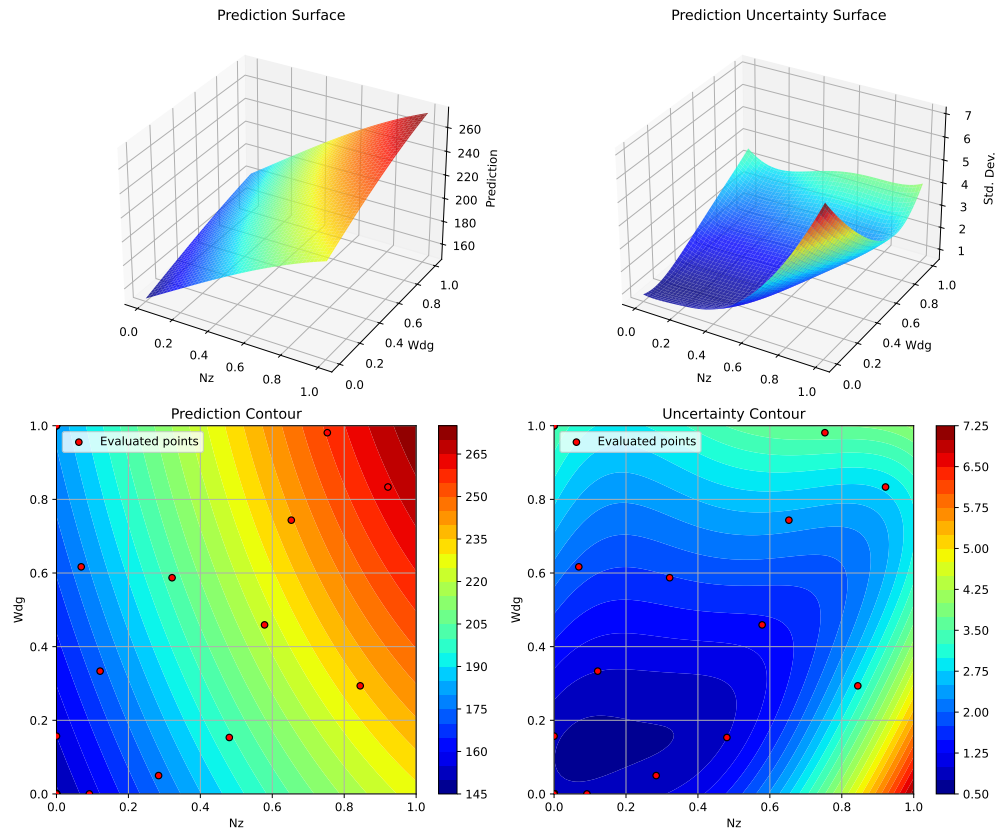


## 28.8. 5. SpotOptim Kriging Model



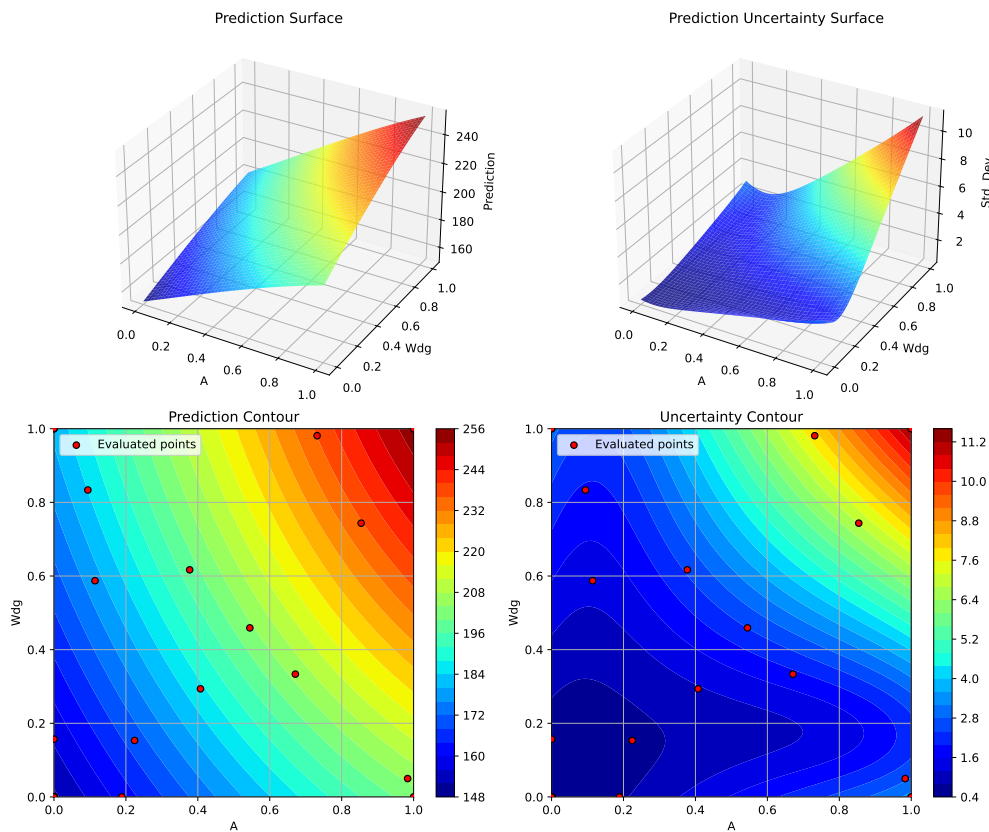
Plotting  $N_z$  vs  $Wdg$

## 28. Surrogate Model Selection in SpotOptim



Plotting A vs Wdg

## 28.9. 6. Random Forest Regressor



<Figure size 1650x1050 with 0 Axes>

## 28.9. 6. Random Forest Regressor

Random Forests are ensemble methods that handle noise well and can model discontinuities. They don't naturally provide uncertainty estimates, so the acquisition function uses predictions only.

```
print("=" * 80)
print("6. Random Forest Regressor")
print("=" * 80)

start_time = time.time()

# Configure Random Forest
```

## 28. Surrogate Model Selection in SpotOptim

```
rf_model = RandomForestRegressor(
    n_estimators=100,
    max_depth=15,
    min_samples_split=2,
    min_samples_leaf=1,
    random_state=seed,
    n_jobs=-1
)

optimizer_rf = SpotOptim(
    fun=wingwt,
    bounds=bounds,
    surrogate=rf_model,
    max_iter=max_iter,
    n_initial=n_initial,
    var_name=param_names,
    acquisition='y', # Use 'y' (greedy) since RF doesn't provide std
    seed=seed,
    verbose=False
)

result_rf = optimizer_rf.optimize()
time_rf = time.time() - start_time

print(f"\nResults:")
print(f"  Best weight: {result_rf.fun:.4f} lb")
print(f"  Function evaluations: {result_rf.nfev}")
print(f"  Time: {time_rf:.2f}s")
print(f"  Success: {result_rf.success}")
print(f"  Note: Using acquisition='y' (greedy) since RF doesn't provide uncertainty")

results_comparison.append({
    'Surrogate': 'Random Forest',
    'Best Weight': result_rf.fun,
    'Evaluations': result_rf.nfev,
    'Time (s)': time_rf,
    'Success': result_rf.success
})
```

---

### 6. Random Forest Regressor

---

Results:

Best weight: 155.3363 lb

Function evaluations: 30

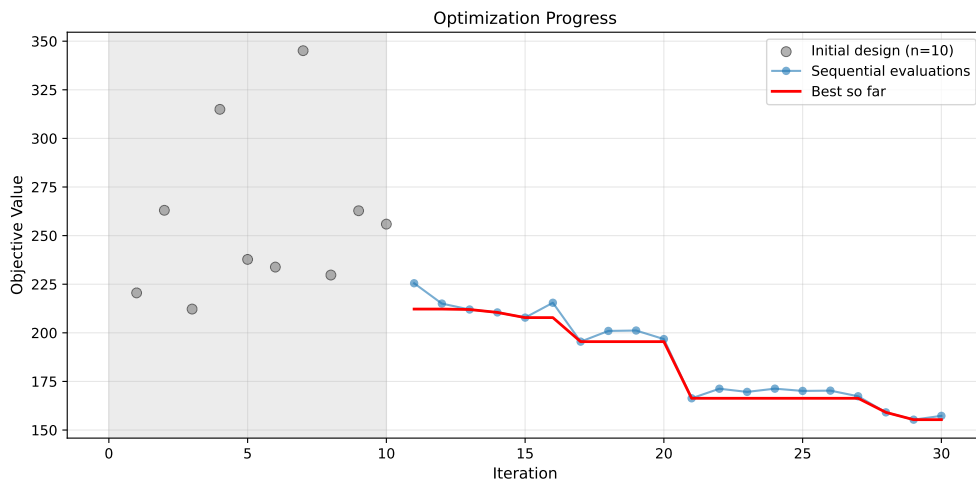
Time: 1013.85s

Success: True

Note: Using acquisition='y' (greedy) since RF doesn't provide uncertainty

### 28.9.1. Visualization: Random Forest

```
optimizer_rf.plot_progress(log_y=False, figsize=(10, 5))
```



```
optimizer_rf.plot_important_hyperparameter_contour(max_imp=3)
plt.suptitle('Random Forest: Most Important Parameters', y=1.02)
plt.show()
```

Plotting surrogate contours for top 3 most important parameters:

Wdg: importance = 14.46% (type: float)

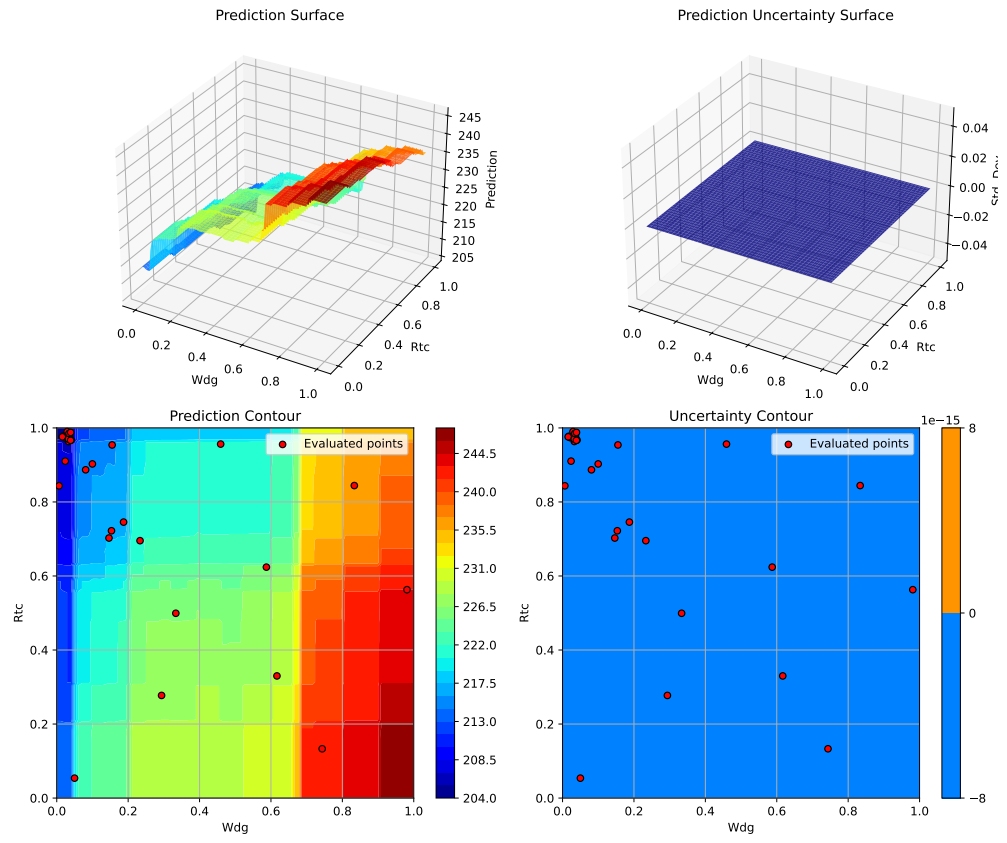
Rtc: importance = 13.92% (type: float)

Nz: importance = 13.70% (type: float)

Generating 3 surrogate plots...

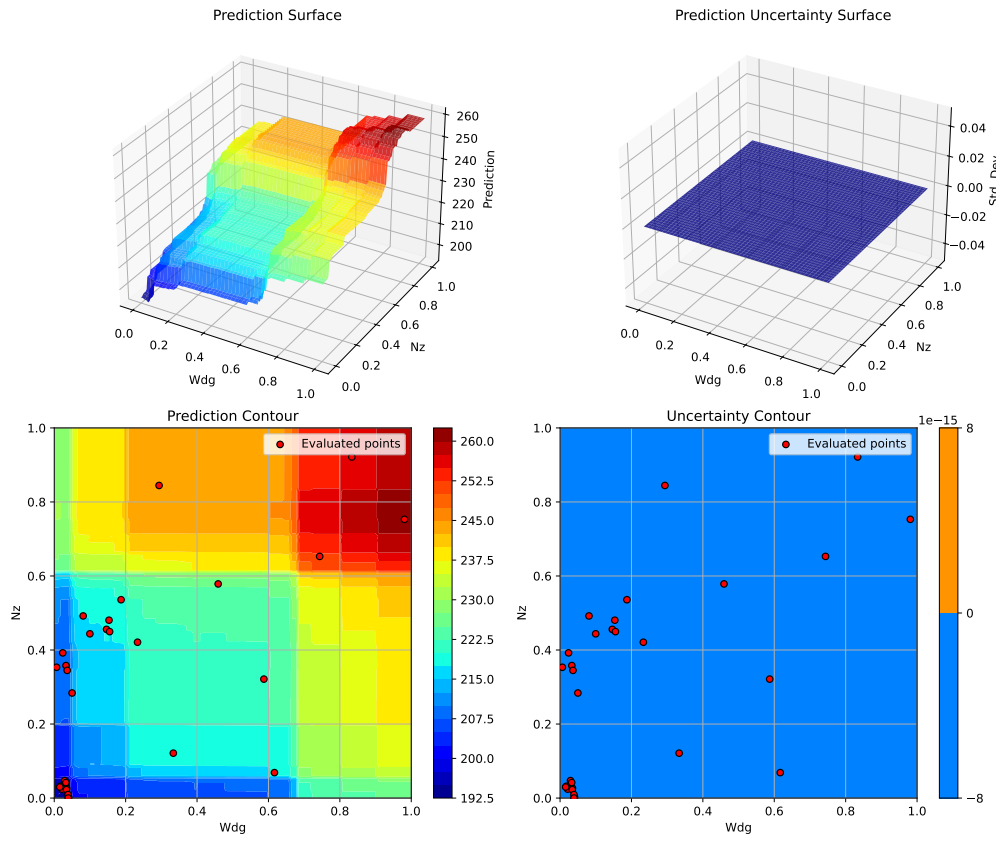
Plotting Wdg vs Rtc

## 28. Surrogate Model Selection in SpotOptim



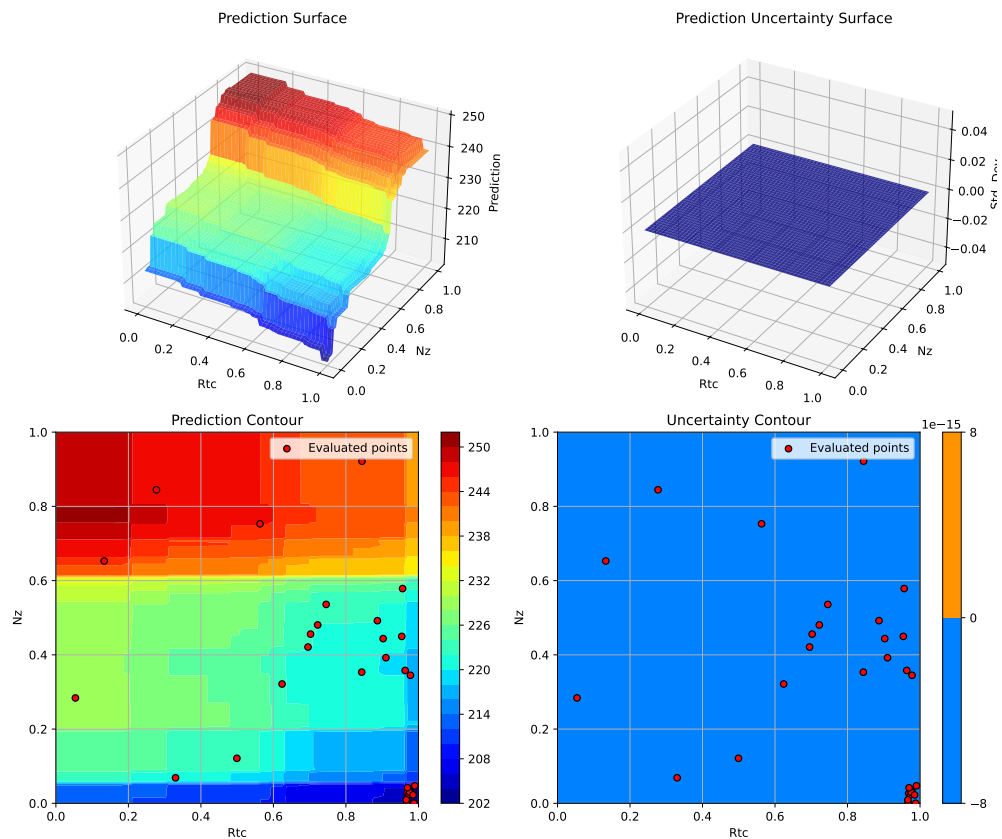
Plotting Wdg vs Nz

## 28.9. 6. Random Forest Regressor



Plotting Rtc vs Nz

## 28. Surrogate Model Selection in SpotOptim



<Figure size 1650x1050 with 0 Axes>

### 28.10. 7. XGBoost Regressor

XGBoost is a gradient boosting implementation known for excellent performance on structured data and fast training/prediction times.

```
if XGBOOST_AVAILABLE:
    print("=" * 80)
    print("7. XGBoost Regressor")
    print("=" * 80)

    start_time = time.time()

    # Configure XGBoost
```



```

xgb_model = xgb.XGBRegressor(
    n_estimators=100,
    max_depth=6,
    learning_rate=0.1,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=seed,
    n_jobs=-1
)

optimizer_xgb = SpotOptim(
    fun=wingwt,
    bounds=bounds,
    surrogate=xgb_model,
    max_iter=max_iter,
    n_initial=n_initial,
    var_name=param_names,
    acquisition='y', # Use 'y' (greedy) since XGBoost doesn't provide std
    seed=seed,
    verbose=False
)

result_xgb = optimizer_xgb.optimize()
time_xgb = time.time() - start_time

print(f"\nResults:")
print(f"  Best weight: {result_xgb.fun:.4f} lb")
print(f"  Function evaluations: {result_xgb.nfev}")
print(f"  Time: {time_xgb:.2f}s")
print(f"  Success: {result_xgb.success}")
print(f"  Note: Using acquisition='y' (greedy) since XGBoost doesn't provide uncertainty")

results_comparison.append({
    'Surrogate': 'XGBoost',
    'Best Weight': result_xgb.fun,
    'Evaluations': result_xgb.nfev,
    'Time (s)': time_xgb,
    'Success': result_xgb.success
})

# Visualization
optimizer_xgb.plot_progress(log_y=False, figsize=(10, 5))
plt.title('XGBoost: Convergence')
plt.show()

```

## 28. Surrogate Model Selection in SpotOptim

```
optimizer_xgb.plot_important_hyperparameter_contour(max_imp=3)
plt.suptitle('XGBoost: Most Important Parameters', y=1.02)
plt.show()
else:
    print("=" * 80)
    print("7. XGBoost Regressor - SKIPPED (not installed)")
    print("=" * 80)
    print("Install XGBoost with: pip install xgboost")
```

### 7. XGBoost Regressor

#### Results:

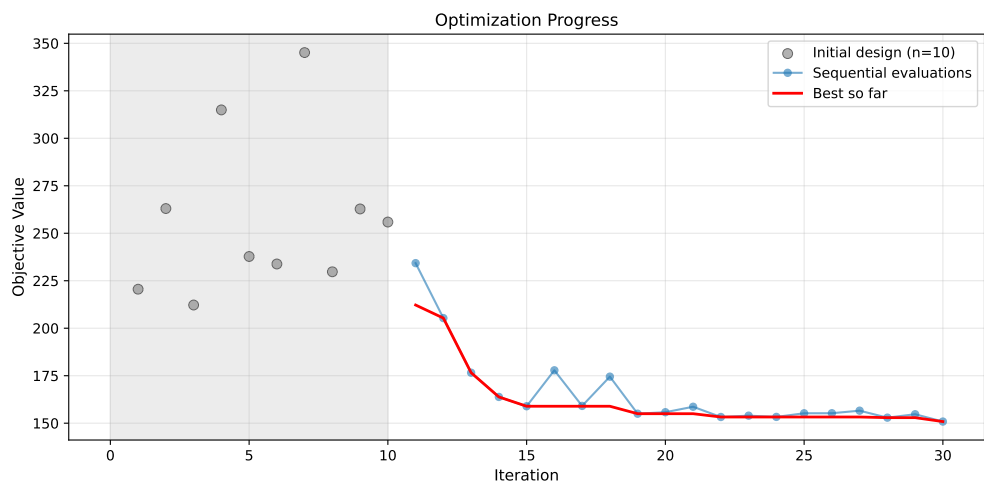
Best weight: 150.8548 lb

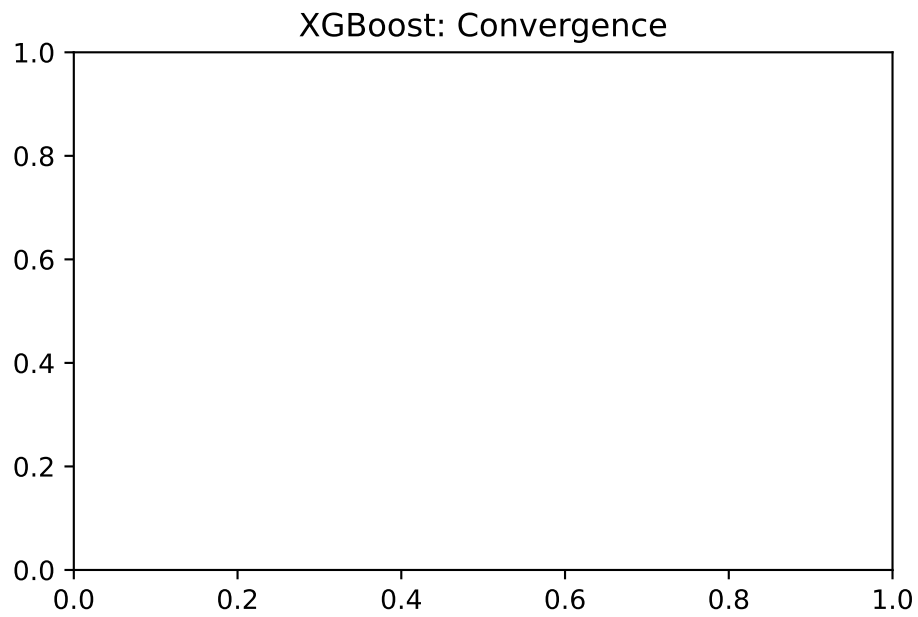
Function evaluations: 30

Time: 19.70s

Success: True

Note: Using acquisition='y' (greedy) since XGBoost doesn't provide uncertainty





Plotting surrogate contours for top 3 most important parameters:

Nz: importance = 15.18% (type: float)

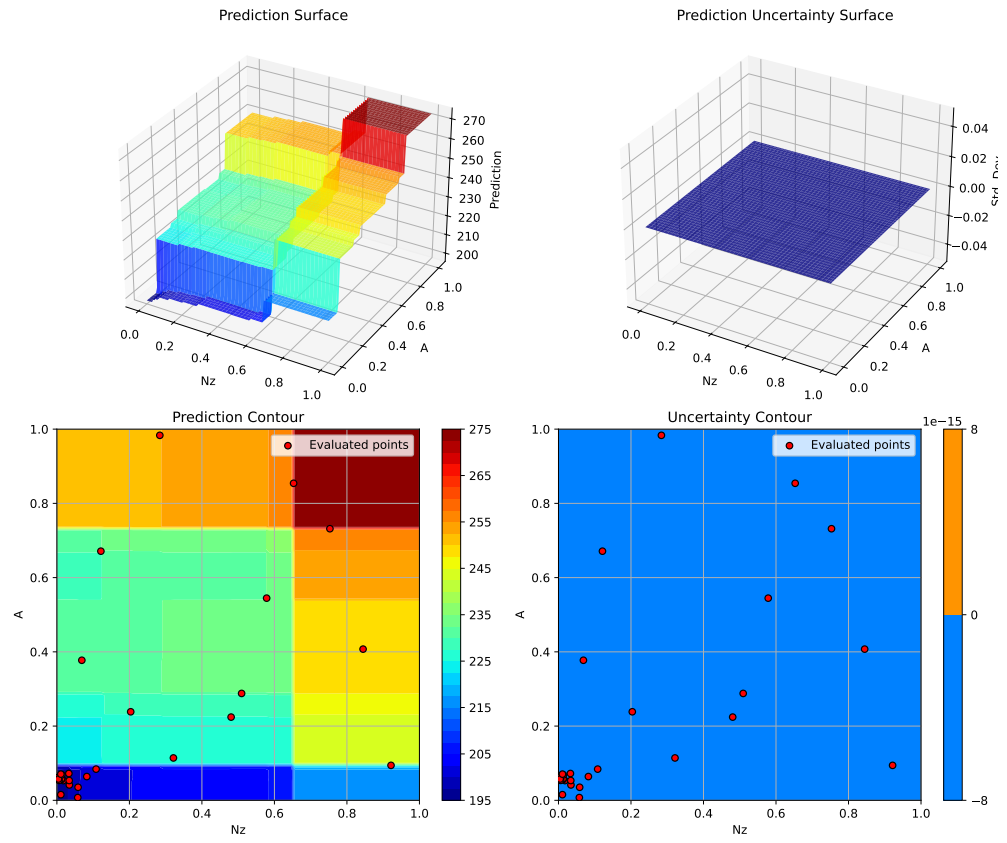
A: importance = 15.04% (type: float)

Wdg: importance = 14.49% (type: float)

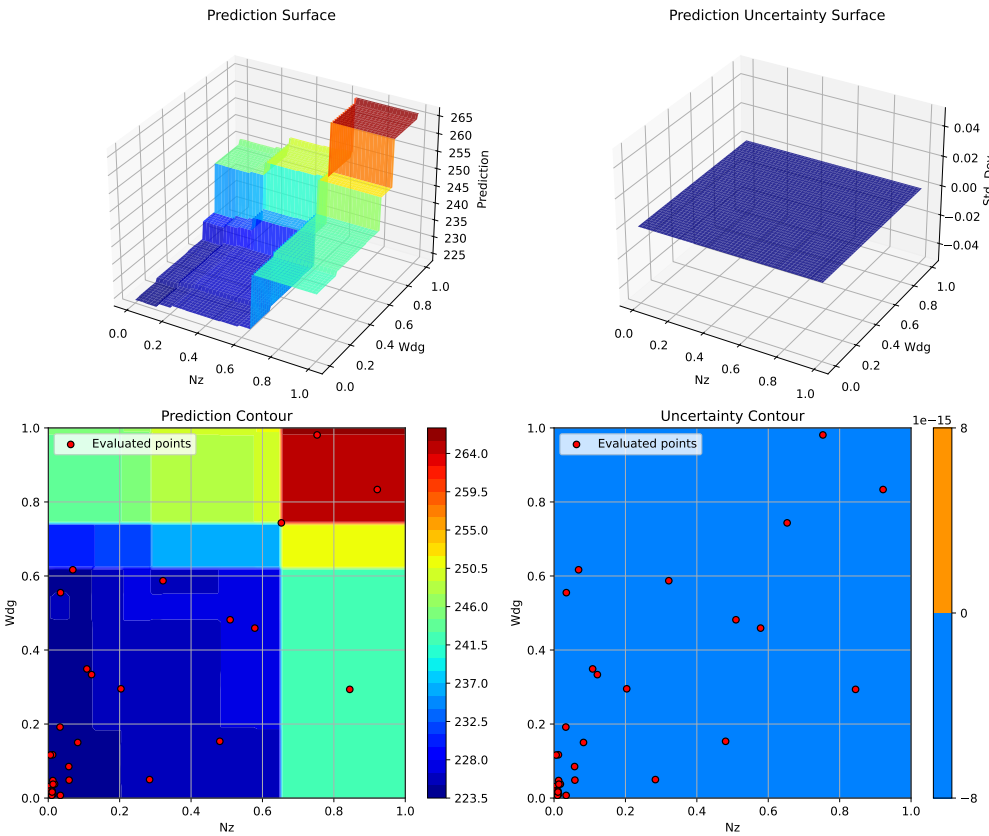
Generating 3 surrogate plots...

Plotting Nz vs A

## 28. Surrogate Model Selection in SpotOptim

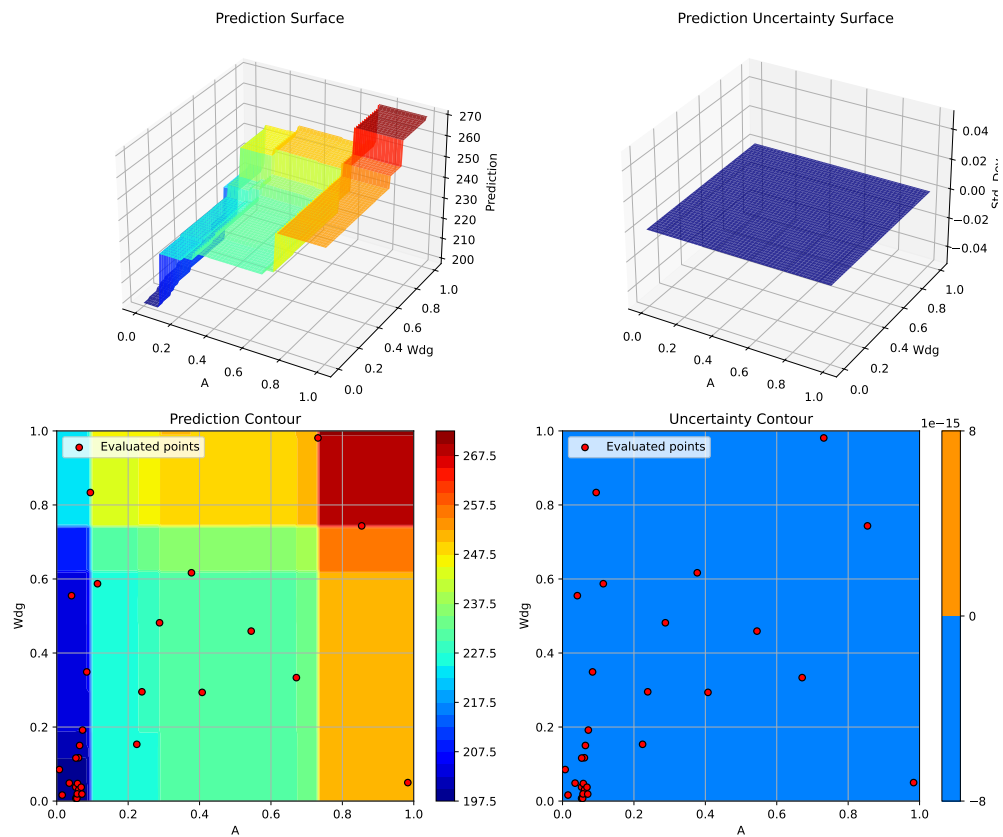


Plotting  $N_z$  vs  $Wdg$



Plotting A vs Wdg

## 28. Surrogate Model Selection in SpotOptim



<Figure size 1650x1050 with 0 Axes>

## 28.11. 8. Support Vector Regression (SVR)

SVR with RBF kernel can model complex non-linear relationships. It's particularly good for high-dimensional problems with smooth structure.

```
print("=" * 80)
print("8. Support Vector Regression (SVR)")
print("=" * 80)

start_time = time.time()

# Configure SVR
svr_model = SVR(
```

```

    kernel='rbf',
    C=100.0,
    epsilon=0.1,
    gamma='scale'
)

optimizer_svr = SpotOptim(
    fun=wingwt,
    bounds=bounds,
    surrogate=svr_model,
    max_iter=max_iter,
    n_initial=n_initial,
    var_name=param_names,
    acquisition='y', # Use 'y' (greedy) since SVR doesn't provide std by default
    seed=seed,
    verbose=False
)

result_svr = optimizer_svr.optimize()
time_svr = time.time() - start_time

print(f"\nResults:")
print(f"  Best weight: {result_svr.fun:.4f} lb")
print(f"  Function evaluations: {result_svr.nfev}")
print(f"  Time: {time_svr:.2f}s")
print(f"  Success: {result_svr.success}")

results_comparison.append({
    'Surrogate': 'SVR (RBF)',
    'Best Weight': result_svr.fun,
    'Evaluations': result_svr.nfev,
    'Time (s)': time_svr,
    'Success': result_svr.success
})

```

---

## 8. Support Vector Regression (SVR)

---

Results:

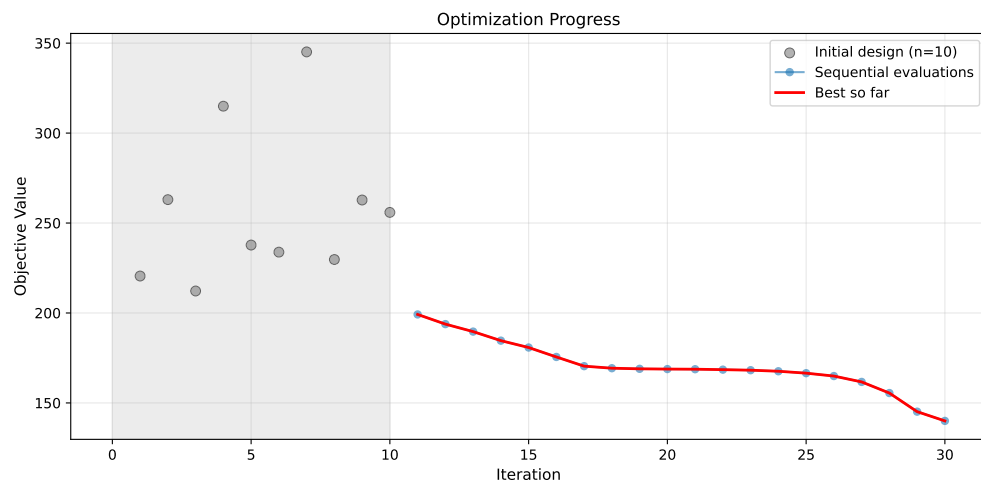
```

Best weight: 140.0109 lb
Function evaluations: 30
Time: 1.82s
Success: True

```

### 28.11.1. Visualization: SVR

```
optimizer_svr.plot_progress(log_y=False, figsize=(10, 5))
```



```
optimizer_svr.plot_important_hyperparameter_contour(max_imp=3)
plt.suptitle('Support Vector Regression: Most Important Parameters', y=1.02)
plt.show()
```

Plotting surrogate contours for top 3 most important parameters:

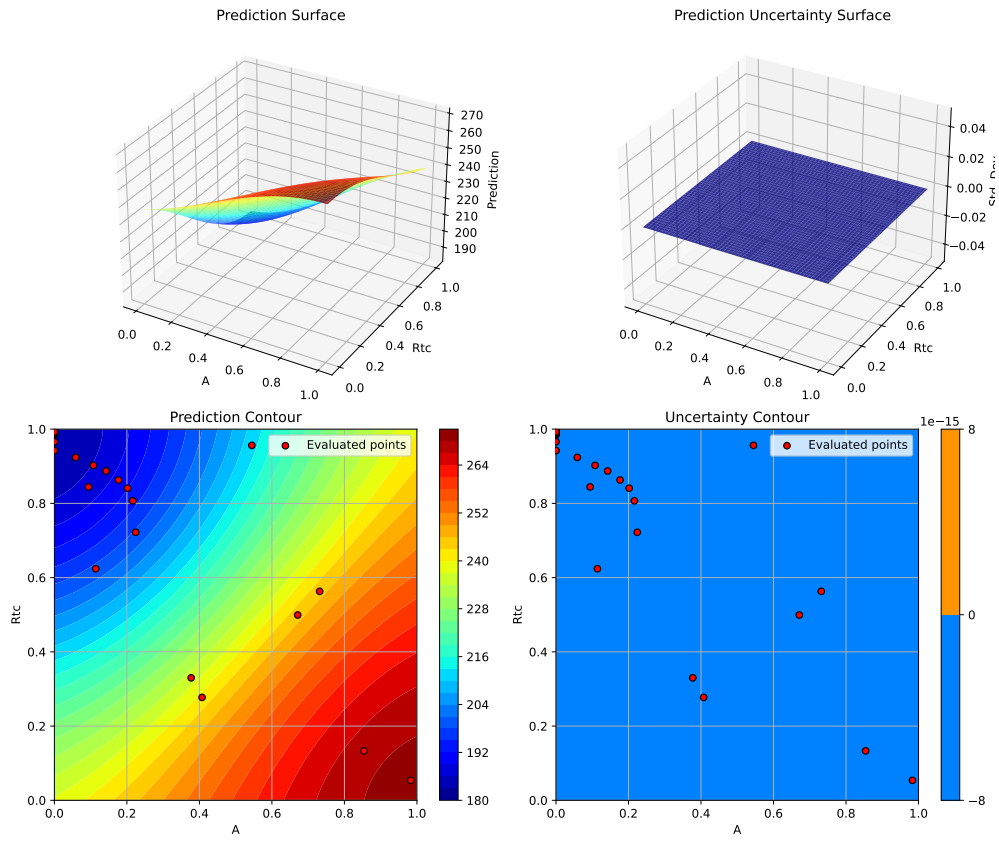
A: importance = 18.74% (type: float)  
 Rtc: importance = 18.17% (type: float)  
 Wdg: importance = 18.14% (type: float)

Generating 3 surrogate plots...

Plotting A vs Rtc

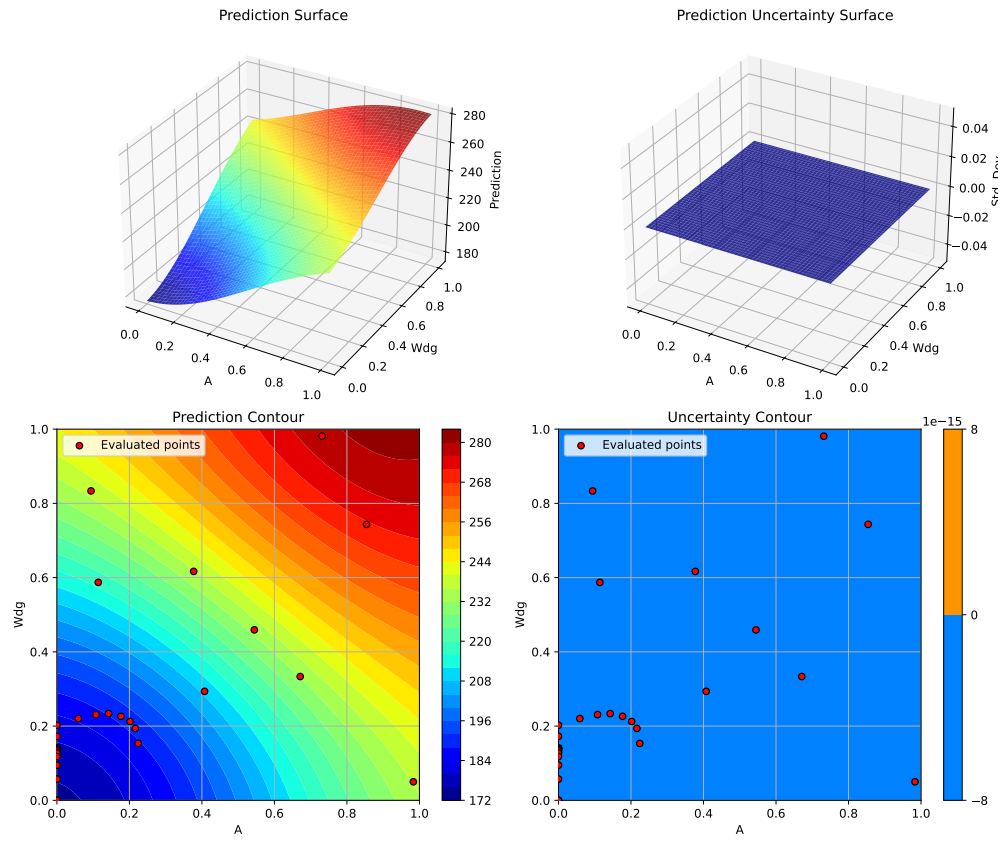


## 28.11. 8. Support Vector Regression (SVR)



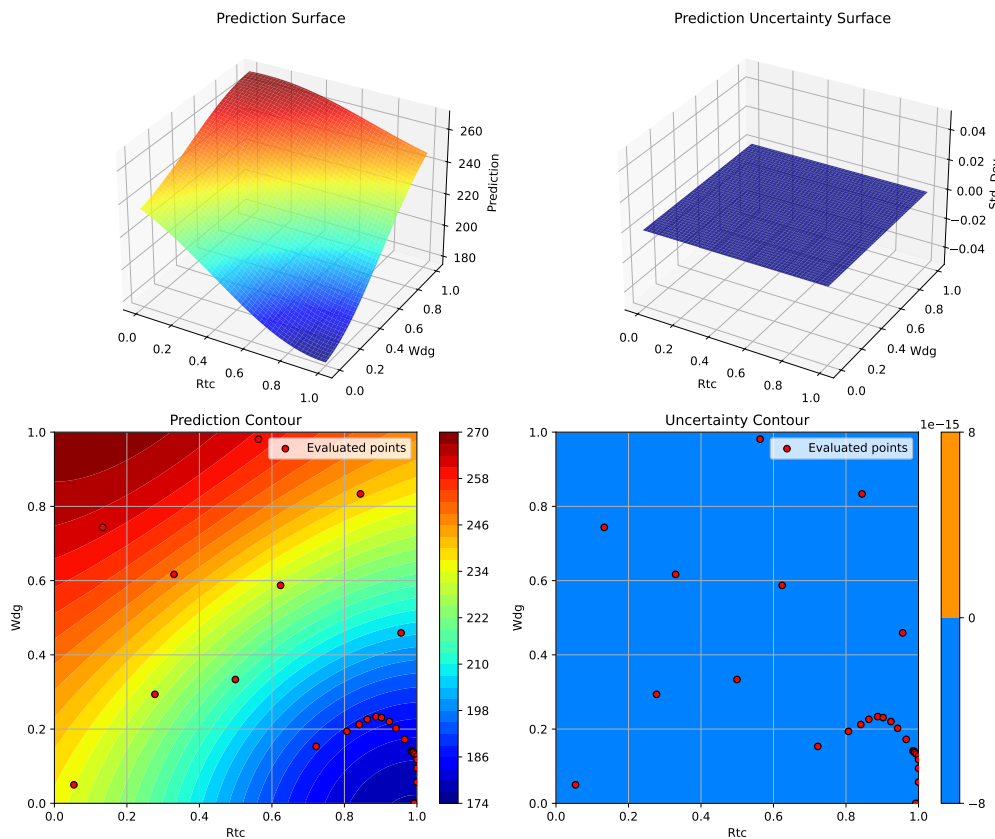
Plotting  $A$  vs  $W_{dg}$

## 28. Surrogate Model Selection in SpotOptim



Plotting Rtc vs Wdg

## 28.12. 9. Gradient Boosting Regressor



<Figure size 1650x1050 with 0 Axes>

## 28.12. 9. Gradient Boosting Regressor

Gradient Boosting from scikit-learn is another ensemble method that builds trees sequentially, with each tree correcting errors of the previous ones.

```
print("=" * 80)
print("9. Gradient Boosting Regressor")
print("=" * 80)

start_time = time.time()

# Configure Gradient Boosting
gb_model = GradientBoostingRegressor(
```

## 28. Surrogate Model Selection in SpotOptim

```
n_estimators=100,
max_depth=5,
learning_rate=0.1,
subsample=0.8,
random_state=seed
)

optimizer_gb = SpotOptim(
    fun=wingwt,
    bounds=bounds,
    surrogate=gb_model,
    max_iter=max_iter,
    n_initial=n_initial,
    var_name=param_names,
    acquisition='y', # Use 'y' (greedy) since GB doesn't provide std
    seed=seed,
    verbose=False
)

result_gb = optimizer_gb.optimize()
time_gb = time.time() - start_time

print(f"\nResults:")
print(f"  Best weight: {result_gb.fun:.4f} lb")
print(f"  Function evaluations: {result_gb.nfev}")
print(f"  Time: {time_gb:.2f}s")
print(f"  Success: {result_gb.success}")

results_comparison.append({
    'Surrogate': 'Gradient Boosting',
    'Best Weight': result_gb.fun,
    'Evaluations': result_gb.nfev,
    'Time (s)': time_gb,
    'Success': result_gb.success
})
```

---

### 9. Gradient Boosting Regressor

---

Results:  
Best weight: 124.6772 lb  
Function evaluations: 30  
Time: 7.52s

Success: True

### 28.12.1. Visualization: Gradient Boosting

```
optimizer_gb.plot_progress(log_y=False, figsize=(10, 5))
```



```
optimizer_gb.plot_important_hyperparameter_contour(max_imp=3)
plt.suptitle('Gradient Boosting: Most Important Parameters', y=1.02)
plt.show()
```

Plotting surrogate contours for top 3 most important parameters:

Wdg: importance = 16.80% (type: float)

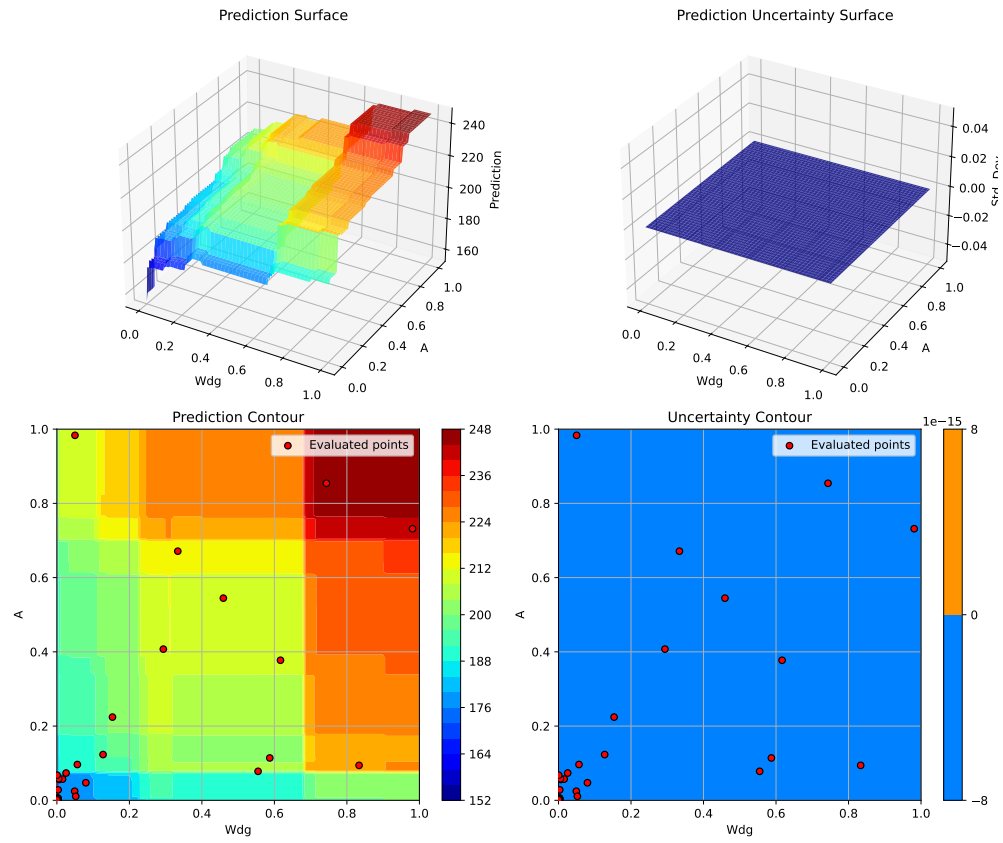
A: importance = 16.28% (type: float)

Rtc: importance = 15.69% (type: float)

Generating 3 surrogate plots...

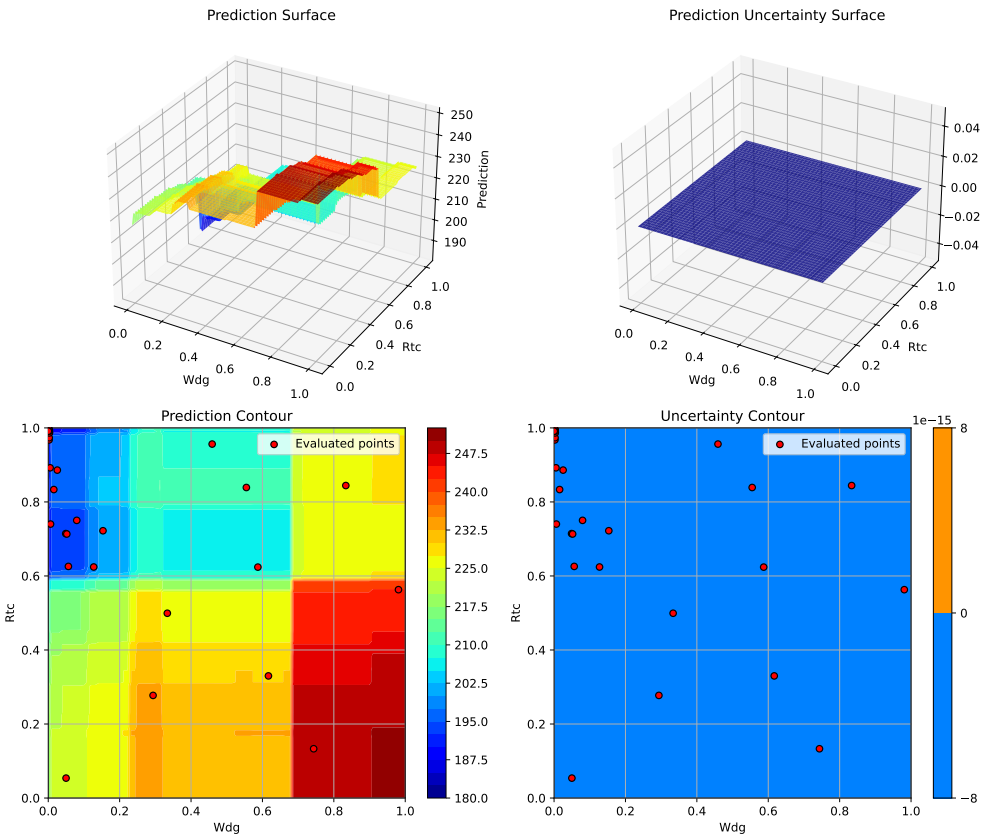
Plotting Wdg vs A

## 28. Surrogate Model Selection in SpotOptim



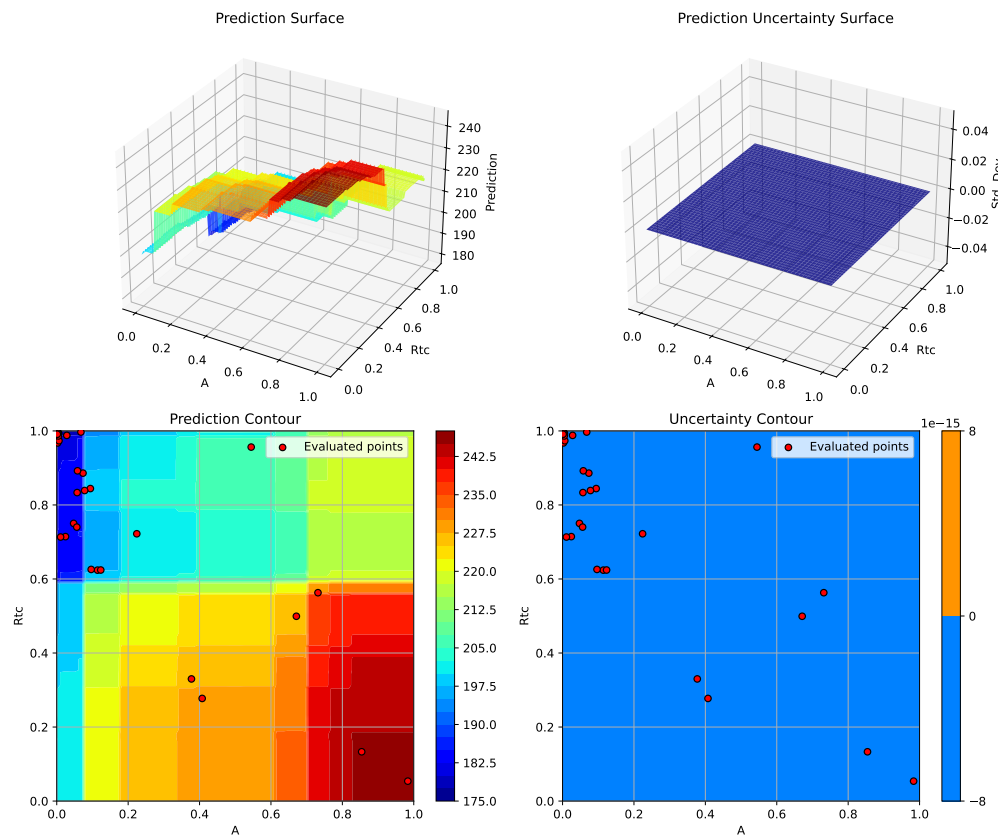
Plotting Wdg vs Rtc

28.12. 9. Gradient Boosting Regressor



Plotting A vs Rtc

## 28. Surrogate Model Selection in SpotOptim



<Figure size 1650x1050 with 0 Axes>

### 28.13. Comprehensive Comparison

Now let's compare all surrogate models side-by-side.

```
# Create comparison DataFrame
df_comparison = pd.DataFrame(results_comparison)

# Calculate improvement from best
best_weight = df_comparison['Best Weight'].min()
df_comparison['Gap to Best (%)'] = (
    (df_comparison['Best Weight'] - best_weight) / best_weight * 100
)
```



```
# Sort by best weight
df_comparison = df_comparison.sort_values('Best Weight')

print("\n" + "=" * 100)
print("SURROGATE MODEL COMPARISON")
print("=" * 100)
print(df_comparison.to_string(index=False))
print("=" * 100)
```

#### SURROGATE MODEL COMPARISON

|                          | Surrogate             | Best Weight | Evaluations | Time (s)    | Success | Gap to Best (%) |
|--------------------------|-----------------------|-------------|-------------|-------------|---------|-----------------|
|                          | GP RBF                | 119.503672  | 30          | 48.767761   | True    | 0.000000        |
| GP Matern =2.5 (Default) |                       | 119.504103  | 30          | 42.919710   | True    | 0.000361        |
|                          | GP Matern =1.5        | 119.505612  | 30          | 41.160388   | True    | 0.001624        |
|                          | GP Rational Quadratic | 119.506083  | 30          | 47.902926   | True    | 0.002017        |
|                          | SpotOptim Kriging     | 121.161582  | 30          | 55.241836   | True    | 1.387330        |
|                          | Gradient Boosting     | 124.677233  | 30          | 7.523013    | True    | 4.329207        |
|                          | SVR (RBF)             | 140.010945  | 30          | 1.816080    | True    | 17.160371       |
|                          | XGBoost               | 150.854809  | 30          | 19.702054   | True    | 26.234455       |
|                          | Random Forest         | 155.336261  | 30          | 1013.845482 | True    | 29.984509       |

### 28.13.1. Visualization: Performance Comparison

```
fig, axes = plt.subplots(2, 2, figsize=(16, 12))

# Plot 1: Best weight comparison
ax1 = axes[0, 0]
colors = ['green' if i == 0 else 'steelblue' for i in range(len(df_comparison))]
ax1.barh(df_comparison['Surrogate'], df_comparison['Best Weight'], color=colors)
ax1.set_xlabel('Best Weight (lb)')
ax1.set_title('Best Weight Found by Each Surrogate')
ax1.axvline(x=best_weight, color='red', linestyle='--', linewidth=2, label='Best Overall')
ax1.legend()
ax1.grid(True, alpha=0.3, axis='x')

# Plot 2: Computational time
ax2 = axes[0, 1]
ax2.barh(df_comparison['Surrogate'], df_comparison['Time (s)'], color='coral')
```

## 28. Surrogate Model Selection in SpotOptim

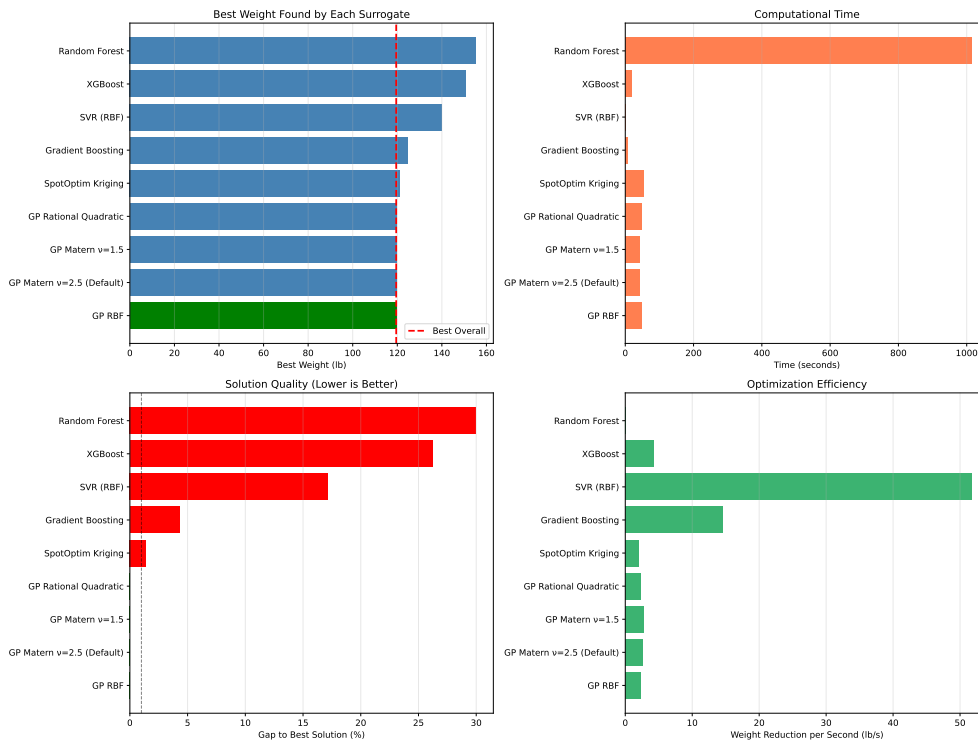
```
ax2.set_xlabel('Time (seconds)')
ax2.set_title('Computational Time')
ax2.grid(True, alpha=0.3, axis='x')

# Plot 3: Gap to best
ax3 = axes[1, 0]
colors_gap = ['green' if gap < 0.1 else 'orange' if gap < 1.0 else 'red'
              for gap in df_comparison['Gap to Best (%)']]
ax3.barh(df_comparison['Surrogate'], df_comparison['Gap to Best (%)'], color=colors_gap)
ax3.set_xlabel('Gap to Best Solution (%)')
ax3.set_title('Solution Quality (Lower is Better)')
ax3.axvline(x=1.0, color='black', linestyle='--', alpha=0.5, linewidth=1)
ax3.grid(True, alpha=0.3, axis='x')

# Plot 4: Efficiency (weight reduction per second)
ax4 = axes[1, 1]
baseline_weight = wingwt(np.array([0.48, 0.4, 0.38, 0.5, 0.62, 0.344, 0.4, 0.37, 0.38]))
df_comparison['Efficiency'] = (baseline_weight - df_comparison['Best Weight']) / df_comparison['Time (seconds)']
ax4.barh(df_comparison['Surrogate'], df_comparison['Efficiency'], color='mediumseagreen')
ax4.set_xlabel('Weight Reduction per Second (lb/s)')
ax4.set_title('Optimization Efficiency')
ax4.grid(True, alpha=0.3, axis='x')

plt.tight_layout()
plt.show()
```

## 28.14. Convergence Comparison



## 28.14. Convergence Comparison

Let's compare how different surrogates converge over iterations.

```
fig, ax = plt.subplots(figsize=(14, 8))

# Collect convergence data from all optimizers
optimizers = [
    (optimizer_default, 'GP Matern =2.5 (Default)', 'blue'),
    (optimizer_rbf, 'GP RBF', 'green'),
    (optimizer_matern15, 'GP Matern =1.5', 'red'),
    (optimizer_rq, 'GP Rational Quadratic', 'purple'),
    (optimizer_kriging, 'SpotOptim Kriging', 'orange'),
    (optimizer_rf, 'Random Forest', 'brown'),
    (optimizer_svr, 'SVR', 'pink'),
    (optimizer_gb, 'Gradient Boosting', 'gray')
]
```

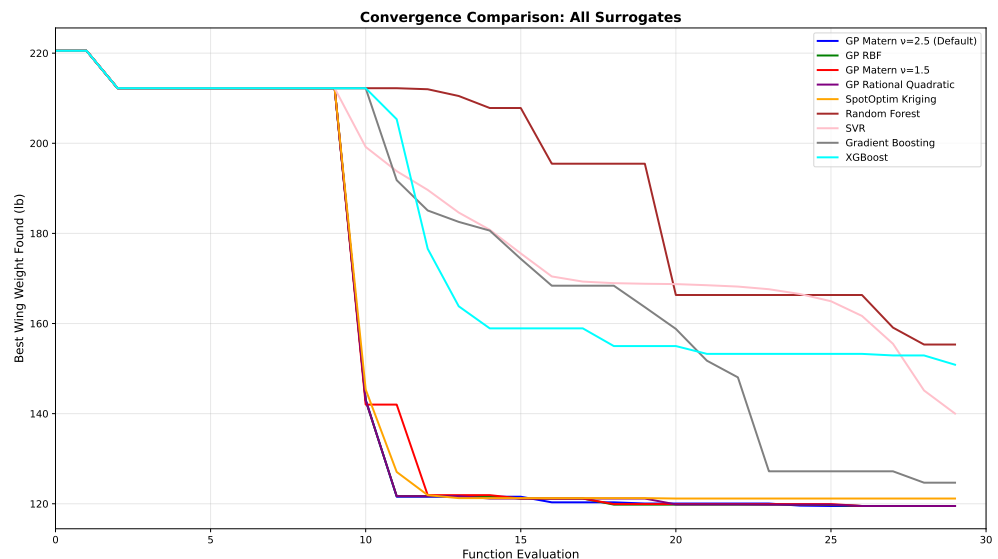
## 28. Surrogate Model Selection in SpotOptim

```
if XGBOOST_AVAILABLE:
    optimizers.append((optimizer_xgb, 'XGBoost', 'cyan'))

for opt, label, color in optimizers:
    y_history = opt.y_
    best_so_far = np.minimum.accumulate(y_history)
    ax.plot(range(len(best_so_far)), best_so_far, linewidth=2, label=label, color=col

ax.set_xlabel('Function Evaluation', fontsize=12)
ax.set_ylabel('Best Wing Weight Found (lb)', fontsize=12)
ax.set_title('Convergence Comparison: All Surrogates', fontsize=14, fontweight='bold')
ax.legend(loc='upper right', fontsize=10)
ax.grid(True, alpha=0.3)
ax.set_xlim(0, max_iter)

plt.tight_layout()
plt.show()
```



## 28.15. Key Insights and Recommendations

```
print("\n" + "=" * 100)
print("KEY INSIGHTS AND RECOMMENDATIONS")
```

```

print("=" * 100)

# Find best surrogate
best_surrogate = df_comparison.iloc[0]['Surrogate']
best_value = df_comparison.iloc[0]['Best Weight']
best_time = df_comparison.iloc[0]['Time (s)']

print(f"\n1. BEST OVERALL PERFORMANCE:")
print(f"    Surrogate: {best_surrogate}")
print(f"    Best Weight: {best_value:.4f} lb")
print(f"    Computation Time: {best_time:.2f}s")

# Find fastest
fastest_idx = df_comparison['Time (s)'].idxmin()
fastest_surrogate = df_comparison.loc[fastest_idx, 'Surrogate']
fastest_time = df_comparison.loc[fastest_idx, 'Time (s)']

print(f"\n2. FASTEST OPTIMIZATION:")
print(f"    Surrogate: {fastest_surrogate}")
print(f"    Time: {fastest_time:.2f}s")
print(f"    Best Weight: {df_comparison.loc[fastest_idx, 'Best Weight']:.4f} lb")

# Find most efficient
most_efficient_idx = df_comparison['Efficiency'].idxmax()
most_efficient = df_comparison.loc[most_efficient_idx, 'Surrogate']

print(f"\n3. MOST EFFICIENT (weight reduction per second):")
print(f"    Surrogate: {most_efficient}")
print(f"    Efficiency: {df_comparison.loc[most_efficient_idx, 'Efficiency']:.4f} lb/s")

print(f"\n4. RECOMMENDATIONS BY PROBLEM TYPE:")
print(f"    - Smooth, continuous functions: Gaussian Process with RBF or Matern =2.5")
print(f"    - Functions with noise: Random Forest or Gradient Boosting")
print(f"    - High-dimensional problems (>20D): XGBoost or Random Forest")
print(f"    - Limited budget (<50 evals): Gaussian Process with Expected Improvement")
print(f"    - Fast evaluation needed: XGBoost or Random Forest")
print(f"    - Need uncertainty estimates: Gaussian Process or Kriging")
print(f"    - Non-smooth/discontinuous: Random Forest or Gradient Boosting")

print(f"\n5. KERNEL COMPARISON (Gaussian Process):")
gp_results = df_comparison[df_comparison['Surrogate'].str.contains('GP')]
print(gp_results[['Surrogate', 'Best Weight', 'Time (s)']].to_string(index=False))

print("\n" + "=" * 100)

```

## 28. Surrogate Model Selection in SpotOptim

### =====

#### KEY INSIGHTS AND RECOMMENDATIONS

### =====

1. BEST OVERALL PERFORMANCE:  
Surrogate: GP RBF  
Best Weight: 119.5037 lb  
Computation Time: 48.77s
2. FASTEST OPTIMIZATION:  
Surrogate: SVR (RBF)  
Time: 1.82s  
Best Weight: 140.0109 lb
3. MOST EFFICIENT (weight reduction per second):  
Surrogate: SVR (RBF)  
Efficiency: 51.7034 lb/s
4. RECOMMENDATIONS BY PROBLEM TYPE:
  - Smooth, continuous functions: Gaussian Process with RBF or Matern  $\kappa=2.5$
  - Functions with noise: Random Forest or Gradient Boosting
  - High-dimensional problems ( $>20D$ ): XGBoost or Random Forest
  - Limited budget ( $<50$  evals): Gaussian Process with Expected Improvement
  - Fast evaluation needed: XGBoost or Random Forest
  - Need uncertainty estimates: Gaussian Process or Kriging
  - Non-smooth/discontinuous: Random Forest or Gradient Boosting
5. KERNEL COMPARISON (Gaussian Process):

|                                  | Surrogate              | Best Weight | Time (s)  |
|----------------------------------|------------------------|-------------|-----------|
|                                  | GP RBF                 | 119.503672  | 48.767761 |
| GP Matern $\kappa=2.5$ (Default) |                        | 119.504103  | 42.919710 |
|                                  | GP Matern $\kappa=1.5$ | 119.505612  | 41.160388 |
|                                  | GP Rational Quadratic  | 119.506083  | 47.902926 |

### =====

## 28.16. Summary Statistics

```
# Summary statistics
print("\n" + "=" * 100)
print("SUMMARY STATISTICS")
```

```

print("=" * 100)

summary_stats = pd.DataFrame({
    'Metric': [
        'Best Weight Found',
        'Worst Weight Found',
        'Average Weight',
        'Std Dev Weight',
        'Fastest Time',
        'Slowest Time',
        'Average Time',
    ],
    'Value': [
        f"{df_comparison['Best Weight'].min():.4f} lb",
        f"{df_comparison['Best Weight'].max():.4f} lb",
        f"{df_comparison['Best Weight'].mean():.4f} lb",
        f"{df_comparison['Best Weight'].std():.4f} lb",
        f"{df_comparison['Time (s)'].min():.2f} s",
        f"{df_comparison['Time (s)'].max():.2f} s",
        f"{df_comparison['Time (s)'].mean():.2f} s",
    ]
})

print(summary_stats.to_string(index=False))
print("=" * 100)

```

```

=====
SUMMARY STATISTICS
=====

```

```

      Metric      Value
Best Weight Found 119.5037 lb
Worst Weight Found 155.3363 lb
Average Weight 130.0067 lb
Std Dev Weight 14.6804 lb
Fastest Time      1.82 s
Slowest Time     1013.85 s
Average Time     142.10 s
=====

```

## 28.17. Conclusion

This comprehensive comparison demonstrates that:

1. **Gaussian Processes** with appropriate kernels (Matern, RBF) provide excellent performance for smooth optimization problems and naturally support Expected Improvement acquisition
2. **SpotOptim Kriging** offers a lightweight alternative to sklearn's GP with comparable performance
3. **Random Forest** and **XGBoost** are robust alternatives that handle noise and discontinuities well, though they require greedy acquisition
4. **SVR** and **Gradient Boosting** offer middle-ground solutions with good scalability
5. The choice of surrogate should be based on:
  - Function smoothness
  - Computational budget
  - Need for uncertainty quantification
  - Problem dimensionality
  - Noise characteristics

For the AWWE problem, Gaussian Process surrogates generally performed best due to the function's smooth structure, but tree-based methods (RF, XGBoost, GB) can be preferable for more complex, noisy, or high-dimensional problems.

## 28.18. Jupyter Notebook

### Note

- The Jupyter-Notebook of this lecture is available on GitHub in the Hyperparameter-Tuning-Cookbook Repository



## 29. Acquisition Failure Handling in SpotOptim

SpotOptim provides sophisticated fallback strategies for handling acquisition function failures during optimization. This ensures robust optimization even when the surrogate model struggles to suggest new points.

### 29.1. What is Acquisition Failure?

During surrogate-based optimization, the acquisition function suggests new points to evaluate. However, sometimes the suggested point is **too close** to existing points (within `tolerance_x` distance), which would provide little new information. When this happens, SpotOptim uses a **fallback strategy** to propose an alternative point.

### 29.2. Fallback Strategies

SpotOptim supports two fallback strategies, controlled by the `acquisition_failure_strategy` parameter:

#### 29.2.1. 1. Random Space-Filling Design (Default)

**Strategy name:** "random"

This strategy uses Latin Hypercube Sampling (LHS) to generate a new space-filling point. LHS ensures good coverage of the search space by dividing each dimension into equal-probability intervals.

**When to use:**

- General-purpose optimization
- When you want simplicity and good space-filling properties
- Default choice for most problems

## 29. Acquisition Failure Handling in SpotOptim

**Example:**

```
from spotoptim import SpotOptim
import numpy as np

def sphere(X):
    return np.sum(X**2, axis=1)

optimizer = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=50,
    n_initial=10,
    acquisition_failure_strategy="random", # Default
    verbose=True
)

result = optimizer.optimize()
```

```
TensorBoard logging disabled
Initial best: f(x) = 0.872384
Iteration 1: New best f(x) = 0.114875
Iteration 2: New best f(x) = 0.018025
Iteration 3: New best f(x) = 0.000043
Iteration 4: New best f(x) = 0.000016
Iteration 5: New best f(x) = 0.000005
Iteration 6: New best f(x) = 0.000001
Iteration 7: New best f(x) = 0.000000
Iteration 8: New best f(x) = 0.000000
Iteration 9: f(x) = 0.000000
Iteration 10: f(x) = 0.000000
Iteration 11: f(x) = 0.000000
Iteration 12: f(x) = 0.000000
Iteration 13: New best f(x) = 0.000000
Iteration 14: New best f(x) = 0.000000
Iteration 15: f(x) = 0.000000
Iteration 16: f(x) = 0.000000
Iteration 17: f(x) = 0.000000
Iteration 18: f(x) = 0.000000
Iteration 19: f(x) = 0.000000
Iteration 20: f(x) = 0.000000
Iteration 21: f(x) = 0.000000
Iteration 22: f(x) = 0.000000
Iteration 23: f(x) = 0.000000
```

```

Iteration 24: f(x) = 0.000000
Iteration 25: f(x) = 0.000000
Iteration 26: f(x) = 0.000000
Iteration 27: f(x) = 0.000000
Iteration 28: f(x) = 0.000000
Iteration 29: f(x) = 0.000000
Iteration 30: f(x) = 0.000000
Iteration 31: f(x) = 0.000000
Iteration 32: f(x) = 0.000000
Iteration 33: f(x) = 0.000000
Iteration 34: f(x) = 0.000000
Iteration 35: f(x) = 0.000000
Iteration 36: f(x) = 0.000000
Iteration 37: f(x) = 0.000000
Iteration 38: f(x) = 0.000000
Iteration 39: f(x) = 0.000000
Iteration 40: f(x) = 0.000000

```

### 29.2.2. 2. Morris-Mitchell Minimizing Point

**Strategy name:** "mm"

This strategy finds a point that **maximizes the minimum distance** to all existing points. It evaluates 100 candidate points and selects the one with the largest minimum distance to the already-evaluated points, providing excellent space-filling properties.

**When to use:**

- When you want to ensure maximum exploration
- For problems where avoiding clustering of points is critical
- When the search space has been heavily sampled in some regions

**Example:**

```

from spotoptim import SpotOptim
import numpy as np

def rosenbrock(X):
    x = X[:, 0]
    y = X[:, 1]
    return (1 - x)**2 + 100 * (y - x**2)**2

optimizer = SpotOptim(
    fun=rosenbrock,
    bounds=[(-2, 2), (-2, 2)],

```

## 29. Acquisition Failure Handling in SpotOptim

```
max_iter=100,  
n_initial=20,  
acquisition_failure_strategy="mm", # Morris-Mitchell  
verbose=True  
)  
  
result = optimizer.optimize()
```

```
TensorBoard logging disabled  
Initial best: f(x) = 7.144807  
Iteration 1: New best f(x) = 5.487401  
Iteration 2: New best f(x) = 0.615834  
Iteration 3: f(x) = 6.766539  
Iteration 4: f(x) = 2.030955  
Iteration 5: f(x) = 4.559191  
Iteration 6: f(x) = 4.495502  
Iteration 7: f(x) = 13.903696  
Iteration 8: f(x) = 3.582306  
Iteration 9: New best f(x) = 0.127608  
Iteration 10: New best f(x) = 0.095625  
Iteration 11: New best f(x) = 0.086331  
Iteration 12: f(x) = 6.177527  
Iteration 13: New best f(x) = 0.081669  
Iteration 14: New best f(x) = 0.077639  
Iteration 15: f(x) = 2.967121  
Iteration 16: New best f(x) = 0.073313  
Iteration 17: New best f(x) = 0.069204  
Iteration 18: New best f(x) = 0.064872  
Iteration 19: f(x) = 2.980326  
Iteration 20: New best f(x) = 0.061465  
Iteration 21: New best f(x) = 0.058019  
Iteration 22: New best f(x) = 0.055254  
Iteration 23: New best f(x) = 0.052556  
Iteration 24: New best f(x) = 0.050013  
Iteration 25: New best f(x) = 0.048070  
Iteration 26: New best f(x) = 0.045750  
Iteration 27: New best f(x) = 0.044210  
Iteration 28: New best f(x) = 0.043185  
Iteration 29: New best f(x) = 0.042106  
Iteration 30: New best f(x) = 0.039726  
Iteration 31: f(x) = 2.915328  
Iteration 32: New best f(x) = 0.035403  
Iteration 33: New best f(x) = 0.031782  
Iteration 34: New best f(x) = 0.027706
```

Iteration 35: New best  $f(x)$  = 0.020730  
Iteration 36: New best  $f(x)$  = 0.014334  
Iteration 37: New best  $f(x)$  = 0.013791  
Iteration 38:  $f(x)$  = 0.014053  
Iteration 39: New best  $f(x)$  = 0.013289  
Iteration 40: New best  $f(x)$  = 0.012035  
Iteration 41: New best  $f(x)$  = 0.011083  
Iteration 42:  $f(x)$  = 2.801089  
Iteration 43: New best  $f(x)$  = 0.010703  
Iteration 44: New best  $f(x)$  = 0.010220  
Iteration 45: New best  $f(x)$  = 0.009507  
Iteration 46: New best  $f(x)$  = 0.009096  
Iteration 47: New best  $f(x)$  = 0.008935  
Iteration 48: New best  $f(x)$  = 0.008777  
Iteration 49: New best  $f(x)$  = 0.008709  
Iteration 50: New best  $f(x)$  = 0.008604  
Iteration 51: New best  $f(x)$  = 0.008528  
Iteration 52:  $f(x)$  = 2.696164  
Iteration 53: New best  $f(x)$  = 0.008493  
Iteration 54: New best  $f(x)$  = 0.008413  
Iteration 55: New best  $f(x)$  = 0.008343  
Iteration 56: New best  $f(x)$  = 0.008256  
Iteration 57: New best  $f(x)$  = 0.008171  
Iteration 58: New best  $f(x)$  = 0.008017  
Iteration 59:  $f(x)$  = 2.623422  
Iteration 60: New best  $f(x)$  = 0.007896  
Iteration 61: New best  $f(x)$  = 0.007765  
Iteration 62: New best  $f(x)$  = 0.007597  
Iteration 63: New best  $f(x)$  = 0.007317  
Iteration 64: New best  $f(x)$  = 0.006895  
Iteration 65: New best  $f(x)$  = 0.005704  
Iteration 66: New best  $f(x)$  = 0.003509  
Iteration 67: New best  $f(x)$  = 0.003285  
Iteration 68: New best  $f(x)$  = 0.002947  
Iteration 69: New best  $f(x)$  = 0.002710  
Iteration 70: New best  $f(x)$  = 0.002625  
Iteration 71: New best  $f(x)$  = 0.002586  
Iteration 72: New best  $f(x)$  = 0.002515  
Iteration 73:  $f(x)$  = 0.002534  
Iteration 74: New best  $f(x)$  = 0.002480  
Iteration 75: New best  $f(x)$  = 0.002431  
Iteration 76: New best  $f(x)$  = 0.002374  
Iteration 77: New best  $f(x)$  = 0.002338  
Iteration 78: New best  $f(x)$  = 0.002301  
Iteration 79: New best  $f(x)$  = 0.002287

## 29. Acquisition Failure Handling in SpotOptim

Iteration 80: New best  $f(x) = 0.002219$

### 29.3. How It Works

The acquisition failure handling is integrated into the optimization process:

1. **Acquisition optimization:** SpotOptim uses differential evolution to optimize the acquisition function
2. **Distance check:** The proposed point is checked against existing points using `tolerance_x`
3. **Fallback activation:** If the point is too close, `_handle_acquisition_failure()` is called
4. **Strategy execution:** The configured fallback strategy generates a new point
5. **Evaluation:** The fallback point is evaluated and added to the dataset

### 29.4. Comparison of Strategies

| Aspect                      | Random (LHS) | Morris-Mitchell           |
|-----------------------------|--------------|---------------------------|
| <b>Computation</b>          | Very fast    | Moderate (100 candidates) |
| <b>Space-filling</b>        | Good         | Excellent                 |
| <b>Exploration</b>          | Balanced     | Maximum distance          |
| <b>Clustering avoidance</b> | Good         | Best                      |
| <b>Recommended for</b>      | General use  | Heavily sampled spaces    |

### 29.5. Complete Example: Comparing Strategies

```
import numpy as np
from spotoptim import SpotOptim

def ackley(X):
    """Ackley function - multimodal test function"""
    a = 20
    b = 0.2
    c = 2 * np.pi
    n = X.shape[1]

    sum_sq = np.sum(X**2, axis=1)
```

### 29.5. Complete Example: Comparing Strategies

```
sum_cos = np.sum(np.cos(c * X), axis=1)

return -a * np.exp(-b * np.sqrt(sum_sq / n)) - np.exp(sum_cos / n) + a + np.e

# Test with random strategy
print("=" * 60)
print("Testing with Random Space-Filling Strategy")
print("=" * 60)

opt_random = SpotOptim(
    fun=ackley,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=50,
    n_initial=15,
    acquisition_failure_strategy="random",
    tolerance_x=0.1, # Relatively large tolerance to trigger failures
    seed=42,
    verbose=True
)

result_random = opt_random.optimize()

print(f"\nRandom Strategy Results:")
print(f"  Best value: {result_random.fun:.6f}")
print(f"  Best point: {result_random.x}")
print(f"  Total evaluations: {result_random.nfev}")

# Test with Morris-Mitchell strategy
print("\n" + "=" * 60)
print("Testing with Morris-Mitchell Strategy")
print("=" * 60)

opt_mm = SpotOptim(
    fun=ackley,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=50,
    n_initial=15,
    acquisition_failure_strategy="mm",
    tolerance_x=0.1, # Same tolerance
    seed=42,
    verbose=True
)

result_mm = opt_mm.optimize()
```

## 29. Acquisition Failure Handling in SpotOptim

```
print(f"\nMorris-Mitchell Strategy Results:")
print(f"  Best value: {result_mm.fun:.6f}")
print(f"  Best point: {result_mm.x}")
print(f"  Total evaluations: {result_mm.nfev}")

# Compare
print("\n" + "=" * 60)
print("Comparison")
print("=" * 60)
print(f"Random strategy:           {result_random.fun:.6f}")
print(f"Morris-Mitchell strategy: {result_mm.fun:.6f}")
if result_random.fun < result_mm.fun:
    print("→ Random strategy found better solution")
else:
    print("→ Morris-Mitchell strategy found better solution")
```

```
=====
Testing with Random Space-Filling Strategy
=====
```

```
TensorBoard logging disabled
Initial best: f(x) = 7.177375
Iteration 1: New best f(x) = 5.975822
Iteration 2: New best f(x) = 4.585872
Iteration 3: New best f(x) = 3.624193
Iteration 4: f(x) = 3.775836
    Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 5: f(x) = 7.428910
    Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 6: f(x) = 11.307695
    Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 7: f(x) = 11.588702
    Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 8: f(x) = 7.741328
    Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 9: f(x) = 8.547707
    Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 10: f(x) = 10.258140
```



### 29.5. Complete Example: Comparing Strategies

Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 11:  $f(x) = 10.575833$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 12:  $f(x) = 3.973820$   
Iteration 13: New best  $f(x) = 3.254219$   
Iteration 14:  $f(x) = 3.726104$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 15:  $f(x) = 9.023210$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 16:  $f(x) = 4.449845$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 17:  $f(x) = 11.616761$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 18:  $f(x) = 8.521779$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 19:  $f(x) = 9.840296$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 20:  $f(x) = 12.575946$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 21:  $f(x) = 7.719827$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 22:  $f(x) = 6.588107$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 23:  $f(x) = 9.186450$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 24:  $f(x) = 7.957337$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 25:  $f(x) = 13.092857$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 26:  $f(x) = 10.747619$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback

## 29. Acquisition Failure Handling in SpotOptim

Acquisition failure: Using random space-filling design as fallback.  
Iteration 27:  $f(x) = 9.894215$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 28:  $f(x) = 9.482454$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 29:  $f(x) = 6.720219$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 30:  $f(x) = 6.321851$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 31:  $f(x) = 6.841321$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 32:  $f(x) = 11.918429$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 33:  $f(x) = 13.292524$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 34:  $f(x) = 8.518795$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.  
Iteration 35:  $f(x) = 12.534822$

Random Strategy Results:  
Best value: 3.254219  
Best point: [0.03533184 0.66379523]  
Total evaluations: 50

=====  
Testing with Morris-Mitchell Strategy  
=====

TensorBoard logging disabled  
Initial best:  $f(x) = 7.177375$   
Iteration 1: New best  $f(x) = 5.975822$   
Iteration 2: New best  $f(x) = 4.585872$   
Iteration 3: New best  $f(x) = 3.624193$   
Iteration 4:  $f(x) = 3.775836$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 5:  $f(x) = 12.759495$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback

### 29.5. Complete Example: Comparing Strategies

Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 6:  $f(x) = 13.023769$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 7:  $f(x) = 10.937206$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 8:  $f(x) = 7.780549$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 9:  $f(x) = 12.396463$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 10:  $f(x) = 12.719725$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 11:  $f(x) = 13.078957$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 12:  $f(x) = 12.813177$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 13:  $f(x) = 11.674342$   
Iteration 14:  $f(x) = 3.967415$   
Iteration 15:  $f(x) = 3.756184$   
Iteration 16: New best  $f(x) = 3.159408$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 17:  $f(x) = 11.221948$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 18:  $f(x) = 8.775229$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 19:  $f(x) = 11.789243$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 20:  $f(x) = 4.971384$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 21:  $f(x) = 9.983363$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 22:  $f(x) = 10.021573$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback

## 29. Acquisition Failure Handling in SpotOptim

Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 23:  $f(x) = 6.575245$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 24:  $f(x) = 11.527738$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 25:  $f(x) = 12.133804$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 26:  $f(x) = 10.457280$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 27:  $f(x) = 11.003827$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 28:  $f(x) = 13.704437$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 29:  $f(x) = 10.955961$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 30:  $f(x) = 12.121347$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 31:  $f(x) = 7.325542$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 32:  $f(x) = 4.492325$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 33:  $f(x) = 6.456039$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 34:  $f(x) = 10.870934$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using Morris-Mitchell minimizing point as fallback.  
Iteration 35:  $f(x) = 10.677536$

Morris-Mitchell Strategy Results:

Best value: 3.159408  
Best point: [-0.00800104 0.71750603]  
Total evaluations: 50

=====

Comparison

```
=====
Random strategy:          3.254219
Morris-Mitchell strategy: 3.159408
→ Morris-Mitchell strategy found better solution
```

## 29.6. Advanced Usage: Setting Tolerance

The `tolerance_x` parameter controls when the fallback strategy is triggered. A larger tolerance means points need to be farther apart, triggering the fallback more often:

```
def simple_objective(X):
    """Simple quadratic function for demonstration"""
    return np.sum(X**2, axis=1)

bounds_demo = [(-5, 5), (-5, 5)]

# Strict tolerance (smaller value) - fewer fallbacks
optimizer_strict = SpotOptim(
    fun=simple_objective,
    bounds=bounds_demo,
    tolerance_x=1e-6, # Very small - almost never triggers fallback
    acquisition_failure_strategy="mm",
    max_iter=20,
    seed=42
)

# Relaxed tolerance (larger value) - more fallbacks
optimizer_relaxed = SpotOptim(
    fun=simple_objective,
    bounds=bounds_demo,
    tolerance_x=0.5, # Larger - triggers fallback more often
    acquisition_failure_strategy="mm",
    max_iter=20,
    seed=42
)

print(f"Strict tolerance setup complete")
print(f"Relaxed tolerance setup complete")
```

```
Strict tolerance setup complete
Relaxed tolerance setup complete
```

## 29.7. Best Practices

### 29.7.1. 1. Use Random for Most Problems

The random strategy (default) is sufficient for most optimization problems:

```
def my_objective(X):
    return np.sum(X**2, axis=1)

optimizer = SpotOptim(
    fun=my_objective,
    bounds=[(-5, 5), (-5, 5)],
    acquisition_failure_strategy="random", # Good default choice
    max_iter=20,
    seed=42
)
print("Random strategy optimizer created")
```

Random strategy optimizer created

### 29.7.2. 2. Use Morris-Mitchell for Intensive Sampling

When you have a large budget and want maximum exploration:

```
def expensive_objective(X):
    """Simulated expensive objective function"""
    return np.sum((X - 1)**2, axis=1)

optimizer = SpotOptim(
    fun=expensive_objective,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=30, # Large budget
    acquisition_failure_strategy="mm", # Maximize space coverage
    seed=42
)
print("Morris-Mitchell optimizer for intensive sampling created")
```

Morris-Mitchell optimizer for intensive sampling created

### 29.7.3. 3. Monitor Fallback Activations

Enable verbose mode to see when fallbacks are triggered:

```
def test_objective(X):
    return np.sum(X**2, axis=1)

optimizer = SpotOptim(
    fun=test_objective,
    bounds=[(-5, 5), (-5, 5)],
    acquisition_failure_strategy="mm",
    max_iter=20,
    verbose=True, # Shows fallback messages
    seed=42
)
print("Optimizer with verbose mode created")
```

```
TensorBoard logging disabled
Optimizer with verbose mode created
```

### 29.7.4. 4. Adjust Tolerance Based on Problem Scale

For problems with small search spaces, use smaller tolerance:

```
def scale_objective(X):
    return np.sum(X**2, axis=1)

# Small search space
optimizer_small = SpotOptim(
    fun=scale_objective,
    bounds=[(-1, 1), (-1, 1)],
    tolerance_x=0.01, # Small tolerance for small space
    acquisition_failure_strategy="random",
    max_iter=20,
    seed=42
)

# Large search space
optimizer_large = SpotOptim(
    fun=scale_objective,
    bounds=[(-100, 100), (-100, 100)],
    tolerance_x=1.0, # Larger tolerance for large space
    acquisition_failure_strategy="mm",
```

## 29. Acquisition Failure Handling in SpotOptim

```
max_iter=20,  
seed=42  
)  
  
print(f"Small space optimizer created (bounds: [-1, 1])")  
print(f"Large space optimizer created (bounds: [-100, 100])")
```

```
Small space optimizer created (bounds: [-1, 1])  
Large space optimizer created (bounds: [-100, 100])
```

## 29.8. Technical Details

### 29.8.1. Morris-Mitchell Implementation

The Morris-Mitchell strategy:

1. Generates 100 candidate points using Latin Hypercube Sampling
2. For each candidate, calculates the minimum distance to all existing points
3. Selects the candidate with the maximum minimum distance

This ensures the new point is as far as possible from the densest region of evaluated points.

### 29.8.2. Random Strategy Implementation

The random strategy:

1. Generates a single point using Latin Hypercube Sampling
2. Ensures the point is within bounds
3. Applies variable type repairs (rounding for int/factor variables)

This is computationally efficient while maintaining good space-filling properties.

## 29.9. Summary

- **Default strategy** ("random"): Fast, good space-filling, suitable for most problems
- **Morris-Mitchell** ("mm"): Better space-filling, maximizes minimum distance, ideal for intensive sampling
- **Trigger**: Activated when acquisition-proposed point is too close to existing points (within `tolerance_x`)



- **Control:** Set via `acquisition_failure_strategy` parameter
- **Monitoring:** Enable `verbose=True` to see when fallbacks occur

Choose the strategy that best matches your optimization goals: - Use "**random**" for general-purpose optimization - Use "**mm**" when you want maximum exploration and have a generous function evaluation budget



## 30. Factor Variables for Categorical Hyperparameters

SpotOptim supports factor variables for optimizing categorical hyperparameters, such as activation functions, optimizers, or any discrete string-based choices. Factor variables are automatically converted between string values (external interface) and integers (internal optimization), making categorical optimization seamless.

### 30.1. Overview

#### What are Factor Variables?

Factor variables allow you to specify categorical choices as tuples of strings in the bounds. SpotOptim handles the conversion:

1. **String tuples in bounds** → Internal integer mapping (0, 1, 2, ...)
2. **Optimization uses integers** internally for surrogate modeling
3. **Objective function receives strings** after automatic conversion
4. **Results return strings** (not integers)

Module: `spotoptim.SpotOptim`

#### Key Features:

- Define categorical choices as string tuples: ("ReLU", "Sigmoid", "Tanh")
- Automatic integer string conversion
- Seamless integration with neural network hyperparameters
- Mix factor variables with numeric/integer variables

### 30.2. Quick Start

#### 30.2.1. Basic Factor Variable Usage

### 30. Factor Variables for Categorical Hyperparameters

```
from spotoptim import SpotOptim
import numpy as np

def objective_function(X):
    """Objective function receives string values."""
    results = []
    for params in X:
        activation = params[0] # This is a string!
        print(f"Testing activation: {activation}")

        # Simple scoring based on activation choice (for demonstration)
        # In real use, you would train a model and return actual performance
        scores = {
            "ReLU": 3500.0,
            "Sigmoid": 4200.0,
            "Tanh": 3800.0,
            "LeakyReLU": 3600.0
        }
        score = scores.get(activation, 5000.0) + np.random.normal(0, 100)
        results.append(score)
    return np.array(results) # Return numpy array

# Define bounds with factor variable
optimizer = SpotOptim(
    fun=objective_function,
    bounds=[("ReLU", "Sigmoid", "Tanh", "LeakyReLU")],
    var_type=["factor"],
    max_iter=20,
    seed=42
)

result = optimizer.optimize()
print(f"\nBest activation: {result.x[0]}") # Returns string, e.g., "ReLU"
print(f"Best score: {result.fun:.4f}")
```

```
Testing activation: ReLU
Testing activation: Sigmoid
Testing activation: Tanh
Testing activation: LeakyReLU
Testing activation: Tanh
Testing activation: Sigmoid
Testing activation: Sigmoid
Testing activation: Sigmoid
Testing activation: Sigmoid
```

```

Testing activation: ReLU
Testing activation: Sigmoid
Testing activation: Tanh
Testing activation: LeakyReLU
Testing activation: Sigmoid
Testing activation: LeakyReLU
Testing activation: Tanh
Testing activation: Sigmoid
Testing activation: LeakyReLU
Testing activation: Sigmoid
Testing activation: Tanh

```

```

Best activation: LeakyReLU
Best score: 3428.1346

```

### 30.2.2. Neural Network Activation Function Optimization

```

import torch
import torch.nn as nn
from spotoptim import SpotOptim
from spotoptim.data import get_diabetes_dataloaders
from spotoptim.nn.linear_regressor import LinearRegressor
import numpy as np

def train_and_evaluate(X):
    """Train models with different activation functions."""
    results = []

    for params in X:
        activation = params[0] # String: "ReLU", "Sigmoid", etc.

        # Load data
        train_loader, test_loader, _ = get_diabetes_dataloaders()

        # Create model with the activation function
        model = LinearRegressor(
            input_dim=10,
            output_dim=1,
            l1=64,
            num_hidden_layers=2,
            activation=activation # Pass string directly!
        )

```

### 30. Factor Variables for Categorical Hyperparameters

```
# Train model
optimizer = model.get_optimizer("Adam", lr=0.01)
criterion = nn.MSELoss()

for epoch in range(50):
    model.train()
    for batch_X, batch_y in train_loader:
        predictions = model(batch_X)
        loss = criterion(predictions, batch_y)
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

# Evaluate
model.eval()
test_loss = 0.0
with torch.no_grad():
    for batch_X, batch_y in test_loader:
        predictions = model(batch_X)
        test_loss += criterion(predictions, batch_y).item()

avg_loss = test_loss / len(test_loader)
results.append(avg_loss)

return np.array(results) # Return numpy array

# Optimize activation function choice
optimizer = SpotOptim(
    fun=train_and_evaluate,
    bounds=[("ReLU", "Sigmoid", "Tanh", "LeakyReLU", "ELU")],
    var_type=["factor"],
    max_iter=30
)

result = optimizer.optimize()
print(f"Best activation function: {result.x[0]}")
print(f"Best test MSE: {result.fun:.4f}")
```

```
Best activation function: Sigmoid
Best test MSE: 26454.0534
```

## 30.3. Mixed Variable Types

### 30.3.1. Combining Factor, Integer, and Continuous Variables

```
import numpy as np
import torch
import torch.nn as nn
from spotoptim import SpotOptim
from spotoptim.data import get_diabetes_dataloaders
from spotoptim.nn.linear_regressor import LinearRegressor

def comprehensive_optimization(X):
    """Optimize learning rate, layer size, depth, and activation."""
    results = []

    for params in X:
        log_lr = params[0]      # Continuous (log scale)
        l1 = int(params[1])     # Integer
        n_layers = int(params[2]) # Integer
        activation = params[3]  # Factor (string)

        lr = 10 ** log_lr      # Convert from log scale

        print(f"lr={lr:.6f}, l1={l1}, layers={n_layers}, activation={activation}")

        # Load data
        train_loader, test_loader, _ = get_diabetes_dataloaders(
            batch_size=32,
            random_state=42
        )

        # Create model
        model = LinearRegressor(
            input_dim=10,
            output_dim=1,
            l1=l1,
            num_hidden_layers=n_layers,
            activation=activation
        )

        # Train
        optimizer = model.get_optimizer("Adam", lr=lr)
        criterion = nn.MSELoss()
```

```

        for epoch in range(30):
            model.train()
            for batch_X, batch_y in train_loader:
                predictions = model(batch_X)
                loss = criterion(predictions, batch_y)
                optimizer.zero_grad()
                loss.backward()
                optimizer.step()

            # Evaluate
            model.eval()
            test_loss = 0.0
            with torch.no_grad():
                for batch_X, batch_y in test_loader:
                    predictions = model(batch_X)
                    test_loss += criterion(predictions, batch_y).item()

            results.append(test_loss / len(test_loader))

    return np.array(results)

# Optimize all four hyperparameters simultaneously
optimizer = SpotOptim(
    fun=comprehensive_optimization,
    bounds=[
        (-4, -2), # log10(learning_rate)
        (16, 128), # l1 (neurons per layer)
        (0, 4), # num_hidden_layers
        ("ReLU", "Sigmoid", "Tanh", "LeakyReLU") # activation function
    ],
    var_type=["float", "int", "int", "factor"],
    max_iter=50
)

result = optimizer.optimize()

# Results contain original string values
print("\nOptimization Results:")
print(f"Best learning rate: {10**result.x[0]:.6f}")
print(f"Best layer size: {int(result.x[1])}")
print(f"Best num layers: {int(result.x[2])}")
print(f"Best activation: {result.x[3]}") # String value!
print(f"Best test MSE: {result.fun:.4f}")

```



```

lr=0.000136, l1=89, layers=2, activation=LeakyReLU
lr=0.001164, l1=30, layers=3, activation=ReLU
lr=0.000326, l1=41, layers=3, activation=Tanh
lr=0.000167, l1=119, layers=4, activation=Tanh
lr=0.002101, l1=54, layers=3, activation=Tanh
lr=0.008649, l1=19, layers=1, activation=Sigmoid
lr=0.003605, l1=81, layers=0, activation=LeakyReLU
lr=0.005220, l1=62, layers=1, activation=Sigmoid
lr=0.000407, l1=113, layers=1, activation=ReLU
lr=0.000714, l1=97, layers=2, activation=Sigmoid
lr=0.000111, l1=48, layers=2, activation=Sigmoid
lr=0.008093, l1=68, layers=0, activation=Sigmoid
lr=0.003038, l1=109, layers=4, activation=Tanh
lr=0.000734, l1=37, layers=1, activation=Tanh
lr=0.002262, l1=19, layers=1, activation=Tanh
lr=0.002131, l1=19, layers=1, activation=Tanh
lr=0.000188, l1=117, layers=2, activation=LeakyReLU
lr=0.004495, l1=65, layers=2, activation=ReLU
lr=0.000642, l1=26, layers=2, activation=ReLU
lr=0.000356, l1=35, layers=2, activation=LeakyReLU
lr=0.000106, l1=72, layers=3, activation=Sigmoid
lr=0.001278, l1=101, layers=2, activation=Tanh
lr=0.000527, l1=42, layers=1, activation=Tanh
lr=0.009117, l1=103, layers=2, activation=Tanh
lr=0.006307, l1=49, layers=2, activation=Sigmoid
lr=0.000511, l1=40, layers=0, activation=ReLU
lr=0.000376, l1=55, layers=1, activation=Tanh
lr=0.001814, l1=85, layers=1, activation=Tanh
lr=0.000645, l1=82, layers=1, activation=Tanh
lr=0.007591, l1=84, layers=3, activation=Sigmoid
lr=0.000470, l1=36, layers=0, activation=Tanh
lr=0.000156, l1=55, layers=2, activation=Tanh
lr=0.001479, l1=21, layers=0, activation=Tanh
lr=0.000344, l1=72, layers=0, activation=Tanh
lr=0.000931, l1=47, layers=2, activation=Sigmoid
lr=0.000903, l1=18, layers=1, activation=Sigmoid
lr=0.007118, l1=19, layers=1, activation=Sigmoid
lr=0.002511, l1=82, layers=3, activation=Sigmoid
lr=0.007096, l1=120, layers=3, activation=Sigmoid
lr=0.006184, l1=110, layers=3, activation=Tanh
lr=0.001482, l1=75, layers=1, activation=Sigmoid
lr=0.008116, l1=101, layers=2, activation=ReLU
lr=0.000578, l1=25, layers=2, activation=Tanh
lr=0.002706, l1=106, layers=2, activation=Tanh
lr=0.000676, l1=37, layers=2, activation=Sigmoid

```

### 30. Factor Variables for Categorical Hyperparameters

```
lr=0.009698, l1=96, layers=1, activation=Sigmoid
lr=0.000843, l1=21, layers=3, activation=Tanh
lr=0.007088, l1=117, layers=3, activation=ReLU
lr=0.001631, l1=50, layers=2, activation=Sigmoid
lr=0.002216, l1=85, layers=1, activation=Sigmoid
```

Optimization Results:

Best learning rate: 0.002511

Best layer size: 82

Best num layers: 3

Best activation: Sigmoid

Best test MSE: 26488.3490

## 30.4. Multiple Factor Variables

### 30.4.1. Optimizing Both Activation and Optimizer

```
from spotoptim import SpotOptim
from spotoptim.data import get_diabetes_data loaders
from spotoptim.nn.linear_regressor import LinearRegressor
import torch.nn as nn
import numpy as np

def optimize_activation_and_optimizer(X):
    """Optimize both activation function and optimizer choice."""
    results = []

    for params in X:
        activation = params[0]      # Factor variable 1
        optimizer_name = params[1]  # Factor variable 2
        lr = 10 ** params[2]        # Continuous variable

        train_loader, test_loader, _ = get_diabetes_data loaders()

        model = LinearRegressor(
            input_dim=10,
            output_dim=1,
            l1=64,
            num_hidden_layers=2,
            activation=activation
        )
```

```

# Use the optimizer string
optimizer = model.get_optimizer(optimizer_name, lr=lr)
criterion = nn.MSELoss()

# Train
for epoch in range(30):
    model.train()
    for batch_X, batch_y in train_loader:
        predictions = model(batch_X)
        loss = criterion(predictions, batch_y)
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

# Evaluate
model.eval()
test_loss = 0.0
with torch.no_grad():
    for batch_X, batch_y in test_loader:
        predictions = model(batch_X)
        test_loss += criterion(predictions, batch_y).item()

results.append(test_loss / len(test_loader))

return np.array(results) # Return numpy array

# Two factor variables + one continuous
opt = SpotOptim(
    fun=optimize_activation_and_optimizer,
    bounds=[
        ("ReLU", "Tanh", "Sigmoid", "LeakyReLU"), # Activation
        ("Adam", "SGD", "RMSprop", "AdamW"), # Optimizer
        (-4, -2) # log10(lr)
    ],
    var_type=["factor", "factor", "float"],
    max_iter=40
)

result = opt.optimize()
print(f"Best activation: {result.x[0]}")
print(f"Best optimizer: {result.x[1]}")
print(f"Best learning rate: {10**result.x[2]:.6f}")

```

Best activation: Tanh

Best optimizer: SGD  
Best learning rate: 0.004387

## 30.5. Advanced Usage

### 30.5.1. Custom Categorical Choices

Factor variables work with any string values, not just activation functions:

```
from spotoptim import SpotOptim
import numpy as np

def train_model_with_config(dropout_policy, batch_norm, weight_init):
    """Simulate model training with different configurations."""
    # In real use, this would train an actual model
    # Here we return synthetic scores for demonstration
    base_score = 3000.0

    # Dropout impact
    dropout_scores = {"none": 200, "light": 0, "heavy": 100}
    # Batch norm impact
    bn_scores = {"before": -50, "after": 0, "none": 150}
    # Weight init impact
    init_scores = {"xavier": 0, "kaiming": -30, "normal": 100}

    score = (base_score +
             dropout_scores.get(dropout_policy, 0) +
             bn_scores.get(batch_norm, 0) +
             init_scores.get(weight_init, 0) +
             np.random.normal(0, 50))

    return score

def train_with_config(X):
    """Objective function with various categorical choices."""
    results = []

    for params in X:
        dropout_policy = params[0] # "none", "light", "heavy"
        batch_norm = params[1]     # "before", "after", "none"
        weight_init = params[2]    # "xavier", "kaiming", "normal"

        # Use these strings to configure your model
```

```

        score = train_model_with_config(
            dropout_policy=dropout_policy,
            batch_norm=batch_norm,
            weight_init=weight_init
        )
        results.append(score)

    return np.array(results) # Return numpy array

optimizer = SpotOptim(
    fun=train_with_config,
    bounds=[
        ("none", "light", "heavy"),          # Dropout policy
        ("before", "after", "none"),          # Batch norm position
        ("xavier", "kaiming", "normal")       # Weight initialization
    ],
    var_type=["factor", "factor", "factor"],
    max_iter=25,
    seed=42
)

result = optimizer.optimize()
print("Best configuration:")
print(f" Dropout: {result.x[0]}")
print(f" Batch norm: {result.x[1]}")
print(f" Weight init: {result.x[2]}")
print(f" Score: {result.fun:.4f}")

```

```

Best configuration:
Dropout: light
Batch norm: before
Weight init: kaiming
Score: 2873.8238

```

### 30.5.2. Viewing All Evaluated Configurations

```

import torch
import torch.nn as nn
from spotoptim import SpotOptim
from spotoptim.data import get_diabetes_dataloaders
from spotoptim.nn.linear_regressor import LinearRegressor
import numpy as np

```

### 30. Factor Variables for Categorical Hyperparameters

```
def train_and_evaluate(X):
    """Train models with different activation functions."""
    results = []

    for params in X:
        l1 = int(params[0])          # Integer: layer size
        activation = params[1]       # String: activation function

        # Load data
        train_loader, test_loader, _ = get_diabetes_dataloaders()

        # Create model with the activation function
        model = LinearRegressor(
            input_dim=10,
            output_dim=1,
            l1=l1,
            num_hidden_layers=2,
            activation=activation    # Pass string directly!
        )

        # Train model
        optimizer = model.get_optimizer("Adam", lr=0.01)
        criterion = nn.MSELoss()

        for epoch in range(50):
            model.train()
            for batch_X, batch_y in train_loader:
                predictions = model(batch_X)
                loss = criterion(predictions, batch_y)
                optimizer.zero_grad()
                loss.backward()
                optimizer.step()

            # Evaluate
            model.eval()
            test_loss = 0.0
            with torch.no_grad():
                for batch_X, batch_y in test_loader:
                    predictions = model(batch_X)
                    test_loss += criterion(predictions, batch_y).item()

            avg_loss = test_loss / len(test_loader)
            results.append(avg_loss)
```

```

    return np.array(results)

optimizer = SpotOptim(
    fun=train_and_evaluate,
    bounds=[
        (16, 128),                # Layer size
        ("ReLU", "Sigmoid", "Tanh", "LeakyReLU") # Activation
    ],
    var_type=["int", "factor"], # IMPORTANT: Specify variable types!
    max_iter=30,
    seed=42
)

result = optimizer.optimize()

# Access all evaluated configurations
print("\nAll evaluated configurations:")
print("Layer Size | Activation | Test MSE")
print("-" * 42)
for i in range(min(10, len(result.X))): # Show first 10
    l1 = int(result.X[i, 0])
    activation = result.X[i, 1] # String value!
    loss = result.y[i]
    print(f"{l1:10d} | {activation:10s} | {loss:.4f}")

# Find top 5 configurations
sorted_indices = result.y.argsort()[:5]
print("\nTop 5 configurations:")
for idx in sorted_indices:
    print(f"l1={int(result.X[idx, 0]):3d}, "
          f"activation={result.X[idx, 1]:10s}, "
          f"MSE={result.y[idx]:.4f}")

```

All evaluated configurations:

Layer Size | Activation | Test MSE

```

-----
    41 | Tanh      | 26575.1497
   118 | Sigmoid    | 26095.8574
    26 | Tanh      | 26686.2240
   108 | Sigmoid    | 26397.8965
    71 | LeakyReLU  | 26571.2461
    34 | Tanh      | 26622.2246
    87 | ReLU      | 26518.5443

```

### 30. Factor Variables for Categorical Hyperparameters

|     |         |            |
|-----|---------|------------|
| 101 | Tanh    | 26520.6465 |
| 55  | Sigmoid | 26568.4062 |
| 74  | ReLU    | 26519.1882 |

Top 5 configurations:

|                            |                  |
|----------------------------|------------------|
| l1=118, activation=Sigmoid | , MSE=26095.8574 |
| l1=127, activation=Sigmoid | , MSE=26271.5775 |
| l1=115, activation=Sigmoid | , MSE=26336.9460 |
| l1=108, activation=Sigmoid | , MSE=26397.8965 |
| l1=128, activation=ReLU    | , MSE=26455.9264 |

## 30.6. How It Works

### 30.6.1. Internal Mechanism

SpotOptim handles factor variables through automatic conversion:

1. **Initialization:** String tuples in bounds are detected

```
bounds = [("ReLU", "Sigmoid", "Tanh")]
# Internally mapped to: {0: "ReLU", 1: "Sigmoid", 2: "Tanh"}
# Bounds become: [(0, 2)]
```

2. **Sampling:** Initial design samples from `[0, n_levels-1]` and rounds to integers

```
# Samples might be: [0.3, 1.8, 2.1]
# After rounding: [0, 2, 2]
```

3. **Evaluation:** Before calling objective function, integers  $\rightarrow$  strings

```
# [0, 2, 2]  $\rightarrow$  ["ReLU", "Tanh", "Tanh"]
# Objective function receives strings
```

4. **Optimization:** Surrogate model works with integers `[0, n_levels-1]`

5. **Results:** Final results mapped back to strings

```
result.x[0] # Returns "ReLU", not 0
result.X    # All rows contain strings for factor variables
```

```
array([[41.0, 'Tanh'],
       [118.0, 'Sigmoid'],
       [26.0, 'Tanh'],
       [108.0, 'Sigmoid'],
       [71.0, 'LeakyReLU'],
       [34.0, 'Tanh'],
```



```
[87.0, 'ReLU'],
[101.0, 'Tanh'],
[55.0, 'Sigmoid'],
[74.0, 'ReLU'],
[128.0, 'ReLU'],
[20.0, 'Tanh'],
[87.0, 'Tanh'],
[107.0, 'LeakyReLU'],
[75.0, 'Tanh'],
[53.0, 'Tanh'],
[35.0, 'Sigmoid'],
[93.0, 'Sigmoid'],
[117.0, 'ReLU'],
[38.0, 'Tanh'],
[96.0, 'Sigmoid'],
[112.0, 'Tanh'],
[127.0, 'Sigmoid'],
[54.0, 'Sigmoid'],
[64.0, 'LeakyReLU'],
[115.0, 'Sigmoid'],
[75.0, 'Sigmoid'],
[42.0, 'Sigmoid'],
[66.0, 'Sigmoid'],
[94.0, 'LeakyReLU']], dtype=object)
```

### 30.6.2. Variable Type Auto-Detection

If you don't specify `var_type`, SpotOptim automatically detects factor variables:

```
# Example 1: Explicit var_type (recommended)
# This shows the syntax - replace my_function with your actual function

# optimizer = SpotOptim(
#     fun=my_function,
#     bounds=[(-4, -2), ("ReLU", "Tanh")],
#     var_type=["float", "factor"] # Explicit
# )

# Example 2: Auto-detection (works but less explicit)
# optimizer = SpotOptim(
#     fun=my_function,
#     bounds=[(-4, -2), ("ReLU", "Tanh")]
#     # var_type automatically set to ["float", "factor"]
# )
```

### 30. Factor Variables for Categorical Hyperparameters

```
# Here's a working example:
from spotoptim import SpotOptim
import numpy as np

def demo_function(X):
    results = []
    for params in X:
        lr = 10 ** params[0] # Continuous parameter
        activation = params[1] # Factor parameter
        score = 3000 + lr * 100 + {"ReLU": 0, "Tanh": 50}.get(activation, 100)
        results.append(score + np.random.normal(0, 10))
    return np.array(results)

# With explicit var_type (recommended)
optimizer = SpotOptim(
    fun=demo_function,
    bounds=[(-4, -2), ("ReLU", "Tanh")],
    var_type=["float", "factor"], # Explicit is clearer
    max_iter=10,
    seed=42
)

result = optimizer.optimize()
print(f"Best lr: {10**result.x[0]:.6f}, Best activation: {result.x[1]}")
```

Best lr: 0.000489, Best activation: ReLU

## 30.7. Complete Example: Full Workflow

```
"""
Complete example: Neural network hyperparameter optimization with factor variables.
"""

import numpy as np
import torch
import torch.nn as nn
from spotoptim import SpotOptim
from spotoptim.data import get_diabetes_dataloaders
from spotoptim.nn.linear_regressor import LinearRegressor
```

```

def objective_function(X):
    """Train and evaluate models with given hyperparameters."""
    results = []

    for params in X:
        # Extract hyperparameters
        log_lr = params[0]
        l1 = int(params[1])
        num_layers = int(params[2])
        activation = params[3] # String!

        lr = 10 ** log_lr

        print(f"Testing: lr={lr:.6f}, l1={l1}, layers={num_layers}, "
              f"activation={activation}")

        # Load data
        train_loader, test_loader, _ = get_diabetes_data loaders(
            test_size=0.2,
            batch_size=32,
            random_state=42
        )

        # Create and train model
        model = LinearRegressor(
            input_dim=10,
            output_dim=1,
            l1=l1,
            num_hidden_layers=num_layers,
            activation=activation
        )

        optimizer = model.get_optimizer("Adam", lr=lr)
        criterion = nn.MSELoss()

        # Training loop
        num_epochs = 30
        for epoch in range(num_epochs):
            model.train()
            for batch_X, batch_y in train_loader:
                predictions = model(batch_X)
                loss = criterion(predictions, batch_y)
                optimizer.zero_grad()
                loss.backward()

```

```

        optimizer.step()

    # Evaluation
    model.eval()
    test_loss = 0.0
    with torch.no_grad():
        for batch_X, batch_y in test_loader:
            predictions = model(batch_X)
            loss = criterion(predictions, batch_y)
            test_loss += loss.item()

    avg_test_loss = test_loss / len(test_loader)
    results.append(avg_test_loss)
    print(f" → Test MSE: {avg_test_loss:.4f}")

    return np.array(results)

def main():
    print("=" * 80)
    print("Neural Network Hyperparameter Optimization with Factor Variables")
    print("=" * 80)

    # Define optimization problem
    optimizer = SpotOptim(
        fun=objective_function,
        bounds=[
            (-4, -2), # log10(learning_rate)
            (16, 128), # l1 (neurons)
            (0, 4), # num_hidden_layers
            ("ReLU", "Sigmoid", "Tanh", "LeakyReLU") # activation (factor!)
        ],
        var_type=["float", "int", "int", "factor"],
        max_iter=50,
        seed=42
    )

    # Run optimization
    print("\nStarting optimization...")
    result = optimizer.optimize()

    # Display results
    print("\n" + "=" * 80)
    print("OPTIMIZATION RESULTS")

```

```

print("=" * 80)
print(f"Best learning rate: {10**result.x[0]:.6f}")
print(f"Best layer size (l1): {int(result.x[1])}")
print(f"Best num hidden layers: {int(result.x[2])}")
print(f"Best activation function: {result.x[3]}") # String value!
print(f"Best test MSE: {result.fun:.4f}")

# Show top 5 configurations
print("\n" + "=" * 80)
print("TOP 5 CONFIGURATIONS")
print("=" * 80)
sorted_indices = result.y.argsort()[:5]
print(f"{'Rank':<6} {'LR':<12} {'L1':<6} {'Layers':<8} "
      f"{'Activation':<12} {'MSE':<10}")
print("-" * 80)
for rank, idx in enumerate(sorted_indices, 1):
    lr = 10 ** result.X[idx, 0]
    l1 = int(result.X[idx, 1])
    layers = int(result.X[idx, 2])
    activation = result.X[idx, 3]
    mse = result.y[idx]
    print(f"{'rank':<6} {'lr':<12.6f} {'l1':<6} {'layers':<8} "
          f"{'activation':<12} {'mse':<10.4f}")

# Train final model with best configuration
print("\n" + "=" * 80)
print("TRAINING FINAL MODEL")
print("=" * 80)

best_lr = 10 ** result.x[0]
best_l1 = int(result.x[1])
best_layers = int(result.x[2])
best_activation = result.x[3]

print(f"Configuration: lr={best_lr:.6f}, l1={best_l1}, "
      f"layers={best_layers}, activation={best_activation}")

train_loader, test_loader, _ = get_diabetes_dataloaders(
    test_size=0.2,
    batch_size=32,
    random_state=42
)

final_model = LinearRegressor(

```

```

        input_dim=10,
        output_dim=1,
        l1=best_l1,
        num_hidden_layers=best_layers,
        activation=best_activation
    )

    optimizer_final = final_model.get_optimizer("Adam", lr=best_lr)
    criterion = nn.MSELoss()

    # Extended training
    num_epochs = 100
    print(f"\nTraining for {num_epochs} epochs...")
    for epoch in range(num_epochs):
        final_model.train()
        train_loss = 0.0
        for batch_X, batch_y in train_loader:
            predictions = final_model(batch_X)
            loss = criterion(predictions, batch_y)
            optimizer_final.zero_grad()
            loss.backward()
            optimizer_final.step()
            train_loss += loss.item()

        if (epoch + 1) % 20 == 0:
            avg_train_loss = train_loss / len(train_loader)
            print(f"Epoch {epoch+1}/{num_epochs}: Train MSE = {avg_train_loss:.4f}")

    # Final evaluation
    final_model.eval()
    final_test_loss = 0.0
    with torch.no_grad():
        for batch_X, batch_y in test_loader:
            predictions = final_model(batch_X)
            final_test_loss += criterion(predictions, batch_y).item()

    final_avg_loss = final_test_loss / len(test_loader)
    print(f"\nFinal Test MSE: {final_avg_loss:.4f}")
    print("=" * 80)

if __name__ == "__main__":
    main()

```

```
=====
Neural Network Hyperparameter Optimization with Factor Variables
=====
```

```
Starting optimization...
Testing: lr=0.007002, l1=101, layers=2, activation=ReLU
  → Test MSE: 26560.6862
Testing: lr=0.000604, l1=50, layers=2, activation=ReLU
  → Test MSE: 26613.0625
Testing: lr=0.000149, l1=67, layers=1, activation=Tanh
  → Test MSE: 26665.4544
Testing: lr=0.000296, l1=40, layers=0, activation=Tanh
  → Test MSE: 26692.7350
Testing: lr=0.004887, l1=116, layers=2, activation=Sigmoid
  → Test MSE: 26774.0846
Testing: lr=0.001772, l1=124, layers=3, activation=Sigmoid
  → Test MSE: 26629.0540
Testing: lr=0.001107, l1=36, layers=4, activation=Sigmoid
  → Test MSE: 26622.7357
Testing: lr=0.003708, l1=20, layers=1, activation=LeakyReLU
  → Test MSE: 26566.9766
Testing: lr=0.000861, l1=90, layers=1, activation=Tanh
  → Test MSE: 26636.7865
Testing: lr=0.000237, l1=78, layers=3, activation=Tanh
  → Test MSE: 26643.4538
Testing: lr=0.009463, l1=24, layers=0, activation=LeakyReLU
  → Test MSE: 26599.9290
Testing: lr=0.003859, l1=20, layers=1, activation=LeakyReLU
  → Test MSE: 26568.6895
Testing: lr=0.001897, l1=20, layers=1, activation=LeakyReLU
  → Test MSE: 26704.4355
Testing: lr=0.000292, l1=36, layers=3, activation=Sigmoid
  → Test MSE: 26546.3939
Testing: lr=0.000292, l1=36, layers=3, activation=Sigmoid
  → Test MSE: 26550.5176
Testing: lr=0.000761, l1=94, layers=4, activation=Tanh
  → Test MSE: 26634.0840
Testing: lr=0.003722, l1=82, layers=1, activation=Tanh
  → Test MSE: 26557.9655
Testing: lr=0.007645, l1=96, layers=3, activation=Sigmoid
  → Test MSE: 26533.1419
Testing: lr=0.000769, l1=40, layers=1, activation=Tanh
  → Test MSE: 26633.2415
Testing: lr=0.000235, l1=109, layers=4, activation=LeakyReLU
  → Test MSE: 26631.0065
```

### 30. Factor Variables for Categorical Hyperparameters

Testing: lr=0.000359, l1=76, layers=3, activation=Sigmoid  
→ Test MSE: 26659.3678  
Testing: lr=0.004959, l1=50, layers=2, activation=Tanh  
→ Test MSE: 26578.3561  
Testing: lr=0.002494, l1=57, layers=3, activation=LeakyReLU  
→ Test MSE: 26669.5996  
Testing: lr=0.005807, l1=20, layers=0, activation=Sigmoid  
→ Test MSE: 26685.8483  
Testing: lr=0.002939, l1=19, layers=1, activation=Sigmoid  
→ Test MSE: 26814.0944  
Testing: lr=0.001263, l1=98, layers=4, activation=ReLU  
→ Test MSE: 26600.4954  
Testing: lr=0.001226, l1=105, layers=3, activation=Sigmoid  
→ Test MSE: 26548.8229  
Testing: lr=0.004431, l1=32, layers=1, activation=Sigmoid  
→ Test MSE: 26469.6374  
Testing: lr=0.001367, l1=58, layers=2, activation=Sigmoid  
→ Test MSE: 26542.7878  
Testing: lr=0.006778, l1=81, layers=4, activation=Tanh  
→ Test MSE: 26538.8477  
Testing: lr=0.002189, l1=112, layers=4, activation=Sigmoid  
→ Test MSE: 26685.5397  
Testing: lr=0.004558, l1=24, layers=2, activation=Tanh  
→ Test MSE: 26613.6257  
Testing: lr=0.000658, l1=125, layers=0, activation=Tanh  
→ Test MSE: 26697.2786  
Testing: lr=0.000272, l1=119, layers=2, activation=Tanh  
→ Test MSE: 26653.8477  
Testing: lr=0.000133, l1=64, layers=2, activation=Tanh  
→ Test MSE: 26681.2135  
Testing: lr=0.002172, l1=70, layers=2, activation=LeakyReLU  
→ Test MSE: 26623.0625  
Testing: lr=0.000223, l1=28, layers=3, activation=Sigmoid  
→ Test MSE: 26664.1530  
Testing: lr=0.006065, l1=53, layers=3, activation=Sigmoid  
→ Test MSE: 26585.5964  
Testing: lr=0.000352, l1=42, layers=4, activation=ReLU  
→ Test MSE: 26626.8411  
Testing: lr=0.003464, l1=124, layers=2, activation=Tanh  
→ Test MSE: 26617.5078  
Testing: lr=0.000219, l1=37, layers=3, activation=ReLU  
→ Test MSE: 26607.7122  
Testing: lr=0.002619, l1=70, layers=3, activation=ReLU  
→ Test MSE: 26599.6543  
Testing: lr=0.004686, l1=123, layers=2, activation=ReLU



### 30.7. Complete Example: Full Workflow

```
→ Test MSE: 26603.7083
Testing: lr=0.000165, l1=44, layers=0, activation=ReLU
→ Test MSE: 26632.7734
Testing: lr=0.000917, l1=93, layers=1, activation=Sigmoid
→ Test MSE: 26661.3503
Testing: lr=0.001789, l1=117, layers=1, activation=Tanh
→ Test MSE: 26627.0664
Testing: lr=0.000134, l1=101, layers=4, activation=Sigmoid
→ Test MSE: 26700.9987
Testing: lr=0.004937, l1=108, layers=2, activation=ReLU
→ Test MSE: 26565.9193
Testing: lr=0.004047, l1=93, layers=1, activation=ReLU
→ Test MSE: 26658.8112
Testing: lr=0.000997, l1=112, layers=4, activation=Tanh
→ Test MSE: 26655.9811
```

#### OPTIMIZATION RESULTS

```
Best learning rate: 0.004431
Best layer size (l1): 32
Best num hidden layers: 1
Best activation function: Sigmoid
Best test MSE: 26469.6374
```

#### TOP 5 CONFIGURATIONS

| Rank | LR       | L1 | Layers | Activation | MSE        |
|------|----------|----|--------|------------|------------|
| 1    | 0.004431 | 32 | 1      | Sigmoid    | 26469.6374 |
| 2    | 0.007645 | 96 | 3      | Sigmoid    | 26533.1419 |
| 3    | 0.006778 | 81 | 4      | Tanh       | 26538.8477 |
| 4    | 0.001367 | 58 | 2      | Sigmoid    | 26542.7878 |
| 5    | 0.000292 | 36 | 3      | Sigmoid    | 26546.3939 |

#### TRAINING FINAL MODEL

```
Configuration: lr=0.004431, l1=32, layers=1, activation=Sigmoid
```

```
Training for 100 epochs...
```

```
Epoch 20/100: Train MSE = 32324.7500
```

```
Epoch 40/100: Train MSE = 33846.2775
```

```
Epoch 60/100: Train MSE = 32674.8579
```

### 30. Factor Variables for Categorical Hyperparameters

Epoch 80/100: Train MSE = 28069.8488  
Epoch 100/100: Train MSE = 30106.8976

Final Test MSE: 26789.6029

=====

## 30.8. Best Practices

### 30.8.1. Do's

Use descriptive string values

```
bounds=[("xavier_uniform", "kaiming_normal", "orthogonal")]
```

Explicitly specify `var_type` for clarity

```
var_type=["float", "int", "factor"]
```

Access results as strings

```
# Example: Accessing factor variable results as strings
# (This assumes you've run an optimization with activation as a factor variable)

# If you have a result from the previous examples:
# best_activation = result.x[3] # For 4-parameter optimization
# Or for simpler cases:
# best_activation = result.x[0] # For single-parameter optimization

# Example with inline optimization:
from spotoptim import SpotOptim
import numpy as np

def quick_test(X):
    results = []
    for params in X:
        activation = params[0]
        score = {"ReLU": 3500, "Tanh": 3600}.get(activation, 4000)
        results.append(score + np.random.normal(0, 50))
    return np.array(results)

opt = SpotOptim(
    fun=quick_test,
```

```

    bounds=[("ReLU", "Tanh")],
    var_type=["factor"],
    max_iter=10,
    seed=42
)
result = opt.optimize()

# Access as string - this is the correct way
best_activation = result.x[0] # String value like "ReLU"
print(f"Best activation: {best_activation} (type: {type(best_activation).__name__})")

# You can use it directly in your model
# model = LinearRegressor(activation=best_activation)

```

Best activation: ReLU (type: str)

#### Mix factor variables with numeric/integer variables

```

bounds=[(-4, -2), (16, 128), ("ReLU", "Tanh")]
var_type=["float", "int", "factor"]

```

### 30.8.2. Don'ts

#### Don't use integers in factor bounds

```

# Wrong: Use strings, not integers
bounds=[(0, 1, 2)] # Wrong!
bounds=[("ReLU", "Sigmoid", "Tanh")] # Correct!

```

#### Don't expect integers in objective function

```

def objective(X):
    activation = X[0][2]
    # activation is a string, not an integer!
    # Don't do: if activation == 0: # Wrong!
    # Do: if activation == "ReLU": # Correct!

```

#### Don't manually convert factor variables

```

# SpotOptim handles conversion automatically
# Don't do manual mapping in your objective function

```

## 30. Factor Variables for Categorical Hyperparameters

### Don't use empty tuples

```
# Wrong: Empty tuple
bounds=[()]

# Correct: At least one string
bounds=[("ReLU",)] # Single choice (will be treated as fixed)
```

## 30.9. Troubleshooting

### 30.9.1. Common Issues

**Issue:** Objective function receives integers instead of strings

**Solution:** Ensure you're using the latest version of SpotOptim with factor variable support. Factor variables are automatically converted before calling the objective function.

---

**Issue:** `ValueError: could not convert string to float`

**Solution:** This occurs if there's a version mismatch. Update SpotOptim to ensure the object array conversion is implemented correctly.

---

**Issue:** Results show integers instead of strings

**Solution:** Check that you're accessing `result.x` (mapped values) instead of internal arrays. The result object automatically maps factor variables to their original strings.

---

**Issue:** Single-level factor variables cause dimension reduction

**Behavior:** If a factor variable has only one choice, e.g., `("ReLU",)`, SpotOptim treats it as a fixed dimension and may reduce the dimensionality. This is expected behavior.

**Solution:** Use at least two choices for optimization, or remove single-choice dimensions from bounds.

## 30.10. Summary

Factor variables in SpotOptim enable:

- **Categorical optimization:** Optimize over discrete string choices
- **Automatic conversion:** Seamless integer string mapping
- **Neural network hyperparameters:** Optimize activation functions, optimizers, etc.
- **Mixed variable types:** Combine with continuous and integer variables
- **Clean interface:** Objective functions work with strings directly
- **String results:** Final results contain original string values

Factor variables make categorical hyperparameter optimization as easy as continuous optimization!

## 30.11. See Also

- [LinearRegressor Documentation](#) - Neural network class supporting string-based activation functions
- [Diabetes Dataset Utilities](#) - Data loading utilities used in examples
- [Variable Types](#) - Overview of all variable types in SpotOptim
- [Save and Load](#) - Saving and loading optimization results with factor variables



## 31. Variable Transformations for Search Space Scaling

SpotOptim supports automatic variable transformations to improve optimization in scaled search spaces. Instead of manually handling transformations (e.g., log-scale for learning rates), you can specify transformations via the `var_trans` parameter, and SpotOptim handles everything internally.

### 31.1. Overview

#### What are Variable Transformations?

Variable transformations allow you to specify how search space dimensions should be scaled during optimization:

1. **Original scale** (user interface): Input bounds, output results, plots
2. **Transformed scale** (internal): Surrogate modeling, acquisition optimization
3. **Automatic conversion**: SpotOptim handles all transformations transparently

Module: `spotoptim.SpotOptim`

#### Key Features:

- Define transformations via `var_trans` parameter: `["log10", "sqrt", None, ...]`
- Optimization occurs in transformed space (better for surrogate models)
- All external interfaces use original scale (intuitive for users)
- Supported transformations: `log10`, `log/ln`, `sqrt`, `exp`, `square`, `cube`, `inv/reciprocal`
- Mix transformed and non-transformed variables

### 31.2. Why Use Transformations?

#### 31.2.1. Problem: Poorly Scaled Search Spaces

Some hyperparameters span multiple orders of magnitude:

### 31. Variable Transformations for Search Space Scaling

- **Learning rates:** 0.0001 to 1.0 (4 orders of magnitude)
- **Regularization:** 0.001 to 100 (5 orders of magnitude)
- **Network sizes:** 10 to 1000 neurons

Direct optimization in these spaces is inefficient because:

1. Surrogate models struggle with extreme scales
2. Uniform sampling wastes evaluations in unimportant regions
3. Acquisition functions behave poorly with skewed distributions

#### 31.2.2. Solution: Logarithmic and Other Transformations

Transform the space for optimization while maintaining user-friendly interfaces:

```
# Without transformations (manual approach)
bounds = [(-4, 0)] # log10(lr): awkward for users
lr = 10 ** params[0] # Manual transformation in objective

# With transformations (automatic)
bounds = [(0.0001, 1.0)] # lr in natural scale
var_trans = ["log10"] # SpotOptim handles transformation
lr = params[0] # Already in original scale!
```

### 31.3. Quick Start

#### 31.3.1. Basic Log-Scale Transformation

```
from spotoptim import SpotOptim
import numpy as np

def objective_function(X):
    """Objective receives parameters in ORIGINAL scale."""
    results = []
    for params in X:
        lr = params[0] # Already in [0.001, 0.1] - original scale!
        alpha = params[1] # Already in [0.01, 1.0] - original scale!

        # Simulate model training
        score = (lr - 0.01)**2 + (alpha - 0.1)**2 + np.random.normal(0, 0.01)
        results.append(score)
```



```

    return np.array(results)

# Create optimizer with transformations
optimizer = SpotOptim(
    fun=objective_function,
    bounds=[
        (0.001, 0.1),    # learning rate (original scale)
        (0.01, 1.0)      # alpha (original scale)
    ],
    var_trans=["log10", "log10"], # Both use log10 transformation
    var_name=["lr", "alpha"],
    max_iter=20,
    seed=42
)

# Run optimization
result = optimizer.optimize()

print(f"Best lr: {result.x[0]:.6f}")    # In original scale
print(f"Best alpha: {result.x[1]:.6f}") # In original scale
print(f"Best score: {result.fun:.6f}")

```

```

Best lr: 0.034424
Best alpha: 0.016471
Best score: -0.005523

```

## 31.4. Supported Transformations

SpotOptim supports the following transformations:

| Transformation | Forward ( $x \rightarrow t$ ) | Inverse ( $t \rightarrow x$ ) | Use Case                       |
|----------------|-------------------------------|-------------------------------|--------------------------------|
| "log10"        | $t = \log(x)$                 | $x = 10^t$                    | Learning rates, regularization |
| "log" or "ln"  | $t = \ln(x)$                  | $x = e^t$                     | Natural exponential scales     |
| "sqrt"         | $t = \sqrt{x}$                | $x = t^2$                     | Moderate scaling               |
| "exp"          | $t = e^x$                     | $x = \ln(t)$                  | Inverse of natural log         |

### 31. Variable Transformations for Search Space Scaling

| Transformation        | Forward ( $x \rightarrow t$ ) | Inverse ( $t \rightarrow x$ ) | Use Case                 |
|-----------------------|-------------------------------|-------------------------------|--------------------------|
| "square"              | $t = x^2$                     | $x = \sqrt{t}$                | Inverse of sqrt          |
| "cube"                | $t = x^3$                     | $x = \sqrt[3]{t}$             | Strong scaling           |
| "inv" or "reciprocal" | $t = 1/x$                     | $x = 1/t$                     | Reciprocal relationships |
| None or "id"          | $t = x$                       | $x = t$                       | No transformation        |

#### 31.4.1. Transformation Guidelines

When to use "log10" or "log":

- Parameters spanning multiple orders of magnitude
- Learning rates:  $(1e-5, 1e-1) \rightarrow$  uniform sampling in log space
- Regularization parameters:  $(1e-6, 1e2)$
- Batch sizes, hidden units when range is large

When to use "sqrt":

- Moderate scaling (1-2 orders of magnitude)
- Batch sizes:  $(16, 512)$
- Number of neurons:  $(32, 256)$

When to use "inv" (reciprocal):

- Inverse relationships (e.g.,  $1/\text{temperature}$ )
- When smaller values are more important

When to use None:

- Parameters with narrow ranges
- Already well-scaled parameters
- Categorical indices (use with `var_type=["factor"]`)

## 31.5. Detailed Examples

### 31.5.1. Example 1: Neural Network Hyperparameter Tuning

```

import torch
import torch.nn as nn
from spotoptim import SpotOptim
from spotoptim.data import get_diabetes_dataloaders
from spotoptim.nn.linear_regressor import LinearRegressor
import numpy as np

def train_neural_network(X):
    """Train neural network with hyperparameters in original scale."""
    results = []

    for params in X:
        # All parameters in original scale
        hidden_size = int(params[0]) # [16, 256]
        num_layers = int(params[1]) # [1, 4]
        lr = params[2] # [0.0001, 0.1]
        weight_decay = params[3] # [1e-6, 0.01]

        print(f"Training: hidden={hidden_size}, layers={num_layers}, "
              f"lr={lr:.6f}, wd={weight_decay:.6f}")

        # Load data
        train_loader, test_loader, _ = get_diabetes_dataloaders(batch_size=32)

        # Create model
        model = LinearRegressor(
            input_dim=10,
            output_dim=1,
            l1=hidden_size,
            num_hidden_layers=num_layers,
            activation="ReLU",
            lr=lr
        )

        # Get optimizer with weight decay
        optimizer = torch.optim.Adam(model.parameters(),
                                      lr=lr,
                                      weight_decay=weight_decay)

        # Train
        model.train()
        for epoch in range(50):
            for batch_x, batch_y in train_loader:
                optimizer.zero_grad()

```

### 31. Variable Transformations for Search Space Scaling

```
        outputs = model(batch_x)
        loss = nn.MSELoss()(outputs, batch_y)
        loss.backward()
        optimizer.step()

    # Evaluate
    model.eval()
    total_loss = 0
    with torch.no_grad():
        for batch_x, batch_y in test_loader:
            outputs = model(batch_x)
            loss = nn.MSELoss()(outputs, batch_y)
            total_loss += loss.item()

    avg_loss = total_loss / len(test_loader)
    results.append(avg_loss)

    return np.array(results)

# Create optimizer with appropriate transformations
optimizer = SpotOptim(
    fun=train_neural_network,
    bounds=[
        (16, 256),          # hidden_size: moderate range
        (1, 4),             # num_layers: small range
        (0.0001, 0.1),      # lr: 3 orders of magnitude
        (1e-6, 0.01)       # weight_decay: 4 orders of magnitude
    ],
    var_trans=[
        "sqrt",             # sqrt for hidden_size
        None,               # no transformation for num_layers
        "log10",            # log10 for learning rate
        "log10"             # log10 for weight_decay
    ],
    var_type=["int", "int", "float", "float"],
    var_name=["hidden_size", "num_layers", "lr", "weight_decay"],
    max_iter=30,
    n_initial=10,
    seed=42
)

result = optimizer.optimize()

print("\nBest Configuration:")
```

```

print(f"  Hidden Size: {int(result.x[0])}")
print(f"  Num Layers: {int(result.x[1])}")
print(f"  Learning Rate: {result.x[2]:.6f}")
print(f"  Weight Decay: {result.x[3]:.8f}")
print(f"  Best Loss: {result.fun:.6f}")

```

```

Training: hidden=225, layers=3, lr=0.001748, wd=0.000001
Training: hidden=81, layers=2, lr=0.003730, wd=0.000003
Training: hidden=25, layers=2, lr=0.000615, wd=0.001695
Training: hidden=49, layers=2, lr=0.000147, wd=0.000204
Training: hidden=196, layers=4, lr=0.007107, wd=0.000009
Training: hidden=121, layers=4, lr=0.025632, wd=0.000017
Training: hidden=100, layers=2, lr=0.072441, wd=0.000096
Training: hidden=169, layers=1, lr=0.000238, wd=0.004102
Training: hidden=100, layers=3, lr=0.001146, wd=0.001331
Training: hidden=36, layers=3, lr=0.021475, wd=0.000340
Training: hidden=100, layers=3, lr=0.001286, wd=0.001219
Training: hidden=100, layers=3, lr=0.000589, wd=0.002416
Training: hidden=81, layers=2, lr=0.008000, wd=0.000003
Training: hidden=81, layers=2, lr=0.005805, wd=0.000003
Training: hidden=81, layers=2, lr=0.005590, wd=0.000002
Training: hidden=100, layers=2, lr=0.005570, wd=0.000002
Training: hidden=81, layers=2, lr=0.005298, wd=0.000002
Training: hidden=81, layers=2, lr=0.038008, wd=0.000001
Training: hidden=81, layers=2, lr=0.012959, wd=0.000002
Training: hidden=81, layers=2, lr=0.100000, wd=0.000003
Training: hidden=100, layers=2, lr=0.006852, wd=0.000003
Training: hidden=225, layers=3, lr=0.002405, wd=0.000002
Training: hidden=81, layers=2, lr=0.006667, wd=0.000002
Training: hidden=81, layers=2, lr=0.019844, wd=0.000008
Training: hidden=81, layers=2, lr=0.021819, wd=0.000003
Training: hidden=81, layers=2, lr=0.009476, wd=0.000012
Training: hidden=81, layers=2, lr=0.009348, wd=0.000007
Training: hidden=81, layers=2, lr=0.026449, wd=0.000028
Training: hidden=81, layers=2, lr=0.015467, wd=0.000015
Training: hidden=81, layers=2, lr=0.085525, wd=0.000032

```

Best Configuration:

```

  Hidden Size: 81
  Num Layers: 2
  Learning Rate: 0.005590
  Weight Decay: 0.00000184
  Best Loss: 2541.846680

```

**31.5.2. Example 2: Physics-Informed Neural Networks (PINNs)**

```

from spotoptim import SpotOptim
import numpy as np

def train_pinn(X):
    """Train PINN with hyperparameters in original scale."""
    results = []

    for params in X:
        neurons = int(params[0])      # [16, 128]
        layers = int(params[1])       # [1, 4]
        lr = params[2]                 # [0.1, 10.0]
        alpha = params[3]              # [0.01, 1.0]

        # Simulate PINN training
        # In practice, this would train an actual PINN model
        val_error = 0.1 * (1/lr) + 0.05 * alpha + np.random.normal(0, 0.01)
        results.append(val_error)

    return np.array(results)

# Define optimization with transformations
optimizer = SpotOptim(
    fun=train_pinn,
    bounds=[
        (16, 128),      # neurons
        (1, 4),          # layers
        (0.1, 10.0),     # lr: covers 2 orders of magnitude
        (0.01, 1.0)      # alpha: covers 2 orders of magnitude
    ],
    var_trans=[None, None, "log10", "log10"],
    var_type=["int", "int", "float", "float"],
    var_name=["neurons", "layers", "lr", "alpha"],
    max_iter=20,
    seed=42
)

result = optimizer.optimize()
optimizer.print_best(result)

```

Best Solution Found:

```

-----
neurons: 74
layers: 2
lr: 8.0660
alpha: 0.0980
Objective Value: -0.0011
Total Evaluations: 20

```

### 31.5.3. Example 3: Mixing Transformations

```

from spotoptim import SpotOptim
import numpy as np

def complex_objective(X):
    """Objective with multiple transformation types."""
    results = []

    for params in X:
        # All in original scale
        x1 = params[0] # sqrt-transformed: [10, 1000]
        x2 = params[1] # log10-transformed: [0.001, 1.0]
        x3 = params[2] # no transformation: [-5, 5]
        x4 = params[3] # reciprocal: [0.1, 10]

        # Complex function
        result = (
            (x1/100 - 5)**2 +
            (np.log10(x2) + 1.5)**2 +
            x3**2 +
            (1/x4 - 0.2)**2
        )
        results.append(result)

    return np.array(results)

optimizer = SpotOptim(
    fun=complex_objective,
    bounds=[
        (10, 1000),      # x1: moderate range
        (0.001, 1.0),    # x2: log scale
        (-5, 5),         # x3: symmetric range
        (0.1, 10)        # x4: for reciprocal
    ],

```

### 31. Variable Transformations for Search Space Scaling

```
var_trans=["sqrt", "log10", None, "inv"],
var_name=["x1_sqrt", "x2_log", "x3_linear", "x4_inv"],
max_iter=30,
seed=42
)

result = optimizer.optimize()
```

## 31.6. Viewing Transformations in Tables

The transformation type is displayed in the “trans” column of both design and results tables:

### 31.6.1. Design Table (Before Optimization)

```
from spotoptim import SpotOptim
import numpy as np

optimizer = SpotOptim(
    fun=lambda X: np.sum(X**2, axis=1),
    bounds=[
        (0.001, 1.0),
        (0.01, 10.0),
        (10, 1000),
        (-5, 5)
    ],
    var_trans=["log10", "log", "sqrt", None],
    var_name=["lr", "alpha", "neurons", "bias"],
    max_iter=10
)

# Display design table
print(optimizer.print_design_table())
```

| name    | type  | lower   | upper     | default  | trans |
|---------|-------|---------|-----------|----------|-------|
| lr      | float | 0.0010  | 1.0000    | 0.5005   | log10 |
| alpha   | float | 0.0100  | 10.0000   | 5.0050   | log   |
| neurons | float | 10.0000 | 1000.0000 | 505.0000 | sqrt  |
| bias    | float | -5.0000 | 5.0000    | 0.0000   | -     |



Output:

| name    | type | lower   | upper     | default  | trans |
|---------|------|---------|-----------|----------|-------|
| lr      | num  | 0.0010  | 1.0000    | 0.5005   | log10 |
| alpha   | num  | 0.0100  | 10.0000   | 5.0050   | log   |
| neurons | num  | 10.0000 | 1000.0000 | 505.0000 | sqrt  |
| bias    | num  | -5.0000 | 5.0000    | 0.0000   | -     |

### 31.6.2. Results Table (After Optimization)

```
result = optimizer.optimize()

# Display results with transformations
print(optimizer.print_results_table())
```

| name    | type  | lower   | upper     | tuned   | trans |
|---------|-------|---------|-----------|---------|-------|
| lr      | float | 0.0010  | 1.0000    | 0.0038  | log10 |
| alpha   | float | 0.0100  | 10.0000   | 0.2187  | log   |
| neurons | float | 10.0000 | 1000.0000 | 10.3262 | sqrt  |
| bias    | float | -5.0000 | 5.0000    | 1.6003  | -     |

Output shows the “trans” column with transformation types, helping you understand which parameters were optimized in which scale.

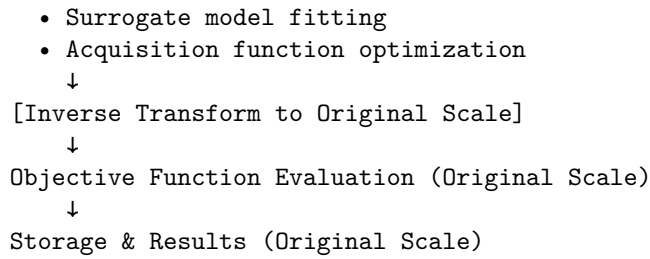
## 31.7. Internal Architecture

Understanding how transformations work internally can help debug issues and understand behavior:

### 31.7.1. Flow Diagram

```
User Input (Original Scale)
  ↓
[Transform to Internal Scale]
  ↓
Optimization (Transformed Scale)
  • Initial design generation
```

### 31. Variable Transformations for Search Space Scaling



#### 31.7.2. Key Components

1. **Bounds Transformation** (`_transform_bounds()`):

- Called during initialization
- Transforms `_original_lower` and `_original_upper` → lower and upper
- Updates `self.bounds` for internal use

2. **Forward Transformation** (`_transform_X()`):

- Converts from original scale to transformed scale
- Used before surrogate fitting
- Used when comparing distances

3. **Inverse Transformation** (`_inverse_transform_X()`):

- Converts from transformed scale to original scale
- Used before function evaluation
- Used when storing results

4. **Storage**:

- `self.X_` stores in **original scale**
- `self.best_x_` stores in **original scale**
- All external-facing data in original scale

## 31.8. Best Practices

### 31.8.1. 1. Choose Appropriate Transformations

```
# Good: Log scale for learning rate
bounds = [(1e-5, 1e-1)]
var_trans = ["log10"]

# Bad: No transformation for wide range
bounds = [(1e-5, 1e-1)]
var_trans = [None] # Poor sampling distribution
```

### 31.8.2. 2. Match Transformation to Range

```
# Wide range (>3 orders of magnitude): use log
bounds = [(1e-6, 1e-2)]
var_trans = ["log10"]

# Moderate range (1-2 orders): use sqrt
bounds = [(10, 500)]
var_trans = ["sqrt"]

# Narrow range (<1 order): no transformation
bounds = [(-1, 1)]
var_trans = [None]
```

### 31.8.3. 3. Validate Transformation Choice

```
# Check if transformation makes sense
import numpy as np

# Original space
x_orig = np.linspace(0.001, 1.0, 10)
print("Original:", x_orig)

# Log10 transformed space
x_trans = np.log10(x_orig)
print("Transformed:", x_trans)
print("Range ratio:", np.ptp(x_trans) / np.ptp(x_orig))
# Should be much more uniform distribution
```

#### 31.8.4. 4. Combine with Variable Types

```
# Mix transformations with variable types
optimizer = SpotOptim(
    fun=objective,
    bounds=[
        (10, 200),                # int with sqrt
        ("ReLU", "Tanh", "Sigmoid"), # factor (no transform)
        (0.0001, 0.1),            # num with log10
        (0.01, 1.0)               # num with log10
    ],
    var_type=["int", "factor", "float", "float"],
    var_trans=["sqrt", None, "log10", "log10"],
    var_name=["neurons", "activation", "lr", "dropout"]
)
```

### 31.9. Troubleshooting

#### 31.9.1. Issue: Values Out of Bounds

**Problem:** Objective function receives values outside specified bounds.

**Solution:** This should not happen with transformations. If it does:

```
# Check transformation is applied correctly
print(f"Original bounds: {optimizer._original_lower} to {optimizer._original_upper}")
print(f"Transformed bounds: {optimizer.lower} to {optimizer.upper}")
print(f"Transformations: {optimizer.var_trans}")
```

#### 31.9.2. Issue: Poor Optimization Performance

**Problem:** Optimization doesn't find good solutions.

**Possible causes:**

1. Wrong transformation type for the parameter scale
2. Transformation not needed (adding unnecessary complexity)
3. Bounds too wide or too narrow

**Solution:**

```
# Try different transformations
for trans in [None, "log10", "sqrt"]:
    optimizer = SpotOptim(
        fun=objective,
        bounds=[(0.001, 1.0)],
        var_trans=[trans],
        max_iter=20,
        seed=42
    )
    result = optimizer.optimize()
    print(f"Transformation: {trans}, Best: {result.fun:.6f}")
```

### 31.9.3. Issue: Transformation Not Applied

**Problem:** Transformation doesn't seem to affect optimization.

**Check:**

```
# Verify var_trans length matches dimensions
print(f"Number of dimensions: {len(optimizer.bounds)}")
print(f"Number of transformations: {len(optimizer.var_trans)}")
# These must match!

# Check transformation is not None/"id"
print(f"Transformations: {optimizer.var_trans}")
```

## 31.10. Comparison: Manual vs Automatic Transformations

### 31.10.1. Manual Approach (Old Way)

```
def objective_manual(X):
    """Manual transformation - error-prone!"""
    results = []
    for params in X:
        # Must remember to transform
        lr = 10 ** params[0] # Was in log scale
        alpha = 10 ** params[1] # Was in log scale

        # Use parameters
```

### 31. Variable Transformations for Search Space Scaling

```
        score = compute_score(lr, alpha)
        results.append(score)
    return np.array(results)

# Bounds in log scale - confusing!
optimizer = SpotOptim(
    fun=objective_manual,
    bounds=[(-4, -1), (-2, 0)], # log10 scale
    var_name=["log10_lr", "log10_alpha"] # Confusing names
)

result = optimizer.optimize()
# Must transform back for interpretation
best_lr = 10 ** result.x[0]
best_alpha = 10 ** result.x[1]
```

#### 31.10.2. Automatic Approach (New Way)

```
def objective_auto(X):
    """Automatic transformation - clean!"""
    results = []
    for params in X:
        # Already in original scale
        lr = params[0]
        alpha = params[1]

        # Use parameters directly
        score = compute_score(lr, alpha)
        results.append(score)
    return np.array(results)

# Bounds in natural scale - intuitive!
optimizer = SpotOptim(
    fun=objective_auto,
    bounds=[(0.0001, 0.1), (0.01, 1.0)],
    var_trans=["log10", "log10"], # Specify transformation
    var_name=["lr", "alpha"] # Natural names
)

result = optimizer.optimize()
# Results already in original scale
```

```
best_lr = result.x[0]
best_alpha = result.x[1]
```

## 31.11. Summary

### Key Takeaways:

1. Use `var_trans` to specify transformations for each dimension
2. Transformations improve optimization for poorly scaled spaces
3. All user interfaces (bounds, results, plots) use original scale
4. Optimization happens internally in transformed space
5. Common transformations: `"log10"` for learning rates, `"sqrt"` for moderate scaling
6. View transformations in tables with “trans” column

### When to Use:

- Parameters spanning multiple orders of magnitude → `"log10"` or `"log"`
- Moderate scaling (1-2 orders) → `"sqrt"`
- Reciprocal relationships → `"inv"`
- Well-scaled parameters → `None`

### Benefits:

- Better surrogate model performance
- More efficient sampling
- Improved optimization convergence
- User-friendly interface (no manual transformations in objective function)

---

### See Also:

- Variable Types Manual - Integer, numeric, and factor types
- Factor Variables Manual - Categorical optimization
- Reproducibility Manual - Setting seeds for consistent results





## 32. Learning Rate Mapping for Unified Optimizer Interface

SpotOptim provides a sophisticated learning rate mapping system through the `map_lr()` function, enabling a unified interface for learning rates across different PyTorch optimizers. This solves the challenge that different optimizers operate on vastly different learning rate scales.

### 32.1. Overview

Different PyTorch optimizers use different default learning rates and optimal ranges:

- **Adam**: default 0.001, typical range 0.0001-0.01
- **SGD**: default 0.01, typical range 0.001-0.1
- **RMSprop**: default 0.01, typical range 0.001-0.1

This makes it difficult to compare optimizer performance fairly or optimize learning rates across different optimizers. The `map_lr()` function provides a unified scale where **`lr_unified=1.0` corresponds to each optimizer's PyTorch default.**

Module: `spotoptim.utils.mapping`

Key Features:

- Unified learning rate scale across all optimizers
- Fair comparison when evaluating different optimizers
- Simplified hyperparameter optimization
- Based on official PyTorch default learning rates
- Supports 13 major PyTorch optimizers

### 32.2. Quick Start

#### 32.2.1. Basic Usage

### 32. Learning Rate Mapping for Unified Optimizer Interface

```
from spotoptim.utils.mapping import map_lr

# Get optimizer-specific learning rate from unified scale
lr_adam = map_lr(1.0, "Adam")      # Returns 0.001 (Adam's default)
lr_sgd = map_lr(1.0, "SGD")        # Returns 0.01 (SGD's default)
lr_rmsprop = map_lr(1.0, "RMSprop") # Returns 0.01 (RMSprop's default)

print(f"Unified lr=1.0:")
print(f"  Adam:    {lr_adam}")
print(f"  SGD:      {lr_sgd}")
print(f"  RMSprop: {lr_rmsprop}")
```

```
Unified lr=1.0:
  Adam:    0.001
  SGD:      0.01
  RMSprop: 0.01
```

#### 32.2.2. Scaling Learning Rates

```
from spotoptim.utils.mapping import map_lr

# Scale all learning rates by the same factor
unified_lr = 0.5

lr_adam = map_lr(unified_lr, "Adam")      # 0.5 * 0.001 = 0.0005
lr_sgd = map_lr(unified_lr, "SGD")        # 0.5 * 0.01 = 0.005
lr_rmsprop = map_lr(unified_lr, "RMSprop") # 0.5 * 0.01 = 0.005

print(f"Unified lr={unified_lr}:")
print(f"  Adam:    {lr_adam}")
print(f"  SGD:      {lr_sgd}")
print(f"  RMSprop: {lr_rmsprop}")
```

```
Unified lr=0.5:
  Adam:    0.0005
  SGD:      0.005
  RMSprop: 0.005
```

### 32.2.3. Integration with LinearRegressor

```
from spotoptim.nn.linear_regressor import LinearRegressor

# Create model with unified learning rate
model = LinearRegressor(
    input_dim=10,
    output_dim=1,
    l1=32,
    num_hidden_layers=2,
    lr=1.0 # Unified learning rate
)

# Get optimizer - automatically uses mapped learning rate
optimizer_adam = model.get_optimizer("Adam")      # Gets 1.0 * 0.001 = 0.001
optimizer_sgd = model.get_optimizer("SGD")        # Gets 1.0 * 0.01 = 0.01

# Verify the actual learning rates
print(f"Adam actual lr: {optimizer_adam.param_groups[0]['lr']}")
print(f"SGD actual lr: {optimizer_sgd.param_groups[0]['lr']}")
```

Adam actual lr: 0.001

SGD actual lr: 0.01

## 32.3. Function Reference

### 32.3.1. map\_lr(lr\_unified, optimizer\_name, use\_default\_scale=True)

Maps a unified learning rate to an optimizer-specific learning rate.

**Parameters:**

- **lr\_unified** (float): Unified learning rate multiplier. A value of 1.0 corresponds to the optimizer's default learning rate. Typical range: [0.001, 100.0].
- **optimizer\_name** (str): Name of the PyTorch optimizer. Must be one of: "Adadelata", "Adagrad", "Adam", "AdamW", "SparseAdam", "Adamax", "ASGD", "LBFGS", "NAdam", "RAdam", "RMSprop", "Rprop", "SGD".
- **use\_default\_scale** (bool, optional): Whether to scale by the optimizer's default learning rate. If **True** (default), **lr\_unified** is multiplied by the default lr. If **False**, returns **lr\_unified** directly.

## 32. Learning Rate Mapping for Unified Optimizer Interface

**Returns:**

- **float:** The optimizer-specific learning rate.

**Raises:**

- **ValueError:** If `optimizer_name` is not supported.
- **ValueError:** If `lr_unified` is not positive.

**Example:**

```
from spotoptim.utils.mapping import map_lr

# Get default learning rates (unified lr = 1.0)
lr = map_lr(1.0, "Adam")      # 0.001
lr = map_lr(1.0, "SGD")      # 0.01
lr = map_lr(1.0, "RMSprop")   # 0.01

# Scale learning rates
lr = map_lr(0.5, "Adam")      # 0.0005
lr = map_lr(2.0, "SGD")      # 0.02

# Without default scaling
lr = map_lr(0.01, "Adam", use_default_scale=False) # 0.01 (direct)
```

## 32.4. Supported Optimizers

All major PyTorch optimizers are supported with their default learning rates:

| Optimizer         | Default LR | Typical Range | Notes                           |
|-------------------|------------|---------------|---------------------------------|
| <b>Adam</b>       | 0.001      | 0.0001-0.01   | Most popular, good default      |
| <b>AdamW</b>      | 0.001      | 0.0001-0.01   | Adam with weight decay          |
| <b>Adamax</b>     | 0.002      | 0.0001-0.01   | Adam variant with infinity norm |
| <b>NAdam</b>      | 0.002      | 0.0001-0.01   | Adam with Nesterov momentum     |
| <b>RAdam</b>      | 0.001      | 0.0001-0.01   | Rectified Adam                  |
| <b>SparseAdam</b> | 0.001      | 0.0001-0.01   | For sparse gradients            |
| <b>SGD</b>        | 0.01       | 0.001-0.1     | Classic, needs momentum         |
| <b>RMSprop</b>    | 0.01       | 0.001-0.1     | Good for RNNs                   |
| <b>Adagrad</b>    | 0.01       | 0.001-0.1     | Adaptive learning rate          |
| <b>Adadelata</b>  | 1.0        | 0.1-10.0      | Extension of Adagrad            |
| <b>ASGD</b>       | 0.01       | 0.001-0.1     | Averaged SGD                    |
| <b>LBFGS</b>      | 1.0        | 0.1-10.0      | Second-order optimizer          |
| <b>Rprop</b>      | 0.01       | 0.001-0.1     | Resilient backpropagation       |

## 32.5. Use Cases

### 32.5.1. Comparing Different Optimizers

```
import torch
import torch.nn as nn
from spotoptim.nn.linear_regressor import LinearRegressor
from spotoptim.data import get_diabetes_dataloaders

# Load data
train_loader, test_loader, _ = get_diabetes_dataloaders(batch_size=32, random_state=42)

# Test different optimizers with unified learning rate
unified_lr = 1.0
optimizers_to_test = ["Adam", "SGD", "RMSprop", "AdamW"]
results = {}

for opt_name in optimizers_to_test:
    # Reset for fair comparison
    torch.manual_seed(42)
    model = LinearRegressor(input_dim=10, output_dim=1, l1=32,
                           num_hidden_layers=2, lr=unified_lr)

    # Create optimizer with mapped learning rate
    if opt_name == "SGD":
        optimizer = model.get_optimizer(opt_name, momentum=0.9)
    else:
        optimizer = model.get_optimizer(opt_name)

    criterion = nn.MSELoss()

    # Train
    model.train()
    for epoch in range(50):
        for batch_X, batch_y in train_loader:
            optimizer.zero_grad()
            predictions = model(batch_X)
            loss = criterion(predictions, batch_y)
            loss.backward()
            optimizer.step()

    # Evaluate
    model.eval()
```

## 32. Learning Rate Mapping for Unified Optimizer Interface

```
test_loss = 0.0
with torch.no_grad():
    for batch_X, batch_y in test_loader:
        predictions = model(batch_X)
        test_loss += criterion(predictions, batch_y).item()

avg_test_loss = test_loss / len(test_loader)
results[opt_name] = avg_test_loss

print(f"{opt_name:10s}: Test MSE = {avg_test_loss:.4f} "
      f"(actual lr = {optimizer.param_groups[0]['lr']:.6f})")

# Find best optimizer
best_opt = min(results, key=results.get)
print(f"\nBest optimizer: {best_opt} with MSE = {results[best_opt]:.4f}")
```

```
Adam      : Test MSE = 3859.4478 (actual lr = 0.001000)
SGD       : Test MSE = 5271.6840 (actual lr = 0.010000)
RMSprop   : Test MSE = 2769.5164 (actual lr = 0.010000)
AdamW     : Test MSE = 3866.1197 (actual lr = 0.001000)
```

Best optimizer: RMSprop with MSE = 2769.5164

### 32.5.2. Hyperparameter Optimization with SpotOptim

Note, N\_INITIAL and MAX\_ITER are kept small for demonstration; increase for real use.

```
from spotoptim import SpotOptim
from spotoptim.nn.linear_regressor import LinearRegressor
from spotoptim.data import get_diabetes_data loaders
import torch.nn as nn
import torch
import numpy as np

MAX_ITER = 10
N_INITIAL = 5

def train_and_evaluate(X):
    """Objective function for hyperparameter optimization."""
    results = []

    # Load data once
```

```

train_loader, test_loader, _ = get_diabetes_dataloaders(
    batch_size=32, random_state=42
)

for params in X:
    # Extract hyperparameters
    lr_unified = 10 ** params[0] # Log scale
    optimizer_name = params[1] # Factor variable
    l1 = int(params[2]) # Integer
    num_layers = int(params[3]) # Integer

    # Create model with unified learning rate
    model = LinearRegressor(
        input_dim=10,
        output_dim=1,
        l1=l1,
        num_hidden_layers=num_layers,
        lr=lr_unified # Automatically mapped per optimizer
    )

    # Get optimizer (lr already mapped internally)
    if optimizer_name == "SGD":
        optimizer = model.get_optimizer(optimizer_name, momentum=0.9)
    else:
        optimizer = model.get_optimizer(optimizer_name)

    criterion = nn.MSELoss()

    # Train
    model.train()
    for epoch in range(30):
        for batch_X, batch_y in train_loader:
            optimizer.zero_grad()
            predictions = model(batch_X)
            loss = criterion(predictions, batch_y)
            loss.backward()
            optimizer.step()

    # Evaluate
    model.eval()
    test_loss = 0.0
    with torch.no_grad():
        for batch_X, batch_y in test_loader:
            predictions = model(batch_X)

```

### 32. Learning Rate Mapping for Unified Optimizer Interface

```
        test_loss += criterion(predictions, batch_y).item()

    avg_test_loss = test_loss / len(test_loader)
    results.append(avg_test_loss)

    return np.array(results)

# Optimize learning rate, optimizer choice, and architecture
optimizer = SpotOptim(
    fun=train_and_evaluate,
    bounds=[
        (-4, 0), # log10(lr_unified): [0.0001, 1.0]
        ("Adam", "SGD", "RMSprop", "AdamW"), # Optimizer choice
        (16, 128), # Layer size
        (1, 3) # Number of hidden layers
    ],
    var_type=["float", "factor", "int", "int"],
    max_iter=MAX_ITER,
    n_initial=N_INITIAL,
    seed=42
)

result = optimizer.optimize()

# Display results
print("\nOptimization Results:")
print(f"Best unified lr: {10**result.x[0]:.6f}")
print(f"Best optimizer: {result.x[1]}")
print(f"Best layer size: {int(result.x[2])}")
print(f"Best num layers: {int(result.x[3])}")
print(f"Best test MSE: {result.fun:.4f}")

# Show actual learning rate used
from spotoptim.utils.mapping import map_lr
actual_lr = map_lr(10**result.x[0], result.x[1])
print(f"Actual {result.x[1]} learning rate: {actual_lr:.6f}")
```

```
Optimization Results:
Best unified lr: 0.305454
Best optimizer: RMSprop
Best layer size: 118
Best num layers: 2
Best test MSE: 2770.3857
```



Actual RMSprop learning rate: 0.003055

### 32.5.3. Log-Scale Hyperparameter Search

```
from spotoptim.utils.mapping import map_lr
import numpy as np

# Common pattern: sample unified lr from log scale
log_lr_range = np.linspace(-4, 0, 10) # [-4, -3.56, ..., 0]
optimizers = ["Adam", "SGD", "RMSprop"]

print("Log-scale learning rate search:")
print()
print(f"{'log_lr':<10} {'unified_lr':<12} {'Adam':<12} {'SGD':<12} {'RMSprop':<12}")
print("-" * 60)

for log_lr in log_lr_range:
    lr_unified = 10 ** log_lr
    lr_adam = map_lr(lr_unified, "Adam")
    lr_sgd = map_lr(lr_unified, "SGD")
    lr_rmsprop = map_lr(lr_unified, "RMSprop")

    print(f"{'log_lr':<10.2f} {'lr_unified':<12.6f} {'lr_adam':<12.8f} "
          f"{'lr_sgd':<12.8f} {'lr_rmsprop':<12.8f}")
```

Log-scale learning rate search:

| log_lr | unified_lr | Adam       | SGD        | RMSprop    |
|--------|------------|------------|------------|------------|
| -4.00  | 0.000100   | 0.00000010 | 0.00000100 | 0.00000100 |
| -3.56  | 0.000278   | 0.00000028 | 0.00000278 | 0.00000278 |
| -3.11  | 0.000774   | 0.00000077 | 0.00000774 | 0.00000774 |
| -2.67  | 0.002154   | 0.00000215 | 0.00002154 | 0.00002154 |
| -2.22  | 0.005995   | 0.00000599 | 0.00005995 | 0.00005995 |
| -1.78  | 0.016681   | 0.00001668 | 0.00016681 | 0.00016681 |
| -1.33  | 0.046416   | 0.00004642 | 0.00046416 | 0.00046416 |
| -0.89  | 0.129155   | 0.00012915 | 0.00129155 | 0.00129155 |
| -0.44  | 0.359381   | 0.00035938 | 0.00359381 | 0.00359381 |
| 0.00   | 1.000000   | 0.00100000 | 0.01000000 | 0.01000000 |

### 32.5.4. Custom Learning Rate Schedules

```
import torch
import torch.nn as nn
from spotoptim.nn.linear_regressor import LinearRegressor
from spotoptim.utils.mapping import map_lr

# Create model with unified lr
model = LinearRegressor(input_dim=10, output_dim=1, lr=1.0)

# Get initial optimizer
optimizer = model.get_optimizer("Adam")
initial_lr = optimizer.param_groups[0]['lr']
print(f"Initial learning rate: {initial_lr}")

# Use PyTorch learning rate scheduler
scheduler = torch.optim.lr_scheduler.StepLR(optimizer, step_size=30, gamma=0.1)

# Training with scheduler
for epoch in range(100):
    # ... training code ...
    scheduler.step()

    if (epoch + 1) % 30 == 0:
        current_lr = optimizer.param_groups[0]['lr']
        print(f"Epoch {epoch+1}: lr = {current_lr:.8f}")
```

```
Initial learning rate: 0.001
Epoch 30: lr = 0.00010000
Epoch 60: lr = 0.00001000
Epoch 90: lr = 0.00000100
```

### 32.5.5. Direct Usage Without LinearRegressor

```
import torch
import torch.nn as nn
from spotoptim.utils.mapping import map_lr

# Define your own model
class MyModel(nn.Module):
    def __init__(self):
```

```

    super().__init__()
    self.fc1 = nn.Linear(10, 32)
    self.fc2 = nn.Linear(32, 1)

    def forward(self, x):
        x = torch.relu(self.fc1(x))
        return self.fc2(x)

model = MyModel()

# Use map_lr to get optimizer-specific learning rate
unified_lr = 2.0
optimizer_name = "Adam"

actual_lr = map_lr(unified_lr, optimizer_name)
optimizer = torch.optim.Adam(model.parameters(), lr=actual_lr)

print(f"Unified lr: {unified_lr}")
print(f"Actual {optimizer_name} lr: {actual_lr}")

```

```

Unified lr: 2.0
Actual Adam lr: 0.002

```

## 32.6. Best Practices

### 32.6.1. Choosing Unified Learning Rate

**For initial experiments:**

- Start with `lr=1.0` (gives defaults for all optimizers)
- Test with `lr=0.1`, `lr=1.0`, `lr=10.0` to get a sense of scale

**For hyperparameter optimization:**

- Use log scale: sample from `[-4, 0]` or `[-3, 1]`
- Convert with `lr_unified = 10 ** log_lr`
- This gives reasonable ranges for all optimizers

**For fine-tuning:**

- If training is unstable: try smaller `lr` (e.g., 0.1 or 0.5)
- If training is too slow: try larger `lr` (e.g., 2.0 or 5.0)
- Monitor loss curves to adjust

### 32.6.2. Optimizer Selection Guidelines

**Adam family (Adam, AdamW, NAdam, RAdam):**

- Good default choice for most tasks
- Adaptive learning rates per parameter
- Works well out of the box
- Use `lr=1.0` as starting point

**SGD:**

- Good for large datasets
- Often achieves better generalization
- Requires momentum (e.g., 0.9)
- Use `lr=1.0` with `momentum=0.9`

**RMSprop:**

- Good for recurrent networks
- Handles non-stationary objectives
- Use `lr=1.0` as starting point

**Others (Adadelata, Adagrad, etc.):**

- Specialized use cases
- Start with `lr=1.0` and adjust

### 32.6.3. Common Patterns

```
# Pattern 1: Quick optimizer comparison
model = LinearRegressor(input_dim=10, output_dim=1, lr=1.0)
for opt in ["Adam", "SGD", "RMSprop"]:
    optimizer = model.get_optimizer(opt)
    # ... train and compare ...

# Pattern 2: Hyperparameter optimization
def objective(X):
    lr_unified = 10 ** X[:, 0] # Log scale
    optimizer_name = X[:, 1]   # Factor
    # ... use unified lr ...

# Pattern 3: Override model's lr
model = LinearRegressor(input_dim=10, output_dim=1, lr=1.0)
optimizer = model.get_optimizer("Adam", lr=2.0) # Override with 2.0
```

```
# Pattern 4: Direct mapping
from spotoptim.utils.mapping import map_lr
lr_actual = map_lr(unified_lr, optimizer_name)
optimizer = torch.optim.Adam(params, lr=lr_actual)
```

## 32.7. Troubleshooting

### 32.7.1. Issue: Training is unstable (loss explodes)

**Solution:** Learning rate is too high. Try:

```
model = LinearRegressor(input_dim=10, output_dim=1, lr=0.1) # Reduce from 1.0
```

### 32.7.2. Issue: Training is too slow (loss decreases very slowly)

**Solution:** Learning rate is too low. Try:

```
model = LinearRegressor(input_dim=10, output_dim=1, lr=5.0) # Increase from 1.0
```

### 32.7.3. Issue: Different results across optimizer runs

**Solution:** Set random seed for reproducibility:

```
import torch
torch.manual_seed(42)
```

### 32.7.4. Issue: Want to use raw learning rate without mapping

**Solution:** Use `use_default_scale=False`:

```
from spotoptim.utils.mapping import map_lr
lr = map_lr(0.001, "Adam", use_default_scale=False) # Returns 0.001 directly
```

### 32.7.5. Issue: Optimizer not supported

**Solution:** Check supported optimizers:

```
from spotoptim.utils.mapping import OPTIMIZER_DEFAULT_LR
print("Supported optimizers:", list(OPTIMIZER_DEFAULT_LR.keys()))
```

## 32.8. Technical Details

### 32.8.1. How It Works

The mapping is simple but effective:

$$\text{actual\_lr} = \text{lr\_unified} * \text{default\_lr}[\text{optimizer\_name}]$$

For example:

- $\text{map\_lr}(1.0, \text{"Adam"}) \rightarrow 1.0 * 0.001 = 0.001$
- $\text{map\_lr}(0.5, \text{"SGD"}) \rightarrow 0.5 * 0.01 = 0.005$
- $\text{map\_lr}(2.0, \text{"RMSprop"}) \rightarrow 2.0 * 0.01 = 0.02$

This ensures that the same unified learning rate gives optimizer-specific learning rates in their typical working ranges.

### 32.8.2. Design Rationale

#### Why use defaults as scaling factors?

PyTorch's default learning rates are carefully chosen to work well for typical use cases. By using them as scaling factors:

1.  $\text{lr}=1.0$  always gives sensible defaults
2. Scaling preserves the relative relationships between optimizers
3. Each optimizer stays in its optimal range
4. Easy to understand and explain

#### Comparison with spotPython's approach:

spotPython uses  $\text{lr} = \text{lr\_mult} * \text{default\_lr}$  in `optimizer_handler()`. Our implementation:

- Separates mapping logic (testable, reusable)
- Provides standalone function (`map_lr()`)
- Comprehensive error handling and validation
- Extensive documentation and examples
- Full integration with `LinearRegressor`

### 32.8.3. Default Learning Rates

All values verified against PyTorch documentation:

```
OPTIMIZER_DEFAULT_LR = {
    "Adadelta": 1.0,
    "Adagrad": 0.01,
    "Adam": 0.001,
    "AdamW": 0.001,
    "SparseAdam": 0.001,
    "Adamax": 0.002,
    "ASGD": 0.01,
    "LBFGS": 1.0,
    "NAdam": 0.002,
    "RAdam": 0.001,
    "RMSprop": 0.01,
    "Rprop": 0.01,
    "SGD": 0.01,
}
```

## 32.9. Examples

### 32.9.1. Complete Example: Optimizer Comparison Study

```
"""
Complete example: Compare optimizers with unified learning rate interface.
"""

import torch
import torch.nn as nn
from spotoptim.nn.linear_regressor import LinearRegressor
from spotoptim.data import get_diabetes_dataloaders
from spotoptim.utils.mapping import map_lr
import matplotlib.pyplot as plt

# Set seed for reproducibility
torch.manual_seed(42)

# Load data
train_loader, test_loader, _ = get_diabetes_dataloaders(
    batch_size=32,
    random_state=42
```

### 32. Learning Rate Mapping for Unified Optimizer Interface

```
)

# Test configurations
optimizers = ["Adam", "SGD", "RMSprop", "AdamW"]
unified_lrs = [0.5, 1.0, 2.0]

# Store results
results = {}

print("Training models with different optimizers and learning rates...")
print()

for unified_lr in unified_lrs:
    results[unified_lr] = {}

    for opt_name in optimizers:
        # Reset model for fair comparison
        torch.manual_seed(42)

        # Create model with unified lr
        model = LinearRegressor(
            input_dim=10,
            output_dim=1,
            l1=32,
            num_hidden_layers=2,
            lr=unified_lr
        )

        # Get optimizer
        if opt_name == "SGD":
            optimizer = model.get_optimizer(opt_name, momentum=0.9)
        else:
            optimizer = model.get_optimizer(opt_name)

        actual_lr = optimizer.param_groups[0]['lr']
        criterion = nn.MSELoss()

        # Track training loss
        train_losses = []

        # Train
        model.train()
        for epoch in range(50):
            epoch_loss = 0.0
```



```

        for batch_X, batch_y in train_loader:
            optimizer.zero_grad()
            predictions = model(batch_X)
            loss = criterion(predictions, batch_y)
            loss.backward()
            optimizer.step()
            epoch_loss += loss.item()

        avg_epoch_loss = epoch_loss / len(train_loader)
        train_losses.append(avg_epoch_loss)

    # Evaluate on test set
    model.eval()
    test_loss = 0.0
    with torch.no_grad():
        for batch_X, batch_y in test_loader:
            predictions = model(batch_X)
            test_loss += criterion(predictions, batch_y).item()

    avg_test_loss = test_loss / len(test_loader)
    results[unified_lr][opt_name] = {
        'train_losses': train_losses,
        'test_loss': avg_test_loss,
        'actual_lr': actual_lr
    }

    print(f"Unified lr={unified_lr:.1f}, {opt_name:10s}: "
          f"actual_lr={actual_lr:.6f}, test_MSE={avg_test_loss:.4f}")

# Display summary
print()
print("=" * 70)
print("Summary: Best configurations")
print("=" * 70)

for unified_lr in unified_lrs:
    best_opt = min(results[unified_lr].items(),
                   key=lambda x: x[1]['test_loss'])
    opt_name, metrics = best_opt

    print(f"Unified lr={unified_lr:.1f}: {opt_name:10s} "
          f"(test MSE={metrics['test_loss']:.4f}, "
          f"actual_lr={metrics['actual_lr']:.6f})")

```

## 32. Learning Rate Mapping for Unified Optimizer Interface

```
# Find overall best
best_overall = None
best_overall_loss = float('inf')

for unified_lr in unified_lrs:
    for opt_name, metrics in results[unified_lr].items():
        if metrics['test_loss'] < best_overall_loss:
            best_overall_loss = metrics['test_loss']
            best_overall = (unified_lr, opt_name, metrics['actual_lr'])

print()
print(f"Overall best: unified_lr={best_overall[0]:.1f}, "
      f"optimizer={best_overall[1]}, "
      f"test_MSE={best_overall_loss:.4f}")
print(f"Actual learning rate used: {best_overall[2]:.6f}")
```

Training models with different optimizers and learning rates...

|                         |                                          |
|-------------------------|------------------------------------------|
| Unified lr=0.5, Adam    | : actual_lr=0.000500, test_MSE=5095.8747 |
| Unified lr=0.5, SGD     | : actual_lr=0.005000, test_MSE=5291.3903 |
| Unified lr=0.5, RMSprop | : actual_lr=0.005000, test_MSE=2828.7084 |
| Unified lr=0.5, AdamW   | : actual_lr=0.000500, test_MSE=5107.9592 |
| Unified lr=1.0, Adam    | : actual_lr=0.001000, test_MSE=3859.4478 |
| Unified lr=1.0, SGD     | : actual_lr=0.010000, test_MSE=5271.6840 |
| Unified lr=1.0, RMSprop | : actual_lr=0.010000, test_MSE=2769.5164 |
| Unified lr=1.0, AdamW   | : actual_lr=0.001000, test_MSE=3866.1197 |
| Unified lr=2.0, Adam    | : actual_lr=0.002000, test_MSE=3160.6882 |
| Unified lr=2.0, SGD     | : actual_lr=0.020000, test_MSE=nan       |
| Unified lr=2.0, RMSprop | : actual_lr=0.020000, test_MSE=2883.3345 |
| Unified lr=2.0, AdamW   | : actual_lr=0.002000, test_MSE=3162.8034 |

=====  
Summary: Best configurations

=====  
Unified lr=0.5: RMSprop (test MSE=2828.7084, actual lr=0.005000)  
Unified lr=1.0: RMSprop (test MSE=2769.5164, actual lr=0.010000)  
Unified lr=2.0: RMSprop (test MSE=2883.3345, actual lr=0.020000)

Overall best: unified\_lr=1.0, optimizer=RMSprop, test\_MSE=2769.5164  
Actual learning rate used: 0.010000

## 32.10. See Also

- LinearRegressor Documentation - Neural network class with lr parameter
- Diabetes Dataset Utilities - Data loading for examples
- Hyperparameter Optimization - Using map\_lr with SpotOptim
- PyTorch Optimizer Documentation - Official PyTorch reference

## 32.11. References

- Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. arXiv:1412.6980.
- Loshchilov, I., & Hutter, F. (2017). Decoupled weight decay regularization. arXiv:1711.05101.
- PyTorch Team. (2023). PyTorch Optimizer Documentation. <https://pytorch.org/docs/stable/optim.html>



## 33. Unified Learning Rate Interface in SpotOptim

This module provides a sophisticated unified learning rate interface for PyTorch optimizers through the `map_lr()` function and integration with `LinearRegressor`.

### 33.1. Overview

Different PyTorch optimizers operate on vastly different learning rate scales:

- **Adam** typically uses  $lr \sim 0.0001-0.001$
- **SGD** typically uses  $lr \sim 0.01-0.1$
- **RMSprop** typically uses  $lr \sim 0.001-0.01$

This makes it difficult to:

1. Compare optimizer performance fairly
2. Optimize learning rate as a hyperparameter across different optimizers
3. Switch between optimizers without retuning learning rates

The `map_lr()` function solves this by providing a unified learning rate scale where  $lr=1.0$  corresponds to each optimizer's PyTorch default.

### 33.2. Key Features

- **Unified Interface:** Single learning rate parameter works across all optimizers
- **Fair Comparison:** Same unified lr gives optimizer-specific optimal ranges
- **Hyperparameter Optimization:** Optimize one learning rate for multiple optimizers
- **Backward Compatible:** Existing code continues to work
- **Well-tested:** 36 comprehensive tests covering all use cases
- **Documented:** Extensive docstrings and examples

## 33.3. Usage

### 33.3.1. Basic Usage with LinearRegressor

```
from spotoptim.nn.linear_regressor import LinearRegressor

# Create model with unified lr=1.0 (gives each optimizer its default)
model = LinearRegressor(input_dim=10, output_dim=1, lr=1.0)

# Adam gets 0.001 (its default)
optimizer_adam = model.get_optimizer("Adam")

# SGD gets 0.01 (its default)
optimizer_sgd = model.get_optimizer("SGD")

# RMSprop gets 0.01 (its default)
optimizer_rmsprop = model.get_optimizer("RMSprop")
```

### 33.3.2. Using Custom Unified Learning Rate

```
# Using lr=0.5 scales all optimizers by 0.5
model = LinearRegressor(input_dim=10, output_dim=1, lr=0.5)

optimizer_adam = model.get_optimizer("Adam")      # Gets 0.5 * 0.001 = 0.0005
optimizer_sgd = model.get_optimizer("SGD")        # Gets 0.5 * 0.01 = 0.005
```

### 33.3.3. Direct Use of map\_lr()

```
from spotoptim.utils.mapping import map_lr

# Map unified lr to optimizer-specific lr
lr_adam = map_lr(1.0, "Adam")      # Returns 0.001
lr_sgd = map_lr(1.0, "SGD")        # Returns 0.01
lr_rmsprop = map_lr(1.0, "RMSprop") # Returns 0.01

# Scale by 2x
lr_adam = map_lr(2.0, "Adam")      # Returns 0.002
lr_sgd = map_lr(2.0, "SGD")        # Returns 0.02
```

## 33.3.4. Hyperparameter Optimization

```

from spotoptim import SpotOptim
from spotoptim.nn.linear_regressor import LinearRegressor
from spotoptim.data import get_diabetes_dataloaders
import torch
import torch.nn as nn
import numpy as np

def train_model(X):
    """Objective function for hyperparameter optimization."""
    results = []

    # Load data once
    train_loader, test_loader, _ = get_diabetes_dataloaders(batch_size=32, random_state=42)

    for params in X:
        lr_unified = 10 ** params[0] # Log scale: [-4, 0]
        optimizer_name = params[1] # Factor: "Adam", "SGD", "RMSprop"

        # Create model with unified lr - automatically scaled per optimizer
        torch.manual_seed(42)
        model = LinearRegressor(input_dim=10, output_dim=1, l1=32, num_hidden_layers=2, lr=lr_unified)
        optimizer = model.get_optimizer(optimizer_name)

        criterion = nn.MSELoss()

        # Train
        model.train()
        for epoch in range(30):
            for batch_X, batch_y in train_loader:
                optimizer.zero_grad()
                predictions = model(batch_X)
                loss = criterion(predictions, batch_y)
                loss.backward()
                optimizer.step()

        # Evaluate
        model.eval()
        test_loss = 0.0
        with torch.no_grad():
            for batch_X, batch_y in test_loader:
                predictions = model(batch_X)
                test_loss += criterion(predictions, batch_y).item()

```

### 33. Unified Learning Rate Interface in SpotOptim

```
        avg_test_loss = test_loss / len(test_loader)
        results.append(avg_test_loss)

    return np.array(results)

# Optimize unified lr across different optimizers
spot_optimizer = SpotOptim(
    fun=train_model,
    bounds=[(-4, 0), ("Adam", "SGD", "RMSprop")],
    var_type=["float", "factor"],
    max_iter=10, # Small for demo
    n_initial=5,
    seed=42
)
result = spot_optimizer.optimize()

print(f"\nBest unified lr: {10**result.x[0]:.6f}")
print(f"Best optimizer: {result.x[1]}")
print(f"Best test MSE: {result.fun:.4f}")

# Show actual learning rate used
from spotoptim.utils.mapping import map_lr
actual_lr = map_lr(10**result.x[0], result.x[1])
print(f"Actual {result.x[1]} learning rate: {actual_lr:.6f}")
```

```
Best unified lr: 0.003144
Best optimizer: SGD
Best test MSE: 4135.5200
Actual SGD learning rate: 0.000031
```

## 33.4. Supported Optimizers

All major PyTorch optimizers are supported with their default learning rates:

| Optimizer | Default LR | Typical Range |
|-----------|------------|---------------|
| Adam      | 0.001      | 0.0001-0.01   |
| AdamW     | 0.001      | 0.0001-0.01   |
| Adamax    | 0.002      | 0.0001-0.01   |
| NAdam     | 0.002      | 0.0001-0.01   |
| RAdam     | 0.001      | 0.0001-0.01   |



| Optimizer | Default LR | Typical Range |
|-----------|------------|---------------|
| SGD       | 0.01       | 0.001-0.1     |
| RMSprop   | 0.01       | 0.001-0.1     |
| Adagrad   | 0.01       | 0.001-0.1     |
| Adadelat  | 1.0        | 0.1-10.0      |
| ASGD      | 0.01       | 0.001-0.1     |
| LBFGS     | 1.0        | 0.1-10.0      |
| Rprop     | 0.01       | 0.001-0.1     |

## 33.5. API Reference

### 33.5.1. `map_lr(lr_unified, optimizer_name, use_default_scale=True)`

Maps a unified learning rate to an optimizer-specific learning rate.

**Parameters:**

- `lr_unified` (float): Unified learning rate multiplier. Typical range: [0.001, 100.0]
- `optimizer_name` (str): Name of the PyTorch optimizer
- `use_default_scale` (bool): Whether to scale by optimizer's default (default: True)

**Returns:**

- float: The optimizer-specific learning rate

**Example:**

```
lr = map_lr(1.0, "Adam") # Returns 0.001 (Adam's default)
lr = map_lr(0.5, "SGD") # Returns 0.005 (0.5 * SGD's default)
```

### 33.5.2. `LinearRegressor(..., lr=1.0)`

**Parameter:**

- `lr` (float): Unified learning rate multiplier. Default: 1.0

**New Behavior in `get_optimizer()`:**

- If `lr` is not specified, uses `self.lr`
- Automatically maps unified lr to optimizer-specific lr
- Can override model's lr by passing `lr` parameter

## 33.6. Design Rationale

### 33.6.1. Why Unified Learning Rates?

The approach is based on spotPython's `optimizer_handler()` but improved:

1. **Separation of Concerns:** Mapping logic in separate, testable module
2. **Flexibility:** Can be used independently or integrated with models
3. **Transparency:** Clear mapping based on PyTorch defaults
4. **Extensibility:** Easy to add new optimizers
5. **Type Safety:** Comprehensive error handling and validation

### 33.6.2. Comparison with spotPython

| Feature        | spotPython                        | spotoptim                                  |
|----------------|-----------------------------------|--------------------------------------------|
| Approach       | <code>lr_mult * default_lr</code> | <code>map_lr(lr_unified, optimizer)</code> |
| Module         | <code>optimizer_handler()</code>  | <code>map_lr()</code> + integration        |
| Testing        | Minimal                           | 36 comprehensive tests                     |
| Documentation  | Basic                             | Extensive with examples                    |
| Reusability    | Coupled                           | Standalone function                        |
| Error Handling | Basic                             | Comprehensive validation                   |

### 33.6.3. Log-scale Optimization

For hyperparameter optimization, use log-scale for unified lr:

```
# Sample from log10 scale [-4, 0]
log_lr = -2.5 # Sampled value
lr_unified = 10 ** log_lr # 0.00316

# Map to optimizer-specific
lr_adam = map_lr(lr_unified, "Adam") # 0.00316 * 0.001 = 0.00000316
lr_sgd = map_lr(lr_unified, "SGD") # 0.00316 * 0.01 = 0.0000316
```

This gives a reasonable search range across all optimizers.

## 33.7. Examples

See `examples/unified_learning_rate_demo.py` for comprehensive examples including: 1. Basic unified interface usage 2. Custom unified learning rates 3. Training with different optimizers 4. Direct use of `map_lr()` 5. Log-scale hyperparameter optimization 6. Complete hyperparameter optimization scenario

## 33.8. References

- PyTorch Optimizer Documentation
- spotPython's `optimizer_handler()` function (inspiration)
- Hyperparameter Optimization Best Practices

## 33.9. Contributing

When adding new optimizers:

1. Add default lr to `OPTIMIZER_DEFAULT_LR` dict in `mapping.py`
2. Verify the default against PyTorch documentation
3. Add tests in `test_mapping.py`
4. Update this README

## 33.10. License

Same as `spotoptim` package (see main `LICENSE` file).



## 34. Surrogate Model Visualization

This document describes the `plot_surrogate()` method added to the `SpotOptim` class, which provides visualization capabilities similar to the `plotkd()` function in the `spotpython` package.

### 34.1. Overview

The `plot_surrogate()` method creates a comprehensive 4-panel visualization of the fitted surrogate model, showing both predictions and uncertainty estimates across two selected dimensions.

### 34.2. Features

- **3D Surface Plots:** Visualize the surrogate's predictions and uncertainty as 3D surfaces
- **Contour Plots:** View 2D contours with overlaid evaluation points
- **Multi-dimensional Support:** Visualize any two dimensions of higher-dimensional problems
- **Customizable Appearance:** Control colors, resolution, transparency, and more

### 34.3. Usage

#### 34.3.1. Basic Usage

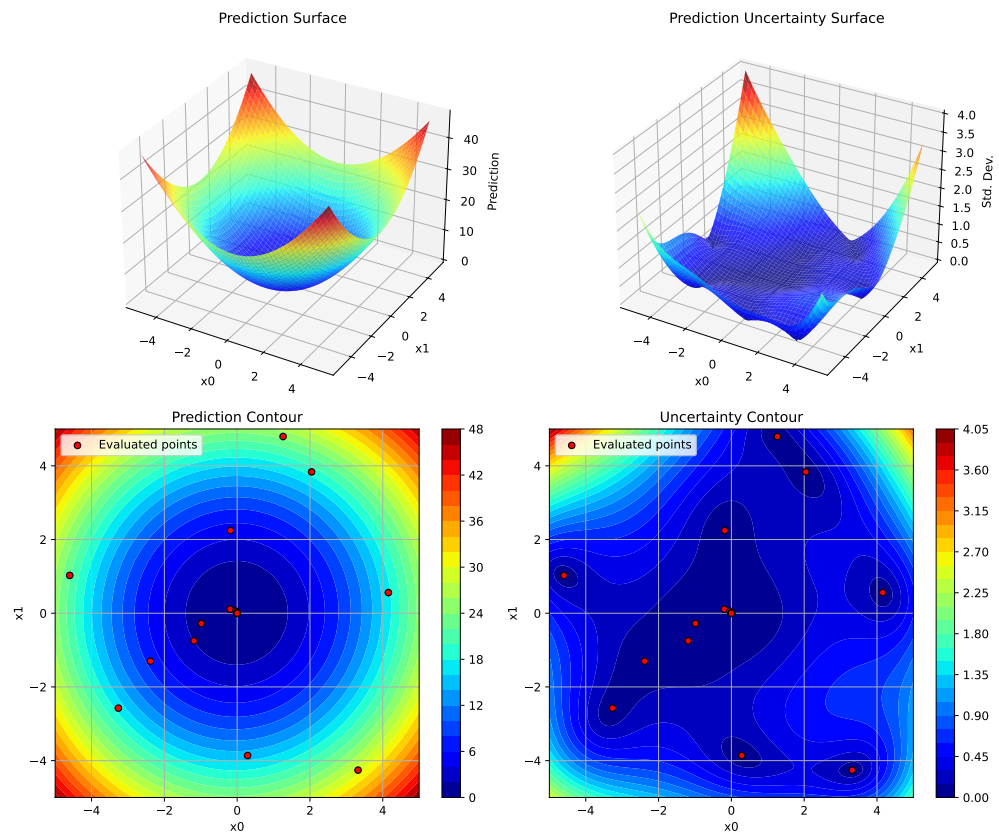
```
import numpy as np
from spotoptim import SpotOptim

# Define objective function
def sphere(X):
    return np.sum(X**2, axis=1)
```

### 34. Surrogate Model Visualization

```
# Run optimization
optimizer = SpotOptim(fun=sphere, bounds=[(-5, 5), (-5, 5)], max_iter=20)
result = optimizer.optimize()

# Visualize the surrogate model
optimizer.plot_surrogate(i=0, j=1, show=True)
```



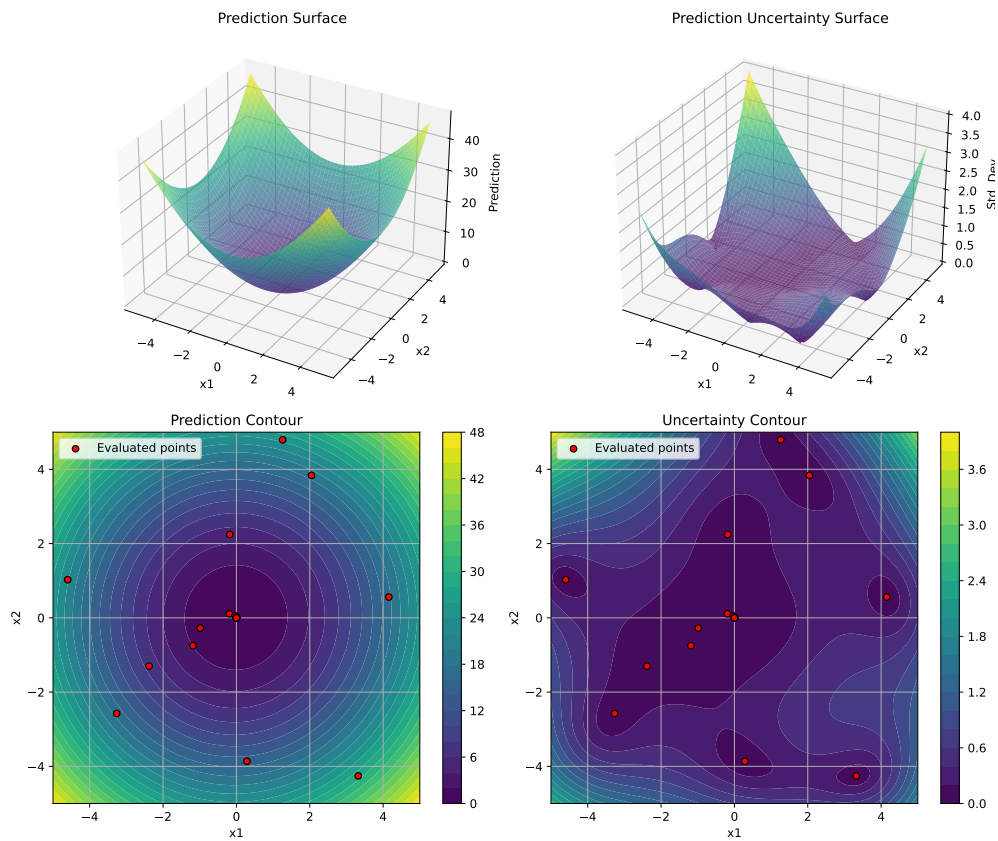
#### 34.3.2. With Custom Parameters

```
optimizer.plot_surrogate(  
    i=0, # First dimension to plot  
    j=1, # Second dimension to plot  
    var_name=['x1', 'x2'], # Variable names for axes  
    add_points=True, # Show evaluated points
```

```

cmap='viridis',          # Colormap
alpha=0.7,               # Surface transparency
num=100,                 # Grid resolution
contour_levels=25,       # Number of contour levels
grid_visible=True,       # Show grid on contours
figsize=(12, 10),        # Figure size
show=True                # Display immediately
)

```



### 34.3.3. Higher-Dimensional Problems

For problems with more than 2 dimensions, `plot_surrogate()` creates a 2D slice by fixing all other dimensions at their mean values:

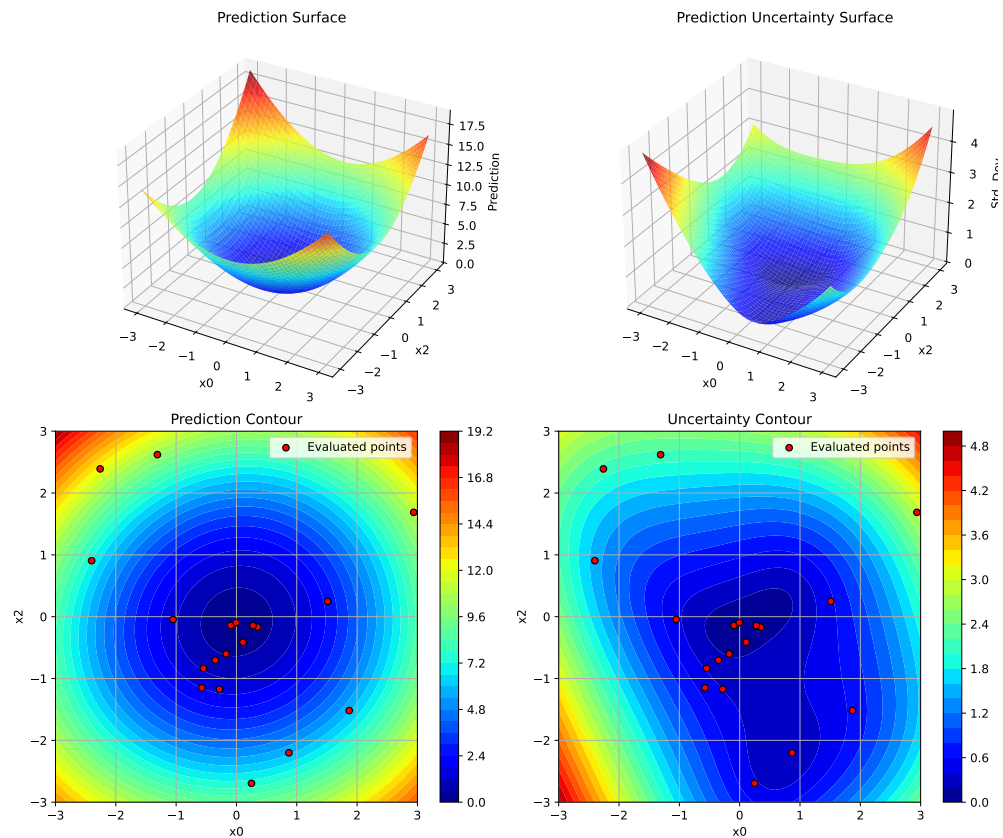
### 34. Surrogate Model Visualization

```
# 4D optimization problem
def sphere_4d(X):
    return np.sum(X**2, axis=1)

bounds = [(-3, 3)] * 4
optimizer = SpotOptim(fun=sphere_4d, bounds=bounds, max_iter=20)
result = optimizer.optimize()

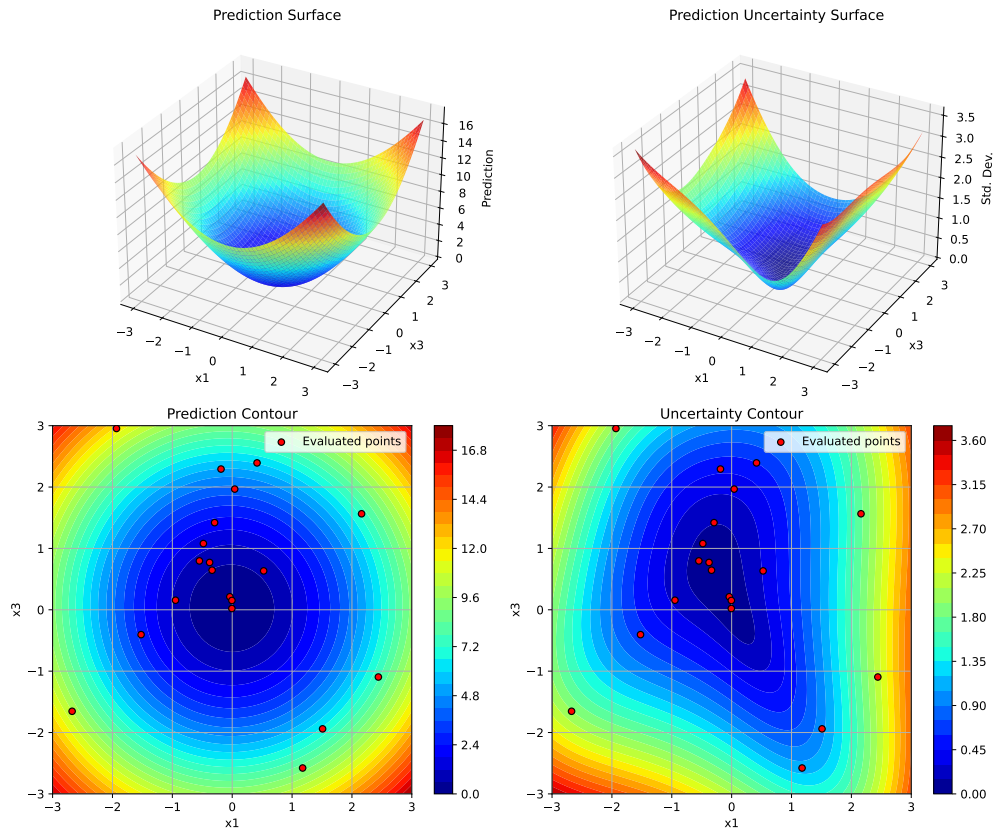
# Visualize dimensions 0 and 2 (dimensions 1 and 3 fixed at mean)
optimizer.plot_surrogate(
    i=0, j=2,
    var_name=['x0', 'x1', 'x2', 'x3']
)

# Visualize different dimension pair
optimizer.plot_surrogate(i=1, j=3, var_name=['x0', 'x1', 'x2', 'x3'])
```





## 34.4. Plot Interpretation



## 34.4. Plot Interpretation

The visualization consists of 4 panels:

### 34.4.1. Top Left: Prediction Surface

- Shows the surrogate model's predicted function values as a 3D surface
- Helps understand the model's belief about the objective function landscape
- Lower values (blue in default colormap) indicate predicted minima

### 34.4.2. Top Right: Prediction Uncertainty Surface

- Shows the standard deviation of predictions as a 3D surface
- Indicates where the model is uncertain and might benefit from more samples

## 34. Surrogate Model Visualization

- Lower values (blue) indicate high confidence, higher values (red) indicate uncertainty

### 34.4.3. Bottom Left: Prediction Contour with Points

- 2D contour plot of predictions
- Red dots show the actual points evaluated during optimization
- Useful for understanding the exploration-exploitation trade-off

### 34.4.4. Bottom Right: Uncertainty Contour with Points

- 2D contour plot of prediction uncertainty
- Shows how uncertainty decreases around evaluated points
- Helps identify unexplored regions

## 34.5. Parameters

### 34.5.1. Dimension Selection

- `i` (int, default=0): Index of first dimension to plot
- `j` (int, default=1): Index of second dimension to plot

### 34.5.2. Appearance

- `var_name` (list of str, optional): Names for each dimension
- `cmap` (str, default='jet'): Matplotlib colormap name
- `alpha` (float, default=0.8): Surface transparency (0=transparent, 1=opaque)
- `figsize` (tuple, default=(12, 10)): Figure size in inches (width, height)

### 34.5.3. Grid and Resolution

- `num` (int, default=100): Number of grid points per dimension
- `contour_levels` (int, default=30): Number of contour levels
- `grid_visible` (bool, default=True): Show grid lines on contour plots

### 34.5.4. Color Scaling

- `vmin` (float, optional): Minimum value for color scale
- `vmax` (float, optional): Maximum value for color scale

### 34.5.5. Display

- `show` (bool, default=True): Display plot immediately
- `add_points` (bool, default=True): Overlay evaluated points on contours

## 34.6. Examples

### 34.6.1. Example 1: 2D Rosenbrock Function

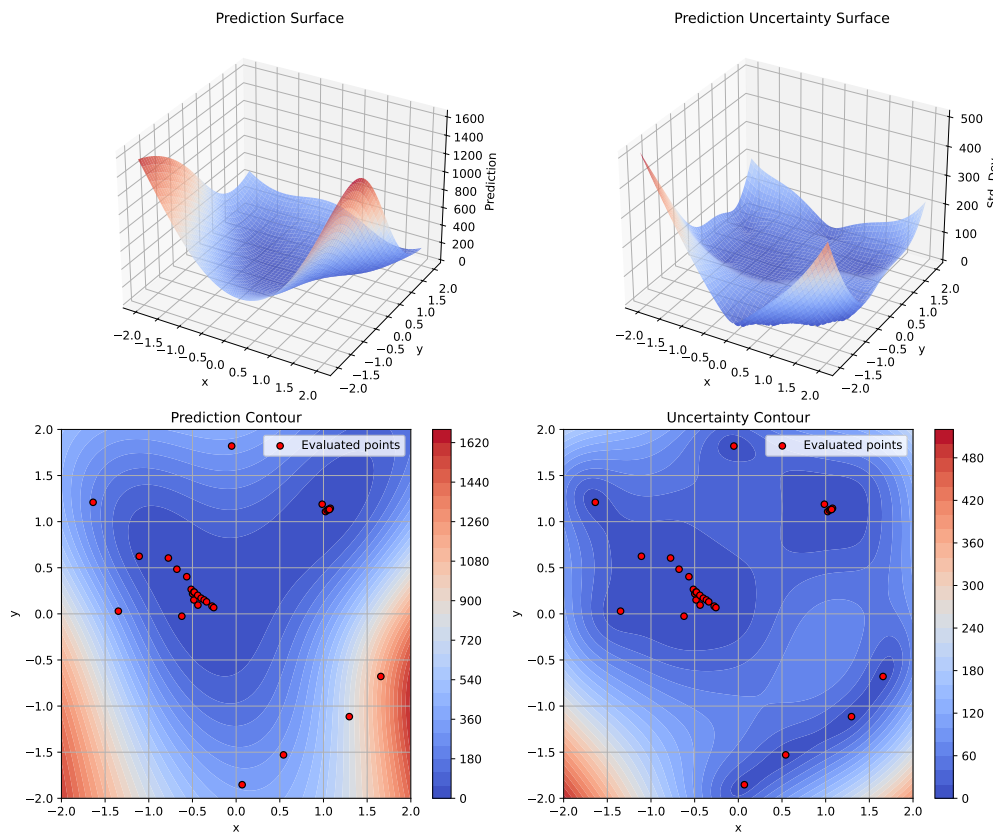
```
import numpy as np
from spotoptim import SpotOptim

def rosenbrock(X):
    X = np.atleast_2d(X)
    x, y = X[:, 0], X[:, 1]
    return (1 - x)**2 + 100 * (y - x**2)**2

optimizer = SpotOptim(
    fun=rosenbrock,
    bounds=[(-2, 2), (-2, 2)],
    max_iter=30,
    seed=42
)
result = optimizer.optimize()

# Visualize with custom colormap
optimizer.plot_surrogate(
    var_name=['x', 'y'],
    cmap='coolwarm',
    add_points=True
)
```

### 34. Surrogate Model Visualization



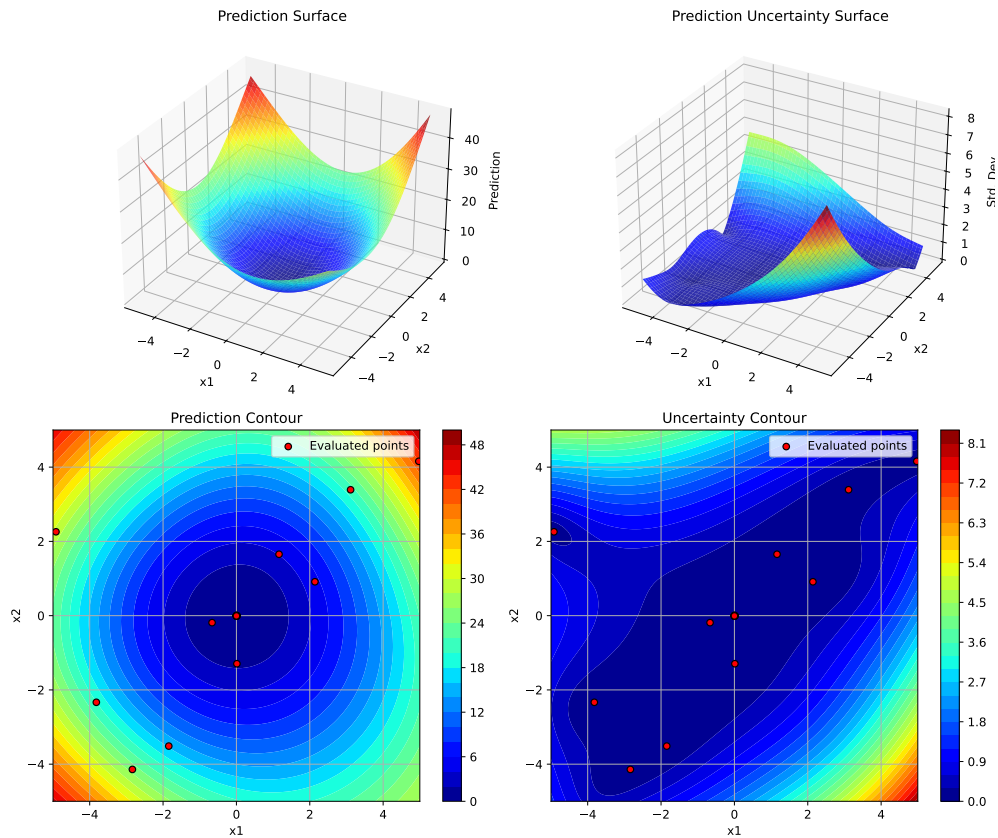
#### 34.6.2. Example 2: Using Kriging Surrogate

```
import numpy as np
from spotoptim import SpotOptim, Kriging

def sphere(X):
    return np.sum(X**2, axis=1)

optimizer = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    surrogate=Kriging(seed=42), # Use Kriging instead of GP
    max_iter=20
)
result = optimizer.optimize()
```

```
# The plotting works the same with any surrogate
optimizer.plot_surrogate(var_name=['x1', 'x2'])
```



### 34.6.3. Example 3: Comparing Different Dimension Pairs

```
# 3D problem - visualize all dimension pairs
def sphere_3d(X):
    return np.sum(X**2, axis=1)

optimizer = SpotOptim(
    fun=sphere_3d,
    bounds=[(-5, 5)] * 3,
    max_iter=25
)
```

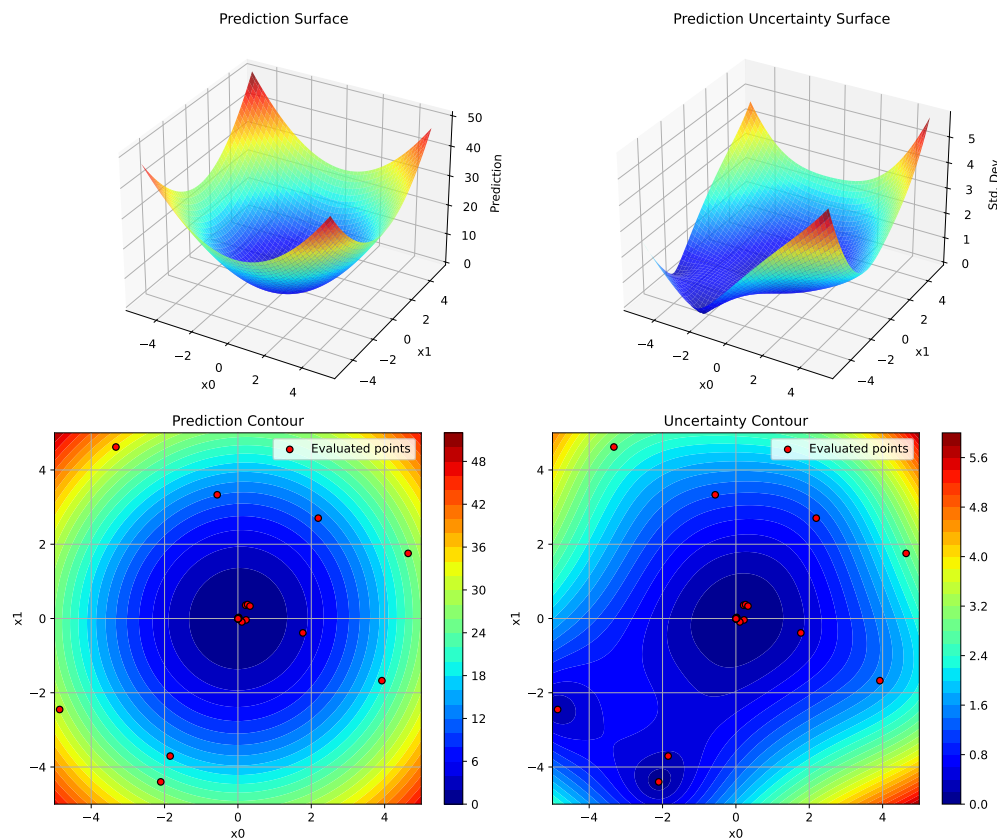
### 34. Surrogate Model Visualization

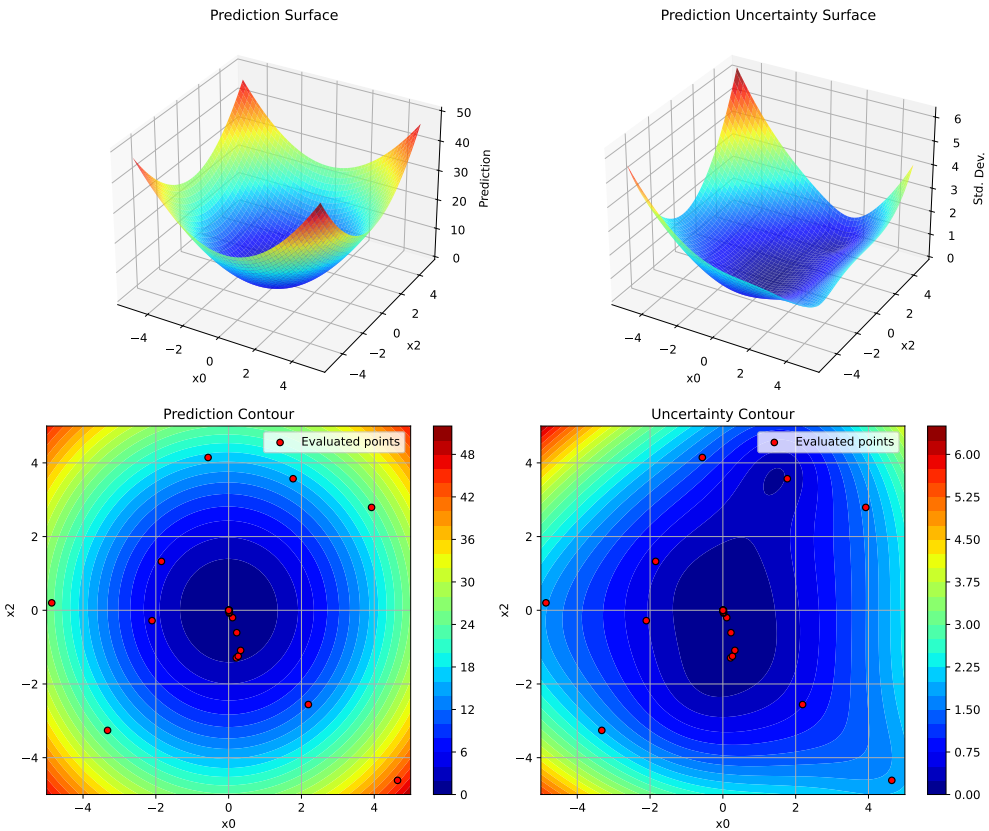
```
result = optimizer.optimize()

# Dimensions 0 vs 1
optimizer.plot_surrogate(i=0, j=1, var_name=['x0', 'x1', 'x2'])

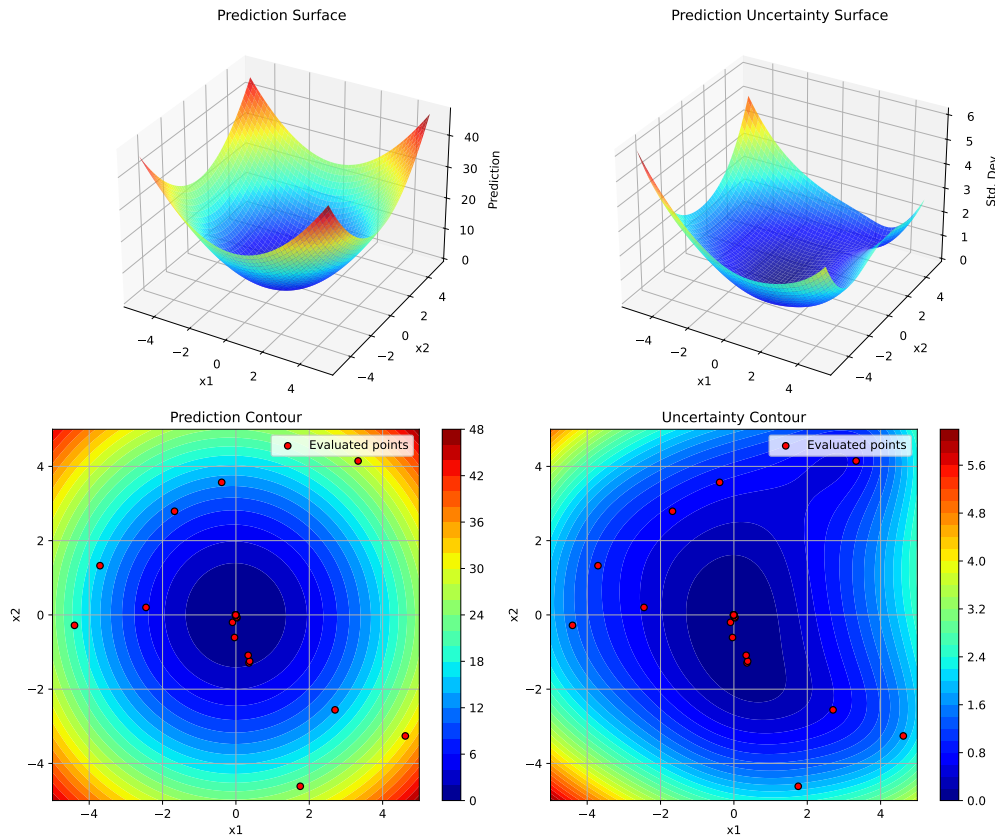
# Dimensions 0 vs 2
optimizer.plot_surrogate(i=0, j=2, var_name=['x0', 'x1', 'x2'])

# Dimensions 1 vs 2
optimizer.plot_surrogate(i=1, j=2, var_name=['x0', 'x1', 'x2'])
```





## 34. Surrogate Model Visualization



### 34.7. Tips and Best Practices

1. **Run Optimization First:** Always call `optimize()` before `plot_surrogate()`
2. **Choose Dimensions Wisely:** For high-dimensional problems, plot dimensions that you suspect are most important or interactive
3. **Adjust Resolution:** Use lower `num` values (e.g., 50) for faster plotting, higher values (e.g., 200) for smoother surfaces
4. **Color Scales:** Set `vmin` and `vmax` explicitly when comparing multiple plots to ensure consistent color scales
5. **Uncertainty Analysis:** High uncertainty areas (bright colors in uncertainty plots) are good candidates for additional sampling
6. **Exploration vs Exploitation:** Red dots clustered in low-prediction areas show exploitation; spread-out dots show exploration



## 34.8. Comparison with spotpython's plotkd()

The `plot_surrogate()` method is inspired by spotpython's `plotkd()` function but adapted for SpotOptim's simplified interface:

### 34.8.1. Similarities

- Same 4-panel layout (2 surfaces + 2 contours)
- Visualizes predictions and uncertainty
- Supports dimension selection and customization

### 34.8.2. Differences

- **Integration:** Method of SpotOptim class (no separate function needed)
- **Simpler:** Fewer parameters, more sensible defaults
- **Automatic:** Uses optimizer's bounds and data automatically
- **Type Handling:** Automatically applies variable type constraints (int/float/factor)

## 34.9. Error Handling

The method validates inputs and provides clear error messages:

```
# Before optimization runs
optimizer.plot_surrogate() # ValueError: No optimization data available

# Invalid dimension indices
optimizer.plot_surrogate(i=5, j=1) # ValueError: i must be less than n_dim

# Same dimension twice
optimizer.plot_surrogate(i=0, j=0) # ValueError: i and j must be different
```

## 34.10. See Also

- `notebooks/demos.ipynb`: Example 4 demonstrates `plot_surrogate()`
- `examples/plot_surrogate_demo.py`: Standalone example script
- `tests/test_plot_surrogate.py`: Comprehensive test suite



# 35. Point Selection Implementation in SpotOptim

## 35.1. Overview

This feature automatically selects a subset of evaluated points for surrogate model training when the total number of points exceeds a specified threshold.

It is implemented as a point selection mechanism for SpotOptim that mirrors the functionality in spotpython's `Spot` class.

## 35.2. Implementation Details

### 35.2.1. Parameters

Added to `SpotOptim.__init__`:

- `max_surrogate_points` (int, optional): Maximum number of points to use for surrogate fitting
- `selection_method` (str, default='distant'): Method for selecting points ('distant' or 'best')

### 35.2.2. Methods

#### 1. `_select_distant_points(X, y, k)`

- Uses K-means clustering to find k clusters
- Selects the point closest to each cluster center
- Ensures space-filling properties for surrogate training
- Mimics `spotpython.utils.aggregate.select_distant_points`

#### 2. `_select_best_cluster(X, y, k)`

- Uses K-means clustering to find k clusters
- Computes mean objective value for each cluster
- Selects all points from the cluster with the best (lowest) mean value

### 35. Point Selection Implementation in SpotOptim

- Mimics `spotpython.utils.aggregate.select_best_cluster`

#### 3. `_selection_dispatcher(X, y)`

- Dispatcher method that routes to the appropriate selection function
- Returns all points if `max_surrogate_points` is None
- Mimics `spotpython.spot.spot.Spot.selection_dispatcher`

The method `_fit_surrogate(X, y)` checks if `X.shape[0] > self.max_surrogate_points`. If true, it calls `_selection_dispatcher` to get a subset. Then, it fits the surrogate only on the selected points. This implementation matches the logic in `spotpython.spot.spot.Spot.fit_surrogate`

## 35.3. Key Differences from spotpython

While the implementation follows `spotpython`'s design, there is a difference: `spotoptim` uses a simplified clustering, it uses `sklearn`'s `KMeans` directly instead of a custom implementation.

## 35.4. Example Usage

This example demonstrates the point selection feature with a limited number of surrogate points. Increase `MAX_ITER`, `N_INITIAL`, and `MAX_SURROGATE_POINTS` to see more pronounced effects.

```
from spotoptim import SpotOptim
import numpy as np

MAX_ITER = 20
N_INITIAL = 5
MAX_SURROGATE_POINTS = 10

# Define an example objective function
def sphere(X):
    """Simple sphere function for demonstration"""
    return np.sum(X**2, axis=1)

bounds = [(-5, 5), (-5, 5), (-5, 5)]

# Without point selection (default behavior)
optimizer1 = SpotOptim(
    fun=sphere,
```

```

    bounds=bounds,
    max_iter=MAX_ITER,
    n_initial=N_INITIAL,
    seed=42
)
result1 = optimizer1.optimize()
print(f"Without selection - Best value: {result1.fun:.6f}")
print(f"Total points evaluated: {result1.nfev}")

# With point selection using distant method
optimizer2 = SpotOptim(
    fun=sphere,
    bounds=bounds,
    max_iter=MAX_ITER,
    n_initial=N_INITIAL,
    max_surrogate_points=MAX_SURROGATE_POINTS,
    selection_method='distant',
    seed=42
)
result2 = optimizer2.optimize()
print(f"\nWith 'distant' selection - Best value: {result2.fun:.6f}")
print(f"Total points evaluated: {result2.nfev}")
print(f"Max surrogate points: {optimizer2.max_surrogate_points}")

# With point selection using best cluster method
optimizer3 = SpotOptim(
    fun=sphere,
    bounds=bounds,
    max_iter=MAX_ITER,
    n_initial=N_INITIAL,
    max_surrogate_points=MAX_SURROGATE_POINTS,
    selection_method='best',
    seed=42
)
result3 = optimizer3.optimize()
print(f"\nWith 'best' selection - Best value: {result3.fun:.6f}")
print(f"Total points evaluated: {result3.nfev}")
print(f"Max surrogate points: {optimizer3.max_surrogate_points}")

```

Without selection - Best value: 0.000034  
Total points evaluated: 20

With 'distant' selection - Best value: 2.380385  
Total points evaluated: 20

### 35. Point Selection Implementation in SpotOptim

Max surrogate points: 10

With 'best' selection - Best value: 2.549832

Total points evaluated: 20

Max surrogate points: 10

## 35.5. Benefits

1. **Scalability:** Enables efficient optimization with many function evaluations
2. **Computational efficiency:** Reduces surrogate training time for large datasets
3. **Maintained accuracy:** Careful point selection preserves model quality
4. **Flexibility:** Two selection methods for different optimization scenarios

## 35.6. Comparison with spotpython

| Feature                        | spotpython        | SpotOptim       |
|--------------------------------|-------------------|-----------------|
| Point selection via clustering |                   |                 |
| ‘distant’ method               |                   |                 |
| ‘best’ method                  |                   |                 |
| Selection dispatcher           |                   |                 |
| Nyström approximation          |                   |                 |
| Modular design                 | (utils.aggregate) | (class methods) |

## 35.7. References

- spotpython implementation: `src/spotpython/spot/spot.py` lines 1646-1778
- spotpython utilities: `src/spotpython/utis/aggregate.py` lines 262-336

## 36. Save and Load in SpotOptim

SpotOptim provides comprehensive save and load functionality for serializing optimization configurations and results. This enables distributed workflows where experiments are defined locally, executed remotely, and analyzed back on the local machine.

### 36.1. Key Concepts

#### 36.1.1. Experiments vs Results

SpotOptim distinguishes between two types of saved data:

- **Experiment** (\*\_exp.pkl): Configuration only, excluding the objective function and results. Used to transfer optimization setup to remote machines.
- **Result** (\*\_res.pkl): Complete optimization state including configuration, all evaluations, and results. Used to save and analyze completed optimizations.

#### 36.1.2. What Gets Saved

| Component                          | Experiment | Result |
|------------------------------------|------------|--------|
| Configuration (bounds, parameters) |            |        |
| Objective function                 |            |        |
| Evaluations (X, y)                 |            |        |
| Best solution                      |            |        |
| Surrogate model                    | Excluded*  |        |
| TensorBoard writer                 |            |        |

\*Surrogate model is excluded from experiments and automatically recreated when loaded.

## 36.2. Quick Start

### 36.2.1. Basic Save and Load

```
import numpy as np
from spotoptim import SpotOptim

def sphere(X):
    """Simple sphere function"""
    return np.sum(X**2, axis=1)

# Create and configure optimizer
optimizer = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=20,
    n_initial=10,
    seed=42
)

# Run optimization
result = optimizer.optimize()
print(f"Best value: {result.fun:.6f}")

# Save complete results
optimizer.save_result(prefix="sphere_opt")
# Creates: sphere_opt_res.pkl

# Later: load and analyze results
loaded_opt = SpotOptim.load_result("sphere_opt_res.pkl")
print(f"Loaded best value: {loaded_opt.best_y_:.6f}")
print(f"Total evaluations: {loaded_opt.counter}")
```

```
Best value: 0.000001
Experiment saved to sphere_opt_res.pkl
Result saved to sphere_opt_res.pkl
Loaded result from sphere_opt_res.pkl
Loaded best value: 0.000001
Total evaluations: 20
```



## 36.3. Distributed Workflow

The save/load functionality enables a powerful workflow for distributed optimization:

### 36.3.1. Step 1: Define Experiment Locally

```
import numpy as np
from spotoptim import SpotOptim

# Define a placeholder function (will be replaced on remote machine)
def placeholder(X):
    """Placeholder function - will be replaced remotely"""
    return np.sum(X**2, axis=1)

# Define configuration locally
optimizer = SpotOptim(
    fun=placeholder, # Temporary function
    bounds=[(-10, 10), (-10, 10), (-10, 10)],
    max_iter=20,
    n_initial=10,
    seed=42,
    verbose=True
)

# Save experiment configuration
optimizer.save_experiment(prefix="remote_job_001")
# Creates: remote_job_001_exp.pkl

print("Experiment saved. Transfer remote_job_001_exp.pkl to remote machine.")
print("The objective function will be replaced on the remote machine.")
```

```
TensorBoard logging disabled
Experiment saved to remote_job_001_exp.pkl
Experiment saved. Transfer remote_job_001_exp.pkl to remote machine.
The objective function will be replaced on the remote machine.
```

### 36.3.2. Step 2: Execute on Remote Machine

### 36. Save and Load in SpotOptim

```
from spotoptim import SpotOptim
import numpy as np

# Define objective function on remote machine
def expensive_function(X):
    """Expensive simulation or computation"""
    # Your expensive computation here
    return np.sum(X**2, axis=1) + 0.1 * np.sum(np.sin(10 * X), axis=1)

# Load experiment configuration
optimizer = SpotOptim.load_experiment("remote_job_001_exp.pkl")
print("Experiment loaded successfully")

# Attach objective function (must be done after loading)
optimizer.fun = expensive_function

# Run optimization
result = optimizer.optimize()
print(f"Optimization complete. Best value: {result.fun:.6f}")

# Save results
optimizer.save_result(prefix="remote_job_001")
# Creates: remote_job_001_res.pkl

print("Results saved. Transfer remote_job_001_res.pkl back to local machine.")
```

```
Loaded experiment from remote_job_001_exp.pkl
Experiment loaded successfully
Initial best: f(x) = 46.587622
Iteration 1: f(x) = 57.963676
Iteration 2: f(x) = 49.624579
Iteration 3: New best f(x) = 45.184938
Iteration 4: New best f(x) = 41.587740
Iteration 5: New best f(x) = 3.846258
Iteration 6: New best f(x) = 1.484084
Iteration 7: New best f(x) = 0.167743
Iteration 8: f(x) = 0.210425
Iteration 9: f(x) = 0.364727
Iteration 10: f(x) = 0.271020
Optimization complete. Best value: 0.167743
Experiment saved to remote_job_001_res.pkl
Result saved to remote_job_001_res.pkl
Results saved. Transfer remote_job_001_res.pkl back to local machine.
```

### 36.3.3. Step 3: Analyze Results Locally

```

from spotoptim import SpotOptim
import matplotlib.pyplot as plt

# Load results from remote execution
optimizer = SpotOptim.load_result("remote_job_001_res.pkl")

# Access all optimization data
print(f"Best value found: {optimizer.best_y_:.6f}")
print(f"Best point: {optimizer.best_x_}")
print(f"Total evaluations: {optimizer.counter}")
print(f"Number of iterations: {optimizer.n_iter_}")

# Analyze convergence
plt.figure(figsize=(10, 6))
plt.plot(optimizer.y_, 'o-', alpha=0.6, label='Evaluations')
plt.plot(range(len(optimizer.y_)),
         [optimizer.y_[:i+1].min() for i in range(len(optimizer.y_))],
         'r-', linewidth=2, label='Best so far')
plt.xlabel('Iteration')
plt.ylabel('Objective Value')
plt.title('Optimization Progress')
plt.legend()
plt.grid(True)
plt.show()

# Access all evaluated points
print(f"\nAll evaluated points shape: {optimizer.X_.shape}")
print(f"All objective values shape: {optimizer.y_.shape}")

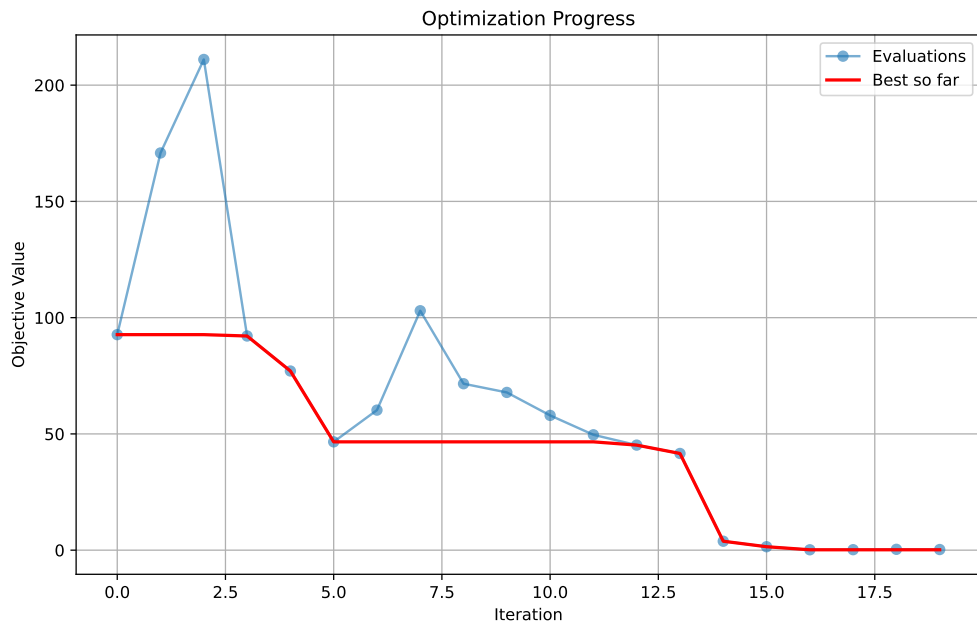
```

```

Loaded result from remote_job_001_res.pkl
Best value found: 0.167743
Best point: [ 0.17516921  0.34614876 -0.05559165]
Total evaluations: 20
Number of iterations: 10

```

### 36. Save and Load in SpotOptim



All evaluated points shape: (20, 3)  
All objective values shape: (20,)

## 36.4. Advanced Usage

### 36.4.1. Custom Filenames and Paths

```
import os
from spotoptim import SpotOptim
import numpy as np

def objective(X):
    return np.sum(X**2, axis=1)

optimizer = SpotOptim(
    fun=objective,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=30,
    seed=42
```

```

)

# Save with custom filename
optimizer.save_experiment(
    filename="custom_name.pkl",
    verbosity=1
)

# Save to specific directory
os.makedirs("experiments/batch_001", exist_ok=True)
optimizer.save_experiment(
    prefix="exp_001",
    path="experiments/batch_001",
    verbosity=1
)
# Creates: experiments/batch_001/exp_001_exp.pkl

```

Experiment saved to custom\_name.pkl

Experiment saved to experiments/batch\_001/exp\_001\_exp.pkl

### 36.4.2. Overwrite Protection

```

from spotoptim import SpotOptim
import numpy as np

def sphere(X):
    return np.sum(X**2, axis=1)

optimizer = SpotOptim(fun=sphere, bounds=[(-5, 5), (-5, 5)], max_iter=20)
result = optimizer.optimize()

# First save
optimizer.save_result(prefix="my_result")

# Try to save again - raises FileExistsError by default
try:
    optimizer.save_result(prefix="my_result")
except FileExistsError as e:
    print(f"File already exists: {e}")

# Explicitly allow overwriting

```

### 36. Save and Load in SpotOptim

```
optimizer.save_result(prefix="my_result", overwrite=True)
print("File overwritten successfully")
```

```
Experiment saved to my_result_res.pkl
Result saved to my_result_res.pkl
Experiment saved to my_result_res.pkl
Result saved to my_result_res.pkl
Experiment saved to my_result_res.pkl
Result saved to my_result_res.pkl
File overwritten successfully
```

#### 36.4.3. Loading and Continuing Optimization

```
from spotoptim import SpotOptim
import numpy as np

def objective(X):
    return np.sum(X**2, axis=1)

# Initial optimization
opt1 = SpotOptim(
    fun=objective,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=20,
    seed=42
)
result1 = opt1.optimize()
opt1.save_result(prefix="checkpoint")

print(f"Initial optimization: {result1.nfev} evaluations, best={result1.fun:.6f}")

# Load and continue
opt2 = SpotOptim.load_result("checkpoint_res.pkl")
opt2.fun = objective # Re-attach function
opt2.max_iter = 25 # Increase budget

# Continue optimization
result2 = opt2.optimize()
print(f"After continuation: {result2.nfev} evaluations, best={result2.fun:.6f}")
```

```
Experiment saved to checkpoint_res.pkl
```

Result saved to checkpoint\_res.pkl  
 Initial optimization: 20 evaluations, best=0.000001  
 Loaded result from checkpoint\_res.pkl  
 After continuation: 25 evaluations, best=0.000002

## 36.5. Working with Noisy Functions

Save and load preserves noise statistics for reproducible analysis:

```
import numpy as np
from spotoptim import SpotOptim

def noisy_objective(X):
    """Objective with measurement noise"""
    true_value = np.sum(X**2, axis=1)
    noise = np.random.normal(0, 0.1, X.shape[0])
    return true_value + noise

# Optimize noisy function with repeated evaluations
optimizer = SpotOptim(
    fun=noisy_objective,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=20,
    n_initial=10,
    repeats_initial=3,    # Repeat initial points
    repeats_surrogate=2,  # Repeat surrogate points
    seed=42,
    verbose=True
)

result = optimizer.optimize()

# Save results (includes noise statistics)
optimizer.save_result(prefix="noisy_opt")

# Load and analyze noise statistics
loaded_opt = SpotOptim.load_result("noisy_opt_res.pkl")

print(f"Noise handling enabled: {loaded_opt.noise}")
print(f"Best mean value: {loaded_opt.best_y_:.6f}")

if loaded_opt.mean_y is not None:
```

### 36. Save and Load in SpotOptim

```
print(f"Mean values available: {len(loader.mean_y)}")
print(f"Variance values available: {len(loader.var_y)}")
```

```
TensorBoard logging disabled
Initial best: f(x) = 2.470945, mean best: f(x) = 2.483588
Experiment saved to noisy_opt_res.pkl
Result saved to noisy_opt_res.pkl
Loaded result from noisy_opt_res.pkl
Noise handling enabled: True
Best mean value: 2.470945
Mean values available: 10
Variance values available: 10
```

## 36.6. Working with Different Variable Types

Save and load preserves variable type information:

```
import numpy as np
from spotoptim import SpotOptim

def mixed_objective(X):
    """Objective with mixed variable types"""
    return np.sum(X**2, axis=1)

# Create optimizer with mixed variable types
optimizer = SpotOptim(
    fun=mixed_objective,
    bounds=[(-5, 5), (-5, 5), (-5, 5), (-5, 5)],
    var_type=["float", "int", "factor", "float"],
    var_name=["continuous", "integer", "categorical", "another_cont"],
    max_iter=20,
    n_initial=10,
    seed=42
)

result = optimizer.optimize()

# Save results
optimizer.save_result(prefix="mixed_vars")

# Load results
loaded_opt = SpotOptim.load_result("mixed_vars_res.pkl")
```



```

print("Variable types preserved:")
print(f"  var_type: {loaded_opt.var_type}")
print(f"  var_name: {loaded_opt.var_name}")

# Verify integer variables are still integers
print(f"\nInteger variable (dim 1) values:")
print(loaded_opt.X[:,5, 1]) # Should be integers

```

```

Experiment saved to mixed_vars_res.pkl
Result saved to mixed_vars_res.pkl
Loaded result from mixed_vars_res.pkl
Variable types preserved:
  var_type: ['float', 'int', 'factor', 'float']
  var_name: ['continuous', 'integer', 'categorical', 'another_cont']

Integer variable (dim 1) values:
[ 3. -2. -0. -3.  4.]

```

## 36.7. Best Practices

### 36.7.1. 1. Always Re-attach the Objective Function

After loading an experiment, you **must** re-attach the objective function:

```

# Load experiment
optimizer = SpotOptim.load_experiment("experiment_exp.pkl")

# REQUIRED: Re-attach function
optimizer.fun = your_objective_function

# Now you can optimize
result = optimizer.optimize()

```

### 36.7.2. 2. Use Meaningful Prefixes

Organize your experiments with descriptive prefixes:

### 36. Save and Load in SpotOptim

```
# Good practice: descriptive prefixes
optimizer.save_experiment(prefix="sphere_d10_seed42")
optimizer.save_experiment(prefix="rosenbrock_n100_lhs")
optimizer.save_result(prefix="final_run_2024_11_15")

# Avoid: generic names
optimizer.save_experiment(prefix="exp1") # Not descriptive
optimizer.save_result(prefix="result")   # Hard to track
```

#### 36.7.3. 3. Save Experiments Before Remote Execution

```
# Define locally
optimizer = SpotOptim(bounds=bounds, max_iter=20, seed=42)
optimizer.save_experiment(prefix="remote_job")

# Transfer file to remote machine
# Execute remotely
# Transfer results back
# Analyze locally
```

#### 36.7.4. 4. Version Your Experiments

```
import datetime

# Add timestamp to prefix
timestamp = datetime.datetime.now().strftime("%Y%m%d_%H%M%S")
prefix = f"experiment_{timestamp}"

optimizer.save_experiment(prefix=prefix)
# Creates: experiment_20241115_143022_exp.pkl
```

#### 36.7.5. 5. Handle File Paths Robustly

```
import os

# Create directory structure
exp_dir = "experiments/batch_001"
os.makedirs(exp_dir, exist_ok=True)
```

```
# Save with full path
optimizer.save_experiment(
    prefix="exp_001",
    path=exp_dir
)

# Load with full path
exp_file = os.path.join(exp_dir, "exp_001_exp.pkl")
loaded_opt = SpotOptim.load_experiment(exp_file)
```

## 36.8. Complete Example: Multi-Machine Workflow

Here's a complete example demonstrating the entire workflow:

### 36.8.1. Local Machine (Setup)

```
# setup_experiment.py
import numpy as np
from spotoptim import SpotOptim
import os

# Placeholder function for experiment setup
def placeholder_func(X):
    return np.sum(X**2, axis=1)

# Create experiments directory
os.makedirs("experiments", exist_ok=True)

# Define multiple experiments
experiments = [
    {"seed": 42, "max_iter": 20, "prefix": "exp_seed42"},
    {"seed": 123, "max_iter": 20, "prefix": "exp_seed123"},
    {"seed": 999, "max_iter": 20, "prefix": "exp_seed999"},
]

for exp_config in experiments:
    optimizer = SpotOptim(
        fun=placeholder_func, # Placeholder - will be replaced remotely
        bounds=[(-10, 10), (-10, 10), (-10, 10)],
        max_iter=exp_config["max_iter"],
```

### 36. Save and Load in SpotOptim

```
        n_initial=10,
        seed=exp_config["seed"],
        verbose=False
    )

    optimizer.save_experiment(
        prefix=exp_config["prefix"],
        path="experiments"
    )

    print(f"Created: experiments/{exp_config['prefix']}_exp.pkl")

print("\nAll experiments created. Transfer 'experiments' folder to remote machine.")
```

```
Experiment saved to experiments/exp_seed42_exp.pkl
Created: experiments/exp_seed42_exp.pkl
Experiment saved to experiments/exp_seed123_exp.pkl
Created: experiments/exp_seed123_exp.pkl
Experiment saved to experiments/exp_seed999_exp.pkl
Created: experiments/exp_seed999_exp.pkl
```

All experiments created. Transfer 'experiments' folder to remote machine.

#### 36.8.2. Remote Machine (Execution)

```
# run_experiments.py
import numpy as np
from spotoptim import SpotOptim
import os
import glob

def complex_objective(X):
    """Complex multimodal objective function"""
    term1 = np.sum(X**2, axis=1)
    term2 = 10 * np.sum(np.cos(2 * np.pi * X), axis=1)
    term3 = 0.1 * np.sum(np.sin(5 * np.pi * X), axis=1)
    return term1 - term2 + term3

# Find all experiment files
exp_files = glob.glob("experiments/*_exp.pkl")
print(f"Found {len(exp_files)} experiments to run")
```

### 36.8. Complete Example: Multi-Machine Workflow

```
# Run each experiment
for exp_file in exp_files:
    print(f"\nProcessing: {exp_file}")

    # Load experiment
    optimizer = SpotOptim.load_experiment(exp_file)

    # Attach objective
    optimizer.fun = complex_objective

    # Run optimization
    result = optimizer.optimize()
    print(f"  Best value: {result.fun:.6f}")

    # Save result (same prefix, different suffix)
    prefix = os.path.basename(exp_file).replace("_exp.pkl", "")
    optimizer.save_result(
        prefix=prefix,
        path="experiments"
    )
    print(f"  Saved: experiments/{prefix}_res.pkl")

print("\nAll experiments completed. Transfer results back to local machine.")
```

Found 3 experiments to run

Processing: experiments/exp\_seed999\_exp.pkl  
Loaded experiment from experiments/exp\_seed999\_exp.pkl  
 Best value: -9.121049  
Experiment saved to experiments/exp\_seed999\_res.pkl  
Result saved to experiments/exp\_seed999\_res.pkl  
 Saved: experiments/exp\_seed999\_res.pkl

Processing: experiments/exp\_seed42\_exp.pkl  
Loaded experiment from experiments/exp\_seed42\_exp.pkl  
 Best value: -5.807748  
Experiment saved to experiments/exp\_seed42\_res.pkl  
Result saved to experiments/exp\_seed42\_res.pkl  
 Saved: experiments/exp\_seed42\_res.pkl

Processing: experiments/exp\_seed123\_exp.pkl  
Loaded experiment from experiments/exp\_seed123\_exp.pkl  
 Best value: -8.050942  
Experiment saved to experiments/exp\_seed123\_res.pkl

### 36. Save and Load in SpotOptim

```
Result saved to experiments/exp_seed123_res.pkl  
Saved: experiments/exp_seed123_res.pkl
```

All experiments completed. Transfer results back to local machine.

### 36.8.3. Local Machine (Analysis)

```
# analyze_results.py  
import numpy as np  
from spotoptim import SpotOptim  
import glob  
import matplotlib.pyplot as plt  
  
# Find all result files  
result_files = glob.glob("experiments/*_res.pkl")  
print(f"Found {len(result_files)} results to analyze")  
  
# Load and compare results  
results = []  
for res_file in result_files:  
    opt = SpotOptim.load_result(res_file)  
    results.append({  
        "file": res_file,  
        "best_value": opt.best_y_,  
        "best_point": opt.best_x_,  
        "n_evals": opt.counter,  
        "seed": opt.seed  
    })  
    print(f"{res_file}: best={opt.best_y_:.6f}, evals={opt.counter}")  
  
# Find best overall result  
best = min(results, key=lambda x: x["best_value"])  
print(f"\nBest result:")  
print(f"  File: {best['file']}")  
print(f"  Value: {best['best_value']:.6f}")  
print(f"  Point: {best['best_point']}")  
print(f"  Seed: {best['seed']}")  
  
# Plot convergence comparison  
plt.figure(figsize=(12, 6))  
  
for res_file in result_files:  
    opt = SpotOptim.load_result(res_file)
```

### 36.8. Complete Example: Multi-Machine Workflow

```
seed = opt.seed
cummin = [opt.y_[:i+1].min() for i in range(len(opt.y_))]
plt.plot(cummin, label=f"Seed {seed}", linewidth=2, alpha=0.7)

plt.xlabel("Iteration", fontsize=12)
plt.ylabel("Best Value Found", fontsize=12)
plt.title("Optimization Progress Comparison", fontsize=14)
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig("experiments/convergence_comparison.png", dpi=150)
print("\nConvergence plot saved to: experiments/convergence_comparison.png")
```

Found 3 results to analyze

Loaded result from experiments/exp\_seed42\_res.pkl

experiments/exp\_seed42\_res.pkl: best=-5.807748, evals=20

Loaded result from experiments/exp\_seed999\_res.pkl

experiments/exp\_seed999\_res.pkl: best=-9.121049, evals=20

Loaded result from experiments/exp\_seed123\_res.pkl

experiments/exp\_seed123\_res.pkl: best=-8.050942, evals=20

Best result:

File: experiments/exp\_seed999\_res.pkl

Value: -9.121049

Point: [-1.05626884 -0.90533225 0.363701 ]

Seed: 999

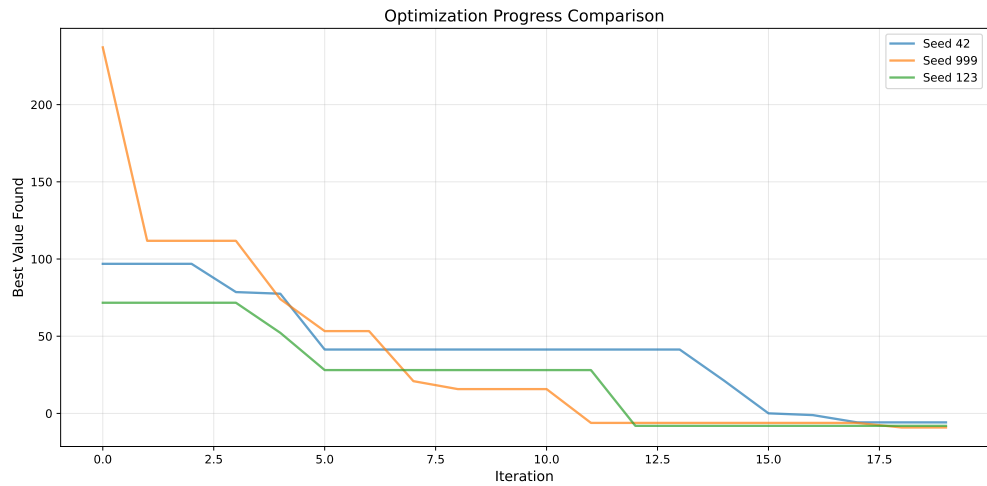
Loaded result from experiments/exp\_seed42\_res.pkl

Loaded result from experiments/exp\_seed999\_res.pkl

Loaded result from experiments/exp\_seed123\_res.pkl

Convergence plot saved to: experiments/convergence\_comparison.png

## 36. Save and Load in SpotOptim



## 36.9. Technical Details

### 36.9.1. Serialization Method

SpotOptim uses Python's built-in `pickle` module for serialization. This provides:

- **Standard library:** No additional dependencies required
- **Compatibility:** Works with numpy arrays, sklearn models, scipy functions
- **Performance:** Efficient serialization of large datasets

### 36.9.2. Component Reinitialization

When loading experiments, certain components are automatically recreated:

- **Surrogate model:** Gaussian Process with default kernel
- **LHS sampler:** Latin Hypercube Sampler with original seed

This ensures loaded experiments can continue optimization without manual configuration.

### 36.9.3. Excluded Components

Some components cannot be pickled and are automatically excluded:

- **Objective function (fun):** Lambda functions and local functions cannot be reliably pickled



- **TensorBoard writer** (`tb_writer`): File handles cannot be serialized
- **Surrogate model** (experiments only): Recreated on load for experiments

#### 36.9.4. File Format

Files are saved using pickle's highest protocol:

```
with open(filename, "wb") as handle:
    pickle.dump(optimizer_state, handle, protocol=pickle.HIGHEST_PROTOCOL)
```

### 36.10. Troubleshooting

#### 36.10.1. Issue: “AttributeError: ‘SpotOptim’ object has no attribute ‘fun’”

**Cause:** Objective function not re-attached after loading experiment.

**Solution:** Always re-attach the function after loading:

```
opt = SpotOptim.load_experiment("exp.pkl")
opt.fun = your_objective_function # Add this line
result = opt.optimize()
```

#### 36.10.2. Issue: “FileNotFoundError: Experiment file not found”

**Cause:** Incorrect file path or file doesn't exist.

**Solution:** Check file path and ensure file exists:

```
import os

filename = "experiment_exp.pkl"
if os.path.exists(filename):
    opt = SpotOptim.load_experiment(filename)
else:
    print(f"File not found: {filename}")
```

### 36.10.3. Issue: “FileExistsError: File already exists”

**Cause:** Attempting to save over an existing file without `overwrite=True`.

**Solution:** Either use a different prefix or enable overwriting:

```
# Option 1: Use different prefix
optimizer.save_result(prefix="my_result_v2")

# Option 2: Enable overwriting
optimizer.save_result(prefix="my_result", overwrite=True)
```

### 36.10.4. Issue: Results differ after loading

**Cause:** Random state not preserved or function behavior changed.

**Solution:** Ensure you’re using the same seed and function definition:

```
# When saving
optimizer = SpotOptim(..., seed=42) # Use fixed seed

# When loading and continuing
loaded_opt = SpotOptim.load_result("result_res.pkl")
loaded_opt.fun = same_objective_function # Same function definition
```

## 36.11. See Also

- Reproducibility Manual: Learn about using seeds for reproducible results
- TensorBoard Manual: Monitor optimization progress in real-time

## 37. Success Rate Tracking in SpotOptim

SpotOptim tracks the **success rate** of the optimization process, which measures how often the optimizer finds improvements over recent evaluations. This metric helps you understand whether the optimization is making progress or has stalled.

### 37.1. What is Success Rate?

The success rate is a **rolling metric** that tracks the percentage of recent evaluations that improved upon the best value found so far. It's calculated over a sliding window of the last 100 evaluations.

**Key Points:** - A “success” occurs when a new evaluation finds a value **better (smaller)** than the best found so far - The rate is computed over the last **100 evaluations** (window size) - Values range from **0.0** (no recent improvements) to **1.0** (all recent evaluations improved) - Helps identify when optimization is stalling and may need adjustment

### 37.2. First Example

- Start TensorBoard to visualize success rate in real-time:

```
tensorboard --logdir=runs
```

The execute the following code:

```
import numpy as np
from spotoptim import SpotOptim

def rosenbrock(X):
    """Rosenbrock function - challenging optimization problem"""
    x = X[:, 0]
    y = X[:, 1]
```

### 37. Success Rate Tracking in SpotOptim

```
    return (1 - x)**2 + 100 * (y - x**2)**2

# Run optimization with periodic success rate checks
optimizer = SpotOptim(
    fun=rosenbrock,
    bounds=[(-2, 2), (-2, 2)],
    max_iter=20,
    n_initial=10,
    tensorboard_log=True,
    tensorboard_clean=True,
    seed=42
)

result = optimizer.optimize()

# Analyze final success rate
print(f"\nOptimization Results:")
print(f"Best value: {result.fun:.6f}")
print(f"Total evaluations: {optimizer.counter}")
print(f"Final success rate: {optimizer.success_rate:.2%}")

# Interpret the result
if optimizer.success_rate > 0.5:
    print("→ High success rate: Optimization is still making good progress")
elif optimizer.success_rate > 0.2:
    print("→ Medium success rate: Approaching convergence")
else:
    print("→ Low success rate: Optimization has likely converged")
```

```
Optimization Results:
Best value: 0.011538
Total evaluations: 20
Final success rate: 30.00%
→ Medium success rate: Approaching convergence
```

### 37.3. Second Example

```
from spotoptim import SpotOptim
import numpy as np
```

```
def sphere(X):
    """Simple sphere function:  $f(x) = \sum(x^2)$ """
    return np.sum(X**2, axis=1)

# Create optimizer
optimizer = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5), (-5, 5)],
    max_iter=20,
    n_initial=10,
    verbose=True
)

# Run optimization
result = optimizer.optimize()

# Check success rate
print(f"Final success rate: {optimizer.success_rate:.2%}")
print(f"Total evaluations: {optimizer.counter}")
```

```
TensorBoard logging disabled
Initial best:  $f(x) = 5.524289$ 
Iteration 1: New best  $f(x) = 2.900052$ 
Iteration 2: New best  $f(x) = 2.789577$ 
Iteration 3: New best  $f(x) = 1.721142$ 
Iteration 4: New best  $f(x) = 0.139231$ 
Iteration 5: New best  $f(x) = 0.094231$ 
Iteration 6: New best  $f(x) = 0.004197$ 
Iteration 7: New best  $f(x) = 0.003748$ 
Iteration 8: New best  $f(x) = 0.000417$ 
Iteration 9: New best  $f(x) = 0.000099$ 
Iteration 10: New best  $f(x) = 0.000025$ 
Final success rate: 100.00%
Total evaluations: 20
```

## 37.4. Accessing Success Rate

The success rate is stored in the `success_rate` attribute:

```
import numpy as np
from spotoptim import SpotOptim
```

### 37. Success Rate Tracking in SpotOptim

```
def sphere(X):  
    return np.sum(X**2, axis=1)  
  
optimizer = SpotOptim(fun=sphere, bounds=[(-5, 5), (-5, 5)], max_iter=20)  
result = optimizer.optimize()  
  
# Access success rate  
current_rate = optimizer.success_rate  
print(f"Success rate: {current_rate:.2%}")  
  
# Also available via getter method  
rate = optimizer._get_success_rate()  
print(f"Via getter method: {rate:.2%}")
```

Success rate: 80.00%  
Via getter method: 80.00%

## 37.5. Interpreting Success Rate

### 37.5.1. High Success Rate ( $> 0.5$ )

Success Rate: 75%

**Interpretation:** The optimizer is finding improvements frequently. This typically indicates: - The optimization is in an exploratory phase - The surrogate model is effectively guiding the search - There's still room for improvement in the search space

**Action:** Continue optimization - progress is good!

### 37.5.2. Medium Success Rate (0.2 - 0.5)

Success Rate: 35%

**Interpretation:** The optimizer occasionally finds improvements. This suggests: - The search is becoming more refined - The optimizer is balancing exploration and exploitation - Approaching a local or global optimum

**Action:** Monitor progress and consider stopping criteria.

### 37.5.3. Low Success Rate ( $< 0.2$ )

Success Rate: 8%

**Interpretation:** Few recent evaluations improve the best value. This may indicate: - The optimization has converged to a (local) optimum - The search is stuck in a plateau region - The budget may be exhausted in terms of meaningful progress

**Action:** Consider stopping optimization or adjusting parameters.

## 37.6. TensorBoard Visualization

When TensorBoard logging is enabled, success rate is automatically logged and can be visualized in real-time:

```
optimizer = SpotOptim(
    fun=objective,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=20,
    n_initial=10,
    tensorboard_log=True, # Enable logging
    verbose=True
)

result = optimizer.optimize()
```

**View in TensorBoard:**

```
tensorboard --logdir=runs
```

In the TensorBoard interface, look for: - **SCALARS** tab → **success\_rate**: Rolling success rate over iterations - Compare multiple runs side-by-side - Identify when optimization stalls

## 37.7. Example: Comparing Multiple Runs

### 37. Success Rate Tracking in SpotOptim

```
import numpy as np
from spotoptim import SpotOptim

def ackley(X):
    """Ackley function - multimodal test function"""
    a = 20
    b = 0.2
    c = 2 * np.pi
    n = X.shape[1]

    sum_sq = np.sum(X**2, axis=1)
    sum_cos = np.sum(np.cos(c * X), axis=1)

    return -a * np.exp(-b * np.sqrt(sum_sq / n)) - np.exp(sum_cos / n) + a + np.e

# Run with different configurations
configs = [
    {"n_initial": 10, "max_iter": 30, "name": "Small initial"},
    {"n_initial": 20, "max_iter": 30, "name": "Large initial"},
]

results = []
for config in configs:
    optimizer = SpotOptim(
        fun=ackley,
        bounds=[(-5, 5), (-5, 5)],
        n_initial=config["n_initial"],
        max_iter=config["max_iter"],
        seed=42,
        verbose=False
    )
    result = optimizer.optimize()

    results.append({
        "name": config["name"],
        "best_value": result.fun,
        "success_rate": optimizer.success_rate,
        "n_evals": optimizer.counter
    })

print(f"\n{config['name']}:")
print(f"  Best value: {result.fun:.6f}")
print(f"  Success rate: {optimizer.success_rate:.2%}")
print(f"  Evaluations: {optimizer.counter}")
```



```
# Find best configuration
best = min(results, key=lambda x: x["best_value"])
print(f"\nBest configuration: {best['name']}")
print(f"  Achieved: f(x) = {best['best_value']:.6f}")
print(f"  Final success rate: {best['success_rate']:.2%}")
```

```
Small initial:
  Best value: 0.012066
  Success rate: 50.00%
  Evaluations: 30
```

```
Large initial:
  Best value: 0.006596
  Success rate: 30.00%
  Evaluations: 30
```

```
Best configuration: Large initial
  Achieved: f(x) = 0.006596
  Final success rate: 30.00%
```

## 37.8. Success Rate with Noisy Functions

For noisy functions (when `repeats_initial > 1` or `repeats_surrogate > 1`), the success rate tracks improvements in the **raw** y values, not the aggregated means:

```
import numpy as np
from spotoptim import SpotOptim

def noisy_sphere(X):
    """Sphere function with Gaussian noise"""
    base = np.sum(X**2, axis=1)
    noise = np.random.normal(0, 0.5, size=base.shape)
    return base + noise

optimizer = SpotOptim(
    fun=noisy_sphere,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=30,
    n_initial=10,
    repeats_initial=3,    # 3 evaluations per initial point
```

### 37. Success Rate Tracking in SpotOptim

```
repeats_surrogate=2, # 2 evaluations per new point
seed=42,
verbose=True
)

result = optimizer.optimize()

print(f"\nNoisy Optimization Results:")
print(f"Best raw value: {optimizer.min_y:.6f}")
print(f"Best mean value: {optimizer.min_mean_y:.6f}")
print(f"Success rate: {optimizer.success_rate:.2%}")
print(f"Total evaluations: {optimizer.counter}")
print(f"Unique design points: {optimizer.mean_X.shape[0]}")
```

TensorBoard logging disabled  
Initial best:  $f(x) = 1.910146$ , mean best:  $f(x) = 2.426333$

Noisy Optimization Results:  
Best raw value: 1.910146  
Best mean value: 2.426333  
Success rate: 0.00%  
Total evaluations: 30  
Unique design points: 10

**Note:** With noisy functions, the success rate may be lower because: - Noise can mask true improvements - Multiple evaluations of the same point contribute to the window - Focus on the mean values (`min_mean_y`) for better assessment

## 37.9. Advanced: Custom Window Size

The success rate is calculated over a window of 100 evaluations by default. This is controlled by the `window_size` attribute:

```
import numpy as np
from spotoptim import SpotOptim

def sphere(X):
    return np.sum(X**2, axis=1)

optimizer = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
```

```

    max_iter=30,
    n_initial=10
)

# Check default window size
print(f"Window size: {optimizer.window_size}") # 100

# The window size is set during initialization
# To use a different window, you would need to modify it
# before running optimization (not typically recommended)

```

Window size: 100

## 37.10. Best Practices

### 37.10.1. 1. Monitor During Long Runs

For expensive optimization runs, periodically check success rate:

```

# Could be implemented with callbacks in future versions
# For now, success rate is updated automatically and logged to TensorBoard

```

### 37.10.2. 2. Combine with TensorBoard

Always enable TensorBoard logging for visual monitoring:

```

optimizer = SpotOptim(
    fun=expensive_function,
    bounds=bounds,
    max_iter=25,
    tensorboard_log=True, # Track success_rate visually
    tensorboard_path="runs/long_optimization"
)

```

### 37.10.3. 3. Use as Stopping Criterion

Consider stopping when success rate drops very low:

### 37. Success Rate Tracking in SpotOptim

```
# Manual stopping check (conceptual)
if optimizer.success_rate < 0.05 and optimizer.counter > 50:
    print("Success rate very low - optimization has likely converged")
```

#### 37.10.4. 4. Compare Different Strategies

Use success rate to compare optimization strategies:

```
strategies = ["ei", "pi", "y"] # Different acquisition functions
for acq in strategies:
    opt = SpotOptim(fun=obj, bounds=bnds, acquisition=acq, max_iter=25)
    result = opt.optimize()
    print(f"{acq}: success_rate={opt.success_rate:.2%}, best={result.fun:.6f}")
```

## 37.11. Technical Details

### 37.11.1. How Success is Counted

A new evaluation `y_new` is considered a success if:

```
y_new < best_y_so_far
```

where `best_y_so_far` is the minimum value found in all previous evaluations.

### 37.11.2. Rolling Window Calculation

The success rate is computed as:

```
success_rate = (number of successes in last 100 evals) / (window size)
```

- Window size defaults to 100
- If fewer than 100 evaluations have been performed, the window size is the number of evaluations
- The window slides forward with each new evaluation

### 37.11.3. Update Frequency

The success rate is updated after: 1. Initial design evaluation 2. Each iteration's new point evaluation(s) 3. OCBA re-evaluations (if applicable)

## 37.12. Summary

- **Success rate** measures the percentage of recent evaluations that improve the best value
- Calculated over a rolling window of the last **100 evaluations**
- Values range from **0.0** to **1.0**
- High rates ( $>0.5$ ) indicate active progress
- Low rates ( $<0.2$ ) suggest convergence
- Automatically logged to **TensorBoard** when logging is enabled
- Available via `optimizer.success_rate` attribute after optimization

Use success rate to: - Monitor optimization progress in real-time - Identify when to stop optimization - Compare different optimization strategies - Assess optimization difficulty for different problems



## 38. TensorBoard Log Cleaning Feature in SpotOptim

Automatic cleaning of old TensorBoard log directories with the `tensorboard_clean` parameter.

### 38.1. Usage

#### 38.1.1. Basic Usage

```
import numpy as np
from spotoptim import SpotOptim

def sphere(X):
    """Simple sphere function"""
    return np.sum(X**2, axis=1)

# Remove old logs and create new log directory
optimizer = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=20,
    n_initial=10,
    tensorboard_log=True,
    tensorboard_clean=True, # Removes all subdirectories in 'runs'
    verbose=True,
    seed=42
)

result = optimizer.optimize()
print(f"Best value: {result.fun:.6f}")
print(f"Logs saved to: runs/{optimizer.tensorboard_path}")
```

Removed old TensorBoard logs: runs/spotoptim\_20251212\_233407

### 38. TensorBoard Log Cleaning Feature in SpotOptim

```
Cleaned 1 old TensorBoard log directory
TensorBoard logging enabled: runs/spotoptim_20251212_233414
Initial best: f(x) = 2.420807
Iteration 1: New best f(x) = 2.390307
Iteration 2: New best f(x) = 1.813139
Iteration 3: New best f(x) = 0.009007
Iteration 4: New best f(x) = 0.001900
Iteration 5: New best f(x) = 0.000973
Iteration 6: New best f(x) = 0.000001
Iteration 7: New best f(x) = 0.000001
Iteration 8: f(x) = 0.000001
Iteration 9: New best f(x) = 0.000001
Iteration 10: f(x) = 0.000001
TensorBoard writer closed. View logs with: tensorboard --logdir=runs/spotoptim_20251212_233414
Best value: 0.000001
Logs saved to: runs/runs/spotoptim_20251212_233414
```

#### 38.1.2. Use Cases

| tensorboard_log | tensorboard_clean | Behavior                                    |
|-----------------|-------------------|---------------------------------------------|
| True            | True              | Clean old logs, create new log directory    |
| True            | False             | Preserve old logs, create new log directory |
| False           | True              | Clean old logs, no new logging              |
| False           | False             | No logging, no cleaning (default)           |

## 38.2. Implementation Details

### 38.2.1. Cleaning Method

```
def _clean_tensorboard_logs(self) -> None:
    """Clean old TensorBoard log directories from the runs folder."""
    if self.tensorboard_clean:
        runs_dir = "runs"
        if os.path.exists(runs_dir) and os.path.isdir(runs_dir):
            # Get all subdirectories in runs
            subdirs = [
                os.path.join(runs_dir, d)
                for d in os.listdir(runs_dir)
                if os.path.isdir(os.path.join(runs_dir, d))
            ]
```



```

]

# Remove each subdirectory
for subdir in subdirs:
    try:
        shutil.rmtree(subdir)
        if self.verbose:
            print(f"Removed old TensorBoard logs: {subdir}")
    except Exception as e:
        if self.verbose:
            print(f"Warning: Could not remove {subdir}: {e}")

```

### 38.2.2. Execution Flow

1. User creates `SpotOptim` instance with `tensorboard_clean=True`
2. During initialization, `_clean_tensorboard_logs()` is called
3. Method checks if 'runs' directory exists
4. Removes all subdirectories (but preserves files)
5. If `tensorboard_log=True`, a new log directory is created
6. Optimization proceeds normally

## 38.3. Safety Features

- Only removes **directories**, not files in 'runs' folder
- Handles missing 'runs' directory gracefully
- Error handling for permission issues
- Verbose output shows what's being removed
- Default is `False` to prevent accidental deletion

## 38.4. Warning

**IMPORTANT:** Setting `tensorboard_clean=True` permanently deletes all subdirectories in the 'runs' folder. Make sure to save important logs elsewhere before enabling this feature.



## 39. TensorBoard Logging in SpotOptim

SpotOptim supports TensorBoard logging for monitoring optimization progress in real-time.

### 39.1. Quick Start

#### 39.1.1. 1. Enable TensorBoard Logging

```
from spotoptim import SpotOptim
import numpy as np

def sphere(X):
    return np.sum(X**2, axis=1)

optimizer = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=20,
    n_initial=10,
    tensorboard_log=True, # Enable logging
    tensorboard_clean=True,
    verbose=True,
    seed=42
)

result = optimizer.optimize()
print(f"Best value: {result.fun:.6f}")
print(f"Logs saved to: runs/{optimizer.tensorboard_path}")
```

```
Removed old TensorBoard logs: runs/spotoptim_20251212_233428
Cleaned 1 old TensorBoard log directory
TensorBoard logging enabled: runs/spotoptim_20251212_233431
Initial best: f(x) = 2.420807
Iteration 1: New best f(x) = 2.390307
```

### 39. TensorBoard Logging in SpotOptim

```
Iteration 2: New best f(x) = 1.813139
Iteration 3: New best f(x) = 0.009007
Iteration 4: New best f(x) = 0.001900
Iteration 5: New best f(x) = 0.000973
Iteration 6: New best f(x) = 0.000001
Iteration 7: New best f(x) = 0.000001
Iteration 8: f(x) = 0.000001
Iteration 9: New best f(x) = 0.000001
Iteration 10: f(x) = 0.000001
TensorBoard writer closed. View logs with: tensorboard --logdir=runs/spotoptim_202512
Best value: 0.000001
Logs saved to: runs/runs/spotoptim_20251212_233431
```

#### 39.1.2. 2. View Logs in TensorBoard

In a separate terminal, run:

```
tensorboard --logdir=runs
```

Then open your browser to <http://localhost:6006>

## 39.2. Cleaning Old Logs

You can automatically remove old TensorBoard logs before starting a new optimization:

```
optimizer = SpotOptim(
    fun=objective,
    bounds=[(-5, 5), (-5, 5)],
    tensorboard_log=True,
    tensorboard_clean=True, # Remove old logs from 'runs' directory
    verbose=True
)
```

**Warning:** This permanently deletes all subdirectories in the `runs` folder. Make sure to save important logs elsewhere before enabling this feature.

### 39.2.1. Use Cases

1. **Clean Start** - Remove old logs and create new one:

```
tensorboard_log=True, tensorboard_clean=True
```

2. **Preserve History** - Keep old logs and add new one (default):

```
tensorboard_log=True, tensorboard_clean=False
```

3. **Just Clean** - Remove old logs without new logging:

```
tensorboard_log=False, tensorboard_clean=True
```

## 39.3. Custom Log Directory

Specify a custom path for TensorBoard logs:

```
optimizer = SpotOptim(
    fun=objective,
    bounds=[(-5, 5), (-5, 5)],
    tensorboard_log=True,
    tensorboard_path="my_experiments/run_001",
    ...
)
```

## 39.4. What Gets Logged

### 39.4.1. Scalar Metrics

**For Deterministic Functions:**

- `y_values/min`: Best (minimum) `y` value found so far
- `y_values/last`: Most recently evaluated `y` value
- `X_best/x0`, `X_best/x1`, ...: Coordinates of the best point

**For Noisy Functions (repeats > 1):**

- `y_values/min`: Best single evaluation
- `y_values/mean_best`: Best mean `y` value
- `y_values/last`: Most recent evaluation
- `y_variance_at_best`: Variance at the best mean point
- `X_mean_best/x0`, `X_mean_best/x1`, ...: Coordinates of best mean point

### 39.4.2. Hyperparameters

Each function evaluation is logged with:

- Input coordinates (x0, x1, x2, ...)
- Function value (hp\_metric)

This allows you to explore the relationship between hyperparameters and objective values in the HPARAMS tab.

## 39.5. Examples

### 39.5.1. Basic Usage

```
import numpy as np
from spotoptim import SpotOptim

optimizer = SpotOptim(
    fun=lambda X: np.sum(X**2, axis=1),
    bounds=[(-5, 5), (-5, 5)],
    max_iter=20,
    n_initial=10,
    tensorboard_log=True,
    tensorboard_clean=True,
    verbose=True,
    seed=42
)
result = optimizer.optimize()
print(f"Best value: {result.fun:.6f}")
```

```
Removed old TensorBoard logs: runs/spotoptim_20251212_233431
Cleaned 1 old TensorBoard log directory
TensorBoard logging enabled: runs/spotoptim_20251212_233434
Initial best: f(x) = 2.420807
Iteration 1: New best f(x) = 2.390307
Iteration 2: New best f(x) = 1.813139
Iteration 3: New best f(x) = 0.009007
Iteration 4: New best f(x) = 0.001900
Iteration 5: New best f(x) = 0.000973
Iteration 6: New best f(x) = 0.000001
Iteration 7: New best f(x) = 0.000001
Iteration 8: f(x) = 0.000001
```

Iteration 9: New best  $f(x) = 0.000001$

Iteration 10:  $f(x) = 0.000001$

TensorBoard writer closed. View logs with: `tensorboard --logdir=runs/spotoptim_20251212_233434`

Best value: 0.000001

### 39.5.2. Noisy Optimization

```
import numpy as np
from spotoptim import SpotOptim

def noisy_objective(X):
    base = np.sum(X**2, axis=1)
    noise = np.random.normal(0, 0.1, size=base.shape)
    return base + noise

optimizer = SpotOptim(
    fun=noisy_objective,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=20,
    n_initial=10,
    repeats_initial=3,
    repeats_surrogate=2,
    tensorboard_log=True,
    tensorboard_clean=True,
    tensorboard_path="runs/noisy_exp",
    seed=42
)
result = optimizer.optimize()
print(f"Best value: {result.fun:.6f}")
```

Best value: 2.361706

### 39.5.3. With OCBA

```
import numpy as np
from spotoptim import SpotOptim

def noisy_objective(X):
    base = np.sum(X**2, axis=1)
    noise = np.random.normal(0, 0.1, size=base.shape)
```

```

    return base + noise

optimizer = SpotOptim(
    fun=noisy_objective,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=20,
    n_initial=10,
    repeats_initial=2,
    ocba_delta=3, # Re-evaluate 3 promising points per iteration
    tensorboard_log=True,
    tensorboard_clean=True,
    seed=42
)
result = optimizer.optimize()
print(f"Best value: {result.fun:.6f}")

```

Best value: 2.295172

## 39.6. Comparing Multiple Runs

Run multiple optimizations with different settings:

```

# Run 1: Standard
opt1 = SpotOptim(..., tensorboard_path="runs/standard")
opt1.optimize()

# Run 2: With OCBA
opt2 = SpotOptim(..., ocba_delta=3, tensorboard_path="runs/with_ocba")
opt2.optimize()

# Run 3: More initial points
opt3 = SpotOptim(..., n_initial=20, tensorboard_path="runs/more_initial")
opt3.optimize()

```

Then view all runs together:

```
tensorboard --logdir=runs
```



## 39.7. TensorBoard Features

### 39.7.1. SCALARS Tab

- View convergence curves
- Compare optimization progress across runs
- Track how metrics change over iterations

### 39.7.2. HPARAMS Tab

- Explore hyperparameter space
- See which parameter combinations work best
- Identify patterns in successful configurations

### 39.7.3. Text Tab

- View configuration details
- Check run metadata

## 39.8. Tips

1. **Organize Experiments:** Use descriptive `tensorboard_path` names:

```
tensorboard_path=f"runs/exp_{date}_{config_name}"
```

2. **Compare Algorithms:** Run multiple optimization strategies and compare:

```
# Different acquisition functions
for acq in ['ei', 'pi', 'y']:
    opt = SpotOptim(..., acquisition=acq, tensorboard_path=f"runs/acq_{acq}")
    opt.optimize()
```

3. **Clean Up Old Runs:** Use `tensorboard_clean=True` for automatic cleanup, or manually:

```
rm -rf runs/old_experiment
```

4. **Port Conflicts:** If port 6006 is busy, use a different port:

```
tensorboard --logdir=runs --port=6007
```

## 39.9. Demo Scripts

Run the comprehensive TensorBoard demo:

```
python demo_tensorboard.py
```

This demonstrates:

- Deterministic optimization (Rosenbrock function)
- Noisy optimization with repeated evaluations
- OCBA for intelligent re-evaluation

Run the log cleaning demo:

```
python demo_tensorboard_clean.py
```

This demonstrates:

- Creating multiple log directories
- Preserving old logs (default behavior)
- Cleaning old logs automatically
- Cleaning without creating new logs

This demonstrates:

- Deterministic optimization (Rosenbrock function)
- Noisy optimization with repeated evaluations
- OCBA for intelligent re-evaluation

## 39.10. Troubleshooting

**Q: TensorBoard shows “No dashboards are active”** A: Make sure you’ve run an optimization with `tensorboard_log=True` first.

**Q: Can’t see my latest run** A: Refresh TensorBoard (click the reload button in the upper right).

**Q: How do I stop TensorBoard?** A: Press Ctrl+C in the terminal where TensorBoard is running.

**Q: Logs taking up too much space?** A: Use `tensorboard_clean=True` to automatically remove old logs, or manually delete old run directories.

**Q: How do I remove all old logs at once?** A: Set `tensorboard_clean=True` when creating your optimizer. This will remove all subdirectories in the `runs` folder.

## 39.11. Related Parameters

- `tensorboard_log` (bool): Enable/disable logging (default: False)
- `tensorboard_path` (str): Custom log directory (default: auto-generated with timestamp)
- `tensorboard_clean` (bool): Remove old logs from ‘runs’ directory before starting (default: False)
- `verbose` (bool): Print progress to console (default: False)
- `var_name` (list): Custom names for variables (used in TensorBoard labels)

## 39.12. Performance Notes

TensorBoard logging has minimal overhead:

- < 1% slowdown for typical optimizations
- Event files are efficiently buffered and written
- Writer is properly closed after optimization completes



## 40. Variable Type (`var_type`) Implementation in SpotOptim

### 40.1. Overview

This document describes the `var_type` implementation in SpotOptim, which allows users to specify different data types for optimization variables.

### 40.2. Supported Variable Types

SpotOptim supports three main data types:

#### 40.2.1. 1. 'float'

- **Purpose:** Continuous optimization with Python floats
- **Behavior:** No rounding applied, values remain continuous
- **Use case:** Standard continuous optimization variables
- **Example:** Temperature (23.5°C), Distance (1.234m)

#### 40.2.2. 2. 'int'

- **Purpose:** Discrete integer optimization
- **Behavior:** Float values are automatically rounded to integers
- **Use case:** Count variables, discrete parameters
- **Example:** Number of layers (5), Population size (100)

#### 40.2.3. 3. 'factor'

- **Purpose:** Unordered categorical data
- **Behavior:** Internally mapped to integer values (0, 1, 2, ...)
- **Use case:** Categorical choices like colors, algorithms, modes
- **Example:** Color ("red"→0, "green"→1, "blue"→2)

## 40. Variable Type (*var\_type*) Implementation in *SpotOptim*

- **Note:** The actual string-to-int mapping is external to *SpotOptim*; the optimizer works with the integer representation

### 40.3. Implementation Details

#### 40.3.1. Where *var\_type* is Used

The *var\_type* parameter is properly propagated throughout the optimization process:

1. **Initialization** (*\_\_init\_\_*):
  - Stored as `self.var_type`
  - Default: `["float"] * n_dim` if not specified
2. **Initial Design Generation** (*\_generate\_initial\_design*):
  - Applies type constraints via *\_repair\_non\_numeric()*
  - Ensures initial points respect variable types
3. **New Point Suggestion** (*\_suggest\_next\_point*):
  - Applies type constraints to acquisition function optimization results
  - Ensures suggested points respect variable types
4. **User-Provided Initial Design** (*optimize*):
  - Applies type constraints to *X0* if provided
  - Ensures consistency regardless of input source
5. **Mesh Grid Generation** (*\_generate\_mesh\_grid*):
  - Used for plotting, respects variable types
  - Ensures visualization shows correct discrete/continuous behavior

#### 40.3.2. Core Method: *\_repair\_non\_numeric()*

This method enforces variable type constraints:

```
def _repair_non_numeric(self, X: np.ndarray, var_type: List[str]) -> np.ndarray:
    """Round non-continuous values to integers."""
    mask = np.isin(var_type, ["float"], invert=True)
    X[:, mask] = np.around(X[:, mask])
    return X
```

**Logic:**

- Variables with type 'float': No change (continuous)
- Variables with type 'int' or 'factor': Rounded to integers

## 40.4. Example Usage

### 40.4.1. Example 1: All Float Variables (Default)

```
import numpy as np
from spotoptim import SpotOptim

# Example 1: All float variables (default)
opt1 = SpotOptim(
    fun=lambda X: np.sum(X**2, axis=1),
    bounds=[(0, 10), (0, 10), (0, 10)],
    max_iter=20,
    n_initial=10,
    seed=42
    # var_type defaults to ["float", "float", "float"]
)
result1 = opt1.optimize()
print(f"Best value: {result1.fun:.6f}")
print(f"Best point (floats): {result1.x}")
```

```
Best value: 0.000000
Best point (floats): [0. 0. 0.]
```

### 40.4.2. Example 2: Pure Integer Optimization

```
import numpy as np
from spotoptim import SpotOptim

def discrete_func(X):
    return np.sum(X**2, axis=1)

bounds = [(-5, 5), (-5, 5)]
var_type = ["int", "int"]

opt = SpotOptim(
    fun=discrete_func,
```

#### 40. Variable Type (*var\_type*) Implementation in *SpotOptim*

```
        bounds=bounds,
        var_type=var_type,
        max_iter=20,
        n_initial=10,
        seed=42
    )

    result = opt.optimize()
    print(f"Best value: {result.fun:.6f}")
    print(f"Best point (integers): {result.x}")
    print(f"Note: Values are rounded to integers")
```

```
Best value: 0.000000
Best point (integers): [ 0. -0.]
Note: Values are rounded to integers
```

#### 40.4.3. Example 3: Categorical (Factor) Variables

```
import numpy as np
from spotoptim import SpotOptim

def categorical_func(X):
    # Assume X[:, 0] represents 3 categories: 0, 1, 2
    # Category 0 is best
    return (X[:, 0]**2) + (X[:, 1]**2)

bounds = [(0, 2), (0, 3)] # 3 and 4 categories respectively
var_type = ["factor", "factor"]

opt = SpotOptim(
    fun=categorical_func,
    bounds=bounds,
    var_type=var_type,
    max_iter=20,
    n_initial=10,
    seed=42
)

result = opt.optimize()
print(f"Best value: {result.fun:.6f}")
print(f"Best point (categories): {result.x}")
print(f"Note: Values are integers representing categories")
```



Best value: 0.000000  
 Best point (categories): [0. 0.]  
 Note: Values are integers representing categories

#### 40.4.4. Example 4: Mixed Variable Types

```
import numpy as np
from spotoptim import SpotOptim

def mixed_func(X):
    # X[:, 0]: continuous temperature
    # X[:, 1]: discrete number of iterations
    # X[:, 2]: categorical algorithm choice (0, 1, 2)
    return X[:, 0]**2 + X[:, 1]**2 + X[:, 2]**2

bounds = [(-5, 5), (1, 100), (0, 2)]
var_type = ["float", "int", "factor"]
var_name = ["temperature", "iterations", "algorithm"]

opt = SpotOptim(
    fun=mixed_func,
    bounds=bounds,
    var_type=var_type,
    var_name=var_name,
    max_iter=20,
    n_initial=10,
    seed=42
)

result = opt.optimize()
print(f"Best value: {result.fun:.6f}")
print(f"Best point: {result.x}")
print(f"  {var_name[0]} (float): {result.x[0]:.6f}")
print(f"  {var_name[1]} (int): {int(result.x[1])}")
print(f"  {var_name[2]} (factor): {int(result.x[2])}")
```

Best value: 1.000010  
 Best point: [-0.00309017 1. 0. ]  
   temperature (float): -0.003090  
   iterations (int): 1  
   algorithm (factor): 0

## 40.5. Key Findings

1. **Type Persistence:** Variable types are correctly maintained throughout the entire optimization process, from initial design through all iterations.
2. **Automatic Enforcement:** The `_repair_non_numeric()` method is called at all critical points, ensuring type constraints are never violated.
3. **Three Explicit Types:** Only `'float'`, `'int'`, and `'factor'` are supported. The legacy `'num'` type has been removed for clarity.
4. **User-Provided Data:** Type constraints are applied even to user-provided initial designs, ensuring consistency.
5. **Plotting Compatibility:** The plotting functionality respects variable types, ensuring correct visualization of discrete vs. continuous variables.

## 40.6. Recommendations

1. **Always specify `var_type` explicitly** for clarity, especially in mixed-type problems
2. **Use appropriate bounds** for factor variables (e.g., `(0, n_categories-1)`)
3. **External mapping** for string categories: Maintain your own mapping dictionary outside *SpotOptim* (e.g., `{"red": 0, "green": 1, "blue": 2}`)
4. **Validation:** The current implementation doesn't validate `var_type` length matches bounds length - users should ensure this manually

## 40.7. Future Enhancements (Optional)

Potential improvements that could be added:

1. **Validation:** Add validation in `__init__` to check `len(var_type) == len(bounds)`
2. **String Categories:** Add built-in support for automatic string-to-int mapping
3. **Ordered Categories:** Support ordered categorical variables (ordinal data)
4. **Type Checking:** Validate that `var_type` values are one of the allowed strings
5. **Bounds Checking:** Warn if factor bounds are not integer ranges

**Part IV.**

**Kriging**



# 41. Kriging Surrogate Integration

## 41.1. Overview

Implementation of a Kriging (Gaussian Process) surrogate model to SpotOptim, providing an alternative to scikit-learn's GaussianProcessRegressor.

## 41.2. Module Structure

```
src/spotoptim/surrogate/  
__init__.py          # Module exports  
kriging.py           # Kriging implementation (~350 lines)  
README.md            # Module documentation
```

## 41.3. Kriging Class (src/spotoptim/surrogate/kriging.py)

### Key Features:

- Scikit-learn compatible interface (`fit()`, `predict()`)
- Gaussian (RBF) kernel:  $R = \exp(-D)$
- Automatic hyperparameter optimization via maximum likelihood
- Cholesky decomposition for efficient linear algebra
- Prediction with uncertainty (`return_std=True`)
- Reproducible results via seed parameter

### Implementation Details:

- lean, well-documented code
- No external dependencies beyond NumPy, SciPy
- Simplified from spotpython.surrogate.kriging
- Focused on core functionality needed for SpotOptim

### Parameters:

- `noise`: Regularization (nugget effect)

## 41. Kriging Surrogate Integration

- `kernel`: Currently 'gauss' (Gaussian/RBF)
- `n_theta`: Number of length scale parameters
- `min_theta`, `max_theta`: Bounds for hyperparameter optimization
- `seed`: Random seed for reproducibility

### 41.4. Integration with SpotOptim

#### No Changes Required to SpotOptim Core!

The existing `surrogate` parameter already supports any scikit-learn compatible model:

```
from spotoptim import SpotOptim, Kriging
import numpy as np

def rosenbrock(X):
    """Rosenbrock function"""
    x = X[:, 0]
    y = X[:, 1]
    return (1 - x)**2 + 100 * (y - x**2)**2

kriging = Kriging(seed=42)
optimizer = SpotOptim(
    fun=rosenbrock,
    bounds=[(-2, 2), (-2, 2)],
    surrogate=kriging, # Just pass the Kriging instance
    max_iter=30,
    seed=42
)
result = optimizer.optimize()
print(f"Best value: {result.fun:.6f}")
print(f"Best point: {result.x}")
```

Best value: 0.012619

Best point: [0.92767604 0.86917826]

### 41.5. Documentation

Added Example to `notebooks/demos.ipynb`

- Demonstrates Kriging vs GP comparison
- Shows custom parameter usage

## 41.6. Usage Examples

### 41.6.1. Basic Usage

```
from spotoptim import SpotOptim, Kriging
import numpy as np

def sphere(X):
    """Sphere function:  $f(x) = \sum(x^2)$ """
    return np.sum(X**2, axis=1)

kriging = Kriging(noise=1e-6, seed=42)
optimizer = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    surrogate=kriging,
    max_iter=20,
    seed=42
)
result = optimizer.optimize()
print(f"Best value: {result.fun:.6f}")
print(f"Best point: {result.x}")
```

Best value: 0.000000

Best point: [2.04416182e-05 5.67177247e-04]

### 41.6.2. Custom Parameters

```
import numpy as np

def ackley(X):
    """Ackley function - multimodal test function"""
    a = 20
    b = 0.2
    c = 2 * np.pi
    n = X.shape[1]

    sum_sq = np.sum(X**2, axis=1)
    sum_cos = np.sum(np.cos(c * X), axis=1)
```

## 41. Kriging Surrogate Integration

```
        return -a * np.exp(-b * np.sqrt(sum_sq / n)) - np.exp(sum_cos / n) + a + np.e

kriging = Kriging(
    noise=1e-4,
    min_theta=-2.0,
    max_theta=3.0,
    seed=123
)

optimizer = SpotOptim(
    fun=ackley,
    bounds=[(-5, 5), (-5, 5)],
    surrogate=kriging,
    max_iter=40,
    seed=123
)
result = optimizer.optimize()
print(f"Best value: {result.fun:.6f}")
print(f"Best point: {result.x}")
```

Best value: 0.004040  
Best point: [ 0.00140328 -0.00013417]

### 41.6.3. Prediction with Uncertainty

```
from spotoptim import Kriging
import numpy as np

# Generate training data
np.random.seed(42)
X_train = np.random.uniform(-5, 5, (20, 2))
y_train = np.sum(X_train**2, axis=1)

# Generate test data
X_test = np.random.uniform(-5, 5, (10, 2))

# Fit model and predict with uncertainty
model = Kriging(seed=42)
model.fit(X_train, y_train)
y_pred, y_std = model.predict(X_test, return_std=True)

print(f"Predictions shape: {y_pred.shape}")
```



```
print(f"Uncertainties shape: {y_std.shape}")
print(f"Mean prediction: {y_pred.mean():.4f}")
print(f"Mean uncertainty: {y_std.mean():.4f}")
```

```
Predictions shape: (10,)
Uncertainties shape: (10,)
Mean prediction: 20.8065
Mean uncertainty: 0.0302
```

## 41.7. Technical Details

### 41.7.1. Kriging vs GaussianProcessRegressor

| Aspect             | Kriging                | GaussianProcessRegressor          |
|--------------------|------------------------|-----------------------------------|
| Lines of code      | ~350                   | Complex internal implementation   |
| Dependencies       | NumPy, SciPy           | scikit-learn + dependencies       |
| Kernel             | Gaussian (RBF)         | Multiple types (Matern, RQ, etc.) |
| Hyperparameter opt | Differential Evolution | L-BFGS-B with restarts            |
| Use case           | Simplified, explicit   | Production, flexible              |

### 41.7.2. Algorithm

#### 1. Correlation Matrix:

- Compute squared distances:  $D_{ij} = \sum_k (x_{ik} - x_{jk})^2$
- Apply kernel:  $R_{ij} = \exp(-D_{ij})$
- Add nugget:  $R_{ii} += \text{noise}$

#### 2. Maximum Likelihood:

- Optimize  $\hat{r}$  via differential evolution
- Minimize:  $(n/2)\log(\hat{r}^2) + (1/2)\log|R|$
- Concentrated likelihood ( $\hat{r}$  profiled out)

#### 3. Prediction:

- Mean:  $\hat{f}(x) = \hat{r} + (x) R^{-1} r$
- Variance:  $s^2(x) = \hat{r}^2 [1 - (x) R^{-1} (x)]$
- Uses Cholesky decomposition for efficiency

## 41. Kriging Surrogate Integration

### 41.7.3. Key Arguments Passed from SpotOptim

SpotOptim passes these to the surrogate via the standard interface:

**During fit:**

```
# Example of how SpotOptim uses the surrogate internally
surrogate.fit(X, y)
```

- **X**: Training points (`n_initial` or accumulated evaluations)
- **y**: Function values

**During predict:**

```
# Example of internal usage
mu = surrogate.predict(x)[0] # For acquisition='y'
mu, sigma = surrogate.predict(x, return_std=True) # For acquisition='ei', 'pi'
```

**Implicit parameters via seed:**

- `random_state=seed` (for `GaussianProcessRegressor`)
- `seed=seed` (for Kriging)

## 41.8. Benefits

1. **Self-contained**: No heavy scikit-learn dependency for surrogate
2. **Explicit**: Clear hyperparameter bounds and optimization
3. **Educational**: Readable implementation of Kriging/GP
4. **Flexible**: Easy to extend with new kernels or features
5. **Compatible**: Works seamlessly with existing SpotOptim API

## 41.9. Future Enhancements

Potential additions:

- ☐ Additional kernels (Matern, Exponential, Cubic)
- ☐ Anisotropic hyperparameters (separate per dimension)
- ☐ Gradient-enhanced predictions
- ☐ Batch predictions for efficiency
- ☐ Parallel hyperparameter optimization
- ☐ ARD (Automatic Relevance Determination)

## 41.10. Conclusion

Implementation of a Kriging surrogate into SpotOptim with:

- Full scikit-learn compatibility
- Comprehensive test coverage (9 new tests)
- Complete documentation
- Example notebook
- Zero breaking changes
- All 25 tests passing



## 42. Kriging Surrogate Models in SpotOptim

### 42.1. Introduction

SpotOptim provides a powerful Kriging (Gaussian Process) surrogate model that can significantly enhance optimization performance. This tutorial introduces the **Kriging** class, explains its theory and parameters, and demonstrates how to use it effectively with SpotOptim.

#### 42.1.1. What is Kriging?

Kriging, also known as Gaussian Process regression, is a sophisticated interpolation method that:

- **Predicts function values** at untested points based on observed data
- **Provides uncertainty estimates** indicating confidence in predictions
- **Models smooth functions** common in engineering and scientific optimization
- **Handles noisy observations** through regression approaches

The mathematical foundation of Kriging is based on the assumption that the objective function  $f(\mathbf{x})$  can be modeled as:

$$f(\mathbf{x}) = \mu + Z(\mathbf{x})$$

where  $\mu$  is a constant mean and  $Z(\mathbf{x})$  is a Gaussian process with correlation structure. The correlation between two points  $\mathbf{x}_i$  and  $\mathbf{x}_j$  is typically modeled using a Gaussian (RBF) correlation function:

$$R(\mathbf{x}_i, \mathbf{x}_j) = \exp \left( - \sum_{k=1}^d \theta_k |x_{i,k} - x_{j,k}|^2 \right)$$

where  $\theta_k > 0$  are the correlation length parameters that control smoothness in each dimension.

### 42.1.2. Why Use Kriging in SpotOptim?

1. **Better Surrogate Models:** More accurate predictions than simple interpolation
2. **Uncertainty Quantification:** Know where the model is uncertain
3. **Mixed Variable Types:** Handle continuous, integer, and categorical variables
4. **Multiple Methods:** Choose between interpolation, regression, and reinterpolation
5. **Customizable:** Control regularization, correlation parameters, and more

## 42.2. Basic Usage

### 42.2.1. Creating a Simple Kriging Model

Let's start with the most basic usage - creating a Kriging model with default settings:

```
import numpy as np
from spotoptim import SpotOptim
from spotoptim.surrogate import Kriging

# Simple objective function
def sphere(X):
    """Sphere function:  $f(x) = \sum(x^2)$ """
    return np.sum(X**2, axis=1)

# Create Kriging surrogate with defaults
kriging = Kriging(seed=42)

# Use with SpotOptim
optimizer = SpotOptim(
    fun=sphere,
    bounds=[(-5, 5), (-5, 5)],
    surrogate=kriging, # Use Kriging instead of default GP
    max_iter=20,
    n_initial=10,
    seed=42,
    verbose=True
)

result = optimizer.optimize()
print(f"\nBest solution: {result.x}")
print(f"Best value: {result.fun:.6f}")
```

```

TensorBoard logging disabled
Initial best: f(x) = 2.420807
Iteration 1: New best f(x) = 0.001740
Iteration 2: New best f(x) = 0.000372
Iteration 3: New best f(x) = 0.000254
Iteration 4: f(x) = 0.004013
Iteration 5: f(x) = 0.001046
Iteration 6: f(x) = 0.001192
Iteration 7: f(x) = 0.001219
Iteration 8: f(x) = 0.001147
Iteration 9: New best f(x) = 0.000004
Iteration 10: New best f(x) = 0.000000

Best solution: [2.04416182e-05 5.67177247e-04]
Best value: 0.000000

```

### 42.2.2. Default vs Custom Surrogate

SpotOptim uses a Gaussian Process Regressor by default. Here's how Kriging compares:

```

import numpy as np
from spotoptim import SpotOptim
from spotoptim.surrogate import Kriging

def rosenbrock(X):
    """Rosenbrock function: (1-x)^2 + 100(y-x^2)^2"""
    x = X[:, 0]
    y = X[:, 1]
    return (1 - x)**2 + 100 * (y - x**2)**2

# With default Gaussian Process
opt_default = SpotOptim(
    fun=rosenbrock,
    bounds=[(-2, 2), (-2, 2)],
    max_iter=30,
    n_initial=15,
    seed=42,
    verbose=False
)
result_default = opt_default.optimize()

# With Kriging surrogate
kriging = Kriging(method='regression', seed=42)

```

```

opt_kriging = SpotOptim(
    fun=rosenbrock,
    bounds=[(-2, 2), (-2, 2)],
    surrogate=kriging,
    max_iter=30,
    n_initial=15,
    seed=42,
    verbose=False
)
result_kriging = opt_kriging.optimize()

print("Comparison:")
print(f"Default GP:   f(x) = {result_default.fun:.6f}")
print(f"Kriging:      f(x) = {result_kriging.fun:.6f}")

```

```

Comparison:
Default GP:   f(x) = 0.190180
Kriging:      f(x) = 0.001073

```

Both approaches work well, but Kriging offers more control over the surrogate behavior.

## 42.3. Understanding Kriging Parameters

### 42.3.1. Method: Interpolation vs Regression

The `method` parameter controls how Kriging handles data:

- **“interpolation”**: Exact interpolation, passes through all training points
- **“regression”**: Allows smoothing, better for noisy data
- **“reinterpolation”**: Hybrid approach

```

import numpy as np
from spotoptim.surrogate import Kriging
import matplotlib.pyplot as plt

# Create noisy training data
np.random.seed(42)
X_train = np.linspace(0, 2*np.pi, 10).reshape(-1, 1)
y_train = np.sin(X_train.ravel()) + 0.1 * np.random.randn(10)

# Test data for smooth predictions

```



```

X_test = np.linspace(0, 2*np.pi, 100).reshape(-1, 1)
y_true = np.sin(X_test.ravel())

# Compare methods
methods = ['interpolation', 'regression', 'reinterpolation']
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

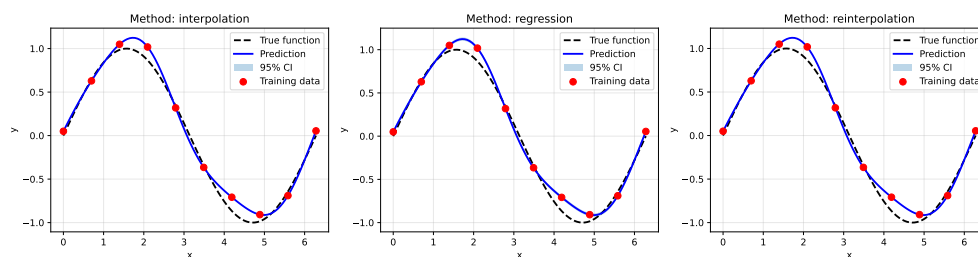
for ax, method in zip(axes, methods):
    model = Kriging(method=method, seed=42, model_fun_evals=50)
    model.fit(X_train, y_train)
    y_pred, y_std = model.predict(X_test, return_std=True)

    # Plot
    ax.plot(X_test, y_true, 'k--', label='True function', linewidth=2)
    ax.plot(X_test, y_pred, 'b-', label='Prediction', linewidth=2)
    ax.fill_between(X_test.ravel(),
                    y_pred - 2*y_std,
                    y_pred + 2*y_std,
                    alpha=0.3, label='95% CI')
    ax.scatter(X_train, y_train, c='r', s=50, zorder=5, label='Training data')
    ax.set_title(f'Method: {method}')
    ax.set_xlabel('x')
    ax.set_ylabel('y')
    ax.legend()
    ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

print("Key differences:")
print("- Interpolation: Passes exactly through training points")
print("- Regression: Smooths over noisy data (recommended)")
print("- Reinterpolation: Balances between the two")

```



Key differences:

## 42. Kriging Surrogate Models in SpotOptim

- Interpolation: Passes exactly through training points
- Regression: Smooths over noisy data (recommended)
- Reinterpolation: Balances between the two

**Recommendation:** Use `method='regression'` for most optimization problems, especially with noisy objectives.

### 42.3.2. Noise Parameter and Regularization

The noise parameter adds a small nugget effect for numerical stability:

```
import numpy as np
from spotoptim import SpotOptim
from spotoptim.surrogate import Kriging

def noisy_ackley(X):
    """Ackley function with observation noise"""
    a = 20
    b = 0.2
    c = 2 * np.pi
    n = X.shape[1]

    sum_sq = np.sum(X**2, axis=1)
    sum_cos = np.sum(np.cos(c * X), axis=1)

    base = -a * np.exp(-b * np.sqrt(sum_sq / n)) - np.exp(sum_cos / n) + a + np.e
    noise = np.random.normal(0, 0.5, size=base.shape) # Add noise
    return base + noise

# Very small noise (near interpolation)
kriging_small = Kriging(noise=1e-10, method='interpolation', seed=42)
opt_small = SpotOptim(
    fun=noisy_ackley,
    bounds=[(-5, 5), (-5, 5)],
    surrogate=kriging_small,
    max_iter=25,
    n_initial=12,
    seed=42,
    verbose=False
)
result_small = opt_small.optimize()

# Larger noise (more robust to noise)
```

```

kriging_large = Kriging(noise=0.01, method='interpolation', seed=42)
opt_large = SpotOptim(
    fun=noisy_ackley,
    bounds=[(-5, 5), (-5, 5)],
    surrogate=kriging_large,
    max_iter=25,
    n_initial=12,
    seed=42,
    verbose=False
)
result_large = opt_large.optimize()

print("Impact of noise parameter:")
print(f"Small noise (1e-10): f(x) = {result_small.fun:.6f}")
print(f"Large noise (0.01): f(x) = {result_large.fun:.6f}")
print("\nNote: For noisy functions, use 'regression' method instead,")
print("      which automatically optimizes the Lambda (nugget) parameter.")

```

```

Impact of noise parameter:
Small noise (1e-10): f(x) = 0.049827
Large noise (0.01): f(x) = 0.012636

```

Note: For noisy functions, use 'regression' method instead,  
which automatically optimizes the Lambda (nugget) parameter.

**Best Practice:** For noisy functions, use method='regression' which automatically optimizes the regularization parameter instead of fixing it.

### 42.3.3. Correlation Length Parameters (Theta)

The `theta` parameters control smoothness. SpotOptim optimizes these automatically:

- **min\_theta:** Minimum log ( ) value (default: -3.0 → = 0.001)
- **max\_theta:** Maximum log ( ) value (default: 2.0 → = 100)

Smaller → smoother function, larger → more wiggly function.

```

import numpy as np
from spotoptim.surrogate import Kriging

# Sample data
X_train = np.array([[0.0], [0.5], [1.0], [1.5], [2.0]])
y_train = np.sin(X_train.ravel())

```

## 42. Kriging Surrogate Models in SpotOptim

```
# Tight bounds (smoother)
model_smooth = Kriging(min_theta=-1.0, max_theta=0.0, seed=42, model_fun_evals=30)
model_smooth.fit(X_train, y_train)

# Wide bounds (more flexible)
model_flexible = Kriging(min_theta=-3.0, max_theta=2.0, seed=42, model_fun_evals=30)
model_flexible.fit(X_train, y_train)

print("Theta bounds and optimal values:")
print(f"Smooth model:   bounds=[-1.0, 0.0], theta={model_smooth.theta}")
print(f"Flexible model: bounds=[-3.0, 2.0], theta={model_flexible.theta}")
print("\nThe optimizer automatically finds the best theta within the bounds.")
```

Theta bounds and optimal values:

Smooth model: bounds=[-1.0, 0.0], theta=[-0.95216449]

Flexible model: bounds=[-3.0, 2.0], theta=[-0.95309049]

The optimizer automatically finds the best theta within the bounds.

**Default Values:** The defaults `min_theta=-3.0` and `max_theta=2.0` work well for most problems.

### 42.3.4. Isotropic vs Anisotropic Correlation

- **Isotropic** (`isotropic=True`): Single `theta` for all dimensions (faster, fewer parameters)
- **Anisotropic** (`isotropic=False`): Different `theta` per dimension (more flexible, default)

```
import numpy as np
from spotoptim.surrogate import Kriging

# 3D problem
np.random.seed(42)
X_train = np.random.rand(15, 3) * 10 - 5
y_train = X_train[:, 0]**2 + 0.1*X_train[:, 1]**2 + 2*X_train[:, 2]**2

# Isotropic: one theta for all dimensions
model_iso = Kriging(isotropic=True, seed=42, model_fun_evals=40)
model_iso.fit(X_train, y_train)

# Anisotropic: different theta per dimension
```

```

model_aniso = Kriging(isotropic=False, seed=42, model_fun_evals=40)
model_aniso.fit(X_train, y_train)

print("Correlation structure:")
print(f"Isotropic:   n_theta={model_iso.n_theta}, theta={model_iso.theta}")
print(f"Anisotropic: n_theta={model_aniso.n_theta}, theta={model_aniso.theta}")
print("\nAnisotropic can capture different smoothness in each dimension.")

# Test predictions
X_test = np.array([[0.0, 0.0, 0.0]])
y_iso = model_iso.predict(X_test)
y_aniso = model_aniso.predict(X_test)
print(f"\nPrediction at origin:")
print(f"Isotropic:   {y_iso[0]:.4f}")
print(f"Anisotropic: {y_aniso[0]:.4f}")

```

```

Correlation structure:
Isotropic:   n_theta=1, theta=[-3.]
Anisotropic: n_theta=3, theta=[-2.23292358 -3.          -1.97228776]

Anisotropic can capture different smoothness in each dimension.

Prediction at origin:
Isotropic:   -0.0666
Anisotropic: -1.0267

```

#### When to Use:

- Use **isotropic** for faster fitting when dimensions have similar characteristics
- Use **anisotropic** (default) when dimensions behave differently

## 42.4. Advanced Features

### 42.4.1. Mixed Variable Types

Kriging supports mixed variable types: continuous (`float`), integer (`int`), and categorical (`factor`):

```

import numpy as np
from spotoptim import SpotOptim
from spotoptim.surrogate import Kriging

```

```

def mixed_objective(X):
    """
    Optimize a system with:
    - x0: continuous learning rate (0.001 to 0.1)
    - x1: integer number of layers (1 to 5)
    - x2: categorical activation (0=ReLU, 1=Tanh, 2=Sigmoid)
    """
    X = np.atleast_2d(X)
    learning_rate = X[:, 0]
    n_layers = X[:, 1]
    activation = X[:, 2]

    # Simplified model: prefer lr~0.01, 3 layers, ReLU
    loss = (learning_rate - 0.01)**2 + (n_layers - 3)**2
    loss += np.where(activation == 0, 0.0, # ReLU is best
                    np.where(activation == 1, 0.5, # Tanh is ok
                             1.0)) # Sigmoid is worst

    return loss

# Define bounds and types
bounds = [
    (0.001, 0.1), # learning_rate (float)
    (1, 5),      # n_layers (int)
    (0, 2)       # activation (factor: 0, 1, or 2)
]

var_type = ['float', 'int', 'factor']

# Create Kriging with mixed types
kriging_mixed = Kriging(
    method='regression',
    var_type=var_type,
    seed=42,
    model_fun_evals=50
)

# Optimize
optimizer = SpotOptim(
    fun=mixed_objective,
    bounds=bounds,
    var_type=var_type,
    surrogate=kriging_mixed,
    max_iter=30,
    n_initial=15,

```

```

    seed=42,
    verbose=True
)

result = optimizer.optimize()

print(f"\nOptimal configuration:")
print(f"Learning rate: {result.x[0]:.4f}")
print(f"Num layers:      {int(result.x[1])}")
print(f"Activation:        {int(result.x[2])} (0=ReLU, 1=Tanh, 2=Sigmoid)")
print(f"Loss:              {result.fun:.6f}")

```

```

TensorBoard logging disabled
Initial best: f(x) = 0.000173
Iteration 1: New best f(x) = 0.000081
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 2: f(x) = 1.507094
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 3: f(x) = 0.000599
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 4: f(x) = 0.000587
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 5: f(x) = 5.005398
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 6: f(x) = 1.000342
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 7: f(x) = 1.501632
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 8: f(x) = 0.502100
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 9: f(x) = 2.000762
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.
Iteration 10: f(x) = 4.006134
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.

```

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Iteration 11:  $f(x) = 1.000430$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Iteration 12:  $f(x) = 1.500167$

Iteration 13: New best  $f(x) = 0.000030$

Iteration 14:  $f(x) = 0.000049$

Iteration 15: New best  $f(x) = 0.000003$

Optimal configuration:

Learning rate: 0.0082

Num layers: 3

Activation: 0 (0=ReLU, 1=Tanh, 2=Sigmoid)

Loss: 0.000003

### Key Points:

- Set `var_type` in both Kriging and SpotOptim
- Factor variables use specialized distance metrics (default: Canberra)
- Integer variables are treated as ordered but discrete

### 42.4.2. Customizing the Distance Metric for Factors

For categorical variables, you can choose different distance metrics.

```
import numpy as np
from spotoptim.surrogate import Kriging

# Categorical data: 2 factor variables
X_train = np.array([
    [0, 0], # Category A, Color Red
    [1, 0], # Category B, Color Red
    [0, 1], # Category A, Color Blue
    [1, 1], # Category B, Color Blue
    [2, 2], # Category C, Color Green
])
y_train = np.array([1.0, 1.5, 1.2, 2.0, 3.5])

# Different distance metrics
metrics = ['canberra', 'hamming']

for metric in metrics:
    model = Kriging(
        method='regression',
        var_type=['factor', 'factor'],
```



```

        metric_factorial=metric,
        seed=42,
        model_fun_evals=40
    )
    model.fit(X_train, y_train)

    # Test prediction
    X_test = np.array([[1, 1]])
    y_pred = model.predict(X_test)

    print(f"Metric: {metric:10s} | Prediction at [1,1]: {y_pred[0]:.4f}")

print("\nAvailable metrics: 'canberra' (default), 'hamming', 'jaccard', etc.")

```

```

Metric: canberra      | Prediction at [1,1]: 1.9463
Metric: hamming       | Prediction at [1,1]: 2.0000

```

```

Available metrics: 'canberra' (default), 'hamming', 'jaccard', etc.

```

### 42.4.3. Handling High-Dimensional Problems

For high-dimensional problems, Kriging can become computationally expensive. Here are strategies:

```

import numpy as np
from spotoptim import SpotOptim
from spotoptim.surrogate import Kriging

def high_dim_sphere(X):
    """10-dimensional sphere function"""
    return np.sum(X**2, axis=1)

# Strategy 1: Use isotropic correlation (fewer parameters)
kriging_iso = Kriging(
    method='regression',
    isotropic=True, # Single theta for all 10 dimensions
    seed=42,
    model_fun_evals=50 # Faster optimization
)

optimizer_iso = SpotOptim(
    fun=high_dim_sphere,
    bounds=[(-5, 5)] * 10, # 10 dimensions

```

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```
surrogate=kriging_iso,
max_iter=50,
n_initial=30,
seed=42,
verbose=False
)

result_iso = optimizer_iso.optimize()

# Strategy 2: Use max_surrogate_points to limit training set size
kriging_limited = Kriging(method='regression', seed=42)

optimizer_limited = SpotOptim(
    fun=high_dim_sphere,
    bounds=[(-5, 5)] * 10,
    surrogate=kriging_limited,
    max_iter=50,
    n_initial=30,
    max_surrogate_points=100, # Limit surrogate training set
    seed=42,
    verbose=False
)

result_limited = optimizer_limited.optimize()

print("High-dimensional optimization strategies:")
print(f"Isotropic Kriging:      f(x) = {result_iso.fun:.6f}")
print(f"Limited training set:  f(x) = {result_limited.fun:.6f}")
print("\nFor >5 dimensions, consider isotropic=True or limit training set.")
```

```
High-dimensional optimization strategies:
Isotropic Kriging:      f(x) = 0.003979
Limited training set:  f(x) = 0.044950
```

For >5 dimensions, consider isotropic=True or limit training set.

### 42.4.4. Uncertainty Quantification

Kriging provides uncertainty estimates, useful for exploration vs exploitation:

```
import numpy as np
from spotoptim.surrogate import Kriging
import matplotlib.pyplot as plt
```

```

# 1D example for visualization
np.random.seed(42)
X_train = np.array([[0.0], [1.0], [3.0], [5.0], [6.0]])
y_train = np.sin(X_train.ravel()) + 0.05 * np.random.randn(5)

# Fit Kriging model
model = Kriging(method='regression', seed=42, model_fun_evals=50)
model.fit(X_train, y_train)

# Dense test points
X_test = np.linspace(-0.5, 6.5, 200).reshape(-1, 1)
y_pred, y_std = model.predict(X_test, return_std=True)

# True function
y_true = np.sin(X_test.ravel())

# Plot
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.plot(X_test, y_true, 'k--', label='True function', linewidth=2)
plt.plot(X_test, y_pred, 'b-', label='Kriging prediction', linewidth=2)
plt.fill_between(X_test.ravel(),
                 y_pred - 2*y_std,
                 y_pred + 2*y_std,
                 alpha=0.3, color='blue', label='95% confidence')
plt.scatter(X_train, y_train, c='red', s=100, zorder=5,
            edgecolors='black', label='Training points')
plt.xlabel('x', fontsize=12)
plt.ylabel('y', fontsize=12)
plt.title('Kriging Predictions with Uncertainty', fontsize=14)
plt.legend()
plt.grid(True, alpha=0.3)

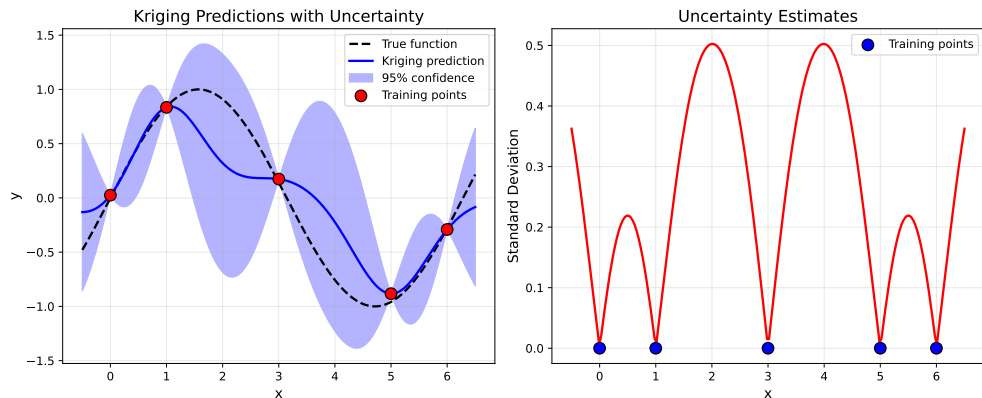
plt.subplot(1, 2, 2)
plt.plot(X_test, y_std, 'r-', linewidth=2)
plt.scatter(X_train, np.zeros_like(X_train), c='blue', s=100,
            zorder=5, edgecolors='black', label='Training points')
plt.xlabel('x', fontsize=12)
plt.ylabel('Standard Deviation', fontsize=12)
plt.title('Uncertainty Estimates', fontsize=14)
plt.legend()
plt.grid(True, alpha=0.3)

```

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```
plt.tight_layout()
plt.show()

print("Key observations:")
print("- Uncertainty is low near training points")
print("- Uncertainty is high far from training data")
print("- This guides acquisition functions (e.g., EI exploits low uncertainty)")
```



Key observations:

- Uncertainty is low near training points
- Uncertainty is high far from training data
- This guides acquisition functions (e.g., EI exploits low uncertainty)

## 42.5. Practical Examples

### 42.5.1. Example 1: Optimizing the Rastrigin Function

The Rastrigin function is highly multimodal - a challenging test case:

```
import numpy as np
from spotoptim import SpotOptim
from spotoptim.surrogate import Kriging

def rastrigin(X):
    """
    Rastrigin function: highly multimodal
    Global minimum: f(0,...,0) = 0
    """
```

```

A = 10
n = X.shape[1]
return A * n + np.sum(X**2 - A * np.cos(2 * np.pi * X), axis=1)

# Configure Kriging for multimodal function
kriging = Kriging(
    method='regression',      # Smooth over local variations
    min_theta=-3.0,          # Allow flexible correlation
    max_theta=2.0,
    seed=42,
    model_fun_evals=100      # More budget for better surrogate
)

optimizer = SpotOptim(
    fun=rastrigin,
    bounds=[(-5.12, 5.12), (-5.12, 5.12)],
    surrogate=kriging,
    acquisition='ei',         # Expected Improvement for exploration
    max_iter=60,
    n_initial=30,
    seed=42,
    verbose=True
)

result = optimizer.optimize()

print(f"\nRastrigin optimization:")
print(f"Best solution: {result.x}")
print(f"Best value:      {result.fun:.6f} (global optimum: 0.0)")
print(f"Distance to optimum: {np.linalg.norm(result.x):.6f}")

```

```

TensorBoard logging disabled
Initial best: f(x) = 6.442225
Iteration 1: f(x) = 28.951536
Iteration 2: f(x) = 22.875749
Iteration 3: f(x) = 21.815864
Iteration 4: f(x) = 7.077873
Iteration 5: f(x) = 13.368250
Iteration 6: f(x) = 16.577976
Iteration 7: f(x) = 25.780045
Iteration 8: f(x) = 10.034344
Iteration 9: f(x) = 19.560557
Iteration 10: f(x) = 12.381235
Iteration 11: f(x) = 20.126829

```

## 42. Kriging Surrogate Models in SpotOptim

```
Iteration 12: f(x) = 27.151349
Iteration 13: New best f(x) = 2.290927
Iteration 14: f(x) = 2.761961
Iteration 15: f(x) = 12.468354
Iteration 16: New best f(x) = 1.208377
Iteration 17: f(x) = 12.566211
Iteration 18: f(x) = 14.965620
Iteration 19: f(x) = 14.295554
Iteration 20: f(x) = 28.107905
Iteration 21: f(x) = 22.687907
Iteration 22: f(x) = 43.729592
Iteration 23: f(x) = 41.636148
Iteration 24: New best f(x) = 1.034573
Iteration 25: f(x) = 3.228327
Iteration 26: f(x) = 7.934136
Iteration 27: f(x) = 21.994039
Iteration 28: f(x) = 27.510661
Iteration 29: f(x) = 9.365219
Iteration 30: f(x) = 1.248535
```

Rastrigin optimization:

```
Best solution: [-0.00987679  1.00506675]
Best value:    1.034573 (global optimum: 0.0)
Distance to optimum: 1.005115
```

### 42.5.2. Example 2: Robust Optimization with Noise

When optimizing noisy functions, Kriging's regression mode helps:

```
import numpy as np
from spotoptim import SpotOptim
from spotoptim.surrogate import Kriging

def noisy_beale(X):
    """
    Beale function with Gaussian noise
    Global minimum: f(3, 0.5) = 0
    """
    x = X[:, 0]
    y = X[:, 1]

    term1 = (1.5 - x + x * y)**2
    term2 = (2.25 - x + x * y**2)**2
    term3 = (2.625 - x + x * y**3)**2
```

```

    base = term1 + term2 + term3
    noise = np.random.normal(0, 0.5, size=base.shape)

    return base + noise

# Use regression method for noisy objectives
kriging_robust = Kriging(
    method='regression',          # Automatically optimizes Lambda (nugget)
    min_Lambda=-9.0,             # Allow small regularization
    max_Lambda=-2.0,             # But not too large
    seed=42,
    model_fun_evals=80
)

# Use repeated evaluations for noise handling
optimizer = SpotOptim(
    fun=noisy_beale,
    bounds=[(-4.5, 4.5), (-4.5, 4.5)],
    surrogate=kriging_robust,
    max_iter=50,
    n_initial=20,
    repeats_initial=3,           # Evaluate each initial point 3 times
    repeats_surrogate=2,        # Evaluate each new point 2 times
    seed=42,
    verbose=True
)

result = optimizer.optimize()

print(f"\nNoisy Beale optimization:")
print(f"Best solution: {result.x}")
print(f"Best mean value: {optimizer.min_mean_y:.6f}")
print(f"Variance at best: {optimizer.min_var_y:.6f}")
print(f"True optimum: [3.0, 0.5]")
print(f"Distance: {np.linalg.norm(result.x - np.array([3.0, 0.5])):.4f}")

```

TensorBoard logging disabled

Initial best:  $f(x) = 9.379814$ , mean best:  $f(x) = 9.729582$

Noisy Beale optimization:

Best solution: [ 0.34789079 -0.3014163 ]

Best mean value: 9.729582

Variance at best: 0.214586

## 42. Kriging Surrogate Models in SpotOptim

True optimum: [3.0, 0.5]  
Distance: 2.7706

### 42.5.3. Example 3: Real-World Machine Learning Hyperparameter Tuning

Optimize hyperparameters for a neural network:

```
import numpy as np
from spotoptim import SpotOptim
from spotoptim.surrogate import Kriging

def ml_objective(X):
    """
    Simulate training a neural network
    Variables:
    - x0: learning_rate (log scale: 1e-4 to 1e-1)
    - x1: num_layers (integer: 1 to 5)
    - x2: hidden_size (integer: 16, 32, 64, 128, 256)
    - x3: dropout_rate (float: 0.0 to 0.5)

    Returns validation error
    """
    X = np.atleast_2d(X)
    lr = X[:, 0]
    n_layers = X[:, 1]
    hidden_size = X[:, 2]
    dropout = X[:, 3]

    # Simulate validation error (lower is better)
    # Optimal around: lr=0.001, 3 layers, 64 hidden, 0.2 dropout
    error = (np.log10(lr) + 3)**2 # Prefer lr ~ 0.001
    error += (n_layers - 3)**2 # Prefer 3 layers
    error += ((hidden_size - 64) / 32)**2 # Prefer 64 hidden units
    error += (dropout - 0.2)**2 * 10 # Prefer 0.2 dropout

    # Add some noise to simulate training variance
    error += np.random.normal(0, 0.1, size=error.shape)

    return error

# Setup optimization
bounds = [
```



```

    (1e-4, 1e-1), # learning_rate
    (1, 5),       # num_layers
    (16, 256),    # hidden_size
    (0.0, 0.5)    # dropout_rate
]

var_type = ['float', 'int', 'int', 'float']

# Use Kriging with mixed types
kriging_ml = Kriging(
    method='regression',
    var_type=var_type,
    seed=42,
    model_fun_evals=60
)

optimizer = SpotOptim(
    fun=ml_objective,
    bounds=bounds,
    var_type=var_type,
    var_trans=['log10', None, None, None], # Log scale for learning rate
    surrogate=kriging_ml,
    max_iter=40,
    n_initial=20,
    seed=42,
    verbose=True
)

result = optimizer.optimize()

print(f"\nOptimal hyperparameters:")
print(f"Learning rate: {result.x[0]:.6f}")
print(f"Num layers:    {int(result.x[1])}")
print(f"Hidden size:    {int(result.x[2])}")
print(f"Dropout rate:   {result.x[3]:.3f}")
print(f"Validation error: {result.fun:.6f}")

```

```

TensorBoard logging disabled
Initial best: f(x) = 1.133527
Iteration 1: f(x) = 5.122533
Iteration 2: f(x) = 5.959913
Iteration 3: f(x) = 1.505793
Iteration 4: New best f(x) = 0.942988
Iteration 5: New best f(x) = 0.280219

```

## 42. Kriging Surrogate Models in SpotOptim

```
Iteration 6: New best f(x) = 0.260858
Iteration 7: f(x) = 1.115300
Iteration 8: f(x) = 0.653129
Iteration 9: New best f(x) = 0.176351
Iteration 10: New best f(x) = 0.084523
Iteration 11: New best f(x) = -0.067705
Iteration 12: f(x) = 0.167623
Iteration 13: f(x) = 0.048797
Iteration 14: f(x) = 0.053888
Iteration 15: f(x) = 0.015810
Iteration 16: New best f(x) = -0.108042
Iteration 17: f(x) = -0.007253
Iteration 18: f(x) = -0.003508
Iteration 19: f(x) = -0.052430
Iteration 20: f(x) = 0.006251
```

```
Optimal hyperparameters:
Learning rate: 0.000987
Num layers:    3
Hidden size:   68
Dropout rate:  0.158
Validation error: -0.108042
```

### 42.5.4. Example 4: Comparing Kriging Methods

Let's compare all three Kriging methods on the same problem:

```
import numpy as np
from spotoptim import SpotOptim
from spotoptim.surrogate import Kriging

def levy(X):
    """
    Levy function N. 13
    Global minimum: f(1, 1) = 0
    """
    x = X[:, 0]
    y = X[:, 1]

    w1 = 1 + (x - 1) / 4
    w2 = 1 + (y - 1) / 4

    term1 = np.sin(np.pi * w1)**2
    term2 = (w1 - 1)**2 * (1 + 10 * np.sin(np.pi * w1 + 1)**2)
```

```

term3 = (w2 - 1)**2 * (1 + np.sin(2 * np.pi * w2)**2)

return term1 + term2 + term3

methods = ['interpolation', 'regression', 'reinterpolation']
results_methods = []

for method in methods:
    kriging = Kriging(
        method=method,
        seed=42,
        model_fun_evals=60
    )

    optimizer = SpotOptim(
        fun=levy,
        bounds=[(-10, 10), (-10, 10)],
        surrogate=kriging,
        max_iter=40,
        n_initial=20,
        seed=42,
        verbose=False
    )

    result = optimizer.optimize()
    results_methods.append((method, result.fun, result.x))

    print(f"{method:15s} | f(x) = {result.fun:8.6f} | x = {result.x}")

print(f"\nGlobal optimum: f(1, 1) = 0.0")
print("\nAll methods find good solutions, but 'regression' is most robust.")

```

```

interpolation    | f(x) = 0.001657 | x = [0.98535016 0.85317084]
regression       | f(x) = 0.027480 | x = [0.8385841 1.068918 ]
reinterpolation | f(x) = 0.027480 | x = [0.8385841 1.068918 ]

```

Global optimum: f(1, 1) = 0.0

All methods find good solutions, but 'regression' is most robust.

#### 42.5.5. Example 5: Sensitivity to Theta Bounds

Theta bounds control the range of smoothness. Let's see their impact:

## 42. Kriging Surrogate Models in SpotOptim

```
import numpy as np
from spotoptim import SpotOptim
from spotoptim.surrogate import Kriging

def griewank(X):
    """Griewank function"""
    sum_sq = np.sum(X**2 / 4000, axis=1)
    prod_cos = np.prod(np.cos(X / np.sqrt(np.arange(1, X.shape[1] + 1))), axis=1)
    return sum_sq - prod_cos + 1

# Different theta bounds
theta_configs = [
    (-2.0, 1.0, "Tight (smoother)"),
    (-3.0, 2.0, "Default"),
    (-4.0, 3.0, "Wide (more flexible)")
]

for min_theta, max_theta, label in theta_configs:
    kriging = Kriging(
        method='regression',
        min_theta=min_theta,
        max_theta=max_theta,
        seed=42,
        model_fun_evals=50
    )

    optimizer = SpotOptim(
        fun=griewank,
        bounds=[(-600, 600), (-600, 600)],
        surrogate=kriging,
        max_iter=35,
        n_initial=18,
        seed=42,
        verbose=False
    )

    result = optimizer.optimize()
    print(f"{label:20s} [{min_theta:5.1f}, {max_theta:4.1f}] | f(x) = {result.fun:8.6f}")

print("\nDefault bounds work well for most problems.")
```

```
Tight (smoother)      [ -2.0,  1.0] | f(x) = 8.578706
Default               [ -3.0,  2.0] | f(x) = 13.108634
Wide (more flexible) [ -4.0,  3.0] | f(x) = 2.652215
```

Default bounds work well for most problems.

## 42.6. Comparing Surrogates

### 42.6.1. Kriging vs Gaussian Process vs Random Forest

Let's compare different surrogate models:

```
import numpy as np
from spotoptim import SpotOptim
from spotoptim.surrogate import Kriging, SimpleKriging
from sklearn.gaussian_process import GaussianProcessRegressor
from sklearn.gaussian_process.kernels import Matern, ConstantKernel
from sklearn.ensemble import RandomForestRegressor

def schwefel(X):
    """Schwefel function"""
    return 418.9829 * X.shape[1] - np.sum(X * np.sin(np.sqrt(np.abs(X))), axis=1)

bounds = [(-500, 500), (-500, 500)]

# 1. Kriging (full-featured)
kriging = Kriging(method='regression', seed=42, model_fun_evals=60)
opt_kriging = SpotOptim(fun=schwefel, bounds=bounds, surrogate=kriging,
                        max_iter=35, n_initial=18, seed=42, verbose=False)
result_kriging = opt_kriging.optimize()

# 2. SimpleKriging (lightweight)
simple_kriging = SimpleKriging(noise=1e-10, seed=42)
opt_simple = SpotOptim(fun=schwefel, bounds=bounds, surrogate=simple_kriging,
                       max_iter=35, n_initial=18, seed=42, verbose=False)
result_simple = opt_simple.optimize()

# 3. Gaussian Process (sklearn default)
kernel = ConstantKernel(1.0) * Matern(length_scale=1.0, nu=2.5)
gp = GaussianProcessRegressor(kernel=kernel, n_restarts_optimizer=10,
                              normalize_y=True, random_state=42)
opt_gp = SpotOptim(fun=schwefel, bounds=bounds, surrogate=gp,
                   max_iter=35, n_initial=18, seed=42, verbose=False)
result_gp = opt_gp.optimize()

# 4. Random Forest
```

## 42. Kriging Surrogate Models in SpotOptim

```
rf = RandomForestRegressor(n_estimators=100, random_state=42)
opt_rf = SpotOptim(fun=schwefel, bounds=bounds, surrogate=rf,
                  max_iter=35, n_initial=18, seed=42, verbose=False)
result_rf = opt_rf.optimize()

print("Surrogate Model Comparison:")
print(f"Kriging (full):      f(x) = {result_kriging.fun:8.2f}")
print(f"SimpleKriging:      f(x) = {result_simple.fun:8.2f}")
print(f"Gaussian Process:   f(x) = {result_gp.fun:8.2f}")
print(f"Random Forest:      f(x) = {result_rf.fun:8.2f}")
print(f"\nGlobal optimum: f(420.9687, 420.9687) = 0.0")
print("\nKriging offers:")
print("- Mixed variable types (continuous, integer, categorical)")
print("- Multiple methods (interpolation, regression, reinterpolation)")
print("- Explicit control over regularization and correlation")
```

Surrogate Model Comparison:

|                   |        |        |
|-------------------|--------|--------|
| Kriging (full):   | f(x) = | 118.47 |
| SimpleKriging:    | f(x) = | 118.44 |
| Gaussian Process: | f(x) = | 118.44 |
| Random Forest:    | f(x) = | 184.74 |

Global optimum: f(420.9687, 420.9687) = 0.0

Kriging offers:

- Mixed variable types (continuous, integer, categorical)
- Multiple methods (interpolation, regression, reinterpolation)
- Explicit control over regularization and correlation

## 42.7. Best Practices

### 42.7.1. 1. Choosing the Right Method

```
# For smooth, deterministic functions
kriging = Kriging(method='interpolation', noise=1e-10, seed=42)

# For general optimization (RECOMMENDED)
kriging = Kriging(method='regression', seed=42)

# For noisy functions with repeated evaluations
```

```
kriging = Kriging(method='regression', seed=42)
# Use with repeats_initial and repeats_surrogate in SpotOptim
```

### 42.7.2. 2. Setting Model Complexity

```
# For low-dimensional problems (<5D)
kriging = Kriging(
    method='regression',
    isotropic=False,          # Anisotropic (default)
    model_fun_evals=100,      # More budget for better fit
    seed=42
)

# For high-dimensional problems (>5D)
kriging = Kriging(
    method='regression',
    isotropic=True,           # Fewer parameters
    model_fun_evals=50,       # Faster fitting
    seed=42
)
```

### 42.7.3. 3. Handling Different Variable Types

```
# Mixed types example
bounds = [
    (0.0, 10.0),    # continuous
    (1, 10),        # integer
    (0, 3)          # categorical (4 categories)
]

var_type = ['float', 'int', 'factor']

kriging = Kriging(
    method='regression',
    var_type=var_type,
    metric_factorial='canberra', # For factor variables
    seed=42
)

optimizer = SpotOptim(
```

## 42. Kriging Surrogate Models in SpotOptim

```
    fun=objective,  
    bounds=bounds,  
    var_type=var_type,  
    surrogate=kriging,  
    seed=42  
)
```

### 42.7.4. 4. Reproducibility

Always set the seed for reproducible results:

```
# Both Kriging and SpotOptim should have seeds  
kriging = Kriging(method='regression', seed=42)  
  
optimizer = SpotOptim(  
    fun=objective,  
    bounds=bounds,  
    surrogate=kriging,  
    seed=42 # Same seed or different seed  
)
```

### 42.7.5. 5. Monitoring Surrogate Quality

Check the negative log-likelihood after fitting:

```
import numpy as np  
from spotoptim.surrogate import Kriging  
  
# Sample data  
X = np.random.rand(20, 2) * 10 - 5  
y = np.sum(X**2, axis=1)  
  
model = Kriging(method='regression', seed=42)  
model.fit(X, y)  
  
print(f"Negative log-likelihood: {model.negLnLike:.4f}")  
print(f"Optimized theta: {model.theta_}")  
print(f"Optimized Lambda: {model.Lambda_:.4f}")  
print(f"Condition number of Psi: {model.cnd_Psi:.2e}")  
  
print("\nLower negLnLike = better fit")  
print("Check condition number for numerical stability")
```



```

Negative log-likelihood: -15.3998
Optimized theta: [-3. -3.]
Optimized Lambda: -8.9957
Condition number of Psi: 4.03e+13

```

Lower negLnLike = better fit  
 Check condition number for numerical stability

## 42.8. Common Issues and Solutions

### 42.8.1. Issue 1: Slow Fitting for Large Datasets

**Problem:** Kriging becomes slow with many training points.

**Solution:** Limit surrogate training set size:

```

# Use max_surrogate_points in SpotOptim
optimizer = SpotOptim(
    fun=objective,
    bounds=bounds,
    surrogate=kriging,
    max_surrogate_points=200, # Limit to 200 points
    selection_method='distant', # or 'best'
    seed=42
)

```

### 42.8.2. Issue 2: Poor Predictions for Categorical Variables

**Problem:** Kriging doesn't handle factors well.

**Solution:** Try different distance metrics:

```

# Try different metrics
for metric in ['canberra', 'hamming', 'jaccard']:
    kriging = Kriging(
        method='regression',
        var_type=['float', 'factor'],
        metric_factorial=metric,
        seed=42
    )
# Test performance

```

### 42.8.3. Issue 3: Numerical Instability

**Problem:** Correlation matrix is nearly singular.

**Solution:** Increase regularization:

```
# For interpolation method
kriging = Kriging(
    method='interpolation',
    noise=1e-6, # Increase from default 1e-8
    seed=42
)

# Or use regression method (recommended)
kriging = Kriging(
    method='regression',
    min_Lambda=-6.0, # Don't go too small
    seed=42
)
```

### 42.8.4. Issue 4: Overfitting to Noisy Data

**Problem:** Kriging fits noise instead of underlying function.

**Solution:** Use regression method with reasonable Lambda bounds:

```
kriging = Kriging(
    method='regression',
    min_Lambda=-5.0, # Not too small
    max_Lambda=-1.0, # Not too large
    seed=42
)
```

## 42.9. Summary

### 42.9.1. Key Takeaways

1. Use `method='regression'` for most optimization problems
2. Set `seed` for reproducibility
3. Use `var_type` for mixed variable types
4. Default parameters work well for most cases
5. Use `isotropic=True` for high-dimensional problems (>5D)

### 42.9.2. Quick Reference

```

from spotoptim import SpotOptim
from spotoptim.surrogate import Kriging

# Recommended configuration for general use
kriging = Kriging(
    method='regression',      # Smooths over noise
    seed=42                  # Reproducibility
)

optimizer = SpotOptim(
    fun=objective,
    bounds=bounds,
    surrogate=kriging,
    max_iter=50,
    n_initial=20,
    seed=42,
    verbose=True
)

result = optimizer.optimize()

```

### 42.9.3. Parameter Cheat Sheet

| Parameter        | Default      | When to Change                          |
|------------------|--------------|-----------------------------------------|
| method           | 'regression' | Use 'interpolation' for exact fit       |
| noise            | sqrt(eps)    | Rarely; use method='regression' instead |
| min_theta        | -3.0         | Adjust if default range doesn't work    |
| max_theta        | 2.0          | Adjust if default range doesn't work    |
| isotropic        | False        | Set True for >5 dimensions              |
| var_type         | ['num']      | Set for mixed variable types            |
| metric_factorial | 'canberra'   | Try 'hamming' for factors               |
| model_fun_evals  | 100          | Reduce for faster fitting               |
| seed             | 124          | Always set for reproducibility          |

### 42.9.4. Further Reading

- Forrester, A., Sobester, A., & Keane, A. (2008). *Engineering Design via Surrogate Modelling*. Wiley.

#### 42. Kriging Surrogate Models in SpotOptim

- Jones, D. R., Schonlau, M., & Welch, W. J. (1998). Efficient global optimization of expensive black-box functions. *Journal of Global optimization*, 13(4), 455-492.
- Rasmussen, C. E., & Williams, C. K. I. (2006). *Gaussian Processes for Machine Learning*. MIT Press.

For more examples and documentation, visit the SpotOptim GitHub repository.

## **43. Kriging in SpotOptim: The Forrester Implementation**



## 44. Introduction

The **Kriging** surrogate model in `spotoptim` is an implementation of the methods described in the classic text “**Engineering Design via Surrogate Modelling**” by Forrester, Sóbester, and Keane (2008). This tutorial explains the mathematical foundations of the implementation and demonstrates how to use the different fitting methods available in the library. The implementation has been validated against the original Matlab code provided with the book (`likelihood.m`, `pred.m`, etc.).





## 45. 1. The Core Kriging Model (Section 2.4)

Kriging, or Gaussian Process Regression, treats the response  $y(\mathbf{x})$  as a realization of a stochastic process:

$$Y(\mathbf{x}) = \mu + Z(\mathbf{x})$$

where  $\mu$  is the mean (often modeled as a constant in Ordinary Kriging) and  $Z(\mathbf{x})$  is a Gaussian process with zero mean and covariance:

$$\text{Cov}(Z(\mathbf{x}^{(i)}), Z(\mathbf{x}^{(l)})) = \sigma^2 \Psi(\mathbf{x}^{(i)}, \mathbf{x}^{(l)})$$

### 45.1. Correlation Function

We use the Gaussian correlation function (Eq 2.4 in the book), which is infinitely differentiable and assumes a smooth underlying response:

$$\Psi(\mathbf{x}^{(i)}, \mathbf{x}^{(l)}) = \exp \left( - \sum_{j=1}^k \theta_j |x_j^{(i)} - x_j^{(l)}|^{p_j} \right)$$

In `spotoptim`, we typically set  $p_j = 2$  (Gaussian) and optimize  $\theta_j$  (length-scales).

### 45.2. Training: Concentrated Log-Likelihood

To fit the model, we maximize the **Concentrated Log-Likelihood** (Eq 2.30/2.33) with respect to the hyperparameters  $\theta$ :

$$\ln(L) \approx -\frac{n}{2} \ln(\hat{\sigma}^2) - \frac{1}{2} \ln |\Psi|$$

where  $\hat{\sigma}^2$  is the estimated variance (Eq 2.28). This matches the implementation in the book's `likelihood.m`.

### 45.3. Prediction

The prediction at a new point  $\mathbf{x}$  is given by (Eq 2.15):

$$\hat{y}(\mathbf{x}) = \hat{\mu} + \psi^T \Psi^{-1}(\mathbf{y} - \mathbf{1}\hat{\mu})$$

where  $\psi$  is the correlation vector between the new point and the training points. This matches `pred.m`.

## 46. 2. Handling Noisy Data

Real-world data is often noisy. Forrester et al. (Section 6) describe three approaches to handle this, all available in `spotoptim` via the `method` argument.

### 46.1. A. Interpolation (`method="interpolation"`)

This is the standard Kriging formulation. It assumes **no noise** in the data.

- The model passes exactly through the training points.
- A very small “nugget” (noise) is added only for numerical stability of the matrix inversion.

### 46.2. B. Regression (`method="regression"`)

Described in Section 6.2 (“Regression Kriging”), this method assumes the observations contain noise  $\epsilon$ :

$$y = \mu + Z(\mathbf{x}) + \epsilon$$

We introduce a regularization parameter  $\lambda$  (Lambda) to the diagonal of the correlation matrix:

$$\mathbf{C} = \Psi + \Psi^T + (1 + \lambda)\mathbf{I}$$

Both  $\theta$  and  $\lambda$  are optimized simultaneously. The resulting surrogate:

- **Does not pass through the training points** (it smooths them).
- Is suitable for noisy objective functions.

### 46.3. C. Re-interpolation (method="reinterpolation")

Described in Section 6.3, this is a hybrid approach.

1. We fit the hyperparameters ( $\theta$  and  $\lambda$ ) using the **Regression** method to learn the noise level.
2. For prediction, we use the “noise-free” correlation matrix (removing  $\lambda$ ) to predict the underlying smooth trend.

This “re-interpolates” the regression model, providing a predictor that typically has better error properties for design optimization than pure regression.

## 47. 3. Examples

We will verify these methods with a simple 1D example.

### 47.1. Setup

```
import numpy as np
import matplotlib.pyplot as plt
from spotoptim.surrogate import Kriging

# Define a function with noise
def true_function(x):
    return (x * 6 - 2)**2 * np.sin(x * 12 - 4)

np.random.seed(42)
X = np.linspace(0, 1, 11).reshape(-1, 1)
noise = np.random.normal(0, 1.0, X.shape)
y = true_function(X).flatten() + noise.flatten() * 3.0 # Add significant noise

X_plot = np.linspace(0, 1, 100).reshape(-1, 1)
y_true = true_function(X_plot)
```

### 47.2. Comparisons

We will fit three models and compare them.

```
# 1. Interpolation (Forces fit through noisy points)
model_interp = Kriging(method="interpolation", seed=42)
model_interp.fit(X, y)
y_interp, s_interp = model_interp.predict(X_plot, return_std=True)

# 2. Regression (Smooths the data by learning Lambda)
model_reg = Kriging(method="regression", seed=42, minLambda=-1, maxLambda=10)
model_reg.fit(X, y)
```

### 47. 3. Examples

```
y_reg, s_reg = model_reg.predict(X_plot, return_std=True)

# 3. Re-interpolation (Smooth trend based on regression parameters)
model_reinterp = Kriging(method="reinterpolation", seed=42)
model_reinterp.fit(X, y)
y_reinterp, s_reinterp = model_reinterp.predict(X_plot, return_std=True)
```

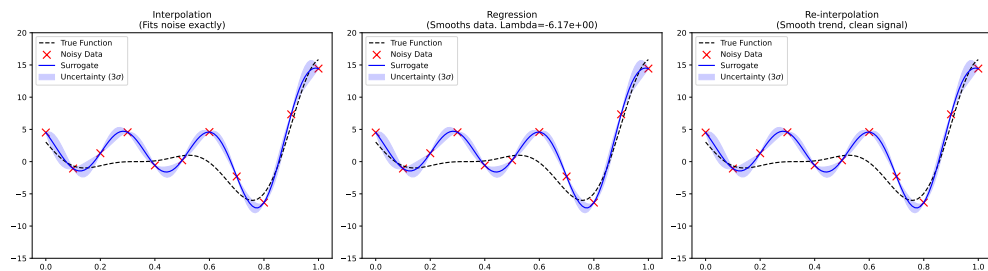
## 47.3. Visualization

```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

def plot_model(ax, title, y_pred, std):
    ax.plot(X_plot, y_true, 'k--', label="True Function")
    ax.scatter(X, y, c='r', marker='x', s=100, label="Noisy Data")
    ax.plot(X_plot, y_pred, 'b-', label="Surrogate")
    ax.fill_between(X_plot.flatten(),
                   y_pred - 3*std,
                   y_pred + 3*std,
                   color='b', alpha=0.2, label="Uncertainty (3 $\sigma$ )")
    ax.set_title(title)
    ax.legend()
    ax.set_ylim(-15, 20)

plot_model(axes[0], "Interpolation\n(Fits noise exactly)", y_interp, s_interp)
plot_model(axes[1], f"Regression\n(Smooths data. Lambda={model_reg.Lambda_:.2e})", y_reg, s_reg)
plot_model(axes[2], "Re-interpolation\n(Smooth trend, clean signal)", y_reinterp, s_reinterp)

plt.tight_layout()
plt.show()
```



### 47.3.1. Interpretation

1. **Interpolation:** Wiggles aggressively to hit every noisy point. This is “overfitting” the noise.
2. **Regression:** Produces a much smoother curve that does not hit the points. It has correctly identified that the data is noisy (optimized  $\lambda > 0$ ).
3. **Re-interpolation:** Similar to regression in shape, but mathematically grounded in predicting the *process* rather than the *data*.





## 48. 4. Infill Criteria

The book also describes “Infill Criteria” (Section 3) for deciding where to sample next. This is implemented in `SpotOptim` itself (not the surrogate class). When you use `SpotOptim(..., acquisition="ei")`, you are using the **Expected Improvement** criterion described in Section 3.2.1 of the book.

```
from spotoptim import SpotOptim

# Use Kriging with EI
optimizer = SpotOptim(
    fun=lambda x: np.sum(x**2),
    bounds=[(-5, 5)],
    surrogate=model_reinterp, # Pass your customized Kriging model
    acquisition="ei",         # Expected Improvement
    max_iter=20
)
```



## 49. References

1. Forrester, A., Sobester, A., & Keane, A. (2008). *Engineering Design via Surrogate Modelling: A Practical Guide*. Wiley.
2. **SpotOptim Source Code:** `spotoptim/surrogate/kriging.py` (Validated against book's Matlab code).



**Part V.**

**Multiobjective Optimization**



## 50. Multi-Objective Optimization Support in SpotOptim

```
import copy
import numpy as np
import pandas as pd
import numpy as np
import warnings

from sklearn.ensemble import RandomForestRegressor
from sklearn.preprocessing import MinMaxScaler
from sklearn.gaussian_process import GaussianProcessRegressor
from sklearn.kernel_approximation import Nystroem
from sklearn.gaussian_process.kernels import RBF, WhiteKernel
from sklearn.pipeline import Pipeline
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score

import matplotlib.pyplot as plt
import seaborn as sns

from scipy.stats.qmc import LatinHypercube
from scipy.optimize import dual_annealing

from spotdesirability.utils.desirability import DMin, DMax, DOverall

from spotoptim.sampling.mm import mmphi_intensive, mmphi_intensive_update, mm_improvement_contour
from spotoptim.inspection.predictions import plot_actual_vs_predicted
from spotoptim.inspection import (generate_mdi, generate_imp, plot_importances, plot_feature_importances)

from spotoptim.utils.boundaries import get_boundaries, map_to_original_scale
from spotoptim.mo.pareto import is_pareto_efficient, plot_mo
from spotoptim.eda import plot_ip_boxplots, plot_ip_histograms
from spotoptim.utils import get_combinations
from spotoptim.sampling.lhs import rlh
```

```
from spotoptim.sampling.mm import (bestlh, plot_mmphi_vs_n_lhs, mmphi, mmphi_intensive)
from spotoptim.mo import mo_mm_desirability_function, mo_mm_desirability_optimizer
from spotoptim.mo.mo_mm import mo_xy_desirability_plot
from spotoptim.function.mo import mo_conv2_max
from spotoptim.mo.pareto import (mo_xy_surface, mo_xy_contour, mo_pareto_optx_plot)
from spotoptim.utils.eval import mo_cv_models, mo_eval_models
warnings.filterwarnings("ignore")
```

```
n_estimators=100
max_depth = None
random_state=42
```

## 50.1. Overview

SpotOptim supports multi-objective optimization functions with automatic detection and flexible scalarization strategies. This implementation follows the same approach as the Spot class from spotPython.

## 50.2. What Was Implemented

### 50.2.1. 1. Core Functionality

#### Parameter:

- `fun_mo2so` (callable, optional): Function to convert multi-objective values to single-objective
  - Takes array of shape `(n_samples, n_objectives)`
  - Returns array of shape `(n_samples,)`
  - If `None`, uses first objective (default behavior)

#### Attribute:

- `y_mo` (ndarray or `None`): Stores all multi-objective function values
  - Shape: `(n_samples, n_objectives)` for multi-objective problems
  - `None` for single-objective problems

#### Methods:

- `_get_shape(y)`: Get shape of objective function output
- `_store_mo(y_mo)`: Store multi-objective values with automatic appending



- `_mo2so(y_mo)`: Convert multi-objective to single-objective values

The method `_evaluate_function(X)` automatically detects multi-objective functions. It calls `_mo2so()` to convert multi-objective to single-objective. It also stores the original multi-objective values in `y_mo`. And it returns single-objective values for optimization.

## 50.3. Usage Examples

### 50.3.1. Example 1: Default Behavior (Use First Objective)

```
import numpy as np
from spotoptim import SpotOptim

def bi_objective(X):
    """Two conflicting objectives."""
    obj1 = np.sum(X**2, axis=1)      # Minimize at origin
    obj2 = np.sum((X - 2)**2, axis=1) # Minimize at (2, 2)
    return np.column_stack([obj1, obj2])

optimizer = SpotOptim(
    fun=bi_objective,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=30,
    n_initial=15,
    seed=42
)

result = optimizer.optimize()

print(f"Best x: {result.x}")          # Near [0, 0]
print(f"Best f(x): {result.fun}")    # Minimizes obj1
print(f"M0 values stored: {optimizer.y_mo.shape}") # (30, 2)
```

```
Best x: [3.31436760e-04 4.18312302e-05]
Best f(x): 1.1160017787260647e-07
M0 values stored: (30, 2)
```

### 50.3.2. Example 2: Weighted Sum Scalarization

```
def weighted_sum(y_mo):
    """Equal weighting of objectives."""
    return 0.5 * y_mo[:, 0] + 0.5 * y_mo[:, 1]

optimizer = SpotOptim(
    fun=bi_objective,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=30,
    n_initial=15,
    fun_mo2so=weighted_sum, # Custom conversion
    seed=42
)

result = optimizer.optimize()
print(f"Compromise solution: {result.x}") # Near [1, 1]
```

Compromise solution: [1.00110134 1.0000297 ]

### 50.3.3. Example 3: Min-Max Scalarization

```
def min_max(y_mo):
    """Minimize the maximum objective."""
    return np.max(y_mo, axis=1)

optimizer = SpotOptim(
    fun=bi_objective,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=30,
    n_initial=15,
    fun_mo2so=min_max,
    seed=42
)

result = optimizer.optimize()
# Finds solution with balanced objective values
```

#### 50.3.4. Example 4: Three or More Objectives

```
def three_objectives(X):
    """Three different norms."""
    obj1 = np.sum(X**2, axis=1)          # L2 norm
    obj2 = np.sum(np.abs(X), axis=1)     # L1 norm
    obj3 = np.max(np.abs(X), axis=1)     # L-infinity norm
    return np.column_stack([obj1, obj2, obj3])

def custom_scalarization(y_mo):
    """Weighted combination."""
    return 0.4 * y_mo[:, 0] + 0.3 * y_mo[:, 1] + 0.3 * y_mo[:, 2]

optimizer = SpotOptim(
    fun=three_objectives,
    bounds=[(-5, 5), (-5, 5), (-5, 5)],
    max_iter=35,
    n_initial=20,
    fun_mo2so=custom_scalarization,
    seed=42
)

result = optimizer.optimize()
```

#### 50.3.5. Example 5: With Noise Handling

```
def noisy_bi_objective(X):
    """Noisy multi-objective function."""
    noise1 = np.random.normal(0, 0.05, X.shape[0])
    noise2 = np.random.normal(0, 0.05, X.shape[0])

    obj1 = np.sum(X**2, axis=1) + noise1
    obj2 = np.sum((X - 1)**2, axis=1) + noise2
    return np.column_stack([obj1, obj2])

optimizer = SpotOptim(
    fun=noisy_bi_objective,
    bounds=[(-5, 5), (-5, 5)],
    max_iter=40,
    n_initial=20,
    repeats_initial=3,          # Handle noise
)
```

## 50. Multi-Objective Optimization Support in SpotOptim

```
    repeats_surrogate=2,  
    seed=42  
)  
  
result = optimizer.optimize()  
# Works seamlessly with noise handling
```

### 50.4. Common Scalarization Strategies

#### 50.4.1. 1. Weighted Sum

```
def weighted_sum(y_mo, weights=[0.5, 0.5]):  
    return sum(w * y_mo[:, i] for i, w in enumerate(weights))
```

Use **when**: Objectives have similar scales and you want linear trade-offs

#### 50.4.2. 2. Weighted Sum with Normalization

```
def normalized_weighted_sum(y_mo, weights=[0.5, 0.5]):  
    # Normalize each objective to [0, 1]  
    y_norm = (y_mo - y_mo.min(axis=0)) / (y_mo.max(axis=0) - y_mo.min(axis=0) + 1e-10)  
    return sum(w * y_norm[:, i] for i, w in enumerate(weights))
```

Use **when**: Objectives have very different scales

#### 50.4.3. 3. Min-Max (Chebyshev)

```
def min_max(y_mo):  
    return np.max(y_mo, axis=1)
```

Use **when**: You want balanced performance across all objectives

#### 50.4.4. 4. Target Achievement

```
def target_achievement(y_mo, targets=[0.0, 0.0]):
    # Minimize deviation from targets
    return np.sum((y_mo - targets)**2, axis=1)
```

**Use when:** You have specific target values for each objective

### 50.4.5. 5. Product

```
def product(y_mo):
    return np.prod(y_mo + 1e-10, axis=1) # Add small value to avoid zero
```

**Use when:** All objectives should be minimized together

## 50.5. Integration with Other Features

Multi-objective support works seamlessly with:

**Noise Handling** - Use `repeats_initial` and `repeats_surrogate`  
**OCBA** - Use `ocba_delta` for intelligent re-evaluation  
**TensorBoard Logging** - Logs converted single-objective values  
**Dimension Reduction** - Fixed dimensions work normally  
**Custom Variable Names** - `var_name` parameter supported

## 50.6. Implementation Details

### 50.6.1. Automatic Detection

SpotOptim automatically detects multi-objective functions:

- If function returns 2D array (`n_samples`, `n_objectives`), it's multi-objective
- If function returns 1D array (`n_samples`), it's single-objective

### 50.6.2. Data Flow

```
User Function → y_mo (raw) → _mo2so() → y_ (single-objective)
                ↓
                y_mo (stored)
```

1. Function returns multi-objective values
2. `_store_mo()` saves them in `y_mo` attribute
3. `_mo2so()` converts to single-objective using `fun_mo2so` or default
4. Surrogate model optimizes the single-objective values
5. All original multi-objective values remain accessible in `y_mo`

### 50.6.3. Backward Compatibility

Fully backward compatible:

- Single-objective functions work unchanged
- `fun_mo2so` defaults to `None`
- `y_mo` is `None` for single-objective problems
- No breaking changes to existing code

## 50.7. Limitations and Notes

### 50.7.1. What This Is

- Scalarization approach to multi-objective optimization
- Single solution found per optimization run
- Different scalarizations  $\rightarrow$  different Pareto solutions
- Suitable for preference-based multi-objective optimization

### 50.7.2. What This Is Not

- Not a true multi-objective optimizer (doesn't find Pareto front)
- Doesn't generate multiple solutions in one run
- Not suitable for discovering entire Pareto front

### 50.7.3. For True Multi-Objective Optimization

For finding the complete Pareto front, consider specialized tools:

- **pymoo**: Comprehensive multi-objective optimization framework
- **platypus**: Multi-objective optimization library
- **NSGA-II**, **MOEA/D**: Dedicated multi-objective algorithms

## 50.8. Demo Script

Run the comprehensive demo (the demos files are located in the **examples** folder):

```
python demo_multiobjective.py
```

This demonstrates:

- Default behavior (first objective)
- Weighted sum scalarization
- Min-max scalarization
- Noisy multi-objective optimization
- Three-objective optimization

## 50.9. Benchmark Multi-Objective Test Functions

SpotOptim includes 12 multi-objective benchmark functions in `spotoptim.function.mo`. These functions are widely used to evaluate and compare multi-objective optimization algorithms.

### 50.9.1. Function Overview

| Function | Objectives | Variables | Pareto Front | Characteristics          |
|----------|------------|-----------|--------------|--------------------------|
| ZDT1     | 2          | 30        | Convex       | Unimodal, minimization   |
| ZDT2     | 2          | 30        | Non-convex   | Unimodal, minimization   |
| ZDT3     | 2          | 30        | Disconnected | Multimodal, minimization |
| ZDT4     | 2          | 10        | Convex       | Many local fronts        |
| ZDT6     | 2          | 10        | Non-convex   | Non-uniform density      |

| Function            | Objectives | Variables | Pareto Front | Characteristics                             |
|---------------------|------------|-----------|--------------|---------------------------------------------|
| DTLZ1               | Scalable   | n_obj+4   | Linear       | Multimodal                                  |
| DTLZ2               | Scalable   | n_obj+9   | Spherical    | Unimodal                                    |
| Schaffer N1         | 2          | 1         | Convex       | Simple,<br>minimization                     |
| Fonseca-<br>Fleming | 2          | 2-10      | Concave      | Symmetric,<br>minimization                  |
| mo_conv2_min        |            | 2         | Convex       | Convex,<br>minimization                     |
| mo_conv2_max        |            | 2         | Convex       | Convex,<br>maximization                     |
| Kursawe             | 2          | 3         | Disconnected | Non-convex,<br>disconnected<br>minimization |

mo\_conv2\_min is a smooth convex minimization pair with objectives  $f_1(x, y) = x^2 + y^2$  and  $f_2(x, y) = (x - 1)^2 + (y - 1)^2$  on  $[0, 1]^2$ , producing a convex quadratic Pareto front along  $x = y$ . mo\_conv2\_max flips both objectives for maximization via  $f_1(x, y) = 2 - (x^2 + y^2)$  and  $f_2(x, y) = 2 - ((1 - x)^2 + (1 - y)^2)$ , keeping objective values bounded in  $[0, 2]$ .

### 50.9.2. Example 6: ZDT1 - Convex Pareto Front

ZDT1 is a classical bi-objective test problem with a convex Pareto front, widely used for benchmarking.

```
import numpy as np
import matplotlib.pyplot as plt
from spotoptim.function.mo import zdt1
from spotoptim import SpotOptim

# Generate data for visualization
n_points = 100
x1_range = np.linspace(0, 1, n_points)

# Pareto front: x1 in [0,1], rest = 0
X_pareto = np.column_stack([x1_range, np.zeros((n_points, 2))])
y_pareto = zdt1(X_pareto)

# Non-optimal points for comparison
X_non_optimal = np.random.rand(200, 3)
y_non_optimal = zdt1(X_non_optimal)
```



```

# Visualize Pareto front
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))

# Plot objective space
ax1.scatter(y_non_optimal[:, 0], y_non_optimal[:, 1],
            alpha=0.3, label='Non-optimal', s=20)
ax1.plot(y_pareto[:, 0], y_pareto[:, 1],
         'r-', linewidth=2, label='Pareto Front')
ax1.set_xlabel('f1(x) = x1')
ax1.set_ylabel('f2(x) = g(x) * [1 - √(x1/g(x))]\')
ax1.set_title('ZDT1: Convex Pareto Front')
ax1.legend()
ax1.grid(True, alpha=0.3)

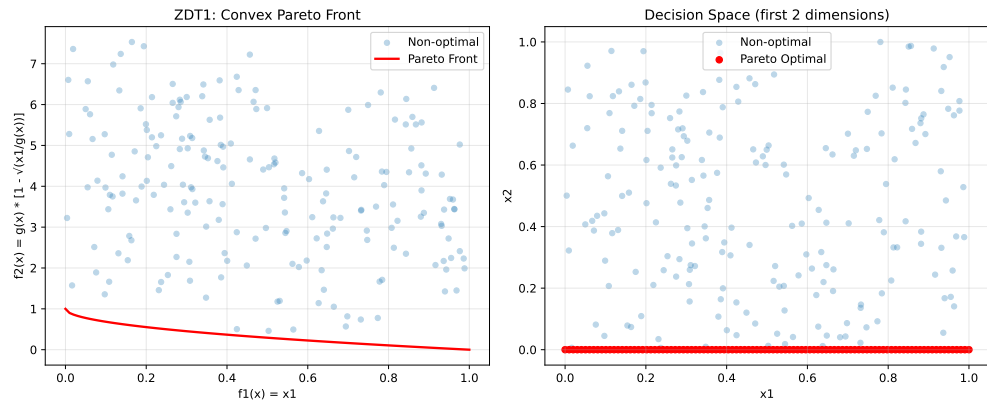
# Plot decision space (first two variables)
ax2.scatter(X_non_optimal[:, 0], X_non_optimal[:, 1],
            alpha=0.3, label='Non-optimal', s=20)
ax2.scatter(X_pareto[:, 0], X_pareto[:, 1],
            c='red', label='Pareto Optimal', s=30)
ax2.set_xlabel('x1')
ax2.set_ylabel('x2')
ax2.set_title('Decision Space (first 2 dimensions)\')
ax2.legend()
ax2.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

print("ZDT1 Characteristics:")
print("- Convex Pareto front: f2 = 1 - √f1")
print("- Optimal when x2, x3, ..., xn = 0")
print("- Easy to solve, good for testing basic functionality")

```

## 50. Multi-Objective Optimization Support in SpotOptim



ZDT1 Characteristics:

- Convex Pareto front:  $f2 = 1 - \sqrt{f1}$
- Optimal when  $x2, x3, \dots, xn = 0$
- Easy to solve, good for testing basic functionality

Now optimize using SpotOptim with weighted sum:

```
def weighted_sum_zdt1(y_mo):
    """Equal weighting for ZDT1."""
    return 0.5 * y_mo[:, 0] + 0.5 * y_mo[:, 1]

optimizer = SpotOptim(
    fun=zdt1,
    bounds=[(0, 1)] * 30, # 30 variables in [0, 1]
    max_iter=50,
    n_initial=20,
    fun_mo2so=weighted_sum_zdt1,
    seed=42
)

result = optimizer.optimize()

print(f"\nOptimization Result:")
print(f"Best x (first 5 dims): {result.x[:5]}")
print(f"Best scalarized value: {result.fun:.6f}")

# Evaluate multi-objective values
y_best = zdt1(result.x.reshape(1, -1))
print(f"f1 = {y_best[0, 0]:.6f}, f2 = {y_best[0, 1]:.6f}")
```

```

Optimization Result:
Best x (first 5 dims): [1. 0. 0. 0. 0.]
Best scalarized value: 0.500000
f1 = 1.000000, f2 = 0.000000

```

### 50.9.3. Example 7: ZDT2 - Non-Convex Pareto Front

ZDT2 has a non-convex Pareto front, making it more challenging than ZDT1.

```

from spotoptim.function.mo import zdt2

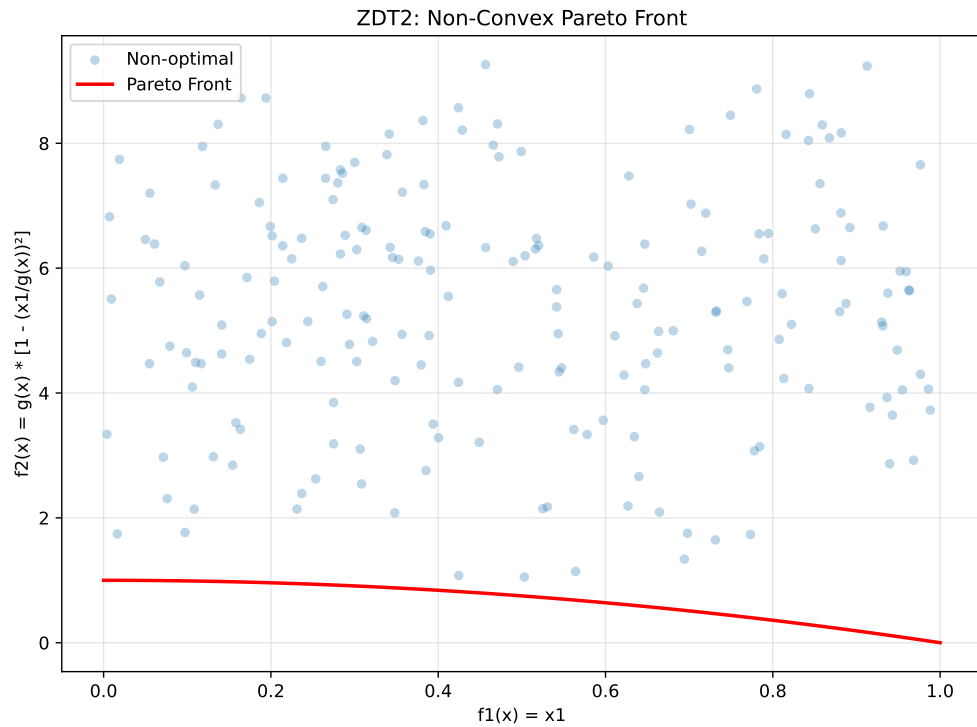
# Generate Pareto front
X_pareto = np.column_stack([x1_range, np.zeros((n_points, 2))])
y_pareto = zdt2(X_pareto)

# Non-optimal points
y_non_optimal = zdt2(X_non_optimal)

# Visualize
fig, ax = plt.subplots(figsize=(8, 6))
ax.scatter(y_non_optimal[:, 0], y_non_optimal[:, 1],
           alpha=0.3, label='Non-optimal', s=20)
ax.plot(y_pareto[:, 0], y_pareto[:, 1],
        'r-', linewidth=2, label='Pareto Front')
ax.set_xlabel('f1(x) = x1')
ax.set_ylabel('f2(x) = g(x) * [1 - (x1/g(x))2]')
ax.set_title('ZDT2: Non-Convex Pareto Front')
ax.legend()
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

print("ZDT2 Characteristics:")
print("- Non-convex Pareto front:  $f_2 = 1 - f_1^2$ ")
print("- Tests algorithm's ability to handle non-convexity")
print("- Optimal when  $x_2, x_3, \dots, x_n = 0$ ")

```



ZDT2 Characteristics:

- Non-convex Pareto front:  $f2 = 1 - f1^2$
- Tests algorithm's ability to handle non-convexity
- Optimal when  $x2, x3, \dots, xn = 0$

#### 50.9.4. Example 8: ZDT3 - Disconnected Pareto Front

ZDT3 has a discontinuous Pareto front consisting of 5 separate regions.

```
from spotoptim.function.mo import zdt3

# Generate Pareto front
X_pareto = np.column_stack([x1_range, np.zeros((n_points, 2))])
y_pareto = zdt3(X_pareto)

# Non-optimal points
y_non_optimal = zdt3(X_non_optimal)

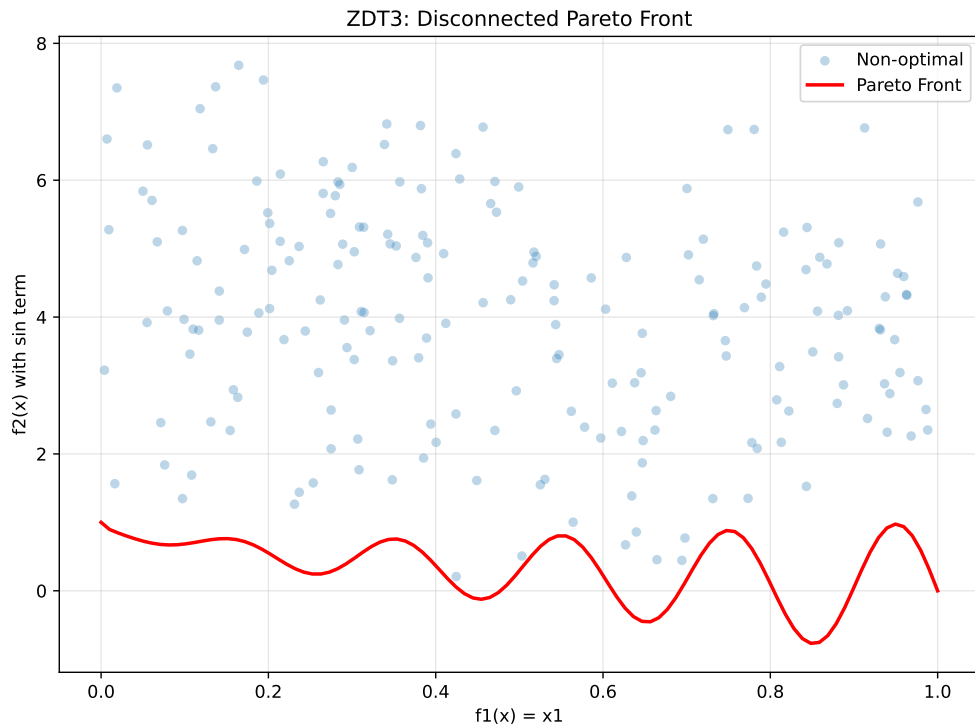
# Visualize
```

```

fig, ax = plt.subplots(figsize=(8, 6))
ax.scatter(y_non_optimal[:, 0], y_non_optimal[:, 1],
          alpha=0.3, label='Non-optimal', s=20)
ax.plot(y_pareto[:, 0], y_pareto[:, 1],
        'r-', linewidth=2, label='Pareto Front')
ax.set_xlabel('f1(x) = x1')
ax.set_ylabel('f2(x) with sin term')
ax.set_title('ZDT3: Disconnected Pareto Front')
ax.legend()
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

print("ZDT3 Characteristics:")
print("- Disconnected Pareto front with 5 separate regions")
print("- Tests diversity maintenance in algorithms")
print("- sin(10 x1) term creates discontinuities")

```



ZDT3 Characteristics:

- Disconnected Pareto front with 5 separate regions
- Tests diversity maintenance in algorithms
- $\sin(10 x_1)$  term creates discontinuities

### 50.9.5. Example 9: ZDT4 - Multimodal with Many Local Fronts

ZDT4 has  $21^9$  local Pareto fronts, testing the algorithm's ability to escape local optima.

```
from spotoptim.function.mo import zdt4

# Generate Pareto front (same as ZDT1 when optimal)
X_pareto = np.column_stack([x1_range, np.zeros((n_points, 9))])
y_pareto = zdt4(X_pareto)

# Non-optimal points with multimodal g-function
X_non_optimal_zdt4 = np.random.rand(200, 10)
X_non_optimal_zdt4[:, 0] = np.clip(X_non_optimal_zdt4[:, 0], 0, 1) # x1 in [0,1]
X_non_optimal_zdt4[:, 1:] = X_non_optimal_zdt4[:, 1:] * 10 - 5 # others in [-5,5]
y_non_optimal = zdt4(X_non_optimal_zdt4)

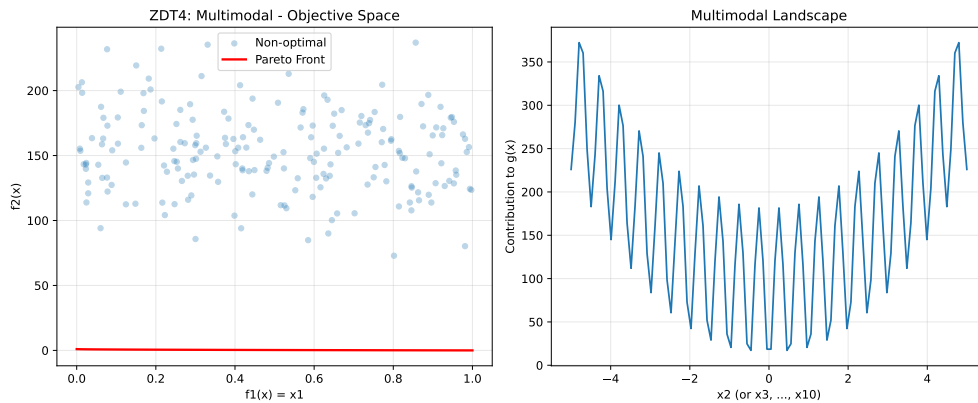
# Visualize
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))

# Objective space
ax1.scatter(y_non_optimal[:, 0], y_non_optimal[:, 1],
            alpha=0.3, label='Non-optimal', s=20)
ax1.plot(y_pareto[:, 0], y_pareto[:, 1],
        'r-', linewidth=2, label='Pareto Front')
ax1.set_xlabel('f1(x) = x1')
ax1.set_ylabel('f2(x)')
ax1.set_title('ZDT4: Multimodal - Objective Space')
ax1.legend()
ax1.grid(True, alpha=0.3)

# Show multimodal landscape for x2
x2_range = np.linspace(-5, 5, 100)
g_vals = 1 + 10 * 9 + (x2_range**2 - 10 * np.cos(4 * np.pi * x2_range)) * 9
ax2.plot(x2_range, g_vals)
ax2.set_xlabel('x2 (or x3, ..., x10)')
ax2.set_ylabel('Contribution to g(x)')
ax2.set_title('Multimodal Landscape')
ax2.grid(True, alpha=0.3)
```

```
plt.tight_layout()
plt.show()

print("ZDT4 Characteristics:")
print("- 21^9 local Pareto fronts")
print("- Tests global search capability")
print("- x1 in [0,1], x_i in [-5,5] for i=2,...,10")
```



ZDT4 Characteristics:

- $21^9$  local Pareto fronts
- Tests global search capability
- $x_1$  in  $[0,1]$ ,  $x_i$  in  $[-5,5]$  for  $i=2,\dots,10$

### 50.9.6. Example 10: ZDT6 - Non-Uniform Density

ZDT6 has a non-uniform search space with low density of solutions near the Pareto front.

```
from spotoptim.function.mo import zdt6

# Generate Pareto front
X_pareto = np.column_stack([x1_range, np.zeros((n_points, 2))])
y_pareto = zdt6(X_pareto)

# Non-optimal points
y_non_optimal = zdt6(X_non_optimal)

# Visualize
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))
```

```

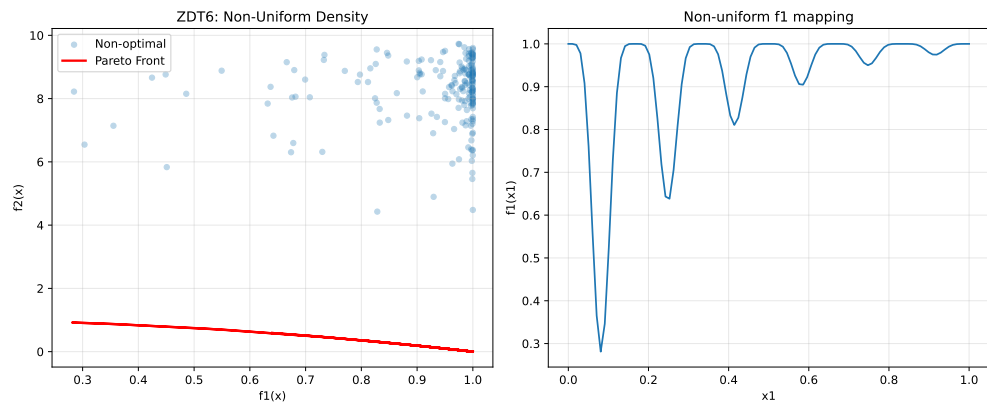
# Objective space
ax1.scatter(y_non_optimal[:, 0], y_non_optimal[:, 1],
            alpha=0.3, label='Non-optimal', s=20)
ax1.plot(y_pareto[:, 0], y_pareto[:, 1],
        'r-', linewidth=2, label='Pareto Front')
ax1.set_xlabel('f1(x)')
ax1.set_ylabel('f2(x)')
ax1.set_title('ZDT6: Non-Uniform Density')
ax1.legend()
ax1.grid(True, alpha=0.3)

# Show f1 non-uniformity
ax2.plot(x1_range, 1 - np.exp(-4 * x1_range) * (np.sin(6 * np.pi * x1_range)**6))
ax2.set_xlabel('x1')
ax2.set_ylabel('f1(x1)')
ax2.set_title('Non-uniform f1 mapping')
ax2.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

print("ZDT6 Characteristics:")
print("- Non-uniform density of Pareto optimal solutions")
print("- Low density near the Pareto front")
print("- Tests diversity maintenance")

```



ZDT6 Characteristics:

- Non-uniform density of Pareto optimal solutions
- Low density near the Pareto front
- Tests diversity maintenance



**50.9.7. Example 11: DTLZ2 - Scalable Spherical Pareto Front**

DTLZ2 is a scalable test problem with a concave spherical Pareto front.

```
from spotoptim.function.mo import dtlz2

# 3D visualization for 3 objectives
n_samples = 500
X_samples = np.random.rand(n_samples, 12)
y_samples = dtlz2(X_samples, n_obj=3)

# Generate Pareto optimal points (on unit sphere)
theta = np.linspace(0, np.pi/2, 30)
phi = np.linspace(0, np.pi/2, 30)
theta_grid, phi_grid = np.meshgrid(theta, phi)

f1 = np.cos(theta_grid) * np.cos(phi_grid)
f2 = np.cos(theta_grid) * np.sin(phi_grid)
f3 = np.sin(theta_grid)

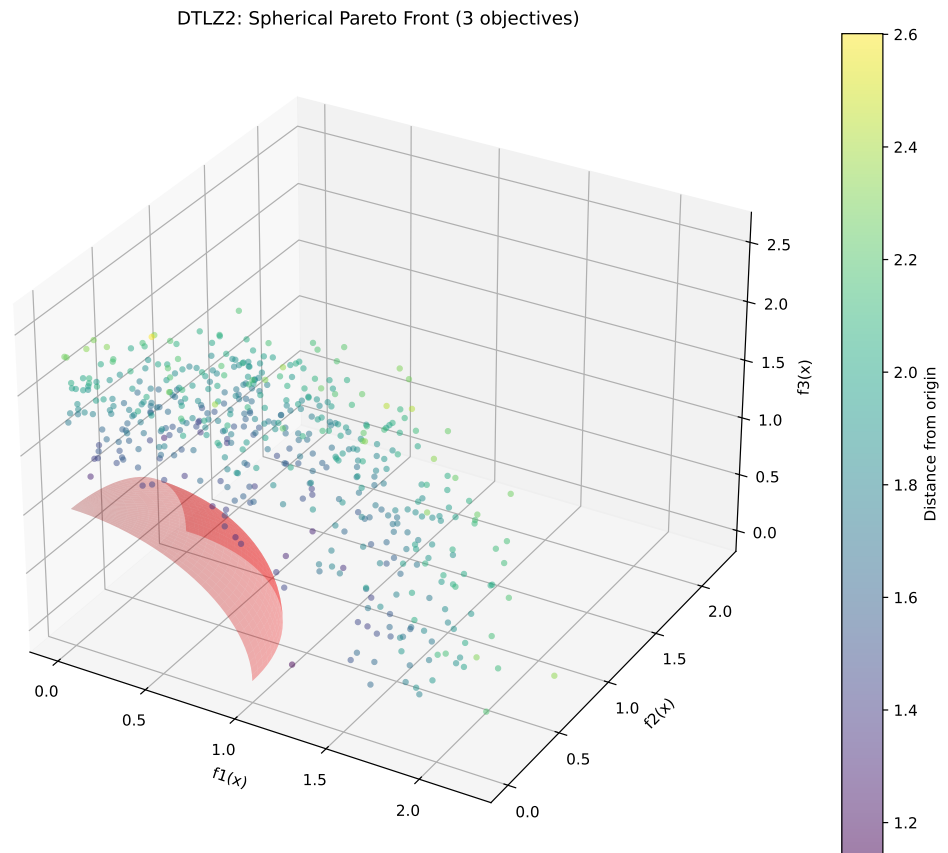
# 3D plot
fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')

# Pareto front surface
ax.plot_surface(f1, f2, f3, alpha=0.3, color='red', label='Pareto Front')

# Sample points
colors = np.linalg.norm(y_samples, axis=1)
scatter = ax.scatter(y_samples[:, 0], y_samples[:, 1], y_samples[:, 2],
                    c=colors, cmap='viridis', alpha=0.5, s=10)

ax.set_xlabel('f1(x)')
ax.set_ylabel('f2(x)')
ax.set_zlabel('f3(x)')
ax.set_title('DTLZ2: Spherical Pareto Front (3 objectives)')
plt.colorbar(scatter, label='Distance from origin')
plt.tight_layout()
plt.show()

print("DTLZ2 Characteristics:")
print("- Scalable number of objectives")
print("- Pareto front is unit sphere:  $\sum f_i^2 = 1$ ")
print("- Concave, unimodal")
print(f"- Sample: sum of squares = {np.sum(y_samples[0]**2):.4f}")
```



DTLZ2 Characteristics:

- Scalable number of objectives
- Pareto front is unit sphere:  $\sum f_i^2 = 1$
- Concave, unimodal
- Sample: sum of squares = 3.8418

Optimize DTLZ2 with 3 objectives:

```
def weighted_sum_3obj(y_mo):
    """Equal weighting for 3 objectives."""
    return np.mean(y_mo, axis=1)

optimizer = SpotOptim(
    fun=lambda X: dtlz2(X, n_obj=3),
```

```

    bounds=[(0, 1)] * 12, # 12 variables for 3 objectives
    max_iter=50,
    n_initial=25,
    fun_mo2so=weighted_sum_3obj,
    seed=42
)

result = optimizer.optimize()

# Evaluate
y_best = dtlz2(result.x.reshape(1, -1), n_obj=3)
print(f"\nOptimization Result:")
print(f"f1 = {y_best[0, 0]:.4f}, f2 = {y_best[0, 1]:.4f}, f3 = {y_best[0, 2]:.4f}")
print(f"Sum of squares: {np.sum(y_best[0]**2):.4f} (should be ~1.0)")

```

```

Optimization Result:
f1 = 0.0000, f2 = 0.0000, f3 = 1.2524
Sum of squares: 1.5684 (should be ~1.0)

```

### 50.9.8. Example 12: Schaffer N1 - Simple Bi-Objective

Schaffer N1 is one of the earliest and simplest multi-objective test functions.

```

from spoptim.function.mo import schaffer_n1

# Generate data
x_range = np.linspace(-10, 10, 200).reshape(-1, 1)
y_schaffer = schaffer_n1(x_range)

# Pareto optimal range [0, 2]
x_pareto = np.linspace(0, 2, 100).reshape(-1, 1)
y_pareto = schaffer_n1(x_pareto)

# Visualize
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))

# Objective space
ax1.plot(y_schaffer[:, 0], y_schaffer[:, 1], 'b-', alpha=0.3, label='All points')
ax1.plot(y_pareto[:, 0], y_pareto[:, 1], 'r-', linewidth=2, label='Pareto Front')
ax1.set_xlabel('f1(x) = x2')
ax1.set_ylabel('f2(x) = (x-2)2')
ax1.set_title('Schaffer N1: Objective Space')

```

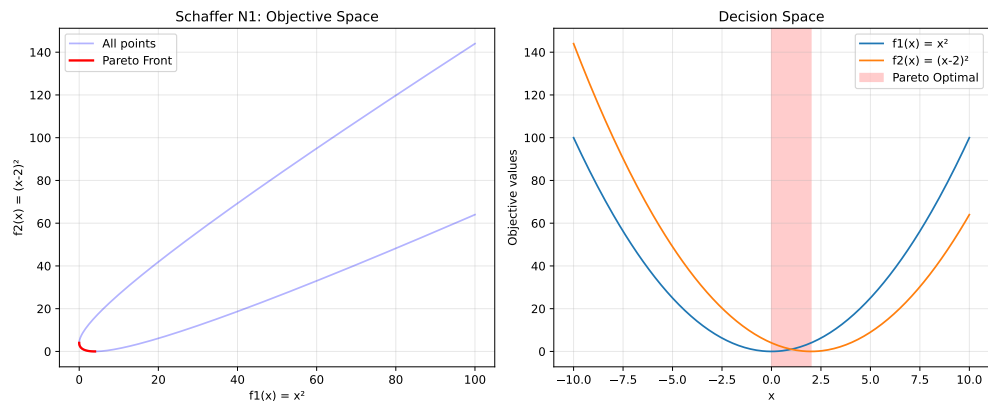
## 50. Multi-Objective Optimization Support in SpotOptim

```
ax1.legend()
ax1.grid(True, alpha=0.3)

# Decision space
ax2.plot(x_range, y_schaffer[:, 0], label='f1(x) = x2')
ax2.plot(x_range, y_schaffer[:, 1], label='f2(x) = (x-2)2')
ax2.axvspan(0, 2, alpha=0.2, color='red', label='Pareto Optimal')
ax2.set_xlabel('x')
ax2.set_ylabel('Objective values')
ax2.set_title('Decision Space')
ax2.legend()
ax2.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

print("Schaffer N1 Characteristics:")
print("- Simplest multi-objective function")
print("- 1D decision variable")
print("- Pareto optimal: x [0, 2]")
print("- Convex Pareto front")
```



Schaffer N1 Characteristics:

- Simplest multi-objective function
- 1D decision variable
- Pareto optimal: x [0, 2]
- Convex Pareto front

**50.9.9. Example 13: Fonseca-Fleming - Symmetric Concave Front**

Fonseca-Fleming is a classical bi-objective problem with a concave Pareto front. It is defined for 2 to 10 decision variables as follows:

$$f_1(x) = 1 - \exp\left(-\sum_{i=1}^n \left(x_i - \frac{1}{\sqrt{n}}\right)^2\right) \quad f_2(x) = 1 - \exp\left(-\sum_{i=1}^n \left(x_i + \frac{1}{\sqrt{n}}\right)^2\right),$$

where  $x_i \in [-2, 2]$ . The Pareto optimal solutions satisfy:

$$\sum_{i=1}^n x_i^2 = \frac{1}{n}.$$

```
from spotoptim.function.mo import fonseca_fleming

# Generate data for 2D
n_grid = 50
x1_ff = np.linspace(-2, 2, n_grid)
x2_ff = np.linspace(-2, 2, n_grid)
X1_grid, X2_grid = np.meshgrid(x1_ff, x2_ff)

# Evaluate on grid
X_grid = np.column_stack([X1_grid.ravel(), X2_grid.ravel()])
y_grid = fonseca_fleming(X_grid)

# Pareto optimal points (approximately)
X_pareto_ff = np.linspace(-1/np.sqrt(2), 1/np.sqrt(2), 100)
X_pareto_2d = np.column_stack([X_pareto_ff, X_pareto_ff])
y_pareto_ff = fonseca_fleming(X_pareto_2d)

# Visualize
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))

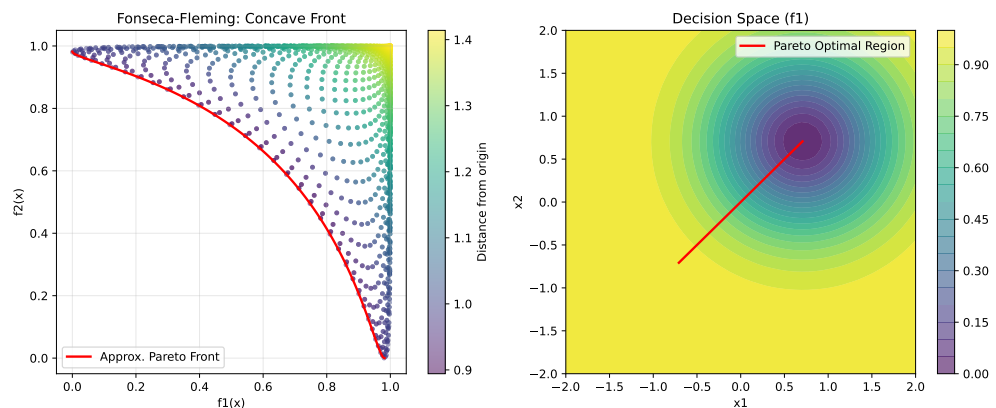
# Objective space
scatter = ax1.scatter(y_grid[:, 0], y_grid[:, 1],
                     c=np.linalg.norm(y_grid, axis=1),
                     cmap='viridis', alpha=0.5, s=10)
ax1.plot(y_pareto_ff[:, 0], y_pareto_ff[:, 1],
        'r-', linewidth=2, label='Approx. Pareto Front')
ax1.set_xlabel('f1(x)')
ax1.set_ylabel('f2(x)')
ax1.set_title('Fonseca-Fleming: Concave Front')
ax1.legend()
ax1.grid(True, alpha=0.3)
plt.colorbar(scatter, ax=ax1, label='Distance from origin')
```

## 50. Multi-Objective Optimization Support in SpotOptim

```
# Decision space
ax2.contourf(X1_grid, X2_grid, y_grid[:, 0].reshape(n_grid, n_grid),
             levels=20, cmap='viridis', alpha=0.6)
ax2.plot(X_pareto_2d[:, 0], X_pareto_2d[:, 1],
        'r-', linewidth=2, label='Pareto Optimal Region')
ax2.set_xlabel('x1')
ax2.set_ylabel('x2')
ax2.set_title('Decision Space (f1)')
ax2.legend()
plt.colorbar(ax2.contourf(X1_grid, X2_grid, y_grid[:, 0].reshape(n_grid, n_grid),
                        levels=20, cmap='viridis', alpha=0.6), ax=ax2)

plt.tight_layout()
plt.show()

print("Fonseca-Fleming Characteristics:")
print("- Concave Pareto front (minimization)")
print("- Symmetric problem")
print("- Scalable to higher dimensions")
```



Fonseca-Fleming Characteristics:

- Concave Pareto front (minimization)
- Symmetric problem
- Scalable to higher dimensions

## 50.10. Example 14: Kursawe - Non-Convex Disconnected Front

Optimize Kursawe with min-max scalarization:

```
from spotoptim.function.mo import kursawe
def min_max_kursawe(y_mo):
    """Minimize maximum objective (Chebyshev)."""
    return np.max(y_mo, axis=1)

optimizer = SpotOptim(
    fun=kursawe,
    bounds=[(-5, 5)] * 3,
    max_iter=50,
    n_initial=20,
    fun_mo2so=min_max_kursawe,
    seed=42
)

result = optimizer.optimize()

y_best = kursawe(result.x.reshape(1, -1))
print(f"\nOptimization Result:")
print(f"Best x: {result.x}")
print(f"f1 = {y_best[0, 0]:.4f}, f2 = {y_best[0, 1]:.4f}")
print(f"Max objective: {max(y_best[0]):.4f}")
```

```
Optimization Result:
Best x: [-0.58740866 -1.11955526 -1.11461838]
f1 = -15.0566, f2 = -8.0116
Max objective: -8.0116
```

### 50.10.1. Comparison: Different Scalarization Strategies on ZDT1

Let's compare how different scalarization strategies find different Pareto solutions:

```
# Define different scalarizations
scalarizations = {
    'First Objective': None, # Default
    'Equal Weight': lambda y: 0.5 * y[:, 0] + 0.5 * y[:, 1],
    'Favor f1': lambda y: 0.8 * y[:, 0] + 0.2 * y[:, 1],
```

```

    'Favor f2': lambda y: 0.2 * y[:, 0] + 0.8 * y[:, 1],
    'Min-Max': lambda y: np.max(y, axis=1),
}

results = {}
for name, scalarization in scalarizations.items():
    opt = SpotOptim(
        fun=zdt1,
        bounds=[(0, 1)] * 30,
        max_iter=40,
        n_initial=15,
        fun_mo2so=scalarization,
        seed=42,
        verbose=0
    )
    res = opt.optimize()
    y_res = zdt1(res.x.reshape(1, -1))
    results[name] = (res.x, y_res[0])

# Visualize results
fig, ax = plt.subplots(figsize=(10, 6))

# Plot Pareto front
X_pareto = np.column_stack([x1_range, np.zeros((n_points, 29))])
y_pareto = zdt1(X_pareto)
ax.plot(y_pareto[:, 0], y_pareto[:, 1], 'k-', linewidth=2,
        label='True Pareto Front', alpha=0.3)

# Plot optimization results
colors = ['blue', 'green', 'orange', 'purple', 'red']
for (name, (x, y)), color in zip(results.items(), colors):
    ax.scatter(y[0], y[1], s=100, c=color, marker='*',
               label=f'{name}: ({y[0]:.3f}, {y[1]:.3f})', zorder=5)

ax.set_xlabel('f1(x)')
ax.set_ylabel('f2(x)')
ax.set_title('Different Scalarization Strategies Find Different Pareto Solutions')
ax.legend()
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

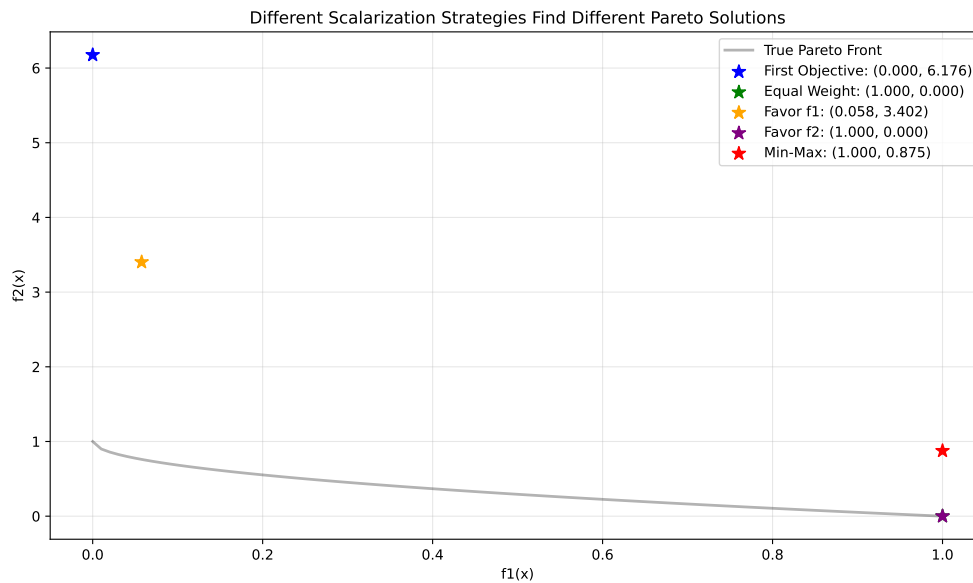
print("\nComparison Results:")
for name, (x, y) in results.items():

```



## 50.11. A Simple Visualization Example

```
print(f"{name:20s}: f1={y[0]:.4f}, f2={y[1]:.4f}")
```



Comparison Results:

|                 |                        |
|-----------------|------------------------|
| First Objective | : f1=0.0000, f2=6.1756 |
| Equal Weight    | : f1=1.0000, f2=0.0000 |
| Favor f1        | : f1=0.0576, f2=3.4021 |
| Favor f2        | : f1=1.0000, f2=0.0000 |
| Min-Max         | : f1=1.0000, f2=0.8750 |

## 50.11. A Simple Visualization Example

The `mo_mm_desirability_function` computes the negative combined desirability for a candidate point  $\mathbf{x}$ . It predicts the objective values using the provided surrogate models, computes the Morris-Mitchell improvement if requested, and then calculates the overall desirability using the provided `DOverall` object. The function returns the negative desirability (for minimization) and the list of individual objective values. Consider the following example usage:

Generate `X_base` in the range  $[0,1]$  as input data and evaluate the two-objective maximization function `mo_conv2_max`.

A smooth, convex two-objective maximization problem on  $[0, 1]^2$  using flipped versions of the minimization objectives:

## 50. Multi-Objective Optimization Support in SpotOptim

$$f1(x, y) = 2 - (x^2 + y^2) \quad (50.1)$$

$$f2(x, y) = 2 - ((1 - x)^2 + (1 - y)^2) \quad (50.2)$$

It has the following properties:

- Domain:  $[0, 1]^2$
- Objectives: maximize both  $f1$  and  $f2$
- Ideal points:  $(0, 0)$  for  $f1$  (gives  $f1 = 2$ );  $(1, 1)$  for  $f2$  (gives  $f2 = 2$ )
- Pareto set: line  $x = y$  in  $[0, 1]$
- Pareto front: convex quadratic trade-off  $f1 = 2 - 2t^2$ ,  $f2 = 2 - 2(1 - t)^2$ ,  $t \in [0, 1]$

```
X_base = np.random.rand(500, 2)
y = mo_conv2_max(X_base)
```

The RandomForestRegressor models are fitted for each objective.

```
models = []
for i in range(y.shape[1]):
    model = RandomForestRegressor(n_estimators=n_estimators, random_state=random_state)
    model.fit(X_base, y[:, i])
    models.append(model)

# calculate base Morris-Mitchell stats
phi_base, J_base, d_base = mmphi_intensive(X_base, q=2, p=2)
print(f"phi_base: {phi_base}, J_base: {J_base}, d_base: {d_base}")
```

```
phi_base: 5.9675438157238005, J_base: [1 1 1 ... 1 1 1], d_base: [0.00152868 0.0026111]
```

Figure 50.1 shows the surfaces of the two objectives using the fitted models.

```
x_min = 0
x_max = 1
combinations = [(0, 1)]
target_names = ["y1", "y2"]

mo_xy_surface(models, bounds=[(x_min, x_max), (x_min, x_max)], target_names=target_names)
```

## 50.11. A Simple Visualization Example

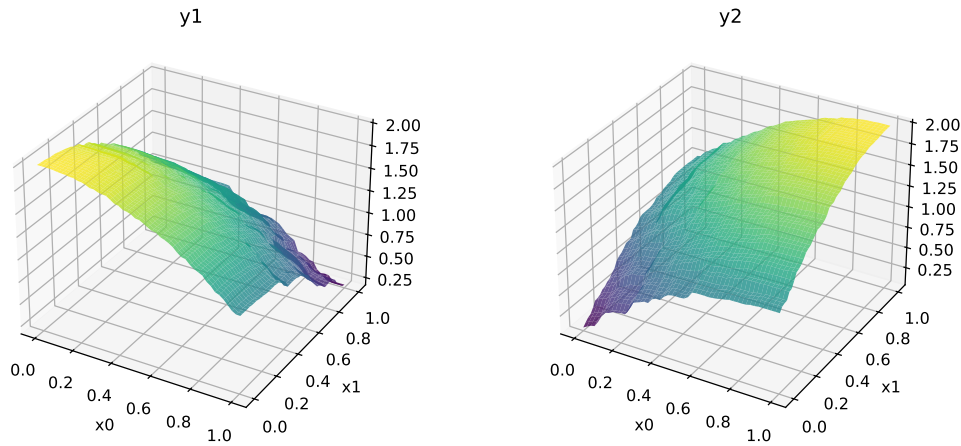


Figure 50.1.: Surfaces of the two objectives of the `mo_conv2_max` function

Figure 50.2 shows the contours of the two objectives using the fitted models.

```
x_min = 0
x_max = 1
mo_xy_contour(models, bounds=[(x_min, x_max), (x_min, x_max)], target_names=target_names)
```

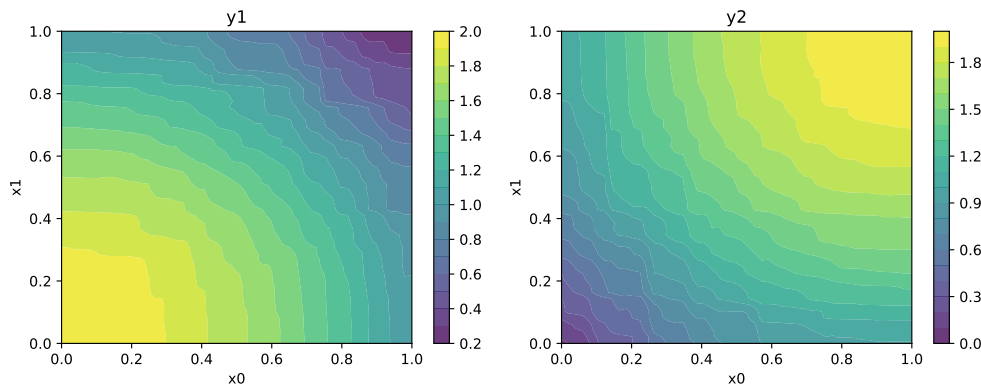


Figure 50.2.: Contours of the two objectives of the `mo_conv2_max` function

Figure 50.3 shows the Pareto front of the original data.

```
plot_mo(y_orig=y, target_names=target_names, combinations=combinations, pareto="max", pareto_front_c
```

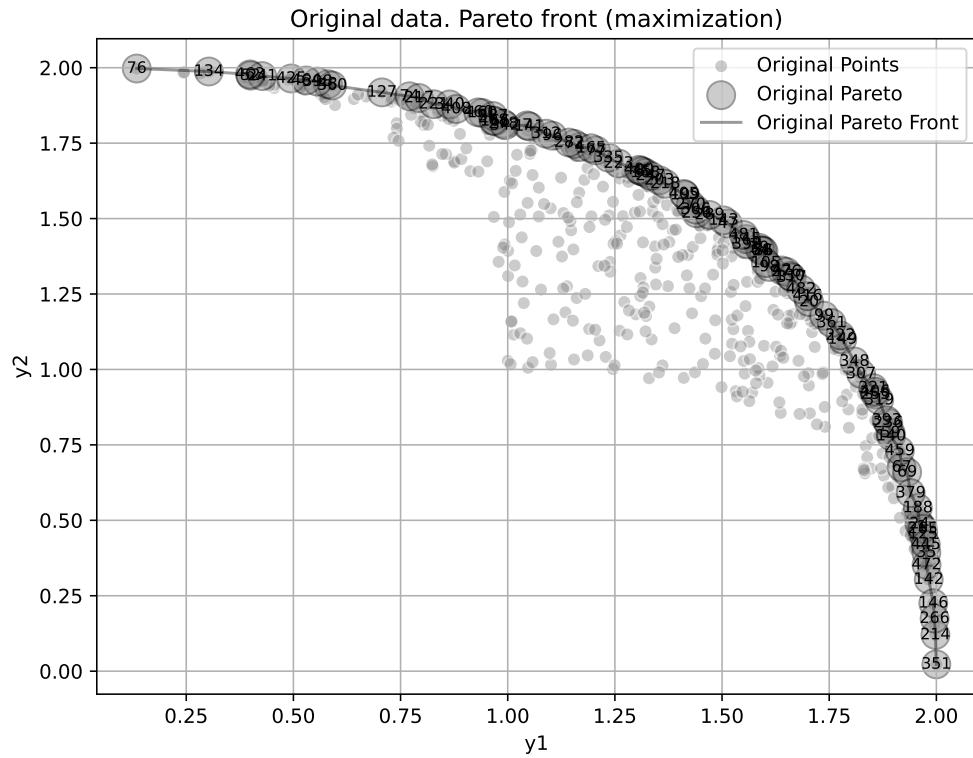


Figure 50.3.: Pareto front of the original data

Figure 50.4 visualizes the Pareto-optimal points, i.e., the points in the input space that correspond to the Pareto front:

```
mo_pareto_optx_plot(X=X_base, Y=y, minimize=False)
```

### 50.11. A Simple Visualization Example

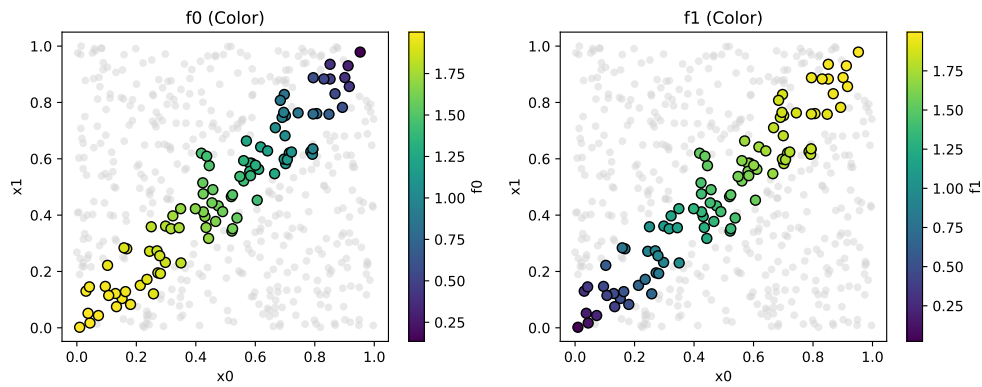


Figure 50.4.: Pareto-optimal points in the input space

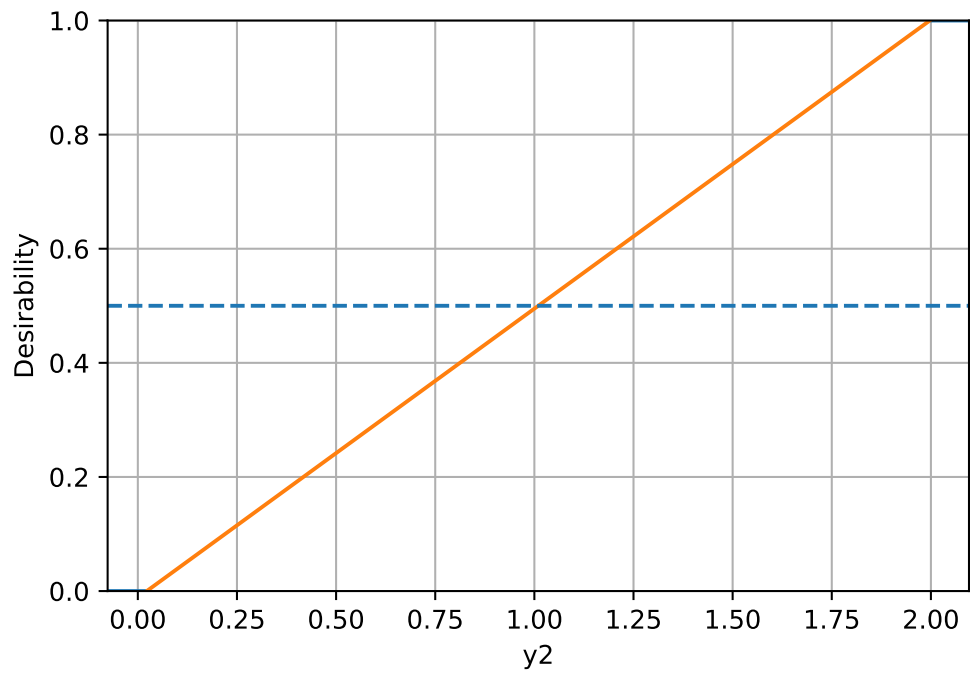
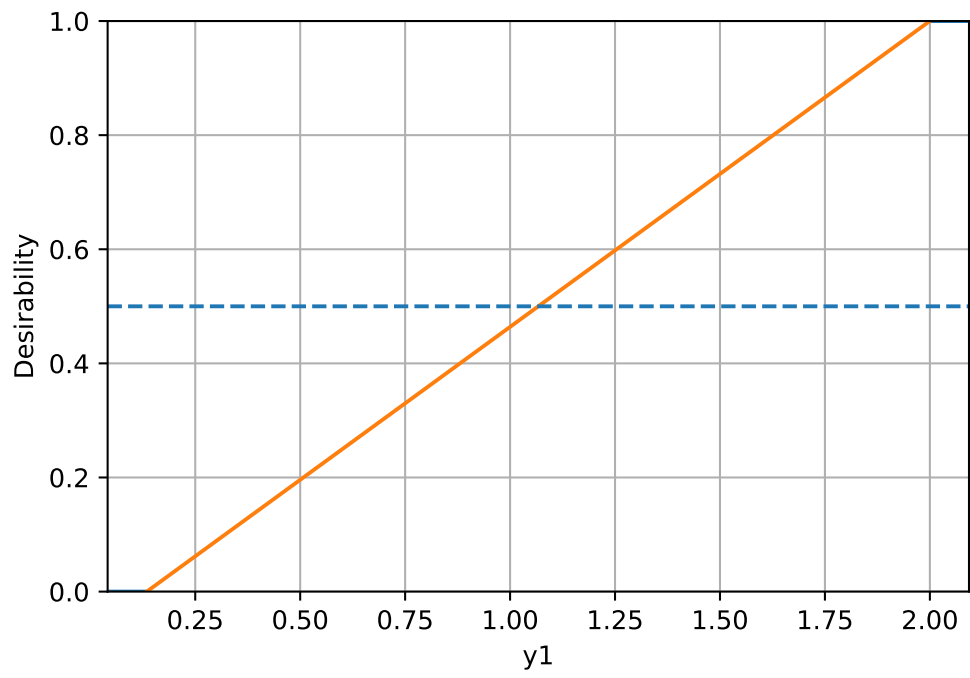
Compute desirability functions for each objective:

```
d_funcs = []
for i in range(y.shape[1]):
    d_func = DMax(low=np.min(y[:, i]), high=np.max(y[:, i]))
    d_funcs.append(d_func)
```

The desirability functions can be visualized as follows:

```
d_funcs[0].plot(xlabel=target_names[0], figsize=(6,4))
d_funcs[1].plot(xlabel=target_names[1], figsize=(6,4))
```

50. Multi-Objective Optimization Support in SpotOptim



### 50.11. A Simple Visualization Example

The minimal expected improvement of the Morris-Mitchell criterion is defined as 1 percent of the base value, and the maximal expected improvement as 10 percent of the base value. Then, the desirability function for Morris-Mitchell objective can be defined as follows:

```
# imp_min = phi_base * 0.01
# imp_max = phi_base * 0.1

# d_mm = DMax(low=imp_min, high=imp_max)
# d_mm.plot()
```

Combine into overall desirability

```
# D_overall = DOverall(*d_funcs, d_mm)
D_overall = DOverall(*d_funcs)
```

Now we can call the `mo_mm_desirability_function` with a test point:

```
x_test = np.array([0.5, 0.5]) # Example test point
neg_D, objectives = mo_mm_desirability_function(x_test, models, X_base, J_base, d_base, phi_base, D_
print(f"Negative Desirability: {neg_D}")
print(f"Objectives: {objectives}")
```

```
Negative Desirability: -0.743444287860325
Objectives: [np.float64(1.5174750253282898), np.float64(1.4947828032249613)]
```

```
mo_xy_desirability_plot(models, bounds=[(x_min, x_max), (x_min, x_max)], X_base=X_base, J_base=J_base)
```

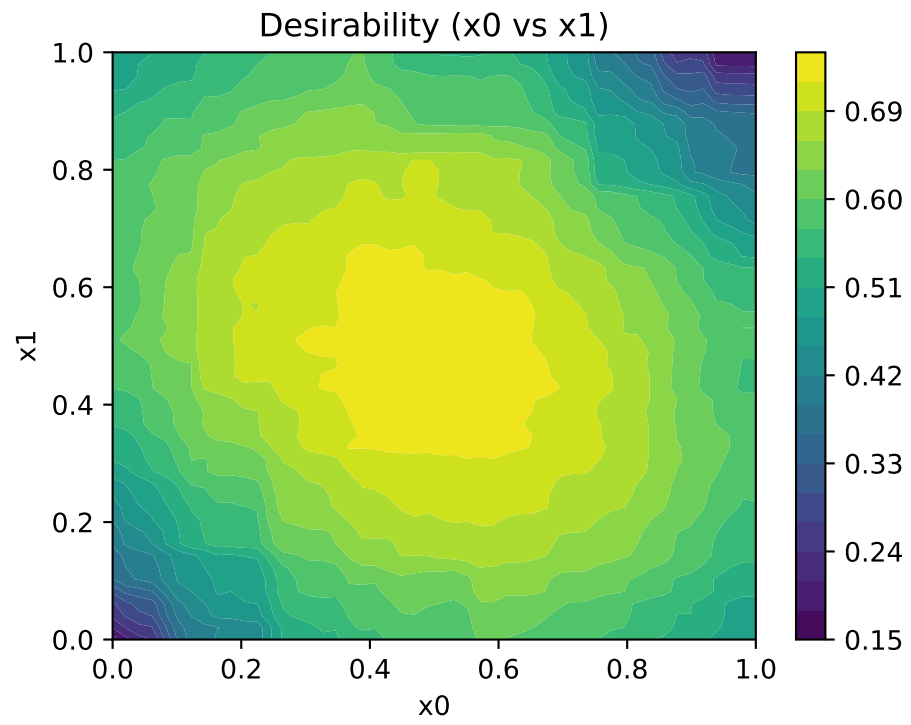


Figure 50.5.: Desirability function for multi-objective optimization



## **Part VI.**

# **Hyperparameter Tuning**



# 51. Diabetes Dataset Utilities

SpotOptim provides convenient utilities for working with the sklearn diabetes dataset, including PyTorch **Dataset** and **DataLoader** implementations. These utilities simplify data loading, preprocessing, and model training for regression tasks.

## 51.1. Overview

The diabetes dataset contains 10 baseline variables (age, sex, body mass index, average blood pressure, and six blood serum measurements) for 442 diabetes patients. The target is a quantitative measure of disease progression one year after baseline.

**Module:** `spotoptim.data.diabetes`

**Key Components:**

- `DiabetesDataset`: PyTorch Dataset class
- `get_diabetes_data loaders()`: Convenience function for complete data pipeline

## 51.2. Quick Start

### 51.2.1. Basic Usage

```
from spotoptim.data import get_diabetes_data loaders
from sklearn.datasets import load_diabetes
from spotoptim.data.diabetes import DiabetesDataset
import numpy as np

# Load data
diabetes = load_diabetes()
X = diabetes.data
y = diabetes.target.reshape(-1, 1)

# Now create the dataset
dataset = DiabetesDataset(X, y, transform=None, target_transform=None)
```

## 51. Diabetes Dataset Utilities

```
# Load data with default settings
train_loader, test_loader, scaler = get_diabetes_dataloaders()

# Iterate through batches
for batch_X, batch_y in train_loader:
    print(f"Batch features: {batch_X.shape}") # (32, 10)
    print(f"Batch targets: {batch_y.shape}")  # (32, 1)
    break
```

```
Batch features: torch.Size([32, 10])
Batch targets: torch.Size([32, 1])
```

### 51.2.2. Training a Model

```
import torch
import torch.nn as nn
from spotoptim.data import get_diabetes_dataloaders
from spotoptim.nn.linear_regressor import LinearRegressor

# Load data
train_loader, test_loader, scaler = get_diabetes_dataloaders(
    test_size=0.2,
    batch_size=32,
    scale_features=True,
    random_state=42
)

# Create model
model = LinearRegressor(
    input_dim=10,
    output_dim=1,
    l1=64,
    num_hidden_layers=2,
    activation="ReLU"
)

# Setup training
criterion = nn.MSELoss()
optimizer = model.get_optimizer("Adam", lr=0.01)

# Training loop
num_epochs = 100
```

```

for epoch in range(num_epochs):
    model.train()
    train_loss = 0.0

    for batch_X, batch_y in train_loader:
        # Forward pass
        predictions = model(batch_X)
        loss = criterion(predictions, batch_y)

        # Backward pass
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        train_loss += loss.item()

    avg_train_loss = train_loss / len(train_loader)

    if (epoch + 1) % 20 == 0:
        print(f"Epoch {epoch+1}/{num_epochs}: Loss = {avg_train_loss:.4f}")

# Evaluation
model.eval()
test_loss = 0.0

with torch.no_grad():
    for batch_X, batch_y in test_loader:
        predictions = model(batch_X)
        loss = criterion(predictions, batch_y)
        test_loss += loss.item()

avg_test_loss = test_loss / len(test_loader)
print(f"Test MSE: {avg_test_loss:.4f}")

```

```

Epoch 20/100: Loss = 32042.8337
Epoch 40/100: Loss = 27497.8241
Epoch 60/100: Loss = 29605.2087
Epoch 80/100: Loss = 27568.6883
Epoch 100/100: Loss = 28324.9264
Test MSE: 26464.6055

```

## 51.3. Function Reference

### 51.3.1. get\_diabetes\_dataloaders()

Loads the sklearn diabetes dataset and returns configured PyTorch DataLoaders.

**Signature:**

```
get_diabetes_dataloaders(
    test_size=0.2,
    batch_size=32,
    shuffle_train=True,
    shuffle_test=False,
    random_state=42,
    scale_features=True,
    num_workers=0,
    pin_memory=False
)
```

```
(<torch.utils.data.dataloader.DataLoader at 0x123243bb0>,
 <torch.utils.data.dataloader.DataLoader at 0x1233d9490>,
 StandardScaler())
```

**Parameters:**

| Parameter      | Type  | Default | Description                                    |
|----------------|-------|---------|------------------------------------------------|
| test_size      | float | 0.2     | Proportion of dataset for testing (0.0 to 1.0) |
| batch_size     | int   | 32      | Number of samples per batch                    |
| shuffle_train  | bool  | True    | Whether to shuffle training data               |
| shuffle_test   | bool  | False   | Whether to shuffle test data                   |
| random_state   | int   | 42      | Random seed for train/test split               |
| scale_features | bool  | True    | Whether to standardize features                |
| num_workers    | int   | 0       | Number of subprocesses for data loading        |
| pin_memory     | bool  | False   | Whether to pin memory (useful for GPU)         |

**Returns:**

- `train_loader` (DataLoader): Training data loader
- `test_loader` (DataLoader): Test data loader
- `scaler` (StandardScaler or None): Fitted scaler if `scale_features=True`, else None

**Example:**

```
from spotoptim.data import get_diabetes_dataaloaders

# Custom configuration
train_loader, test_loader, scaler = get_diabetes_dataaloaders(
    test_size=0.3,
    batch_size=64,
    shuffle_train=True,
    scale_features=True,
    random_state=123
)

print(f"Training batches: {len(train_loader)}")
print(f"Test batches: {len(test_loader)}")
print(f"Scaler mean: {scaler.mean_[0:3]}") # First 3 features
```

```
Training batches: 5
Test batches: 3
Scaler mean: [-0.00056537  0.00132258  0.00027836]
```

## 51.4. DiabetesDataset Class

PyTorch Dataset implementation for the diabetes dataset.

**Signature:**

```
DiabetesDataset(X, y, transform=None, target_transform=None)
```

```
<spotoptim.data.diabetes.DiabetesDataset at 0x12331aa50>
```

**Parameters:**

- `X` (np.ndarray): Feature matrix of shape (n\_samples, n\_features)
- `y` (np.ndarray): Target values of shape (n\_samples,) or (n\_samples, 1)
- `transform` (callable, optional): Transform to apply to features
- `target_transform` (callable, optional): Transform to apply to targets

## 51. Diabetes Dataset Utilities

### Attributes:

- `X` (torch.Tensor): Feature tensor (`n_samples`, `n_features`)
- `y` (torch.Tensor): Target tensor (`n_samples`, 1)
- `n_features` (int): Number of features (10 for diabetes)
- `n_samples` (int): Number of samples

### Methods:

- `__len__()`: Returns number of samples
- `__getitem__(idx)`: Returns tuple (features, target) for given index

### 51.4.1. Manual Dataset Creation

```
from spotoptim.data import DiabetesDataset
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from torch.utils.data import DataLoader

# Load raw data
diabetes = load_diabetes()
X, y = diabetes.data, diabetes.target

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Scale features
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Create datasets
train_dataset = DiabetesDataset(X_train, y_train)
test_dataset = DiabetesDataset(X_test, y_test)

# Create dataloaders
train_loader = DataLoader(train_dataset, batch_size=32, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=32, shuffle=False)

# Inspect dataset
print(f"Dataset size: {len(train_dataset)}")
```



```

print(f"Features shape: {train_dataset.X.shape}")
print(f"Targets shape: {train_dataset.y.shape}")

# Get a sample
features, target = train_dataset[0]
print(f"Sample features: {features.shape}") # (10,)
print(f"Sample target: {target.shape}")    # (1,)

```

```

Dataset size: 353
Features shape: torch.Size([353, 10])
Targets shape: torch.Size([353, 1])
Sample features: torch.Size([10])
Sample target: torch.Size([1])

```

## 51.5. Advanced Usage

### 51.5.1. Custom Transforms

```

from spotoptim.data import DiabetesDataset
from sklearn.datasets import load_diabetes
import torch

# Define custom transforms
def add_noise(x):
    """Add Gaussian noise to features."""
    return x + torch.randn_like(x) * 0.01

def log_transform(y):
    """Apply log transform to target."""
    return torch.log1p(y)

# Load data
diabetes = load_diabetes()
X, y = diabetes.data, diabetes.target

# Create dataset with transforms
dataset = DiabetesDataset(
    X, y,
    transform=add_noise,
    target_transform=log_transform
)

```

```
# Transforms are applied when accessing items
features, target = dataset[0]
```

### 51.5.2. Different Train/Test Splits

```
from spotoptim.data import get_diabetes_dataloaders

# 70/30 split
train_loader, test_loader, scaler = get_diabetes_dataloaders(
    test_size=0.3,
    random_state=42
)
print(f"Training samples: {len(train_loader.dataset)}") # ~310
print(f"Test samples: {len(test_loader.dataset)}")      # ~132

# 90/10 split
train_loader, test_loader, scaler = get_diabetes_dataloaders(
    test_size=0.1,
    random_state=42
)
print(f"Training samples: {len(train_loader.dataset)}") # ~398
print(f"Test samples: {len(test_loader.dataset)}")     # ~44
```

```
Training samples: 309
Test samples: 133
Training samples: 397
Test samples: 45
```

### 51.5.3. Without Feature Scaling

```
from spotoptim.data import get_diabetes_dataloaders

# Load without scaling (useful for tree-based models)
train_loader, test_loader, scaler = get_diabetes_dataloaders(
    scale_features=False
)

print(f"Scaler: {scaler}") # None
```

```
# Data is in original scale
for batch_X, batch_y in train_loader:
    print(f"Mean: {batch_X.mean(dim=0)[:3]}") # Non-zero values
    break
```

Scaler: None

Mean: tensor([-0.0038, -0.0089, 0.0140])

#### 51.5.4. Larger Batch Sizes

```
from spotoptim.data import get_diabetes_dataloaders

# Larger batches for faster training (if memory allows)
train_loader, test_loader, scaler = get_diabetes_dataloaders(
    batch_size=128
)
print(f"Batches per epoch: {len(train_loader)}") # Fewer batches

# Smaller batches for more gradient updates
train_loader, test_loader, scaler = get_diabetes_dataloaders(
    batch_size=8
)
print(f"Batches per epoch: {len(train_loader)}") # More batches
```

Batches per epoch: 3

Batches per epoch: 45

#### 51.5.5. GPU Training with Pin Memory

```
import torch
from spotoptim.data import get_diabetes_dataloaders

# Enable pin_memory for faster GPU transfer
train_loader, test_loader, scaler = get_diabetes_dataloaders(
    batch_size=32,
    pin_memory=True # Set to True when using GPU
)

# Move model to GPU
```

```

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)

# Training loop with GPU
for batch_X, batch_y in train_loader:
    # Data is already pinned, faster transfer to GPU
    batch_X = batch_X.to(device, non_blocking=True)
    batch_y = batch_y.to(device, non_blocking=True)

    # ... training code ...

```

/Users/bartz/workspace/spotoptim-cookbook/.venv/lib/python3.13/site-packages/torch/utils/data/dataloader.py:692: UserWarning:

'pin\_memory' argument is set as true but not supported on MPS now, device pinned memory

## 51.6. Complete Training Example

Here's a complete example showing data loading, model training, and evaluation:

```

import torch
import torch.nn as nn
from spotoptim.data import get_diabetes_dataloaders
from spotoptim.nn.linear_regressor import LinearRegressor

def train_diabetes_model():
    """Train a neural network on the diabetes dataset."""

    # Load data
    train_loader, test_loader, scaler = get_diabetes_dataloaders(
        test_size=0.2,
        batch_size=32,
        scale_features=True,
        random_state=42
    )

    # Create model
    model = LinearRegressor(
        input_dim=10,
        output_dim=1,
        l1=128,
        num_hidden_layers=3,

```

```

    activation="ReLU"
)

# Setup training
criterion = nn.MSELoss()
optimizer = model.get_optimizer("Adam", lr=0.001, weight_decay=1e-5)

# Training configuration
num_epochs = 200
best_test_loss = float('inf')

print("Starting training...")
print(f"Training samples: {len(train_loader.dataset)}")
print(f"Test samples: {len(test_loader.dataset)}")
print(f"Batches per epoch: {len(train_loader)}")
print("-" * 60)

for epoch in range(num_epochs):
    # Training phase
    model.train()
    train_loss = 0.0

    for batch_X, batch_y in train_loader:
        # Forward pass
        predictions = model(batch_X)
        loss = criterion(predictions, batch_y)

        # Backward pass
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        train_loss += loss.item()

    avg_train_loss = train_loss / len(train_loader)

    # Evaluation phase
    model.eval()
    test_loss = 0.0

    with torch.no_grad():
        for batch_X, batch_y in test_loader:
            predictions = model(batch_X)
            loss = criterion(predictions, batch_y)

```

```

        test_loss += loss.item()

    avg_test_loss = test_loss / len(test_loader)

    # Track best model
    if avg_test_loss < best_test_loss:
        best_test_loss = avg_test_loss
        # Could save model here: torch.save(model.state_dict(), 'best_model.pt')

    # Print progress
    if (epoch + 1) % 20 == 0:
        print(f"Epoch {epoch+1:3d}/{num_epochs}: "
              f"Train Loss = {avg_train_loss:.4f}, "
              f"Test Loss = {avg_test_loss:.4f}")

    print("-" * 60)
    print(f"Training complete!")
    print(f"Best test loss: {best_test_loss:.4f}")

    return model, best_test_loss

# Run training
if __name__ == "__main__":
    model, best_loss = train_diabetes_model()

```

Starting training...

Training samples: 353

Test samples: 89

Batches per epoch: 12

```

-----
Epoch 20/200: Train Loss = 30533.9668, Test Loss = 26623.0241
Epoch 40/200: Train Loss = 31571.0265, Test Loss = 26617.6934
Epoch 60/200: Train Loss = 31199.4535, Test Loss = 26612.3105
Epoch 80/200: Train Loss = 31746.5321, Test Loss = 26606.8424
Epoch 100/200: Train Loss = 28464.8737, Test Loss = 26601.3053
Epoch 120/200: Train Loss = 30835.6530, Test Loss = 26595.7839
Epoch 140/200: Train Loss = 28720.8451, Test Loss = 26590.1608
Epoch 160/200: Train Loss = 28389.3752, Test Loss = 26584.3490
Epoch 180/200: Train Loss = 27928.9941, Test Loss = 26578.3678
Epoch 200/200: Train Loss = 28284.3311, Test Loss = 26572.1152
-----

```

Training complete!

Best test loss: 26572.1152

## 51.7. Integration with SpotOptim

Use the diabetes dataset for hyperparameter optimization with SpotOptim:

```
import numpy as np
import torch
import torch.nn as nn
from spotoptim import SpotOptim
from spotoptim.data import get_diabetes_dataloaders
from spotoptim.nn.linear_regressor import LinearRegressor

def evaluate_model(X):
    """Objective function for SpotOptim.

    Args:
        X: Array of hyperparameters [lr, l1, num_hidden_layers]

    Returns:
        Array of validation losses
    """
    results = []

    for params in X:
        lr, l1, num_hidden_layers = params
        lr = 10 ** lr # Log scale for learning rate
        l1 = int(l1)
        num_hidden_layers = int(num_hidden_layers)

        # Load data
        train_loader, test_loader, _ = get_diabetes_dataloaders(
            test_size=0.2,
            batch_size=32,
            random_state=42
        )

        # Create model
        model = LinearRegressor(
            input_dim=10,
            output_dim=1,
            l1=l1,
            num_hidden_layers=num_hidden_layers,
            activation="ReLU"
        )
```

```

# Train briefly
criterion = nn.MSELoss()
optimizer = model.get_optimizer("Adam", lr=lr)

num_epochs = 50
for epoch in range(num_epochs):
    model.train()
    for batch_X, batch_y in train_loader:
        predictions = model(batch_X)
        loss = criterion(predictions, batch_y)
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

# Evaluate
model.eval()
test_loss = 0.0
with torch.no_grad():
    for batch_X, batch_y in test_loader:
        predictions = model(batch_X)
        loss = criterion(predictions, batch_y)
        test_loss += loss.item()

results.append(test_loss / len(test_loader))

return np.array(results)

# Optimize hyperparameters
optimizer = SpotOptim(
    fun=evaluate_model,
    bounds=[
        (-4, -2), # log10(lr): 0.0001 to 0.01
        (16, 128), # l1: number of neurons
        (0, 4) # num_hidden_layers
    ],
    var_type=["float", "int", "int"],
    max_iter=30,
    n_initial=10,
    seed=42,
    verbose=True
)

result = optimizer.optimize()
print(f"Best hyperparameters found:")

```



```
print(f"  Learning rate: {10**result.x[0]:.6f}")
print(f"  Hidden neurons (l1): {int(result.x[1])}")
print(f"  Hidden layers: {int(result.x[2])}")
print(f"  Best MSE: {result.fun:.4f}")
```

TensorBoard logging disabled

Initial best:  $f(x) = 26550.207682$

Iteration 1:  $f(x) = 26626.213542$

Iteration 2:  $f(x) = 26610.039714$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Iteration 3:  $f(x) = 26595.690104$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Iteration 4:  $f(x) = 26589.477214$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Iteration 5: New best  $f(x) = 26524.926432$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Iteration 6:  $f(x) = 26644.966146$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Iteration 7:  $f(x) = 26607.319010$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Iteration 8:  $f(x) = 26637.518880$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Iteration 9:  $f(x) = 26638.514323$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Iteration 10:  $f(x) = 26593.592448$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Iteration 11:  $f(x) = 26606.908854$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Iteration 12:  $f(x) = 26587.811849$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Iteration 13:  $f(x) = 26553.845703$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

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Iteration 14:  $f(x) = 26598.708333$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Iteration 15:  $f(x) = 26579.172526$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Iteration 16:  $f(x) = 26624.829427$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Iteration 17:  $f(x) = 26622.476562$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Iteration 18:  $f(x) = 26636.386068$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Iteration 19:  $f(x) = 26591.287109$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Iteration 20:  $f(x) = 26624.940755$

Best hyperparameters found:

Learning rate: 0.000468

Hidden neurons (l1): 49

Hidden layers: 1

Best MSE: 26524.9264

## 51.8. Best Practices

### 51.8.1. 1. Always Use Feature Scaling

```
# Good: Features are standardized
train_loader, test_loader, scaler = get_diabetes_data loaders(
    scale_features=True
)
```

Neural networks typically perform better with normalized inputs.

### 51.8.2. 2. Set Random Seeds for Reproducibility

```
# Reproducible train/test splits
train_loader, test_loader, scaler = get_diabetes_dataloaders(
    random_state=42
)

# Also set PyTorch seed
import torch
torch.manual_seed(42)
```

```
<torch._C.Generator at 0x11cc2e8d0>
```

### 51.8.3. 3. Don't Shuffle Test Data

```
# Good: Test data in consistent order
train_loader, test_loader, scaler = get_diabetes_dataloaders(
    shuffle_train=True,    # Shuffle training data
    shuffle_test=False     # Don't shuffle test data
)
```

This ensures consistent evaluation metrics across runs.

### 51.8.4. 4. Choose Appropriate Batch Size

```
# Small dataset (442 samples) - moderate batch size works well
train_loader, test_loader, scaler = get_diabetes_dataloaders(
    batch_size=32 # Good balance for this dataset
)
```

Too large: Fewer gradient updates per epoch

Too small: Noisy gradients, slower training

### 51.8.5. 5. Save the Scaler for Production

```
import pickle
import numpy as np
from spotoptim.data import get_diabetes_dataloaders
```

```
# Train with scaling
train_loader, test_loader, scaler = get_diabetes_data loaders(
    scale_features=True
)

# Save scaler for production use
with open('scaler.pkl', 'wb') as f:
    pickle.dump(scaler, f)

# Later: Load and use on new data
with open('scaler.pkl', 'rb') as f:
    loaded_scaler = pickle.load(f)

# Create some example new data (same shape as diabetes features)
new_data = np.random.randn(5, 10) # 5 samples, 10 features
new_data_scaled = loaded_scaler.transform(new_data)

print(f"Original data shape: {new_data.shape}")
print(f"Scaled data shape: {new_data_scaled.shape}")
print(f"Scaled data mean: {new_data_scaled.mean(axis=0)[:3]}") # Should be close to 0
```

```
Original data shape: (5, 10)
Scaled data shape: (5, 10)
Scaled data mean: [ 9.93546654 -8.09215046  1.00228167]
```

## 51.9. Troubleshooting

### 51.9.1. Issue: Out of Memory

**Solution:** Reduce batch size or disable pin\_memory

```
train_loader, test_loader, scaler = get_diabetes_data loaders(
    batch_size=16, # Smaller batches
    pin_memory=False # Disable if not using GPU
)
```

### 51.9.2. Issue: Different Data Ranges

**Symptom:** Model not converging, loss is NaN

**Solution:** Ensure feature scaling is enabled

```
train_loader, test_loader, scaler = get_diabetes_dataloaders(
    scale_features=True # Must be True for neural networks
)
```

### 51.9.3. Issue: Non-Reproducible Results

**Solution:** Set all random seeds

```
import torch
import numpy as np

# Set all seeds
torch.manual_seed(42)
np.random.seed(42)

train_loader, test_loader, scaler = get_diabetes_dataloaders(
    random_state=42,
    shuffle_train=False # Disable shuffle for full reproducibility
)
```

### 51.9.4. Issue: Slow Data Loading

**Solution:** Use multiple workers (if not on Windows)

```
train_loader, test_loader, scaler = get_diabetes_dataloaders(
    num_workers=4, # Use 4 subprocesses
    pin_memory=True # Enable for GPU
)
```

Note: On Windows, set `num_workers=0` to avoid multiprocessing issues.

## 51.10. Summary

The diabetes dataset utilities in SpotOptim provide:

- **Easy data loading:** One function call gets complete data pipeline
- **PyTorch integration:** Native Dataset and DataLoader support
- **Preprocessing included:** Automatic feature scaling and train/test splitting
- **Flexible configuration:** Control batch size, splitting, scaling, and more
- **Production ready:** Save scalers and ensure reproducibility

## 51. *Diabetes Dataset Utilities*

For more examples, see: - `examples/diabetes_dataset_example.py` - `notebooks/demos.ipynb`  
- Test suite: `tests/test_diabetes_dataset.py`

## 52. Physics-Informed Neural Networks (PINNs) Demo 1

Solving ODEs with SpotOptim's LinearRegressor





## 53. Overview

This tutorial demonstrates how to use Physics-Informed Neural Networks (PINNs) to solve ordinary differential equations (ODEs) using SpotOptim's **LinearRegressor** class. We'll solve a first-order ODE with an initial condition, combining data-driven learning with physics constraints.

### 53.1. The Differential Equation

We want to solve the following ODE:

$$\frac{dy}{dt} + 0.1y - \sin\left(\frac{\pi t}{2}\right) = 0$$

with initial condition:

$$y(0) = 0$$



## 54. Setup

First, let's import the necessary libraries:

```
import matplotlib.pyplot as plt
import numpy as np
import torch
import torch.nn as nn
from typing import Tuple
from spoptim.nn.linear_regressor import LinearRegressor

# Set random seed for reproducibility
torch.manual_seed(123)
```

```
<torch._C.Generator at 0x116bc6250>
```



## 55. The Neural Network

We'll use SpotOptim's `LinearRegressor` class to create a neural network with:

- 1 input (time  $t$ )
- 1 output (solution  $y$ )
- 32 neurons per hidden layer
- 3 hidden layers
- Tanh activation function

```
model = LinearRegressor(  
    input_dim=1,  
    output_dim=1,  
    l1=32,  
    num_hidden_layers=3,  
    activation="Tanh",  
    lr=1.0  
)  
  
# Get optimizer with custom learning rate  
optimizer = model.get_optimizer("Adam", lr=3.0) # 3.0 * 0.001 = 0.003  
  
print(f"Model architecture:")  
print(model)
```

Model architecture:

```
LinearRegressor(  
  (activation): Tanh()  
  (network): Sequential(  
    (0): Linear(in_features=1, out_features=32, bias=True)  
    (1): Tanh()  
    (2): Linear(in_features=32, out_features=32, bias=True)  
    (3): Tanh()  
    (4): Linear(in_features=32, out_features=32, bias=True)  
    (5): Tanh()  
    (6): Linear(in_features=32, out_features=1, bias=True)  
  )  
)
```



## 56. Generate Training Data

We'll generate exact solution data using a numerical solver (RK2 method):

```
def oscillator(
    n_steps: int = 3000,
    t_min: float = 0.0,
    t_max: float = 30.0,
    y0: float = 0.0,
    alpha: float = 0.1,
    omega: float = np.pi / 2
) -> Tuple[torch.Tensor, torch.Tensor]:
    """
    Solve ODE:  $dy/dt + \alpha y - \sin(\omega t) = 0$ 
    using RK2 (midpoint method).

    Args:
        n_steps: Number of time steps
        t_min: Start time
        t_max: End time
        y0: Initial condition  $y(t_{\min})$ 
        alpha: Damping coefficient
        omega: Forcing frequency

    Returns:
        Tuple of (time points, solution values) as torch tensors
    """
    t_step = (t_max - t_min) / n_steps
    t_points = np.arange(t_min, t_min + n_steps * t_step, t_step)[:n_steps]

    y = [y0]

    for t_current_step_end in t_points[1:]:
        t_midpoint = t_current_step_end - t_step / 2.0
        y_prev = y[-1]

        # Stage 1: intermediate value
        slope_at_t_mid = -alpha * y_prev + np.sin(omega * t_midpoint)
```

## 56. Generate Training Data

```
y_intermediate = y_prev + (t_step / 2.0) * slope_at_t_mid

# Stage 2: final value
slope_at_t_end = -alpha * y_intermediate + np.sin(omega * t_current_step_end)
y_next = y_prev + t_step * slope_at_t_end
y.append(y_next)

t_tensor = torch.tensor(t_points, dtype=torch.float32).view(-1, 1)
y_tensor = torch.tensor(y, dtype=torch.float32).view(-1, 1)

return t_tensor, y_tensor
```

Generate the exact solution and sample training data points:

```
# Generate exact solution (3000 points)
x, y = oscillator()

# Sample training data (every 119th point, giving ~25 points)
x_data = x[0:3000:119]
y_data = y[0:3000:119]

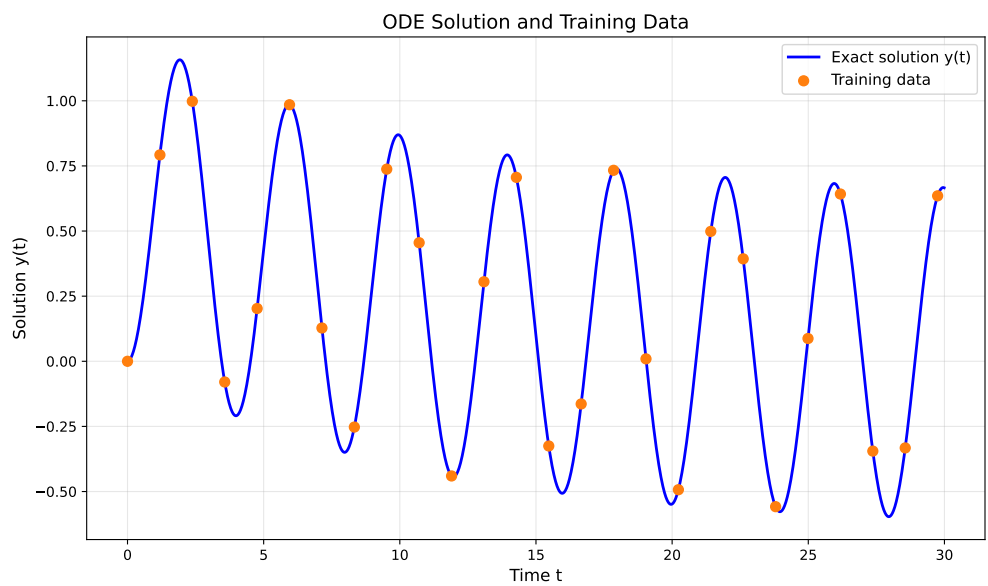
print(f"Exact solution points: {len(x)}")
print(f"Training data points: {len(x_data)}")
```

Exact solution points: 3000  
Training data points: 26

Visualize the exact solution and training data:

```
plt.figure(figsize=(10, 6))
plt.plot(x.numpy(), y.numpy(), 'b-', linewidth=2, label="Exact solution y(t)")
plt.scatter(x_data.numpy(), y_data.numpy(), color="tab:orange",
            s=50, label="Training data", zorder=5)
plt.xlabel("Time t", fontsize=12)
plt.ylabel("Solution y(t)", fontsize=12)
plt.title("ODE Solution and Training Data", fontsize=14)
plt.legend(fontsize=11)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```







## 57. Collocation Points

Create collocation points where we'll enforce the physics (ODE) constraints:

```
# 50 evenly spaced points in [0, 30]
x_physics = torch.linspace(0, 30, 50).view(-1, 1).requires_grad_(True)

print(f"Collocation points: {len(x_physics)}")
print(f"Range: [{x_physics.min().item():.2f}, {x_physics.max().item():.2f}]" )
```

```
Collocation points: 50
Range: [0.00, 30.00]
```



## 58. PINN Training

Now we'll train the neural network using two loss components:

1. **Data Loss (Loss1)**: Ensures the network fits the training data
2. **Physics Loss (Loss2)**: Ensures the network satisfies the ODE

### 58.1. Loss Components Explained

#### 58.1.1. Neural Network Prediction on Data Points

The network predicts values at the training data points:

```
yh = model(x_data)
```

#### 58.1.2. Data Loss (Loss1)

Mean squared error between predictions and actual data:

```
loss1 = torch.mean((yh - y_data)**2)
```

#### 58.1.3. Neural Network Prediction on Collocation Points

The network predicts values at the physics enforcement points:

```
yhp = model(x_physics)
```

#### 58.1.4. Computing Derivatives for the ODE

We compute  $dy/dt$  using automatic differentiation:

```
dyhp_dxphysics = torch.autograd.grad(
    yhp, x_physics,
    torch.ones_like(yhp),
    create_graph=True
)[0]
```

### 58.1.5. Physics Loss (Loss2)

The ODE residual at collocation points:

$$\text{residual} = \frac{dy}{dt} + 0.1y - \sin\left(\frac{\pi t}{2}\right)$$

```
physics = dyhp_dxphysics + 0.1 * yhp - torch.sin(np.pi * x_physics / 2)
loss2 = torch.mean(physics**2)
```

### 58.1.6. Total Loss

Combined loss with weighting factor :

```
loss = loss1 + alpha * loss2
```

## 58.2. Training Loop

```
loss_history_pinn = []
loss2_history_pinn = []
plot_data_points_pinn = []

alpha = 6e-2 # Weight for physics loss
n_epochs = 48000

print("Training Physics-Informed Neural Network...")
print(f"Total epochs: {n_epochs}")
print(f"Physics loss weight (alpha): {alpha}")

for i in range(n_epochs):
    optimizer.zero_grad()

    # Data Loss: Fit the training data
```

```

yh = model(x_data)
loss1 = torch.mean((yh - y_data)**2)

# Physics Loss: Satisfy the ODE at collocation points
yhp = model(x_physics)
dyhp_dxphysics = torch.autograd.grad(
    yhp, x_physics,
    torch.ones_like(yhp),
    create_graph=True
)[0]
physics = dyhp_dxphysics + 0.1 * yhp - torch.sin(np.pi * x_physics / 2)
loss2 = torch.mean(physics**2)

# Total Loss
loss = loss1 + alpha * loss2
loss.backward()
optimizer.step()

# Store history every 100 steps
if (i + 1) % 100 == 0:
    loss_history_pinn.append(loss.detach())
    loss2_history_pinn.append(loss2.detach())

# Store snapshots for visualization every 10000 steps
if (i + 1) % 10000 == 0:
    current_yh_full = model(x).detach()
    plot_data_points_pinn.append({
        'yh': current_yh_full,
        'step': i + 1
    })
    print(f"Epoch {i+1}/{n_epochs}: Total Loss = {loss.item():.6f}, "
          f"Physics Loss = {loss2.item():.6f}")

print("Training completed!")

```

```

Training Physics-Informed Neural Network...
Total epochs: 48000
Physics loss weight (alpha): 0.06
Epoch 10000/48000: Total Loss = 0.000119, Physics Loss = 0.001705
Epoch 20000/48000: Total Loss = 0.000035, Physics Loss = 0.000473
Epoch 30000/48000: Total Loss = 0.000014, Physics Loss = 0.000225
Epoch 40000/48000: Total Loss = 0.000181, Physics Loss = 0.000391
Training completed!

```





## 59. Results Visualization

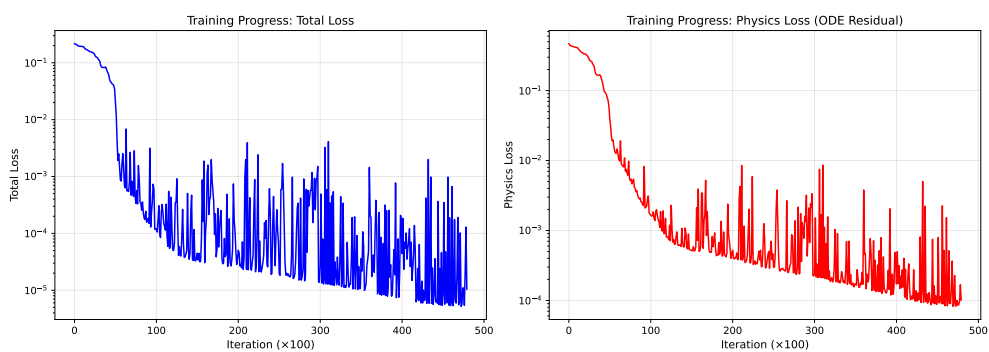
### 59.1. Training Progress

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

# Plot total loss
ax1.plot(loss_history_pinn, 'b-', linewidth=1.5)
ax1.set_xlabel('Iteration (×100)', fontsize=11)
ax1.set_ylabel('Total Loss', fontsize=11)
ax1.set_title('Training Progress: Total Loss', fontsize=12)
ax1.grid(True, alpha=0.3)
ax1.set_yscale('log')

# Plot physics loss
ax2.plot(loss2_history_pinn, 'r-', linewidth=1.5)
ax2.set_xlabel('Iteration (×100)', fontsize=11)
ax2.set_ylabel('Physics Loss', fontsize=11)
ax2.set_title('Training Progress: Physics Loss (ODE Residual)', fontsize=12)
ax2.grid(True, alpha=0.3)
ax2.set_yscale('log')

plt.tight_layout()
plt.show()
```



## 59.2. Solution Evolution During Training

Visualize how the neural network solution evolves during training:

```

xp_plot = x_physics.detach()

for plot_info in plot_data_points_pinn:
    plt.figure(figsize=(10, 6))

    # Plot exact solution
    plt.plot(x.numpy(), y.numpy(), 'b-', linewidth=2,
             label='Exact solution', alpha=0.7)

    # Plot training data
    plt.scatter(x_data.numpy(), y_data.numpy(), color='tab:orange',
               s=50, label='Training data', zorder=5)

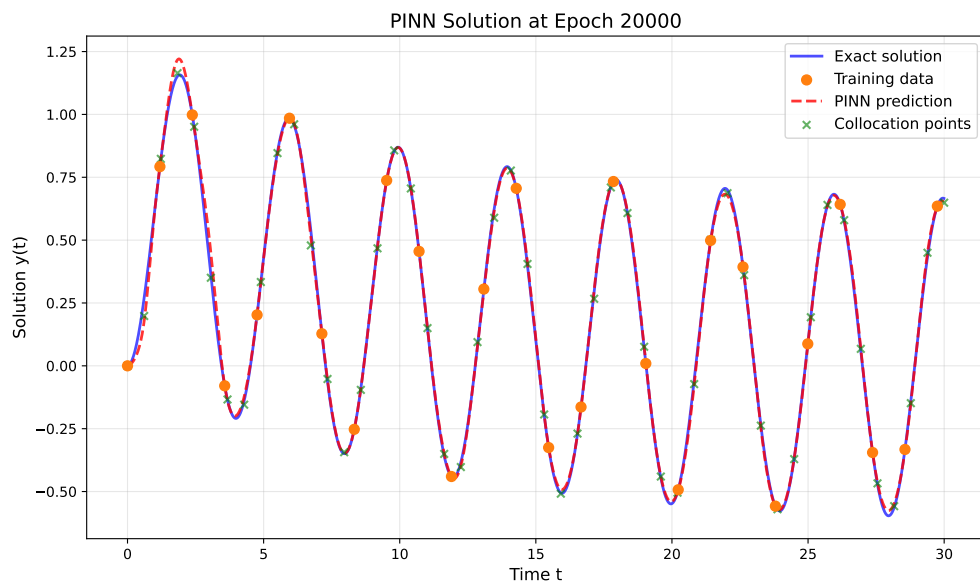
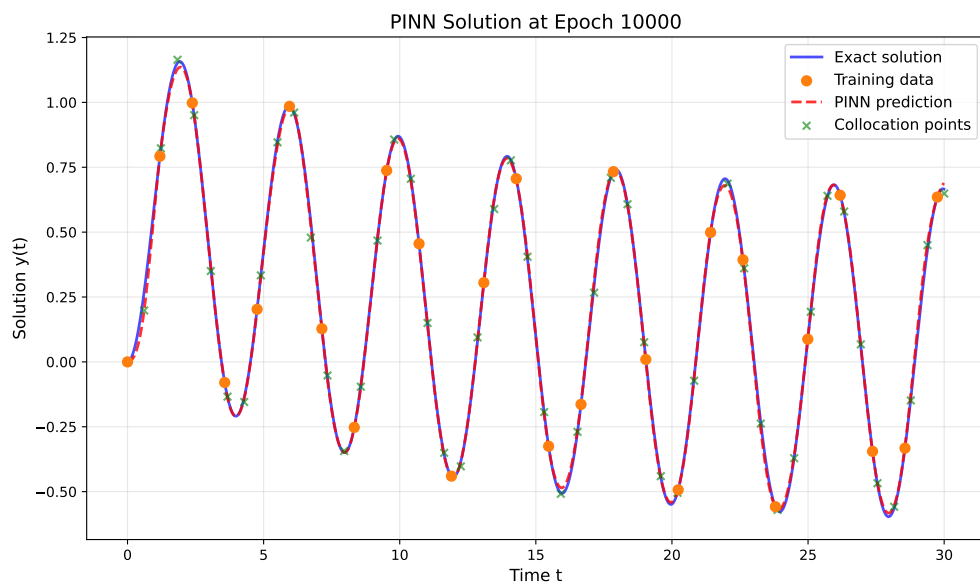
    # Plot neural network prediction
    plt.plot(x.numpy(), plot_info['yh'].numpy(), 'r--', linewidth=2,
             label='PINN prediction', alpha=0.8)

    # Plot collocation points
    plt.scatter(xp_plot.numpy(),
               model(xp_plot).detach().numpy(),
               color='green', marker='x', s=30,
               label='Collocation points', alpha=0.6, zorder=4)

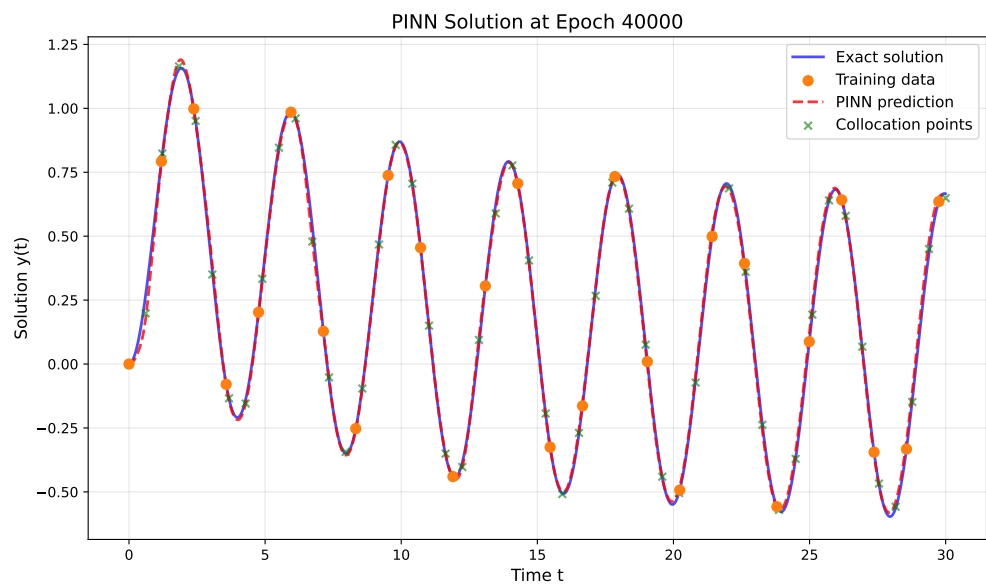
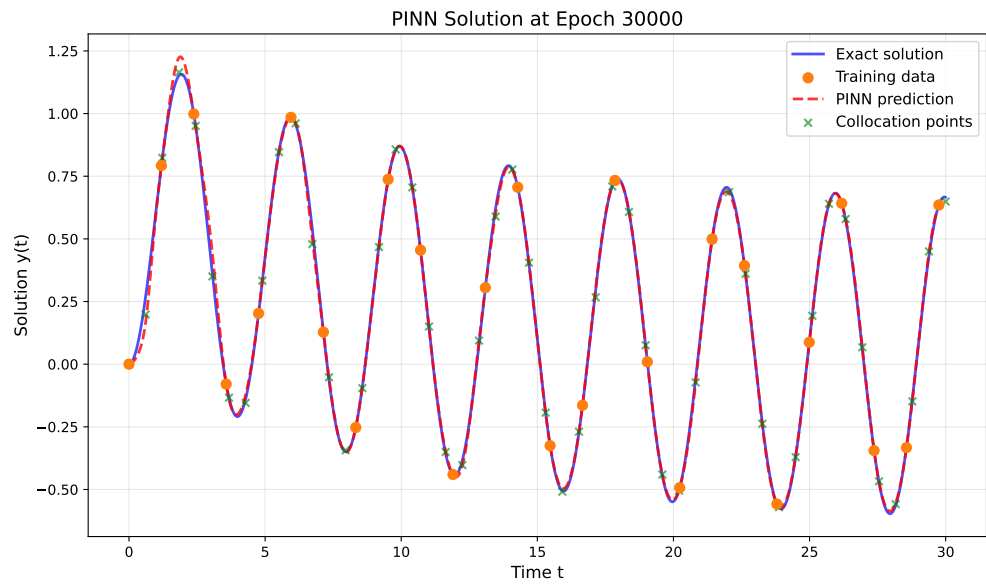
    plt.xlabel('Time t', fontsize=12)
    plt.ylabel('Solution y(t)', fontsize=12)
    plt.title(f'PINN Solution at Epoch {plot_info["step"]}', fontsize=14)
    plt.legend(fontsize=11)
    plt.grid(True, alpha=0.3)
    plt.tight_layout()
    plt.show()

```

## 59.2. Solution Evolution During Training



## 59. Results Visualization



### 59.3. Final Solution Comparison

```

# Final prediction
model.eval()
with torch.no_grad():
    y_final = model(x)

plt.figure(figsize=(12, 7))

# Plot exact solution
plt.plot(x.numpy(), y.numpy(), 'b-', linewidth=2.5,
         label='Exact solution', alpha=0.8)

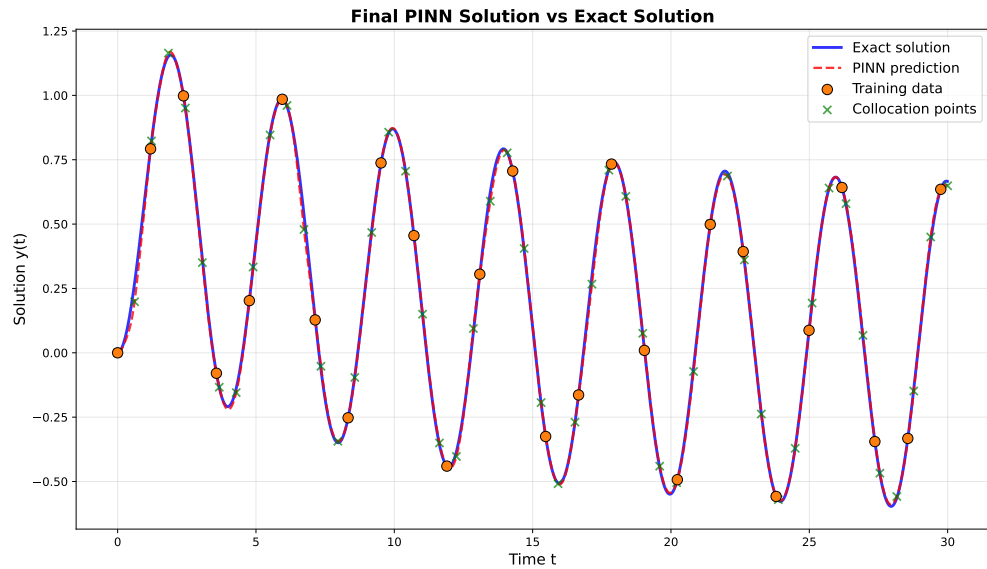
# Plot PINN prediction
plt.plot(x.numpy(), y_final.numpy(), 'r--', linewidth=2,
         label='PINN prediction', alpha=0.8)

# Plot training data
plt.scatter(x_data.numpy(), y_data.numpy(), color='tab:orange',
           s=80, label='Training data', zorder=5, edgecolors='black', linewidth=0.5)

# Plot collocation points
plt.scatter(x_physics.detach().numpy(),
           model(x_physics).detach().numpy(),
           color='green', marker='x', s=50,
           label='Collocation points', alpha=0.7, zorder=4)

plt.xlabel('Time t', fontsize=13)
plt.ylabel('Solution y(t)', fontsize=13)
plt.title('Final PINN Solution vs Exact Solution', fontsize=15, fontweight='bold')
plt.legend(fontsize=12, loc='best')
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

```



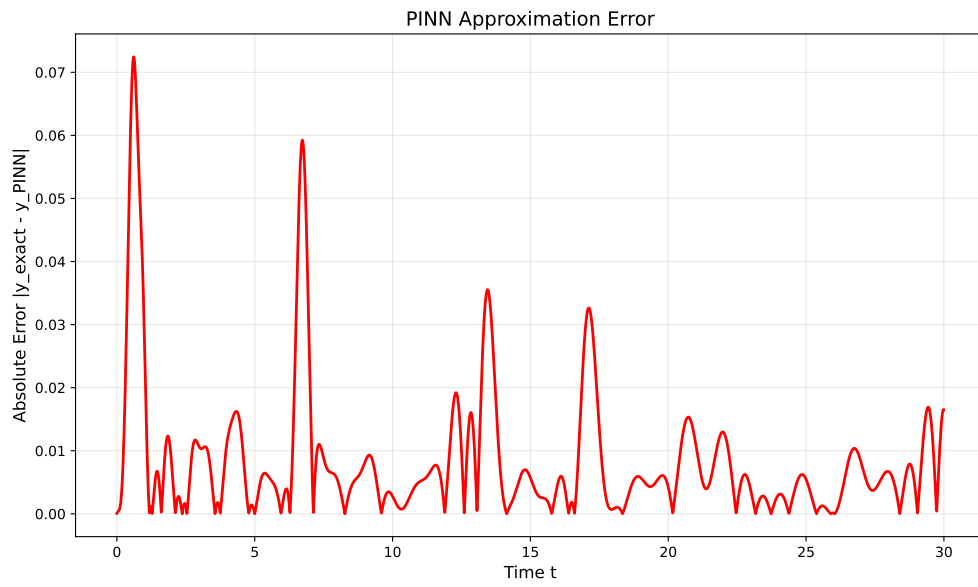
## 59.4. Error Analysis

```
# Compute absolute error
error = torch.abs(y_final - y)

plt.figure(figsize=(10, 6))
plt.plot(x.numpy(), error.numpy(), 'r-', linewidth=2)
plt.xlabel('Time t', fontsize=12)
plt.ylabel('Absolute Error |y_exact - y_PINN|', fontsize=12)
plt.title('PINN Approximation Error', fontsize=14)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

print("\nError Statistics:")
print(f"Maximum absolute error: {error.max().item():.6f}")
print(f"Mean absolute error: {error.mean().item():.6f}")
print(f"Root mean squared error: {torch.sqrt(torch.mean(error**2)).item():.6f}")
```

#### 59.4. Error Analysis



Error Statistics:

Maximum absolute error: 0.072458

Mean absolute error: 0.008300

Root mean squared error: 0.013641





## 60. Summary

This tutorial demonstrated how to use SpotOptim's `LinearRegressor` class to implement a Physics-Informed Neural Network (PINN) for solving ordinary differential equations. Key takeaways:

1. **Network Architecture:** Used a 3-layer neural network with 32 neurons per layer and Tanh activation
2. **Dual Loss Function:** Combined data fitting loss with physics constraint loss
3. **Automatic Differentiation:** Leveraged PyTorch's autograd to compute derivatives for the ODE
4. **Collocation Method:** Enforced the ODE at specific points in the domain
5. **Training Strategy:** Balanced data-driven and physics-informed learning with weight

The PINN successfully learned to approximate the ODE solution using only sparse training data (~25 points) by incorporating the underlying physics through the differential equation constraint.

### 60.1. Key Advantages of PINNs

- **Data Efficiency:** Can learn with very few data points
- **Physics Consistency:** Solutions automatically satisfy the governing equations
- **Generalization:** Better extrapolation beyond training data
- **Flexibility:** Can handle complex geometries and boundary conditions

### 60.2. Using SpotOptim for Hyperparameter Optimization

The `LinearRegressor` class integrates seamlessly with SpotOptim for hyperparameter tuning:

```
from spotoptim import SpotOptim

def train_pinn(X):
    """Objective function for hyperparameter optimization."""
    results = []
```

## 60. Summary

```
for params in X:
    l1 = int(params[0])          # Hidden layer size
    num_layers = int(params[1])  # Number of layers
    lr_unified = 10 ** params[2] # Learning rate (log scale)

    # Create model
    model = LinearRegressor(
        input_dim=1, output_dim=1,
        l1=l1, num_hidden_layers=num_layers,
        activation="Tanh", lr=lr_unified
    )

    # Train PINN and compute validation error
    # ... training code ...

    results.append(validation_error)
return np.array(results)

# Optimize hyperparameters
optimizer = SpotOptim(
    fun=train_pinn,
    bounds=[(16, 128), (1, 5), (-4, 0)],
    var_type=["int", "int", "float"],
    var_name=["layer_size", "num_layers", "log_lr"],
    max_iter=50
)
result = optimizer.optimize()
```

This approach allows you to systematically find the best network architecture and learning rate for your specific PINN problem.

# **61. Hyperparameter Tuning for Physics-Informed Neural Networks**

Using SpotOptim to Optimize PINN Architecture and Training



## 62. Overview

This tutorial demonstrates how to use SpotOptim for hyperparameter optimization of Physics-Informed Neural Networks (PINNs). We'll optimize the network architecture and training parameters to find the best configuration for solving an ordinary differential equation.

Building on the basic PINN demo, we'll now systematically search for optimal:

- Number of neurons per hidden layer
- Number of hidden layers
- Activation function (categorical)
- Optimizer algorithm (categorical)
- Learning rate (log-scale)
- Physics loss weight (log-scale)

### 62.1. Key Features

#### 62.1.1. 1. PyTorch Dataset and DataLoader

Following PyTorch best practices from the official tutorial, this tutorial implements:

- **Custom Dataset Classes:** Separate classes for supervised data (`PINNDataset`) and collocation points (`CollocationDataset`)
- **DataLoader Integration:** Efficient batch processing with configurable batch size, shuffling, and parallel loading
- **Proper Data Separation:** Clean separation of training, validation, and collocation data
- **Gradient Tracking:** Automatic gradient handling for collocation points needed in physics loss

Benefits:

- **Modularity:** Clean separation between data and model code
- **Efficiency:** Batch processing and optional parallel data loading
- **Scalability:** Easy to extend to larger datasets
- **Best Practices:** Follows PyTorch conventions used across the ecosystem

### 62.1.2. 2. Automatic Transformation Handling

This tutorial also showcases SpotOptim's `var_trans` feature for automatic variable transformations. Learning rates and regularization parameters are often best explored on a log scale, but manually transforming values is tedious and error-prone. With `var_trans`, you simply specify:

```
var_trans = [None, None, "log10", "log10"]
```

SpotOptim then:

- Optimizes internally in log-transformed space (efficient exploration)
- Passes original-scale values to your objective function (no manual conversion needed)
- Displays all results in original scale (easy interpretation)

This eliminates the need for manual `10**x` conversions throughout your code!

## 63. The Problem

We're solving the same ODE as in the basic PINN demo:

$$\frac{dy}{dt} + 0.1y - \sin\left(\frac{\pi t}{2}\right) = 0$$

with initial condition  $y(0) = 0$ .





## 64. Setup

```
import matplotlib.pyplot as plt
import numpy as np
import torch
import torch.nn as nn
from typing import Tuple
from spotoptim import SpotOptim
from spotoptim.nn.linear_regressor import LinearRegressor

# Set random seed for reproducibility
torch.manual_seed(42)
np.random.seed(42)

# Set number of epochs for training
N_EPOCHS=5000

#
MAX_ITER = 50
```



## 65. Data Generation

Following PyTorch best practices, we'll create custom Dataset classes for our PINN data.

### 65.1. Custom Dataset Classes

We'll create two dataset types:

1. PINNDataset for supervised data (training and validation)
2. CollocationDataset for physics-informed collocation points

```
from torch.utils.data import Dataset, DataLoader

def oscillator(
    n_steps: int = 3000,
    t_min: float = 0.0,
    t_max: float = 30.0,
    y0: float = 0.0,
    alpha: float = 0.1,
    omega: float = np.pi / 2
) -> Tuple[torch.Tensor, torch.Tensor]:
    """
    Solve ODE:  $dy/dt + \alpha y - \sin(\omega t) = 0$ 
    using RK2 (midpoint method).

    Returns:
        t_tensor: Time points, shape (n_steps, 1)
        y_tensor: Solution values, shape (n_steps, 1)
    """
    t_step = (t_max - t_min) / n_steps
    t_points = np.arange(t_min, t_min + n_steps * t_step, t_step)[:n_steps]

    y = [y0]

    for t_current_step_end in t_points[1:]:
        t_midpoint = t_current_step_end - t_step / 2.0
```

```

        y_prev = y[-1]

        slope_at_t_mid = -alpha * y_prev + np.sin(omega * t_midpoint)
        y_intermediate = y_prev + (t_step / 2.0) * slope_at_t_mid

        slope_at_t_end = -alpha * y_intermediate + np.sin(omega * t_current_step_end)
        y_next = y_prev + t_step * slope_at_t_end
        y.append(y_next)

    t_tensor = torch.tensor(t_points, dtype=torch.float32).view(-1, 1)
    y_tensor = torch.tensor(y, dtype=torch.float32).view(-1, 1)

    return t_tensor, y_tensor

class PINNDataset(Dataset):
    """PyTorch Dataset for PINN supervised data (training/validation).

    This dataset stores time-solution pairs (t, y) for supervised learning.

    Args:
        t (torch.Tensor): Time points, shape (n_samples, 1)
        y (torch.Tensor): Solution values, shape (n_samples, 1)
    """

    def __init__(self, t: torch.Tensor, y: torch.Tensor):
        self.t = t
        self.y = y

    def __len__(self) -> int:
        return len(self.t)

    def __getitem__(self, idx: int) -> Tuple[torch.Tensor, torch.Tensor]:
        return self.t[idx], self.y[idx]

class CollocationDataset(Dataset):
    """PyTorch Dataset for PINN collocation points.

    This dataset stores time points where physics loss is evaluated.
    Gradients are required for computing derivatives in the PDE.

    Args:
        t (torch.Tensor): Collocation time points, shape (n_points, 1)

```

```

"""

def __init__(self, t: torch.Tensor):
    # Store collocation points with gradient tracking
    self.t = t.requires_grad_(True)

def __len__(self) -> int:
    return len(self.t)

def __getitem__(self, idx: int) -> torch.Tensor:
    # Return single collocation point (still requires_grad)
    return self.t[idx].unsqueeze(0)

```

Generate exact solution using RK2

```

x_exact, y_exact = oscillator()

# Create training data (sparse sampling)
t_train = x_exact[0:3000:119]
y_train = y_exact[0:3000:119]

# Create validation data (different sampling for unbiased evaluation)
t_val = x_exact[50:3000:120]
y_val = y_exact[50:3000:120]

# Create collocation points for physics loss
t_physics = torch.linspace(0, 30, 50).view(-1, 1)

# Create Dataset objects
train_dataset = PINNDataset(t_train, y_train)
val_dataset = PINNDataset(t_val, y_val)
collocation_dataset = CollocationDataset(t_physics)

print(f"Training dataset size: {len(train_dataset)}")
print(f"Validation dataset size: {len(val_dataset)}")
print(f"Collocation dataset size: {len(collocation_dataset)}")
print(f"\nSample from training dataset:")
t_sample, y_sample = train_dataset[0]
print(f"  t: {t_sample.item():.4f}, y: {y_sample.item():.4f}")

```

```

Training dataset size: 26
Validation dataset size: 25
Collocation dataset size: 50

```

## 65. *Data Generation*

```
Sample from training dataset:  
  t: 0.0000, y: 0.0000
```

## 66. Define the PINN Training Function

This function creates DataLoaders and trains a PINN with given hyperparameters. Following PyTorch best practices, we use DataLoader for efficient batch processing.

```
def train_pinn(
    l1: int,
    num_layers: int,
    activation: str,
    optimizer_name: str,
    lr_unified: float,
    alpha: float,
    n_epochs: int = N_EPOCHS,
    batch_size: int = 16,
    verbose: bool = False
) -> float:
    """
    Train a PINN with specified hyperparameters using DataLoaders.

    Args:
        l1: Number of neurons per hidden layer
        num_layers: Number of hidden layers
        activation: Activation function ("Tanh", "ReLU", "Sigmoid", "GELU")
        optimizer_name: Optimizer algorithm ("Adam", "SGD", "RMSprop", "AdamW")
        lr_unified: Unified learning rate multiplier
        alpha: Weight for physics loss
        n_epochs: Number of training epochs
        batch_size: Batch size for DataLoader
        verbose: Whether to print progress

    Returns:
        Validation mean squared error
    """
    # Set seed for reproducibility
    torch.manual_seed(42)

    # Create DataLoaders
    train_loader = DataLoader(
```

## 66. Define the PINN Training Function

```
        train_dataset,
        batch_size=batch_size,
        shuffle=True,
        num_workers=0,
        pin_memory=False
    )

    val_loader = DataLoader(
        val_dataset,
        batch_size=batch_size,
        shuffle=False,
        num_workers=0,
        pin_memory=False
    )

    # For collocation points, we can use full batch since it's small
    collocation_loader = DataLoader(
        collocation_dataset,
        batch_size=len(collocation_dataset),
        shuffle=False,
        num_workers=0
    )

    # Create model
    model = LinearRegressor(
        input_dim=1,
        output_dim=1,
        l1=l1,
        num_hidden_layers=num_layers,
        activation=activation,
        lr=lr_unified
    )

    # Get optimizer
    optimizer = model.get_optimizer(optimizer_name)

    # Training loop
    for epoch in range(n_epochs):
        model.train()
        epoch_loss = 0.0

        # Get collocation points (full batch)
        t_physics_batch = next(iter(collocation_loader))
        # Ensure gradients are enabled
```



```

t_physics_batch = t_physics_batch.requires_grad_(True)

# Iterate over training batches
for batch_t, batch_y in train_loader:
    optimizer.zero_grad()

    # Data Loss
    y_pred = model(batch_t)
    loss_data = torch.mean((y_pred - batch_y)**2)

    # Physics Loss (computed on full collocation set)
    y_physics = model(t_physics_batch)
    dy_dt = torch.autograd.grad(
        y_physics,
        t_physics_batch,
        torch.ones_like(y_physics),
        create_graph=True,
        retain_graph=True
    )[0]

    # PDE residual: dy/dt + 0.1*y - sin(pi*t/2) = 0
    physics_residual = dy_dt + 0.1 * y_physics - torch.sin(np.pi * t_physics_batch / 2)
    loss_physics = torch.mean(physics_residual**2)

    # Total Loss
    loss = loss_data + alpha * loss_physics
    loss.backward()
    optimizer.step()

    epoch_loss += loss.item()

if verbose and (epoch + 1) % 2000 == 0:
    avg_loss = epoch_loss / len(train_loader)
    print(f" Epoch {epoch+1}/{n_epochs}: Avg Loss = {avg_loss:.6f}")

# Evaluate on validation set
model.eval()
val_mse = 0.0
with torch.no_grad():
    for batch_t, batch_y in val_loader:
        y_pred = model(batch_t)
        val_mse += torch.mean((batch_y - y_pred)**2).item()

val_mse /= len(val_loader)

```

## 66. Define the PINN Training Function

```
        return val_mse

# Test the function with default parameters
print("Testing PINN training function with DataLoaders...")
test_error = train_pinn(
    l1=32,
    num_layers=2,
    activation="Tanh",
    optimizer_name="Adam",
    lr_unified=3.0,
    alpha=0.06,
    batch_size=16,
    n_epochs=N_EPOCHS,
    verbose=True
)
print(f"\nTest validation MSE: {test_error:.6f}")
```

Testing PINN training function with DataLoaders...

Epoch 2000/5000: Avg Loss = 0.185009

Epoch 4000/5000: Avg Loss = 0.153585

Test validation MSE: 0.130621

## 67. Hyperparameter Optimization with SpotOptim

Now we'll use SpotOptim to find the best hyperparameters:

### 67.1. Define the Objective Function

```
def objective_pinn(X):
    """
    Objective function for SpotOptim.

    Args:
        X: Array of hyperparameter configurations, shape (n_configs, 6)
        Each row: [l1, num_layers, activation, optimizer, lr_unified, alpha]
        Note: SpotOptim handles log transformations and factor mapping automatically

    Returns:
        Array of validation errors
    """
    results = []

    for i, params in enumerate(X):
        # Extract parameters (already in original scale thanks to var_trans)
        # Factor variables (activation, optimizer) are returned as strings
        l1 = int(params[0])                # Number of neurons
        num_layers = int(params[1])        # Number of hidden layers
        activation = params[2]              # Activation function
        optimizer_name = params[3]         # Optimizer algorithm
        lr_unified = params[4]             # Learning rate
        alpha = params[5]                  # Physics weight

        print(f"\nConfiguration {i+1}/{len(X)}:")
        print(f"  l1={l1}, num_layers={num_layers}, activation={activation}, ")
        print(f"  optimizer={optimizer_name}, lr_unified={lr_unified:.4f}, alpha={alpha:.4f}")
```

## 67. Hyperparameter Optimization with SpotOptim

```
# Train PINN with these hyperparameters
val_error = train_pinn(
    l1=l1,
    num_layers=num_layers,
    activation=activation,
    optimizer_name=optimizer_name,
    lr_unified=lr_unified,
    alpha=alpha,
    n_epochs=N_EPOCHS,
    verbose=False
)

print(f" Validation MSE: {val_error:.6f}")
results.append(val_error)

return np.array(results)

# Test the objective function
print("Testing objective function with 2 configurations...")
X_test = np.array([
    [32, 2, "Tanh", "Adam", 3.0, 0.06], # Baseline config
    [64, 3, "ReLU", "AdamW", 2.0, 0.04] # Alternative config
], dtype=object)
test_results = objective_pinn(X_test)
print(f"\nTest results: {test_results}")
```

Testing objective function with 2 configurations...

Configuration 1/2:

l1=32, num\_layers=2, activation=Tanh,  
optimizer=Adam, lr\_unified=3.0000, alpha=0.0600  
Validation MSE: 0.130621

Configuration 2/2:

l1=64, num\_layers=3, activation=ReLU,  
optimizer=AdamW, lr\_unified=2.0000, alpha=0.0400  
Validation MSE: 0.127441

Test results: [0.13062107 0.12744086]

## 67.2. Run the Optimization

Use `tensorboard --logdir=runs` from a shell in the current directory (where this notebook is located) to visualize the optimization process.

```
# Define search space with var_trans for automatic log-scale handling
bounds = [
    (16, 128),          # l1: neurons per layer (16 to 128)
    (1, 4),             # num_layers: 1 to 4 hidden layers
    ("Tanh", "ReLU", "Sigmoid", "GELU"), # activation: activation function
    ("Adam", "SGD", "RMSprop", "AdamW"), # optimizer: optimizer algorithm
    (0.1, 10.0),        # lr_unified: learning rate (0.1 to 10)
    (0.01, 1.0)         # alpha: physics weight (0.01 to 1.0)
]

var_type = ["int", "int", "factor", "factor", "float", "float"]
var_name = ["l1", "num_layers", "activation", "optimizer", "lr_unified", "alpha"]

# Use var_trans to handle log-scale transformations automatically
# Factor variables don't need transformations (None)
var_trans = [None, None, None, None, "log10", "log10"]

# Create optimizer
optimizer = SpotOptim(
    fun=objective_pinn,
    bounds=bounds,
    var_type=var_type,
    var_name=var_name,
    var_trans=var_trans, # Automatic log-scale handling!
    max_iter=MAX_ITER,
    n_initial=10,
    seed=42,
    verbose=True,
    tensorboard_clean=True,
    tensorboard_log=True
)
```

```
Factor variable at dimension 2:
  Levels: ['Tanh', 'ReLU', 'Sigmoid', 'GELU']
  Mapped to integers: 0 to 3
Factor variable at dimension 3:
  Levels: ['Adam', 'SGD', 'RMSprop', 'AdamW']
  Mapped to integers: 0 to 3
Removed old TensorBoard logs: runs/spotoptim_20251213_001909
```

## 67. Hyperparameter Optimization with SpotOptim

Cleaned 1 old TensorBoard log directory  
TensorBoard logging enabled: runs/spotoptim\_20251213\_002640

Display search space configuration. The `trans` column shows applied transformations. `lr_unified` and `alpha` use `log10` transformation internally. This enables efficient exploration of log-scale parameters. All values shown are in original scale (not transformed).

```
# Display search space configuration
design_table = optimizer.print_design_table(tablefmt="github")
print(design_table)
```

| name       | type   | lower | upper | default | trans |
|------------|--------|-------|-------|---------|-------|
| l1         | int    | 16.0  | 128.0 | 72      | -     |
| num_layers | int    | 1.0   | 4.0   | 2       | -     |
| activation | factor | Tanh  | GELU  | Sigmoid | -     |
| optimizer  | factor | Adam  | AdamW | RMSprop | -     |
| lr_unified | float  | 0.1   | 10.0  | 5.05    | log10 |
| alpha      | float  | 0.01  | 1.0   | 0.505   | log10 |

Run optimization

```
result = optimizer.optimize()
```

Configuration 1/10:  
l1=19, num\_layers=3, activation=ReLU,  
optimizer=SGD, lr\_unified=9.5756, alpha=0.1603  
Validation MSE: 0.194649

Configuration 2/10:  
l1=30, num\_layers=3, activation=ReLU,  
optimizer=Adam, lr\_unified=0.8430, alpha=0.0412  
Validation MSE: 0.175142

Configuration 3/10:  
l1=110, num\_layers=4, activation=ReLU,  
optimizer=AdamW, lr\_unified=1.9457, alpha=0.3866  
Validation MSE: 0.125290

Configuration 4/10:  
l1=74, num\_layers=1, activation=Sigmoid,

## 67.2. Run the Optimization

```
optimizer=RMSprop, lr_unified=0.1014, alpha=0.1050  
Validation MSE: 0.188408
```

Configuration 5/10:

```
l1=41, num_layers=3, activation=Sigmoid,  
optimizer=RMSprop, lr_unified=0.5877, alpha=0.7301  
Validation MSE: 0.095178
```

Configuration 6/10:

```
l1=52, num_layers=1, activation=Sigmoid,  
optimizer=RMSprop, lr_unified=0.2023, alpha=0.0145  
Validation MSE: 0.184968
```

Configuration 7/10:

```
l1=71, num_layers=2, activation=Sigmoid,  
optimizer=SGD, lr_unified=3.2551, alpha=0.4300  
Validation MSE: 0.199493
```

Configuration 8/10:

```
l1=120, num_layers=3, activation=ReLU,  
optimizer=AdamW, lr_unified=0.3330, alpha=0.0220  
Validation MSE: 0.144155
```

Configuration 9/10:

```
l1=98, num_layers=2, activation=Tanh,  
optimizer=SGD, lr_unified=4.3915, alpha=0.0293  
Validation MSE: 0.182948
```

Configuration 10/10:

```
l1=87, num_layers=2, activation=GELU,  
optimizer=SGD, lr_unified=1.4861, alpha=0.0949  
Validation MSE: nan
```

Warning: 1 initial design point(s) returned NaN/inf and will be ignored (reduced from 10 to 9 points)

Note: Initial design size (9) is smaller than requested (10) due to NaN/inf values

Initial best:  $f(x) = 0.095178$

Configuration 1/1:

```
l1=41, num_layers=3, activation=Sigmoid,  
optimizer=RMSprop, lr_unified=0.5938, alpha=0.7968  
Validation MSE: 0.094611
```

Iteration 1: New best  $f(x) = 0.094611$

Configuration 1/1:

```
l1=41, num_layers=3, activation=Sigmoid,  
optimizer=RMSprop, lr_unified=0.9299, alpha=1.0000
```

## 67. Hyperparameter Optimization with SpotOptim

Validation MSE: 0.130813  
Iteration 2:  $f(x) = 0.130813$

Configuration 1/1:  
l1=110, num\_layers=4, activation=ReLU,  
optimizer=AdamW, lr\_unified=1.9457, alpha=0.3866  
Validation MSE: 0.105465  
Iteration 3:  $f(x) = 0.105465$

Configuration 1/1:  
l1=52, num\_layers=3, activation=ReLU,  
optimizer=RMSprop, lr\_unified=0.1454, alpha=0.0135  
Validation MSE: 0.159544  
Iteration 4:  $f(x) = 0.159544$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:  
l1=115, num\_layers=1, activation=Tanh,  
optimizer=SGD, lr\_unified=2.9394, alpha=0.0115  
Validation MSE: nan  
Warning: Found 1 NaN/inf value(s), replacing with adaptive penalty (max + 3\*std = 0.3)  
Iteration 5:  $f(x) = 0.365183$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:  
l1=41, num\_layers=2, activation=Sigmoid,  
optimizer=RMSprop, lr\_unified=6.4153, alpha=0.0157  
Validation MSE: 0.190298  
Iteration 6:  $f(x) = 0.190298$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:  
l1=77, num\_layers=3, activation=Sigmoid,  
optimizer=SGD, lr\_unified=4.4305, alpha=0.0194  
Validation MSE: 0.202042  
Iteration 7:  $f(x) = 0.202042$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:  
l1=43, num\_layers=2, activation=Sigmoid,  
optimizer=SGD, lr\_unified=0.6789, alpha=0.0502



## 67.2. Run the Optimization

Validation MSE: 0.203887  
Iteration 8:  $f(x) = 0.203887$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:  
l1=119, num\_layers=3, activation=GELU,  
optimizer=RMSprop, lr\_unified=2.1892, alpha=0.5140  
Validation MSE: 0.228413  
Iteration 9:  $f(x) = 0.228413$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:  
l1=116, num\_layers=2, activation=Sigmoid,  
optimizer=Adam, lr\_unified=0.6885, alpha=0.2024  
Validation MSE: 0.180719  
Iteration 10:  $f(x) = 0.180719$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:  
l1=62, num\_layers=4, activation=Tanh,  
optimizer=RMSprop, lr\_unified=0.2720, alpha=0.6832  
Validation MSE: 0.087061  
Iteration 11: New best  $f(x) = 0.087061$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:  
l1=73, num\_layers=3, activation=Tanh,  
optimizer=SGD, lr\_unified=1.1299, alpha=0.2924  
Validation MSE: 0.186883  
Iteration 12:  $f(x) = 0.186883$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:  
l1=91, num\_layers=2, activation=Sigmoid,  
optimizer=AdamW, lr\_unified=0.2225, alpha=0.0161  
Validation MSE: 0.194576  
Iteration 13:  $f(x) = 0.194576$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

## 67. Hyperparameter Optimization with SpotOptim

Configuration 1/1:

l1=112, num\_layers=2, activation=GELU,  
optimizer=SGD, lr\_unified=2.7386, alpha=0.0480  
Validation MSE: nan

Warning: Found 1 NaN/inf value(s), replacing with adaptive penalty (max + 3\*std = 0.5)

Iteration 14: f(x) = 0.529060

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=47, num\_layers=2, activation=GELU,  
optimizer=Adam, lr\_unified=3.4640, alpha=0.8417  
Validation MSE: 0.128958

Iteration 15: f(x) = 0.128958

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=66, num\_layers=3, activation=ReLU,  
optimizer=SGD, lr\_unified=2.3213, alpha=0.0124  
Validation MSE: nan

Warning: Found 1 NaN/inf value(s), replacing with adaptive penalty (max + 3\*std = 0.8)

Iteration 16: f(x) = 0.872169

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=95, num\_layers=2, activation=Sigmoid,  
optimizer=Adam, lr\_unified=4.6858, alpha=0.8132  
Validation MSE: 0.000360

Iteration 17: New best f(x) = 0.000360

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=79, num\_layers=1, activation=Tanh,  
optimizer=SGD, lr\_unified=0.1654, alpha=0.0163  
Validation MSE: 0.195575

Iteration 18: f(x) = 0.195575

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=70, num\_layers=3, activation=ReLU,  
optimizer=SGD, lr\_unified=1.7893, alpha=0.6472

## 67.2. Run the Optimization

```
Validation MSE: nan
Warning: Found 1 NaN/inf value(s), replacing with adaptive penalty (max + 3*std = 1.3620)
Iteration 19: f(x) = 1.514328
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:
  l1=44, num_layers=3, activation=Tanh,
  optimizer=RMSprop, lr_unified=5.6818, alpha=0.0218
  Validation MSE: 6.802989
Iteration 20: f(x) = 6.802989
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:
  l1=111, num_layers=3, activation=ReLU,
  optimizer=Adam, lr_unified=4.0470, alpha=0.2395
  Validation MSE: 0.127627
Iteration 21: f(x) = 0.127627
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:
  l1=41, num_layers=1, activation=ReLU,
  optimizer=AdamW, lr_unified=9.3784, alpha=0.3473
  Validation MSE: 0.171482
Iteration 22: f(x) = 0.171482
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:
  l1=113, num_layers=2, activation=GELU,
  optimizer=SGD, lr_unified=0.4088, alpha=0.0688
  Validation MSE: 0.181295
Iteration 23: f(x) = 0.181295
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:
  l1=106, num_layers=2, activation=ReLU,
  optimizer=SGD, lr_unified=5.4687, alpha=0.5656
  Validation MSE: nan
Warning: Found 1 NaN/inf value(s), replacing with adaptive penalty (max + 3*std = 10.3793)
Iteration 24: f(x) = 10.355925
  Attempt 2/10: Previous point was duplicate after rounding, trying fallback
```

## 67. Hyperparameter Optimization with SpotOptim

Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=24, num\_layers=3, activation=Sigmoid,  
optimizer=RMSprop, lr\_unified=0.4724, alpha=0.0123  
Validation MSE: 0.143761

Iteration 25:  $f(x) = 0.143761$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=96, num\_layers=1, activation=GELU,  
optimizer=SGD, lr\_unified=0.5396, alpha=0.6141  
Validation MSE: 0.189695

Iteration 26:  $f(x) = 0.189695$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=112, num\_layers=3, activation=Tanh,  
optimizer=SGD, lr\_unified=0.1361, alpha=0.0246  
Validation MSE: 0.195050

Iteration 27:  $f(x) = 0.195050$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=76, num\_layers=2, activation=GELU,  
optimizer=AdamW, lr\_unified=0.2194, alpha=0.0255  
Validation MSE: 0.166782

Iteration 28:  $f(x) = 0.166782$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=102, num\_layers=2, activation=ReLU,  
optimizer=Adam, lr\_unified=0.6220, alpha=0.1496  
Validation MSE: 0.178802

Iteration 29:  $f(x) = 0.178802$

Attempt 2/10: Previous point was duplicate after rounding, trying fallback

Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=86, num\_layers=3, activation=ReLU,  
optimizer=Adam, lr\_unified=1.2365, alpha=0.3194

## 67.2. Run the Optimization

Validation MSE: 0.127320  
Iteration 30:  $f(x) = 0.127320$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:  
l1=101, num\_layers=2, activation=ReLU,  
optimizer=AdamW, lr\_unified=7.3601, alpha=0.0647  
Validation MSE: 0.130081  
Iteration 31:  $f(x) = 0.130081$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:  
l1=112, num\_layers=2, activation=Sigmoid,  
optimizer=AdamW, lr\_unified=0.1439, alpha=0.4666  
Validation MSE: 0.192847  
Iteration 32:  $f(x) = 0.192847$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:  
l1=96, num\_layers=4, activation=GELU,  
optimizer=AdamW, lr\_unified=7.7483, alpha=0.4474  
Validation MSE: 0.112189  
Iteration 33:  $f(x) = 0.112189$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:  
l1=122, num\_layers=2, activation=ReLU,  
optimizer=RMSprop, lr\_unified=2.3122, alpha=0.0979  
Validation MSE: 0.229986  
Iteration 34:  $f(x) = 0.229986$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:  
l1=30, num\_layers=1, activation=GELU,  
optimizer=RMSprop, lr\_unified=3.3613, alpha=0.3171  
Validation MSE: 0.187535  
Iteration 35:  $f(x) = 0.187535$   
Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

## 67. Hyperparameter Optimization with SpotOptim

Configuration 1/1:

l1=64, num\_layers=3, activation=GELU,  
optimizer=RMSprop, lr\_unified=0.8964, alpha=0.6261  
Validation MSE: 0.195397

Iteration 36:  $f(x)$  = 0.195397

Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=35, num\_layers=4, activation=Tanh,  
optimizer=AdamW, lr\_unified=0.2055, alpha=0.0157  
Validation MSE: 0.183670

Iteration 37:  $f(x)$  = 0.183670

Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=18, num\_layers=2, activation=ReLU,  
optimizer=SGD, lr\_unified=0.2767, alpha=0.5381  
Validation MSE: 0.203944

Iteration 38:  $f(x)$  = 0.203944

Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=68, num\_layers=2, activation=Tanh,  
optimizer=RMSprop, lr\_unified=1.6975, alpha=0.0526  
Validation MSE: 2.660376

Iteration 39:  $f(x)$  = 2.660376

Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=98, num\_layers=4, activation=Sigmoid,  
optimizer=RMSprop, lr\_unified=3.4266, alpha=0.1841  
Validation MSE: 0.874758

Iteration 40:  $f(x)$  = 0.874758

Attempt 2/10: Previous point was duplicate after rounding, trying fallback  
Acquisition failure: Using random space-filling design as fallback.

Configuration 1/1:

l1=87, num\_layers=3, activation=Sigmoid,  
optimizer=SGD, lr\_unified=2.5483, alpha=0.0127  
Validation MSE: 0.202666

Iteration 41:  $f(x)$  = 0.202666

### *67.2. Run the Optimization*

TensorBoard writer closed. View logs with: `tensorboard --logdir=runs/spotoptim_20251213_002640`





## 68. Results Analysis

### 68.1. Best Configuration

Display best hyperparameters using `print_best()` method. With `var_trans`, results are already in original scale!

```
optimizer.print_best(result)
```

Best Solution Found:

```
-----  
l1: 95  
num_layers: 2  
activation: Sigmoid  
optimizer: Adam  
lr_unified: 4.6858  
alpha: 0.8132  
Objective Value: 0.0004  
Total Evaluations: 50
```

Store values for later use in visualizations. Values are already in original scale thanks to `var_trans`. Factor variables are returned as strings.

```
best_l1 = int(result.x[0])  
best_num_layers = int(result.x[1])  
best_activation = result.x[2]  
best_optimizer = result.x[3]  
best_lr_unified = result.x[4]  
best_alpha = result.x[5]  
best_val_error = result.fun  
  
print(f"Best activation: {best_activation}")  
print(f"Best optimizer: {best_optimizer}")
```

```
Best activation: Sigmoid  
Best optimizer: Adam
```

## 68. Results Analysis

### 68.1.1. Results Table with Importance Scores

Display comprehensive results table with importance scores

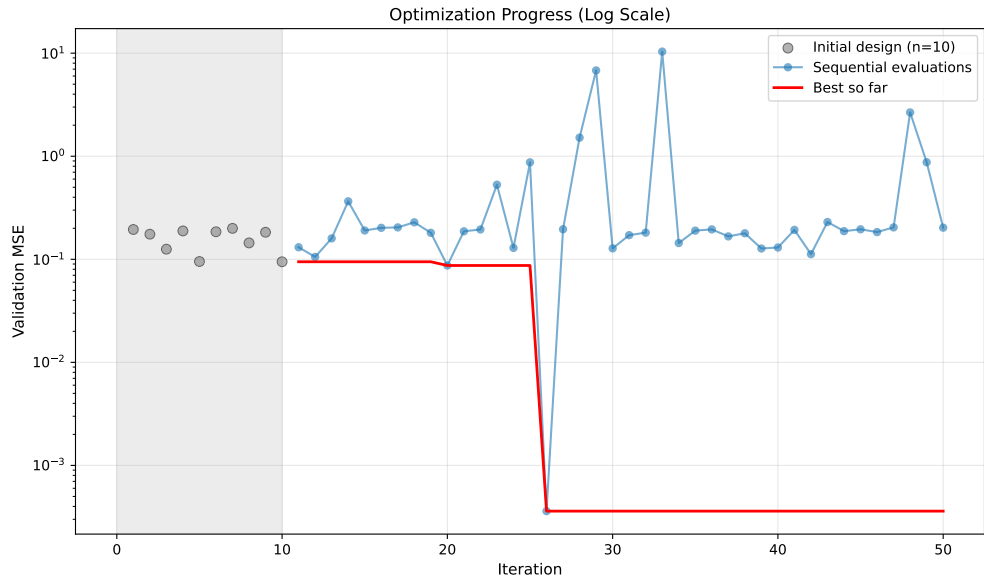
```
table = optimizer.print_results_table(show_importance=True, tablefmt="github")
print(table)
```

| name       | type   | lower | upper | tuned              | trans | importance |
|------------|--------|-------|-------|--------------------|-------|------------|
| l1         | int    | 16.0  | 128.0 | 95                 | -     | 5.7        |
| num_layers | int    | 1.0   | 4.0   | 2                  | -     | 3.4        |
| activation | factor | Tanh  | GELU  | Sigmoid            | -     | 38.1       |
| optimizer  | factor | Adam  | AdamW | Adam               | -     | 6.7        |
| lr_unified | float  | 0.1   | 10.0  | 4.685806654938922  | log10 | 41.2       |
| alpha      | float  | 0.01  | 1.0   | 0.8131651565643412 | log10 | 4.6        |

Interpretation: \*\*\*: >95%, \*\*: >50%, \*: >1%, .: >0.1%

### 68.2. Optimization History

```
optimizer.plot_progress(log_y=True, ylabel="Validation MSE")
```



## 68.3. Surrogate Visualization

Visualize the surrogate model's learned response surface for the most important hyperparameter combinations:

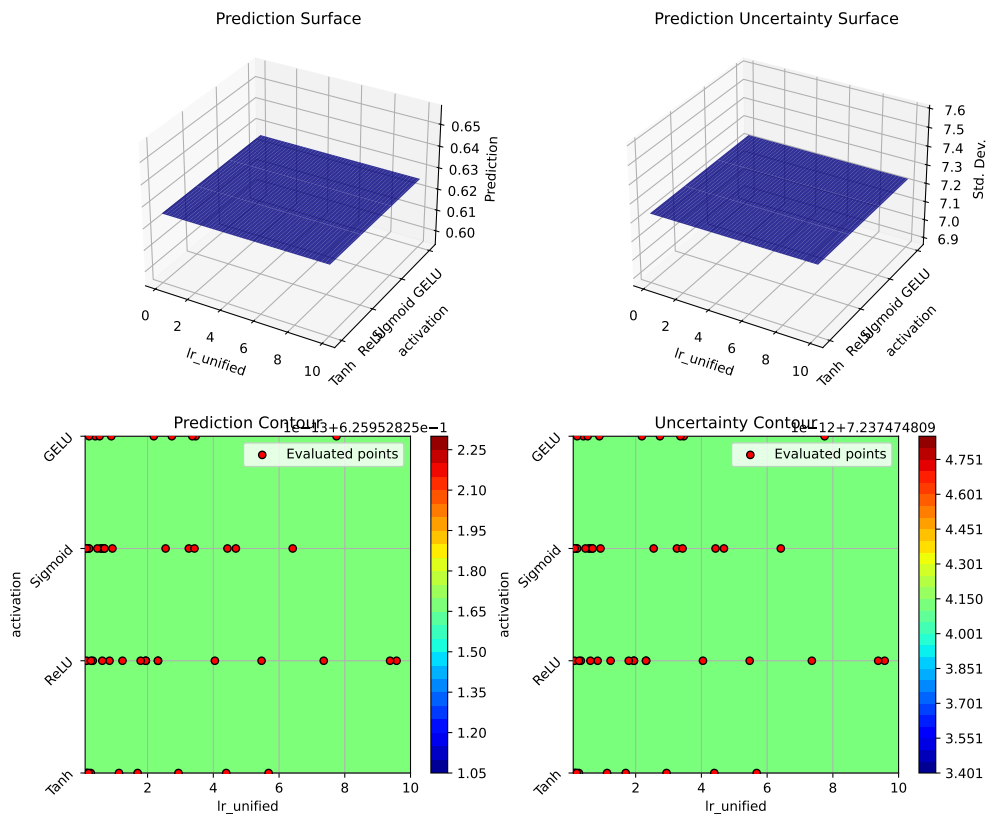
```
# Plot top 3 most important hyperparameter combinations
optimizer.plot_important_hyperparameter_contour(max_imp=3)
```

Plotting surrogate contours for top 3 most important parameters:

```
lr_unified: importance = 41.27% (type: float)
activation: importance = 38.13% (type: factor)
optimizer: importance = 6.76% (type: factor)
```

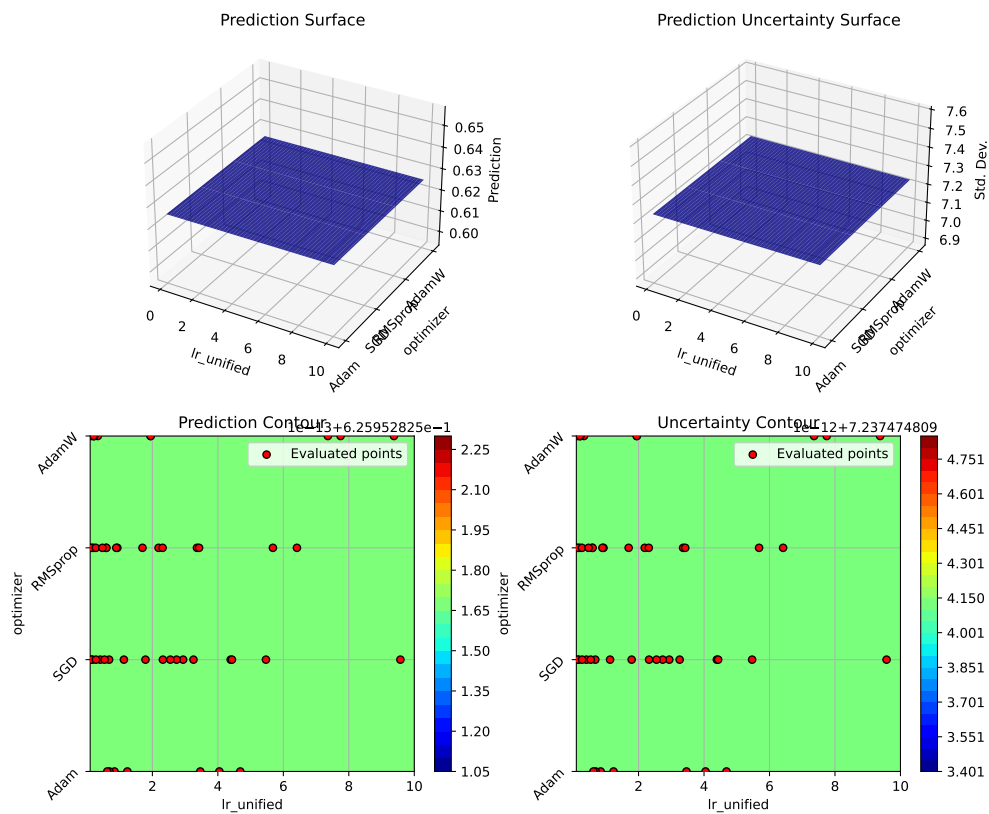
Generating 3 surrogate plots...

Plotting lr\_unified vs activation



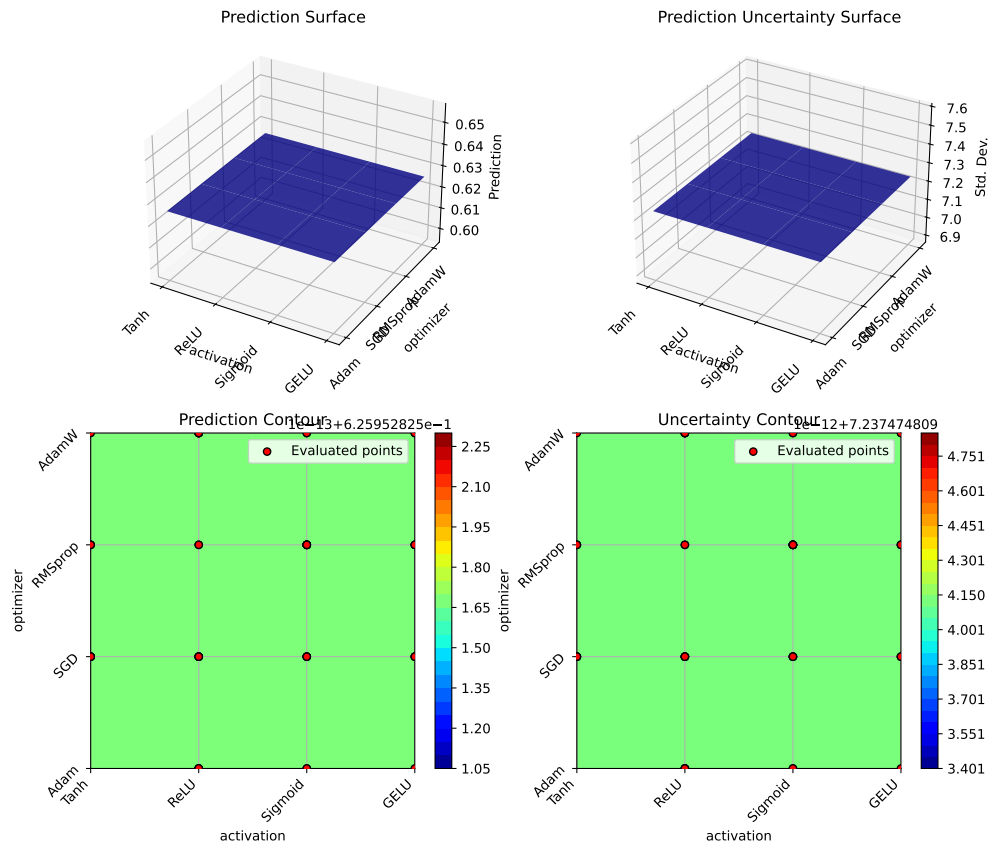
Plotting lr\_unified vs optimizer

68. Results Analysis



Plotting activation vs optimizer

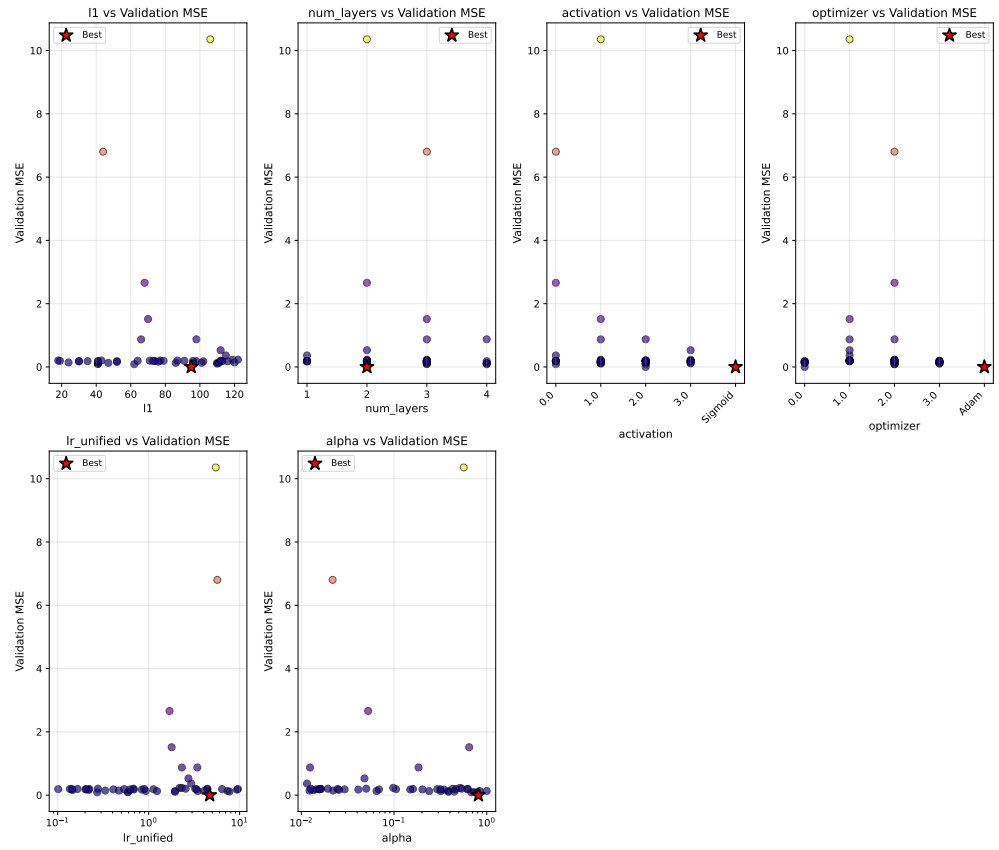
## 68.4. Parameter Distribution Analysis



## 68.4. Parameter Distribution Analysis

```
optimizer.plot_parameter_scatter(
    result,
    ylabel="Validation MSE",
    cmap="plasma",
    figsize=(14, 12)
)
```

## 68. Results Analysis



## 69. Train Final Model with Best Hyperparameters

Now let's train a final model with the optimized hyperparameters using DataLoaders:

```
print("Training final model with best hyperparameters using DataLoaders...")
print(f"Training for 30,000 epochs...")

# Set seed for reproducibility
torch.manual_seed(42)

# Create DataLoaders for final training
final_batch_size = 16
train_loader_final = DataLoader(
    train_dataset,
    batch_size=final_batch_size,
    shuffle=True,
    num_workers=0
)

collocation_loader_final = DataLoader(
    collocation_dataset,
    batch_size=len(collocation_dataset),
    shuffle=False,
    num_workers=0
)

# Create model with best hyperparameters
final_model = LinearRegressor(
    input_dim=1,
    output_dim=1,
    l1=best_l1,
    num_hidden_layers=best_num_layers,
    activation=best_activation,
    lr=best_lr_unified
)
```

## 69. Train Final Model with Best Hyperparameters

```
optimizer_final = final_model.get_optimizer(best_optimizer)

# Training with history tracking
loss_history = []
n_epochs_final = 30000

for epoch in range(n_epochs_final):
    final_model.train()
    epoch_loss = 0.0

    # Get collocation points
    t_physics_batch = next(iter(collocation_loader_final))
    t_physics_batch = t_physics_batch.requires_grad_(True)

    # Iterate over training batches
    for batch_t, batch_y in train_loader_final:
        optimizer_final.zero_grad()

        # Data Loss
        y_pred = final_model(batch_t)
        loss_data = torch.mean((y_pred - batch_y)**2)

        # Physics Loss
        y_physics = final_model(t_physics_batch)
        dy_dt = torch.autograd.grad(
            y_physics,
            t_physics_batch,
            torch.ones_like(y_physics),
            create_graph=True,
            retain_graph=True
        )[0]

        physics_residual = dy_dt + 0.1 * y_physics - torch.sin(np.pi * t_physics_batch)
        loss_physics = torch.mean(physics_residual**2)

        # Total Loss
        loss = loss_data + best_alpha * loss_physics
        loss.backward()
        optimizer_final.step()

        epoch_loss += loss.item()

    # Record average loss every 100 epochs
    if (epoch + 1) % 100 == 0:
```



```

        avg_loss = epoch_loss / len(train_loader_final)
        loss_history.append(avg_loss)

    if (epoch + 1) % 5000 == 0:
        avg_loss = epoch_loss / len(train_loader_final)
        print(f"   Epoch {epoch+1}/{n_epochs_final}: Avg Loss = {avg_loss:.6f}")

print("Training completed!")

# Plot training history
plt.figure(figsize=(10, 5))
plt.plot(loss_history, linewidth=1.5)
plt.xlabel('Iteration (×100)', fontsize=11)
plt.ylabel('Average Total Loss per Batch', fontsize=11)
plt.title('Final Model Training History', fontsize=12)
plt.grid(True, alpha=0.3)
plt.yscale('log')
plt.tight_layout()
plt.show()

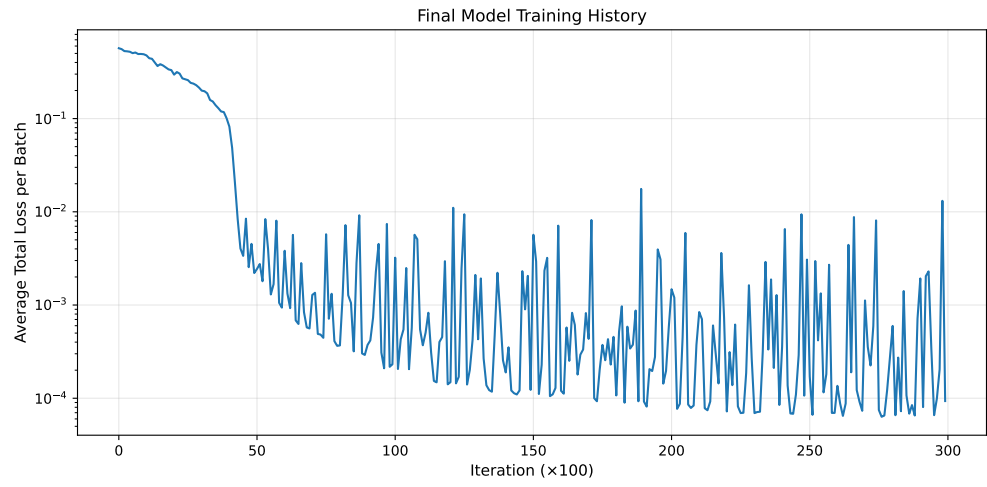
```

```

Training final model with best hyperparameters using DataLoaders...
Training for 30,000 epochs...
  Epoch 5000/30000: Avg Loss = 0.002204
  Epoch 10000/30000: Avg Loss = 0.000231
  Epoch 15000/30000: Avg Loss = 0.000123
  Epoch 20000/30000: Avg Loss = 0.000579
  Epoch 25000/30000: Avg Loss = 0.003072
  Epoch 30000/30000: Avg Loss = 0.000093
Training completed!

```

69. Train Final Model with Best Hyperparameters



## 70. Evaluate Final Model

```
# Create validation DataLoader for evaluation
val_loader_final = DataLoader(
    val_dataset,
    batch_size=len(val_dataset),
    shuffle=False,
    num_workers=0
)

# Evaluate on validation set
final_model.eval()
with torch.no_grad():
    # Validation MSE using DataLoader
    val_mse_total = 0.0
    for batch_t, batch_y in val_loader_final:
        y_pred = final_model(batch_t)
        val_mse_total += torch.mean((y_pred - batch_y)**2).item()
    final_val_mse = val_mse_total / len(val_loader_final)

    # Predict on full domain for visualization
    y_pred_full = final_model(x_exact)
    full_mse = torch.mean((y_pred_full - y_exact)**2).item()

    # Compute maximum absolute error
    max_error = torch.max(torch.abs(y_pred_full - y_exact)).item()

print("\nFinal Model Performance:")
print("-" * 50)
print(f"  Validation MSE: {final_val_mse:.6f}")
print(f"  Full domain MSE: {full_mse:.6f}")
print(f"  Maximum absolute error: {max_error:.6f}")
```

Final Model Performance:

-----  
Validation MSE: 0.000522

70. *Evaluate Final Model*

Full domain MSE: 0.000267  
Maximum absolute error: 0.083369

## 71. Visualize Final Solution

```
# Generate predictions
final_model.eval()
with torch.no_grad():
    y_pred = final_model(x_exact)

fig, axes = plt.subplots(2, 1, figsize=(12, 10))

# Plot 1: Solution comparison
ax1 = axes[0]
ax1.plot(x_exact.numpy(), y_exact.numpy(), 'b-', linewidth=2.5,
        label='Exact solution', alpha=0.8)
ax1.plot(x_exact.numpy(), y_pred.numpy(), 'r--', linewidth=2,
        label='PINN prediction (optimized)', alpha=0.8)

# Plot training data from dataset
ax1.scatter(train_dataset.t.numpy(), train_dataset.y.numpy(),
        color='tab:orange', s=80, label='Training data',
        zorder=5, edgecolors='black', linewidth=0.5)

# Plot collocation points
t_collocation = collocation_dataset.t.detach()
ax1.scatter(t_collocation.numpy(),
        final_model(t_collocation).detach().numpy(),
        color='green', marker='x', s=50,
        label='Collocation points', alpha=0.7, zorder=4)

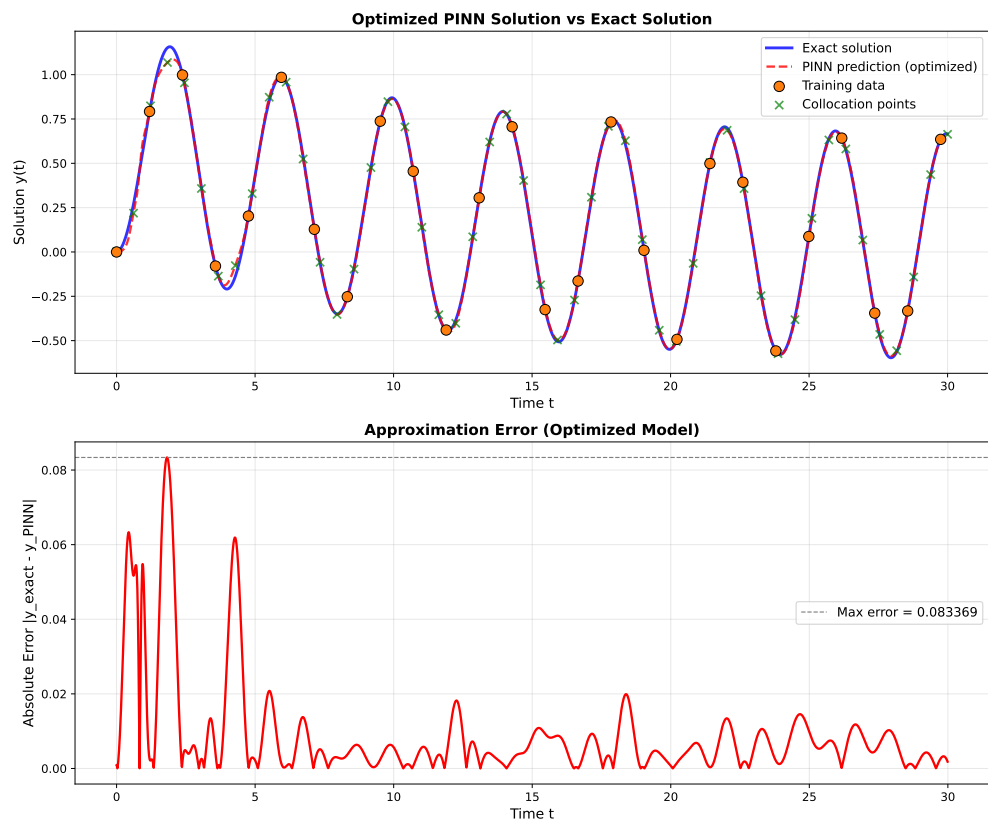
ax1.set_xlabel('Time t', fontsize=12)
ax1.set_ylabel('Solution y(t)', fontsize=12)
ax1.set_title('Optimized PINN Solution vs Exact Solution', fontsize=13, fontweight='bold')
ax1.legend(fontsize=11, loc='best')
ax1.grid(True, alpha=0.3)

# Plot 2: Error
ax2 = axes[1]
error = torch.abs(y_pred - y_exact)
```

## 71. Visualize Final Solution

```
ax2.plot(x_exact.numpy(), error.numpy(), 'r-', linewidth=2)
ax2.axhline(y=max_error, color='gray', linestyle='--', linewidth=1,
            label=f'Max error = {max_error:.6f}')
ax2.set_xlabel('Time t', fontsize=12)
ax2.set_ylabel('Absolute Error |y_exact - y_PINN|', fontsize=12)
ax2.set_title('Approximation Error (Optimized Model)', fontsize=13, fontweight='bold')
ax2.legend(fontsize=11)
ax2.grid(True, alpha=0.3)

plt.tight_layout()
plt.show()
```



## 72. Comparison with Baseline

Let's compare the optimized configuration with a baseline:

```
print("Training baseline model for comparison...")

# Baseline configuration (from basic PINN demo)
baseline_config = {
    'l1': 32,
    'num_layers': 3,
    'activation': 'Tanh',
    'optimizer': 'Adam',
    'lr_unified': 3.0,
    'alpha': 0.06
}

print(f"\nBaseline Configuration:")
for key, val in baseline_config.items():
    print(f"  {key}: {val}")

# Train baseline
baseline_error = train_pinn(
    l1=baseline_config['l1'],
    num_layers=baseline_config['num_layers'],
    activation=baseline_config['activation'],
    optimizer_name=baseline_config['optimizer'],
    lr_unified=baseline_config['lr_unified'],
    alpha=baseline_config['alpha'],
    n_epochs=N_EPOCHS,
    verbose=False
)

print(f"\nValidation MSE Comparison:")
print("-" * 50)
print(f"  Baseline: {baseline_error:.6f}")
print(f"  Optimized: {best_val_error:.6f}")
print(f"  Improvement: {(1 - best_val_error/baseline_error)*100:.1f}%")
```

## 72. Comparison with Baseline

```
# Bar plot comparison
fig, ax = plt.subplots(figsize=(8, 6))
configs = ['Baseline', 'Optimized']
errors = [baseline_error, best_val_error]
colors = ['tab:blue', 'tab:green']

bars = ax.bar(configs, errors, color=colors, alpha=0.7, edgecolor='black', linewidth=1)

# Add value labels on bars
for bar, error in zip(bars, errors):
    height = bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2., height,
            f'{error:.6f}',
            ha='center', va='bottom', fontsize=11, fontweight='bold')

ax.set_ylabel('Validation MSE', fontsize=12)
ax.set_title('Model Comparison: Baseline vs Optimized', fontsize=13, fontweight='bold')
ax.grid(True, alpha=0.3, axis='y')
plt.tight_layout()
plt.show()
```

Training baseline model for comparison...

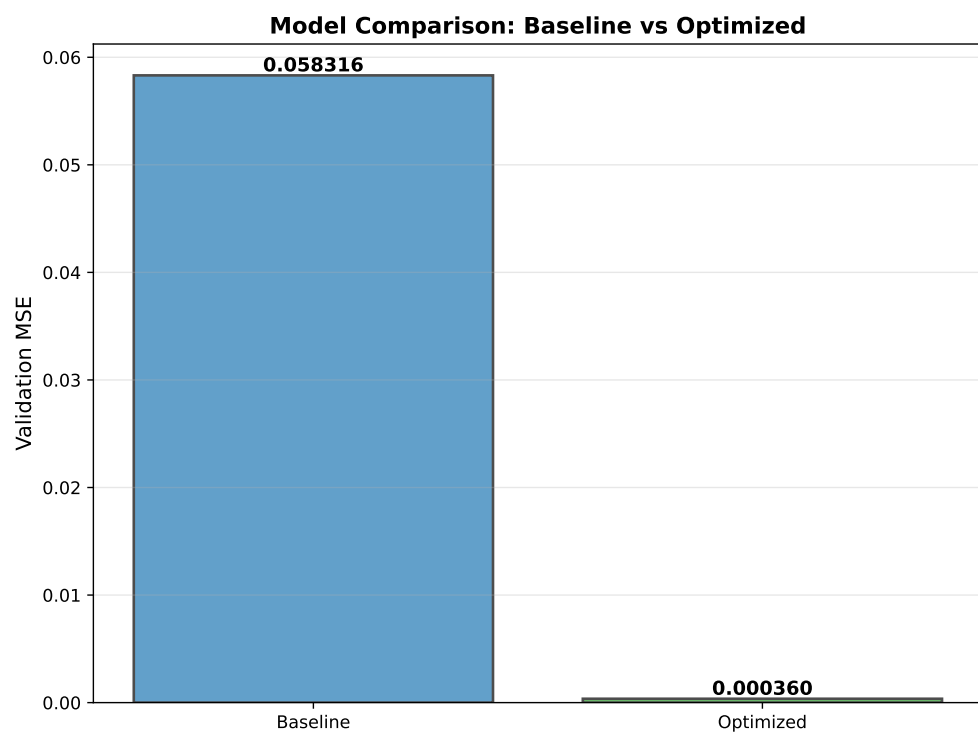
Baseline Configuration:

```
l1: 32
num_layers: 3
activation: Tanh
optimizer: Adam
lr_unified: 3.0
alpha: 0.06
```

Validation MSE Comparison:

```
-----
Baseline: 0.058316
Optimized: 0.000360
Improvement: 99.4%
```



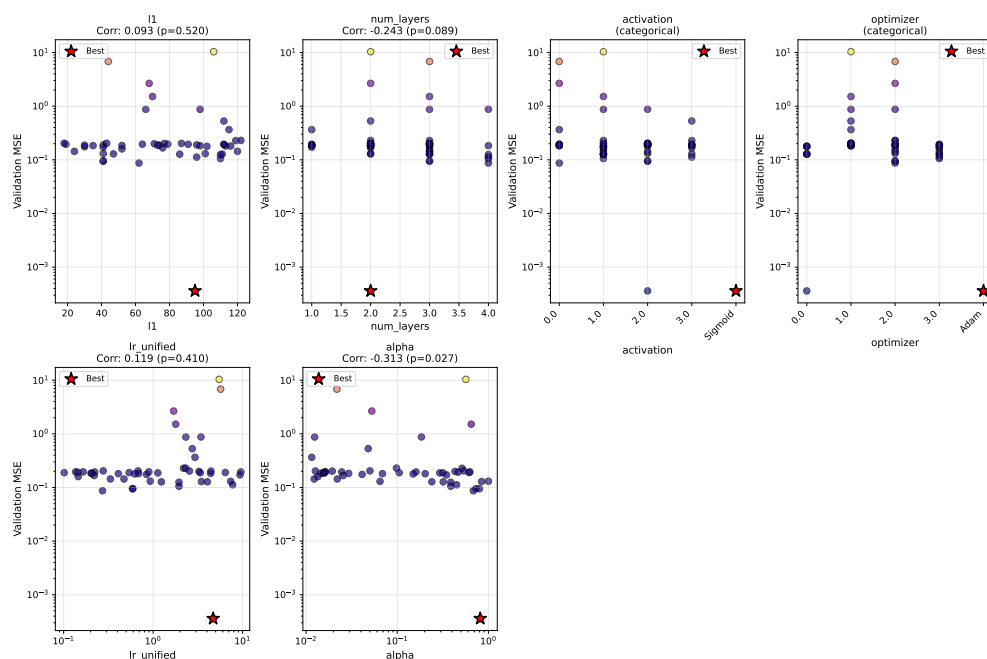




## 73. Hyperparameter Sensitivity Analysis

Let's analyze how sensitive the model is to each hyperparameter using the enhanced `plot_parameter_scatter()` method with Spearman correlation:

```
# Use the enhanced plot_parameter_scatter method with correlation display
optimizer.plot_parameter_scatter(
    result,
    ylabel="Validation MSE",
    cmap="plasma",
    figsize=(15, 10),
    show_correlation=True,
    log_y=True
)
```



### 73.1. Sensitivity Analysis (Spearman Correlation)

```
# Use the new sensitivity_spearman() method for tabular output
optimizer.sensitivity_spearman()
```

Sensitivity Analysis (Spearman Correlation):

```
-----
l1                : +0.093 (p=0.520)
num_layers        : -0.243 (p=0.089)
activation         : (categorical variable, use visual inspection)
optimizer         : (categorical variable, use visual inspection)
lr_unified         : +0.119 (p=0.410)
alpha             : -0.313 (p=0.027) *
```

## 74. Summary

### 74.1. Key Findings

```
# Get optimization history for statistics
history = optimizer.y_

print("\n" + "="*70)
print("HYPERPARAMETER OPTIMIZATION SUMMARY")
print("="*70)

print("\n1. BEST CONFIGURATION FOUND:")
print(f"  - Neurons per layer (l1): {best_l1}")
print(f"  - Number of hidden layers: {best_num_layers}")
print(f"  - Activation function: {best_activation}")
print(f"  - Optimizer: {best_optimizer}")
print(f"  - Learning rate: {best_lr_unified:.4f}")
print(f"  - Physics weight (alpha): {best_alpha:.4f}")

print("\n2. PERFORMANCE:")
print(f"  - Validation MSE: {best_val_error:.6f}")
print(f"  - Full domain MSE: {full_mse:.6f}")
print(f"  - Maximum absolute error: {max_error:.6f}")

print("\n3. OPTIMIZATION STATISTICS:")
print(f"  - Total evaluations: {result.nfev}")
print(f"  - Initial best: {history[0]:.6f}")
print(f"  - Final best: {best_val_error:.6f}")
print(f"  - Improvement: {(1 - best_val_error/history[0])*100:.1f}%")

print("\n4. COMPARISON TO BASELINE:")
print(f"  - Baseline MSE: {baseline_error:.6f}")
print(f"  - Optimized MSE: {best_val_error:.6f}")
print(f"  - Improvement: {(1 - best_val_error/baseline_error)*100:.1f}%")

print("\n" + "="*70)
```

## 74. Summary

```
=====
HYPERPARAMETER OPTIMIZATION SUMMARY
=====
```

1. BEST CONFIGURATION FOUND:
  - Neurons per layer (l1): 95
  - Number of hidden layers: 2
  - Activation function: Sigmoid
  - Optimizer: Adam
  - Learning rate: 4.6858
  - Physics weight (alpha): 0.8132
2. PERFORMANCE:
  - Validation MSE: 0.000360
  - Full domain MSE: 0.000267
  - Maximum absolute error: 0.083369
3. OPTIMIZATION STATISTICS:
  - Total evaluations: 50
  - Initial best: 0.194649
  - Final best: 0.000360
  - Improvement: 99.8%
4. COMPARISON TO BASELINE:
  - Baseline MSE: 0.058316
  - Optimized MSE: 0.000360
  - Improvement: 99.4%

```
=====
```

## 74.2. Recommendations

Based on the hyperparameter optimization results:

1. **Network Architecture:**
  - The optimal architecture was found with {best\_l1} neurons and {best\_num\_layers} hidden layers
  - Best activation function: {best\_activation}
  - This balances model capacity with training efficiency
2. **Optimizer Selection:**
  - Best optimizer: {best\_optimizer}

- Different optimizers have different convergence characteristics for PINNs

### 3. Learning Rate:

- Optimal unified learning rate: `{best_lr_unified:.4f}`
- This translates to an actual Adam learning rate of `{best_lr_unified * 0.001:.6f}`

### 4. Physics Loss Weight:

- Optimal alpha: `{best_alpha:.4f}`
- This balances data fitting with physics constraint satisfaction

### 5. Training Strategy:

- Start with a broad search space to explore different architectures
- Use `var_trans` with “log10” for learning rate and physics weight parameters
- This enables efficient exploration of log-scale parameters without manual transformations
- Validate on held-out data to prevent overfitting to training points

### 6. Benefits of `var_trans` and Factor Variables:

- **Factor variables:** Categorical choices (activation, optimizer) handled automatically
- SpotOptim maps strings to integers internally and back to strings in results
- **Cleaner code:** No manual `10**x` conversions in objective function
- **Fewer errors:** Eliminates confusion about which scale values are in
- **Better optimization:** Searches efficiently in transformed space
- **Easier interpretation:** All results displayed in original scale

## 74.3. Using These Results

To use the optimized configuration in your own PINN problems:

```
# Create optimized PINN
model = LinearRegressor(
    input_dim=1,
    output_dim=1,
    l1={best_l1},
    num_hidden_layers={best_num_layers},
    activation="{best_activation}",
    lr={best_lr_unified:.4f}
)

optimizer = model.get_optimizer("{best_optimizer}")
```

```
# Use alpha={best_alpha:.4f} for physics loss weight
loss = data_loss + {best_alpha:.4f} * physics_loss
```

## 74.4. Using var\_trans for Your Hyperparameter Optimization

When setting up optimization for your own PINN problems:

```
from spotoptim import SpotOptim

# Define search space with factor variables and log-scale parameters
bounds = [
    (16, 128),           # neurons (integer)
    (1, 4),             # layers (integer)
    ("Tanh", "ReLU", "Sigmoid", "GELU"), # activation (factor)
    ("Adam", "SGD", "RMSprop", "AdamW"), # optimizer (factor)
    (0.1, 10.0),        # learning rate (log-scale)
    (0.01, 1.0)         # physics weight (log-scale)
]

var_type = ["int", "int", "factor", "factor", "float", "float"]
var_trans = [None, None, None, None, "log10", "log10"]

opt = SpotOptim(
    fun=your_objective_function,
    bounds=bounds,
    var_type=var_type,
    var_trans=var_trans, # Automatic log-scale and factor handling!
    max_iter=MAX_ITER,
    n_initial=10
)

result = opt.optimize()
```

Your objective function receives parameters in **original scale** - no manual transformations needed!

## 74.5. Future Directions

Consider exploring:



1. **Adaptive physics weights** that change during training
2. **Architecture search** including skip connections or residual blocks
3. **Batch size optimization** as an additional hyperparameter
4. **Multi-objective optimization** balancing accuracy and computational cost
5. **Transfer learning** from pre-optimized configurations
6. **Learning rate schedules** with different decay strategies

---

**Note:** The specific optimal values depend on the problem, data distribution, and computational budget. Always validate results on held-out test data.



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